

PEF screening report of electricity from photovoltaic panels in the context of the EU Product Environmental Footprint Category Rules (PEFCR) Pilots

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Disclaimer

Within the Environmental Footprint (EF) pilot phase normalisation and equal weighting were foreseen to be used in the EF screenings to identify the most relevant impact categories. The use of normalisation and weighting for this purpose remains the objective for the EF pilots and beyond. However, currently PEF screening results after the normalisation and equal weighting present some inconsistencies stemming from errors at various levels of the assessment. Therefore, screening results after normalisation and equal weighting are not sufficiently robust to apply for product comparisons in an automatic and mandatory way in the Environmental Footprint (EF) pilots, e.g. to identify the most relevant impact categories. The interpretation of the results reflects these limitations.

To avoid potential misinterpretation and misuse of the EF screening results we highlight that the results after normalisation and equal weighting, - without further error checking and possibly corrections, - are likely to overestimate or underestimate especially the relevance of the potential impacts related to the categories Human toxicity - cancer effect, Human toxicity - non-cancer effect, Ecotoxicity for aquatic fresh water, water depletion, resource depletion, ionizing radiation and land use.

Table of Contents

1.	Introduction.....	25
2.	Goal of the study.....	25
3.	Scope of the screening	26
3.1	Introduction.....	26
3.2	Function, functional unit and reference flow	26
3.3	System boundaries and system boundary diagram	28
3.4	Assumptions and value judgments.....	31
3.5	Treatment of multi-functionality.....	31
3.6	Information about the data used and data gaps	32
3.7	Impact categories, models and indicators;	33
3.8	Normalization and weighting factors.....	36
4.	Compiling and recording the life cycle inventory analysis.....	38
4.1	Description and documentation of all unit process data	38
4.2	Cadmium-Telluride PV technology	39
4.3	Copper-Indium-Selenium (CIS) PV technology	47
4.4	Micromorphous-Si PV technology	54
4.5	Multicrystalline-Si PV technology.....	59
4.6	Monocrystalline-Si PV technology	66
4.7	Life cycle inventory of the electric installation and the mounting structure.....	72
4.8	Data collection procedures.....	73
4.9	Methodological assumptions used in the screening	74
4.10	Life cycle inventory results.....	78
5.	Calculating PEF impact assessment results	79
5.1	Introduction.....	79
5.2	Data and indicator results prior to normalization.....	80
5.3	Overall results normalized	93
5.4	Overall results weighted.....	93
5.5	Results CdTe PV technology	94
5.6	Results CIS PV technology.....	102
5.7	Results micromorphous-Si PV technology	110
5.8	Results multicrystalline-Si PV technology	118
5.9	Results monocrystalline-Si PV technology.....	125
5.10	Electricity production with average PV module	132
5.11	Sensitivity analyses	134

6.	Interpretation.....	139
6.1	Overview.....	139
6.2	Most relevant impact categories.....	139
6.3	Environmental hotspots.....	140
6.4	Most relevant life cycle stages.....	148
6.5	Most relevant processes.....	152
6.6	Most relevant elementary flows.....	166
6.7	Matching of the ILCD method with elementary flows.....	172
7.	Benchmark.....	172
8.	Summary and Conclusion.....	173
	Sources.....	176
	ANNEX I.....	181
A.1.	Data quality assessment.....	181
A.1.1.	Overview.....	181
A.1.2.	Data quality assessment of electricity production with CdTe PV panels.....	181
A.1.3.	Data quality assessment of electricity production with CIS PV panels.....	184
A.1.4.	Data quality assessment of electricity production with micromorphous-Si PV panels ..	187
A.1.5.	Data quality assessment of electricity production with multicrystalline-Si PV panels...	190
A.1.6.	Data quality assessment of electricity production with monocrystalline-Si PV panels ..	193
	ANNEX II.....	197
A.1.	Results CdTe PV technology.....	197
A.1.1.	Overview.....	197
A.1.2.	Results prior to normalization.....	197
A.1.3.	Results normalized.....	199
A.1.4.	Results weighted.....	200
A.2.	Results CIS PV technology.....	202
A.2.1.	Overview.....	202
A.2.2.	Results prior to normalization.....	202
A.2.3.	Results normalized.....	204
A.2.4.	Results weighted.....	205
A.3.	Results micromorphous-Si PV technology.....	207
A.3.1.	Overview.....	207
A.3.2.	Results prior to normalization.....	207
A.3.3.	Results normalized.....	209
A.3.4.	Results weighted.....	210
A.4.	Results multicrystalline-Si PV technology.....	212

A.4.1.	Overview	212
A.4.2.	Results prior to normalization.....	212
A.4.3.	Results normalized.....	214
A.4.4.	Results weighted	215
A.5.	Results monocrystalline-Si PV technology.....	217
A.5.1.	Overview	217
A.5.2.	Results prior to normalization.....	217
A.5.3.	Results normalized.....	219
A.5.4.	Results weighted	220
A.6.	Results prior to normalization of electricity production with average PV module.....	222
A.7.	Environmental impacts of waste material disposal	223

EXECUTIVE SUMMARY

This study encompasses the life cycle assessment of direct current electricity produced with five different photovoltaic (PV) modules (CdTe, CIS, micromorphous-Si, multicrystalline-Si, monocrystalline-Si) installed on three different mounting types (integrated in roof, mounted on roof and open ground) and two system scales (3 kWp residential scale and 570 kWp large scale). The PV technologies differ in material consumption and module efficiency but also in their environmental performance. The following life cycle stages are included in the product system of PV electricity: product stage, construction stage, operation stage and end-of-life stage. Potential benefits for recycling are granted and allocated 50:50 to the electricity production on the one hand and the products made of recycled material on the other.

This screening LCA report aims at comparing five different PV technologies in terms of their environmental performance and at identifying the most relevant impact categories, life cycle stages, processes and elementary flows as well as the environmental for each technology. Preliminary information about important life cycle impact categories are gained that are used for the definition of data quality requirements. This screening report serves as a basis for the assessment of representative technology specific products as well as a representative (virtual) product, composed of market-share weighted averages of the different technologies. This screening report intends to be a starting point for further research within the PEFCR rather than it tries to make statements about the product group environmental impacts as such. It is not the purpose of this report to be used in the context of comparison or for comparative assertions to be disclosed to the public. Furthermore, as the inverter is not part of the basic assessment, the results of this screening LCA shall not be compared to the environmental impacts of electricity produced with other technologies such as hard coal, nuclear, hydroelectric or wind power plants.

The unit of analysis is defined as 1 kWh (kilowatt hour) of DC (direct current) electricity generated by a PV module. The reference flow is the PV module, measured in kWp (kilowatt peak), the maximum power output of a module under standard conditions.

The main assumptions and value judgments are related to the annual yield, the service life of the PV system and the degradation of the panel. The annual yield of the photovoltaic system depends on the location, the orientation and the degradation. This dependence is very similar for all technologies analysed. In this screening study the average yield of an optimally oriented PV panel installed in Europe is chosen, namely 975 kWh/kWp (including degradation effects). The degradation rate (0.7 % per year) and the expected life time (30 years) are identical for all technologies, even if in reality these may vary depending on the technology applied. Further value judgments are involved in the treatment of multi-functionality and of recycling. Different allocation approaches for quantifying the potential benefits for recycling are assessed in a sensitivity analysis. The production of the inverter is not part of the base case, but its influence on the environmental impacts of PV electricity is shown in a sensitivity analysis.

The main limitations are given by limited data availability of direct, process-specific emissions, in particular regarding processes taking place in the Asian region, and scarce data regarding the mass and energy flows of the end-of-life stage due to yet too little volumes and experiences with industrialized panel takeback and treatment (except CdTe).

In the screening study, a minimum “fair” quality data rating is required for data contributing to at least 90 % of the impact estimated for each EF impact category, as assessed via a qualitative expert

judgment. All datasets used scored good quality or higher. A minimum “fair” quality was met and even excelled in all cases.

15 default environmental footprint impact category indicators are used. Three additional indicators are proposed to complement the environmental profile: renewable cumulative energy demand, non-renewable cumulative energy demand and nuclear waste (quantifying its radiotoxicity). These additional impact categories are not included in the normalized and weighted results.

The environmental performance of 1 kWh of DC electricity produced with the average PV panel mix in Europe is mainly influenced by the production of the panels with exception of human toxicity cancer effects, freshwater eutrophication and ecotoxicity as well as renewable cumulative energy demand (see Fig. S.1). The use phase is of no noteworthy importance except for the last-mentioned impact category. The end-of-life stage has impacts of 0 % to 5 % while potential benefits for recycling can add up to -17 %.

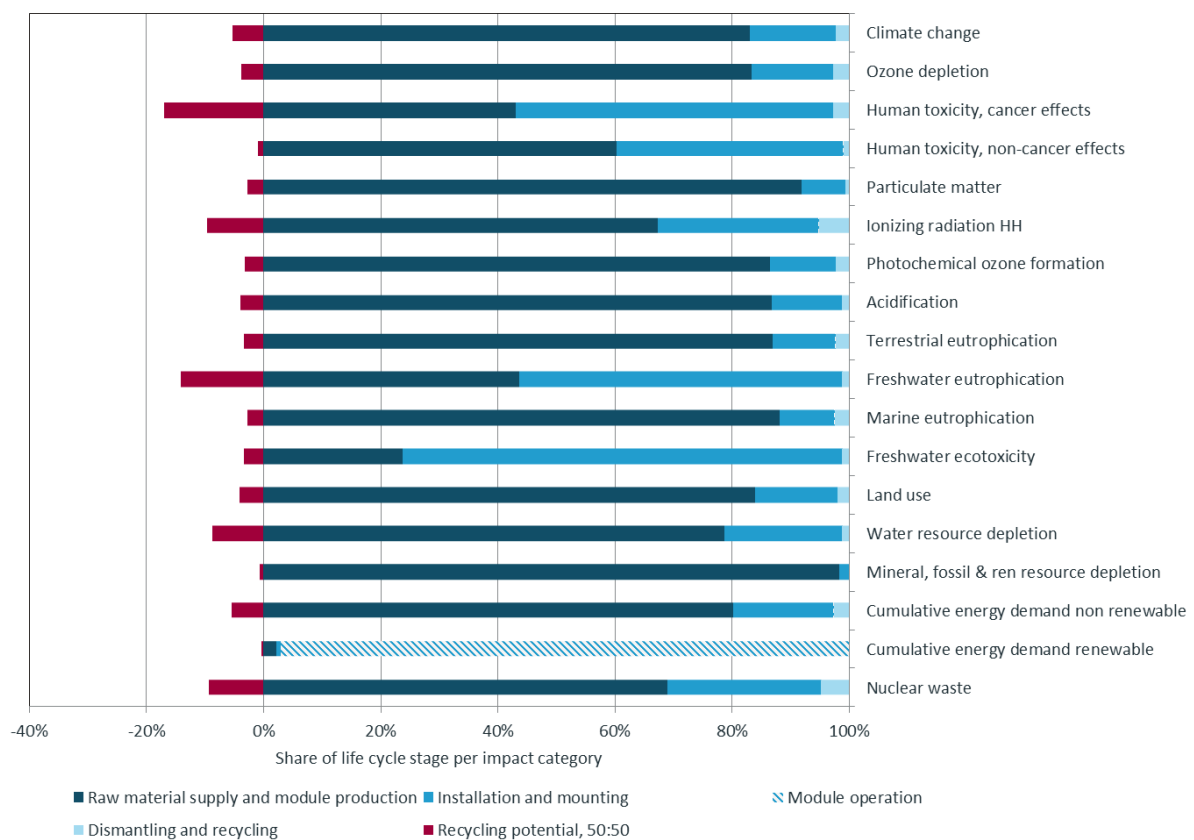


Fig. S.1 Environmental impact results (characterized, indexed to 100 %) of 1 kWh of DC electricity produced with a residential scale (3 kWp) PV system with average PV panels mounted on a slanted roof. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.

The production of 1 kWh DC electricity with an average residential scale PV system mounted on a slanted rooftop causes on average 64.3 grams of CO₂-eq and requires 0.784 MJ of non-renewable primary energy. The particulate matter emissions and the acidification impacts of 1 kWh DC electricity amount to 86.3 mg and 0.566 mmol H⁺-eq, respectively. 1 kWh of DC electricity produced with an average PV module requires 32.1 mg Sb-eq of abiotic resources and consumes 72.1 g water-eq of water. The human toxicity impacts are $1.09 \cdot 10^{-9}$ CTU_{h,c} (cancer effects) and $1.13 \cdot 10^{-8}$ CTU_{h,n-c} (non cancer effects) per kWh of PV electricity and the freshwater ecotoxicity impacts amount to $9.75 \cdot 10^{-2}$ CTU_e/kWh.

Depending on the PV technology the weighted environmental impacts vary and range from $8.37 \cdot 10^{-6}$ pt/kWh (CdTe PV modules integrated in roof) to $3.57 \cdot 10^{-4}$ pt/kWh (CIS PV panels on open ground system). Within each technology, the integrated systems cause the lowest impacts per kWh of electricity produced, curtly followed by the mounted systems. The open ground systems cause the highest environmental impact of the systems analysed. These differences are due to the land use, the mounting system and the electric installation. The ranking of the environmental performance of the different PV technologies for the mounted residential-scale systems is: the smallest impacts are caused by electricity production with CdTe PV modules, followed by micromorphous-Si panels, monocrystalline-Si panels, multicrystalline-Si panels and finally CIS PV panels.

Regarding the relative importance of the different life cycle stages, it becomes clear that the product stage and the construction stage (mounting and installation) cause the major part of the environmental impacts. The share of these life cycle stages varies between the different PV technologies, whereby the product stage is always more important than the construction stage when considering the weighted results of the residential scale PV systems mounted on a slanted roof. The most important impact categories are mineral, fossil and renewable resource depletion, human toxicity (cancer and non cancer effects), freshwater ecotoxicity, particulate matter potential and acidification potential.

With regard to the mineral, fossil and renewable resource depletion impacts, the supply chains of semiconductor materials (cadmium, tellurium, indium), silver (mainly used in the metallization paste for multicrystalline-Si and monocrystalline-Si PV modules), copper (mainly used in the electric installation) and zinc (used in various processes such as secondary aluminium production) are identified as process hotspots. Human toxicity cancer effects are caused by the disposal of redmud from bauxite digestion (supply chain of primary aluminium) and by the disposal of slag generated in the production of unalloyed electric steel. The substance hotspots in this impact category are chromium VI emitted to water and chromium emissions to air. Both of these elementary flows are primarily associated with the supply chain of steel production. The non cancer effects on human toxicity are mainly caused by the production of primary copper and zinc, by indium rich leaching residues as well as by the disposal of hard coal ash. The most relevant elementary flows are zinc and mercury emitted to air mainly in the process of unalloyed electric steel production. Another substance hotspot identified in the impact category human toxicity non cancer effects is the emission of arsenic to water. The processes causing the major part of these emissions are natural gas extraction and the beneficiation of iron ore. The most relevant processes and elementary flows in the impact category freshwater ecotoxicity are identical for the five PV technologies analysed, namely the waste incineration of plastic components of the electric installation and the disposal of redmud from bauxite digestion. Antimony emissions to water occur during the extraction of natural gas and chromiumVI is emitted to water primarily during the beneficiation of iron ore. The particulate matter impacts are particularly relevant for the silicon-based PV technologies and dominated by electricity production in Chinese hard coal power plants. The emissions of PM2.5 and sulfur dioxide to air mainly occur during aluminium electrolysis and flat glass production. The production of MG-silicon is another important process with regard to emissions of sulfur dioxide to air. The acidification potential is among the five most relevant impact categories for the CdTe, CIS and mono-Si PV technologies. In this impact category, the operation of transoceanic freight ships, the production of flat glass and the electricity generation in hard coal power plants are identified as process hotspots. The most relevant elementary flows are emissions of sulfur dioxide and nitrogen oxides to air.

Data quality was assessed according to the requirements of the European Commission. The minimum “fair” required data rating was achieved for all data used. In fact, all datasets used scored good quality or higher. The main information gaps occur with regard to the environmental efficiency of panel and

module production in Asia as well as with regard to the end-of-life treatment efforts and emissions of all types of panels except CdTe. These gaps are closed by assuming the same environmental efficiency of panel production (same process specific emissions) but adjusting the mix of electricity used in manufacturing to the average Chinese and Malaysian situation. Data about end-of-life treatment of CdTe PV modules is extrapolated to all other PV technologies. The estimated recycling efforts have turned out to be conservative compared to data provided by recyclers taking part in the PV CYCLE programme. Data about end-of-life treatment of CIS, micromorphous-Si, multicrystalline-Si and monocrystalline-Si PV panels would enhance the quality of further assessments.

Benchmarking could be achieved when, as a first approach, different technologies for electricity production are compared on basis of their full life cycle. To succeed in establishing a sustainable energy market, fossil and nuclear energy sources are to be replaced by renewable energy sources such as PV, wind, biomass or water. Hence as a starting point for benchmarking, these PV technologies could be compared with the electricity production from coal, natural gas and uranium.

List of figures

Fig. 3.1	Product system of electricity produced with photovoltaic modules; Processes of the foreground and background system are marked with blue and red color, respectively.....	29
Fig. 3.2	Excerpt of the mono-Si, multi-Si, a-Si / micro-Si, CdTe and CIS / CIGS supply chain; Processes of the foreground and background system are marked with blue and red color, respectively.....	30
Fig. 4.1	Schematic illustration about the modeled life cycle chain of CdTe PV panels.....	39
Fig. 4.2	Schematic illustration about the modeled life cycle chain of CIS PV panels.....	47
Fig. 4.3	Schematic illustration about the modeled life cycle chain of micro-Si PV panels.....	54
Fig. 4.4	Schematic illustration about the modeled life cycle chain of multicrystalline silicon PV panels.....	60
Fig. 4.5	Schematic illustration about the modeled life cycle chain of monocrystalline silicon PV panels.....	66
Fig. 5.1	Climate change results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels	83
Fig. 5.2	Ozone depletion potential results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels.....	83
Fig. 5.3	Human toxicity cancer effects results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels.....	84
Fig. 5.4	Human toxicity non cancer effects results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels	85
Fig. 5.5	Particulate matter results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels	85
Fig. 5.6	Ionizing radiation HH results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels	86
Fig. 5.7	Photochemical ozone formation potential results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels	87
Fig. 5.8	Acidification potential results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels	87

Fig. 5.9	Terrestrial eutrophication potential results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels	88
Fig. 5.10	Freshwater eutrophication potential results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels	88
Fig. 5.11	Marine eutrophication potential results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels.....	89
Fig. 5.12	Freshwater ecotoxicity potential results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels.....	90
Fig. 5.13	Land use results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof (blue scale on the x-axis), respectively, and produced with a 570 kWp open ground power system (orange scale on the x-axis) using CdTe, CIS, micro-Si, multi- and mono-Si PV panels.....	90
Fig. 5.14	Water resource depletion potential results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels	91
Fig. 5.15	Mineral, fossil and renewable resource depletion potential results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels	91
Fig. 5.16	Cumulative energy demand, renewable and non-renewable, results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels	92
Fig. 5.17	Nuclear waste results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels	93
Fig. 5.18	Environmental impact results (characterized, indexed to 100 %) of 1 kWh of DC electricity produced with a residential scale (3 kWp) PV system with CdTe PV panels mounted on a slanted roof. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.....	95
Fig. 5.19	Share per impact category on the weighted result for the production of 1 kWh of DC electricity produced with a residential scale CdTe PV system mounted on a slanted roof.....	101

Fig. 5.20	Environmental impact results (weighted) of 1 kWh of DC electricity produced with a residential scale (3 kWp) PV system with CdTe PV panels mounted on and integrated in a slanted roof as well as produced with a large scale open ground system. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.	102
Fig. 5.21	Environmental impact results (characterized, indexed to 100 %) of 1 kWh of DC electricity produced with a 3 kWp power system with CIS PV panels mounted on a slanted roof. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.	103
Fig. 5.22	Share per impact category on the weighted result for the production of 1 kWh of DC electricity produced with a residential scale CIS PV system mounted on a slanted roof.	109
Fig. 5.23	Environmental impact results (weighted) of 1 kWh of DC electricity produced with a residential scale (3 kWp) PV system with CIS PV panels mounted on and integrated in a slanted roof as well as produced with a large scale open ground system. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.	110
Fig. 5.24	Environmental impact results (characterized, indexed to 100 %) of 1 kWh of DC electricity produced with a residential scale micro-Si PV system mounted on a slanted roof. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.	111
Fig. 5.25	Share per impact category on the weighted result for the production of 1 kWh of DC electricity produced with a residential scale micro-Si PV system mounted on a slanted roof.	117
Fig. 5.26	Environmental impact results (weighted) of 1 kWh of DC electricity produced with a residential scale (3 kWp) PV system with micro-Si PV panels mounted on and integrated in a slanted roof as well as produced with a large scale open ground system. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.	118
Fig. 5.27	Environmental impact results (characterized, indexed to 100 %) of 1 kWh of DC electricity produced with a residential scale multi-Si PV system mounted on a slanted roof. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.	119
Fig. 5.28	Share per impact category on the weighted result for the production of 1 kWh of DC electricity produced with a residential scale multi-Si PV system mounted on a slanted roof.	124
Fig. 5.29	Environmental impact results (weighted) of 1 kWh of DC electricity produced with a residential scale (3 kWp) PV system with multi-Si PV panels mounted on and integrated in a slanted roof as well as produced with a large scale open ground system. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.	125
Fig. 5.30	Environmental impact results (characterized, indexed to 100 %) of 1 kWh of DC electricity produced with a residential scale mono-Si PV system mounted on a slanted roof. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.	126
Fig. 5.31	Share per impact category on the weighted result for the production of 1 kWh of DC electricity produced with a residential scale mono-Si PV system mounted on a slanted roof.	131

Fig. 5.32	Environmental impact results (weighted) of 1 kWh of DC electricity produced with a residential scale (3 kWp) PV system with mono-Si PV panels mounted on and integrated in a slanted roof as well as produced with a large scale open ground system. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.	132
Fig. 5.33	Environmental impact results (characterized, indexed to 100 %) of 1 kWh of DC electricity produced with a residential scale (3 kWp) PV system with average PV panels mounted on a slanted roof. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.....	133
Fig. 5.34	Environmental impact (indexed to 100 %) of 1 kWh of DC electricity production (all life cycle stages included) with the average PV module, mounted on a slanted residential roof in Europe.	134
Fig. 5.35	Relative results (characterized) of the sensitivity analyses of allocation of benefits for recycling at the end-of-life stage when producing 1 kWh DC electricity with a residential scale CdTe PV system mounted on a slanted roof top. The allocation 50:50 is the base case and its impacts are indexed to 100 %. The results of the other allocation approaches are shown in relation to the base case.....	136
Fig. 5.36	Relative results (characterized) of the sensitivity analyses when producing 1 kWh of DC electricity including and excluding the inverter with a residential scale multi-Si PV system mounted on a slanted roof top. The base case represents the situation without inverter. Results are indexed to the base case (100 %).	138
Fig. 5.37	Relative results (characterized) of the sensitivity analyses when producing 1 kWh of DC electricity applying a degradation rate of 0.5 % (2 % in the first year) and a life span of 25 years with a residential scale multi-Si PV system mounted on a slanted roof top. The base case represents the situation with a degradation rate of 0.7 % and a life span of 30 years. All results are indexed to the base case (100 %).	139

List of tables

Tab. 3.1	Bill of materials of different types of photovoltaic modules (laminate/unframed and panel/framed) based on the input materials excluding working materials, given in relative terms. The total mass of the input materials of the unframed module (laminate) corresponds to 100 % Sources: KBOB (2014), Frischknecht et al. (2015a), Jungbluth et al. (2012), de Wild-Scholten (2014), Flury et al. (2012), First Solar (Itten & Frischknecht 2014).....	28
Tab. 3.2	Overview of data quality (scores and level) of electricity production of all PV technologies per life cycle stage, mounted on a slanted roof top, residential scale. Regarded in this overview are the processes causing 90 % of the impact per impact category. Data quality scores were assessed according to requirements of the European Commission (2013).....	33
Tab. 3.3	Radiotoxicity index (RTI) per m ³ of waste and characterization factors	34
Tab. 3.4	List of life cycle impact assessment indicators required by the PEF Guide and considered relevant for this PEF CR	35
Tab. 3.5	Normalization factors of the impact category indicators used in the PEF screening LCA.....	37
Tab. 4.1	Characteristics of CdTe PV panels.....	39

Tab. 4.2	unit process inventory data for the production of cadmium-telluride photovoltaic panels in the United States of America and in Malaysia according to First Solar	40
Tab. 4.3	unit process inventory data for cadmium-telluride photovoltaic panels at the European regional storage	41
Tab. 4.4	unit process inventory data for the production site for cadmium-telluride photovoltaic panels according to First Solar for the production site in the United States of America	42
Tab. 4.5	unit process inventory data for the mounting cadmium-telluride photovoltaic panels integrated in a slanted roof top	43
Tab. 4.6	unit process inventory data for the mounting cadmium-telluride photovoltaic panels mounted on a slanted roof top	43
Tab. 4.7	unit process inventory data for the mounting cadmium-telluride photovoltaic panels open ground installation	44
Tab. 4.8	unit process inventory data for the electricity production with cadmium-telluride photovoltaic panels at a 3 kWp roof installation, mounted and integrated, and a 570 kWp open ground installation	44
Tab. 4.9	unit process inventory data for the recycling and the potential benefits and additional loads beyond the system boundary by recycling cadmium-telluride photovoltaic panels, cabling and mounting structures.....	46
Tab. 4.10	Characteristics of CIS PV panels	47
Tab. 4.11	unit process inventory data for the production of CIS photovoltaic panels for the production in Germany	48
Tab. 4.12	unit process inventory data for CIS photovoltaic panels at the European regional storage	49
Tab. 4.13	unit process inventory data for the mounting CIS photovoltaic panels integrated and mounted on a slanted roof top as well as the open ground system.....	50
Tab. 4.14	unit process inventory data for the electricity production with CIS photovoltaic panels at a 3 kWp roof installation, integrated and mounted, and a 570 kWp open ground installation.....	51
Tab. 4.15	unit process inventory data for the recycling and the potential benefits and additional loads beyond the system boundary by recycling CIS photovoltaic panels, cabling and mounting structures	53
Tab. 4.16	Characteristics of micro-Si PV panels.....	54
Tab. 4.17	unit process inventory data for the production of micro-Si photovoltaic panels for the production in China	55
Tab. 4.18	unit process inventory data for micro-Si photovoltaic panels at the European regional storage	56
Tab. 4.19	unit process inventory data for the mounting structure and electrical equipment of micro-Si photovoltaic panels integrated and mounted on a slanted roof top as well as the open ground system.....	57
Tab. 4.20	unit process inventory data for the electricity production with micro-Si photovoltaic panels at a 3 kWp roof installation, integrated and mounted, and a 570 kWp open ground installation	58
Tab. 4.21	unit process inventory data for the recycling and the potential benefits and additional loads beyond the system boundary by recycling micro-Si photovoltaic panels, cabling and mounting structures	59
Tab. 4.22	Characteristics of multi-Si PV panels.....	60

Tab. 4.23	unit process inventory data for the production of multi-Si photovoltaic panels produced in Europe, China and Asia and Pacific region.....	61
Tab. 4.24	unit process inventory data for multicrystalline-Si photovoltaic panels at the European regional storage	62
Tab. 4.25	unit process inventory data for the mounting multicrystalline-Si photovoltaic panels integrated and mounted on a slanted roof top as well as the open ground system.....	63
Tab. 4.26	unit process inventory data for the electricity production with multicrystalline-Si photovoltaic panels at a 3 kWp roof installation, integrated and mounted, and a 570 kWp open ground installation	64
Tab. 4.27	unit process inventory data for the recycling and the potential benefits and additional loads beyond the system boundary by recycling multicrystalline-Si photovoltaic panels, cabling and mounting structures.....	65
Tab. 4.28	Characteristics of mono-Si PV panels.....	66
Tab. 4.29	unit process inventory data for the production of mono-Si photovoltaic panels produced in Europe, China and Asia and Pacific region.....	67
Tab. 4.30	unit process inventory data for monocrystalline-Si photovoltaic panels at the European regional storage	68
Tab. 4.31	unit process inventory data for the mounting monocrystalline Si photovoltaic panels integrated and mounted on a slanted roof top as well as the open ground system.....	69
Tab. 4.32	unit process inventory data for the electricity production with monocrystalline-Si photovoltaic panels at a 3 kWp roof installation, integrated and mounted, and a 570 kWp open ground installation	70
Tab. 4.33	unit process inventory data for the recycling and the potential benefits and additional loads beyond the system boundary by recycling monocrystalline-Si photovoltaic panels, cabling and mounting structures.....	71
Tab. 4.34	unit process inventory data for the electric installation of the 3 kWp PV systems (integrated and mounted on roof), (Jungbluth et al. 2012).....	72
Tab. 4.35	unit process inventory data for the electric installation of the 570 kWp PV system (open ground), (Jungbluth et al. 2012)	72
Tab. 4.36	unit process inventory data for the mounting structure for integrated, mounted and open ground PV systems (Jungbluth et al. 2012)	73
Tab. 4.37	Means of transport and corresponding transport distance in km for the import of mono- and multicrystalline PV modules from the relevant regions of the world.	75
Tab. 4.38	Cumulative installed photovoltaic power in Europe in 2012 according to IEA-PVPS (2013) and country specific average annual yield in kWh/kWp at optimal angle in urban areas according to EPIA (2012), and average solar irradiation at optimal angle based on data retrieved from PVGIS; degradation is not included.....	76
Tab. 4.39	Values of the parameters R1, R2 and R3 and datasets used for the calculation of avoided material uses and emissions at the end-of-life stage of PV panels. The variables correspond to the denotations introduced in the end-of-life formula recommended by the European Commission (2013).	78
Tab. 5.1	Results prior to normalization per impact category of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels. The contribution of potential benefits for recycling is not included in these results.	81

Tab. 5.2	Minimum and maximum results and bandwidth (maximum minus minimum, divided by minimum) per impact category of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels	82
Tab. 5.3	Normalized results per impact category of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels. The contribution of potential benefits for recycling is not included in these results.....	93
Tab. 5.4	Weighted results per impact category of 1 kWh of DC electricity produced with a residential scale (3 kWp) PV system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a large scale (570 kWp) open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels. The contribution of potential benefits for recycling is not included in these results.	94
Tab. 5.5	Environmental impact results (characterized) of 1 kWh of DC electricity produced with a residential scale (3 kWp) PV system with CdTe PV panels mounted on a slanted roof.	95
Tab. 5.6	Environmental impact results (normalized) of 1 kWh of DC electricity produced with a 3 kWp power system with CdTe PV panels mounted on a slanted roof.	100
Tab. 5.7	Environmental impact results (weighted) of 1 kWh of DC electricity produced with a 3 kWp power system with CdTe PV panels mounted on a slanted roof.	100
Tab. 5.8	Environmental impact results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with CIS PV panels mounted on a slanted roof.	103
Tab. 5.9	Environmental impact results (normalized) of 1 kWh of DC electricity produced with a residential scale CIS PV system mounted on a slanted roof.....	108
Tab. 5.10	Environmental impact results (weighted) of 1 kWh of DC electricity produced with a 3 kWp power system with CIS PV panels mounted on a slanted roof.	109
Tab. 5.11	Environmental impact results (characterized) of 1 kWh of DC electricity produced with a residential scale micro-Si PV system mounted on a slanted roof.....	111
Tab. 5.12	Environmental impact results (normalized) of 1 kWh of DC electricity produced with a 3 kWp power system with micro-Si PV panels mounted on a slanted roof.	116
Tab. 5.13	Environmental impact results (weighted) of 1 kWh of DC electricity produced with a residential scale micro-Si PV system mounted on a slanted roof.....	117
Tab. 5.14	Environmental impact results (characterized) of 1 kWh of DC electricity produced with a residential scale multi-Si PV system mounted on a slanted roof.....	119
Tab. 5.15	Environmental impact results (normalized) of 1 kWh of DC electricity produced with a residential scale multi-Si PV system mounted on a slanted roof.....	123
Tab. 5.16	Environmental impact results (weighted) of 1 kWh of DC electricity produced with a residential scale multi-Si PV system mounted on a slanted roof.....	124
Tab. 5.17	Environmental impact results (characterized) of 1 kWh of DC electricity produced with a residential scale mono-Si PV system mounted on a slanted roof.	126
Tab. 5.18	Environmental impact results (normalized) of 1 kWh of DC electricity produced with a 3 kWp power system with mono-Si PV panels mounted on a slanted roof.....	130
Tab. 5.19	Environmental impact results (weighted) of 1 kWh of DC electricity produced with a residential scale mono-Si PV system mounted on a slanted roof.	131

Tab. 5.20	Global PV module production in Gigawatt peak (GWp, the maximum power output under standard conditions of the modules) by technology, reference year 2012, based on FHI-ISE (2013) ¹⁰	132
Tab. 5.21	Environmental impact results (characterized) of 1 kWh of DC electricity produced with a residential scale (3 kWp) PV system with average PV panels mounted on a slanted roof.	133
Tab. 6.1	Identification of the two most relevant life cycle stages, processes and elementary flows for each impact category for 1 kWh of DC electricity produced with a residential scale CdTe PV system mounted on a slanted roof.....	143
Tab. 6.2	Identification of the two most relevant life cycle stages, processes and elementary flows for each impact category for 1 kWh of DC electricity produced with a residential scale CIS PV system mounted on a slanted roof.	144
Tab. 6.3	Identification of the two most relevant life cycle stages, processes and elementary flows for each impact category for 1 kWh of DC electricity produced with a residential scale micro-Si PV system mounted on a slanted roof.....	145
Tab. 6.4	Identification of the two most relevant life cycle stages, processes and elementary flows for each impact category for 1 kWh of DC electricity produced with a residential scale multi-Si PV system mounted on a slanted roof.....	146
Tab. 6.5	Identification of the two most relevant life cycle stages, processes and elementary flows for each impact category for 1 kWh of DC electricity produced with a residential scale mono-Si PV system mounted on a slanted roof.....	147
Tab. 6.6	Identification of the most relevant life cycle stages for each impact category for 1 kWh of DC electricity produced with a residential scale CdTe PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant life cycle stages to the characterized environmental impacts is reached.	148
Tab. 6.7	Identification of the most relevant life cycle stages for each impact category for 1 kWh of DC electricity produced with a residential scale CIS PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant life cycle stages to the characterized environmental impacts is reached.	149
Tab. 6.8	Identification of the most relevant life cycle stages for each impact category for 1 kWh of DC electricity produced with a residential scale micro-Si PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant life cycle stages to the characterized environmental impacts is reached.	150
Tab. 6.9	Identification of the most relevant life cycle stages for each impact category for 1 kWh of DC electricity produced with a residential scale multi-Si PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant life cycle stages to the characterized environmental impacts is reached.	151
Tab. 6.10	Identification of the most relevant life cycle stages for each impact category for 1 kWh of DC electricity produced with a residential scale mono-Si PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant life cycle stages to the characterized environmental impacts is reached.	152
Tab. 6.11	Identification of the most relevant processes for each impact category and each life cycle stage for 1 kWh of DC electricity produced with a residential scale CdTe PV	

	system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant processes to the characterized environmental impacts is reached.	153
Tab. 6.12	Identification of the most relevant processes for each impact category and each life cycle stage for 1 kWh of DC electricity produced with a residential scale CIS PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant processes to the characterized environmental impacts is reached.	156
Tab. 6.13	Identification of the most relevant processes for each impact category and each life cycle stage for 1 kWh of DC electricity produced with a residential scale micro-Si PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant processes to the characterized environmental impacts is reached.	159
Tab. 6.14	Identification of the most relevant processes for each impact category and each life cycle stage for 1 kWh of DC electricity produced with a residential scale multi-Si PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant processes to the characterized environmental impacts is reached.	161
Tab. 6.15	Identification of the most relevant processes for each impact category and each life cycle stage for 1 kWh of DC electricity produced with a residential scale mono-Si PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant processes to the characterized environmental impacts is reached.	164
Tab. 6.16	Identification of the most relevant elementary flows for each impact category for 1 kWh of DC electricity produced with a residential scale CdTe PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant elementary flows to the characterized environmental impacts is reached.	167
Tab. 6.17	Identification of the most relevant elementary flows for each impact category for 1 kWh of DC electricity produced with a 3 kWp power system with CIS PV panels mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant elementary flows to the characterized environmental impacts is reached.	168
Tab. 6.18	Identification of the most relevant elementary flows for each impact category for 1 kWh of DC electricity produced with a 3 kWp power system with micro-Si PV panels mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant elementary flows to the characterized environmental impacts is reached.	169
Tab. 6.19	Identification of the most relevant elementary flows for each impact category for 1 kWh of DC electricity produced with a 3 kWp power system with multi-Si PV panels mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant elementary flows to the characterized environmental impacts is reached.	170
Tab. 6.20	Identification of the most relevant elementary flows for each impact category for 1 kWh of DC electricity with a 3 kWp power system with mono-Si PV panels mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant elementary flows to the characterized environmental impacts is reached.	171

List of Tables ANNEX I and ANNEX II

Tab. A. 1	Overview of data quality (scores and level) of electricity production of all PV technologies per life cycle stage, mounted on a slanted roof top, residential scale. Regarded in this overview are the processes causing 90 % of the impact per impact category. Data quality scores were assessed according to requirements of the European Commission (2013).....	181
Tab. A. 2	data quality assessment according to the European Commission (2013) of the CdTe PV panel mix at the regional storage (product stage) of electricity production with CdTe PV technology	182
Tab. A. 3	data quality assessment according to the European Commission (2013) of the CdTe PV panel produced in Malaysia.....	182
Tab. A. 4	data quality assessment according to the European Commission (2013) of the construction stage of electricity production with CdTe PV technology.....	183
Tab. A. 5	data quality assessment according to the European Commission (2013) of the use stage of electricity production with CdTe PV technology	183
Tab. A. 6	data quality assessment according to the European Commission (2013) of the end-of-life stage of electricity production with CdTe PV technology.....	184
Tab. A. 7	data quality assessment according to the European Commission (2013) of the potential benefits for recycling (50/50) from CdTe panel recycling	184
Tab. A. 8	data quality assessment according to the European Commission (2013) of the CIS PV panel mix at the regional storage (product stage) of electricity production with CIS PV technology	185
Tab. A. 9	data quality assessment according to the European Commission (2013) of the CIS laminate produced in Germany	185
Tab. A. 10	data quality assessment according to the European Commission (2013) of the construction stage of electricity production with CIS PV technology	186
Tab. A. 11	data quality assessment according to the European Commission (2013) of the use stage of electricity production with CIS PV technology	186
Tab. A. 12	data quality assessment according to the European Commission (2013) of the end-of-life stage of electricity production with CIS PV technology	186
Tab. A. 13	data quality assessment according to the European Commission (2013) of the potential benefits for recycling (50/50) from CIS panel recycling.....	187
Tab. A. 14	data quality assessment according to the European Commission (2013) of the micro-Si PV panel mix at the regional storage (product stage) of electricity production with CIS PV technology.....	188
Tab. A. 15	data quality assessment according to the European Commission (2013) of the construction stage of electricity production with micro-Si PV technology.....	188
Tab. A. 16	data quality assessment according to the European Commission (2013) of the use stage of electricity production with micro-Si PV technology	189
Tab. A. 17	data quality assessment according to the European Commission (2013) of the end-of-life stage of electricity production with micro-Si PV technology	189
Tab. A. 18	data quality assessment according to the European Commission (2013) of the potential benefits for recycling (50/50) from micro-Si panel recycling.....	190

Tab. A. 19	data quality assessment according to the European Commission (2013) of the multicrystalline-Si PV panel mix at the regional storage (product stage) of electricity production with multicrystalline PV technology	190
Tab. A. 20	data quality assessment according to the European Commission (2013) of the multicrystalline PV panel produced in China.....	191
Tab. A. 21	data quality assessment according to the European Commission (2013) of the construction stage of electricity production with multicrystalline-Si PV technology	191
Tab. A. 22	data quality assessment according to the European Commission (2013) of the use stage of electricity production with multicrystalline PV technology	192
Tab. A. 23	data quality assessment according to the European Commission (2013) of the end-of-life stage of electricity production with multicrystalline PV technology	192
Tab. A. 24	data quality assessment according to the European Commission (2013) of the potential benefits for recycling (50/50) from multicrystalline -Si panel recycling	193
Tab. A. 25	data quality assessment according to the European Commission (2013) of the monocrystalline-Si PV panel mix at the regional storage (product stage) of electricity production with monocrystalline-Si PV technology.....	193
Tab. A. 26	data quality assessment according to the European Commission (2013) of the monocrystalline-Si PV panel produced in China	194
Tab. A. 27	data quality assessment according to the European Commission (2013) of the construction stage of electricity production with monocrystalline-Si PV technology ...	194
Tab. A. 28	data quality assessment according to the European Commission (2013) of the use stage of electricity production with monocrystalline-Si PV technology	195
Tab. A. 29	data quality assessment according to the European Commission (2013) of the end-of-life stage of electricity production with monocrystalline-Si PV technology	195
Tab. A. 30	data quality assessment according to the European Commission (2013) of the potential benefits for recycling (50/50) from monocrystalline-Si panel recycling.....	196
Tab. A. 31	Environmental impact results (characterized) of 1 kWh of electricity produced with a 3 kWp power system with CdTe PV panels integrated in a slanted roof.	198
Tab. A. 32	Environmental impact results (characterized) of 1 kWh of electricity produced with a 570 kWp power system with CdTe PV panels on an open ground system.	199
Tab. A. 33	Environmental impact results (normalized) of 1 kWh of electricity produced with a 3 kWp power system with CdTe PV panels integrated in a slanted roof.	200
Tab. A. 34	Environmental impact results (normalized) of 1 kWh of electricity produced with a 570 kWp power system with CdTe PV panels on an open ground system.	200
Tab. A. 35	Environmental impact results (weighted) of 1 kWh of electricity produced with a 3 kWp power system with CdTe PV panels integrated in a slanted roof.	201
Tab. A. 36	Environmental impact results (weighted) of 1 kWh of electricity produced with a 570 kWp power system with CdTe PV panels on an open ground system.	202
Tab. A. 37	Environmental impact results (characterized) of 1 kWh of electricity produced with a 3 kWp power system with CIS PV panels integrated in a slanted roof.	203
Tab. A. 38	Environmental impact results (characterized) of 1 kWh of electricity produced with a 570 kWp power system with CIS PV panels on an open ground system.	204
Tab. A. 39	Environmental impact results (normalized) of 1 kWh of electricity produced with a 3 kWp power system with CIS PV panels integrated in a slanted roof.	205
Tab. A. 40	Environmental impact results (normalized) of 1 kWh of electricity produced with a 570 kWp power system with CIS PV panels on an open ground system.....	205
Tab. A. 41	Environmental impact results (weighted) of 1 kWh of electricity produced with a 3 kWp power system with CIS PV panels integrated in a slanted roof.	206

Tab. A. 42	Environmental impact results (weighted) of 1 kWh of electricity produced with a 570 kWp power system with CIS PV panels on an open ground system.....	207
Tab. A. 43	Environmental impact results (characterized) of 1 kWh of electricity produced with a 3 kWp power system with micro-Si PV panels integrated in a slanted roof.	208
Tab. A. 44	Environmental impact results (characterized) of 1 kWh of electricity produced with a 570 kWp power system with micro-Si PV panels on an open ground system.....	209
Tab. A. 45	Environmental impact results (normalized) of 1 kWh of electricity produced with a 3 kWp power system with micro-Si PV panels integrated in a slanted roof.	210
Tab. A. 46	Environmental impact results (normalized) of 1 kWh of electricity produced with a 570 kWp power system with micro-Si PV panels on an open ground system.	210
Tab. A. 47	Environmental impact results (weighted) of 1 kWh of electricity produced with a 3 kWp power system with micro-Si PV panels integrated in a slanted roof.	211
Tab. A. 48	Environmental impact results (weighted) of 1 kWh of electricity produced with a 570 kWp power system with micro-Si PV panels on an open ground system.	212
Tab. A. 49	Environmental impact results (characterized) of 1 kWh of electricity produced with a 3 kWp power system with multi-Si PV panels integrated in a slanted roof.	213
Tab. A. 50	Environmental impact results (characterized) of 1 kWh of electricity produced with a 570 kWp power system with multi-Si PV panels on an open ground system.	214
Tab. A. 51	Environmental impact results (normalized) of 1 kWh of electricity produced with a 3 kWp power system with multi-Si PV panels integrated in a slanted roof.	215
Tab. A. 52	Environmental impact results (normalized) of 1 kWh of electricity produced with a 570 kWp power system with multi-Si PV panels on an open ground system.	215
Tab. A. 53	Environmental impact results (weighted) of 1 kWh of electricity produced with a 3 kWp power system with multi-Si PV panels integrated in a slanted roof.	216
Tab. A. 54	Environmental impact results (weighted) of 1 kWh of electricity produced with a 570 kWp power system with multi-Si PV panels on an open ground system.	217
Tab. A. 55	Environmental impact results (characterized) of 1 kWh of electricity produced with a 3 kWp power system with mono-Si PV panels integrated in a slanted roof.	218
Tab. A. 56	Environmental impact results (characterized) of 1 kWh of electricity produced with a 570 kWp power system with mono-Si PV panels on an open ground system. ...	219
Tab. A. 57	Environmental impact results (normalized) of 1 kWh of electricity produced with a 3 kWp power system with mono-Si PV panels integrated in a slanted roof.	220
Tab. A. 58	Environmental impact results (normalized) of 1 kWh of electricity produced with a 570 kWp power system with mono-Si PV panels on an open ground system.	220
Tab. A. 59	Environmental impact results (weighted) of 1 kWh of electricity produced with a 3 kWp power system with mono-Si PV panels integrated in a slanted roof.	221
Tab. A. 60	Environmental impact results (weighted) of 1 kWh of electricity produced with a 570 kWp power system with mono-Si PV panels on an open ground system.	222
Tab. A. 61	Environmental impact results (characterized per technology) of 1 kWh of electricity produced with a 3 kWp power system with average PV panels mounted on a slanted roof.	223
Tab. A. 62	Environmental impact results (characterized) of 1 kWh of electricity produced with a 3 kWp power system with average PV panels mounted on a slanted roof.....	223
Tab. A. 63	Environmental impact results (characterized, shown as percentages of the characterized environmental impacts of 1 kWh electricity generated) of the disposal of waste materials for a 3 kWp power system with CdTe PV panels mounted on a slanted roof.	224

Abbreviations and Units

a-Si	Amorphous silicon
AC	Alternating current
ADEME	French Environment and Energy Management Agency
AE	Accumulated exceedance
BOM	Bill of materials
Bq	Becquerel, unit of radioactivity
CdTe	Cadmium-Telluride
CED	Cumulative Energy Demand
CH	Switzerland
CIGS	Copper-Indium-Gallium-Selenide
CIS	Copper-Indium-Selenide
CTUe	Comparative toxic unit, ecosystems
CTUh	Comparative toxic unit, humans
DC	Direct current
EF	Environmental footprint
EFTA	European Free Trade Association
EPD	Environmental product declaration
EoL	End-of-Life
EU	European Union
GLO	Global average
GWP	Global warming potential
GWp	Gigawattpeak
HDPE	High density polyethylene
Hz	Hertz
IEA	International energy agency
IEA-PVPS	International energy agency photovoltaic power systems program
KBOB	Public coordination conference of construction services and buildings
kWh	Kilowatt hour
kWp	Kilowatt peak
LCA	Life cycle assessment
LCI	Life cycle inventory analysis
LCIA	Life cycle impact assessment
LDPE	Low-density polyethylene
mSv	Milli-Sievert
micro-Si	Micromorphous silicon
mono-Si	Monocrystalline silicon
multi-Si	Multicrystalline silicon
N	Nitrogen
ODP	Ozone depletion potential
P	Phosphorus
PCR	Product category rules

PE	Polyethylene
PEF	Product environmental footprint
PEFCR	Product environmental footprint category rules
PET	Polyethylene terephthalate
PM	Particulate matter
PV	Photovoltaic
Ribbon-Si	Ribbon silicon
RER	Europe
RTI	Radiotoxicity index
Sv	Sievert, unit of ionizing radiation
tkm	Ton kilometer, unit for transport services
U235	Uranium 235
V	Volt
W	Watt
WEEE	Directive waste of electrical and electronic equipment

1. Introduction

Solar photovoltaic (PV) technology has developed strongly in a relatively short time. Many different technologies for panel production exist today. They differ in material consumption, efficiencies, life expectancies and costs but also in their environmental performance. In this study the inventory and the impact assessment of the production of direct current (DC) electricity with five different PV module technologies are described and compared. The following PV technologies are assessed in this screening study:

- Cadmium-Telluride photovoltaic panels (CdTe panels)
- Copper-Indium-Gallium-Selenide photovoltaic panels (CIS panels)
- micromorphous silicon photovoltaic panels (micro-Si)
- multicrystalline silicon photovoltaic panels (multi-Si)
- monocrystalline silicon photovoltaic panels (mono-Si)

The life cycle inventories include the manufacturing, the operation and dismantling of the photovoltaic modules as well as the use of production equipment, facilities and the supply chain of the materials used. Details of the inventory are found in chapter 4 whereas impact assessment results are described in chapter 5.

The PV panels are electrical equipment and thus classified in the NACE/CPA class 27.90 “Manufacturing of other electrical equipment”¹. They are compared on the basis of 1 kWh (kilowatt hour) of DC electricity generated by a photovoltaic module (unit of analysis). The reference flow is the photovoltaic module, measured in kWp (kilowatt peak), the maximum power output under standard conditions of a module. The screening LCA were calculated with the software Simapro 8.0.6 (PRé Consultants 2015) using ecoinvent data version 2.2+ (KBOB et al. 2014). The default 15 environmental footprint impact category indicators are complemented with three additional indicators: renewable cumulative energy demand, non-renewable cumulative energy demand and nuclear waste.

The screening of electricity from photovoltaic modules represents Task 4 in the PEF Pilot Project “Photovoltaic Electricity Generation” and is in line with the PEFCR electricity generation (Frischknecht & Itten 2014) which is corresponding to the Communication of the European Commission on the single market for green products and the product environmental footprint (PEF) pilot². The screening study was conducted according to the requirements of the PEF Guide (European Commission 2013) and the Product Environmental Footprint Pilot Guidance (version 5.2, European Commission 2015). This PEF screening LCA study is commissioned by the Technical Secretariat of the PEF pilot on PV electricity.

2. Goal of the study

This screening LCA report aims at comparing five different photovoltaic technologies in terms of their environmental performance and at identifying the most relevant life cycle stages and hot spots as well as the most important impact categories for each technology. Preliminary information about important life cycle impact categories will be gained, while furthermore requirements in data quality are identified. This screening report serves as a basis for the assessment of a representative (virtual) product, composed of market-share weighted averages of the different technologies. The screening should help identifying hotspots which are specific to the technologies analyzed and ensure that these hotspots are not overlooked.

¹ http://epp.eurostat.ec.europa.eu/portal/page/portal/cpa_2008/documents/CPA_2008_structure_EN.pdf, accessed on 8.1.2015

² <http://ec.europa.eu/environment/eussd/smgp/>, accessed on 29.04.2014

Based on the preliminary results, technology specific benchmarks are proposed. This screening report is intended as a starting point for further research within the PEFCR and not to make statements about the product group environmental impacts as such, nor is it intended to be used in the context of comparison or for comparative assertions to be disclosed to the public.

Reduced data availability of direct production emissions and of the recycling stage are the main limitations. Avancis, data provider for the supply chain of CIS panels, is no longer producing panels. However it is assumed that their data are still representative. Furthermore as the inverter is not part of the basic assessment, the results of this screening LCA shall not be compared to the environmental impacts of electricity produced with other technologies such as hard coal, nuclear, hydroelectric or wind power plants.

The product system includes the mounting on rooftops (small scale systems) and on open ground (large scale systems). The inverter however is excluded. As with this limitation the PV system is not fully complete, reference will be made to the PV system (consisting of the PV panels and the mounting structure as its main parts) and not to a PV plant.

3. Scope of the screening

3.1 Introduction

The PEF screening affects data collection activities and data quality by setting priorities for the PEFCR supporting study. The objective of the screening is to pre-identify the following key information:

- Most relevant life cycle stages
- Most relevant processes
- Most relevant elementary flows
- Preliminary indication about the most relevant life cycle impact categories
- Data quality needs
- Preliminary indication about the definition of the benchmark for the product category/sub-categories in scope

The PEF screening of PV panels is based on available producer specific data and generic data fulfilling the data quality requirements as defined in the most updated version of the PEF Guide. For the screening step a minimum “fair” quality data rating is required for data contributing to at least 90 % of the environmental impact estimated for each EF impact category, as assessed via a qualitative expert judgment. The results of the screening are analyzed in a sensitivity analysis and are part of the PEFCR review process.

3.2 Function, functional unit and reference flow

The unit of analysis is defined as 1 kWh (kilowatt hour) of DC electricity generated by a PV module. This is fully compliant with the goal of this study, to assess and compare different PV systems in their environmental performance of electricity production over all their life cycle stages. The reference flow is the PV module, measured in kWp (kilowatt peak), the maximum power output under standard conditions of a module.

It is defined according to the following four criteria listed in the PEF Guide.

The function(s) / service(s) provided (“what”):

- DC electrical energy measured in kWh (provided power times unit of time)

The magnitude of the function or service (“how much”)

- 1 kWh of DC electrical energy

The expected level of quality (“how well”)

- DC electrical energy at the photovoltaic panel at a given voltage level

The amount of service provided over the life time (“how long”)

- DC electrical energy at the photovoltaic panel during the service life of 30 years

When referred to a PV system, the PV panels, mounting structure and DC electrical installation are included. As it is not a fully functional PV plant (inverter is excluded), the term “plant” is avoided, but PV system used for clarifying this difference.

Any potential further functions of the PV system, such as weather protection of building integrated PV systems, are disregarded. The main performance characteristics are the annual yield, the service life and the degradation rate of the panels. They are independent of the panel technologies analyzed. The performance characteristics are described in chapter 4.

The materials photovoltaic modules usually consist of are compiled in Tab. 3.1. The average module per technology is modelled using specific data from de Wild-Scholten (2014, mono-Si, multi-Si and CIS), Flury et al. (2012, micro-Si), First Solar (CdTe) (Itten & Frischknecht 2014). In addition, generic data available from ecoinvent data v2.2+ (KBOB et al. 2014), Frischknecht et al. (2015a) and Jungbluth et al. (2012) was used.

The bill of materials is shown per square meter of module and relative to the total mass of the input materials of the unframed laminate. It is based on the weight of the materials used in manufacturing PV modules and listed in the life cycle inventories (LCI) of the different photovoltaic technologies. The losses (due to breakage, cutting) are included but not explicitly listed in the LCIs. Working material inputs are not included in the bill of materials.

The bill of materials shown in Tab. 3.1 differs from the one reported in Frischknecht and Itten (2014). The weight of unframed mono-Si (11.7 kg), multi-Si (11.2 kg) and CIS (14.9 kg) PV modules decreased mainly as a result of a lower input of solar glass, whereas the weight of micro-Si (13.9 kg) and CdTe (17.0 kg) did not change. For the micro-Si PV module, the weight of the aluminium frame is newly added to the bill of materials.

The share of the wafer in the total weight of a mono-Si PV module (10.2 %) increased due to a higher consumption of silicon in the production of Czochralski (CZ) mono crystalline silicon. In contrast, the silicon input decreased in the production of multi-Si PV modules. This results in a lower share of the wafer in the module weight (6.2 %). The weight percentages of materials in both micro-Si and CdTe unframed PV modules is identical to the bill of materials published by Frischknecht and Itten (2014). The share of the semiconductor in CIS modules (0.1 %) decreased due to much lower inputs of cadmium sulphide, gallium, selenium and indium. The share of plastics in the total weight of CIS PV modules increased, however.

The rated power per square meter of module is the same as reported in Frischknecht and Itten (2014) with the exception of CdTe PV modules, where an increase from 134 Wp/m² to 140 Wp/m² is

registered. The efficiency assumed for crystalline silicon PV modules reflects the situation in the year 2011. Up to the time this study was carried out, progress was made in increasing the module conversion efficiency.

Tab. 3.1 Bill of materials of different types of photovoltaic modules (laminated/unframed and panel/framed) based on the input materials excluding working materials, given in relative terms. The total mass of the input materials of the unframed module (laminated) corresponds to 100 % Sources: KBOB (2014), Frischknecht et al. (2015a), Jungbluth et al. (2012), de Wild-Scholten (2014), Flury et al. (2012), First Solar (Itten & Frischknecht 2014)

Material				Photovoltaic module (laminat ^e / unframed and panel / framed)				
				Mono-Si	Multi-Si	Micro-Si	CdTe	Cl(G)S
Source				Frischknecht et al. 2015	Frischknecht et al. 2015	Flury et al. 2012	Frischknecht et al. 2015	Frischknecht et al. 2015
Laminat ^e / unframed	Subtotal wafer / semiconductor			10.2%	6.2%	0.2%	0.2%	0.1%
	Wafer / semiconductor	silicon, production mix, photovoltaics, at plant	silicon for photovoltaics	10.2%	6.2%	0.0%	0.0%	0.0%
		silane, at plant	silane	0.0%	0.0%	0.2%	0.0%	0.0%
		indium, at regional storage	indium	0.0%	0.0%	0.0%	0.0%	0.0%
		cadmium telluride, semiconductor-grade, at plant	cadmium telluride	0.0%	0.0%	0.0%	0.1%	0.0%
		cadmium sulphide, semiconductor-grade, at plant	cadmium sulphide	0.0%	0.0%	0.0%	0.0%	0.0%
		gallium, semiconductor-grade, at regional storage	gallium	0.0%	0.0%	0.0%	0.0%	0.0%
		selenium, at plant	selenium	0.0%	0.0%	0.0%	0.0%	0.0%
		Further semiconductor materials (to be specified)	nn	0.0%	0.0%	0.0%	0.0%	0.0%
	Subtotal metals			1.5%	1.5%	0.1%	0.1%	0.5%
	Metals	aluminium, primary, at plant	aluminium	0.4%	0.4%	0.0%	0.0%	0.3%
		copper, at regional storage	copper	0.9%	0.9%	0.1%	0.1%	0.1%
		lead, at regional storage	lead	0.0%	0.0%	0.0%	0.0%	0.0%
		molybdenum, at regional storage	molybdenum	0.0%	0.0%	0.0%	0.0%	0.0%
		silver, at regional storage	silver	0.1%	0.1%	0.0%	0.0%	0.0%
		tin, at regional storage	tin	0.1%	0.1%	0.0%	0.0%	0.1%
		zinc oxide, at plant	zinc oxide	0.0%	0.0%	0.0%	0.0%	0.1%
		further metals (to be specified)	nn	0.0%	0.0%	0.0%	0.0%	0.0%
	Subtotal plastics			12.7%	13.2%	3.4%	3.4%	12.2%
	Plastics	ethylvinylacetate, foil, at plant	ethylvinylacetate	7.5%	7.8%	3.2%	2.8%	5.0%
		polyvinylfluoride film, at plant	polyvinylfluoride film	1.0%	1.0%	0.0%	0.0%	0.0%
		polyvinylbutyral foil, at plant	polyvinylbutyral foil	0.0%	0.0%	0.0%	0.0%	1.3%
		polyphenylene sulfide, at plant	polyphenylene sulfide	0.0%	0.0%	0.0%	0.0%	0.6%
		polyethylene terephthalate, granulate, amorphous, at plant	polyethylene terephthalate, PET	3.0%	3.1%	0.0%	0.0%	2.3%
		polyethylene, HDPE, granulate, at plant	polyethylene, HDPE	0.2%	0.2%	0.1%	0.0%	0.3%
		glass fibre reinforced plastic, polyamide, injection moulding, at plant	glass fibre reinforced plastic, polyamide	0.0%	0.0%	0.0%	0.6%	0.0%
		silicone product, at plant	silicone product	1.0%	1.1%	0.1%	0.0%	2.7%
		Further plastics (to be specified)	nn	0.0%	0.0%	0.0%	0.0%	0.0%
	Subtotal solar glass			75.6%	79.0%	96.3%	96.3%	87.2%
	Solar glass	flat glass, uncoated, at plant	flat glass	0.0%	0.0%	0.0%	47.6%	35.4%
		solar glass, low-iron, at regional storage	solar glass	75.6%	79.0%	96.3%	48.7%	51.8%
	Further materials (to be specified)			0.0%	0.0%	0.0%	0.0%	0.0%
Panel / framed	Subtotal metals panel			18.2%	19.0%	19.2%	n.a.	14.8%
	Metals	aluminium alloy, AlMg3, at plant	aluminium alloy	18.2%	19.0%	19.2%	n.a.	14.8%
		further metals (to be specified)	nn	0.0%	0.0%	0.0%	0.0%	0.0%
Total laminat ^e /unframed				100.0%	100.0%	100.0%	100.0%	100.0%
Total panel/framed				118.2%	119.0%	119.2%	n.a.	114.8%
Total weight in kg per square meter of unframed module				11.7	11.2	13.9	17.1	14.9
Rated power in Wp per square meter of module				151	147	100	140	108

3.3 System boundaries and system boundary diagram

3.3.1 Overview

The product system of the representative product consists of the three stages manufacturing, use and end-of-life (see Fig. 3.1). The manufacturing of PV modules includes five supply chains. They cover raw material extraction to wafer, cell and module production of crystalline silicon. They include the supply chain of semi-conductors (micromorphous silicon, cadmium sulphide, cadmium telluride, gallium and other materials used in thin film technologies), and the supply chain of carrier and connection materials (such as glass and silver). The system also includes the mounting system, which can be open ground, building integrated or building mounted, but not the inverter nor the alternating current (AC) cabling. The use phase includes electricity production and the maintenance efforts (cleaning). The end-of-life covers the dismantling of the modules including transport to a recycling facility and the recycling process itself (see also section 3.3.4).

The product system is divided into foreground processes (i.e. core processes in the product life cycle for which direct access to information is available) and background processes (i.e. those processes in

the product life cycle for which no direct access to information is possible). The processes of the foreground and background system are marked with blue and red color, respectively in Fig. 3.1.

The processes shown in Fig. 3.1, which are part of the foreground system, are considered the minimum set of company specific information. It is recommended to use company specific data in the supply chain of photovoltaic modules if available.

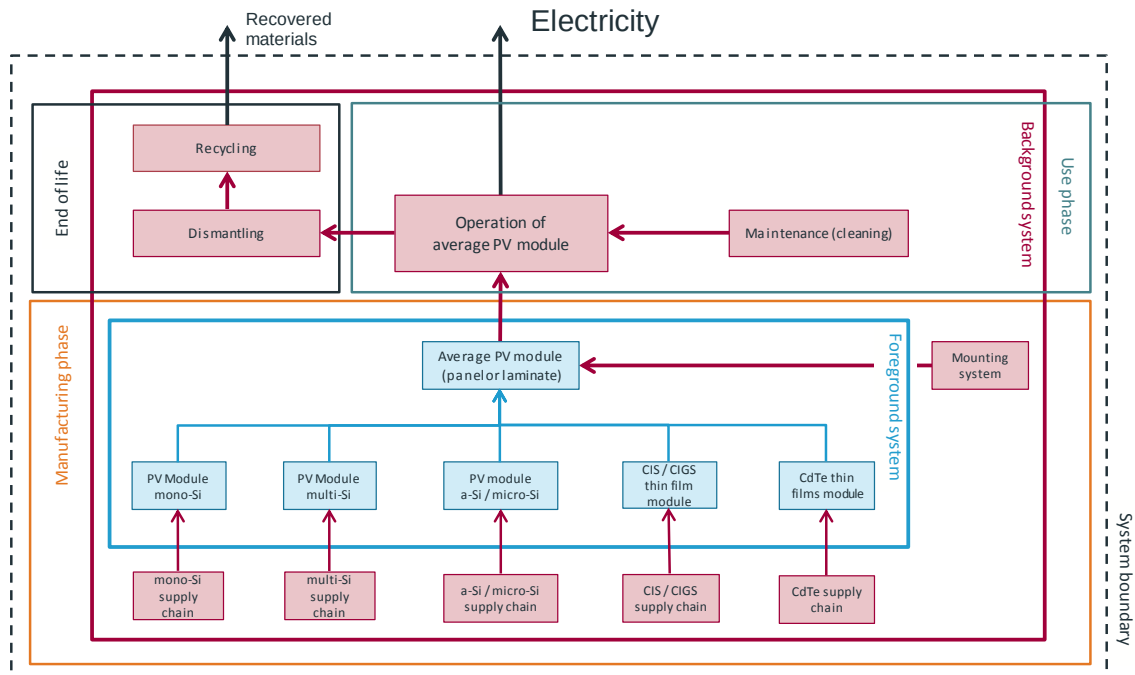


Fig. 3.1 Product system of electricity produced with photovoltaic modules; Processes of the foreground and background system are marked with blue and red color, respectively

3.3.2 Manufacturing stage

The manufacturing of the modules includes the production of the cells and modules themselves and the supply chain of the required materials. The supply chain of the photovoltaic modules is different depending on the photovoltaic technology. Fig. 3.2 shows the processes of manufacturing and the supply chain of the five different photovoltaic technologies (mono-Si, multi-Si, micro-Si, CdTe and CIS) described in this PEFCR. The processes of the foreground and background system are marked with blue and red color, respectively.

Fig. 3.2 focuses on the technology specific processes in the supply chain. The supply chains of other components made of metals or plastics and of the working materials, which are used in the different processes of the supply chain, are not shown but included as well. The supply chain of crystalline silicon PV modules differentiates between four world regions (China, Europe, Asia and Pacific (excluding China), and the Americas).

The mounting structures (used in building integrated, building mounted, open ground PV systems) are included in the system. The inverters and the electric installations for AC cabling are out of scope and thus not part of the product system.

The infrastructure (manufacture equipment and facilities) used in the supply chain of the raw materials (solar grade silicon, CdTe, etc.) and in the production of photovoltaic cells and modules is included. The contribution of the infrastructure to the overall environmental impacts is evaluated during the screening. Where relevant, it is highlighted in the results discussion and the modelling

assumptions regarding manufacturing equipment and facilities will be revisited and possibly adjusted in the PEF supporting studies.

The representative product is defined on the level of the PV module (not on cell, wafer or ingot level). The supply chain of each technology is modelled with individual unit processes as shown in Fig. 3.2. Each technology (and in the PEF supporting studies each PV product) has its own supply chain.

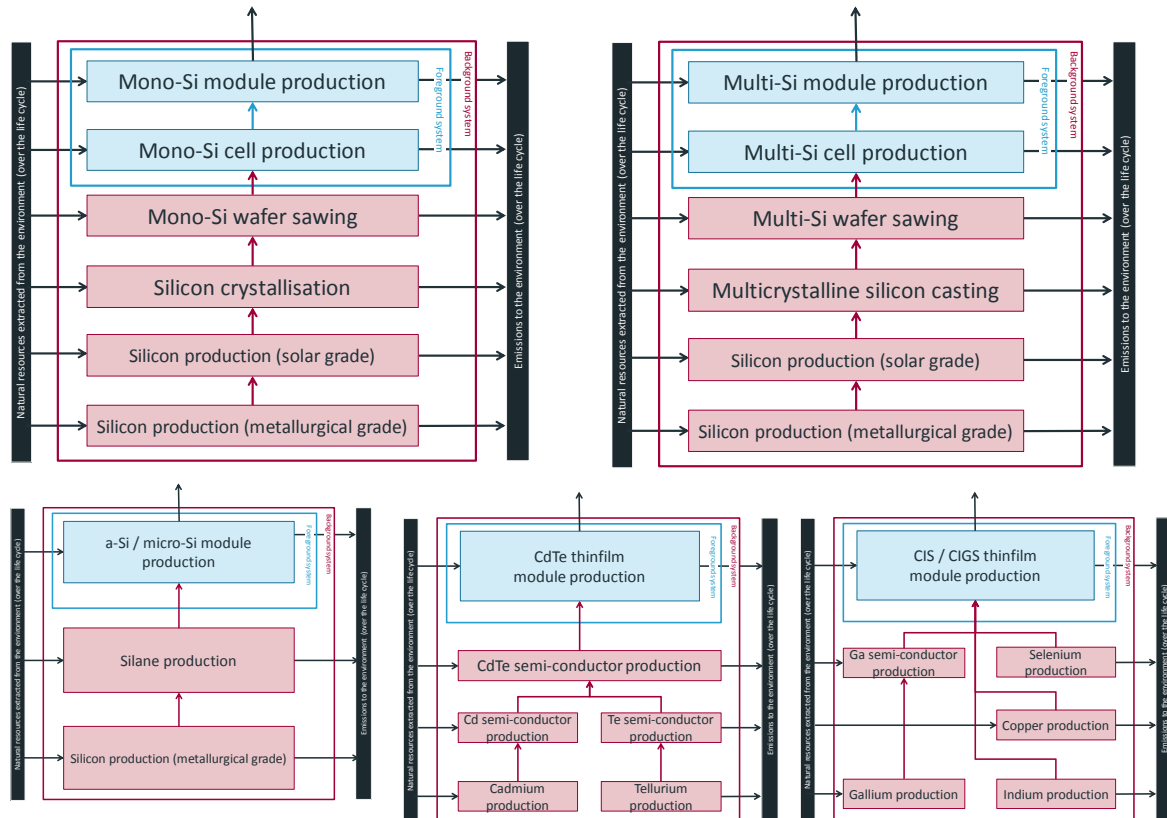


Fig. 3.2 Excerpt of the mono-Si, multi-Si, a-Si / micro-Si, CdTe and CIS / CIGS supply chain; Processes of the foreground and background system are marked with blue and red color, respectively

3.3.3 Use stage

The operation of the photovoltaic system represents the working stage of the panels. It includes the conversion of solar energy to electrical energy and the maintenance of the photovoltaic power system (cleaning). There are no significant direct emissions or significant processes during the operation of the photovoltaic system.

The annual average yield of optimally oriented modules in Europe, weighted according to the cumulative installed photovoltaic power corresponds to 1090 kWh/kWp when excluding degradation effects (Frischknecht & Itten 2014, see also section 4.9.7). According to producer consensus and applicable performance warranties as given on today's commercially available products, the expected life time of the photovoltaic panels is assumed to be 30 years. If, according to Frischknecht and Itten (Frischknecht & Itten 2014), a degradation of 0.7 % is applied, the performance is 21 % lower during the last year of the operation compared to the first year of operation. Hence the effective, weighted performance of the panels is 10.5 % below the average annual yield and results in 975 kWh/kWp, which is used in this screening LCA.

According to Jungbluth et al. (2012), 2 % of the modules are replaced and 1 % are rejects. This corresponds to a rate of failure of 3 % during the lifetime of the photovoltaic power system. The

manufacture of substitute panels is taken into account. According to First Solar CdTe panels show a breakage rate of 1 %³ (Sinha et al. 2012).

3.3.4 End-of-Life stage

The end-of-life stage includes the dismantling of the photovoltaic module and the mounting structure including the transport to a recycling plant. PV CYCLE is the leading European organization offering EU-law compliant take-back and recycling solutions for all commercially available PV modules. The recycling process is modelled in accordance with the WEEE requirements (EU Parliament 2012), which require minimum recovery and recycling rates. Photovoltaic modules are classified in category 4 “consumer equipment and photovoltaic panels”. The recovery rate shall be 75 % until August 2015 and increase to 85 % beyond August 2018. The recycling rate shall be 65 % until August 2015 and increase to 80 % (being prepared for re-use and recycling). During the recycling process bulk materials such as aluminium (frame) and glass are recovered to meet these minimum recovery targets. As there is significant uncertainty around the regulation of reuse of electricity generating equipment (especially in view of product and consumer safety), reuse of PV modules will not be assessed in the pilot.

3.4 Assumptions and value judgments

The main assumptions and value judgments are related to the annual yield, the service life of the PV system and the degradation of the panel.

The annual yield of the photovoltaic system depends on the location, the orientation, the degradation as well as the mounting. This dependence is very similar for all technologies analyzed. In this screening the average yield of an optimally oriented PV panel installed in Europe is chosen. This assumption determines the absolute environmental impacts per kWh of PV electricity but not the values relative to a reference PV technology.

In this screening LCA the degradation rate and the expected life time are the same for all technologies, even if in reality these may vary depending on the technology applied. Differences in published degradation rates are low and are based on historical modules (Jordan & Kurtz 2012) so they are not varied in this analysis. An agreed definition of the service life of panels does not exist so the service life is not varied among technologies. Further value judgments are involved in the treatment of multi-functionality and of recycling (see subchapter 3.5).

3.5 Treatment of multi-functionality

If a process or facility provides more than one function, i.e. it delivers several goods or services (“co-products”), it is “multifunctional”. In these situations, all inputs and emissions linked to the process must be partitioned between the product of interest and the other co-products in a consistent manner (European Commission 2013).

In this study multi-functionality occurs for the residential scale PV system, integrated in a slanted roof top. The PV panels fulfill simultaneously the function of roof tiles, as well as of a power plant. In this study no multi-functionality is accounted for. All resource requirements and emissions of manufacturing, operating and dismantling the PV system are attributed to the electricity produced. The roof tile function of building integrated PV systems bears no environmental impacts nor are any credits given for savings in roof tiles. As this report does not deal with comparisons of complete buildings, this simplification is assumed to be appropriate.

³

E-mail communication with First Solar, Mr. Parikhith Sinha, 9.9.2014

Multi-functionality of recycling processes is accounted for. It is assumed that the PV modules and mounting systems are recycled at the end-of-life (see section 3.3.4). The recycling processes modelled in the screening LCA and in the PEF supporting studies comply with the WEEE directive on the disposal of electrical and electronic equipment (EU Parliament 2012). Potential future environmental benefits which result from recycling are allocated according to the formula provided in the recommendation of the European Commission (2013). 50 % of the potential future environmental benefits are allocated to the PV system delivering the goods for recycling, the remaining 50 % are allocated to the product system reusing the recycled goods in the future. This represents the base case according to the PEF Guide.

In the sensitivity analyses two different allocation approaches are applied (see section 5.11.2):

- recycled content approach (100 % / 0 %)
- the avoided burden approach (0 % / 100 %)

Both alternative approaches are modelled according to the formulas provided in the PEF guidance document (European Commission 2015).

3.6 Information about the data used and data gaps

The inventory data are highly accurate as the data for the production of the PV panels and its supply chain mainly stem from producers. Inventory data describing the supply chain of the monocrystalline-Si, and multicrystalline-Si PV panels were provided by leading manufacturers. Inventory data describing the supply chain of thin film PV panels stem from FirstSolar (CdTe), Oerlikon Solar (now TEL, micromorphous silicon) and Avancis and Solar Frontier (CIS). The CIS inventory data are from 2010 and published by SmartGreenScans in 2014 (de Wild-Scholten 2014). All data come with uncertainty information.

All data used in this study meet “true and fair” quality standards. “True and fair” is understood as a general guideline to ensure that all data and information used are representative for reality and hence are not influenced by prejudices, personal views or deliberate deceptions (true). Furthermore the data and information used are free of financial or other interconnections or dependencies and are transparent and creditable (fair).

Data quality was assessed according to requirements of the European Commission (2013). Six quality criteria were adopted for PEF studies, five relating to the data and one to the method. These were summarized in the characteristics of completeness, precision, technological, geographical and time-related representativeness, which describe to what degree the processes and products selected are depicting the system analyzed.

In the screening step a minimum “fair” quality data rating is required for data contributing to at least 90 % of the impact estimated for each EF impact category, as assessed via a qualitative expert judgment (European Commission 2013). The parameter uncertainty was difficult to obtain. It was assumed that the uncertainty is low (10 %-20 % uncertainty according to the requirements of the European Commission (2013)). However even if the data uncertainty would be very high, all data would fulfill the required fair quality level.

All datasets used scored good quality or higher. A minimum “fair” quality was met and excelled in all cases. The following Tab. 3.2 shows an overview of the data quality rating achieved for all life cycle stages of the solar electricity production and all PV technologies for the mounted installation. For the

overview the data quality of the processes causing 90 % of the environmental impact per category is considered.

Tab. 3.2 Overview of data quality (scores and level) of electricity production of all PV technologies per life cycle stage, mounted on a slanted roof top, residential scale. Regarded in this overview are the processes causing 90 % of the impact per impact category. Data quality scores were assessed according to requirements of the European Commission (2013)

electricity production with:	Product stage		Construction stage		Use stage		End-of-life stage		Potential benefits due to recycling	
	data quality scores	data quality level	data quality scores	data quality level	data quality scores	data quality level	data quality scores	data quality level	data quality scores	data quality level
CdTe PV technology	1.5	Excellent data quality	1.7	Very good data quality	1.7	Very good data quality	1.8	Very good data quality	1.9	Very good data quality
CIS PV technology	1.9	Very good data quality	1.8	Very good data quality	1.7	Very good data quality	1.9	Very good data quality	2.0	Very good data quality
micro-Si PV technology	2.3	Good data quality	1.7	Very good data quality	1.5	Excellent data quality	1.9	Very good data quality	2.0	Very good data quality
multi-Si PV technology	2.5	Good data quality	1.7	Good data quality	1.5	Excellent data quality	1.9	Very good data quality	2.0	Very good data quality
mono-Si PV technology	2.5	Good data quality	1.7	Good data quality	1.5	Excellent data quality	1.9	Very good data quality	2.0	Very good data quality

Detailed data quality scores for all PV technologies per life cycle stage are listed in ANNEX I. Data quality was assessed in detail only for the electricity production of PV systems mounted on roof tops, residential scale. The data quality for the integrated system and the open ground system is comparable to the data quality for the mounted system.

3.7 Impact categories, models and indicators;

The default 15 environmental footprint impact category indicators are used (see Tab. 3.4). Long-term emissions are excluded. Three additional indicators are proposed to complement the environmental profile:

- renewable cumulative energy demand
- non-renewable cumulative energy demand
- nuclear waste

The indicators covering the cumulative energy demand are required by international guidelines regarding photovoltaic electricity production by ADEME (Payet et al. 2013) and IEA-PVPS (Frischknecht et al. 2015b), and by the PCR of the International EPD System (PCR CPC 171 & 173 2013). They are also part of the indicators of the CEN standard EN 15804:2012+A1:2013 on core rules for the product category of construction products of environmental product declarations (EN 15804 2013). In this study, the renewable cumulative energy demand is determined by the harvested solar energy by the photovoltaic panel. However, a second approach suggested by VDI (2012) quantifies the solar energy radiated onto the panel and thereby takes the efficiency of the PV panel into account. The sensitivity of the renewable cumulative energy demand to the choice of either of these two concepts is demonstrated in subchapter 5.2.

The amount of nuclear waste is also mandatory in the IEA-PVPS methodological guidelines and in the EN 15804 (2013). However, the amount (mass or volume) of nuclear wastes does not consider the radiotoxicity of the different types of nuclear waste.

The hazardousness of radioactive wastes depends on their persistence (half-life), and on the type and intensity of their radiation.

The radiotoxicity of the individual isotopes contained in the various types of radioactive waste is quantified using the internationally discussed and accepted radiotoxicity index (RTI) of radioactive wastes (see e.g. Miyahara et al. 2001; OECD 2007). It is defined as

$$RTI = \sum_i A_i \times DF_i/DL$$

with:

A_i = activity of the nuclide i in Bq,

DF_i = dose factor for the ingestion of the nuclide i in Sv/Bq, dose limit value for the release from a deep geological repository

$DL = 0.1$ mSv/a (NAGRA 2008).

The specific RTI of the different standard types of nuclear wastes is used as the basis for characterization. The high level wastes are used as the reference substance (see Tab. 3.3).

Tab. 3.3 Radiotoxicity index (RTI) per m^3 of waste and characterization factors

	RTI per m^3 (m^{-3})	Characterization factor (cm^3 HLW-eq/ cm^3)
Low-level and medium-level wastes (LMLW)	$4.1 * 10^7$	0.000045
Alpha-toxic wastes (ATW)	$1.4 * 10^9$	0.0015
High-level wastes and spent fuel (HLW & SF)	$9.3 * 10^{11}$	1.0
High-level and alpha-toxic wastes (HLW, SF & ATW)	$7.1 * 10^{11}$	0.76

The impact category indicator radiotoxicity index representing the impact category nuclear waste has been developed within the ecological scarcity method 2013 (Frischknecht & Büsser Knöpfel 2013; 2014).

Tab. 3.4 List of life cycle impact assessment indicators required by the PEF Guide and considered relevant for this PEFCR

Impact category	Indicator	Source
Indicators required according to the PEF guide		
Climate change	Radiative forcing as Global Warming Potential (GWP100) [kg CO ₂ eq.]	IPCC 2007
Ozone depletion	Ozone Depletion Potential (ODP) [kg CFC-11 eq.]	WMO 1999
Human toxicity, cancer effects	Comparative Toxic Unit for humans [CTUh, c]	Rosenbaum et al. 2008
Human toxicity, non-cancer effects	Comparative Toxic Unit for humans [CTUh, nc]	Rosenbaum et al. 2008
Particulate matter / respiratory effects	Intake fraction for fine particles [kg PM _{2.5} eq.]	Humbert 2009
Ionizing radiation, human health	Human exposure efficiency relative to U ²³⁵ [kBq U ²³⁵ eq.]	Frischknecht et al. 2000
Photochemical ozone formation	Tropospheric ozone concentration increase [kg NMVOC eq.]	Van Zelm et al. 2008 as applied in ReCiPe
Acidification	Accumulated Exceedance (AE) [mol H ⁺ eq.]	Seppälä et al. 2006, Posch et al. 2008
Eutrophication, terrestrial	Accumulated Exceedance (AE) [mol N eq.]	Seppälä et al. 2006, Posch et al. 2008
Eutrophication, freshwater	Fraction of nutrients reaching freshwater end compartment (P) [kg P eq.]	Struijs et al. 2009as implemented in ReCiPe
Eutrophication, marine	Fraction of nutrients reaching marine end compartment (N) [kg N eq.]	Struijs et al. 2009as implemented in ReCiPe
Ecotoxicity, freshwater	Comparative Toxic Unit for ecosystems [CTUe]	Rosenbaum et al. 2008
Land use	Soil Organic Matter [kg C deficit]	Milà i Canals et al. 2007
Resource depletion, water	Water abstraction related to local scarcity of water [m ³ water eq.]	Frischknecht et al. 2008
Resource depletion, mineral, fossil, renewable	Scarcity [kg Sb eq.]	Guinée et al. 2001
Additional indicators		
Cumulative energy demand, renewable	Gross energy content of primary energy resources [MJ oil eq.]	Frischknecht et al. 2007
Cumulative energy demand, non-renewable	Gross energy content of primary energy resources [MJ oil eq.]	Frischknecht et al. 2007
Nuclear waste	Radiotoxicity index, RTI [m ³ HAA eq.]	Frischknecht & Büsser Knöpfel 2013; 2014

3.8 Normalization and weighting factors

The updated recommended normalization factors and equal weighting factors are the basis to determine the environmental relevance of the different environmental impacts (Benini et al. 2014, p. 92).

The normalization and weighting factors of the impact category indicators analysed in this study are compiled in Tab. 3.5. The additional indicators are not taken into account in the normalized and weighted results (i.e., they have normalization and weighting factors equal to 0).

Tab. 3.5 Normalization factors of the impact category indicators used in the PEF screening LCA

Impact category	Indicator	Normalization factor	Weighting factor
Indicators required according to the PEF guide			
Climate change	Radiative forcing as Global Warming Potential (GWP100)	9.22E+03 kg CO ₂ eq.	1/15
Ozone depletion	Ozone Depletion Potential (ODP)	2.16E-02 kg CFC-11 eq.	1/15
Human toxicity, cancer effects	Comparative Toxic Unit for humans (CTUh)	3.69E-05 CTUh, c	1/15
Human toxicity, non-cancer effects	Comparative Toxic Unit for humans (CTUh)	5.33E-04 CTUh, n-c	1/15
Particulate matter / respiratory effects	Intake fraction for fine particles (kg PM _{2.5} -eq/kg)	3.80E+00 kg PM _{2.5} eq.	1/15
Ionising radiation, human health	Human exposure efficiency relative to U ²³⁵	1.13E+03 kBq U ²³⁵ eq. (to air)	1/15
Photochemical ozone formation	Tropospheric ozone concentration increase	3.17E+01 kg NMVOC eq.	1/15
Acidification	Accumulated Exceedance (AE)	4.73E+01 mol H ⁺ eq.	1/15
Eutrophication, terrestrial	Accumulated Exceedance (AE)	1.76E+02 mol N eq.	1/15
Eutrophication, freshwater	Fraction of nutrients reaching freshwater end compartment (P)	1.48E+00 kg P eq.	1/15
Eutrophication, marine	Fraction of nutrients reaching marine end compartment (N)	1.69E+01 kg N eq.	1/15
Ecotoxicity, freshwater	Comparative Toxic Unit for ecosystems (CTUe)	8.74E+03 CTUe	1/15
Land use	Soil Organic Matter	7.48E+04 kg C deficit	1/15
Resource depletion, water	Water abstraction related to local scarcity of water	8.14E+01 m ³ water eq.	1/15
Resource depletion, mineral, fossil, renewable	Scarcity	1.01E-01 kg Sb eq.	1/15
Additional indicators			
Cumulative energy demand, renewable	Gross energy content of total primary energy resources	0 MJ oil eq.	0
Cumulative energy demand, non-renewable	Gross energy content of total primary energy resources	0 MJ oil eq.	0
Nuclear waste	Radiotoxicity index, RTI	0 cm ³ HLW eq.	0

4. Compiling and recording the life cycle inventory analysis

4.1 Description and documentation of all unit process data

In the following subchapters the different technologies for producing PV panels, the installations for the PV systems as well as gaining electricity thereof are described in detail. The inventories are subdivided into production of the PV laminates and panels, the installation and mounting as well as the electricity generation and the end-of-life of the PV system. The inventories are listed as unit process data. Subchapter 4.2 describes the CdTe PV technology while in subchapter 4.3 CIS PV technology, in subchapter 4.4 micromorphous Si PV technology, in subchapter 4.5 multicrystalline Si PV and in subchapter 4.6 monocrystalline Si technology are listed. In subchapter 4.7, the life cycle inventories of the electric installation and the mounting structure are described. Subchapter 4.8 shows data collection procedures and in subchapter 4.9 are assumptions and methodological topics addressed.

The colored tables presented in the following subchapters describe the life cycle inventory datasets developed for this screening LCA. They refer to the EcoSpold v1 data format.

How to read an EcoSpold table:

The **light green fields** describe the name of the product/process, its region (e.g. RER stands for Europe) and the unit data it refers to. It is the output product (the reference output) of the process and always equal to '1'. The **yellow fields** show the inputs and outputs of the respective processes. The **grey fields** specify whether it is an input from or an output to nature or technosphere and the compartment to which a pollutant is emitted. For each product, additional descriptive information is given in separate tables.

The location codes (an extended ISO alpha-2 code-set) have the following meaning:

Regions:

<i>APAC</i>	Asia Pacific
<i>ENTSO</i>	European electricity network
<i>GLO</i>	Global
<i>OCE</i>	Oceanic
<i>RER</i>	Europe

Countries:

<i>AU</i>	Australia	<i>JP</i>	Japan
<i>CH</i>	Switzerland	<i>KR</i>	South Korea
<i>CN</i>	China	<i>NO</i>	Norway
<i>DE</i>	Germany	<i>NZ</i>	New Zealand
<i>ES</i>	Spain	<i>US</i>	United States of America

4.2 Cadmium-Telluride PV technology

4.2.1 Overview

The CdTe PV chain is built up according to the following structure:

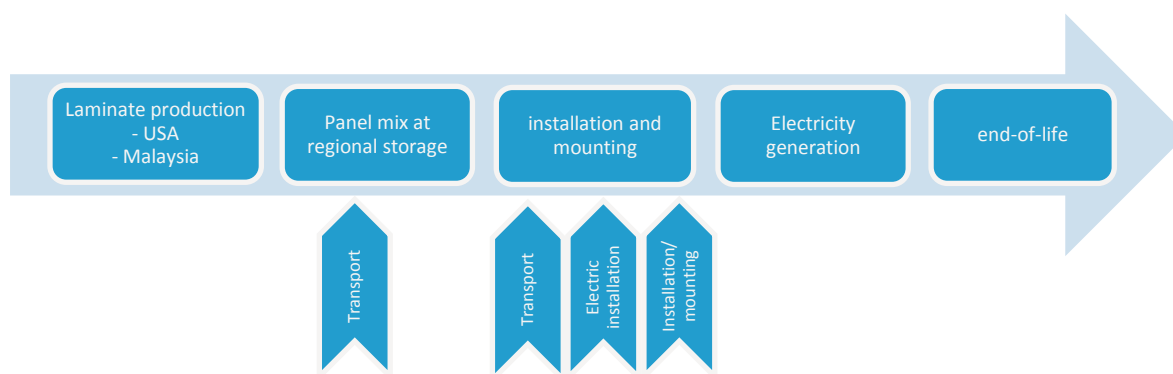


Fig. 4.1 Schematic illustration about the modeled life cycle chain of CdTe PV panels

The processes are described in the following sections 4.2.2 to 4.2.5. The following Tab. 4.1 summarizes the main characteristics of the CdTe PV panel.

Tab. 4.1 Characteristics of CdTe PV panels

	Unit	Amount/value
Panel size	m ² /panel	0.72
Panel weight (unframed)	kg/m ²	17.1
Module conversion efficiency	%	14
Wafer thickness	μm	2.5
Main data source / year	-	First Solar (2014)

4.2.2 Laminate and panel production

In the field of CdTe PV laminates only a few market players exist. Data are provided by First Solar. These data are considered representative for products from other CdTe PV producers (Seitz et al. 2013). CdTe PV laminates from First Solar are produced at two production facilities located in the United States of America and in Malaysia. The German plant is no longer producing CdTe PV laminates. The raw material inputs and working material as well as energy needed for the production of the laminates are considered. The amounts of waste and waste water and their fate (treatment) are considered as well. Direct airborne emissions of cadmium, copper, lead, nitric acid and waste-heat as well as waterborne emissions of cadmium, copper, lead and nitrate are taken into account. The emission data stem from First Solar US and is adopted for the production facility in Malaysia. Transport covers the logistics of supplying the raw materials to the production facility. Transport from the facility to Europe and the construction site is considered in downstream processes. The panel factory is modeled based on information for the US site and taken for the production site in Malaysia as well. The unit process inventory data for the production factory is described in Tab. 4.4.

The following Tab. 4.2 shows the unit process inventory data for the production of CdTe laminates in the US and Malaysia.

Tab. 4.2 unit process inventory data for the production of cadmium-telluride photovoltaic panels in the United States of America and in Malaysia according to First Solar

	Name	Location	Category	SubCategory	Infrastructure-Process	Unit	photovoltaic laminate, CdTe, at plant	photovoltaic laminate, CdTe, at plant	uncertainty Type	Standard Deviation 95%	GeneralComment
	Location						MY	US			
	InfrastructureProcess						1 m2	1 m2			
product	photovoltaic laminate, CdTe, at plant	MY	-	-	1	m2	1	0			
	photovoltaic laminate, CdTe, at plant	US	-	-	1	m2	0	1			
energy	electricity, medium voltage, at grid	MY	-	-	0	kWh	3.02E+1	-	1	1.07	(1,1,1,1,1,3,BU:1.05); 2010 data for First Solar in Malaysia
	electricity, medium voltage, at grid	US	-	-	0	kWh	-	2.95E+1	1	1.07	(1,1,1,1,1,3,BU:1.05); 2011 data for First Solar in US
	natural gas, burned in boiler modulating >100kW	RER	-	-	0	MJ	-	1.16E+1	1	1.07	(1,1,1,1,1,3,BU:1.05); 2010 data for First Solar in US
infrastructure	Photovoltaic panel factory CdTe	US	-	-	1	unit	4.00E-6	4.00E-6	1	3.04	(2,2,1,1,1,3,BU:3); 2013 data for First Solar in US
materials	cadmium telluride, semiconductor-grade, at plant	US	-	-	0	kg	2.34E-2	2.58E-2	1	1.07	(1,1,1,1,1,3,BU:1.05); 2010 data for First Solar in US
	cadmium sulphide, semiconductor-grade, at plant	US	-	-	0	kg	3.52E-3	3.52E-3	1	1.16	(1,4,3,3,1,3,BU:1.05); Fthenakis, literature, incl. Part of Cd compound powder
	copper, at regional storage	RER	-	-	0	kg	1.16E-2	1.10E-2	1	1.07	(1,1,1,1,1,3,BU:1.05); 2010 data for First Solar in US
	flat glass, uncoated, at plant	RER	-	-	0	kg	8.13E+0	8.25E+0	1	1.07	(1,1,1,1,1,3,BU:1.05); 2010 data for First Solar in US
	silicone product, at plant	RER	-	-	0	kg	3.07E-3	3.07E-3	1	1.08	(1,2,2,3,1,3,BU:1.05); Fthenakis, literature
	ethylvinylacetate, foil, at plant	RER	-	-	0	kg	4.86E-1	4.86E-1	1	1.07	(1,1,1,1,1,3,BU:1.05); 2010 data for First Solar in US
	glass fibre reinforced plastic, polyamide, injection moulding, at plant	RER	-	-	0	kg	1.08E-1	1.08E-1	1	1.16	(1,4,3,3,1,3,BU:1.05); Fthenakis, literature, sum up of several materials
	solar glass, low-iron, at regional storage	RER	-	-	0	kg	8.38E+0	8.47E+0	1	1.07	(1,1,1,1,1,3,BU:1.05); 2010 data for First Solar in US
	tempering, flat glass	RER	-	-	0	kg	8.38E+0	8.47E+0	1	1.07	(1,1,1,1,1,3,BU:1.05); 2010 data for First Solar in US
	tap water, at user	RER	-	-	0	kg	2.11E+2	1.32E+2	1	1.07	(1,1,1,1,1,3,BU:1.05); 2010 data for First Solar in US
auxiliaries	silica sand, at plant	DE	-	-	0	kg	4.68E-2	4.68E-2	1	1.16	(1,4,3,3,1,3,BU:1.05); Fthenakis, literature
	nitric acid, 50% in H2O, at plant	RER	-	-	0	kg	5.72E-2	5.72E-2	1	1.16	(1,4,3,3,1,3,BU:1.05); Fthenakis, literature
	sulphuric acid, liquid, at plant	RER	-	-	0	kg	3.93E-2	3.93E-2	1	1.16	(1,4,3,3,1,3,BU:1.05); Fthenakis, literature
	sodium chloride, powder, at plant	RER	-	-	0	kg	4.53E-2	4.53E-2	1	1.16	(1,4,3,3,1,3,BU:1.05); Fthenakis, literature
	sodium hydroxide, 50% in H2O, production mix, at plant	RER	-	-	0	kg	4.93E-2	4.93E-2	1	1.16	(1,4,3,3,1,3,BU:1.05); Fthenakis, literature
	hydrogen peroxide, 50% in H2O, at plant	RER	-	-	0	kg	1.67E-2	1.67E-2	1	1.16	(1,4,3,3,1,3,BU:1.05); Fthenakis, literature
	isopropanol, at plant	RER	-	-	0	kg	2.08E-3	2.08E-3	1	1.16	(1,4,3,3,1,3,BU:1.05); Fthenakis, literature
	chemicals inorganic, at plant	GLO	-	-	0	kg	3.76E-2	3.76E-2	1	1.07	(1,1,1,1,1,3,BU:1.05); 2010 data for First Solar in US
	chemicals organic, at plant	GLO	-	-	0	kg	9.74E-3	9.74E-3	1	1.16	(1,4,3,3,1,3,BU:1.05); Fthenakis, literature, sum up of several chemicals
	nitrogen, liquid, at plant	RER	-	-	0	kg	7.32E-2	7.32E-2	1	1.16	(1,4,3,3,1,3,BU:1.05); Fthenakis, literature
	helium, at plant	GLO	-	-	0	kg	3.64E-2	3.64E-2	1	1.16	(1,4,3,3,1,3,BU:1.05); Fthenakis, literature
	corrugated board, mixed fibre, single wall, at plant	RER	-	-	0	kg	5.22E-1	5.22E-1	1	1.07	(1,1,1,1,1,3,BU:1.05); 2010 data for First Solar in US
transport	transport, lorry >16t, fleet average	RER	-	-	0	tkm	4.13E-1	7.75E+0	1	2.00	(1,1,1,1,1,3,BU:2); 2010 data for First Solar in US
	transport, freight, rail	RER	-	-	0	tkm	5.35E+0	-	1	2.00	(1,1,1,1,1,3,BU:2); 2010 data for First Solar in Malaysia
	transport, transoceanic freight ship	OCE	-	-	0	tkm	2.31E+2	-	1	2.00	(1,1,1,1,1,3,BU:2); 2010 data for First Solar in Malaysia
disposal	disposal, municipal solid waste, 22.9% water, to municipal incineration	CH	-	-	0	kg	3.00E-2	3.00E-2	1	1.16	(1,4,3,3,1,3,BU:1.05); Alsema (personal communication) 2007, production waste
	disposal, plastics, mixture, 15.3% water, to municipal incineration	CH	-	-	0	kg	7.08E-1	7.08E-1	1	1.16	(1,4,3,3,1,3,BU:1.05); Calculation
	treatment, sewage, unpolluted, to wastewater treatment, class 3	CH	-	-	0	m3	-	6.16E-2	1	1.07	(1,1,1,1,1,3,BU:1.05); 2010 data for First Solar in US
emissions air	Heat, waste	-	air	high population density	-	MJ	2.09E+2	2.09E+2	1	1.29	(3,4,3,3,1,5,BU:1.05); Calculation
	Cadmium	-	air	high population density	-	kg	5.34E-9	5.34E-9	1	5.00	(1,1,1,1,1,3,BU:5); 2010 data for First Solar in US
	Copper	-	air	high population density	-	kg	7.56E-9	7.56E-9	1	5.00	(1,1,1,1,1,3,BU:5); 2010 data for First Solar in US
	Lead	-	air	high population density	-	kg	4.44E-9	4.44E-9	1	5.00	(1,1,1,1,1,3,BU:5); 2010 data for First Solar in US
	Nitric acid	-	air	high population density	-	kg	4.30E-5	4.30E-5	1	5.00	(1,1,1,1,1,3,BU:5); 2010 data for First Solar in US
emissions water	Cadmium	-	water	unspecified	-	kg	4.43E-7	4.43E-7	1	3.00	(1,1,1,1,1,3,BU:3); 2010 data for First Solar in US
	Copper	-	water	unspecified	-	kg	1.92E-6	1.92E-6	1	3.00	(1,1,1,1,1,3,BU:3); 2010 data for First Solar in US
	Lead	-	water	unspecified	-	kg	1.72E-7	1.72E-7	1	3.00	(1,1,1,1,1,3,BU:3); 2010 data for First Solar in US
	Nitrate	-	water	unspecified	-	kg	1.50E-2	1.50E-2	1	3.00	(1,1,1,1,1,3,BU:3); 2010 data for First Solar in US

Photovoltaic CdTe panels on the European market are transported from the US and Malaysia by freight ship, rail and lorry to the European regional storage. Around 15 % are imported from the US, the remaining 85 % are imported from Malaysia. It is assumed that Rotterdam is the import-port for Europe. From Rotterdam the panels are transported by lorry to the regional storage. The three main countries producing solar energy in Europe are Germany (60 %), Italy (31 %) and Spain (9 %). The average weighted transport distance to these countries is about 950 km. The unit process inventory data for the CdTe laminates at the European regional storage are listed in Tab. 4.3.

Tab. 4.3 unit process inventory data for cadmium-telluride photovoltaic panels at the European regional storage

	Name	Location	Infrastructure	Process	Unit	photovoltaic laminate, CdTe, mix, at regional storage	UncertaintyType	StandardDeviation95%	GeneralComment
	Location								
	Infrastructure	Process							
	Unit								
product	photovoltaic laminate, CdTe, mix, at regional storage	RER	1	m2	1				
materials	photovoltaic laminate, CdTe, at plant	US	1	m2	1.55E-1	1	3.03	(1,1,1,1,1,3); 2013 CdTe module production from US	
	photovoltaic laminate, CdTe, at plant	MY	1	m2	8.45E-1	1	3.03	(1,1,1,1,1,3); 2013 CdTe module production from Malaysia	
transport	transport, transoceanic freight ship	OCE	0	tkm	2.49E+2	1	2.09	(4,5,na,na,na,na); Import of modules from the US 6469 km, from Malaysia 14783 km	
	transport, lorry >16t, fleet average	RER	0	tkm	1.56E+1	1	2.09	(4,5,na,na,na,na); average transport distance from Rotterdam to Europe is 943 km.	

The unit process inventory data for the production site in the US (infrastructure) was updated in 2013 by First Solar and is described in Tab. 4.4. Due to the production philosophy applied in the factory (identical production lines), this panel factory is also appropriate for the Malaysian production site.

Tab. 4.4 unit process inventory data for the production site for cadmium-telluride photovoltaic panels according to First Solar for the production site in the United States of America

	Name	Location	Infrastructure	Process	Unit	photovoltaic panel factory CdTe			UncertaintyType		StandardDeviation95%	GeneralComment
	Location					US						
	Infrastructure					1						
	Unit					unit						
product	Photovoltaic panel factory CdTe	US	1	unit	1.00E+0							
infrastructure	Building, hall	CH	1	m2	7.99E+2	1	2.00	(1,2,1,1,1,3); 2013 data for First Solar in US				
	Metal working machine, unspecified, at plant	RER	1	kg	4.00E+3	1	2.00	(5,5,1,1,1,5); rough assumption, 4t weight per laminator				
resources	Occupation, industrial area, built up	-	0	m2a	4.69E+3	1	1.51	(1,2,1,1,1,3); 25a occupation				
	Occupation, industrial area, vegetation	-	0	m2a	1.01E+3	1	1.51	(1,2,1,1,1,3); 25a occupation				
	Occupation, traffic area, road network	-	0	m2a	3.63E+3	1	1.51	(1,2,1,1,1,3); 25a occupation				
	Transformation, from unknown	-	0	m2	3.73E+2	1	2.00	(1,2,1,1,1,3); 2013 data for First Solar in US				
	Transformation, to industrial area, built up	-	0	m2	1.88E+2	1	2.00	(1,2,1,1,1,3); 2013 data for First Solar in US				
	Transformation, to industrial area, vegetation	-	0	m2	4.00E+1	1	2.00	(1,2,1,1,1,3); 2013 data for First Solar in US				
	Transformation, to traffic area, road network	-	0	m2	1.45E+2	1	2.00	(1,2,1,1,1,3); 2013 data for First Solar in US				

4.2.3 Installation and mounting

The CdTe panels are mounted on three different systems: 3 kWp installation, integrated in a slanted roof-top, 3 kWp installation, mounted on a slanted roof-top and a 570 kWp installation on open ground.

The mounting of the panels consists of the electric installation, the construction for mounting, the panels as well as electricity consumption during installation (screwdrivers, lifts). The electric installation and the construction are similar for all PV technologies and are described in sections 4.7.1 and 4.7.2.

The panel and construction material demand for all three installations is calculated based on an efficiency of 14 % (FirstSolar 2014). According to First Solar the panels show 1 % of field breakage over a period of 25 years (Sinha et al. 2012). This is considered when calculating the panel demand per system. The roof systems need 21.4 m² of installation area, while the open ground system consists of 4'070 m² module area. The following Tab. 4.5 to Tab. 4.7 describe the inventory for each photovoltaic system with CdTe PV panels. 100 km lorry transport of the panels, the electric installation and the mounting structure from the regional storage to the construction site is considered.

Tab. 4.5 unit process inventory data for the mounting cadmium-telluride photovoltaic panels integrated in a slanted roof top

	Name	Location	Category	SubCategory	InfrastructureProcess	Unit	3kWp slanted-roof installation, CdTe, laminated, integrated, on roof	UncertaintyType	StandardDeviation95%	GeneralComment
	Location									
	InfrastructureProcess									
	Unit						RER 1 unit			
product	3kWp slanted-roof installation, CdTe, laminated, integrated, on roof	RER	-	-	1	unit	1.00E+0			
energy	electricity, low voltage, production ENTSO, at grid	ENTSO	-	-	0	kWh	4.00E-2	1	1.28	(3,4,3,1,1,5); Energy use for erection of 3kWp plant
materials	electric installation, photovoltaic plant, at plant	CH	-	-	1	unit	1.00E+0	1	2.09	(3,4,3,1,1,5); Literature
	slanted-roof construction, integrated, on roof	RER	-	-	1	m2	2.14E+1	1	1.23	(3,1,1,1,1,na); New estimation with mean value of frame weights, correction for panel area
	photovoltaic laminate, CdTe, mix, at regional storage	RER	-	-	1	m2	2.16E+1	1	1.36	(1,1,1,1,1,3); Calculation, 1% rejects
transport	transport, lorry 3.5-16t, fleet average	RER	-	-	0	tkm	4.25E+1	1	2.09	(3,4,3,1,1,5); 100km for transport of panels from regional storage, electrical installatio and the mounting structure to construction site
emissions air	Heat, waste	-	air	high population density	-	MJ	1.44E-1	1	1.28	(3,4,3,1,1,5); calculated with electricity use

Tab. 4.6 unit process inventory data for the mounting cadmium-telluride photovoltaic panels mounted on a slanted roof top

	Name	Location	Category	SubCategory	InfrastructureProcess	Unit	3kWp slanted-roof installation, CdTe, panel, mounted, on roof	UncertaintyType	StandardDeviation95%	GeneralComment
	Location									
	InfrastructureProcess									
	Unit						RER 1 unit			
product	3kWp slanted-roof installation, CdTe, panel, mounted, on roof	RER	-	-	1	unit	1.00E+0			
energy	electricity, low voltage, production ENTSO, at grid	ENTSO	-	-	0	kWh	4.00E-2	1	1.28	(3,4,3,1,1,5); Energy use for erection of 3kWp plant
materials	electric installation, photovoltaic plant, at plant	CH	-	-	1	unit	1.00E+0	1	2.09	(3,4,3,1,1,5); Literature
	slanted-roof construction, mounted, on roof	RER	-	-	1	m2	2.14E+1	1	1.23	(3,1,1,1,1,na); New estimation with mean value of frame weights, correction for panel area
	photovoltaic laminate, CdTe, mix, at regional storage	RER	-	-	1	m2	2.16E+1	1	1.36	(1,1,1,1,1,3); Calculation, 1% rejects
transport	transport, lorry 3.5-16t, fleet average	RER	-	-	0	tkm	4.66E+1	1	2.09	(3,4,3,1,1,5); 100km for transport of panels from regional storage, electrical installatio and the mounting structure to construction site
emissions air	Heat, waste	-	air	high population density	-	MJ	1.44E-1	1	1.28	(3,4,3,1,1,5); calculated with electricity use

Tab. 4.7 unit process inventory data for the mounting cadmium-telluride photovoltaic panels open ground installation

	Name	Location	Category	Subcategory	Infrastructure	Process	Unit	570 kWp open ground installation, CdTe, on open ground	Uncertainty	Type	Standard Deviation	5%	General Comment
	Location							RER					
	Infrastructure							1 unit					
product	570 kWp open ground installation, CdTe, on open ground	RER	-	-	1	unit	1.00E+0						
energy	electricity, low voltage, production	ENTSO	-	-	0	kWh	3.60E+1	1	1.43	(3,4,3,1,1,5); scaled from a 3kWp plant over capacity			
	Diesel, burned in building machine	GLO	-	-	0	MJ	7.67E+3	1	1.41	(3,4,1,1,1,5); Leit-Ramm; Energy use for foundation piling and wheel loader			
materials	electric installation, 570 kWp photovoltaic plant, at plant	ES	-	-	1	unit	1.00E+0	1	2.16	(3,4,1,1,1,5);			
	open ground construction, on ground	RER	-	-	1	m2	4.07E+3	1	2.16	(3,4,1,1,1,5); calculation per m2 panel			
	photovoltaic laminate, CdTe, mix, at regional storage	RER	-	-	1	m2	4.11E+3	1	1.47	(1,1,1,1,1,3); Calculation, 1% rejects			
transport	transport, lorry 3.5-16t, fleet average	RER	-	-	0	tkm	1.14E+4	1	2.18	(3,4,3,1,1,5); 100km for transport of panels from regional storage, electrical installatio and the mounting structure to construction site			
emissions air	Heat, waste	-	air	high population density	-	MJ	1.30E+2	1	1.29	(1,na,na,na,na,na); calculated with electricity use			

4.2.4 Electricity generation

The 3 kWp roof installations and the 570 kWp open ground installation have an annual yield of 975 kWh/kWp (see also section 3.3.3). The expected life time is 30 years. Hence the overall output of electricity is 87.8 MWh for the roof installations and 16.7 GWh for the open ground system.

It is assumed that 20 liters of water are used per square meter for washing the modules (Jungbluth et al. 2012). This results in 4.88 ml of water per kWh electricity. The washing water is discharged into the sewer and to the municipal waste water treatment plant. No cleaning agents are considered.

The solar energy harvested by the PV system to produce electricity is calculated from the energy transformed to electricity fed into the inverter. Hence, 3.6 MJ solar energy are harvested to generate 1 kWh electricity. No direct emissions occur during the use stage of the panels.

The land occupation of the open ground PV system is taken into account in the mounting structure datasets. The land occupied by building roof integrated and mounted systems is fully attributed to the building. The following Tab. 4.8 shows the unit process inventory data of the electricity production with CdTe PV systems.

Tab. 4.8 unit process inventory data for the electricity production with cadmium-telluride photovoltaic panels at a 3 kWp roof installation, mounted and integrated, and a 570 kWp open ground installation

	Name	Location	Category	Sub-category	Process	Unit	electricity, PV, at 3kWp slanted-roof, CdTe, laminated, integrated			electricity, PV, at 3kWp slanted-roof, CdTe, panel, mounted			electricity, PV, at 570 kWp open ground, CdTe			Uncertainty	Type	Standard deviation 95%	General Comment
							RER			RER			RER						
							0			0			0						
							kWh			kWh			kWh						
	Location																		
	Infrastructure	Unit	Process																
product	electricity, PV, at 3kWp slanted-roof, CdTe, laminated, integrated	RER	-	-	0	kWh	1	0	0										
	electricity, PV, at 3kWp slanted-roof, CdTe, panel, mounted	RER	-	-	0	kWh	0	1	0										
	electricity, PV, at 570 kWp open ground, CdTe	RER	-	-	0	kWh	0	0	1										
infrastructure	3kWp slanted-roof installation, CdTe, laminated, integrated, on roof	RER	-	-	1	unit	1.14E-5			1	3.00	(1,1,1,1,1,1,BU-3); infrastructure; calculation							
	3kWp slanted-roof installation, CdTe, panel, mounted, on roof	RER	-	-	1	unit		1.14E-5		1	1.24	(2,3,1,1,3,5,BU-3); infrastructure; calculation							
	570 kWp open ground installation, CdTe, on open ground	RER	-	-	1	unit			6.00E-8	1	1.24	(2,3,1,1,3,5,BU-3); infrastructure; calculation							
auxiliaries	tap water, at user	RER	-	-	0	kg	4.88E-3	4.88E-3	4.88E-3	1	1.05	(1,1,1,1,1,1,BU-1.05); water for washing, estimation 20 l per m2 panel; calculation							
disposal	treatment, sewage, from residence, to wastewater treatment, class 2	CH	-	-	0	m3	4.88E-6	4.88E-6	4.88E-6	1	1.05	(1,2,1,1,1,1,BU-1.05); waste water treatment of washing water; calculation							
resource in air	Energy, solar, converted	-	resource	in air	-	MJ	3.60E+0	3.60E+0	3.60E+0	1	1.20	(2,3,1,1,3,5,BU-1.05); solar energy needed for the production of 1kWh electricity (direct current) ; equivalent to 2							

4.2.5 *End-of-life and benefits for recycling*

At the end-of-life the PV panels, the electric installation and mounting structures are collected and transported to recycling plants. There the panels are shredded and dissolved in a chemical bath. The materials gained are sorted and prepared for recycling. This process requires electricity. The waste water and the waste material are disposed of in a waste water treatment plant and in a municipal incineration plant or inert material landfill, respectively.

The largest share of recycled materials is obtained from the flat glass and the solar glass, which is used as cullet input in recycled foam glass production. The benefits for glass recycling are calculated based on the production of flat glass since it is not possible to clearly identify the material inputs replaced by glass cullets compared to the production of primary foam glass. 11.4 kg glass cullets are recovered per panel (0.72 m^2). The difference to the mass of glass used in the production of CdTe PV modules according to the bill of materials (Tab. 3.1) represents manufacturing losses, which amount to approximately 3.8 %⁴.

The efficiency of glass recycling is 90 %⁵. Using recycled glass cullets in glass production avoids the consumption of primary materials such as limestone, silica sand and soda powder. Hence the geogenic CO₂-emissions caused by these raw materials in flat glass production are prevented (Held & Ilg 2011), which amount to 0.208 kg CO₂ eq/kg flat glass (Kellenberger et al. 2007). This results in avoided emissions of 2.97 kg CO₂ eq per m² recycled PV module. Because the melting of the glass cullets takes less energy than the melting of limestone and silica sand, heavy fuel oil and natural gas can additionally be saved. Held and Ilg (2011) assume an energy reduction potential in the melting process of around 30 % when replacing 100 % input material by glass cullets.

1 kg of unrefined semiconductor materials (USM) is recovered per 16.5 m² of module recycled. The molecular weights of cadmium and tellurium are used to assess the shares of each metal per kg USM.

Aluminium and steel used in the mounting structures as well as the copper and steel contained in the electric installation are likewise recycled. In line with Jungbluth et al. (2012) it is assumed that only aluminium, copper and steel are recycled but not the plastics. Being large construction parts, a recycling efficiency of 100 % is assumed.

Efforts to produce secondary metals from scrap are taken into account for aluminium and copper. These secondary metals are potentially avoiding primary aluminium and primary copper production, respectively. Steel scrap is potentially avoiding pig iron feedstock in converter steel production. Benefits for recycling are granted only for the net surplus amount of recycled material, which leaves the PV system (see also EN 15804 2013). According to the ecoinvent data v2.2+ the recycled content of the metals used in panels, cabling and mounting structures is 32 % for aluminium, 44 % for copper and 37 % for steel (KBOB et al. 2014). The application of the end-of-life formula is described in more detail in section 4.9.9.

The unit process inventory data of CdTe module takeback and recycling, including benefits and additional loads beyond the system boundary, are listed in Tab. 4.9.

⁴ Personal email communication with First Solar, Parikhith Sinha, 12.2.2016

⁵ Personal email communication with First Solar, Parikhith Sinha, 6.10.2014

Tab. 4.9 unit process inventory data for the recycling and the potential benefits and additional loads beyond the system boundary by recycling cadmium-telluride photovoltaic panels, cabling and mounting structures

	Name	Location	Category	SubCategory	InfrastructureProcess	Unit	CdTe PV module takeback + recycling	Avoided burden from recycling, CdTe PV module, integrated construction	Avoided burden from recycling, CdTe PV module, mounted construction	Avoided burden from recycling, CdTe PV module, open ground construction	Uncertainty Type	StandardDeviation5%	GeneralComment
							RER	RER	RER	RER			
							0 m2	0 m2	0 m2	0 m2			
product	CdTe PV module takeback + recycling	RER	-	-	0	m2	1.00E+0	0	0	0			
	Avoided burden from recycling, CdTe PV module, integrated construction	RER	-	-	0	m2	0	1.00E+0	0	0			
	Avoided burden from recycling, CdTe PV module, mounted construction	RER	-	-	0	m2	0	0	1.00E+0	0			
	Avoided burden from recycling, CdTe PV module, open ground construction	RER	-	-	0	m2	0	0	0	1.00E+0			
energy	Electricity, medium voltage, at grid	DE	-	-	0	kWh	4.38E+0				1	1.07	(2.4.1.1.1.3); Sinha, End-of-Life CdTe PV Recycling with Semiconductor Refining
auxiliaries	Water, deionised, at plant	CH	-	-	0	kg	5.42E+0				1	1.07	(2.4.1.1.1.3); Sinha, End-of-Life CdTe PV Recycling with Semiconductor Refining
	Sulphuric acid, liquid, at plant	RER	-	-	0	kg	8.33E-2				1	1.07	(2.4.1.1.1.3); Sinha, End-of-Life CdTe PV Recycling with Semiconductor Refining
	Hydrogen peroxide, 50% in H2O, at plant	RER	-	-	0	kg	5.71E-1				1	1.07	(2.4.1.1.1.3); Sinha, End-of-Life CdTe PV Recycling with Semiconductor Refining
	Sodium hydroxide, 50% in H2O, production mix, at plant	RER	-	-	0	kg	1.04E-1				1	1.07	(2.4.1.1.1.3); Sinha, End-of-Life CdTe PV Recycling with Semiconductor Refining
transport	Transport, lorry >16t, fleet average	RER	-	-	0	tkm	1.25E+1				1	2.00	(2.4.1.1.1.3); Sinha, End-of-Life CdTe PV Recycling with Semiconductor Refining. Assumption: transport of mounting and electrical parts for 200 km
disposal	Treatment, PV cell production effluent, to wastewater treatment, class 3	CH	-	-	0	m3	4.79E-3				1	1.23	(2.4.1.1.1.3); Sinha, End-of-Life CdTe PV Recycling with Semiconductor Refining
	Disposal, plastics, mixture, 15.3% water, to municipal incineration	CH	-	-	0	kg	6.16E-1				1	1.23	(2.4.1.1.1.3); Sinha, End-of-Life CdTe PV Recycling with Semiconductor Refining
	Disposal, inert waste, 5% water, to inert material landfill	CH	-	-	0	kg	1.28E-1				1	1.23	(2.4.1.1.1.3); Sinha, End-of-Life CdTe PV Recycling with Semiconductor Refining
emissions air	Cadmium	-	air	unspecif	-	kg	5.89E-9				1	1.28	(2.4.1.1.1.3); Sinha, End-of-Life CdTe PV Recycling with Semiconductor Refining
emissions water	Cadmium	-	water	unspecif	-	kg	8.92E-8				1	2.09	(2.4.1.1.1.3); Sinha, End-of-Life CdTe PV Recycling with Semiconductor Refining
avoided energy	natural gas, burned in industrial furnace >100kW	RER	-	-	-	MJ		-1.96E+01	-1.96E+01	-1.96E+01	1	2.00	(2.4.1.1.1.3); Avoided primary glass production materials; Held and Ilg, Update of environmental indicators and energy payback time of CdTe PV systems in Europe
	heavy fuel oil, burned in industrial furnace 1MW, non-modulating	RER	-	-	-	MJ		-1.27E+01	-1.27E+01	-1.27E+01	1	2.00	(2.4.1.1.1.3); Avoided primary glass production materials; Held and Ilg, Update of environmental indicators and energy payback time of CdTe PV systems in Europe
avoided materials	silica sand, at plant	DE	-	-	-	kg		-8.27E+00	-8.27E+00	-8.27E+00	1	2.00	(2.4.1.1.1.3); Avoided primary glass production materials; Held and Ilg, Update of environmental indicators and energy payback time of CdTe PV systems in Europe
	soda, powder, at plant	RER	-	-	-	kg		-3.28E+00	-3.28E+00	-3.28E+00	1	2.00	(2.4.1.1.1.3); Avoided primary glass production materials; Held and Ilg, Update of environmental indicators and energy payback time of CdTe PV systems in Europe
	limestone, milled, packed, at plant	CH	-	-	-	kg		-5.72E+00	-5.72E+00	-5.72E+00	1	2.00	(2.4.1.1.1.3); Avoided primary glass production materials; Held and Ilg, Update of environmental indicators and energy payback time of CdTe PV systems in Europe
	copper, at regional storage	RER	-	-	-	kg		-3.84E-01	-3.84E-01	-1.02E-01	1	2.00	(2.4.1.1.1.3); Avoided primary copper production materials; Held and Ilg, Update of environmental indicators and energy payback time of CdTe PV systems in Europe; plus the copper avoided by recycling the electric installation. Recycling content of input copper is 44 % according to KBOB-list
	copper, secondary, at refinery	RER	-	-	-	kg		3.84E-01	3.84E-01	1.02E-01	1	2.00	(2.4.1.1.1.3); efforts for making secondary copper from scrap
	aluminium, primary, at plant	RER	-	-	-	kg		-1.53E+00	-1.93E+00	-2.70E+00	1	2.00	(2.4.1.1.1.3); Avoided primary aluminium production due to recycling of the mounting structure
	aluminium, secondary, from old scrap, at plant	RER	-	-	-	kg		1.53E+00	1.93E+00	2.70E+00	1	2.00	(2.4.1.1.1.3); Avoided primary aluminium production due to recycling of the mounting structure
	pig iron, at plant	GLO	-	-	-	kg		-1.52E-01	-9.70E-01	-4.55E+00	1	2.00	(2.4.1.1.1.3); Avoided pig iron production due to recycling of mounting structure and the electric installation
	cadmium sludge, from zinc electrolysis, at plant	GLO	-	-	-	kg		-2.84E-02	-2.84E-02	-2.84E-02	1	2.00	(2.4.1.1.1.3); Avoided unrefined semiconductor materials; Sinha, End-of-Life CdTe PV Recycling with Semiconductor Refining
	copper telluride cement, from copper production	GLO	-	-	-	kg		-3.22E-02	-3.22E-02	-3.22E-02	1	2.00	(2.4.1.1.1.3); Avoided unrefined semiconductor materials; Sinha, End-of-Life CdTe PV Recycling with Semiconductor Refining
avoided emissions air	Carbon dioxide, fossil	-	air	unspecif	-	kg		-2.97E+0	-2.97E+0	-2.97E+0	1	2.00	(2.4.1.1.1.3); Avoided CO2 in glass melting process; Held and Ilg, Update of environmental indicators and energy payback time of CdTe PV systems in Europe

4.3 Copper-Indium-Selenium (CIS) PV technology

4.3.1 Overview

The CIS PV chain is modeled according to the following structure:

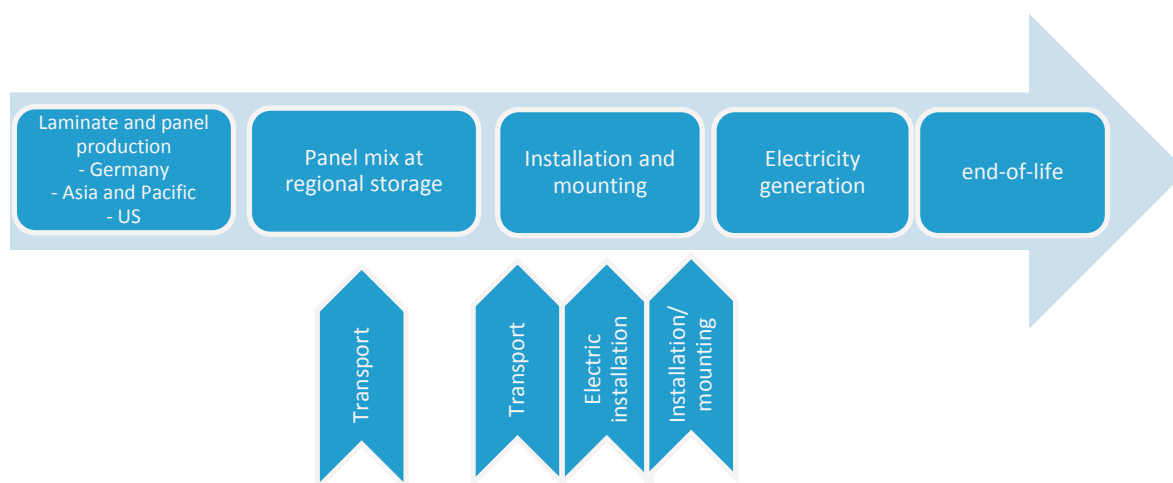


Fig. 4.2 Schematic illustration about the modeled life cycle chain of CIS PV panels

The processes are described in the following sections 4.3.2 to 4.3.5. The following Tab. 4.10 summarizes the main characteristics of the CIS PV panel.

Tab. 4.10 Characteristics of CIS PV panels

	Unit	Amount/value
Panel size	m ² /panel	0.72
Panel weight (unframed)	kg/m ²	14.9
Module conversion efficiency	%	10.8
Wafer thickness	μm	2.0
Main data source / year	-	de Wild-Scholten (2014)

4.3.2 Laminate and panel production

Copper-Indium-Selenium (CIS) PV laminates and panels are today produced commercially in US, France (Nexcis, expected 2015), Germany (Avancis ceased production in 2013) and in Japan (by Solar Frontier). Best available data came from Avancis, Germany, which ceased production in 2013, and Solar Frontier, Japan. The data are still considered as accurate. The inventory data are from 2010 and published by SmartGreenScans in 2014.

The feedstock inputs and material as well as energy needed for the production of the laminates are considered. The laminates are framed with an AlMg₃ alloy frame. 2.2 kg of metal are used per square meter of panel. Electricity (for laminate production) and light fuel oil (for panel production) are the energy carriers consumed in the laminate and panel production. Wastes and waste water and their appropriate treatment are taken into account. Direct airborne emissions of cadmium are taken into account. Transport covers the logistics of the raw materials to the production facility. The following Tab. 4.11 shows the unit process inventory data for the production of CIS laminates and panels.

Tab. 4.11 unit process inventory data for the production of CIS photovoltaic panels for the production in Germany

	Name	Location	Category	SubCategory	Infrastructure	Process	Unit	photovoltaic laminate, CIS, at plant	photovoltaic panel, CIS, at plant	Uncertainty Type	Standard Deviation 95%	General Comment
product	photovoltaic laminate, CIS, at plant	DE	-	-	1	m2	1.00E+0	0				
	photovoltaic panel, CIS, at plant	DE	-	-	1	m2	0	1.00E+0				
energy	electricity, medium voltage, at grid	DE	-	-	0	kWh	4.47E+1	-	1	1.07	(1,1,1,1,1,3); company information, coating, air-conditioning, water purification, etc.	
	light fuel oil, burned in industrial furnace 1MW, non-modulating	RER	-	-	0	MJ	-	1.55E+1	1	1.07	(1,1,1,1,1,3); Raugel, literature	
infrastructure	photovoltaic panel factory	GLO	-	-	1	unit	4.00E-6	-	1	3.02	(1,4,1,3,1,3); Assumption	
materials	photovoltaic laminate, CIS, at plant	DE	-	-	1	m2	-	1.00E+0	1	3.00	(1,1,1,1,1,3); Assumption	
	aluminium alloy, AlMg3, at plant	RER	-	-	0	kg	-	2.20E+0	1	1.07	(1,1,1,1,1,3); company information	
	copper, at regional storage	RER	-	-	0	kg	9.77E-3	-	1	1.07	(1,1,1,1,1,3); company information	
	wire drawing, copper	RER	-	-	0	kg	9.77E-3	-	1	1.07	(1,1,1,1,1,3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)	
	aluminium, production mix, at plant	RER	-	-	0	kg	4.44E-2	-	1	1.07	(1,1,1,1,1,3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)	
	flat glass, uncoated, at plant	RER	-	-	0	kg	5.27E+0	-	1	1.07	(1,1,1,1,1,3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)	
	diode, unspecified, at plant	GLO	-	-	0	kg	1.44E-3	-	1	1.07	(1,1,1,1,1,3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)	
	silicone product, at plant	RER	-	-	0	kg	4.04E-1	-	1	1.07	(1,1,1,1,1,3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)	
coating	molybdenum, at regional storage	RER	-	-	0	kg	6.06E-3	-	1	1.13	(3,2,2,1,1,3); company information and assumption for share of metals	
	indium, at regional storage	RER	-	-	0	kg	2.82E-3	-	1	1.13	(3,2,2,1,1,3); company information and assumption for share of metals	
	cadmium sulphide, semiconductor-grade, at plant	US	-	-	0	kg	2.69E-4	-	1	1.07	(1,1,1,1,1,3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)	
	gallium, semiconductor-grade, at regional storage	RER	-	-	0	kg	8.99E-4	-	1	1.13	(3,2,2,1,1,3); company information and assumption for share of metals	
	selenium, at plant	RER	-	-	0	kg	5.60E-3	-	1	1.13	(3,2,2,1,1,3); company information and assumption for share of metals	
	zinc, primary, at regional storage	RER	-	-	0	kg	-	-	1	1.13	(3,2,2,1,1,3); company information and assumption for share of metals	
	tin, at regional storage	RER	-	-	0	kg	1.23E-2	-	1	1.13	(3,2,2,1,1,3); company information and assumption for share of metals	
	solar glass, low-iron, at regional storage	RER	-	-	0	kg	7.70E+0	-	1	1.07	(1,1,1,1,1,3); company information	
	tempering, flat glass	RER	-	-	0	kg	7.70E+0	-	1	1.07	(1,1,1,1,1,3); Assumption	
	glass fibre reinforced plastic, polyamide, injection moulding, at plant	RER	-	-	0	kg	-	4.00E-2	1	1.07	(1,1,1,1,1,3); Raugel, literature	
	ethylvinylacetate, foil, at plant	RER	-	-	0	kg	7.51E-1	-	1	1.07	(1,1,1,1,1,3); company information	
	flux wave soldering, at plant	GLO	-	-	0	kg	1.23E-2	-	1	1.07	(1,1,1,1,1,3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)	
	zinc oxide, at plant	RER	-	-	0	kg	9.09E-3	-	1	1.07	(1,1,1,1,1,3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)	
	polyethylene terephthalate, granulate, amorphous, at plant	RER	-	-	0	kg	3.36E-1	-	1	1.07	(1,1,1,1,1,3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)	
	polyethylene, HDPE, granulate, at plant	RER	-	-	0	kg	4.84E-2	-	1	1.07	(1,1,1,1,1,3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)	
	polyvinylbutyral foil, at plant	RER	-	-	0	kg	1.89E-1	-	1	1.07	(1,1,1,1,1,3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)	
	polyphenylene sulfide, at plant	GLO	-	-	0	kg	8.59E-2	-	1	1.07	(1,1,1,1,1,3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)	
auxiliaries	tap water, at user	RER	-	-	0	kg	1.31E+2	-	1	1.07	(1,1,1,1,1,3); company information	
	argon, liquid, at plant	RER	-	-	0	kg	1.90E-2	-	1	1.07	(1,1,1,1,1,3); protection gas, company information	
	butyl acrylate, at plant	RER	-	-	0	kg	1.01E-1	-	1	1.07	(1,1,1,1,1,3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)	
	diborane, at plant	GLO	-	-	0	kg	2.01E-4	-	1	1.07	(1,1,1,1,1,3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)	
	sulphuric acid, liquid, at plant	RER	-	-	0	kg	3.31E-2	-	1	1.07	(1,1,1,1,1,3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)	
	hydrogen sulphide, H2S, at plant	RER	-	-	0	kg	1.91E-1	-	1	1.07	(1,1,1,1,1,3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)	
	sodium hydroxide, 50% in H2O, production mix, at plant	RER	-	-	0	kg	3.34E-2	-	1	1.07	(1,1,1,1,1,3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)	
	hydrogen peroxide, 50% in H2O, at plant	RER	-	-	0	kg	2.31E-2	-	1	1.07	(1,1,1,1,1,3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)	
	hydrochloric acid, 30% in H2O, at plant	RER	-	-	0	kg	9.94E-2	-	1	1.07	(1,1,1,1,1,3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)	
	nitrogen, liquid, at plant	RER	-	-	0	kg	1.57E+1	-	1	1.07	(1,1,1,1,1,3); protection gas, company information	
	ammonia, liquid, at regional storehouse	RER	-	-	0	kg	9.29E-2	-	1	1.07	(1,1,1,1,1,3); dip coating for CdS, company information	
	urea, as N, at regional storehouse	RER	-	-	0	kg	1.15E-3	-	1	1.16	(3,1,3,1,1,3); dip coating for CdS, Ampenberg 1998	
	EUR-flat pallet	RER	-	-	0	unit	5.00E-2	-	1	1.07	(1,1,1,1,1,3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)	
transport	transport, lorry >16t, fleet average	RER	-	-	0	tkm	1.70E+1	2.25E-1	1	2.09	(4,5,n,a,n,a,n,a); Standard distance 100km	
	transport, freight, rail	RER	-	-	0	tkm	1.02E+2	1.34E+0	1	2.09	(4,5,n,a,n,a,n,a); Standard distance 600km	
disposal	disposal, waste, Si waferprod., inorg. 9.4% water, to residual material landfill	CH	-	-	0	kg	2.02E-2	-	1	1.24	(3,1,1,1,3,3); company information, amount of deposited waste, own estimation for type	
	disposal, plastics, mixture, 15.3% water, to municipal incineration	CH	-	-	0	kg	7.51E-1	4.00E-2	1	1.07	(1,1,1,1,1,3); Calculation for plastic parts burned after recycling	
	disposal, inert waste, 5% water, to inert material landfill	CH	-	-	0	kg	6.50E-1	-	1	1.07	(1,1,1,1,1,3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)	
	disposal, glass, 0% water, to municipal incineration	CH	-	-	0	kg	3.44E+0	-	1	1.07	(1,1,1,1,1,3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)	
	treatment, sewage, unpolluted, to wastewater treatment, class 3	CH	-	-	0	m3	1.31E-1	-	1	1.07	(1,1,1,1,1,3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)	
emissions air	Heat, waste	-	air	high population density	-	MJ	1.61E+2	-	1	1.07	(1,1,1,1,1,3); Calculation	
	Cadmium	-	air	high population density	-	kg	2.10E-8	-	1	5.09	(3,4,3,3,1,5); Rough estimation	

To cope with the fact that most CIS PV panels are produced nowadays not only in Europe but in Asia and the US, a transport distance for the import to Europe was considered (Tab. 4.12). They are transported by freight ship and lorry to the European regional storage. An average transport distance of about 15'000 km from US and Asia to Europe is included. It is assumed that Rotterdam is the point of import for Europe. From Rotterdam the panels are transported by lorry to the regional storage. The three main countries producing solar energy in Europe are Germany (60 %), Italy (31 %) and Spain (9 %). The average weighted transport distance to these countries is about 950 km.

Tab. 4.12 unit process inventory data for CIS photovoltaic panels at the European regional storage

	Name	Location	Category	SubCategory	Infrastructure	Process	Unit	photovoltaic panel, CIS, at regional storage	UncertaintyTyp	StandardDeviation95%	GeneralComment
	Location							RER			
	InfrastructureProcess							1			
	Unit						m2				
product	photovoltaic panel, CIS, at regional storage	RER	-	-	1	m2	1.00E+0				
materials	photovoltaic panel, CIS, at plant	DE	-	-	1	m2	1.00E+0	1	3.00	(1,1,1,1,1,3);	
transport	transport, lorry >16t, fleet average	RER	-	-	0	tkm	1.56E+1	1	2.09	(4,5,na,na,na,na); transport from Rotterdam to regional storage	
	transport, transoceanic freight ship	OCE	-	-	0	tkm	2.46E+2	1	2.09	(4,5,na,na,na,na); import from US and Asia to Rotterdam	

4.3.3 Installation and mounting

The CIS panels are mounted on three different systems: 3 kWp installation, integrated in a slanted roof-top, 3 kWp installation, mounted on a slanted roof-top and a 570 kWp installation on open ground. The panel and construction material demand for all three installations is calculated based on an efficiency of 10.8 % (Frischknecht & Itten 2014). It is assumed that the panels have 1 % of rejects and 2 % need to be replaced during the life time. This is considered when calculating the panel demand per system. The roof systems need 27.8 m² of installation area, while the open ground system consists of 5'280 m² module area. Electricity is used for the installation of the modules.

The electric installation and supporting structure are modeled according to Jungbluth et. al (2012). The unit process inventory data are described in sections 4.7.1 and 4.7.2.

The panels, the electric installation and the mounting structure are transported 100 km from the regional storage to the construction site. Inside Europe the means of transport are lorries. The following Tab. 4.13 describes the inventory per each photovoltaic system with CIS PV panels.

Tab. 4.13 unit process inventory data for the mounting CIS photovoltaic panels integrated and mounted on a slanted roof top as well as the open ground system

	Name	Location	Infrastructure	Process	Unit	3kWp slanted-roof installation, CIS, laminated, integrated, on roof	3kWp slanted-roof installation, CIS, panel, mounted, on roof	570 kWp open ground installation, CIS, on open ground	Uncertainty Type	Standard Deviation 95 %	General Comment
	Location					RER 1 unit	RER 1 unit	RER 1 unit			
	Infrastructure Process Unit										
product	3kWp slanted-roof installation, CIS, laminated, integrated, on roof	RER	1	unit	1.00E+0	0	0				
	3kWp slanted-roof installation, CIS, panel, mounted, on roof	RER	1	unit	0	1.00E+0	0				
	570 kWp open ground installation, CIS, on open ground	RER	1	unit	0	0	1.00E+0				
energy	electricity, low voltage, production ENTSO, at grid	ENTSO	0	kWh	4.00E-2	4.00E-2	3.60E+1	1	1.07	(1,1,1,1,1,3,BU:1.05); Energy use for erection of plant	
	diesel, burned in building machine	GLO	0	MJ			7.67E+3	1	1.07	(1,1,1,1,1,3,BU:1.05); Energy use for foundation piling and wheel loader	
materials	electric installation, photovoltaic plant, at plant	CH	1	unit	1.00E+0	1.00E+0		1	3.00	(1,1,1,1,1,3,BU:3); Literature	
	electric installation, 570 kWp photovoltaic plant, at plant	ES	1	unit			1.00E+0	1	3.00	(1,1,1,1,1,3,BU:3); Literature	
	slanted-roof construction, integrated, on roof	RER	1	m2	2.78E+1			1	3.00	(1,1,1,1,1,3,BU:3); panel demand based on yield and efficiency	
	slanted-roof construction, mounted, on roof	RER	1	m2		2.78E+1		1	3.00	(1,1,1,1,1,3,BU:3); panel demand based on yield and efficiency	
	open ground construction, on ground	RER	1	m2			5.28E+3	1	3.00	(1,1,1,1,1,3,BU:3); panel demand based on yield and efficiency	
	photovoltaic panel, CIS, at regional storage	RER	1	m2	2.86E+1	2.86E+1	5.44E+3	1	3.02	(1,4,1,3,1,3,BU:3); panel demand based on yield and efficiency. 1% rejects and 2% replacement considered	
transport	transport, lorry 3.5-16t, fleet average	RER	0	tkm	5.58E+1	6.11E+1	1.50E+4	1	2.09	(4,5,na,na,na,na,BU:2); transport panel, electrical installation and mounting structure, 100km to construction place	
emissions air	Heat, waste	-	-	MJ	1.44E-1	1.44E-1	1.30E+2	1	1.07	(1,1,1,1,1,3,BU:1.05); calculated with electricity use	

4.3.4 Electricity generation

The 3 kWp roof installations and the 570 kWp open ground installation have an annual yield of 975 kWh/kWp (see also section 3.3.3). The expected life time is 30 years. Hence the overall output of electricity is 87.8 MWh for the roof installations and 16.7 GWh for the open ground system.

It is assumed that 20 liter of water are used per square meter for washing the modules (Jungbluth et al. 2012). This results in 6.33 ml of water per kWh electricity. The washing water is discharged into the sewer and to the municipal waste water treatment plant. No cleaning agents are considered.

The solar energy harvested by the PV system to produce electricity is calculated from the energy transformed to electricity fed into the inverter. Hence, 3.6 MJ solar energy are harvested to generate 1 kWh electricity. No direct emissions occur during the use stage of the panels.

The land occupation of the open ground PV system is taken into account in the mounting structure datasets. The land occupied by building roof integrated and mounted systems is fully attributed to the building. The following Tab. 4.14 shows the unit process inventory data of the electricity production from CIS PV systems.

Tab. 4.14 unit process inventory data for the electricity production with CIS photovoltaic panels at a 3 kWp roof installation, integrated and mounted, and a 570 kWp open ground installation

	Name	Location	Category	SubCategory	InfrastructureProcess	Unit	electricity, PV, at 3kWp slanted-roof, CIS, laminated, integrated	electricity, PV, at 3kWp slanted-roof, CIS, panel, mounted	electricity, PV, at 570kWp open ground, CIS	UncertaintyType	StandardDeviation5%	GeneralComment
	Location						RER	RER	RER			
	InfrastructureProcess						0 kWh	0 kWh	0 kWh			
product	electricity, PV, at 3kWp slanted-roof, CIS, laminated, integrated	RER	-	-	0 kWh	1	0	0	0			
	electricity, PV, at 3kWp slanted-roof, CIS, panel, mounted	RER	-	-	0 kWh	0	1	0	0			
	electricity, PV, at 570kWp open ground, CIS	RER	-	-	0 kWh	0	0	1	1			
infrastructure	3kWp slanted-roof installation, CIS, laminated, integrated, on roof	RER	-	-	1 unit	1.14E-5				1	3.00	(1.1.1.1.1.BU-3); infrastructure; calculation
	3kWp slanted-roof installation, CIS, panel, mounted, on roof	RER	-	-	1 unit		1.14E-5			1	3.00	(1.1.1.1.1.BU-3); infrastructure; calculation
	570 kWp open ground installation, CIS, on open ground	RER	-	-	1 unit			6.00E-8		1	1.24	(2.3.1.1.3.5.BU-3); infrastructure; calculation
auxiliaries	tap water, at user	RER	-	-	0 kg	6.33E-3	6.33E-3	6.33E-3		1	1.05	(1.1.1.1.1.BU-1.05); water for washing, estimation 20 l per m2 panel; calculation
disposal	treatment, sewage, from residence, to wastewater treatment, class 2	CH	-	-	0 m3	6.33E-6	6.33E-6	6.33E-6		1	1.05	(1.2.1.1.1.BU-1.05); waste water treatment of washing water; calculation
resource in air	Energy, solar, converted	-	resource	in air	- MJ	3.60E+0	3.60E+0	3.60E+0		1	1.20	(2.3.1.1.3.5.BU-1.05); solar energy needed for the production of 1kWh electricity (direct current) ; ecoinvent v2.2

4.3.5 End-of-life and benefits for recycling

At the end-of-life the PV panels are collected and transported to the recycling plant. Large scale CIS recycling has not been established yet. When recycling CIS panels today, the frame is removed and materials are shredded and used in downstream processes. Because no primary data on CIS panel recycling are available, the recycling process is modeled using data about recycling of CdTe modules (see section 4.2.5). The specific recycling efforts per kg of unframed CdTe PV modules recycled are adapted with a factor of 1.5. Copper, selenium and cadmium emissions are accounted for based on a rough estimation. The emission rate of cadmium into water and air is calculated for the CdTe recycling process as the ratio of cadmium emissions to the cadmium content in the PV module. The emission rate into water and air is 3.27 µg and 0.22 µg per kg cadmium contained in the panel, respectively. These emission rates are applied to copper, selenium and cadmium emissions into water and air for the recycling of CIS PV panels and scaled by a factor 1.5.

The largest share of recycled materials is flat glass and solar glass, which is used as cullets input in recycled foam glass production. The benefits for glass recycling are calculated based on the production of flat glass since it is not possible to clearly identify the material inputs replaced by glass cullets compared to the production of primary foam glass. According to the bill of materials of CIS PV modules (Tab. 3.1), the mass of glass per m² panel is 13.0 kg. This amount is corrected in order to account for manufacturing losses, which are estimated to about 3.8 % based on data for CdTe PV module production. The efficiency of glass recycling is assumed to be equal to the CdTe panel recycling, namely 90 %.

Similar to the recycling of CdTe PV modules, the consumption of primary materials used in glass production is avoided when using the glass cullets from recycled CIS PV panels. Hence, the geogenic CO₂-emissions caused by primary raw materials are prevented (Held & Ilg 2011), which amount to 0.208 kg CO₂/kg flat glass (Kellenberger et al. 2007). This results in avoided emissions of 2.33 kg CO₂ per m² recycled PV module. Because the melting of the glass cullet takes less energy than the melting of limestone and silica sand, heavy fuel oil and natural gas can additionally be saved. Held and Ilg (2011) assume an energy reduction potential in the melting process of around 30 % when replacing 100 % input material with glass cullets.

None of the semiconductor materials (cadmium, indium, selenium and gallium) is assumed to be recovered.

Aluminium and steel used in the mounting structures as well as the copper and steel contained in the electric installation are likewise recycled. In line with Jungbluth et al. (2012) it is assumed that only aluminium, copper and steel are recycled but not the plastics. Being large construction parts, a recycling efficiency of 100 % is assumed.

Efforts to produce secondary metals from scrap are taken into account for aluminium and copper. These secondary metals are potentially avoiding primary aluminium and primary copper production, respectively. Steel scrap is potentially avoiding pig iron feedstock in converter steel production. Benefits for recycling are granted only for the net surplus amount of recycled material, which leaves the PV system (see also EN 15804 2013). According to the ecoinvent data v2.2+ the recycled content of the metals used in panels, cabling and mounting structures is 32 % for aluminium, 77 % for the AlMg₃ alloy used in the panel frame, 44 % for copper and 37 % for steel (KBOB et al. 2014). The application of the end-of-life formula is described in more detail in section 4.9.9.

The unit process inventory data of CIS module takeback and recycling, including benefits and additional loads beyond the system boundary, are listed in Tab. 4.15.

Tab. 4.15 unit process inventory data for the recycling and the potential benefits and additional loads beyond the system boundary by recycling CIS photovoltaic panels, cabling and mounting structures

	Name	Location	Category	SubCategory	InfrastructureProcess	Unit	CIS PV module takeback + recycling	Avoided burden from recycling, CIS PV module, integrated construction	Avoided burden from recycling, CIS PV module, mounted construction	Avoided burden from recycling, CIS PV module, open ground construction	Uncertainty Type	StandardDeviation5%	GeneralComment
							RER	RER	RER	RER			
	Location						0	0	0	0			
	InfrastructureProcess						m2	m2	m2	m2			
product	CIS PV module takeback + recycling	RER	-	-	0	m2	1.00E+0	0	0	0			
	Avoided burden from recycling, CIS PV module, integrated construction	RER	-	-	0	m2	0	1.00E+0	0	0			
	Avoided burden from recycling, CIS PV module, mounted construction	RER	-	-	0	m2	0	0	1.00E+0	0			
	Avoided burden from recycling, CIS PV module, open ground construction	RER	-	-	0	m2	0	0	0	1.00E+0			
energy	electricity, medium voltage, production ENTSO, at grid	ENTSO	-	-	-	kWh	5.72E+0				1	1.07	(1.1.1.1.1.3.BU1.05); according to CdTe panel recycling, First Solar
auxiliaries	water, deionised, at plant	CH	-	-	-	kg	7.08E+0				1	1.13	(1.4.1.3.1.3.BU1.05); according to CdTe panel recycling, First Solar
	sulphuric acid, liquid, at plant	RER	-	-	-	kg	1.09E-1				1	1.07	(1.1.1.1.1.3.BU1.05); according to CdTe panel recycling, First Solar
	hydrogen peroxide, 50% in H2O, at plant	RER	-	-	-	kg	7.46E-1				1	1.07	(1.1.1.1.1.3.BU1.05); according to CdTe panel recycling, First Solar
	sodium hydroxide, 50% in H2O, production mix, at plant	RER	-	-	-	kg	1.36E-1				1	1.07	(1.1.1.1.1.3.BU1.05); according to CdTe panel recycling, First Solar
transport	transport, lorry >16t, fleet average	RER	-	-	-	tkm	1.63E+1				1	2.00	(1.1.1.1.1.3.BU2); according to CdTe panel recycling, First Solar. Assumption: materials for electric installation and mounting is transported 200 km.
disposal	treatment, PV cell production effluent, to wastewater treatment, class 3	CH	-	-	-	m3	6.26E-3				1	1.07	(1.1.1.1.1.3.BU1.05); according to CdTe panel recycling, First Solar
	disposal, plastics, mixture, 15.3% water, to municipal incineration	CH	-	-	-	kg	8.05E-1				1	1.07	(1.1.1.1.1.3.BU1.05); according to CdTe panel recycling, First Solar
	disposal, inert waste, 5% water, to inert material landfill	CH	-	-	-	kg	1.68E-1				1	1.07	(1.1.1.1.1.3.BU1.05); according to CdTe panel recycling, First Solar
emissions air	Copper	-	air	high population density	-	kg	3.16E-9				1	5.00	(1.1.1.1.1.3.BU5); Emissionrate equal to cadmium emission from CdTe panel recycling
	Selenium	-	air	high population density	-	kg	1.81E-9				1	5.00	(1.1.1.1.1.3.BU5); Emissionrate equal to cadmium emission from CdTe panel recycling
	Cadmium	-	air	low population density	-	kg	8.71E-11				1	5.00	(1.1.1.1.1.3.BU5); Emissionrate equal to cadmium emission from CdTe panel recycling
emissions water	Copper	-	water	unspecified	-	kg	1.38E-8				1	3.00	(1.1.1.1.1.3.BU3); Emissionrate equal to cadmium emission from CdTe panel recycling
	Selenium	-	water	unspecified	-	kg	4.79E-8				1	5.00	(1.1.1.1.1.3.BU5); Emissionrate equal to cadmium emission from CdTe panel recycling
	Cadmium, ion	-	water	unspecified	-	kg	2.75E-8				1	3.00	(1.1.1.1.1.3.BU3); Emissionrate equal to cadmium emission from CdTe panel recycling
avoided energy	natural gas, burned in industrial furnace >100kW	RER	-	-	-	MJ	-1.54E+01	-1.54E+01	-1.54E+01		1	1.14	(2.4.1.1.1.3.BU1.05); natural gas consumption avoided by the recycling of PV glass
	heavy fuel oil, burned in industrial furnace 1MW, non-modulating	RER	-	-	-	MJ	-9.95E+00	-9.95E+00	-9.95E+00		1	1.14	(2.4.1.1.1.3.BU1.05); heavy fuel oil consumption avoided by the recycling of PV glass
avoided materials	silica sand, at plant	DE	-	-	-	kg	-6.49E+00	-6.49E+00	-6.49E+00		1	1.14	(2.4.1.1.1.3.BU1.05); silica sand consumption avoided by the recycling of the PV glass
	soda, powder, at plant	RER	-	-	-	kg	-2.57E+00	-2.57E+00	-2.57E+00		1	1.14	(2.4.1.1.1.3.BU1.05); soda consumption avoided by the recycling of the PV glass
	limestone, milled, packed, at plant	CH	-	-	-	kg	-4.49E+00	-4.49E+00	-4.49E+00		1	1.14	(2.4.1.1.1.3.BU1.05); limestone consumption avoided by the recycling of PV glass
	copper, at regional storage	RER	-	-	-	kg	-2.96E-01	-2.96E-01	-7.86E-02		1	1.14	(2.4.1.1.1.3.BU1.05); primary copper avoided by the recycling of wires and the electric installation
	copper, secondary, at refinery	RER	-	-	-	kg	2.96E-01	2.96E-01	7.86E-02		1	1.14	(2.4.1.1.1.3.BU1.05); efforts for producing secondary copper from scrap
	aluminium, primary, at plant	RER	-	-	-	kg	-2.03E+00	-2.43E+00	-3.21E+00		1	1.14	(2.4.1.1.1.3.BU1.05); primary aluminium avoided by the recycling of frame and mounting structure
	aluminium, secondary, from old scrap, at plant	RER	-	-	-	kg	2.03E+00	2.43E+00	3.21E+00		1	1.14	(2.4.1.1.1.3.BU1.05); efforts for producing secondary aluminium from scrap
	pig iron, at plant	GLO	-	-	0	kg	-1.46E-01	-9.64E-01	-4.55E+00		1	2.00	(2.4.1.1.1.3); Avoided primary steel production due to recycling of mounting structure and the electric installation
avoided emissions air	Carbon dioxide, fossil	-	air	unspecified	-	kg	-2.33E+00	-2.33E+00	-2.33E+00		1	1.14	(2.4.1.1.1.3.BU1.05); geogenic greenhouse gas emissions avoided by the recycling of PV glass

4.4 Micromorphous-Si PV technology

4.4.1 Overview

The micromorphous-Si PV chain is modeled according to the following structure:

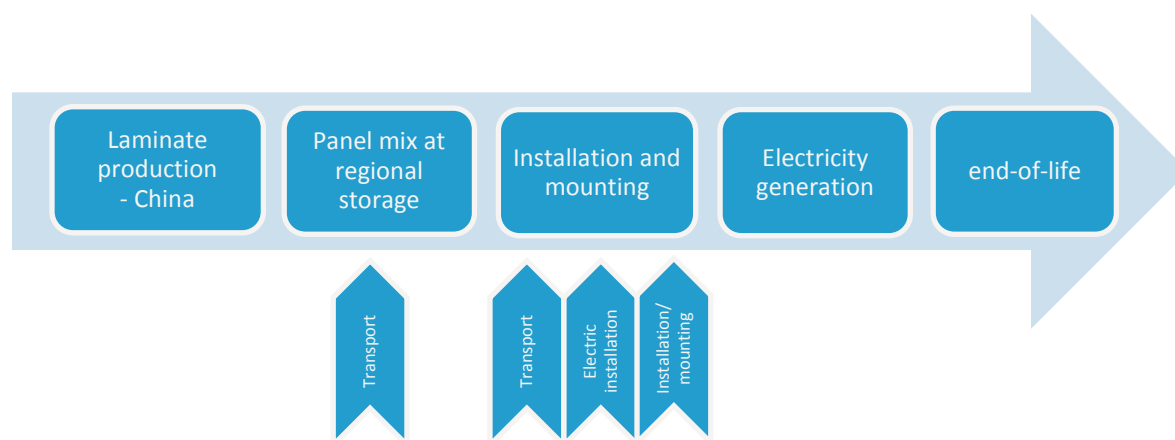


Fig. 4.3 Schematic illustration about the modeled life cycle chain of micro-Si PV panels

The processes are described in the sections 4.4.2 to 4.4.5. The following Tab. 4.16 summarizes the main characteristics of the micromorphous-Si PV panel.

Tab. 4.16 Characteristics of micro-Si PV panels

	Unit	Amount/value
Panel size	m ² /panel	1.4
Panel weight (unframed)	kg/m ²	13.9
Module conversion efficiency	%	10.0
Wafer thickness	μm	10-20
Main data source / year	-	Flury (2012)

4.4.2 Laminate and panel production

The production of micromorphous-Si laminates takes place in China. The data stem from Oerlikon Solar representing the situation in 2012. All relevant material and energy inputs are accounted for. Natural gas and electricity are the energy sources consumed in the product stage. Emissions into water and air are considered. Further processes and data are described in detail in Flury et al. (2012). A tempering process of solar glass, which was not accounted for in the original inventory, has been included in order to harmonize the life cycle inventories of the different PV technologies investigated. The panel frame, which was originally modelled in the inventory of the installation, is now included in the life cycle inventory of the micro-Si panel. The material of the frame has been changed to AlMg₃ alloy as it is used for all other PV technologies. The following Tab. 4.17 shows the unit process inventory data for the production of micromorphous-Si PV panels.

Tab. 4.17 unit process inventory data for the production of micro-Si photovoltaic panels for the production in China

	Name	Location	Category	SubCategory	InfrastructureProcess	Unit	photovoltaic panel, micro-Si, at plant	UncertaintyType	StandardDeviation95%	GeneralComment
	Location						CN			
	InfrastructureProcess						1			
	Unit						m2			
product	photovoltaic panel, micro-Si, at plant	CN	-	-	1	m2	1.00E+0			
energy	electricity, high voltage, at grid	CN	-	-	-	kWh	4.29E+1	1	1.30	(4.1.1.1.1.5, BU:1.05); Electricity (equipment & operation of the facility); Oerlikon Solar (2011): General properties THINFAB - 120 MW
infrastructure	solar collector factory	RER	-	-	1	unit	2.80E-8	1	3.14	(4.1.1.1.3.5, BU:3); Infrastructure building, incl. land use; Oerlikon Solar (2011): General properties THINFAB - 120 MW; extrapolated based on assumptions
	metal working machine, unspecified, at plant	RER	-	-	1	kg	2.17E-2	1	3.14	(4.1.1.1.3.5, BU:3); Infrastructure machines; Oerlikon Solar (2011): General properties THINFAB - 120 MW
materials	silicone product, at plant	RER	-	-	-	kg	1.68E-2	1	1.30	(4.1.1.1.1.5, BU:1.05); Silicon pottant & sealant; Oerlikon Solar (2011): General properties THINFAB - 120 MW
	silver, at regional storage	RER	-	-	-	kg	1.64E-4	1	1.30	(4.1.1.1.1.5, BU:1.05); Contacting Glue; Oerlikon Solar (2011): General properties THINFAB - 120 MW
	copper, at regional storage	RER	-	-	-	kg	9.10E-3	1	1.30	(4.1.1.1.1.5, BU:1.05); Solder Strings Long & Cross contacting ribbon & Junction Box; Oerlikon Solar (2011): General properties THINFAB - 120 MW
	wire drawing, copper	RER	-	-	-	kg	9.10E-3	1	1.30	(4.1.1.1.1.5, BU:1.05); Solder Strings Long & Cross contacting ribbon & Junction Box; Oerlikon Solar (2011): General properties THINFAB - 120 MW
	cable, three-conductor cable, at plant	GLO	-	-	-	m	2.03E-1	1	1.30	(4.1.1.1.1.5, BU:1.05); Junction Box; Oerlikon Solar (2011): General properties THINFAB - 120 MW
	adhesive for metals, at plant	DE	-	-	-	kg	3.60E-3	1	1.30	(4.1.1.1.1.5, BU:1.05); Fixation pad for solder strings long & Glue pad; Oerlikon Solar (2011): General properties THINFAB - 120 MW
	bisphenol A, powder, at plant	RER	-	-	-	kg	1.10E-4	1	1.30	(4.1.1.1.1.5, BU:1.05); Contacting Glue; Oerlikon Solar (2011): General properties THINFAB - 120 MW
	solar glass, low-iron, at regional storage	RER	-	-	-	kg	1.34E+1	1	1.30	(4.1.1.1.1.5, BU:1.05); Front- and Back-Glass, incl. wastes; Oerlikon Solar (2011): General properties THINFAB - 120 MW
	tempering, flat glass	RER	-	-	-	kg	1.34E+1	1	1.30	(4.1.1.1.1.5, BU:1.05); Front- and Back-Glass, incl. wastes; Oerlikon Solar (2011): General properties THINFAB - 120 MW
	aluminium alloy, AlMg3, at plant	RER	-	-	-	kg	2.67E+0	1	1.12	(2.1.1.1.1.4, BU:1.05); Frame and clips; Oerlikon Solar (2011): Recommendation to customer
	adhesive for metals, at plant	DE	-	-	-	kg	1.85E-4	1	1.12	(2.1.1.1.1.4, BU:1.05); Adhesive for clips on frame; Oerlikon Solar (2011): Recommendation to customer
	ethylvinylacetate, foil, at plant	RER	-	-	-	kg	4.52E-1	1	1.30	(4.1.1.1.1.5, BU:1.05); Encapsulation; Oerlikon Solar (2011): General properties THINFAB - 120 MW
	polyethylene, HDPE, granulate, at plant	RER	-	-	-	kg	1.07E-2	1	1.30	(4.1.1.1.1.5, BU:1.05); Junction Box; Oerlikon Solar (2011): General properties THINFAB - 120 MW
	injection moulding	RER	-	-	-	kg	1.07E-2	1	1.30	(4.1.1.1.1.5, BU:1.05); Junction Box; Oerlikon Solar (2011): General properties THINFAB - 120 MW
auxiliaries	Water, unspecified natural origin, CN	-	resource	in water	-	m3	2.68E+0	1	1.30	(4.1.1.1.1.5, BU:1.05); Cooling water (chilled and tempered); Oerlikon Solar (2011): General properties THINFAB - 120 MW
	water, deionised, at plant	CH	-	-	-	kg	8.25E+1	1	1.30	(4.1.1.1.1.5, BU:1.05); Water deionised; Oerlikon Solar (2011): General properties THINFAB - 120 MW
	water, decarbonised, at plant	RER	-	-	-	kg	4.37E+1	1	1.30	(4.1.1.1.1.5, BU:1.05); Water demineralised; Oerlikon Solar (2011): General properties THINFAB - 120 MW
	zeolite, slurry, 50% in H2O, at plant	RER	-	-	-	kg	3.24E-2	1	1.30	(4.1.1.1.1.5, BU:1.05); Detergent; Oerlikon Solar (2011): General properties THINFAB - 120 MW
	chemicals organic, at plant	GLO	-	-	-	kg	3.97E-2	1	1.30	(4.1.1.1.1.5, BU:1.05); Diethylzinc (DEZ); Oerlikon Solar (2011): General properties THINFAB - 120 MW
	compressed air, average generation, >30kW, 7 bar gauge, at compressor	RER	-	-	-	m3	1.23E+0	1	1.30	(4.1.1.1.1.5, BU:1.05); compressed dried air (CDA); Oerlikon Solar (2011): General properties THINFAB - 120 MW
	oxygen, liquid, at plant	RER	-	-	-	kg	2.76E-1	1	1.30	(4.1.1.1.1.5, BU:1.05); Oxygen Technical; Oerlikon Solar (2011): General properties THINFAB - 120 MW
	hydrogen, liquid, at plant	RER	-	-	-	kg	3.39E-2	1	1.30	(4.1.1.1.1.5, BU:1.05); Hydrogen; Oerlikon Solar (2011): General properties THINFAB - 120 MW
	nitrogen, liquid, at plant	RER	-	-	-	kg	5.11E+0	1	1.30	(4.1.1.1.1.5, BU:1.05); Nitrogen, technical & process; Oerlikon Solar (2011): General properties THINFAB - 120 MW
	carbon dioxide liquid, at plant	RER	-	-	-	kg	1.91E-3	1	1.30	(4.1.1.1.1.5, BU:1.05); Carbon dioxide; Oerlikon Solar (2011): General properties THINFAB - 120 MW
	argon, liquid, at plant	RER	-	-	-	kg	2.21E-2	1	1.30	(4.1.1.1.1.5, BU:1.05); Argon; Oerlikon Solar (2011): General properties THINFAB - 120 MW
	silane, at plant	RER	-	-	-	kg	2.12E-2	1	1.30	(4.1.1.1.1.5, BU:1.05); Silane; Oerlikon Solar (2011): General properties THINFAB - 120 MW
	phosphane, at plant	GLO	-	-	-	kg	2.35E-5	1	1.30	(4.1.1.1.1.5, BU:1.05); Phosphine; Oerlikon Solar (2011): General properties THINFAB - 120 MW
	trimethyl borate, at plant	GLO	-	-	-	kg	8.30E-6	1	1.30	(4.1.1.1.1.5, BU:1.05); Trimethylboron 2% in Hydrogen; Oerlikon Solar (2011): General properties THINFAB - 120 MW
	nitrogen trifluoride, at plant	RER	-	-	-	kg	5.92E-2	1	1.30	(4.1.1.1.1.5, BU:1.05); Hydrogen trifluoride; Oerlikon Solar (2011): General properties THINFAB - 120 MW
	diborane, at plant	GLO	-	-	-	kg	1.70E-5	1	1.30	(4.1.1.1.1.5, BU:1.05); Diborane - 2% in Hydrogen; Oerlikon Solar (2011): General properties THINFAB - 120 MW
	natural gas, high pressure, at consumer	CH	-	-	-	MJ	4.38E+0	1	1.30	(4.1.1.1.1.5, BU:1.05); Natural gas; Oerlikon Solar (2011): General properties THINFAB - 120 MW

Tab. 4.17 unit process inventory data for the production of micro-Si photovoltaic panels for the production in China (continued)

transport	transport, lorry 16-32t, EURO3	RER	-	-	-	tkm	4.83E+1	1	2.09	(4,5,na,na,na,na,BU:2); Waste glass and waste foil, returned to supplier & waste glass to recycling; own assumption
	transport, lorry 20-28t, fleet average	CH	-	-	-	tkm	1.74E+0	1	2.09	(4,5,na,na,na,na,BU:2); Transport of the materials and consumables; own assumption
	transport, transoceanic freight ship	OCE	-	-	-	tkm	2.17E+1	1	2.09	(4,5,na,na,na,na,BU:2); Shipping of the equipment from Europe to China; own assumption
disposal	treatment, sewage, unpolluted, from residence, to wastewater treatment, class 2	CH	-	-	-	m3	1.26E-1	1	1.30	(4,1,1,1,1,5,BU:1.05); Waste water flow (substances specified separately); Oerlikon Solar (2011); General properties THINFAB - 120 MW
	disposal, solvents mixture, 16.5% water, to hazardous waste incineration	CH	-	-	-	kg	6.06E-5	1	1.30	(4,1,1,1,1,5,BU:1.05); Solvent mixture from EVA foil; Oerlikon Solar (2011); General properties THINFAB - 120 MW
	disposal, glass, 0% water, to municipal incineration	CH	-	-	-	kg	1.33E+1	1	1.30	(4,1,1,1,1,5,BU:1.05); Waste glasses, laminated (disposal / incineration / reuse as fibre glass / recycling) & disposal of the module at the end of its lifetime; Oerlikon Solar (2011); General properties THINFAB - 120 MW
	disposal, rubber, unspecified, 0% water, to municipal incineration	CH	-	-	-	kg	4.16E-4	1	1.30	(4,1,1,1,1,5,BU:1.05); Hotmelt (synthetic rubber, no hazardous component); Oerlikon Solar (2011); General properties THINFAB - 120 MW
emissions air	Heat, waste	-	air	high population density	-	MJ	4.09E+1	1	1.30	(4,1,1,1,1,5,BU:1.05); Waste heat; Oerlikon Solar (2011); General properties THINFAB - 120 MW
	Water, CN	-	air	unspecified	-	kg	1.34E+2	1	1.62	(4,1,1,1,1,5,BU:1.5); Evaporated water; assumption
	Particulates, > 10 um	-	air	high population density	-	kg	2.96E-6	1	1.62	(4,1,1,1,1,5,BU:1.5); Dust (SiO ₂ , ZnO); Oerlikon Solar (2011); General properties THINFAB - 120 MW
	NM VOC, non-methane volatile organic compounds, unspecified origin	-	air	high population density	-	kg	3.80E-4	1	1.62	(4,1,1,1,1,5,BU:1.5); Volatile organic compounds (VOC); Oerlikon Solar (2011); General properties THINFAB - 120 MW
	Nitrogen oxides	-	air	high population density	-	kg	6.40E-4	1	1.62	(4,1,1,1,1,5,BU:1.5); NO _x ; Oerlikon Solar (2011); General properties THINFAB - 120 MW
	Carbon monoxide, fossil	-	air	high population density	-	kg	1.08E-4	1	5.10	(4,1,1,1,1,5,BU:5); CO; Oerlikon Solar (2011); General properties THINFAB - 120 MW
	Carbon dioxide, fossil	-	air	high population density	-	kg	2.45E-1	1	1.62	(4,1,1,1,1,5,BU:1.05); CO ₂ ; assumption
	Nitrogen fluoride	-	air	high population density	-	kg	1.08E-6	1	1.62	(4,1,1,1,1,5,BU:1.5); NF ₃ ; Oerlikon Solar (2011); General properties THINFAB - 120 MW
	Hydrogen fluoride	-	air	high population density	-	kg	1.24E-7	1	1.62	(4,1,1,1,1,5,BU:1.5); HF / F ₂ ; Oerlikon Solar (2011); General properties THINFAB - 120 MW
	Water	-	water	unspecified	-	kg	2.54E+3	1	1.62	(4,1,1,1,1,5,BU:1.5); Cooling water Released ; Assumption
emissions water	Suspended solids, unspecified	-	water	unspecified	-	kg	2.27E-2	1	1.62	(4,1,1,1,1,5,BU:1.5); Total suspended solids (TSS); Oerlikon Solar (2011); General properties THINFAB - 120 MW
	COD, Chemical Oxygen Demand	-	water	unspecified	-	kg	5.97E-3	1	1.62	(4,1,1,1,1,5,BU:1.5); COD, after on-site waste water treatment; Oerlikon Solar (2011); General properties THINFAB - 120 MW
	Fluoride	-	water	unspecified	-	kg	1.89E-3	1	1.62	(4,1,1,1,1,5,BU:1.5); Fluoride, after on-site waste water treatment; Oerlikon Solar (2011); General properties THINFAB - 120 MW
	Phosphorus	-	water	unspecified	-	kg	5.05E-5	1	1.62	(4,1,1,1,1,5,BU:1.5); Phosphorus, after on-site waste water treatment; Oerlikon Solar (2011); General properties THINFAB - 120 MW
	Nitrogen	-	water	unspecified	-	kg	2.02E-4	1	1.62	(4,1,1,1,1,5,BU:1.5); Nitrogen, after on-site waste water treatment; Oerlikon Solar (2011); General properties THINFAB - 120 MW
	Boron	-	water	unspecified	-	kg	2.52E-5	1	5.10	(4,1,1,1,1,5,BU:5); Boron, after on-site waste water treatment; Oerlikon Solar (2011); General properties THINFAB - 120 MW
	Zinc, ion	-	water	unspecified	-	kg	1.26E-5	1	5.10	(4,1,1,1,1,5,BU:5); Zinc, after on-site waste water treatment; Oerlikon Solar (2011); General properties THINFAB - 120 MW
	Chloride	-	water	unspecified	-	kg	1.74E-1	1	3.09	(4,1,1,1,1,5,BU:3); Chloride, after on-site waste water treatment; Oerlikon Solar (2011); General properties THINFAB - 120 MW
	Calcium, ion	-	water	unspecified	-	kg	8.83E-5	1	3.09	(4,1,1,1,1,5,BU:3); Calcium, after on-site waste water treatment; Oerlikon Solar (2011); General properties THINFAB - 120 MW

The following Tab. 4.18 shows the inventory data for micromorphous-Si PV panels at the regional storage. The panels are imported from China to Rotterdam (about 20'000 km). From Rotterdam the panels are transported by lorry to the regional storage. The three main countries producing solar energy in Europe are Germany (60 %), Italy (31 %) and Spain (9 %). The average weighted transport distance to these countries is about 950 km.

Tab. 4.18 unit process inventory data for micro-Si photovoltaic panels at the European regional storage

	Name	Location	Category	SubCategory	InfrastructureProcess	Unit	photovoltaic panel, micro-Si, at regional storage	UncertaintyType	StandardDeviation95%	GeneralComment
product	photovoltaic panel, micro-Si, at regional storage	RER	-	-	1	m2	1.00E+0			
technosphere	photovoltaic panel, micro-Si, at plant	CN	-	-	1	m2	1.00E+0	1	3.00	(1,1,1,1,1,1,BU:3); Module; Oerlikon Solar (2011); General properties THINFAB - 120 MW
transport	transport, lorry >16t, fleet average	RER	-	-	-	tkm	1.52E+1	1	2.09	(4,5,na,na,na,na,BU:2); Transport from Rotterdam to regional storage;
	transport, transoceanic freight ship	OCE	-	-	-	tkm	3.21E+2	1	2.09	(4,5,na,na,na,na,BU:2); Import from Shanghai to Rotterdam;

4.4.3 Installation and mounting

The micromorphous-Si PV panels are installed at three systems: 3 kWp installation, integrated in a slanted roof-top, 3 kWp installation, mounted on a slanted roof-top and a 570 kWp installation on open ground. The panel and construction material demand for all three installations is calculated

based on an efficiency of 10.0 % (Frischknecht & Itten 2014). It is assumed that the panels have 1 % of rejects and 2 % need to be replaced during the life time. This was considered when calculating the panel demand per system. The roof systems need 30.0 m² of installation area, while the open ground system consists of 5'700 m² module area. Electricity is used for the mounting of the modules.

The electric installation and the mounting structure are modeled according Jungbluth et. al (2012). The unit process inventory data are described in sections 4.7.1 and 4.7.2.

Transport from the regional storage to the construction site is also considered within the lorry-transport (100 km). The following Tab. 4.19 describes the unit process inventory data for the mounting structure and electrical equipment for each photovoltaic system with micromorphous-Si PV panels.

Tab. 4.19 unit process inventory data for the mounting structure and electrical equipment of micro-Si photovoltaic panels integrated and mounted on a slanted roof top as well as the open ground system

	Name	Location	Category	SubCategory	InfrastructureProcess	Unit	3kWp slanted-roof installation, micro-Si, laminated, integrated, on roof	3kWp slanted-roof installation, micro-Si, panel, mounted, on roof	570 kWp open ground installation, micro-Si, on open ground	UncertaintyType	StandardDeviation95%	GeneralComment
	Location						RER	RER	RER			
	InfrastructureProcess						1 unit	1 unit	1 unit			
	Unit											
product	3kWp slanted-roof installation, micro-Si, laminated, integrated, on roof	RER	-	-	1	unit	1.00E+0	0	0			
	3kWp slanted-roof installation, micro-Si, panel, mounted, on roof	RER	-	-	1	unit	0	1.00E+0	0			
	570 kWp open ground installation, micro-Si, on open ground	RER	-	-	1	unit	0	0	1.00E+0			
energy	electricity, low voltage, production ENTSO, at grid	ENTSO	-	-	-	kWh	4.00E-2	4.00E-2	3.60E+1	1	1.28	(3,4,3,1,1,5,BU:1.05); Energy for mounting; Energy use for erection of 3kWp plant
	diesel, burned in building machine	GLO	-	-	-	MJ			7.67E+3	1	1.28	(3,4,3,1,1,5,BU:1.05); Energy use for foundation piling and wheel loader, analogous to single-Si;
materials	electric installation, photovoltaic plant, at plant	CH	-	-	1	unit	1.00E+0	1.00E+0		1	3.08	(3,4,3,1,1,5,BU:3); cabling, lightning protection and fuse box; Literature
	electric installation, 570 kWp photovoltaic plant, at plant	ES	-	-	1	unit			1.00E+0	1	3.08	(3,4,3,1,1,5,BU:3); cabling, lightning protection and fuse box; Literature
	slanted-roof construction, integrated, on roof	RER	-	-	1	m2	3.00E+1			1	3.02	(2,1,1,1,1,4,BU:3); installation;
	slanted-roof construction, mounted, on roof	RER	-	-	1	m2		3.00E+1		1	3.02	(2,1,1,1,1,4,BU:3); installation;
	open ground construction, on ground	RER	-	-	1	m2			5.70E+3	1	3.08	(3,4,3,1,1,5,BU:3); installation;
	photovoltaic panel, micro-Si, at regional storage	RER	-	-	1	m2	3.09E+1	3.09E+1	5.87E+3	1	3.02	(2,1,1,1,1,4,BU:3); Module; Oerlikon Solar (2011); General properties THINFAB - 120 MW
transport	transport, lorry 3.5-16t, fleet average	RER	-	-	-	tkm	5.86E+1	6.42E+1	1.59E+4	1	2.09	(4,5,na,na,na,na,BU:2); Transport of module and material (panel, electrical installation and mounting structure) 100 km from the regional storage to the construction site; Assumption
emissions air	Heat, waste	-	air	high population density	-	MJ	1.44E-1	1.44E-1	1.30E+2	1	1.16	(3,4,2,1,1,1,BU:1.05); calculated with electricity use

4.4.4 Electricity generation

The three PV systems analyzed, 3 kWp roof installations (mounted and integrated) and the 570 kWp open ground installation, have an annual yield of 975 kWh/kWp (see also section 3.3.3). The expected life time is 30 years. Hence the overall output of electricity is 87.8 MWh for the roof installations and 16.7 GWh for the open ground system.

It is assumed that 20 liter of water are used per square meter for washing the modules (Jungbluth et al. 2012). This results in 6.84 ml of water per kWh electricity. The washing water is discharged into the sewer and to the municipal waste water treatment plant. No cleaning agents are considered.

The solar energy harvested by the PV system to produce electricity is calculated from the energy transformed to electricity fed into the inverter. Hence, 3.6 MJ solar energy are harvested to generate 1 kWh electricity. No direct emissions occur during the use stage of the panels.

The land occupation of the open ground PV system is taken into account in the mounting structure datasets. The land occupied by building roof integrated and mounted systems is fully attributed to the building. The following Tab. 4.20 shows the life cycle inventory data of the electricity production from micromorphous-Si PV systems.

	Name	Location	Category	SubCategory	InfrastructureProcess	Unit	electricity, PV, at 3kWp slanted-roof, micro-Si, laminated, integrated			electricity, PV, at 3kWp slanted-roof, micro-Si, panel, mounted	electricity, PV, at 570 kWp open ground, micro-Si	UncertaintyType	StandardDeviation5%	GeneralComment
							Location							
	InfrastructureProcess	Unit					RER 0 kWh	RER 0 kWh	RER 0 kWh					
product	electricity, PV, at 3kWp slanted-roof, micro-Si, laminated, integrated	RER	-	-	0	kWh	1	0	0					
	electricity, PV, at 3kWp slanted-roof, micro-Si, panel, mounted	RER	-	-	0	kWh	0	1	0					
	electricity, PV, at 570 kWp open ground, micro-Si	RER	-	-	0	kWh	0	0	1					
infrastructure	3kWp slanted-roof installation, micro-Si, laminated, integrated, on roof	RER	-	-	1	unit	1.14E-5			1	3.00	(1,1,1,1,1,1,BU-3); infrastructure; calculation		
	3kWp slanted-roof installation, micro-Si, panel, mounted, on roof	RER	-	-	1	unit		1.14E-5		1	1.15	(2,3,1,1,3,5,BU-3); infrastructure; calculation		
	570 kWp open ground installation, micro-Si, on open ground	RER	-	-	1	unit			6.00E-8	1	1.24	(2,3,1,1,3,5,BU-3); infrastructure; calculation		
auxiliaries	tap water, at user	RER	-	-	0	kg	6.84E-3	6.84E-3	6.84E-3	1	1.05	(1,1,1,1,1,1,BU-1.05); water for washing, estimation 20 l per m2 panel; calculation		
disposal	treatment sewage, from residence, to wastewater treatment, class 2	CH	-	-	0	m3	6.84E-6	6.84E-6	6.84E-6	1	1.05	(1,2,1,1,1,1,BU-1.05); waste water treatment or washing water; calculation		
resource in air	Energy, solar, converted	-	resource	in air	-	MJ	3.60E+0	3.60E+0	3.60E+0	1	1.20	(2,3,1,1,3,5,BU-1.05); solar energy needed for the production of 1kWh electricity (direct energy); equipment /Q?		

After the use stage the panels and installations are deconstructed and disposed of. The PV panels are collected and transported to the recycling plant. The recycling is modeled according to the recycling of CdTe panels (see section 4.2.5). The specific recycling efforts per kg of unframed CdTe PV modules recycled are adapted with a factor of 1.5.

Similar to the recycling of CdTe, the consumption of primary materials used in glass production is avoided when using the glass cullets from recycled CIS PV panels. Hence, the geogenic CO₂-emissions caused by primary raw materials are prevented (Held & Ilg 2011), which amount to 0.208 kg CO₂/kg flat glass (Kellenberger et al. 2007). This results in avoided emission of 2.41 kg CO₂ eq per m² recycled micromorphous-Si PV module. As the melting of the glass cullet takes less energy than the melting of limestone and silica sand, heavy fuel oil and natural gas can additionally be saved. Held and Ilg (2011) assume an energy reduction potential in the melting process in the range of around 30 % when replacing 100 % input material with glass cullet.

Efforts to produce secondary metals from scrap are taken into account for aluminium and copper. These secondary metals are potentially avoiding primary aluminium and primary copper production, respectively. Steel scrap is potentially avoiding pig iron feedstock in converter steel production. Benefits for recycling are granted only for the net surplus amount of recycled material, which leaves the PV system (see also EN 15804 2013). According to the ecoinvent data v2.2+ the recycled content of the metals used in panels, cabling and mounting structures is 32 % for aluminium, 77 % for the

AlMg₃ alloy used in the panel frame, 44 % for copper and 37 % for steel (KBOB et al. 2014). The application of the end-of-life formula is described in more detail in section 4.9.9.

The unit process inventory data of micromorphous-Si PV module takeback and recycling, including benefits and additional loads beyond the system boundary, are listed in Tab. 4.21.

Tab. 4.21 unit process inventory data for the recycling and the potential benefits and additional loads beyond the system boundary by recycling micro-Si photovoltaic panels, cabling and mounting structures

	Name	Location	Category	SubCategory	InfrastructureProcess	Unit	micro-Si PV module takeback + recycling	Avoided burden from recycling, micro-Si PV module, integrated construction	Avoided burden from recycling, micro-Si PV module, mounted construction	Avoided burden from recycling, micro-Si PV module, open ground construction	UncertaintyType	StandardDeviation95%	GeneralComment
	Location						RER	RER	RER	RER			
	InfrastructureProcess						0 m2	0 m2	0 m2	0 m2			
	Unit												
product	micro-Si PV module takeback + recycling	RER	-	-	0	m2	1.00E+0	0	0	0			
	Avoided burden from recycling, micro-Si PV module, integrated construction	RER	-	-	0	m2	0	1.00E+0	0	0			
	Avoided burden from recycling, micro-Si PV module, mounted construction	RER	-	-	0	m2	0	0	1.00E+0	0			
	Avoided burden from recycling, micro-Si PV module, open ground construction	RER	-	-	0	m2	0	0	0	1.00E+0			
energy	electricity, medium voltage, production ENTSO, at grid	ENTS O	-	-	-	kWh	5.33E+0				1	1.07	(1.1.1.1.3.BU:1.05); according to CdTe panel recycling, First Solar
auxiliaries	water, deionised, at plant	CH	-	-	-	kg	6.60E+0				1	1.13	(1.1.1.1.3.BU:1.05); according to CdTe panel recycling, First Solar
	sulphuric acid, liquid, at plant	RER	-	-	-	kg	1.02E-1				1	1.07	(1.1.1.1.3.BU:1.05); according to CdTe panel recycling, First Solar
	hydrogen peroxide, 50% in H ₂ O, at plant	RER	-	-	-	kg	6.95E-1				1	1.07	(1.1.1.1.3.BU:1.05); according to CdTe panel recycling, First Solar
	sodium hydroxide, 50% in H ₂ O, production mix, at plant	RER	-	-	-	kg	1.27E-1				1	1.07	(1.1.1.1.3.BU:1.05); according to CdTe panel recycling, First Solar
transport	transport, lorry >16t, fleet average	RER	-	-	-	tkm	1.52E+1				1	2.00	(1.1.1.1.3.BU:2); according to CdTe panel recycling, First Solar. Assumption: material for electric installation and mounting system has a transport of 200 km.
disposal	treatment, PV cell production effluent, to wastewater treatment, class 3	CH	-	-	-	m3	5.84E-3				1	1.07	(1.1.1.1.3.BU:1.05); according to CdTe panel recycling, First Solar
	disposal, plastics, mixture, 15.3% water, to municipal incineration	CH	-	-	-	kg	7.50E-1				1	1.07	(1.1.1.1.3.BU:1.05); according to CdTe panel recycling, First Solar
	disposal, inert waste, 5% water, to inert material landfill	CH	-	-	-	kg	1.57E-1				1	1.07	(1.1.1.1.3.BU:1.05); according to CdTe panel recycling, First Solar
avoided energy	natural gas, burned in industrial furnace >100kW	RER	-	-	-	MJ	-1.59E+01	-1.59E+01	-1.59E+01		1	1.14	(2.4.1.1.3.BU:1.05); natural gas consumption avoided by the recycling of PV glass
	heavy fuel oil, burned in industrial furnace 1MW, non-modulating	RER	-	-	-	MJ	-1.03E+01	-1.03E+01	-1.03E+01		1	1.14	(2.4.1.1.3.BU:1.05); heavy fuel oil consumption avoided by the recycling of PV glass
avoided materials	silica sand, at plant	DE	-	-	-	kg	-6.71E+00	-6.71E+00	-6.71E+00		1	1.14	(2.4.1.1.3.BU:1.05); silica sand consumption avoided by the recycling of the PV glass
	soda, powder, at plant	RER	-	-	-	kg	-2.66E+00	-2.66E+00	-2.66E+00		1	1.14	(2.4.1.1.3.BU:1.05); soda consumption avoided by the recycling of the PV glass
	limestone, milled, packed, at plant	CH	-	-	-	kg	-4.64E+00	-4.64E+00	-4.64E+00		1	1.14	(2.4.1.1.3.BU:1.05); limestone consumption avoided by the recycling of PV glass
	copper, at regional storage	RER	-	-	-	kg	-2.74E-01	-2.74E-01	-7.28E-02		1	1.14	(2.4.1.1.3.BU:1.05); primary copper avoided by the recycling of wires and the electric installation
	copper, secondary, at refinery	RER	-	-	-	kg	2.74E-01	2.74E-01	7.28E-02		1	1.14	(2.4.1.1.3.BU:1.05); efforts for producing secondary copper from scrap
	aluminium, primary, at plant	RER	-	-	-	kg	-2.14E+00	-2.54E+00	-3.32E+00		1	1.14	(2.4.1.1.3.BU:1.05); primary aluminium avoided by the recycling of frame and mounting structure
	aluminium, secondary, from old scrap, at plant	RER	-	-	-	kg	2.14E+00	2.54E+00	3.32E+00		1	1.14	(2.4.1.1.3.BU:1.05); efforts for producing secondary aluminium from scrap
	pig iron, at plant	GLO	-	-	-	kg	-1.44E-01	-9.63E-01	-4.55E+00		1	1.14	(2.4.1.1.3.BU:1.05); primary steel avoided by the recycling of the electric installation and the mounting system
avoided emissions air	Carbon dioxide, fossil	-	air	unspecified	-	kg	-2.41E+00	-2.41E+00	-2.41E+00		1	1.14	(2.4.1.1.3.BU:1.05); geogenic greenhouse gas emissions avoided by the recycling of PV glass

4.5 Multicrystalline-Si PV technology

4.5.1 Overview

The multicrystalline silicon PV chain is modeled according to the following structure:

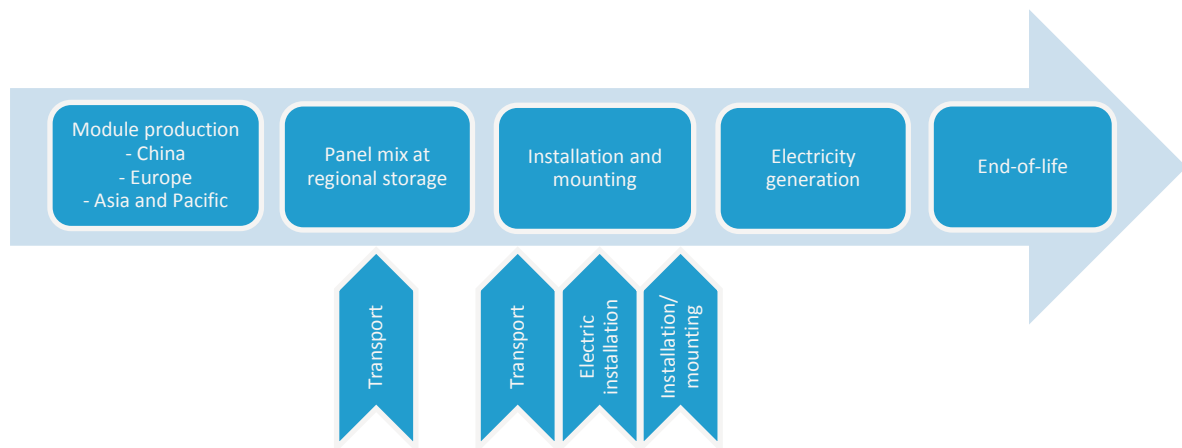


Fig. 4.4 Schematic illustration about the modeled life cycle chain of multicrystalline silicon PV panels

The life cycle inventories of the panel production, the mounting structure, the electricity production and the end-of-life treatment are described in sections 4.5.2 to 4.5.5. The following Tab. 4.22 summarizes the main characteristics of the multicrystalline-Si PV panel. The efficiency assumed for multi-Si PV modules (14.7 %) reflects the situation in the year 2011 and has increased since then.

Tab. 4.22 Characteristics of multi-Si PV panels

	Unit	Amount/value
Panel size	m ² /panel	1.6
Panel weight (unframed)	kg/m ²	11.2
Average module conversion efficiency	%	14.7
Wafer thickness	μm	200
Cell size	mm ²	156 · 156
Technology	-	Average technology mix of front/back cell connection, diffusion and front collection grid
Main data source / year	-	de Wild-Scholten (2014)

4.5.2 *Laminate and panel production*

The LCI data on material and energy consumption as well as emissions are updated based on LCI data of multi-crystalline modules published by de Wild-Scholten (2014). The inventories and detailed explanations are found in Itten & Frischknecht (2014), where, among others, the following life cycle inventories are compiled: photovoltaic cells at plant (Tab. 3.15), photovoltaic laminate and panel at plant (Tab. 3.19), photovoltaic laminate and panel at regional storage (Tab. 3.20).

The production of the multi-Si PV panels, which are used in Europe, takes place in Europe (15 %), in China (79 %), and in Asia and Pacific region (6 %) (Itten & Frischknecht 2014). The following Tab. 4.23 shows the unit process inventory data for the production of the multi-crystalline laminates and panels.

Tab. 4.23 unit process inventory data for the production of multi-Si photovoltaic panels produced in Europe, China and Asia and Pacific region

	Name	Location	Category	SubCategory	InfrastructureProcess	Unit	photovoltaic panel, multi-Si, at plant	photovoltaic laminate, multi-Si, at plant	photovoltaic panel, multi-Si, at plant	photovoltaic laminate, multi-Si, at plant	photovoltaic panel, multi-Si, at plant	photovoltaic laminate, multi-Si, at plant	Uncertainty Type	Standard Deviation 95%	GeneralComment
							CN	CN	APAC	APAC	RER	RER			
							1	1	1	1	1	1			
							m2	m2	m2	m2	m2	m2			
product	photovoltaic panel, multi-Si, at plant	CN	-	-	1	m2	1	0	0	0	0	0			
	photovoltaic laminate, multi-Si, at plant	CN	-	-	1	m2	0	1	0	0	0	0			
	photovoltaic panel, multi-Si, at plant	APAC	-	-	1	m2	0	0	1	0	0	0			
	photovoltaic laminate, multi-Si, at plant	APAC	-	-	1	m2	0	0	0	1	0	0			
	photovoltaic panel, multi-Si, at plant	RER	-	-	1	m2	0	0	0	0	1	0			
	photovoltaic laminate, multi-Si, at plant	RER	-	-	1	m2	0	0	0	0	0	1			
energy	electricity, medium voltage, production ENTSO, at grid	ENTSO	-	-	0	kWh	-	-	-	-	3.73E+0	3.73E+0	1	1.14	(3.3.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	electricity, medium voltage, at grid	CN	-	-	0	kWh	3.73E+0	3.73E+0	-	-	-	-	1	1.14	(3.3.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	electricity, medium voltage, at grid	JP	-	-	0	kWh	-	-	3.73E+0	3.73E+0	-	-	1	1.14	(3.3.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	diesel, burned in building machine	GLO	-	-	0	MJ	8.75E-3	8.75E-3	8.75E-3	8.75E-3	8.75E-3	8.75E-3	1	1.29	(3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
infrastructure	photovoltaic panel factory	GLO	-	-	1	unit	4.00E-6	4.00E-6	4.00E-6	4.00E-6	4.00E-6	4.00E-6	1	3.02	(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
materials	photovoltaic cell, multi-Si, at plant	RER	-	-	0	m2	-	-	-	-	9.35E-1	9.35E-1	1	1.13	(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	photovoltaic cell, multi-Si, at plant	CN	-	-	0	m2	9.35E-1	9.35E-1	-	-	-	-	1	1.13	(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	photovoltaic cell, multi-Si, at plant	APAC	-	-	0	m2	-	-	9.35E-1	9.35E-1	-	-	1	1.13	(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	aluminium alloy, AlMg3, at plant	RER	-	-	0	kg	2.13E+0	-	2.13E+0	-	2.13E+0	-	1	1.13	(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	copper, at regional storage	RER	-	-	0	kg	1.03E-1	1.03E-1	1.03E-1	1.03E-1	1.03E-1	1.03E-1	1	1.13	(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	wire drawing, copper	RER	-	-	0	kg	1.03E-1	1.03E-1	1.03E-1	1.03E-1	1.03E-1	1.03E-1	1	1.13	(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	diode, unspecified, at plant	GLO	-	-	0	kg	2.81E-3	2.81E-3	2.81E-3	2.81E-3	2.81E-3	2.81E-3	1	1.29	(3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	silicone product, at plant	RER	-	-	0	kg	1.22E-1	1.22E-1	1.22E-1	1.22E-1	1.22E-1	1.22E-1	1	1.13	(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	tin, at regional storage	RER	-	-	0	kg	1.29E-2	1.29E-2	1.29E-2	1.29E-2	1.29E-2	1.29E-2	1	1.29	(3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	lead, at regional storage	RER	-	-	0	kg	7.25E-4	7.25E-4	7.25E-4	7.25E-4	7.25E-4	7.25E-4	1	1.29	(3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	solar glass, low-iron, at regional storage	RER	-	-	0	kg	8.81E+0	8.81E+0	8.81E+0	8.81E+0	8.81E+0	8.81E+0	1	1.24	(1.4.1.3.3.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	tempering, flat glass	RER	-	-	0	kg	8.81E+0	8.81E+0	8.81E+0	8.81E+0	8.81E+0	8.81E+0	1	1.13	(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	glass fibre reinforced plastic, polyamide, injection moulding, at plant	RER	-	-	0	kg	2.95E-1	2.95E-1	2.95E-1	2.95E-1	2.95E-1	2.95E-1	1	1.13	(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	polyethylene terephthalate, granulate, amorphous, at plant	RER	-	-	0	kg	3.46E-1	3.46E-1	3.46E-1	3.46E-1	3.46E-1	3.46E-1	1	1.13	(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	polyethylene, HDPE, granulate, at plant	RER	-	-	0	kg	2.38E-2	2.38E-2	2.38E-2	2.38E-2	2.38E-2	2.38E-2	1	1.29	(3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	ethylvinylacetate, foil, at plant	RER	-	-	0	kg	8.75E-1	8.75E-1	8.75E-1	8.75E-1	8.75E-1	8.75E-1	1	1.13	(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	polyvinylfluoride film, at plant	US	-	-	0	kg	1.12E-1	1.12E-1	1.12E-1	1.12E-1	1.12E-1	1.12E-1	1	1.13	(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)

Tab. 4.23 unit process inventory data for the production of multi-Si photovoltaic panels produced in Europe, China and Asia and Pacific region (continued)

auxiliaries	tap water, at user	RER	-	-	0	kg	5.03E+0	5.03E+0	5.03E+0	5.03E+0	5.03E+0	5.03E+0	1	1.13	(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	hydrogen fluoride, at plant	GLO	-	-	0	kg	6.24E-2	6.24E-2	6.24E-2	6.24E-2	6.24E-2	6.24E-2	1	1.29	(3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	1-propanol, at plant	RER	-	-	0	kg	1.59E-2	1.59E-2	1.59E-2	1.59E-2	1.59E-2	1.59E-2	1	1.13	(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	isopropanol, at plant	RER	-	-	0	kg	1.47E-4	1.47E-4	1.47E-4	1.47E-4	1.47E-4	1.47E-4	1	1.29	(3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	potassium hydroxide, at regional storage	RER	-	-	0	kg	5.14E-2	5.14E-2	5.14E-2	5.14E-2	5.14E-2	5.14E-2	1	1.29	(3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	soap, at plant	RER	-	-	0	kg	1.16E-2	1.16E-2	1.16E-2	1.16E-2	1.16E-2	1.16E-2	1	1.29	(3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	corrugated board, mixed fibre, single wall, at plant	RER	-	-	0	kg	7.63E-1	7.63E-1	7.63E-1	7.63E-1	7.63E-1	7.63E-1	1	1.13	(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	EUR-flat pallet	RER	-	-	0	unit	5.00E-2	5.00E-2	5.00E-2	5.00E-2	5.00E-2	5.00E-2	1	1.29	(3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
transport	transport, lorry>16t, fleet average	RER	-	-	0	tkm	5.85E+0	5.64E+0	5.85E+0	5.64E+0	5.85E+0	5.64E+0	1	2.09	(4.5.na.na.na.na); Standard distance 100km, cells 500km
	transport, freight, rail	RER	-	-	0	tkm	4.25E+1	4.12E+1	4.25E+1	4.12E+1	4.25E+1	4.12E+1	1	2.09	(4.5.na.na.na.na); Standard distance 600km
disposal	disposal, municipal solid waste, 22.9% water, to municipal incineration	CH	-	-	0	kg	3.00E-2	3.00E-2	3.00E-2	3.00E-2	3.00E-2	3.00E-2	1	1.13	(1.4.1.3.1.3); Alsema (personal communication) 2007, production waste
	disposal, polyvinylfluoride, 0.2% water, to municipal incineration	CH	-	-	0	kg	1.12E-1	1.12E-1	1.12E-1	1.12E-1	1.12E-1	1.12E-1	1	1.13	(1.4.1.3.1.3); Calculation, including disposal of the panel after life time
	disposal, plastics, mixture, 15.3% water, to municipal incineration	CH	-	-	0	kg	1.64E+0	1.64E+0	1.64E+0	1.64E+0	1.64E+0	1.64E+0	1	1.13	(1.4.1.3.1.3); Calculation, including disposal of the panel after life time
	disposal, used mineral oil, 10% water, to hazardous waste incineration	CH	-	-	0	kg	1.61E-3	1.61E-3	1.61E-3	1.61E-3	1.61E-3	1.61E-3	1	1.13	(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	treatment, sewage, from residence, to wastewater treatment, class 2	CH	-	-	0	m3	5.03E-3	5.03E-3	5.03E-3	5.03E-3	5.03E-3	5.03E-3	1	1.13	(1.4.1.3.1.3); Calculation, water use
emissions air	Heat, waste	-	air	high population density	-	MJ	1.34E+1	1.34E+1	1.34E+1	1.34E+1	1.34E+1	1.34E+1	1	1.29	(3.4.3.3.1.5); Calculation, electricity use
	NM VOC, non-methane volatile organic compounds, unspecified origin	-	air	high population density	-	kg	8.06E-3	8.06E-3	8.06E-3	8.06E-3	8.06E-3	8.06E-3	1	1.61	(3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	Carbon dioxide, fossil	-	air	high population density	-	kg	2.18E-2	2.18E-2	2.18E-2	2.18E-2	2.18E-2	2.18E-2	1	1.29	(3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)

The following Tab. 4.24 shows the inventory data for multicrystalline-Si PV panels at the regional storage. The panels are imported from China and Malaysia to Rotterdam (about 20'000 km and 15'550 km respectively). From Rotterdam the panels are transported by lorry to the regional storage. The three main countries producing solar energy in Europe are Germany (60 %), Italy (31 %) and Spain (9 %). The average weighted transport distance to these countries is about 950 km.

Tab. 4.24 unit process inventory data for multicrystalline-Si photovoltaic panels at the European regional storage

	Name	Location	Category	SubCategory	InfrastructureProcess	Unit	photovoltaic panel, multi-Si, at regional storage	UncertaintyType	StandardDeviation95%	GeneralComment
	Location						RER			
	InfrastructureProcess						1			
	Unit						m2			
product	photovoltaic panel, multi-Si, at regional storage	RER	-	-	1	m2	1.00E+0			
materials	photovoltaic panel, multi-Si, at plant	RER	-	-	1	m2	1.45E-1	1	3.27	(5,1,1,1,1,5); modules produced in Europe
	photovoltaic panel, multi-Si, at plant	CN	-	-	1	m2	7.96E-1	1	3.27	(5,1,1,1,1,5); module import from China
	photovoltaic panel, multi-Si, at plant	APAC	-	-	1	m2	5.88E-2	1	3.27	(5,1,1,1,1,5); module import from APAC
transport	transport, transoceanic freight ship	OCE	-	-	0	tkm	1.82E+2	1	2.09	(4,5,na,na,na,na); Import of modules from China: 19994 km and Malaysia: 15549 km
	transport, lorry>16t, fleet average	RER	-	-	0	tkm	1.02E+1	1	2.09	(4,5,na,na,na,na); distance 950 km to regional storage from Rotterdam

4.5.3 Installation and mounting

The multicrystalline-Si PV panels are installed at three systems: 3 kWp installation, integrated in a slanted roof-top, 3 kWp installation, mounted on a slanted roof-top and a 570 kWp installation on open ground. The panel and construction material demand for all three installations is calculated based on an efficiency of 14.7 % (Frischknecht & Itten 2014). It is assumed that the panels have 1 % of rejects and 2 % need to be replaced during the life time. This was considered when calculating the panel demand per system. The roof systems need 20.4 m² of installation area, while the open ground

system consists of 3'880 m² module area. The electric installation and mounting structure is modeled according Jungbluth et. al (2012). The unit process inventory data are described in sections 4.7.1 and 4.7.2.

Transport from the regional storage to the construction site is also considered (100 km). The following Tab. 4.25 describes the unit process inventory data for each photovoltaic system with multicrystalline-Si PV panels.

Tab. 4.25 unit process inventory data for the mounting multicrystalline-Si photovoltaic panels integrated and mounted on a slanted roof top as well as the open ground system

	Name	Location	Category	SubCategory	InfrastructureProcess	Unit	3kWp slanted-roof installation, multi-Si, laminated, integrated, on roof	3kWp slanted-roof installation, multi-Si, panel, mounted, on roof	570 kWp open ground installation, multi-Si, on open ground	UncertaintyType	StandardDeviation95%	GeneralComment
	Location						RER	RER	RER			
	InfrastructureProcess						1 unit	1 unit	1 unit			
product	3kWp slanted-roof installation, multi-Si, laminated, integrated, on roof	RER	-	-	1	unit	1.00E+0	0	0			
	3kWp slanted-roof installation, multi-Si, panel, mounted, on roof	RER	-	-	1	unit	0	1.00E+0	0			
	570 kWp open ground installation, multi-Si, on open ground	RER	-	-	1	unit	0	0	1.00E+0			
energy	electricity, low voltage, production ENTSO, at grid	ENTSO	-	-	0	kWh	2.30E-1	2.30E-1	3.60E+1	1	1.28	(3,4,3,1,1,5,BU:1.05); Energy use for erection of plant
	diesel, burned in building machine	GLO	-	-	0	MJ			7.67E+3	1	1.28	(3,4,3,1,1,5,BU:1.05); Energy use for foundation piling and wheel loader
materials	electric installation, photovoltaic plant, at plant	CH	-	-	1	unit	1.00E+0	1.00E+0		1	3.08	(3,4,3,1,1,5,BU:3);
	electric installation, 570 kWp photovoltaic plant, at plant	ES	-	-	1	unit			1.00E+0	1	3.08	(3,4,3,1,1,5,BU:3);
	slanted-roof construction, integrated, on roof	RER	-	-	1	m2	2.04E+1			1	3.08	(3,4,3,1,1,5,BU:3); calculation per m2 panel
	slanted-roof construction, mounted, on roof	RER	-	-	1	m2		2.04E+1		1	3.08	(3,4,3,1,1,5,BU:3); calculation per m2 panel
	open ground construction, on ground	RER	-	-	1	m2			3.88E+3	1	3.08	(3,4,3,1,1,5,BU:3);
	photovoltaic panel, multi-Si, at regional storage	RER	-	-	1	m2	2.10E+1	2.10E+1	3.99E+3	1	3.09	(4,5,na,na,na,na,BU:3); panel demand
transport	transport, lorry 3.5-16t, fleet average	RER	-	-	0	tkm	3.38E+1	3.76E+1	9.59E+3	1	2.09	(3,4,3,1,1,5,BU:2); electric parts and panel 100km to construction place
emissions air	Heat, waste	-	air	high population density	-	MJ	8.28E-1	8.28E-1	1.30E+2	1	1.28	(3,4,3,1,1,5,BU:1.05); calculated with electricity use

4.5.4 Electricity generation

The three PV systems analyzed, 3 kWp roof installations (mounted and integrated) and the 570 kWp open ground installation, have an annual yield of 975 kWh/kWp (see also section 3.3.3). The expected life time is 30 years. Hence the overall output of electricity is 87.8 MWh for the roof installations and 16.7 GWh for the open ground system.

It is assumed that 20 liter of water are used per square meter for washing the modules (Jungbluth et al. 2012). This results in 4.65 ml of water per kWh electricity. The washing water is discharged into the sewer and to the municipal waste water treatment plant. No cleaning agents are considered.

The solar energy harvested by the PV system to produce electricity is calculated from the energy transformed to electricity fed into the inverter. Hence, 3.6 MJ solar energy are harvested to generate 1 kWh electricity. No direct emissions occur during the use stage of the panels.

The land occupation of the open ground PV system is taken into account in the mounting structure datasets. The land occupied by building roof integrated and mounted systems is fully attributed to the building. The following Tab. 4.26 shows the unit process inventory data of the electricity production with multicrystalline-Si PV systems.

Tab. 4.26 unit process inventory data for the electricity production with multicrystalline-Si photovoltaic panels at a 3 kWp roof installation, integrated and mounted, and a 570 kWp open ground installation

	Name	Location	Category	SubCategory	InfrastructureProcess	Unit	electricity, PV, at 3kWp slanted-roof, multi-Si, laminated, integrated	electricity, PV, at 3kWp slanted-roof, multi-Si, panel, mounted	electricity, PV, at 570kWp open ground, multi-Si	Uncertainty Type	StandardDeviation5%	GeneralComment
	Location						RER	RER	RER			
	InfrastructureProcess						0 kWh	0 kWh	0 kWh			
product	electricity, PV, at 3kWp slanted-roof, multi-Si, laminated, integrated	RER	-	-	0	kWh	1	0	0			
	electricity, PV, at 3kWp slanted-roof, multi-Si, panel, mounted	RER	-	-	0	kWh	0	1	0			
	electricity, PV, at 570kWp open ground, multi-Si	RER	-	-	0	kWh	0	0	1			
infrastructure	3kWp slanted-roof installation, multi-Si, laminated, integrated, on roof	RER	-	-	1	unit	1.14E-5			1	1.24	(2,3,1,1,3,5,BU:3); infrastructure; calculation
	3kWp slanted-roof installation, multi-Si, panel, mounted, on roof	RER	-	-	1	unit		1.14E-5		1	1.24	(2,3,1,1,3,5,BU:3); infrastructure; calculation
	570 kWp open ground installation, multi-Si, on open ground	RER	-	-	1	unit			6.00E-8	1	1.24	(2,3,1,1,3,5,BU:3); infrastructure; calculation
auxiliaries	tap water, at user	RER	-	-	0	kg	4.65E-3	4.65E-3	4.65E-3	1	1.05	(1,1,1,1,1,1,BU:1.05); water for washing, estimation 20 l per m2 panel; calculation
disposal	treatment, sewage, from residence, to wastewater treatment, class 2	CH	-	-	0	m3	4.65E-6	4.65E-6	4.65E-6	1	1.05	(1,2,1,1,1,1,BU:1.05); waste water treatment of washing water; calculation
resource in air	Energy, solar, converted	-	resource	in air	-	MJ	3.60E+0	3.60E+0	3.60E+0	1	1.20	(2,3,1,1,3,5,BU:1.05); solar energy needed for the production of 1kWh electricity (direct current); ecoinvent v2.2

4.5.5 End-of-life and benefits of recycling

After the use stage the panels and installations are deconstructed and disposed of. The PV panels are collected and transported to the recycling plant. As available data on the recycling of multicrystalline-Si PV panels is scarce, the recycling is modeled according to the recycling of CdTe panels (see section 4.2.5). The specific recycling efforts per kg of unframed CdTe PV modules recycled are adapted with a factor of 1.5. The estimated recycling efforts have been compared to data provided by recyclers taking part in the PV CYCLE programme and turned out to be conservative with regard to energy and material consumption. The WEEE target of a recycling rate higher than 80 % (by weight) until 2018 (EU Parliament 2012) can be achieved by recycling the aluminium frame, the junction box and glass. The frame and the junction box are manually disassembled. The remaining PV module is going to a flat glass recycling line, where the broken glass is separated. Assumed is that 90 % of the glass can replace raw materials in glass manufacturing (silicon sand, limestone and soda). The solar cells with polymer foils attached are currently stored at the glass recycler (like Maltha) and not treated. In this analysis we assume that the plastics are incinerated, the cells are landfilled and the copper is recycled. The avoided products are raw material feedstock in glass production (90 %), aluminium and copper.

Solar glass represents the largest share of recycled material and is used as cullet input in recycled foam glass production. The benefits for glass recycling are calculated based on the production of flat glass since it is not possible to clearly identify the material inputs replaced by glass cullets compared to the production of primary foam glass. According to the bill of materials of multicrystalline-Si PV modules (Tab. 3.1), the mass of glass per m² panel is 8.81 kg. This amount is corrected in order to account for manufacturing losses, which are estimated to about 3.8 % based on data for CdTe PV module production. The efficiency of glass recycling is assumed to be equal to the CdTe panel recycling, namely 90 %.

Similar to the recycling of CdTe, the consumption of primary materials used in glass production is avoided when using the glass cullets from recycled multicrystalline-Si PV panels. Hence, the geogenic CO₂-emissions caused by primary raw materials are prevented (Held & Ilg 2011), which amount to 0.208 kg CO₂/kg flat glass (Kellenberger et al. 2007). This results in avoided emissions of 1.58 kg CO₂ eq per m² recycled multicrystalline-Si PV module. As the melting of the glass cullet takes less energy than the melting of limestone and silica sand, heavy fuel oil and natural gas can additionally be saved. Held and Ilg (2011) assume an energy reduction potential in the melting process in the range of around 30 % when replacing 100 % input material with glass cullet.

Aluminium and steel used in the mounting structures as well as copper and steel contained in the electric installation are likewise recycled. In line with Jungbluth et al. (2012) it is assumed that only aluminium, copper and steel are recycled but not the plastics. Being large construction parts, a recycling efficiency of 100 % is assumed.

Efforts to produce secondary metals from scrap are taken into account for aluminium and copper. These secondary metals are potentially avoiding primary aluminium and primary copper production, respectively. Steel scrap is potentially avoiding pig iron feedstock in converter steel production. Benefits for recycling are granted only for the net surplus amount of recycled material, which leaves the PV system (see also EN 15804 2013). According to the ecoinvent data v2.2+ the recycled content of the metals used in panels, cabling and mounting structures is 32 % for aluminium, 77 % for the AlMg3 alloy used in the panel frame, 44 % for copper and 37 % for steel (KBOB et al. 2014). The application of the end-of-life formula is described in more detail in section 4.9.9.

The unit process inventory data of multicrystalline-Si PV module takeback and recycling, including benefits and additional loads beyond the system boundary, are listed in Tab. 4.27.

Tab. 4.27 unit process inventory data for the recycling and the potential benefits and additional loads beyond the system boundary by recycling multicrystalline-Si photovoltaic panels, cabling and mounting structures

	Name	Location	Category	SubCategory	InfrastructureProcess	Unit	multi-Si PV module takeback + recycling	Avoided burden from recycling, multi-Si PV module, integrated construction	Avoided burden from recycling, multi-Si PV module, mounted construction	Avoided burden from recycling, multi-Si PV module, open ground construction	UncertaintyType	StandardDeviation5%	GeneralComment
	Location						RER	RER	RER	RER			
	InfrastructureProcess						0 m2	0 m2	0 m2	0 m2			
product	multi-Si PV module takeback + recycling	RER	-	-	-	m2	1.00E+0	0	0	0			
	Avoided burden from recycling, multi-Si PV module, integrated construction	RER	-	-	-	m2	0	1.00E+0	0	0			
	Avoided burden from recycling, multi-Si PV module, mounted construction	RER	-	-	-	m2	0	0	1.00E+0	0			
	Avoided burden from recycling, multi-Si PV module, open ground construction	RER	-	-	-	m2	0	0	0	1.00E+0			
energy	electricity, medium voltage, production ENTSO, at grid	ENTSO	-	-	-	kWh	4.30E+0				1	1.07	(1.1.1.1.3.BU:1.05); according to CdTe panel recycling, First Solar
auxiliaries	water, deionised, at plant	CH	-	-	-	kg	5.33E+0				1	1.13	(1.4.1.3.1.3.BU:1.05); according to CdTe panel recycling, First Solar
	sulphuric acid, liquid, at plant	RER	-	-	-	kg	8.20E-2				1	1.07	(1.1.1.1.3.BU:1.05); according to CdTe panel recycling, First Solar
	hydrogen peroxide, 50% in H2O, at plant	RER	-	-	-	kg	5.62E-1				1	1.07	(1.1.1.1.3.BU:1.05); according to CdTe panel recycling, First Solar
	sodium hydroxide, 50% in H2O, production mix, at plant	RER	-	-	-	kg	1.02E-1				1	1.07	(1.1.1.1.3.BU:1.05); according to CdTe panel recycling, First Solar
transport	transport, lorry >16t, fleet average	RER	-	-	-	tkm	1.23E+1				1	2.00	(1.1.1.1.3.BU:2); according to CdTe panel recycling, First Solar. Assumption: material for electric installation and mounting system has a transport of 200 km.
disposal	treatment, PV cell production effluent, to wastewater treatment, class 3	CH	-	-	-	m3	4.71E-3				1	1.07	(1.1.1.1.3.BU:1.05); according to CdTe panel recycling, First Solar
	disposal, plastics, mixture, 15.3% water, to municipal incineration	CH	-	-	-	kg	6.06E-1				1	1.07	(1.1.1.1.3.BU:1.05); according to CdTe panel recycling, First Solar
	disposal, inert waste, 5% water, to inert material landfill	CH	-	-	-	kg	1.26E-1				1	1.07	(1.1.1.1.3.BU:1.05); according to CdTe panel recycling, First Solar
avoided energy	natural gas, burned in industrial furnace >100kW	RER	-	-	-	MJ		-1.04E+01	-1.04E+01	-1.04E+01	1	1.14	(2.4.1.1.1.3.BU:1.05); natural gas consumption avoided by the recycling of PV glass
	heavy fuel oil, burned in industrial furnace 1MW, non-modulating	RER	-	-	-	MJ		-6.76E+00	-6.76E+00	-6.76E+00	1	1.14	(2.4.1.1.1.3.BU:1.05); heavy fuel oil consumption avoided by the recycling of PV glass
avoided materials	silica sand, at plant	DE	-	-	-	kg		-4.41E+00	-4.41E+00	-4.41E+00	1	1.14	(2.4.1.1.1.3.BU:1.05); silica sand consumption avoided by the recycling of the PV glass
	soda, powder, at plant	RER	-	-	-	kg		-1.75E+00	-1.75E+00	-1.75E+00	1	1.14	(2.4.1.1.1.3.BU:1.05); soda consumption avoided by the recycling of the PV glass
	limestone, milled, packed, at plant	CH	-	-	-	kg		-3.05E+00	-3.05E+00	-3.05E+00	1	1.14	(2.4.1.1.1.3.BU:1.05); limestone consumption avoided by the recycling of PV glass
	copper, at regional storage	RER	-	-	-	kg		-4.03E-01	-4.03E-01	-1.07E-01	1	1.14	(2.4.1.1.1.3.BU:1.05); primary copper avoided by the recycling of wires and the electric installation
	copper, secondary, at refinery	RER	-	-	-	kg		4.03E-01	4.03E-01	1.07E-01	1	1.14	(2.4.1.1.1.3.BU:1.05); efforts for producing secondary copper from scrap
	aluminium, primary, at plant	RER	-	-	-	kg		-2.02E+00	-2.42E+00	-3.19E+00	1	1.14	(2.4.1.1.1.3.BU:1.05); primary aluminium avoided by the recycling of frame and mounting structure
	aluminium, secondary, from old scrap, at plant	RER	-	-	-	kg		2.02E+00	2.42E+00	3.19E+00	1	1.14	(2.4.1.1.1.3.BU:1.05); efforts for producing secondary aluminium from scrap
	pig iron, at plant	GLO	-	-	-	kg		-1.53E-01	-9.71E-01	-4.55E+00	1	1.14	(2.4.1.1.1.3.BU:1.05); primary steel avoided by the recycling of the electric installation and the mounting system
avoided emissions air	Carbon dioxide, fossil	-	air	unspecified	-	kg		-1.58E+00	-1.58E+00	-1.58E+00	1	1.14	(2.4.1.1.3.BU:1.05); geogenic greenhouse gas emissions avoided by the recycling of PV glass

4.6 Monocrystalline-Si PV technology

4.6.1 Overview

The monocrystalline silicon PV chain is modeled according to the following structure:

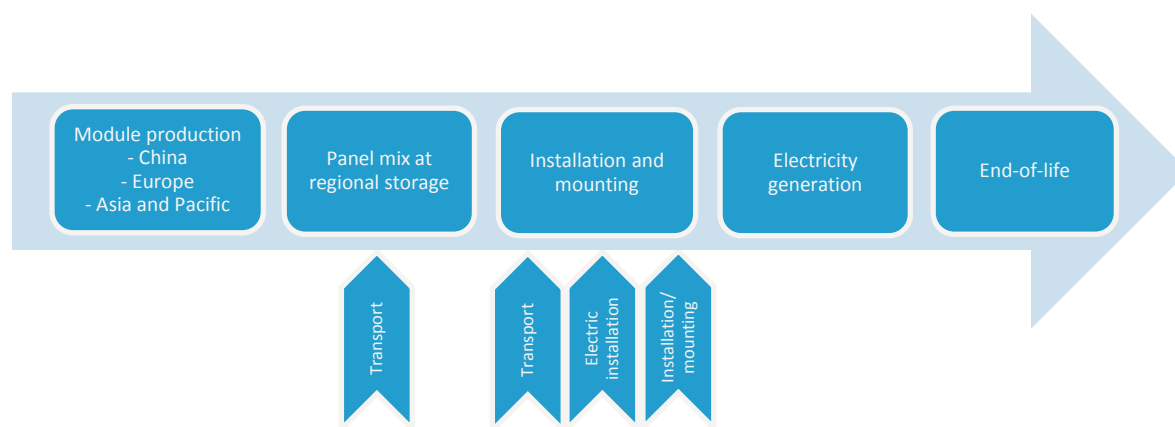


Fig. 4.5 Schematic illustration about the modeled life cycle chain of monocrystalline silicon PV panels

The processes are described in the following sections 4.6.2 to 4.6.5. The following Tab. 4.28 summarizes the main characteristics of the monocrystalline-Si PV panel. The efficiency assumed for mono-Si PV modules (15.1 %) reflects the situation in the year 2011 and has increased since then.

Tab. 4.28 Characteristics of mono-Si PV panels

	Unit	Amount/value
Panel size	m ² /panel	1.6
Panel weight (unframed)	kg/m ²	11.7
Module conversion efficiency	%	15.1
Wafer thickness	μm	190
Cell size	mm ²	156 · 156
Technology		Average technology mix of front/back cell connection, diffusion and front collection grid
Main data source / year	-	de Wild-Scholten (2014)

4.6.2 Laminate and panel production

The LCI data on material and energy consumption as well as emissions are updated based on LCI data of monocrystalline-Si modules published by de Wild-Scholten (2014). The inventories and detailed explanations are found in Itten & Frischknecht (2014), where, among others, the following life cycle inventories are compiled: photovoltaic cells at plant (Tab. 3.15), photovoltaic laminate and panel at plant (Tab. 3.19), photovoltaic laminate and panel at regional storage (Tab. 3.20).

The production of the monocrystalline-Si PV panels, which are used in Europe, takes place in Europe (15 %), China (79 %), and in the Asia and Pacific region (6 %) (Itten & Frischknecht 2014). The following Tab. 4.29 shows the unit process inventory data for the production of the monocrystalline laminates and panels.

Tab. 4.29 unit process inventory data for the production of mono-Si photovoltaic panels produced in Europe, China and Asia and Pacific region

	Name	Location	Category	SubCategory	Infrastructure	Unit	photovoltaic laminate, single-Si, at plant	photovoltaic panel, single-Si, at plant	photovoltaic panel, single-Si, at plant	photovoltaic laminate, single-Si, at plant	photovoltaic panel, single-Si, at plant	photovoltaic laminate, single-Si, at plant	Uncertainty/Type	Standard Deviation	95%	GeneralComment
product	Location						CN	CN	APAC	APAC	RER	RER				
	Infrastructure/Process						1	1	1	1	1	1				
	Unit						m2	m2	m2	m2	m2	m2				
energy	photovoltaic laminate, single-Si, at plant	CN	-	-	1	m2	1	0	0	0	0	0				
	photovoltaic panel, single-Si, at plant	CN	-	-	1	m2	0	1	0	0	0	0				
	photovoltaic panel, single-Si, at plant	APAC	-	-	1	m2	0	0	1	0	0	0				
	photovoltaic laminate, single-Si, at plant	APAC	-	-	1	m2	0	0	0	1	0	0				
	photovoltaic panel, single-Si, at plant	RER	-	-	1	m2	0	0	0	0	1	0				
	photovoltaic laminate, single-Si, at plant	RER	-	-	1	m2	0	0	0	0	0	1				
energy	electricity, medium voltage, production	ENTSO	-	-	0	kWh	-	-	-	-	3.73E+0	3.73E+0	1	1.14		(3.3.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	electricity, medium voltage, at grid	CN	-	-	0	kWh	3.73E+0	3.73E+0	-	-	-	-	1	1.14		(3.3.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	electricity, medium voltage, at grid	JP	-	-	0	kWh	-	-	3.73E+0	3.73E+0	-	-	1	1.14		(3.3.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	diesel, burned in building machine	GLO	-	-	0	MJ	8.75E-3	8.75E-3	8.75E-3	8.75E-3	8.75E-3	8.75E-3	1	1.29		(3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
infrastructure	photovoltaic panel factory	GLO	-	-	1	unit	4.00E-6	4.00E-6	4.00E-6	4.00E-6	4.00E-6	4.00E-6	1	3.02		(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
materials	photovoltaic cell, single-Si, at plant	RER	-	-	0	m2	-	-	-	-	9.35E-1	9.35E-1	1	1.13		(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	photovoltaic cell, single-Si, at plant	CN	-	-	0	m2	9.35E-1	9.35E-1	-	-	-	-	1	1.13		(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	photovoltaic cell, single-Si, at plant	APAC	-	-	0	m2	-	-	9.35E-1	9.35E-1	-	-	1	1.13		(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	aluminium alloy, AlMg3, at plant	RER	-	-	0	kg	-	2.13E+0	2.13E+0	-	2.13E+0	-	1	1.13		(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	copper, at regional storage	RER	-	-	0	kg	1.03E-1	1.03E-1	1.03E-1	1.03E-1	1.03E-1	1.03E-1	1	1.13		(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	wire drawing, copper	RER	-	-	0	kg	1.03E-1	1.03E-1	1.03E-1	1.03E-1	1.03E-1	1.03E-1	1	1.13		(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	diode, unspecified, at plant	GLO	-	-	0	kg	2.81E-3	2.81E-3	2.81E-3	2.81E-3	2.81E-3	2.81E-3	1	1.29		(3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	silicone product, at plant	RER	-	-	0	kg	1.22E-1	1.22E-1	1.22E-1	1.22E-1	1.22E-1	1.22E-1	1	1.13		(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	tin, at regional storage	RER	-	-	0	kg	1.29E-2	1.29E-2	1.29E-2	1.29E-2	1.29E-2	1.29E-2	1	1.29		(3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	lead, at regional storage	RER	-	-	0	kg	7.25E-4	7.25E-4	7.25E-4	7.25E-4	7.25E-4	7.25E-4	1	1.29		(3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	solar glass, low-iron, at regional storage	RER	-	-	0	kg	8.81E+0	8.81E+0	8.81E+0	8.81E+0	8.81E+0	8.81E+0	1	1.24		(1.4.1.3.3.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	tempering, flat glass	RER	-	-	0	kg	8.81E+0	8.81E+0	8.81E+0	8.81E+0	8.81E+0	8.81E+0	1	1.13		(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	glass fibre reinforced plastic, polyamide, injection moulding, at plant	RER	-	-	0	kg	2.95E-1	2.95E-1	2.95E-1	2.95E-1	2.95E-1	2.95E-1	1	1.13		(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	polyethylene terephthalate, granulate, amorphous, at plant	RER	-	-	0	kg	3.46E-1	3.46E-1	3.46E-1	3.46E-1	3.46E-1	3.46E-1	1	1.13		(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	polyethylene, HDPE, granulate, at plant	RER	-	-	0	kg	2.38E-2	2.38E-2	2.38E-2	2.38E-2	2.38E-2	2.38E-2	1	1.29		(3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	ethylvinylacetate, foil, at plant	RER	-	-	0	kg	8.75E-1	8.75E-1	8.75E-1	8.75E-1	8.75E-1	8.75E-1	1	1.13		(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	polyvinylfluoride film, at plant	US	-	-	0	kg	1.12E-1	1.12E-1	1.12E-1	1.12E-1	1.12E-1	1.12E-1	1	1.13		(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)

Tab. 4.29 unit process inventory data for the production of mono-Si photovoltaic panels produced in Europe, China and Asia and Pacific region (continued)

auxiliaries	tap water, at user	RER	-	-	0	kg	5.03E+0	5.03E+0	5.03E+0	5.03E+0	5.03E+0	5.03E+0	1	1.13	(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	hydrogen fluoride, at plant	GLO	-	-	0	kg	6.24E-2	6.24E-2	6.24E-2	6.24E-2	6.24E-2	6.24E-2	1	1.29	(3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	1-propanol, at plant	RER	-	-	0	kg	1.59E-2	1.59E-2	1.59E-2	1.59E-2	1.59E-2	1.59E-2	1	1.13	(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	isopropanol, at plant	RER	-	-	0	kg	1.47E-4	1.47E-4	1.47E-4	1.47E-4	1.47E-4	1.47E-4	1	1.29	(3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	potassium hydroxide, at regional storage	RER	-	-	0	kg	5.14E-2	5.14E-2	5.14E-2	5.14E-2	5.14E-2	5.14E-2	1	1.29	(3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	soap, at plant	RER	-	-	0	kg	1.16E-2	1.16E-2	1.16E-2	1.16E-2	1.16E-2	1.16E-2	1	1.29	(3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	corrugated board, mixed fibre, single wall, at plant	RER	-	-	0	kg	7.63E-1	7.63E-1	7.63E-1	7.63E-1	7.63E-1	7.63E-1	1	1.13	(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
transport	EUR-flat pallet	RER	-	-	0	unit	5.00E-2	5.00E-2	5.00E-2	5.00E-2	5.00E-2	5.00E-2	1	1.29	(3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	transport, lorry>16t, fleet average	RER	-	-	0	tkm	5.64E+0	5.85E+0	5.85E+0	5.64E+0	5.85E+0	5.64E+0	1	2.09	(4.5.na.na.na.na); Standard distance 100km, cells 500km
disposal	transport, freight, rail	RER	-	-	0	tkm	4.12E+1	4.25E+1	4.25E+1	4.12E+1	4.25E+1	4.12E+1	1	2.09	(4.5.na.na.na.na); Standard distance 600km
	disposal, municipal solid waste, 22.9% water, to municipal incineration	CH	-	-	0	kg	3.00E-2	3.00E-2	3.00E-2	3.00E-2	3.00E-2	3.00E-2	1	1.13	(1.4.1.3.1.3); Asema (personal communication) 2007, production waste
	disposal, polyvinylfluoride, 0.2% water, to municipal incineration	CH	-	-	0	kg	1.12E-1	1.12E-1	1.12E-1	1.12E-1	1.12E-1	1.12E-1	1	1.13	(1.4.1.3.1.3); Calculation, including disposal of the panel after life time
	disposal, plastics, mixture, 15.3% water, to municipal incineration	CH	-	-	0	kg	1.64E+0	1.64E+0	1.64E+0	1.64E+0	1.64E+0	1.64E+0	1	1.13	(1.4.1.3.1.3); Calculation, including disposal of the panel after life time
	disposal, used mineral oil, 10% water, to hazardous waste incineration	CH	-	-	0	kg	1.61E-3	1.61E-3	1.61E-3	1.61E-3	1.61E-3	1.61E-3	1	1.13	(1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	treatment, sewage, from residence, to wastewater treatment, class 2	CH	-	-	0	m3	5.03E-3	5.03E-3	5.03E-3	5.03E-3	5.03E-3	5.03E-3	1	1.13	(1.4.1.3.1.3); Calculation, water use
emissions air	Heat, waste	-	air	high population density	-	MJ	1.34E+1	1.34E+1	1.34E+1	1.34E+1	1.34E+1	1.34E+1	1	1.29	(3.4.3.3.1.5); Calculation, electricity use
	NMOC, non-methane volatile organic compounds, unspecified origin	-	air	high population density	-	kg	8.06E-3	8.06E-3	8.06E-3	8.06E-3	8.06E-3	8.06E-3	1	1.61	(3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)
	Carbon dioxide, fossil	-	air	high population density	-	kg	2.18E-2	2.18E-2	2.18E-2	2.18E-2	2.18E-2	2.18E-2	1	1.29	(3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)

The following Tab. 4.30 shows the inventory data for monocrystalline-Si PV panels at the regional storage. The panels are imported from China and Malaysia to Rotterdam (about 20'000 km and 15'550 km respectively). From Rotterdam the panels are transported by lorry to the regional storage. The three main countries producing solar energy in Europe are Germany (60 %), Italy (31 %) and Spain (9 %). The average weighted transport distance to these countries is about 950 km.

Tab. 4.30 unit process inventory data for monocrystalline-Si photovoltaic panels at the European regional storage

	Name	Location	Category	SubCategory	InfrastructureProcess	Unit	photovoltaic panel, single-Si, at regional storage			GeneralComment		
	Location						UncertaintyType			StandardDeviation95%		
	InfrastructureProcess						RER					
	Unit						1			m2		
product	photovoltaic panel, single-Si, at regional storage	RER	-	-	1	m2	1.00E+0					
materials	photovoltaic panel, single-Si, at plant	RER	-	-	1	m2	1.45E-1	1	3.27	(5,1,1,1,1,5);	modules produced in Europe	
	photovoltaic panel, single-Si, at plant	CN	-	-	1	m2	7.96E-1	1	3.27	(5,1,1,1,1,5);	module import from China	
	photovoltaic panel, single-Si, at plant	APAC	-	-	1	m2	5.88E-2	1	3.27	(5,1,1,1,1,5);	module import from APAC	
transport	transport, transoceanic freight ship	OCE	-	-	0	tkm	1.91E+2	1	2.09	(4,5,na,na,na,na);	Import of modules from China: 19994 km and Malaysia: 15549 km	
	transport, lorry >16t, fleet average	RER	-	-	0	tkm	1.07E+1	1	2.09	(4,5,na,na,na,na);	distance 950 km to regional storage from Rotterdam	

4.6.3 Installation and mounting

The monocrystalline-Si PV panels are installed at three systems: 3 kWp installation, integrated in a slanted roof-top, 3 kWp installation, mounted on a slanted roof-top and a 570 kWp installation on open ground. The panel and construction material demand for all three installations is calculated based on an efficiency of 15.1 % (Frischknecht & Itten 2014). It is assumed that the panels have 1 %

of rejects and 2 % need to be replaced during the life time. This was considered when calculating the panel demand per system. The roof systems need 19.9 m² of installation area, while the open ground system consists of 3'770 m² installation area. Electricity is used for the mounting of the modules.

The electric installation and construction is modeled according Jungbluth et. al (2012). The unit process inventory data are described in sections 4.7.1 and 4.7.2.

Transport from the regional storage to the construction site is considered with 100 km lorry-transport. The following Tab. 4.31 describes the unit process inventory data for each photovoltaic system with mono-crystalline-Si PV panels.

Tab. 4.31 unit process inventory data for the mounting monocrystalline Si photovoltaic panels integrated and mounted on a slanted roof top as well as the open ground system

	Name	Location	Category	SubCategory	InfrastructureProcess	Unit	3kWp slanted-roof installation, mono-Si, laminated, integrated, on roof	3kWp slanted-roof installation, mono-Si, panel, mounted, on roof	570 kWp open ground installation, mono-Si, on open ground	UncertaintyType	StandardDeviation5%	GeneralComment
	Location						RER	RER	RER			
	InfrastructureProcess						1 unit	1 unit	1 unit			
	Unit											
product	3kWp slanted-roof installation, mono-Si, laminated, integrated, on roof	RER	-	-	1	unit	1.00E+0	0	0			
	3kWp slanted-roof installation, mono-Si, panel, mounted, on roof	RER	-	-	1	unit	0	1.00E+0	0			
	570 kWp open ground installation, mono-Si, on open ground	RER	-	-	1	unit	0	0	1.00E+0			
energy	electricity, low voltage, production ENTSO, at grid	ENTSO	-	-	0	kWh	2.30E-1	2.30E-1	3.60E+1	1	1.28	(3,4,3,1,1,5,BU:1.05); Energy use for erection of plant
	diesel, burned in building machine	GLO	-	-	0	MJ			7.67E+3	1	1.28	(3,4,3,1,1,5,BU:1.05); Energy use for foundation piling and wheel loader
materials	electric installation, photovoltaic plant, at plant	CH	-	-	1	unit	1.00E+0	1.00E+0		1	3.08	(3,4,3,1,1,5,BU:3);
	electric installation, 570 kWp photovoltaic plant, at plant	ES	-	-	1	unit			1.00E+0	1	3.08	(3,4,3,1,1,5,BU:3);
	slanted-roof construction, integrated, on roof	RER	-	-	1	m2	1.99E+1			1	3.08	(3,4,3,1,1,5,BU:3); calculation per m2 panel
	slanted-roof construction, mounted, on roof	RER	-	-	1	m2		1.99E+1		1	3.08	(3,4,3,1,1,5,BU:3); calculation per m2 panel
	open ground construction, on ground	RER	-	-	1	m2			3.77E+3	1	3.08	(3,4,3,1,1,5,BU:3);
	photovoltaic panel, single-Si, at regional storage	RER	-	-	1	m2	2.05E+1	2.05E+1	3.89E+3	1	3.08	(3,4,3,1,1,5,BU:2); Calculation, 2% of modules repaired in the life time, 1% rejects
transport	transport, lorry 3.5-16t, fleet average	RER	-	-	0	tkm	3.39E+1	3.77E+1	9.53E+3	1	2.09	(3,4,3,1,1,5,BU:2); electric parts and panel 100km to construction place
emissions air	Heat, waste	-	air	high population density	-	MJ	8.28E-1	8.28E-1	1.30E+2	1	1.28	(3,4,3,1,1,5,BU:1.05); calculated with electricity use

4.6.4 Electricity generation

The three PV systems analyzed, 3 kWp roof installations (mounted and integrated) and the 570 kWp open ground installation, have an annual yield of 975 kWh/kWp (see also section 3.3.3). The expected life time is 30 years. Hence the overall output of electricity is 87.8 MWh for the roof installations and 16.7 GWh for the open ground system.

It is assumed that 20 liter of water are used per square meter for washing the modules (Jungbluth et al. 2012). This results in 4.53 ml of water per kWh electricity. The washing water is discharged into the sewer and to the municipal waste water treatment plant. No cleaning agents are considered.

The solar energy harvested by the PV system to produce electricity is calculated from the energy transformed to electricity fed into the inverter. Hence, 3.6 MJ solar energy are harvested to generate 1 kWh electricity. No direct emissions occur during the use stage of the panels.

The land occupation of the open ground PV system is taken into account in the mounting structure datasets. The land occupied by building roof integrated and mounted systems is fully attributed to the building. The following Tab. 4.32 shows the inventory of the electricity production from monocrystalline-Si PV systems.

Tab. 4.32 unit process inventory data for the electricity production with monocrystalline-Si photovoltaic panels at a 3 kWp roof installation, integrated and mounted, and a 570 kWp open ground installation

	Name	Location	Category	SubCategory	InfrastructureProcess	Unit	electricity, PV, at 3kWp slanted-roof, mono-Si, laminated, integrated	electricity, PV, at 3kWp slanted-roof, mono-Si, panel, mounted	electricity, PV, at 570kWp open ground, mono-Si	Uncertainty Type	StandardDeviation5%	GeneralComment
	Location						RER	RER	RER			
	InfrastructureProcess						0 kWh	0 kWh	0 kWh			
product	electricity, PV, at 3kWp slanted-roof, mono-Si, laminated, integrated	RER	-	-	0	kWh	1	0	0			
	electricity, PV, at 3kWp slanted-roof, mono-Si, panel, mounted	RER	-	-	0	kWh	0	1	0			
	electricity, PV, at 570kWp open ground, mono-Si	RER	-	-	0	kWh	0	0	1			
infrastructure	3kWp slanted-roof installation, mono-Si, laminated, integrated, on roof	RER	-	-	1	unit	1.14E-5			1	1.24	(2,3,1,1,3,5,BU:3); infrastructure; calculation
	3kWp slanted-roof installation, mono-Si, panel, mounted, on roof	RER	-	-	1	unit		1.14E-5		1	1.24	(2,3,1,1,3,5,BU:3); infrastructure; calculation
	570 kWp open ground installation, mono-Si, on open ground	RER	-	-	1	unit			6.00E-8	1	1.24	(2,3,1,1,3,5,BU:3); infrastructure; calculation
auxiliaries	tap water, at user	RER	-	-	0	kg	4.53E-3	4.53E-3	4.53E-3	1	1.05	(1,1,1,1,1,1,BU:1.05); water for washing, estimation 20 l per m2 panel; calculation
disposal	treatment, sewage, from residence, to wastewater treatment, class 2	CH	-	-	0	m3	4.53E-6	4.53E-6	4.53E-6	1	1.05	(1,2,1,1,1,1,BU:1.05); waste water treatment of washing water; calculation
resource in air	Energy, solar, converted	-	resource	in air	-	MJ	3.60E+0	3.60E+0	3.60E+0	1	1.20	(2,3,1,1,3,5,BU:1.05); solar energy needed for the production of 1kWh electricity (direct current); ecoinvent v2.2

4.6.5 End-of-life and benefits for recycling

After the use stage the panels and installations are deconstructed and disposed of. The PV panels are collected and transported to the recycling plant. As available data on the recycling of monocrystalline-Si PV panels is scarce, the recycling is modeled according to the recycling of CdTe panels (see section 4.2.5). The specific recycling efforts per kg of unframed CdTe PV modules recycled are adapted with a factor of 1.5. The estimated recycling efforts have been compared to data provided by recyclers taking part in the PV CYCLE programme and turned out to be conservative with regard to energy and material consumption. The WEEE target of a recycling rate higher than 80 % (by weight) until 2018 (EU Parliament 2012) can be achieved by recycling the aluminium frame, the junction box and glass. The frame and the junction box are manually disassembled. The remaining PV module is going to a flat glass recycling line, where the broken glass is separated. Assumed is that 90 % of the glass can replace raw materials in glass manufacturing (silicon sand, limestone and soda). The solar cells with polymer foils attached are currently stored at the glass recycler (like Maltha) and not treated. In this analysis we assume that the plastics are incinerated, the cells are landfilled and the copper is recycled. The avoided products are raw material feedstock in glass production (90 %), aluminium and copper.

Solar glass represents the largest share of recycled material and is used as cullet input in recycled foam glass production. The benefits for glass recycling are calculated based on the production of flat glass since it is not possible to clearly identify the material inputs replaced by glass cullets compared to the production of primary foam glass. According to the bill of materials of monocrystalline-Si PV modules (Tab. 3.1), the mass of glass per m² panel is 8.81 kg. This amount is corrected in order to account for manufacturing losses, which are estimated to about 3.8 % based on data for CdTe PV module production. The efficiency of glass recycling is assumed to be equal to the CdTe panel recycling, namely 90 %.

Similar to the recycling of CdTe, the consumption of primary materials used in glass production is avoided when using the glass cullets from recycled monocrystalline-Si PV panels. Hence, the geogenic CO₂-emissions caused by primary raw materials are prevented (Held & Ilg 2011), which amount to 0.208 kg CO₂/kg flat glass (Kellenberger et al. 2007). This results in avoided emissions of 1.58 kg CO₂ eq per m² recycled monocrystalline-Si PV module. As the melting of the glass cullet takes less energy than the melting of limestone and silica sand, heavy fuel oil and natural gas can additionally be saved. Held and Ilg (2011) assume an energy reduction potential in the melting process of around 30 % when replacing 100 % input material with glass cullet. Aluminium and steel used in the mounting structures as well as the copper and steel contained in the electric installation are

likewise recycled. In line with Jungbluth et al. (2012) it is assumed that only aluminium, copper and steel are recycled but not the plastics. Being large construction parts, a recycling efficiency of 100 % is assumed.

Efforts to produce secondary metals from scrap are taken into account for aluminium and copper. These secondary metals are potentially avoiding primary aluminium and primary copper production, respectively. Steel scrap is potentially avoiding pig iron feedstock in converter steel production. Benefits for recycling are granted only for the net surplus amount of recycled material, which leaves the PV system (see also EN 15804 2013). According to the ecoinvent data v2.2+ the recycled content of the metals used in panels, cabling and mounting structures is 32 % for aluminium, 77 % for the AlMg₃ alloy used in the panel frame, 44 % for copper and 37 % for steel (KBOB et al. 2014). The application of the end-of-life formula is described in more detail in section 4.9.9.

The unit process inventory data of monocrystalline-Si PV module takeback and recycling, including benefits and additional loads beyond the system boundary, are listed in Tab. 4.33.

Tab. 4.33 unit process inventory data for the recycling and the potential benefits and additional loads beyond the system boundary by recycling monocrystalline-Si photovoltaic panels, cabling and mounting structures

	Name	Location	Category	SubCategory	InfrastructureProcess	Unit	mono-Si PV module takeback + recycling	Avoided burden from recycling, mono-Si PV module, integrated construction	Avoided burden from recycling, mono-Si PV module, mounted construction	Avoided burden from recycling, mono-Si PV module, open ground construction	UncertaintyType	StandardDeviation5%	GeneralComment
							RER 0 m2	RER 0 m2	RER 0 m2	RER 0 m2			
	Location												
	InfrastructureProcess												
product	mono-Si PV module takeback + recycling	RER	-	-	-	m2	1.00E+0	0	0	0			
	Avoided burden from recycling, mono-Si PV module, integrated construction	RER	-	-	-	m2	0	1.00E+0	0	0			
	Avoided burden from recycling, mono-Si PV module, mounted construction	RER	-	-	-	m2	0	0	1.00E+0	0			
	Avoided burden from recycling, mono-Si PV module, open ground construction	RER	-	-	-	m2	0	0	0	1.00E+0			
energy	electricity, medium voltage, production ENTSO, at grid	ENTSO	-	-	-	kWh	4.50E+0				1	1.07	(1.1.1.1.3.BU.1.05); according to CdTe panel recycling
auxiliaries	water, deionised, at plant	CH	-	-	-	kg	5.57E+0				1	1.13	(1.4.1.3.1.3.BU.1.05); according to CdTe panel recycling
	sulphuric acid, liquid, at plant	RER	-	-	-	kg	8.57E-2				1	1.07	(1.1.1.1.3.BU.1.05); according to CdTe panel recycling
	hydrogen peroxide, 50% in H ₂ O, at plant	RER	-	-	-	kg	5.87E-1				1	1.07	(1.1.1.1.3.BU.1.05); according to CdTe panel recycling
	sodium hydroxide, 50% in H ₂ O, production mix, at plant	RER	-	-	-	kg	1.07E-1				1	1.07	(1.1.1.1.3.BU.1.05); according to CdTe panel recycling
transport	transport, lorry >16t, fleet average	RER	-	-	-	tkm	1.29E+1				1	2.00	(1.1.1.1.3.BU.2); according to CdTe panel recycling, F
disposal	treatment, PV cell production effluent, to wastewater treatment, class 3	CH	-	-	-	m3	4.93E-3				1	1.07	(1.1.1.1.3.BU.1.05); according to CdTe panel recycling
	disposal, plastics, mixture, 15.3% water, to municipal incineration	CH	-	-	-	kg	6.34E-1				1	1.07	(1.1.1.1.3.BU.1.05); according to CdTe panel recycling
	disposal, inert waste, 5% water, to inert material landfill	CH	-	-	-	kg	1.32E-1				1	1.07	(1.1.1.1.3.BU.1.05); according to CdTe panel recycling
avoided energy	natural gas, burned in industrial furnace >100kW	RER	-	-	-	MJ		-1.04E+01	-1.04E+01	-1.04E+01	1	1.14	(2.4.1.1.3.BU.1.05); natural gas consumption avoided by the recycling of PV glass
	heavy fuel oil, burned in industrial furnace 1MW, non-modulating	RER	-	-	-	MJ		-6.76E+00	-6.76E+00	-6.76E+00	1	1.14	(2.4.1.1.3.BU.1.05); heavy fuel oil consumption avoided by the recycling of PV glass
avoided materials	silica sand, at plant	DE	-	-	-	kg		-4.41E+00	-4.41E+00	-4.41E+00	1	1.14	(2.4.1.1.3.BU.1.05); silica sand consumption avoided by the recycling of the PV glass
	soda, powder, at plant	RER	-	-	-	kg		-1.75E+00	-1.75E+00	-1.75E+00	1	1.14	(2.4.1.1.3.BU.1.05); soda consumption avoided by the recycling of the PV glass
	limestone, milled, packed, at plant	CH	-	-	-	kg		-3.05E+00	-3.05E+00	-3.05E+00	1	1.14	(2.4.1.1.3.BU.1.05); limestone consumption avoided by the recycling of PV glass
	copper, at regional storage	RER	-	-	-	kg		-4.14E-01	-4.14E-01	-1.10E-01	1	1.14	(2.4.1.1.3.BU.1.05); primary copper avoided by the recycling of wires and the electric installation
	copper, secondary, at refinery	RER	-	-	-	kg		4.14E-01	4.14E-01	1.10E-01	1	1.14	(2.4.1.1.3.BU.1.05); efforts for producing secondary copper from scrap
	aluminium, primary, at plant	RER	-	-	-	kg		-2.02E+00	-2.42E+00	-3.19E+00	1	1.14	(2.4.1.1.3.BU.1.05); primary aluminium avoided by the recycling of frame and mounting structure
	aluminium, secondary, from old scrap, at plant	RER	-	-	-	kg		2.02E+00	2.42E+00	3.19E+00	1	1.14	(2.4.1.1.3.BU.1.05); efforts for producing secondary aluminium from scrap
	pig iron, at plant	GLO	-	-	-	kg		-1.54E-01	-9.72E-01	-4.55E+00	1	1.14	(2.4.1.1.3.BU.1.05); primary steel avoided by the recycling of the electric installation and the mounting system
avoided emissions air	Carbon dioxide, fossil	-	air	unspecified	-	kg		-1.58E+00	-1.58E+00	-1.58E+00	1	1.14	(2.4.1.1.3.BU.1.05); geogenic greenhouse gas emissions avoided by the recycling of PV glass

4.7 Life cycle inventory of the electric installation and the mounting structure

4.7.1 Electric installation

The life cycle inventories for electric installation are taken from Jungbluth et al. (2012). The unit process inventory data are shown in Tab. 4.34 (3 kWp systems, roof construction) and Tab. 4.35 (570 kWp system, open ground construction).

Tab. 4.34 unit process inventory data for the electric installation of the 3 kWp PV systems (integrated and mounted on roof), (Jungbluth et al. 2012)

	Name	Location	Infrastructure	Process	Unit	electric installation, photovoltaic plant, at plant	Uncertainty Type	Standard Deviation	95%	General Comment
	Location					CH				
	Infrastructure					1				
	Unit					unit				
product	electric installation, photovoltaic plant, at plant	CH	1	unit	1.00E+0					
materials	copper, at regional storage	RER	0	kg	1.47E+1	1	1.24	(2.1,3.1,1.5); Literature, recycled after use		
	wire drawing, copper	RER	0	kg	1.47E+1	1	1.24	(2.1,3.1,1.5); Assumption		
	brass, at plant	CH	0	kg	2.00E-2	1	1.24	(2.1,3.1,1.5); Literature		
	zinc, primary, at regional storage	RER	0	kg	4.00E-2	1	1.24	(2.1,3.1,1.5); Literature		
	steel, low-alloyed, at plant	RER	0	kg	8.60E-1	1	1.24	(2.1,3.1,1.5); Literature, recycled after use		
	nylon 6, at plant	RER	0	kg	2.30E-1	1	1.24	(2.1,3.1,1.5); Literature		
	polyethylene, HDPE, granulate, at plant	RER	0	kg	1.44E+1	1	1.24	(2.1,3.1,1.5); Literature incl. different plastics and Radox insulation		
	polyvinylchloride, bulk polymerised, at plant	RER	0	kg	2.13E+0	1	1.24	(2.1,3.1,1.5); Literature		
	polycarbonate, at plant	RER	0	kg	2.00E-1	1	1.24	(2.1,3.1,1.5); Literature		
	epoxy resin, liquid, at plant	RER	0	kg	2.00E-3	1	1.24	(2.1,3.1,1.5); Literature		
transport	transport, lorry 20-28t, fleet average	CH	0	tkm	2.84E+0	1	2.09	(4.5,na,na,na,na); Standard distance 60km incl. disposal		
	transport, freight, rail	CH	0	tkm	2.16E+1	1	2.09	(4.5,na,na,na,na); Standard distances 200km (metals 600km)		
disposal	disposal, plastic, industr. electronics, 15.3% water, to municipal incineration	CH	0	kg	1.70E+1	1	1.24	(2.1,3.1,1.5); Estimation		
	disposal, building, electric wiring, to final disposal	CH	0	kg	6.00E-2	1	1.24	(2.1,3.1,1.5); Estimation		

Tab. 4.35 unit process inventory data for the electric installation of the 570 kWp PV system (open ground), (Jungbluth et al. 2012)

	Name	Location	Infrastructure	Process	Unit	electric installation, 570 kWp photovoltaic plant, at plant	Uncertainty Type	Standard Deviation	95%	General Comment
	Location					ES				
	Infrastructure					1				
	Unit					unit				
product	electric installation, 570 kWp photovoltaic plant, at plant	ES	1	unit	1.00E+0					
materials	copper, at regional storage	RER	0	kg	7.41E+2	1	1.36	(2.1,3.1,1.5); Scaled from smaller plants over cabling length and fuse box weight		
	wire drawing, copper	RER	0	kg	7.41E+2	1	1.36	(2.1,3.1,1.5); Scaled from smaller plants over cabling length and fuse box weight		
	brass, at plant	CH	0	kg	1.36E+0	1	1.36	(2.1,3.1,1.5); Scaled from smaller plants over cabling length and fuse box weight		
	zinc, primary, at regional storage	RER	0	kg	2.73E+0	1	1.36	(2.1,3.1,1.5); Scaled from smaller plants over cabling length and fuse box weight		
	steel, low-alloyed, at plant	RER	0	kg	5.29E+1	1	1.36	(2.1,3.1,1.5); Scaled from smaller plants over cabling length and fuse box weight		
	nylon 6, at plant	RER	0	kg	1.57E+1	1	1.36	(2.1,3.1,1.5); Scaled from smaller plants over cabling length and fuse box weight		
	polyethylene, HDPE, granulate, at plant	RER	0	kg	7.09E+2	1	1.36	(2.1,3.1,1.5); Scaled from smaller plants over cabling length and fuse box weight		
	polyvinylchloride, bulk polymerised, at plant	RER	0	kg	4.49E+1	1	1.36	(2.1,3.1,1.5); Scaled from smaller plants over cabling length and fuse box weight		
	polycarbonate, at plant	RER	0	kg	1.36E-1	1	1.36	(2.1,3.1,1.5); Scaled from smaller plants over cabling length and fuse box weight		
	epoxy resin, liquid, at plant	RER	0	kg	1.36E-1	1	1.36	(2.1,3.1,1.5); Scaled from smaller plants over cabling length and fuse box weight		
transport	transport, lorry >16t, fleet average	RER	0	tkm	4.96E+1	1	2.16	(4.5,na,na,na,na); Standard distance 60km incl. disposal		
	transport, freight, rail	RER	0	tkm	1.85E+2	1	2.16	(4.5,na,na,na,na); Standard distances 200km (metals 600km)		
disposal	disposal, plastic, industr. electronics, 15.3% water, to municipal incineration	CH	0	kg	7.70E+2	1	1.36	(2.1,3.1,1.5); Estimation		
	disposal, building, electric wiring, to final disposal	CH	0	kg	4.09E+0	1	1.36	(2.1,3.1,1.5); Estimation		

4.7.2 Slanted roof and open ground mounting structures

The construction of the mounting structure for the roof installations and the open ground construction is modeled based on Jungbluth et al. (2012) and is independent of the PV technology applied. The life cycle inventory of the mounting structure per m² remains unchanged whereas the total area of mounting structure required changes depending on the system's efficiency and hence the panel area. The unit process inventory data for the mounting structure of the integrated and mounted as well as

the open ground systems are described in Tab. 4.36. These datasets include the material consumption, transport processes, land transformation and occupation and end-of-life waste disposal.

Tab. 4.36 unit process inventory data for the mounting structure for integrated, mounted and open ground PV systems (Jungbluth et al. 2012)

	Name	Location	Infrastructure	Process	Unit	slanted-roof construction, mounted, on roof			slanted-roof construction, integrated, on roof			open ground construction, on ground			Uncertainty Type	Standard Deviation 95%	General Comment
	Location					RER		RER		RER							
	Infrastructure					1		1		1							
	Unit					m2		m2		m2							
product	slanted-roof construction, mounted, on roof	RER	1	m2		1.00E+0		0		0							
	slanted-roof construction, integrated, on roof	RER	1	m2		-		1.00E+0		0							
	open ground construction, on ground	RER	1	m2		-		0		1.00E+0							
materials	aluminium, production mix, wrought alloy, at plant	RER	0	kg		2.84E+0		2.25E+0		3.98E+0		1	2.05	(1,2,1,1,1,na); Literature and own estimations			
	corrugated board, mixed fibre, single wall, at plant	RER	0	kg		1.33E-1		1.14E-1		8.64E-2		1	2.18	(3,4,3,1,3,5); Schwarz et al. 1992			
	polyethylene, HDPE, granulate, at plant	RER	0	kg		1.40E-3		2.82E-2		9.09E-4		1	2.05	(1,2,1,1,1,na); Literature and own estimations, recycled PE			
	polystyrene, high impact, HIPS, at plant	RER	0	kg		7.02E-3		6.02E-3		4.55E-3		1	2.18	(3,4,3,1,3,5); Schwarz et al. 1992			
	polyurethane, flexible foam, at plant	RER	0	kg		-		1.84E-2		-		1	2.05	(1,2,1,1,1,na); Literature and own estimations			
	synthetic rubber, at plant	RER	0	kg		-		1.24E+0		-		1	2.05	(1,2,1,1,1,na); Literature and own estimations			
	steel, low-alloyed, at plant	RER	0	kg		1.50E+0		2.00E-1		-		1	2.05	(1,2,1,1,1,na); Literature and own estimations			
	chromium steel 18/8, at plant	RER	0	kg		-		-		2.47E-1		1	2.10	(2,3,1,1,1,5); Literature and own estimations			
	reinforcing steel, at plant	RER	0	kg		-		-		7.21E+0		1	2.10	(2,3,1,1,1,5); Literature and own estimations			
	concrete, normal, at plant	CH	0	m3		-		-		5.37E-4		1	2.18	(3,4,3,1,3,5); Fence foundation			
	section bar extrusion, aluminium	RER	0	kg		2.84E+0		2.25E+0		3.98E+0		1	2.18	(3,4,3,1,3,5); Estimation			
	sheet rolling, steel	RER	0	kg		1.50E+0		-		-		1	2.18	(3,4,3,1,3,5); Estimation			
	section bar rolling, steel	RER	0	kg		-		2.00E-1		6.15E+0		1	2.18	(3,4,3,1,3,5); Brunschweiler 1993			
	wire drawing, steel	RER	0	kg		-		-		1.06E+0		1	2.18	(3,4,3,1,3,5); Mesh wire fence			
	zinc coating, pieces	RER	0	m2		-		-		1.56E-1		1	2.18	(3,4,3,1,3,5); Estimation			
	zinc coating, coils	RER	0	m2		-		-		1.09E-1		1	2.18	(3,4,3,1,3,5); Fence			
transport	transport, lorry >16t, fleet average	RER	0	tkm		2.25E-1		2.07E-1		2.17E-1		1	2.14	(4,5,na,na,na,na); Standard distance 50km			
	transport, freight, rail	RER	0	tkm		1.50E+0		8.52E-1		5.14E+0		1	2.14	(4,5,na,na,na,na); Standard distances 200km, 600km			
	transport, van <3.5t	RER	0	tkm		4.34E-1		3.75E-1		1.14E+0		1	2.18	(3,4,3,1,3,5); 100km to construction place			
disposal	disposal, packaging cardboard, 19.6% water, to municipal incineration	CH	0	kg		1.33E-1		1.14E-1		8.64E-2		1	2.18	(3,4,3,1,3,5); Calculated with use			
	disposal, building, polyethylene/polypropylene products, to final disposal	CH	0	kg		1.40E-3		1.29E+0		9.09E-4		1	2.18	(3,4,3,1,3,5); Disposal of plastics parts at end of life			
	disposal, building, polystyrene isolation, flame-retardant, to final disposal	CH	0	kg		7.02E-3		6.02E-3		4.55E-3		1	2.18	(3,4,3,1,3,5); Disposal of plastics parts at end of life			
resource land	Transformation, from pasture and meadow	-	-	m2		-		-		4.72E+0		1	2.18	(3,4,3,1,3,5); Tucson Electric Power			
	Transformation, to industrial area, built up	-	-	m2		-		-		1.50E+0		1	2.15	(1,3,2,3,3,5); Literature and own estimations			
	Transformation, to industrial area, vegetation	-	-	m2		-		-		3.22E+0		1	2.16	(3,3,2,3,3,5); Literature and own estimations			
	Occupation, industrial area, built up	-	-	m2a		-		-		4.50E+1		1	2.16	(3,3,2,3,3,5); Assumed life time: 30 a			
	Occupation, industrial area, vegetation	-	-	m2a		-		-		9.66E+1		1	2.16	(3,3,2,3,3,5); Assumed life time: 30 a			

4.8 Data collection procedures

Life cycle inventory data for CdTe PV panels were provided by First Solar for the product stage, the end-of-life stage as well as the potential benefits for recycling. Even though there are other producers beside First Solar, the data are regarded as accurate for all CdTe panels and are approved of by other producers⁶.

Life cycle inventory data of CIS PV panels were collected from Avancis, Germany in 2010 and Solar Frontier, Japan in 2010. They were published by de Wild-Scholten (2014).

Life cycle inventory data for micromorphous-Si PV systems were provided by Oerlikon Solar (Flury et al. 2012), representing the situation in 2011. The production data from Oerlikon Solar cover the consumption of feedstock materials, energy and water as well as process specific emissions and are considered reliable (Flury et al. 2012). The original inventory data were slightly adapted for this study

⁶ Personal tel.-communication Calyxo, Hr. Jochen Fritsche, 19.11.2014

in order to harmonize the production process of micromorphous-Si PV systems with the other technologies (see section 4.4.2).

Life cycle inventory data for multi- and monocrystalline-Si PV systems were collected from several manufacturers and the ITRPV roadmap (de Wild-Scholten 2014), representing the situation in 2011. The production data cover the consumption of feedstock materials, energy and water are considered reliable (de Wild-Scholten 2014) except for the shares of different types of poly-silicon used (electronic grade, electronic grade-off-grade, solar grade-modified Siemens, etc.). These shares are based on information from Rogol (2005).

Data about the end-of-life treatment of systems other than CdTe were not available. The recycling of these systems was modeled according to the recycling of the CdTe panels. An extrapolation (safety) factor of 1.5 is applied on the recycling efforts.

The construction as well as the use stages are modelled identically for all five PV technologies. The data used describe an average situation with regard to mounting structure, cabling and operation.

Hence, most of the data representing PV panel production and its supply chains are based on industry data collected from important or even leading companies. Given the large number of PV system installations, data on mounting and construction are based on industry surveys combined with selective analyses of individual mounting systems. The yield during the operation of PV systems is based on statistical data about the country of installation (IEA-PVPS 2013) and the average country specific yield of optimally oriented panels as reported in EPIA (2012).

The main information gaps occur with regard to the environmental efficiency of panel and module production in Asia as well as with regard to the end-of-life treatment efforts and emissions of all types of panels except CdTe. These gaps are closed by assuming the same environmental efficiency of panel production (same process specific emissions) but adjusting the mix of electricity used in manufacturing (which, regarding polysilicon production, in Europe and the US mainly stems from dedicated sources such as hydroelectric power and combined heat and power production) to the average Chinese and Malaysian situation. Data about end-of-life treatment of CdTe PV panels is adopted for the treatment of all other panels.

4.9 Methodological assumptions used in the screening

4.9.1 Overview

This subchapter describes in detail the assumptions and approaches followed in the screening LCAs. It covers the topics carbon storage, land use change and delayed emissions, electricity mixes, transports, treatment of multi-functionality (allocation), use stage scenario, re-use and end-of-life scenario.

4.9.2 Carbon storage

The products under analysis, photovoltaic panels do not contain biogenic carbon. Thus, the carbon storage topic does not apply to the life cycle assessments of electricity from photovoltaic panels.

4.9.3 Land use change and delayed emissions

Dealing with industrial products (and not agricultural products), land use change is of only minor importance.

Delayed emissions (i.e. emissions occurring 100 years and more after manufacturing the product at issue) may occur during the end-of-life stage of photovoltaic panels when landfilling the panels as

well as from overburdens of lignite and coal mines (supply chain of national electricity mixes) and from overburdens of metal mines. The used panels will be treated according to the WEEE directive (EU Parliament 2012). Hence, the amount of landfilled waste will be rather small. Furthermore, the quantification of delayed emissions is subject to substantial uncertainty. That is why delayed emissions are excluded from the environmental impact assessment.

4.9.4 Electricity mixes

The life cycle inventory models use country specific, average annual electricity mixes if no further information is provided by the companies. The national electricity mixes are based on OECD statistics, refer to the year 2011 and include the imports from other countries. Process specific electricity mixes are applied, when companies report the purchase of specific electricity (natural gas combined heat and power production, hydroelectric power).

4.9.5 Transports

Transport services between the different process steps are included. The transport of the product from the manufacturing site(s) to the regional storage or delivery centre is included. The current life cycle inventory of the supply chain of crystalline silicon PV panels installed in Europe relies on four world regions and thus four transport routes from those regions to Europe⁷ (see Tab. 4.37). CdTe PV panels include transportation from America to Europe (6500 km).

Tab. 4.37 Means of transport and corresponding transport distance in km for the import of mono- and multicrystalline PV modules from the relevant regions of the world.

Means of transport	Transport distance in km			
	China	APAC	America	Europe
Regional storage				
Lorry	940	940	940	940
Freight ship	20'000	15'550	0 ¹⁾	0
Construction 3kWp power system				
Lorry	100	100	100	100

¹⁾ In the current LCI the share of imported mono- and multicrystalline PV modules from Americas to Europe is 0 %

According to the information from world leading manufacturers, photovoltaic modules are never shipped with air cargo. Hence, the transport scenarios do not include air transportation.

The shipping of all PV components from the regional storage and delivery centre to the production site and finally to the customer is included as well. At the end-of-life, transports from the place of operation (customer) to the end-of-life treatment facility are included. If no specific information on the transport distances is available, standard distances based on expert judgments are used.

4.9.6 Treatment of Multi-functionality (allocation)

There are two major instances where multi-functionality is of importance:

- use stage of roof integrated photovoltaic panels
- recycled crystalline silicon feedstock

⁷ This modelling approach of transportation processes is considered sufficiently accurate for the screening study. A more detailed assessment of transports may be necessary in the full study.

Firstly, all roof integrated photovoltaic panels serve the two purposes weather protection and electricity generation. All environmental impacts are attributed to the electricity generation, which can be considered a conservative assumption.

Secondly, cuttings (circular segments) from mono crystalline wafer production are fed into multi crystalline silicon casting. In this case the cradle to gate efforts and environmental impacts of the supply of solar grade silicon used in Czochralski mono crystalline production are fully allocated to the mono crystalline silicon wafers. The (internal) recycling efforts and environmental impacts required to prepare the cuttings for an input into the multi crystalline casting process are fully allocated to the multi crystalline silicon wafers.

Further multi-functional processes occur in the background system of the five technologies. Usually either economic or physical causality criteria are used to allocate the environmental impacts of co-products. The details are documented in the respective ecoinvent reports and in the meta-information of the unallocated ecoinvent unit process datasets.

4.9.7 Use-stage scenario

The use stage is characterized by the following three main parameters: annual yield, degradation rate and life time.

The annual yield depends on the location of installation, the mounting and orientation of the modules (façade versus roof top, inclination and orientation) and the degradation. Tab. 4.38 shows the cumulative installed photovoltaic power in Europe according to IEA-PVPS (2013) and the country specific average yield at optimal angle in urban areas according to EPIA (2012). The annual average yield of optimally oriented modules in Europe weighted according to the cumulative installed photovoltaic power corresponds to 1'090 kWh/kWp (excluding degradation effects).

Tab. 4.38 Cumulative installed photovoltaic power in Europe in 2012 according to IEA-PVPS (2013) and country specific average annual yield in kWh/kWp at optimal angle in urban areas according to EPIA (2012), and average solar irradiation at optimal angle based on data retrieved from PVGIS⁸; degradation is not included.

Country	Cumulative installed power (MW)	Share	average yield at optimal angle in urban areas (kWh/kWp)	average solar irradiation at optimal angle (kWh/m ²)
Austria	363	0.6%	1'027	1'314
Belgium	2'698	4.2%	930	1'100
Germany	32'462	51.1%	936	1'147
Denmark	332	0.5%	945	1'130
Spain	4'706	7.4%	1'471	1'812
France	4'033	6.3%	1'117	1'386
United Kingdom	1'901	3.0%	920	1'111
Italy	16'450	25.9%	1'326	1'611
Netherlands	345	0.5%	933	1'112
Portugal	210	0.3%	1'494	1'840
Sweden	24	0.0%	826	1'101
Europe (PVPS members)	63'524	100.0%	1'090	1'331

⁸ <http://re.jrc.ec.europa.eu/pvgis/> (accessed on 29.04.2014)

In line with the IEA PVPS methodology guidelines (Frischknecht et al. 2015b) and the ADEME methodology guidelines (Payet et al. 2013), the lifetime of the photovoltaic power systems is assumed to be 30 years. A degradation of 0.7 % per year is applied leading to a loss in yield of 21 % during the last year of operation. Hence, the weighted average yield of a PV module installed in Europe and operated during 30 years is 10.5 % below the average yield shown in Tab. 4.38. The screening LCA of DC electricity from a PV module will thus be modelled with an annual yield of 975 kWh per kWp.

Lifetime as well as degradation rate of a particular PV module may differ from the values used in the screening LCA. According to Jordan & Kurtz (2012) degradation rates are higher for thin film PV modules (1.0 % mean; 1.5 % average) than for crystalline modules (0.5 % mean, 0.7 % average). Lifetime and degradation rate may be determined individually by TS partners within PEF Supporting Studies based on scientific evidence and/or verified test results.

The operation of the photovoltaic system includes the maintenance of the photovoltaic power system (cleaning) and the conversion of solar energy to electrical energy. There are no significant direct emissions or significant processes during the operation of the photovoltaic system.

According to Jungbluth et al. (2012), 2 % of the modules are replaced and 1 % are rejects. This corresponds to a rate of failure of 3 % during the lifetime of the photovoltaic power system. According to First Solar, CdTe panels have a replacement and reject rate of 1 %. The manufacture of 1 % and 3 % substitute panels for CdTe PV systems and all other PV systems, respectively, is taken into account in the inventory.

4.9.8 *Re-use*

After the service life of 30 years 100 % photovoltaic panels will enter an end-of-life treatment. None of the panels will be reused. Under the current situation no requirements for reuse are defined. Upcoming targets under WEEE may change this.

4.9.9 *End-of-life scenario*

Spent or defective PV modules have to be treated according to the WEEE directive and thus will be recycled. The recycling processes modelled in the screening LCA and in the PEF supporting studies comply with the WEEE directive on the disposal of electrical and electronic equipment (EU Parliament 2012). Average European LCI on WEEE compliant handling is established based on information provided by one member of the Technical Secretariat. The estimated recycling efforts have turned out to be conservative compared to data provided by recyclers taking part in the PV CYCLE programme. Country specific end-of-life treatment of PV modules is disregarded because the volumes of spent modules is yet rather small and thus only few recycling facilities are in operation today. Furthermore, recycling and disposal efforts are expected to be of little importance in terms of overall environmental impacts.

Recycling and the materials recovered are modelled according to different approaches

- Base case: The formula provided in the recommendation of the European Commission (2013) is applied
- Sensitivity analyses: the alternative formulas provided in the PEF guidance document (European Commission 2015), recycled content (100 %/0 %), avoided burden (0 %/100 %) are applied

Benefits for recycling are relevant for glass, copper, aluminium, AlMg₃, steel, copper telluride and cadmium. The recycled content of the materials (R1) was taken from ecoinvent v2.2+ (KBOB et al. 2014). The fraction of the materials that is recycled at the end-of-life of the PV panels (R2) was

determined based on literature information (Held & Ilg 2011; Jungbluth et al. 2012). Recovery of energy (R3) from components of used PV panels is not occurring. The parameter values and datasets used for the calculation of recycling benefits at the end-of-life stage of PV panels are compiled in Tab. 4.39.

Tab. 4.39 Values of the parameters R1, R2 and R3 and datasets used for the calculation of avoided material uses and emissions at the end-of-life stage of PV panels. The variables correspond to the denotations introduced in the end-of-life formula recommended by the European Commission (2013).⁹

		Glass	Copper	Aluminium	AlMg3 alloy	Steel	Copper Telluride	Cadmium
Values	R1	0%	44%	32%	77%	37%	0%	0%
	R2	90%	100%	100%	100%	100%	88%	88%
	R3	0	0	0	0	0	0	0
Datasets	E _v	- solar glass, low-iron, at regional storage - flat glass, uncoated, at plant	- copper, primary, at refinery, RER - copper, primary, at refinery, RLA - copper, primary, at refinery, ID - copper concentrate, at beneficiation, RER - copper concentrate, at beneficiation, RLA - copper concentrate, at beneficiation, ID - copper, from imported concentrates, at refinery, DE	- aluminium, primary, at plant	- aluminium, primary, at plant	- pig iron, at plant	- Copper telluride cement, from copper production	- Cadmium sludge, from zinc electrolysis, at plant
	E _{recycled}	na	- copper, secondary, at refinery	- aluminium, secondary, from new scrap, at plant - aluminium, secondary, from old scrap, at plant	- aluminium, secondary, from new scrap, at plant - aluminium, secondary, from old scrap, at plant	- iron scrap, at plant	na	na
	E _{RecyclingEOL}	- CdTe PV module takeback + recycling - CIS PV module takeback + recycling - micro-Si PV module takeback + recycling - mono-Si PV module takeback + recycling - multi-Si PV module takeback + recycling	- CdTe PV module takeback + recycling - CIS PV module takeback + recycling - micro-Si PV module takeback + recycling - mono-Si PV module takeback + recycling - multi-Si PV module takeback + recycling - copper, secondary, at refinery	- CdTe PV module takeback + recycling - CIS PV module takeback + recycling - micro-Si PV module takeback + recycling - mono-Si PV module takeback + recycling - multi-Si PV module takeback + recycling - aluminium, secondary, from old scrap, at plant	- CIS PV module takeback + recycling - micro-Si PV module takeback + recycling - mono-Si PV module takeback + recycling - multi-Si PV module takeback + recycling - aluminium, secondary, from old scrap, at plant	- CdTe PV module takeback + recycling - CIS PV module takeback + recycling - micro-Si PV module takeback + recycling - mono-Si PV module takeback + recycling - multi-Si PV module takeback + recycling	- CdTe PV module takeback + recycling - CIS PV module takeback + recycling - micro-Si PV module takeback + recycling - mono-Si PV module takeback + recycling - multi-Si PV module takeback + recycling	- CdTe PV module takeback + recycling - CIS PV module takeback + recycling - micro-Si PV module takeback + recycling - mono-Si PV module takeback + recycling - multi-Si PV module takeback + recycling
	E _{*v}	- Carbon dioxide to air - Silica sand, at plant - Soda, powder, at plant - Limestone, milled, packed, at plant - Natural gas, burned in industrial furnace >100kW - Heavy fuel oil, burned in industrial furnace 1MW, non-modulating	- copper, at regional storage	- aluminium, primary, at plant	- aluminium, primary, at plant	- pig iron, at plant	- Copper telluride cement, from copper production	- Cadmium sludge, from zinc electrolysis, at plant
	E _D	negligible	negligible	negligible	negligible	negligible	negligible	negligible
	E _{*D}	negligible	negligible	negligible	negligible	negligible	negligible	negligible

4.10 Life cycle inventory results

All unit process life cycle inventory data for all five PV technologies are listed in subchapters 4.2 to 4.6.

⁹ The cradle to gate environmental impacts of the disposal of waste materials (E_D and E_{*D}) per kWh of DC electricity produced is below 1 % for all PV systems. Tab. A. 63 in ANNEX II shows the characterized environmental impacts of the disposal of waste materials in relation to the total impacts caused by electricity generation for a CdTe PV system mounted on a slanted roof.

5. Calculating PEF impact assessment results

5.1 Introduction

In the following subchapters the environmental impacts of electricity produced with the five different photovoltaic panel technologies cadmium-telluride (CdTe), copper indium selenide (CIS), micromorphous silicon (micro-Si), multicrystalline silicon (multi-Si) and monocrystalline silicon (mono-Si) are shown. The unit of analysis is defined as 1 kWh (kilowatt hour) of DC electricity generated by a PV module. The reference flow is the PV module, measured in kWp (kilowatt peak), the maximum power output under standard conditions of a module. The panels are used in three different configurations: residential scale power system, mounted on slanted roofs, residential scale power system integrated in slanted roofs and large scale open ground power systems. The most relevant life cycle stages and hot spots as well as the most important impact categories for each technology are identified. These preliminary findings are a basis for more detailed assessments in the PEF supporting studies. Special needs for data quality are identified and highlighted. Furthermore indications about a possible definition of a benchmark are discussed in chapter 7.

Subchapter 5.2 contains a description of the overall results prior to normalization and weighting, covering all panel technologies and power plant configurations. The main drivers are explained and hotspots highlighted. Subchapter 5.3 and 5.4 show the normalized and weighted overall results.

Subchapters 5.5 to 5.9 evaluate the environmental impacts of producing 1 kWh DC electricity with the five PV panel technologies used in residential scale photovoltaic power plants mounted on slanted roofs. The environmental impacts of electricity produced with residential scale photovoltaic power plants integrated in slanted roofs and with large scale open ground installations are discussed in ANNEX II.

The main limitations are given by little data availability of direct, process-specific emissions, in particular regarding processes taking place in Asian region, and little data availability regarding mass and energy flows of the end-of-life stage due to yet too little volumes and experiences with industrialized panel take back and treatment. Furthermore the inverter is not part of the basic assessment, but is elaborated on in the sensitivity analysis. Regarding the LCI of CIS panels Avancis, data provider for CIS panels is no longer producing panels (see also chapter 2).

A commercial LCA software (Simapro 8.0.6) is used to model the product system, to calculate the life cycle inventory and impact assessment results and to document the data (PRé Consultants 2015). Background data are represented by ecoinvent data v2.2+ (ecoinvent Centre 2010; KBOB et al. 2014). Present technologies and their environmental impacts are used in the LCA models. The life cycle inventories are established according to the requirements of the PEF Guide (European Commission 2013) as well as the Product Environmental Footprint Pilot Guidance (version 5.2, European Commission 2015).

The environmental footprint is quantified using the 15 environmental indicators proposed by the European Union (European Commission 2013).

The life cycle indicators have been designed to provide information on the environmental impacts linked to European consumption and production. The ‘apparent consumption’ approach is adopted by accounting for both the domestic extractions of resources and emissions in the EU27 as well as the impacts due to international trade (both imports and exports) (European Commission 2015). These default environmental footprint impact category indicators are completed with three additional

indicators: renewable cumulative energy demand, non-renewable cumulative energy demand and nuclear waste, characterized by its volume. The indicators covering the cumulative energy demand are required by international guidelines regarding photovoltaic electricity production by ADEME (Payet et al. 2013) and IEA-PVPS (Frischknecht et al. 2015b), and by the PCR of the International EPD System (PCR CPC 171 & 173 2013). They are also part of the indicators of the CEN standard EN 15804:2012+A1:2013 on core rules for the product category of construction products of environmental product declarations (EN 15804 2013). The amount of nuclear waste is also mandatory in the IEA-PVPS methodological guidelines and in the EN 15804 (2013). The impact category indicator nuclear waste has been developed within ecological scarcity method 2013 (Frischknecht & Büsser Knöpfel 2013; 2014). Further details regarding its characterization factors can be found in subchapter 3.7.

The impact assessment indicators cover a large range of human and ecological toxicity effects, environmental, resource-specific and energetic parameters. As the inventory is assessed on process level, the environmental impacts can be identified per each process step. Most, but not all of the impact indicators analyzed in this screening study show certain significance in the results. It is assumed that all relevant environmental impacts are identified with the 18 impact indicators addressed.

The results of the environmental footprint are relative expressions. They do not predict impacts on endpoint level nor the exceeding of thresholds, safety or risks margins of any kind.

5.2 Data and indicator results prior to normalization

In this subchapter, first the overall results of all impact categories, all technologies and all mounting systems are shown, followed by the minimum and maximum results and the bandwidth of the results. Then, more detailed results per each impact category are shown for all technologies and mounting structures. The results prior to normalization shown in this section include all life cycle stages (i.e., product, operation, use and end-of-life stage) but do not contain the potential benefits for recycling of PV panels. The benefits for recycling are included in the results presented in the subchapters 5.5 to 5.9 for the individual PV technologies.

The following Tab. 5.1 shows the overall results for the environmental impacts of all five PV technologies and the three systems compared.

Tab. 5.1 Results prior to normalization per impact category of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels. The contribution of potential benefits for recycling is not included in these results.

	unit	1 kWh electricity, produced at a 3 kWp PV plant, CdTe panels, integrated in slanted roof	1 kWh electricity, produced at a 3 kWp PV plant, CdTe panels, mounted on slanted roof	1 kWh electricity, produced at a 570 kWp PV plant, CdTe panels, open ground installation	1 kWh electricity, produced at a 3 kWp PV plant, CIS panels, integrated in slanted roof	1 kWh electricity, produced at a 3 kWp PV plant, CIS panels, mounted on slanted roof	1 kWh electricity, produced at a 570 kWp PV plant, CIS panels, open ground installation	1 kWh electricity, produced at a 3 kWp PV plant, micro-Si panels, integrated in slanted roof	1 kWh electricity, produced at a 3 kWp PV plant, micro-Si panels, mounted on slanted roof	1 kWh electricity, produced at a 570 kWp PV plant, micro-Si panels, open ground installation	1 kWh electricity, produced at a 3 kWp PV plant, multi-Si panels, integrated in slanted roof	1 kWh electricity, produced at a 3 kWp PV plant, multi-Si panels, mounted on slanted roof	1 kWh electricity, produced at a 570 kWp PV plant, multi-Si panels, open ground installation	1 kWh electricity, produced at a 3 kWp PV plant, mono-Si panels, integrated in slanted roof	1 kWh electricity, produced at a 3 kWp PV plant, mono-Si panels, mounted on slanted roof	1 kWh electricity, produced at a 570 kWp PV plant, mono-Si panels, open ground installation
Climate change	kg CO ₂ eq	2.47E-02	2.52E-02	3.03E-02	4.09E-02	4.16E-02	4.85E-02	5.64E-02	5.71E-02	6.47E-02	5.58E-02	5.63E-02	6.11E-02	9.04E-02	9.09E-02	9.56E-02
Ozone depletion	kg CFC-11 eq	9.55E-10	1.01E-09	1.21E-09	1.74E-09	1.81E-09	2.07E-09	1.15E-09	1.23E-09	1.51E-09	1.74E-09	1.79E-09	1.98E-09	2.38E-09	2.43E-09	2.62E-09
Human toxicity, cancer effects	CTUh, c	5.75E-10	8.42E-10	1.68E-09	9.55E-10	1.30E-09	2.40E-09	1.08E-09	1.46E-09	2.64E-09	1.01E-09	1.27E-09	2.06E-09	1.17E-09	1.42E-09	2.19E-09
Human toxicity, non-cancer effects	CTUh, n-c	5.06E-09	5.39E-09	8.09E-09	8.74E-09	9.17E-09	1.33E-08	9.43E-09	9.90E-09	1.45E-08	1.03E-08	1.06E-08	1.31E-08	1.33E-08	1.36E-08	1.59E-08
Particulate matter	kg PM _{2.5} eq	1.28E-05	1.34E-05	1.52E-05	1.76E-05	1.83E-05	2.12E-05	5.86E-05	5.93E-05	6.26E-05	6.78E-05	6.84E-05	7.01E-05	1.32E-04	1.33E-04	1.34E-04
Ionizing radiation HH	kBq U235 eq	1.12E-03	1.27E-03	1.72E-03	3.33E-03	3.53E-03	4.12E-03	2.50E-03	2.72E-03	3.36E-03	2.77E-03	2.92E-03	3.35E-03	3.67E-03	3.82E-03	4.23E-03
Photochemical ozone formation	kg NMVOC eq	9.18E-05	9.40E-05	1.04E-04	1.19E-04	1.22E-04	1.36E-04	2.08E-04	2.11E-04	2.28E-04	1.95E-04	1.97E-04	2.07E-04	3.25E-04	3.27E-04	3.36E-04
Acidification	mol H ⁺ eq	1.84E-04	1.91E-04	2.07E-04	2.26E-04	2.35E-04	2.63E-04	4.52E-04	4.61E-04	4.93E-04	4.75E-04	4.81E-04	4.96E-04	8.10E-04	8.17E-04	8.30E-04
Terrestrial eutrophication	mol N eq	3.42E-04	3.53E-04	4.28E-04	4.32E-04	4.46E-04	5.49E-04	7.54E-04	7.69E-04	8.81E-04	6.95E-04	7.05E-04	7.76E-04	1.14E-03	1.15E-03	1.22E-03
Freshwater eutrophication	kg P eq	4.54E-06	4.61E-06	2.35E-06	6.58E-06	6.67E-06	4.51E-06	6.00E-06	6.10E-06	3.97E-06	7.06E-06	7.13E-06	4.84E-06	7.71E-06	7.78E-06	5.48E-06
Marine eutrophication	kg N eq	3.03E-05	3.12E-05	3.48E-05	3.73E-05	3.85E-05	4.36E-05	6.72E-05	6.84E-05	7.41E-05	6.37E-05	6.46E-05	6.80E-05	1.14E-04	1.15E-04	1.18E-04
Freshwater ecotoxicity	CTUe	7.90E-02	8.09E-02	4.38E-02	9.02E-02	9.27E-02	5.96E-02	9.30E-02	9.56E-02	6.39E-02	9.90E-02	1.01E-01	6.30E-02	1.04E-01	1.05E-01	6.72E-02
Land use	kg C deficit	2.78E-02	2.90E-02	8.93E+00	5.56E-02	5.71E-02	1.16E+01	4.94E-02	5.10E-02	1.25E+01	6.62E-02	6.73E-02	8.54E+00	8.37E-02	8.47E-02	8.34E+00
Water resource depletion	m ³ water eq	1.84E-05	2.14E-05	2.86E-05	3.98E-05	4.37E-05	5.31E-05	8.30E-05	8.73E-05	9.75E-05	6.04E-05	6.33E-05	7.01E-05	1.05E-04	1.07E-04	1.14E-04
Mineral, fossil & ren resource depletion	kg Sb eq	7.08E-06	7.09E-06	7.07E-06	5.12E-04	5.12E-04	5.12E-04	1.50E-06	1.51E-06	1.61E-06	1.93E-05	1.93E-05	1.93E-05	1.26E-05	1.26E-05	1.25E-05
Cumulative energy demand non renewable	MJ oil eq	3.53E-01	3.59E-01	4.31E-01	6.16E-01	6.23E-01	7.22E-01	6.54E-01	6.62E-01	7.70E-01	7.00E-01	7.05E-01	7.74E-01	1.07E+00	1.08E+00	1.14E+00
Cumulative energy demand renewable	MJ oil eq	3.63E+00	3.64E+00	3.65E+00	3.67E+00	3.68E+00	3.70E+00	3.67E+00	3.68E+00	3.70E+00	3.70E+00	3.70E+00	3.71E+00	3.74E+00	3.74E+00	3.75E+00
Nuclear waste	m ³ HAA eq	2.60E-11	2.94E-11	3.92E-11	7.86E-11	8.30E-11	9.58E-11	5.51E-11	5.99E-11	7.38E-11	6.30E-11	6.63E-11	7.55E-11	8.54E-11	8.86E-11	9.76E-11

The life cycle based greenhouse gas emissions amount to between 24.7 and 95.6 grams of CO₂-eq and the non-renewable cumulative energy demand to between 0.353 to 1.14 MJ per kWh. The particulate matter emissions vary between 12.8 and 134 mg per kWh. The generation of 1 kWh electricity with PV panels requires between 1.50 mg Sb-eq and 512 mg Sb-eq of abiotic resources and consumes between 18.4 and 114 g water-eq of water.

For most impact categories the electricity production with CdTe PV panels and an integrated mounting structure leads to the smallest impacts of the systems compared. This is shown in the following Tab. 5.2 where the minimum and maximum impacts per category and per technology are listed. In the impact categories freshwater eutrophication and freshwater ecotoxicity potential electricity produced with CdTe PV panels on the open ground system causes the lowest impacts. Within the impact category mineral, fossil and renewable resource depletion potential electricity produced with micromorphous-Si PV systems integrated in roofs causes the fewest impacts.

For most of the impact categories the maximum impact is caused by the electricity production with monocrystalline silicon PV panels on the open ground system. Exceptions are the impact categories human toxicity cancer effects and land use, where the highest impact is caused by the electricity production with micromorphous-Si PV panels on the open ground system. Electricity production with monocrystalline Si PV panels mounted on a slanted roof causes the highest freshwater eutrophication and freshwater ecotoxicity potential. Mineral, fossil and renewable resource depletion is highest for the electricity production with CIS PV panels on the open ground system.

Tab. 5.2 Minimum and maximum results and bandwidth (maximum minus minimum, divided by minimum) per impact category of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels

Indicator	Unit	Min	Max	Bandwidth	Technology and mounting structure causing minimum impact	Technology and mounting structure causing maximum impact
Climate change	kg CO ₂ eq	2.47E-02	9.56E-02	287%	CdTe panels, integrated in slanted roof	mono-Si panels, open grond installation
Ozone depletion	kg CFC-11 eq	9.55E-10	2.62E-09	174%	CdTe panels, integrated in slanted roof	mono-Si panels, open grond installation
Human toxicity, cancer effects	CTUh, c	5.75E-10	2.64E-09	360%	CdTe panels, integrated in slanted roof	micro-Si panels, open grond installation
Human toxicity, non-cancer effects	CTUh, n-c	5.06E-09	1.59E-08	215%	CdTe panels, integrated in slanted roof	mono-Si panels, open grond installation
Particulate matter	kg PM _{2.5} eq	1.28E-05	1.34E-04	945%	CdTe panels, integrated in slanted roof	mono-Si panels, open grond installation
Ionizing radiation HH	kBq U235 eq	1.12E-03	4.23E-03	278%	CdTe panels, integrated in slanted roof	mono-Si panels, open grond installation
Photochemical ozone formation	kg NMVOC eq	9.18E-05	3.36E-04	266%	CdTe panels, integrated in slanted roof	mono-Si panels, open grond installation
Acidification	mol H ⁺ eq	1.84E-04	8.30E-04	351%	CdTe panels, integrated in slanted roof	mono-Si panels, open grond installation
Terrestrial eutrophication	mol N eq	3.42E-04	1.22E-03	257%	CdTe panels, integrated in slanted roof	mono-Si panels, open grond installation
Freshwater eutrophication	kg P eq	2.35E-06	7.78E-06	231%	CdTe panels, open grond installation	mono-Si panels, mounted on slanted roof
Marine eutrophication	kg N eq	3.03E-05	1.18E-04	290%	CdTe panels, integrated in slanted roof	mono-Si panels, open grond installation
Freshwater ecotoxicity	CTUe	4.38E-02	1.05E-01	141%	CdTe panels, open grond installation	mono-Si panels, mounted on slanted roof
Land use	kg C deficit	2.78E-02	1.25E+01	44846%	CdTe panels, integrated in slanted roof	micro-Si panels, open grond installation
Water resource depletion	m ³ water eq	1.84E-05	1.14E-04	519%	CdTe panels, integrated in slanted roof	mono-Si panels, open grond installation
Mineral, fossil & ren resource depletion	kg Sb eq	1.50E-06	5.12E-04	33989%	micro-Si panels, integrated in slanted roof	CIS panels, open grond installation
Cumulative energy demand non renewable	MJ oil eq	3.53E-01	1.14E+00	224%	CdTe panels, integrated in slanted roof	mono-Si panels, open grond installation
Cumulative energy demand renewable	MJ oil eq	3.63E+00	3.75E+00	3%	CdTe panels, integrated in slanted roof	mono-Si panels, open grond installation
Nuclear waste	m ³ HAA eq	2.60E-11	9.76E-11	275%	CdTe panels, integrated in slanted roof	mono-Si panels, open grond installation

The bandwidth between the smallest and the highest impact differs widely between the different impact categories. The smallest spread (3 %) is found in the impact category cumulative energy demand renewable, followed by the impact categories freshwater ecotoxicity potential (141 %) and the ozone depletion potential (174 %). The largest bandwidths are found for the impact categories particulate matter (945 %), mineral, fossil and renewable resource depletion potential (34'000 %) and land use (44'800 %). The large differences in the category land use are due to the fact that for the residential scale PV systems, mounted on roof tops the land use is allocated to the building and not to the PV system whereas the open ground PV installation causes land use impacts directly associated with the PV system. The indium consumption in the CIS PV panels causes high impacts in the category mineral, fossil and renewable resource depletion potential. Hence, the spectrum within this impact category becomes very wide.

The impact category **climate change** leads to the smallest environmental impacts for the electricity production with CdTe PV panels, integrated in a slanted roof followed by the electricity production with CIS panels (see Fig. 5.1). The climate change impacts for electricity production with micromorphous-Si and multicrystalline-Si are approximately in the middle between the CdTe and the monocrystalline-Si PV panels. The application of monocrystalline-Si panels for electricity production leads to the highest climate change impacts of the technologies and systems analyzed. The large difference between monocrystalline-Si PV panels and the CdTe PV panels comes from the energy consumption within the laminate production and the power plant mix (hard coal burned in Chinese power plants for the monocrystalline-Si PV panels compared to natural gas power plants in the Malaysian electricity mix for the production of CdTe PV panels).

Generally, the integrated panel systems lead to the smaller impacts, followed by the mounted panel system. The open ground systems cause the highest greenhouse gas emissions in each technology. The impacts of the integrated PV systems are 1 to 2 % lower than those of the mounted systems whereas the impacts of the open ground systems are between 5 and 20 % higher than those of the mounted system.

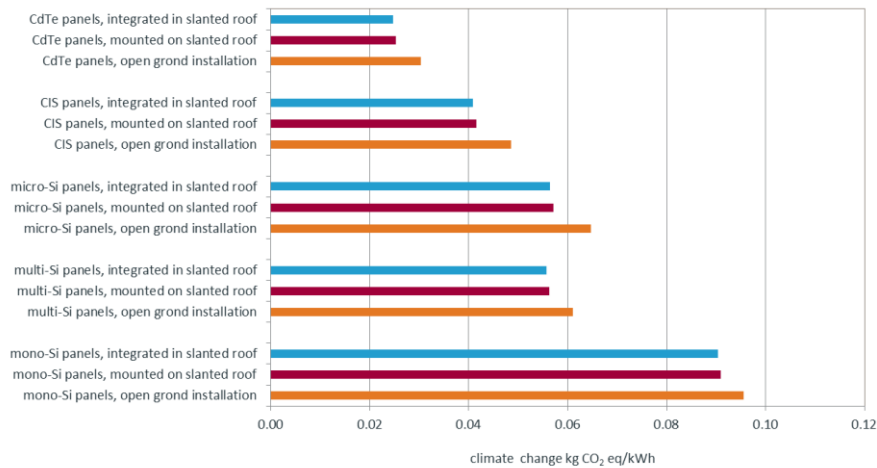


Fig. 5.1 Climate change results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels

Producing electricity with CdTe panels shows the lowest **ozone depletion potential**, followed by the electricity production with micromorphous-Si PV panels. The CIS system and the multicrystalline-Si system have similar ozone depletion potentials. The monocrystalline-Si system causes the highest impacts in this impact category. Fig. 5.2 shows the results of the electricity production with the five technologies and the three mounting systems. The mounted panel systems lead to slightly higher impacts than the integrated system. The open ground systems have the highest impacts in each technology.

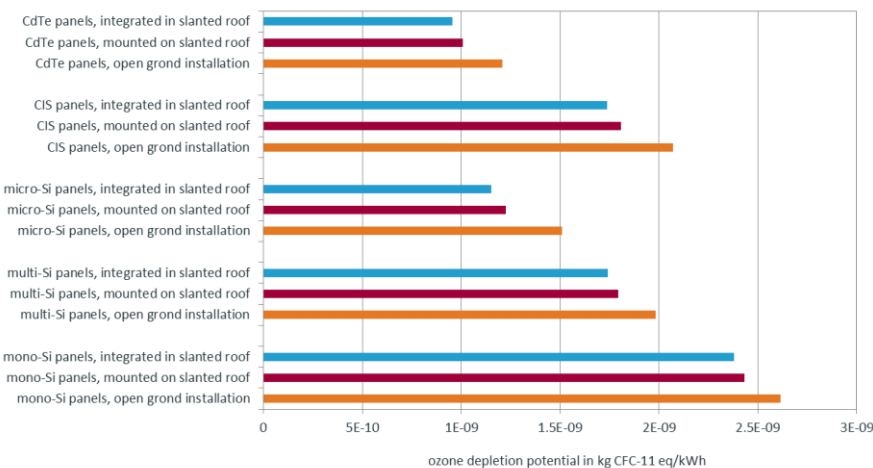


Fig. 5.2 Ozone depletion potential results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels

The **human toxicity cancer effects**, shows higher variances between the different mounting systems than between the different PV technologies (see Fig. 5.3). The integrated systems cause the smallest impacts in all PV technologies, whereas the open ground systems cause impacts being two or three times higher. The mounted roof system leads to impacts 21-46 % higher compared to the integrated system. The high impacts of the open ground system are mainly caused by the aluminium and steel needed in the mounting structure. Around 30 % of the total impacts are caused each by the supply chain of aluminium and steel (see also chapter 6).

The systems with CdTe PV panels cause the lowest impacts among the five PV technologies. The CIS-, micromorphous-Si, multicrystalline-Si and monocrystalline-Si PV panels lead to comparable results, which are typically about 50 % higher than the human toxicity, cancer effects, caused by electricity generated by CdTe PV systems.

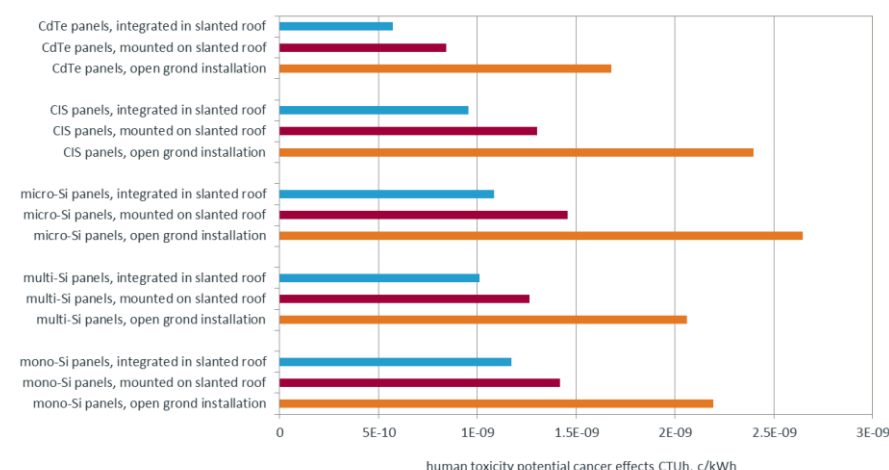


Fig. 5.3 Human toxicity cancer effects results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels

The impact category **human toxicity non cancer effects**, shows similar results to the impact category human toxicity, cancer effects. The mounting structures are more influential than the PV technology. The open ground systems cause the highest impact. The mounted roof structure has slightly higher impacts on the human toxicity, non cancer effects, than the integrated system. The electricity production with CdTe PV panels leads to the smallest impacts in this category. The impacts of electricity production are increasing from CIS-, micromorphous-Si and multicrystalline-Si PV panels but are still in a similar range. The systems with monocrystalline-Si PV panels cause the highest impacts of the technologies compared. Fig. 5.4 shows the human toxicity non cancer effects caused by the electricity production with the five PV technologies and the three mounting systems.

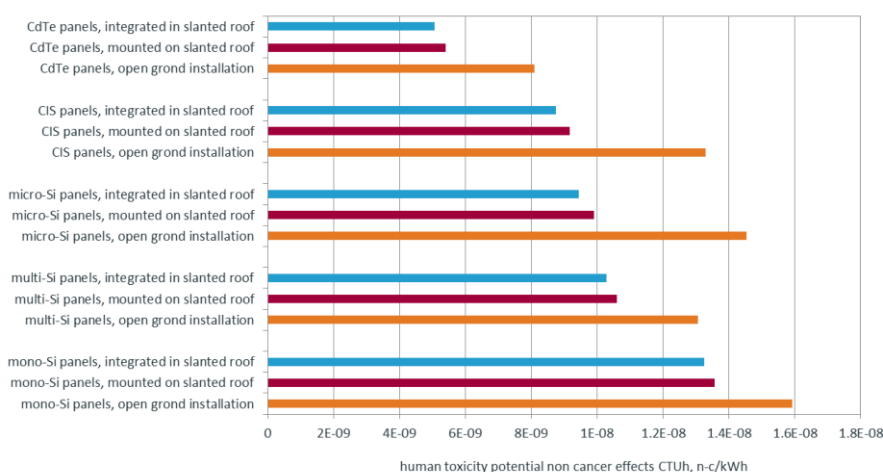


Fig. 5.4 Human toxicity non cancer effects results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels

The impact category **particulate matter** (PM) is dominated by the PV technology used for the electricity production rather than by the mounting structure (see Fig. 5.5). The electricity production with CdTe PV panels leads to the lowest impacts regarding PM, followed by the production with CIS PV panels, which cause only slightly higher PM emissions. Micromorphous-Si and multicrystalline-Si PV panels cause impacts about three to four times higher than when using CdTe PV panels. The electricity production with monocrystalline-Si PV panels leads to nine fold impacts on PM emissions compared to the CdTe PV technology. Within each PV technology the differences between the three mounting systems are small.

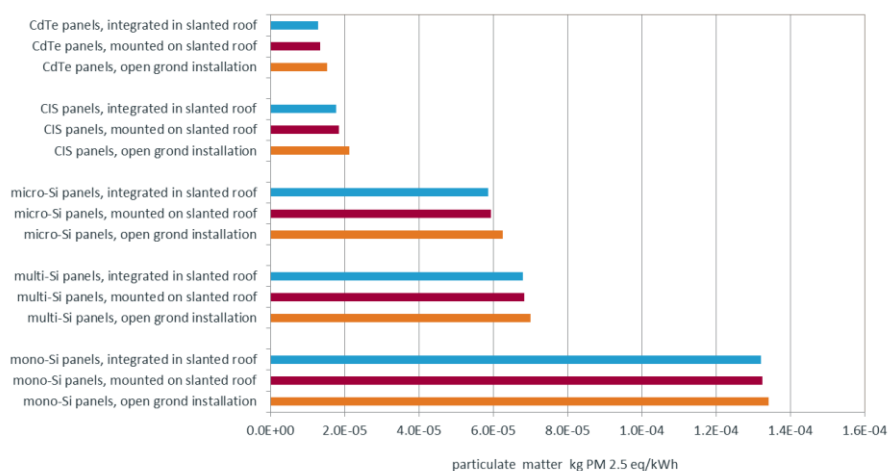


Fig. 5.5 Particulate matter results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels

The **ionizing radiation HH** shows differences between the different PV technologies but also between the different mounting structures (see Fig. 5.6). Electricity production with CdTe PV panels leads to the lowest environmental impacts, which are about half or a third of the impact level compared to the other technologies. The ranking of the ionizing radiation HH impacts of the other PV technologies is: micromorphous-Si, multicrystalline-Si, CIS, monocrystalline-Si.

Within each technology, the impacts caused by the integrated system are between 4 % and 12 % lower compared to the impacts of the mounted system. The impacts caused by the electricity production on the open ground system are 11 % to 35 % higher compared to those of electricity production with the mounted system.

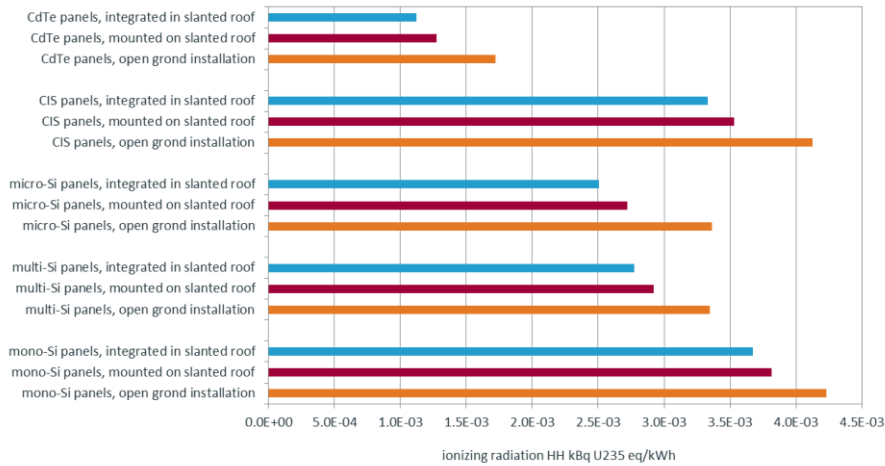


Fig. 5.6 Ionizing radiation HH results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels

The impact categories **photochemical ozone formation potential** and **acidification potential** show very similar patterns (see Fig. 5.7 and Fig. 5.8). They are dominated by the PV technology used for the electricity production and not by the mounting structure. The electricity production with CdTe PV panels leads to the lowest impacts, followed by the production with CIS PV panels, which have a slightly higher photochemical ozone formation potential and acidification potential. Micromorphous-Si and multicrystalline-Si cause impacts about twice as high as CdTe PV panels. Electricity production with monocrystalline-Si PV panels leads to fourfold impacts compared to the CdTe PV technology. Within each PV technology the differences between the mounting systems are small. The integrated systems lead to impacts 1-4 % below the impacts of the mounted systems, whereas the open ground systems cause environmental impacts 2-12 % higher than those of the mounted systems.

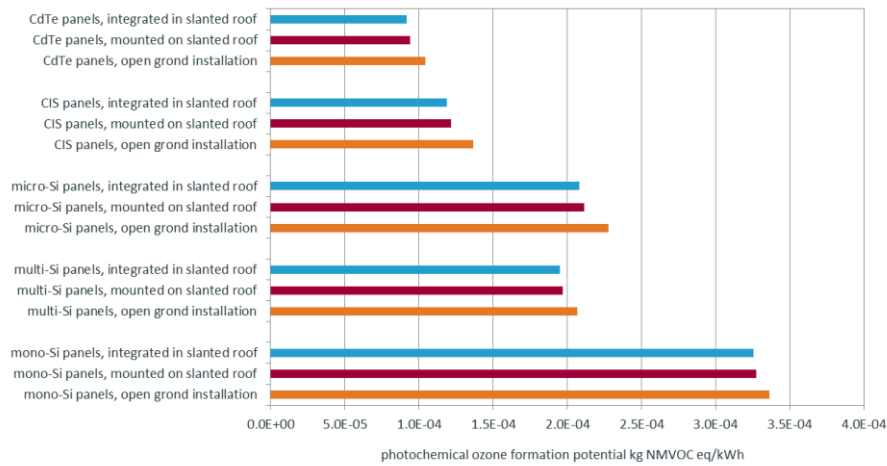


Fig. 5.7 Photochemical ozone formation potential results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels

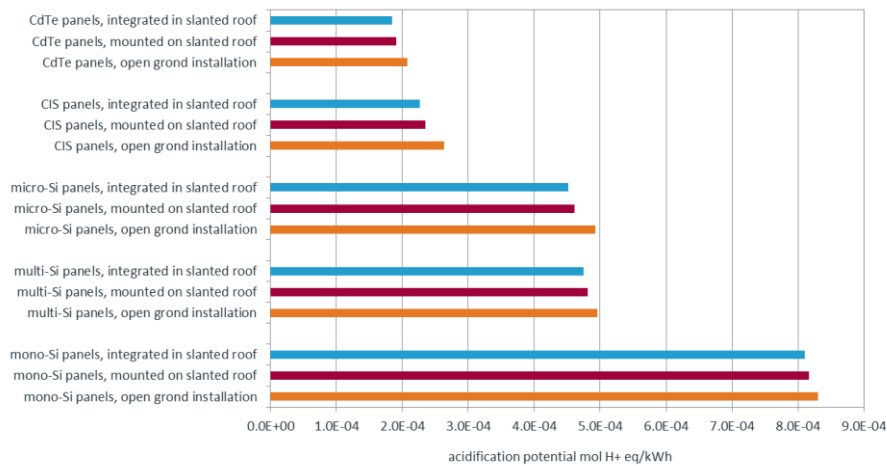


Fig. 5.8 Acidification potential results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels

The **terrestrial eutrophication potential** shows differences between the different PV technologies but also between the different mounting systems (see Fig. 5.9). Electricity production with CdTe and CIS PV panels leads to the lowest environmental impacts, which are about half of the impact level of the micromorphous-Si and multicrystalline-Si PV technologies. Electricity production with monocrystalline-Si causes impacts about three times higher compared to CdTe or CIS PV systems.

Within each technology, the impacts caused by the integrated systems are between 1 % and 3 % lower compared to those of the mounted systems. The impacts caused by the electricity production with the open ground systems are 6 % to 23 % higher compared to those of the mounted systems.

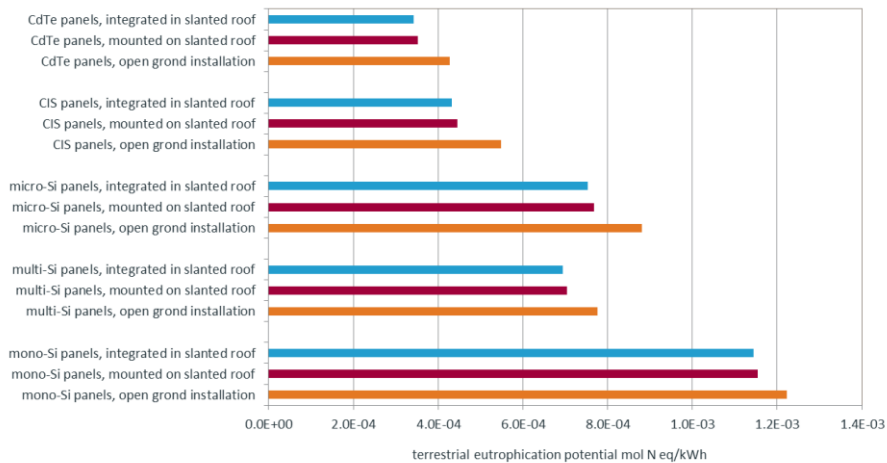


Fig. 5.9 Terrestrial eutrophication potential results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels

The impact category **freshwater eutrophication potential** is one of two impact category where the electricity production with open ground systems leads to smaller impacts compared to the systems installed on roofs. Fig. 5.10 shows the impacts per technology and per mounting system. The impacts of electricity from open ground systems are up to 49 % lower than the impacts of electricity production with the mounted roof systems. This difference can be attributed to the fact that the residential scale roof systems need more copper per electricity produced than the open ground system (within the electric installation). Hence, more sulphidic tailings are produced in the supply chain of copper used in the mounted system compared to the open ground system. The impacts of the mounted and the integrated systems are comparable.

The electricity production with CdTe panels causes the lowest emissions. The freshwater eutrophication potential then increases from micromorphous-Si PV panels to CIS and multicrystalline-Si PV panels and is highest for electricity generated in monocrystalline-Si systems.

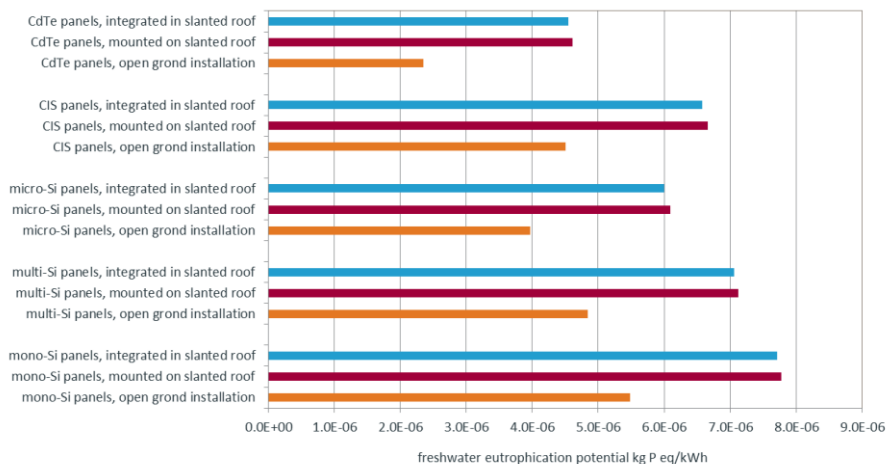


Fig. 5.10 Freshwater eutrophication potential results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels

The **marine eutrophication potential** shows large differences between the different PV technologies and small differences between the different mounting structures, which is illustrated in Fig. 5.11.

Electricity production with CdTe and CIS PV panels leads to the lowest marine eutrophication potential, which is about half of the impact level of the micromorphous-Si and multicrystalline-Si PV technologies. Electricity production with monocrystalline-Si causes impacts about four times higher compared to CdTe or CIS PV systems.

Within each technology, the impacts caused by the integrated systems are between 1 % and 3 % lower compared to those of the mounted systems. The impacts caused by the electricity production on the open ground systems are 3 % to 13 % higher compared to those of the mounted systems.

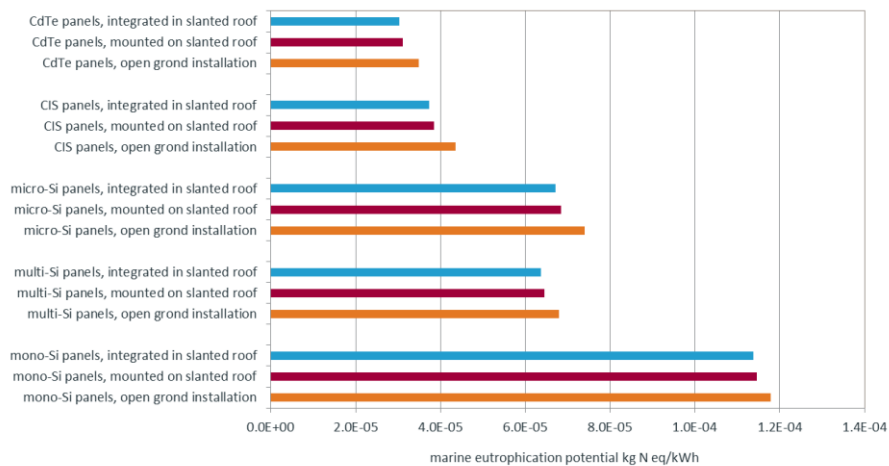


Fig. 5.11 Marine eutrophication potential results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels

The impact category **freshwater ecotoxicity** is the second impact category where the electricity production with open ground systems leads to smaller impacts compared to the systems mounted on or integrated in roofs. Fig. 5.12 shows the impacts per technology and per mounting system. The freshwater ecotoxicity caused by open ground systems are up to 46 % lower than the impacts of electricity production with the mounted roof system. This difference can be attributed to the fact that the residential scale roof systems need more electric installation and more mounting structure per electricity produced than the open ground system. The impacts of the mounted and the integrated systems are comparable, with the latter being 2-3 % lower.

For the integrated and the mounted systems, the electricity production with CdTe PV panels causes the lowest impacts, which rise from production with CIS-, micromorphous-Si, multicrystalline-Si to production with monocrystalline-Si PV panels. The order of freshwater ecotoxicity is the same for open ground PV systems but the impacts of micromorphous-Si and multicrystalline-Si PV panels are almost equal.

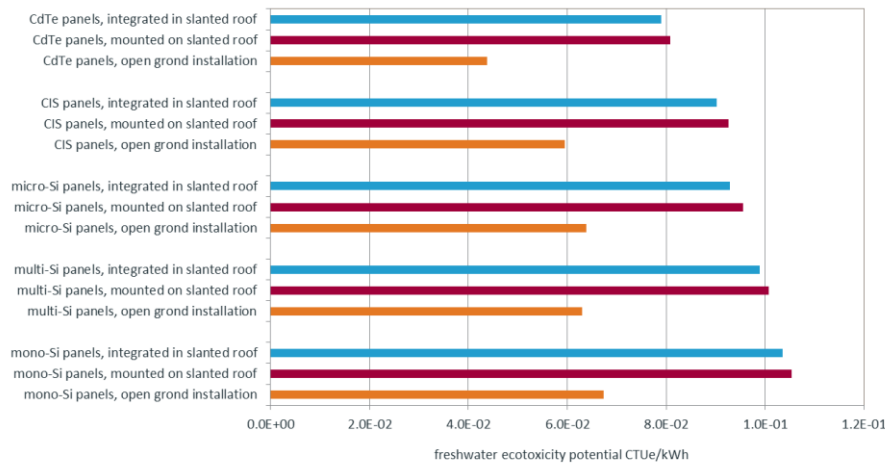


Fig. 5.12 Freshwater ecotoxicity potential results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels

Land use impacts are dominated by the mounting structure, i.e. the land use of the open ground systems. The PV systems on the roof (mounted and integrated) do not cause (additional) direct land use, as it is allocated to the building on which the PV systems are installed. In contrast, the open ground systems cause direct land use, which is the main contribution to this impact category. The supply chain contributions to land use are very small (see Fig. 5.13). That is why the difference between open ground and roof systems is that distinct. The differences between the technologies are caused by the difference in module efficiency.

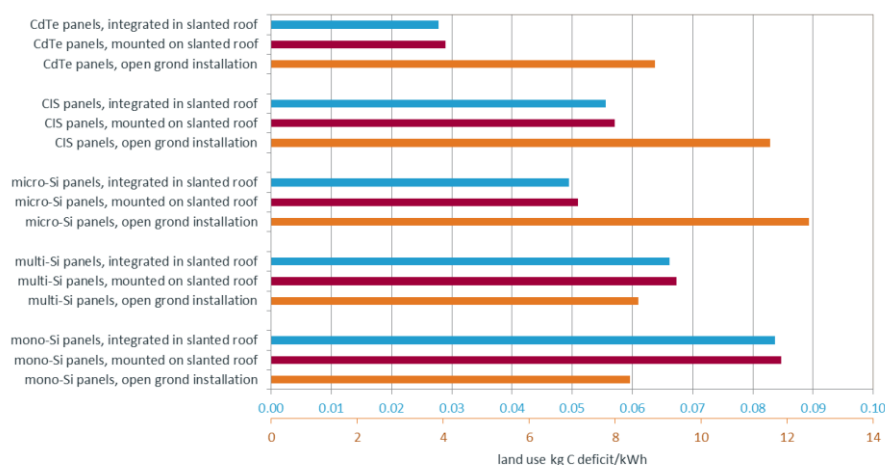


Fig. 5.13 Land use results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof (blue scale on the x-axis), respectively, and produced with a 570 kWp open ground power system (orange scale on the x-axis) using CdTe, CIS, micro-Si, multi- and mono-Si PV panels

Water resource depletion potential is on one hand influenced by the PV technology and on the other hand by the mounting structure. Electricity production with CdTe PV panels causes the smallest impacts, followed by the production with CIS, multicrystalline-Si and micromorphous-Si PV panels. Electricity production with monocrystalline-Si PV panels causes the highest impacts in terms of the water resource depletion potential (see Fig. 5.14).

Within each technology the integrated systems cause impacts 3 to 14 % lower than those of the mounted systems. The impacts of open ground systems are about 6 to 33 % higher than those from the mounted systems.

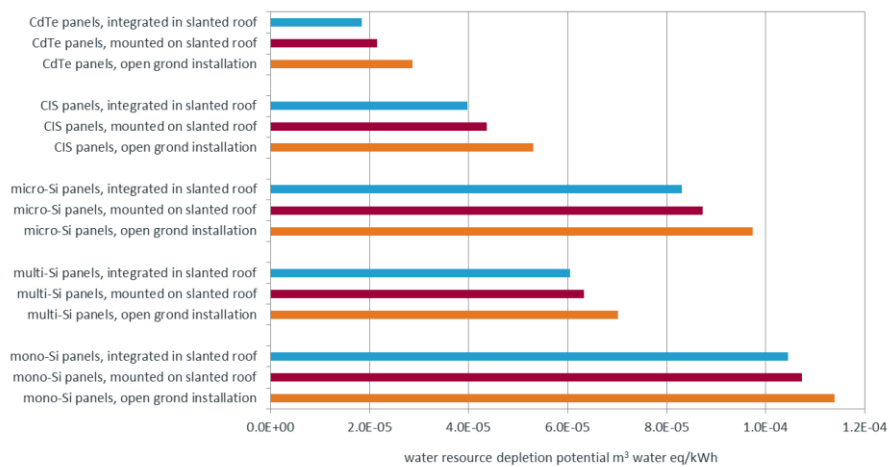


Fig. 5.14 Water resource depletion potential results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels

Mineral, fossil and renewable resource depletion is dominated by the PV technology. The mounting system has hardly any influence. This is illustrated in Fig. 5.15. The electricity production with CIS panels causes impacts being approximately 340 times higher than the smallest impacts. This is due to the indium consumption and its supply chain. Electricity produced with micromorphous-Si PV panels causes the smallest impacts, followed by the electricity production with CdTe PV panels. Electricity production with monocrystalline-Si PV panels leads to lower impacts on mineral, fossil and renewable resource depletion than when producing electricity with multicrystalline-Si PV panels. The influence of the mounting type is negligible for this impact category.

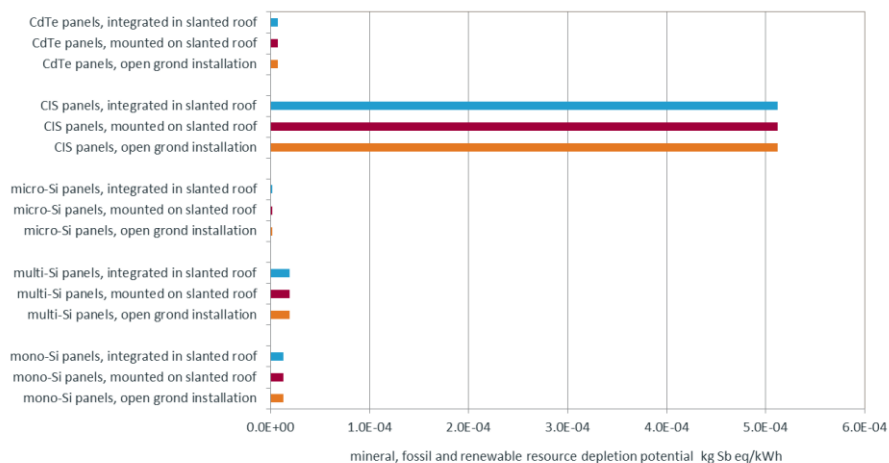


Fig. 5.15 Mineral, fossil and renewable resource depletion potential results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels

The impact indicator **cumulative energy demand (CED)** differs between the renewable and the non-renewable energy sources. The share of CED renewable is clearly higher than the share of CED non-

renewable. This is illustrated in Fig. 5.16. The CED renewable mainly reflects the harvested amount of solar energy needed to produce electricity with the PV panel, which is equal for all technologies and mounting systems analyzed (3.6 MJ per kWh DC electricity produced) and thus hardly varies. The share of non-renewable cumulative energy demand is dominated by the energy consumption during the production of the panels, of the electric installation and of the mounting system. Therefore, the non-renewable CED differs more distinctly between the five PV technologies. Electricity production with CdTe PV panels requires less non-renewable primary energy resources than the electricity production with the other technologies. Electricity production with CIS-, micromorphous-Si and multicrystalline-Si PV panels require similar amounts of non-renewable primary energy resources, whereas the monocrystalline-Si PV technology requires the most. Within each technology, the mounted systems require 1-2 % more non-renewable primary energy resources than the integrated systems and between 6-17 % less energy than the open ground systems.

The CED renewable as determined by the method proposed by VDI (2012) is about a factor of 10 higher than the 3.6 MJ provided above, which is due to the consideration of the PV panel efficiency. When all life cycle stages are considered and no recycling benefits are taken into account, the ranking of the five PV technologies mounted on a slanted roof is as follows: mono-Si (24.0 MJ), multi-Si (24.6 MJ), CdTe (25.8 MJ), CIS (33.4 MJ), micro-Si (36.1 MJ). Hence, the range increases if the renewable CED is quantified with the alternative approach and the results mirror the efficiency of the different PV technologies.

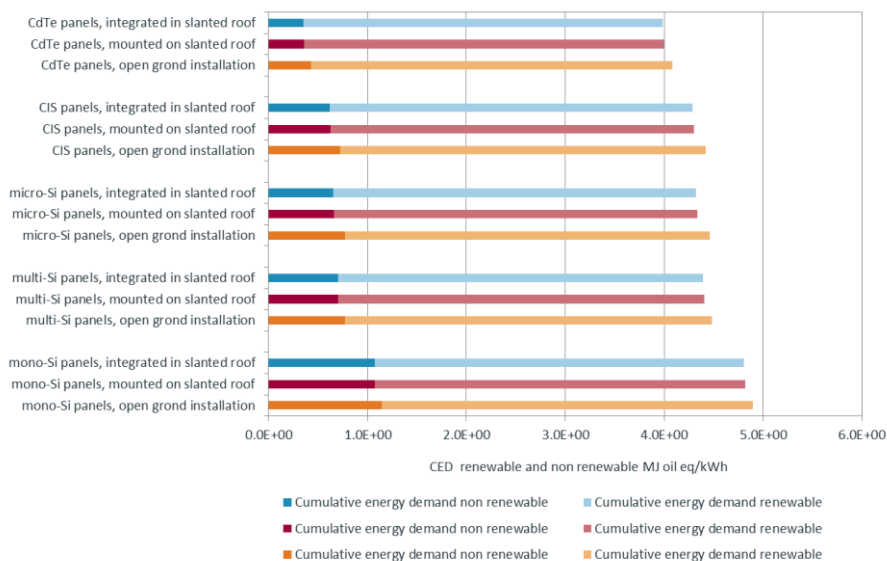


Fig. 5.16 Cumulative energy demand, renewable and non-renewable, results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels

The impact category **nuclear waste** shows differences between the PV technologies but also between the three mounting systems (see Fig. 5.17). Electricity production with CdTe PV panels causes the lowest nuclear waste impacts. Electricity production with micromorphous-Si and multicrystalline-Si lead to similar results, with the latter being slightly higher. CIS- and monocrystalline-Si PV technologies cause comparable amounts of nuclear wastes, which are higher than those of the other three PV technologies.

The integrated systems cause 4-12 % lower amounts of nuclear wastes than the mounted systems, which in turn cause 10-33 % lower amounts than the open ground systems.

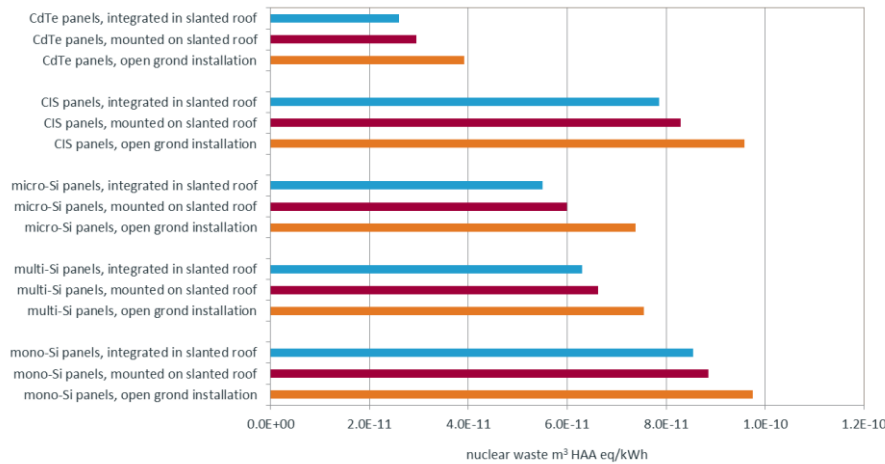


Fig. 5.17 Nuclear waste results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels

5.3 Overall results normalized

The following Tab. 5.3 shows the normalized environmental impacts of 1 kWh of DC electricity produced by all five PV technologies and the three mounting systems. The three additional impact categories cumulative energy demand non-renewable, cumulative energy demand renewable and nuclear waste are not subject to normalization. The normalized results shown in this section include all life cycle stages (i.e., product, operation, use and end-of-life stage) but do not contain the potential benefits for recycling of PV panels. The benefits for recycling are included in the results presented in the subchapters 5.5 to 5.9 for the individual PV technologies.

Tab. 5.3 Normalized results per impact category of 1 kWh of DC electricity produced with a 3 kWp power system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a 570 kWp open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels. The contribution of potential benefits for recycling is not included in these results.

	unit	1 kWh electricity, produced at a 3 kWp PV plant, CdTe panels, integrated in slanted roof	1 kWh electricity, produced at a 3 kWp PV plant, CdTe panels, mounted on slanted roof	1 kWh electricity, produced at a 570 kWp PV plant, CdTe panels, open ground installation	1 kWh electricity, produced at a 3 kWp PV plant, CIS panels, integrated in slanted roof	1 kWh electricity, produced at a 3 kWp PV plant, CIS panels, mounted on slanted roof	1 kWh electricity, produced at a 570 kWp PV plant, CIS panels, open ground installation	1 kWh electricity, produced at a 3 kWp PV plant, micro-Si panels, integrated in slanted roof	1 kWh electricity, produced at a 3 kWp PV plant, micro-Si panels, mounted on slanted roof	1 kWh electricity, produced at a 570 kWp PV plant, micro-Si panels, open ground installation	1 kWh electricity, produced at a 3 kWp PV plant, multi-Si panels, integrated in slanted roof	1 kWh electricity, produced at a 3 kWp PV plant, multi-Si panels, mounted on slanted roof	1 kWh electricity, produced at a 570 kWp PV plant, multi-Si panels, open ground installation	1 kWh electricity, produced at a 3 kWp PV plant, mono-Si panels, integrated in slanted roof	1 kWh electricity, produced at a 3 kWp PV plant, mono-Si panels, mounted on slanted roof	1 kWh electricity, produced at a 570 kWp PV plant, mono-Si panels, open ground installation
Climate change		2.68E-06	2.73E-06	3.29E-06	4.44E-06	4.51E-06	5.27E-06	6.11E-06	6.19E-06	7.02E-06	6.05E-06	6.10E-06	6.63E-06	9.80E-06	9.86E-06	1.04E-05
Ozone depletion		4.42E-08	4.67E-08	5.59E-08	8.05E-08	8.37E-08	9.59E-08	5.33E-08	5.68E-08	7.00E-08	8.06E-08	8.30E-08	9.18E-08	1.10E-07	1.13E-07	1.21E-07
Human toxicity, cancer effects		1.56E-05	2.28E-05	4.55E-05	2.59E-05	3.53E-05	6.50E-05	2.94E-05	3.95E-05	7.17E-05	2.74E-05	3.43E-05	5.59E-05	3.18E-05	3.85E-05	5.94E-05
Human toxicity, non-cancer effects		9.49E-06	1.01E-05	1.52E-05	1.64E-05	1.72E-05	2.49E-05	1.77E-05	1.86E-05	2.72E-05	1.93E-05	1.99E-05	2.45E-05	2.49E-05	2.54E-05	2.99E-05
Particulate matter		3.38E-06	3.52E-06	4.01E-06	4.64E-06	4.83E-06	5.58E-06	1.54E-05	1.56E-05	1.65E-05	1.79E-05	1.80E-05	1.84E-05	3.48E-05	3.49E-05	3.53E-05
Ionizing radiation HH		9.91E-07	1.13E-06	1.52E-06	2.95E-06	3.13E-06	3.65E-06	2.22E-06	2.41E-06	2.97E-06	2.45E-06	2.58E-06	2.96E-06	3.25E-06	3.38E-06	3.74E-06
Photochemical ozone formation		2.90E-06	2.97E-06	3.29E-06	3.75E-06	3.84E-06	4.30E-06	6.57E-06	6.67E-06	7.19E-06	6.15E-06	6.22E-06	6.52E-06	1.03E-05	1.03E-05	1.06E-05
Acidification		3.89E-06	4.03E-06	4.39E-06	4.79E-06	4.97E-06	5.56E-06	9.55E-06	9.75E-06	1.04E-05	1.00E-05	1.02E-05	1.05E-05	1.71E-05	1.73E-05	1.76E-05
Terrestrial eutrophication		1.95E-06	2.00E-06	2.43E-06	2.46E-06	2.53E-06	3.12E-06	4.28E-06	4.37E-06	5.01E-06	3.95E-06	4.00E-06	4.41E-06	6.51E-06	6.56E-06	6.95E-06
Freshwater eutrophication		3.07E-06	3.12E-06	1.59E-06	4.44E-06	4.51E-06	3.05E-06	4.05E-06	4.12E-06	2.69E-06	4.77E-06	4.82E-06	3.27E-06	5.21E-06	5.25E-06	3.71E-06
Marine eutrophication		1.79E-06	1.85E-06	2.03E-06	2.21E-06	2.28E-06	2.58E-06	3.97E-06	4.05E-06	4.38E-06	3.77E-06	3.82E-06	4.02E-06	6.73E-06	6.78E-06	6.98E-06
Freshwater ecotoxicity		9.04E-06	9.25E-06	5.01E-06	1.03E-05	1.06E-05	6.81E-06	1.06E-05	1.09E-05	7.31E-06	1.13E-05	1.15E-05	7.21E-06	1.19E-05	1.21E-05	7.70E-06
Land use		3.72E-07	3.87E-07	1.19E-04	7.43E-07	7.63E-07	1.35E-04	6.61E-07	6.82E-07	1.67E-04	8.85E-07	9.00E-07	1.14E-04	1.12E-03	1.13E-03	1.11E-04
Water resource depletion		2.26E-07	2.63E-07	3.52E-07	4.89E-07	5.37E-07	6.52E-07	1.02E-06	1.07E-06	1.20E-06	7.42E-07	7.78E-07	8.62E-07	1.28E-06	1.32E-06	1.40E-06
Mineral, fossil & ren resource depletion		7.01E-05	7.02E-05	7.00E-05	5.07E-05	5.07E-05	5.07E-05	1.49E-05	1.50E-05	1.60E-05	1.91E-04	1.91E-04	1.91E-04	1.24E-04	1.25E-04	1.24E-04

5.4 Overall results weighted

The following Tab. 5.4 shows the overall weighted environmental impacts of 1 kWh of DC electricity produced by all five PV technologies and the three mounting systems. The three additional impact

categories cumulative energy demand non-renewable, cumulative energy demand renewable and nuclear waste are not included in the weighted results. The weighted results shown in this section include all life cycle stages (i.e., product, operation, use and end-of-life stage) but do not contain the potential benefits for recycling of PV panels. The benefits for recycling are included in the results presented in the subchapters 5.5 to 5.9 for the individual PV technologies.

Tab. 5.4 Weighted results per impact category of 1 kWh of DC electricity produced with a residential scale (3 kWp) PV system with PV panels integrated and mounted on a slanted roof, respectively, and produced with a large scale (570 kWp) open ground power system using CdTe, CIS, micro-Si, multi- and mono-Si PV panels. The contribution of potential benefits for recycling is not included in these results.

	unit	1 kWh electricity, produced at a 3 kWp PV plant, CdTe panels, integrated in slanted roof	1 kWh electricity, produced at a 3 kWp PV plant, CdTe panels, mounted on slanted roof	1 kWh electricity, produced at a 570 kWp PV plant, CdTe panels, open ground installation	1 kWh electricity, produced at a 3 kWp PV plant, CIS panels, integrated in slanted roof	1 kWh electricity, produced at a 3 kWp PV plant, CIS panels, mounted on slanted roof	1 kWh electricity, produced at a 570 kWp PV plant, CIS panels, open ground installation	1 kWh electricity, produced at a 3 kWp PV plant, micro-Si panels, integrated in slanted roof	1 kWh electricity, produced at a 3 kWp PV plant, micro-Si panels, mounted on slanted roof	1 kWh electricity, produced at a 570 kWp PV plant, micro-Si panels, open ground installation	1 kWh electricity, produced at a 3 kWp PV plant, multi-Si panels, integrated in slanted roof	1 kWh electricity, produced at a 3 kWp PV plant, multi-Si panels, mounted on slanted roof	1 kWh electricity, produced at a 570 kWp PV plant, multi-Si panels, open ground installation	1 kWh electricity, produced at a 3 kWp PV plant, mono-Si panels, integrated in slanted roof	1 kWh electricity, produced at a 3 kWp PV plant, mono-Si panels, mounted on slanted roof	1 kWh electricity, produced at a 570 kWp PV plant, mono-Si panels, open ground installation
Total	Pt	8.37E-06	8.96E-06	1.85E-05	3.44E-04	3.44E-04	3.57E-04	8.43E-06	9.26E-06	2.31E-05	2.04E-05	2.10E-05	3.00E-05	1.93E-05	1.98E-05	2.86E-05
Climate change	Pt	1.79E-07	1.82E-07	2.19E-07	2.96E-07	3.01E-07	3.51E-07	4.08E-07	4.13E-07	4.68E-07	4.03E-07	4.07E-07	4.42E-07	6.54E-07	6.57E-07	6.91E-07
Ozone depletion	Pt	2.95E-09	3.11E-09	3.73E-09	5.36E-09	5.58E-09	6.39E-09	3.55E-09	3.79E-09	4.66E-09	5.38E-09	5.53E-09	6.12E-09	7.35E-09	7.50E-09	8.07E-09
Human toxicity, cancer effects	Pt	1.04E-06	1.52E-06	3.03E-06	1.73E-06	2.35E-06	4.33E-06	1.96E-06	2.63E-06	4.78E-06	1.83E-06	2.29E-06	3.72E-06	2.12E-06	2.56E-06	3.96E-06
Human toxicity, non-cancer effects	Pt	6.33E-07	6.74E-07	1.01E-06	1.09E-06	1.15E-06	1.66E-06	1.18E-06	1.24E-06	1.82E-06	1.29E-06	1.32E-06	1.63E-06	1.66E-06	1.70E-06	1.99E-06
Particulate matter	Pt	2.25E-07	2.35E-07	2.67E-07	3.10E-07	3.22E-07	3.72E-07	1.03E-06	1.04E-06	1.10E-06	1.19E-06	1.20E-06	1.23E-06	2.23E-06	2.33E-06	2.35E-06
Ionizing radiation HH	Pt	6.60E-08	7.51E-08	1.02E-07	1.97E-07	2.08E-07	2.43E-07	1.48E-07	1.60E-07	1.98E-07	1.64E-07	1.72E-07	1.97E-07	2.17E-07	2.25E-07	2.50E-07
Photochemical ozone formation	Pt	1.93E-07	1.98E-07	2.19E-07	2.50E-07	2.56E-07	2.87E-07	4.38E-07	4.45E-07	4.79E-07	4.10E-07	4.15E-07	4.35E-07	6.84E-07	6.89E-07	7.08E-07
Acidification	Pt	2.59E-07	2.69E-07	2.92E-07	3.19E-07	3.31E-07	3.71E-07	6.37E-07	6.50E-07	6.95E-07	6.69E-07	6.78E-07	6.99E-07	1.14E-06	1.15E-06	1.17E-06
Terrestrial eutrophication	Pt	1.30E-07	1.34E-07	1.62E-07	1.64E-07	1.69E-07	2.08E-07	2.86E-07	2.91E-07	3.34E-07	2.63E-07	2.67E-07	2.94E-07	4.34E-07	4.37E-07	4.63E-07
Freshwater eutrophication	Pt	2.05E-07	2.08E-07	1.06E-07	2.96E-07	3.00E-07	2.03E-07	2.70E-07	2.75E-07	1.79E-07	3.18E-07	3.21E-07	2.18E-07	3.47E-07	3.50E-07	2.47E-07
Marine eutrophication	Pt	1.19E-07	1.23E-07	1.37E-07	1.47E-07	1.52E-07	1.72E-07	2.65E-07	2.70E-07	2.92E-07	2.51E-07	2.55E-07	2.68E-07	4.49E-07	4.52E-07	4.65E-07
Freshwater ecotoxicity	Pt	6.02E-07	6.17E-07	3.34E-07	6.88E-07	7.07E-07	4.54E-07	7.09E-07	7.29E-07	4.87E-07	7.55E-07	7.69E-07	4.81E-07	7.90E-07	8.04E-07	5.13E-07
Land use	Pt	2.48E-08	2.58E-08	7.96E-08	4.95E-08	5.09E-08	1.03E-05	4.40E-08	4.55E-08	1.12E-05	5.90E-08	6.00E-08	7.61E-06	7.46E-08	7.55E-08	7.43E-06
Water resource depletion	Pt	1.51E-08	1.76E-08	2.34E-08	3.26E-08	3.58E-08	4.35E-08	6.80E-08	7.15E-08	7.98E-08	4.95E-08	5.18E-08	5.74E-08	8.56E-08	8.79E-08	9.33E-08
Mineral, fossil & ren resource depletion	Pt	4.67E-06	4.68E-06	4.67E-06	3.38E-04	3.38E-04	3.38E-04	9.92E-07	9.99E-07	1.06E-06	1.27E-05	1.27E-05	1.27E-05	8.30E-06	8.30E-06	8.28E-06

5.5 Results CdTe PV technology

5.5.1 Overview

In this subchapter the results of electricity production with a residential scale CdTe PV system, mounted on a slanted roof, prior to normalization (section 5.5.2) as well as the normalized (section 5.5.3) and the weighted results (section 5.5.4) are described.

5.5.2 Results prior to normalization

The results prior to normalization of the production of 1 kWh of electricity with a residential scale CdTe PV system, mounted on a slanted roof are summarized in Tab. 5.5. The life cycle stages analyzed are the product stage, the construction stage, the operation stage and the end-of-life stage. Net benefits caused by end-of-life recycling are allocated to the production of electricity with a share of 50 %. The other half is allocated to the life cycle of products potentially using the recycled materials (see also subchapter 3.5).

Tab. 5.5 Environmental impact results (characterized) of 1 kWh of DC electricity produced with a residential scale (3 kWp) PV system with CdTe PV panels mounted on a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, mounted with CdTe PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change	kg CO ₂ eq	1.35E-02	1.01E-02	3.45E-06	1.63E-03	2.52E-02	-3.36E-03	2.19E-02
Ozone depletion	kg CFC-11 eq	6.73E-10	2.83E-10	2.13E-13	5.24E-11	1.01E-09	-7.52E-11	9.33E-10
Human toxicity, cancer effects	CTUh, c	8.10E-11	7.26E-10	5.25E-13	3.42E-11	8.42E-10	-1.83E-10	6.58E-10
Human toxicity, non-cancer effects	CTUh, n-c	8.36E-10	4.44E-09	8.15E-12	1.03E-10	5.39E-09	-2.18E-10	5.17E-09
Particulate matter	kg PM _{2.5} eq	6.25E-06	6.76E-06	1.89E-09	3.70E-07	1.34E-05	-2.25E-06	1.11E-05
Ionizing radiation HH	kBq U235 eq	2.84E-04	8.95E-04	6.40E-07	9.43E-05	1.27E-03	-2.55E-04	1.02E-03
Photochemical ozone formation	kg NMVOC eq	6.14E-05	2.75E-05	1.25E-08	5.09E-06	9.40E-05	-7.20E-06	8.68E-05
Acidification	mol H ⁺ eq	1.14E-04	7.15E-05	2.59E-08	5.38E-06	1.91E-04	-2.22E-05	1.69E-04
Terrestrial eutrophication	mol N eq	2.41E-04	9.35E-05	6.78E-08	1.85E-05	3.53E-04	-2.83E-05	3.25E-04
Freshwater eutrophication	kg P eq	5.32E-07	3.96E-06	4.57E-09	1.22E-07	4.61E-06	-1.04E-06	3.58E-06
Marine eutrophication	kg N eq	2.14E-05	7.84E-06	1.00E-07	1.83E-06	3.12E-05	-2.13E-06	2.91E-05
Freshwater ecotoxicity	CTUe	3.80E-03	7.59E-02	2.03E-05	1.12E-03	8.09E-02	-3.14E-03	7.77E-02
Land use	kg C deficit	1.75E-02	1.01E-02	1.42E-05	1.32E-03	2.90E-02	-2.85E-03	2.61E-02
Water resource depletion	m ³ water eq	4.18E-06	1.61E-05	9.14E-08	1.09E-06	2.14E-05	-5.66E-06	1.58E-05
Mineral, fossil & ren resource depletion	kg Sb eq	6.57E-06	5.19E-07	2.81E-11	6.06E-09	7.09E-06	-2.84E-06	4.25E-06
Cumulative energy demand non renewable	MJ oil eq	1.92E-01	1.45E-01	5.29E-05	2.15E-02	3.59E-01	-4.07E-02	3.18E-01
Cumulative energy demand renewable	MJ oil eq	8.05E-03	2.85E-02	3.60E+00	1.04E-03	3.64E+00	-9.43E-03	3.63E+00
Nuclear waste	m ³ HAA eq	7.25E-12	1.97E-11	1.19E-14	2.49E-12	2.94E-11	-5.69E-12	2.37E-11

The following Fig. 5.18 illustrates the shares of environmental impacts per life cycle stage relative to the overall results per impact category, from production to end-of-life stage. The potential benefits due to recycling are also expressed relative to the overall results. The product stage and the construction stage are most important for most of the impact indicators. The end-of-life stage is much less important (8 % maximum) and the operation stage is insignificant with the exception of the renewable cumulative energy demand.

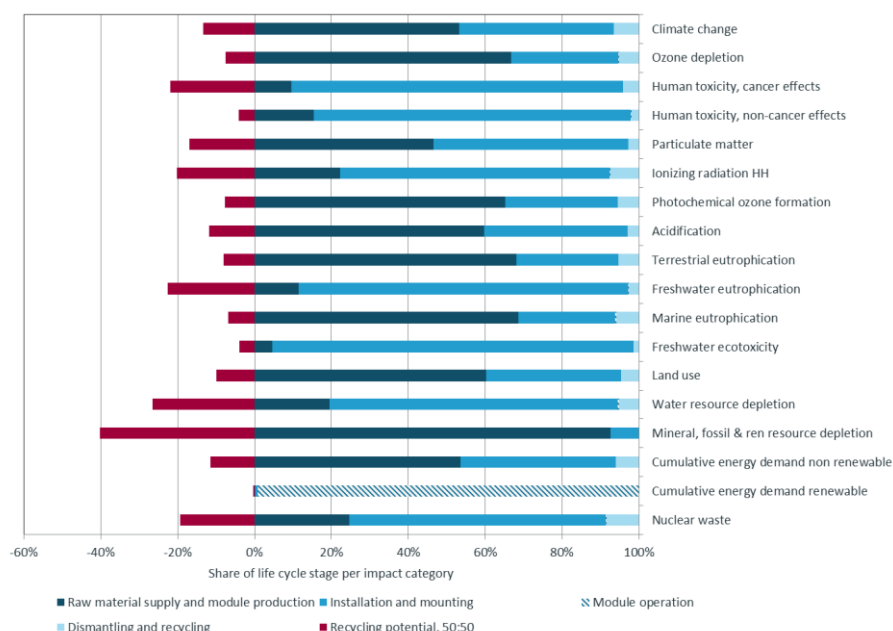


Fig. 5.18 Environmental impact results (characterized, indexed to 100 %) of 1 kWh of DC electricity produced with a residential scale (3 kWp) PV system with CdTe PV panels mounted on a slanted roof. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.

The impact category **climate change** is mainly influenced by the production and construction stages. They contribute 53 % and 40 % to the overall greenhouse gas emissions, respectively. The operation

stage (<1 %) and the end-of-life stage (6 %) are of minor importance with regard to climate change impacts. Within the product stage the panel production (electricity consumption) and the production of the raw materials (flat and solar glass) cause the major share of climate change impacts. The construction stage is dominated by the manufacture of the mounting system (aluminium consumption). The potential benefits due to recycling at the end-of-life sum up to about -13 % of the overall greenhouse gas emissions and are mainly owed to the recycling of the aluminium, copper and steel fractions within the mounting structure and the electric installation. Furthermore the recycling of glass leads to potential savings of mining raw materials (soda powder and silica sand) and fossil fuels (natural gas) because the melting of glass cullets in flat glass production needs less energy than the melting of silica sand. However the credits due to the recycling of aluminium are substantially higher than the credits for recycling glass.

The product stage contributes about 67 % to the **ozone depletion potential** impact, mainly due to the electricity consumption and its supply chain. The construction stage is responsible for the second highest share (28 %), which is due to the aluminium consumption within the slanted roof mounting system. The operation stage is negligible contributing less than 1 %. The contribution of the end-of-life stage (5 %) is influenced most by the electricity consumption during panel recycling. The potential benefits due to recycling at the end-of-life sum up to about -7 %. They are mainly due to the savings of primary aluminium from the mounting structure as well as savings in fossil fuels (mainly natural gas) when producing flat glass from recycled glass cullets instead of primary mineral raw materials.

The product stage (mainly the electricity consumption within the panel production) causes a share of 10 % to the impact category **human toxicity cancer effects**, whereas the construction stage causes 86 % of the impacts mainly due to its aluminium and steel consumption. The electric installation and the copper used for wiring cause the second largest share. The metal production leads to emissions into air and water especially when disposing slag in landfills (redmud when digesting bauxite in the supply chain of aluminium production and slags from unalloyed electric steel production). Substances emitted from the landfills such as heavy metals (chromium VI, arsenic, mercury) lead to human cancer effects. The contribution of the use stage is negligible (<1 %) and the end-of-life stage has a share of 4 % in the total impacts in the category human toxicity, cancer effects. The potential benefits when recycling sum up to -22 % and are mainly owed to the recycling of aluminium from the mounting structure.

The **human toxicity non-cancer effects** are likewise dominated by the construction stage of the PV system (82 %) and therein by the impacts of metals mining (copper for the electric installation and aluminium for the slanted roof mounting system). The production stage causes 16 % of the human toxicity, non-cancer effects and the end-of-life stage contributes only a minor share (2 %) to the total impacts. The potential benefits when recycling the PV system at the end-of-life amount to -4 % and are mainly owed to the recovered copper (and thus to the avoided mining and refining of primary copper) but also to the savings of producing copper telluride cement.

The product stage as well as the construction stage contribute about the same shares, namely 47 % and 51 %, respectively, to the overall **particulate matter (PM)** impacts. Within the product stage, the production of flat glass and the supply chain of electricity consumed in the production of the PV panels are the main sources of PM. Also the transoceanic transport of PV panels from Asia to Europe and of materials causes noteworthy PM emissions. The construction stage is dominated by the slanted roof mounting system (aluminium consumption) and the electric installation (copper consumption). The operation stage is negligible and contributes less than 1 %. The end-of-life stage causes a share of

3 % on the overall PM impacts, mainly caused by the transportation of the panels with lorries, but also due to the supply chain of electricity consumed during panel recycling. The potential benefits from recycling sum up to -17 % of the overall PM impacts and come from replacing primary aluminium when recycling aluminium.

The impact category **ionizing radiation HH** is mainly influenced by the construction stage. This stage accounts for 70 % of the overall impact, whereas the product stage accounts for 22 %. Within the construction stage the slanted roof mounting system causes the largest impact (supply chain of aluminium). The product stage is dominated by the supply chain of the panel production (electricity consumption within the flat glass and solar glass production). But also transoceanic transport and therein the electricity consumption at the ports contribute to the ionizing radiation HH in the product stage. The contribution of the use stage is negligible (<1 %). The end-of-life accounts for 7 % of the overall impact with the electricity consumption for recycling processes being the driving process. The main reason for the large contribution of electricity to the ionizing radiation HH is the nuclear fuel reprocessing within the supply chain of nuclear power production and its significant emissions of carbon-14 to air. Potential benefits due to recycling contribute to -20 % relative to the overall impact and are mainly due to the recycling of aluminium.

The product stage is the most dominant one with regard to the **photochemical ozone formation potential** (65 %). On one hand the solar glass and flat glass production (air emissions from fossil fuels when producing glass and its upstream raw materials) cause a large impact. On the other hand, the electricity consumption (and therein the use of fossil fuels in thermal power plants) when producing the CdTe laminates in the US and Malaysia is important. Furthermore the use of heavy fuel oil in transoceanic transport contributes to the photochemical ozone formation potential of the product stage. The construction stage contributes the second largest share (29 %) and is dominated by the slanted roof mounting system (aluminium production and electricity and fuel oil consumption therein). The contribution of the use stage is negligible (<1 %). The end-of-life accounts for 5 % of the overall impact. The most important process is the transportation by lorry and its diesel supply chain. The potential benefits due to recycling of about -8 % are mainly owed to the recycling of aluminium and copper,.

Acidification is likewise dominated by the product stage (60 %). Sulfur dioxide is emitted in the supply chain of CdTe laminates, namely in the production of flat glass and along the supply chain of electricity as well as in transoceanic transportation. The construction stage causes 37 % of the overall acidification impacts. The slanted roof mounting system as well as the electric installation contribute with the use of aluminium and copper, respectively. The contribution of the use stage is negligible (<1 %). The end-of-life stage accounts for 3 % of the overall impact with the lorry transports of the used materials being the most important process. The potential benefits for recycling contribute with -12 % to the acidification potential. The savings when replacing raw materials such as primary aluminium and copper with recycled aluminium and copper are the main drivers.

The impact category **terrestrial eutrophication potential** is dominated by the product stage (68 %). The emissions of nitrogen oxides into the air cause the main impact. On the one hand nitrogen oxides are emitted in the production of flat glass (supply chain of CdTe laminates) and on the other hand during transoceanic transportation. The construction stage contributes 27 % to the overall terrestrial eutrophication potential. The slanted roof mounting system and the electric installations (electricity consumption within the aluminium supply chain and blasting in the copper supply chain) cause the main impact. The contribution of the use stage is negligible (<1 %). The end-of-life stage causes a share of 5 % and is dominated by the nitrogen oxide emissions into air from road transport. The

potential benefits from recycling sum up to -8 % and are due to avoiding primary aluminium and primary copper production as well as due to soda powder being replaced by recycled glass cullets.

The **freshwater eutrophication potential** impact category is dominated by the construction stage (86 %). Phosphate emitted into groundwater bodies is the most relevant substance in terms of freshwater eutrophication. The electric installation (copper beneficiation) causes the main emissions of phosphate into water bodies. The contribution to the impact of the product stage amounts to 12 % and is mainly due to the consumption of fossil energy carriers in the supply chain of electricity in the laminate production process. The contribution of the use stage is negligible (<1 %). The impacts due to takeback and recycling at the end-of-life (3 %) are caused likewise by the consumption of fossil fuels (lignite and hard coal) in the supply chain of electricity production. Potential benefits from recycling (-22 %) are owed to avoiding primary copper production and thus avoiding emissions in the copper supply chain (savings in the disposal of sulphidic tailings).

The **marine eutrophication potential** is primarily determined by the emissions of nitrogen oxides into air. The same explanations apply as for the terrestrial eutrophication potential. The contributions of the life cycle stages to the total impacts are: product stage 69 %, construction stage 25 %, use stage <1 %, end-of-life stage 6 %. The potential benefits due to recycling amount to -7 %.

Emissions of heavy metals into air and water are the main drivers for the **freshwater ecotoxicity** impacts. The major share of these pollutants is emitted in the construction stage (94 %), and the product stage contributes 5 %. The most relevant processes are the incineration of waste plastics from the electric installation and the disposal of redmud from bauxite digestion, which is a process in the supply chain of aluminium used in the mounting structure. Antimony and chromium VI are the most relevant heavy metals emitted to water. The contribution of the use stage is negligible (<1 %) and the end-of-life stage accounts for 1 % of the overall impacts. The largest potential benefits due to recycling (-4 %) are gained in avoiding primary aluminium and copper production.

Land use impacts are mainly caused by the production and the construction stages (60 % and 35 %, respectively). They are mainly caused by the supply chain of oil and natural gas (wells for exploration and production) and by bauxite mines. In addition, land used by production facilities contributes significant shares to the overall total. The contribution of the use stage is negligible (<1 %). The land use of the end-of-life stage (5 %) is dominated by the production of fossil fuels used in transportation. Potential benefits from recycling amount to around -10 % and are owed to the avoided production of primary aluminium.

The impact category **water resource depletion** is mainly caused by the product stage (20 %) and by the construction stage (75 %). The contribution of the use stage is negligible (<1 %) and the end-of-life stage causes 5 % of the impacts. Within all three stages, the supply chain of electricity consumed in the production and processing of metals (and therein the water evaporated from hydroelectric power plants) is the main source. Potential benefits due to recycling (-26 %) are mainly owed to the savings in replacing primary aluminium with secondary aluminium.

Mineral, fossil and renewable resource depletion is dominated by the product stage (93 %), namely by the extraction of cadmium and tellurium. The construction stage contributes 7 % to the total impacts. The shares of the use and the end-of-life stage are smaller than 1 %. The potential benefits due to recycling are around -40 % and are likewise owed to the savings of producing primary cadmium and tellurium.

The **cumulative energy demand non-renewable** (CED nr) is on one hand caused by the product stage (54 %, electricity consumption based on fossil fuels when manufacturing the PV panels) and on the other hand by the slanted roof mounting system (40 %, aluminium supply chain). The same explanation about electricity applies for the end-of-life stage, which accounts for 6 % to the overall CED nr. The contribution of the use stage is negligible (<1 %). Potential benefits of recycling (-11 %) are owed to the savings when replacing primary aluminium with secondary aluminium, respectively.

The indicator **cumulative energy demand renewable** (CED r) is the only indicator for which the major contribution is in the operation stage (99 %). It reflects the harvested amount of solar energy needed to produce electricity with the PV panel. The remaining life cycle stages have shares below 1 % in the renewable CED and the potential benefits due to recycling are negligible.

The impact category **nuclear waste** is mainly influenced by the installation and mounting (construction stage), and, within this stage, by the electricity production from nuclear power. This stage accounts for 67 % and the product stage accounts for 25 % of the overall impact. The product stage is dominated by the panel production (electricity consumption of the flat glass and solar glass production, electricity consumption during production of the panel and in the ports required by transoceanic transport). Within the installation the slanted roof mounting system causes the largest impact (electricity consumption in the aluminium supply chain). The contribution of the use stage is negligible (<1 %). The end-of-life stage accounts for 8 % of the overall impact with the electricity supply chain being the driving process. Potential benefits due to recycling contribute -19 % to the total impact and are due to savings in the supply chain of aluminium (electricity consumption).

5.5.3 Results normalized

The normalized environmental impact results of the electricity production are listed in Tab. 5.6. The three additional impact categories cumulative energy demand non-renewable, cumulative energy demand renewable and nuclear waste are not subject to normalization.

The results show clearly that the production of 1 kWh of DC electricity contributes the largest share to the mineral, fossil and renewable resource depletion of an average European citizen. Producing 1 kWh of DC electricity generated with CdTe PV panels causes 0.04 ‰ of the mineral, fossil and renewable resource depletion potential per person in Europe. An average European person consumes around 5'000 kWh of electricity. Assuming it would all be produced with such a residential scale CdTe PV system, it would sum up to 21 % of the present personal annual budget of resource depletion.

The second largest contribution to the per capita impacts is observed in the impact category human toxicity, cancer effects, followed by the human toxicity, non-cancer effects.

Tab. 5.6 Environmental impact results (normalized) of 1 kWh of DC electricity produced with a 3 kWp power system with CdTe PV panels mounted on a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, mounted with CdTe PV technology, normalized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change	-	1.46E-06	1.10E-06	3.74E-10	1.77E-07	2.73E-06	-3.64E-07	2.37E-06
Ozone depletion	-	3.12E-08	1.31E-08	9.85E-12	2.42E-09	4.67E-08	-3.48E-09	4.32E-08
Human toxicity, cancer effects	-	2.20E-06	1.97E-05	1.42E-08	9.26E-07	2.28E-05	-4.97E-06	1.78E-05
Human toxicity, non-cancer effects	-	1.57E-06	8.34E-06	1.53E-08	1.93E-07	1.01E-05	-4.08E-07	9.71E-06
Particulate matter	-	1.64E-06	1.78E-06	4.97E-10	9.74E-08	3.52E-06	-5.92E-07	2.93E-06
Ionizing radiation HH	-	2.51E-07	7.92E-07	5.66E-10	8.34E-08	1.13E-06	-2.26E-07	9.01E-07
Photochemical ozone formation	-	1.94E-06	8.67E-07	3.93E-10	1.61E-07	2.97E-06	-2.27E-07	2.74E-06
Acidification	-	2.41E-06	1.51E-06	5.48E-10	1.14E-07	4.03E-06	-4.70E-07	3.56E-06
Terrestrial eutrophication	-	1.37E-06	5.32E-07	3.85E-10	1.05E-07	2.00E-06	-1.61E-07	1.84E-06
Freshwater eutrophication	-	3.59E-07	2.67E-06	3.08E-09	8.24E-08	3.12E-06	-7.01E-07	2.42E-06
Marine eutrophication	-	1.27E-06	4.64E-07	5.93E-09	1.08E-07	1.85E-06	-1.26E-07	1.72E-06
Freshwater ecotoxicity	-	4.35E-07	8.69E-06	2.32E-09	1.28E-07	9.25E-06	-3.59E-07	8.89E-06
Land use	-	2.34E-07	1.36E-07	1.90E-10	1.76E-08	3.87E-07	-3.81E-08	3.49E-07
Water resource depletion	-	5.14E-08	1.98E-07	1.12E-09	1.33E-08	2.63E-07	-6.96E-08	1.94E-07
Mineral, fossil & ren resource depletion	-	6.50E-05	5.14E-06	2.77E-10	6.00E-08	7.02E-05	-2.81E-05	4.21E-05

5.5.4 Results weighted

Tab. 5.7 shows the weighted environmental impact results (equal weighting) of the electricity production with CdTe PV panels. The blue bars within the columns “Total” illustrate the importance of each impact category in the total of impacts. They are discussed in further detail in the following paragraphs and shown in Fig. 5.19. The three additional impact categories cumulative energy demand non-renewable, cumulative energy demand renewable and nuclear waste are not included in the weighted results.

Tab. 5.7 Environmental impact results (weighted) of 1 kWh of DC electricity produced with a 3 kWp power system with CdTe PV panels mounted on a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, mounted with CdTe PV technology, weighted		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Total	Pt	5.35E-06	3.46E-06	3.01E-09	1.51E-07	8.96E-06	-2.46E-06	6.50E-06
Climate change	Pt	9.73E-08	7.32E-08	2.49E-11	1.18E-08	1.82E-07	-2.43E-08	1.58E-07
Ozone depletion	Pt	2.08E-09	8.72E-10	6.57E-13	1.62E-10	3.11E-09	-2.32E-10	2.88E-09
Human toxicity, cancer effects	Pt	1.46E-07	1.31E-06	9.48E-10	6.17E-08	1.52E-06	-3.31E-07	1.19E-06
Human toxicity, non-cancer effects	Pt	1.05E-07	5.56E-07	1.02E-09	1.29E-08	6.74E-07	-2.72E-08	6.47E-07
Particulate matter	Pt	1.10E-07	1.19E-07	3.32E-11	6.50E-09	2.35E-07	-3.94E-08	1.95E-07
Ionizing radiation HH	Pt	1.67E-08	5.28E-08	3.78E-11	5.56E-09	7.51E-08	-1.51E-08	6.01E-08
Photochemical ozone formation	Pt	1.29E-07	5.78E-08	2.62E-11	1.07E-08	1.98E-07	-1.51E-08	1.83E-07
Acidification	Pt	1.61E-07	1.01E-07	3.65E-11	7.58E-09	2.69E-07	-3.14E-08	2.38E-07
Terrestrial eutrophication	Pt	9.12E-08	3.54E-08	2.57E-11	6.99E-09	1.34E-07	-1.07E-08	1.23E-07
Freshwater eutrophication	Pt	2.39E-08	1.78E-07	2.06E-10	5.49E-09	2.08E-07	-4.67E-08	1.61E-07
Marine eutrophication	Pt	8.45E-08	3.09E-08	3.96E-10	7.21E-09	1.23E-07	-8.41E-09	1.15E-07
Freshwater ecotoxicity	Pt	2.90E-08	5.79E-07	1.55E-10	8.53E-09	6.17E-07	-2.40E-08	5.93E-07
Land use	Pt	1.56E-08	9.04E-09	1.26E-11	1.17E-09	2.58E-08	-2.54E-09	2.33E-08
Water resource depletion	Pt	3.43E-09	1.32E-08	7.49E-11	8.89E-10	1.76E-08	-4.64E-09	1.29E-08
Mineral, fossil & ren resource depletion	Pt	4.33E-06	3.42E-07	1.85E-11	4.00E-09	4.68E-06	-1.88E-06	2.80E-06

The environmental impacts of the production of 1 kWh of DC electricity with a residential scale CdTe PV system mounted on a slanted roof, are mainly driven by mineral, fossil and renewable resource depletion (43 %), human toxicity cancer effects (18 %), human toxicity non cancer effects (10 %) and by freshwater ecotoxicity (9 %). Within the impact category mineral, fossil and renewable resource

depletion the extraction of cadmium and tellurium are the driving forces. Even though potential benefits due to recycling are around -40 % and are hence quite high, the weighted impacts remain the largest of all impact categories. Human toxicity cancer effects and human toxicity non cancer effects are dominated by the installation and mounting system and the supply chains of aluminium, copper and steel therein. The freshwater ecotoxicity impacts are mainly caused by the disposal of plastics from industrial electronic waste, which is associated with the production of the electric installation.

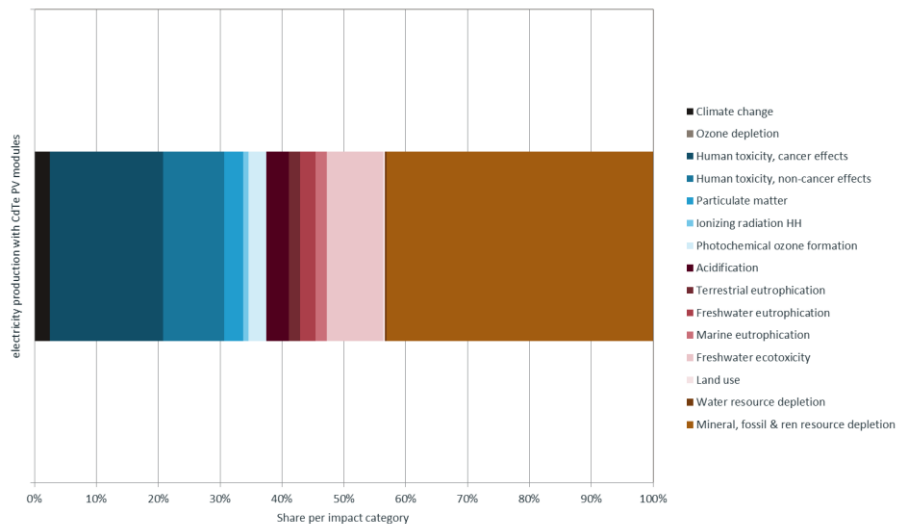


Fig. 5.19 Share per impact category on the weighted result for the production of 1 kWh of DC electricity produced with a residential scale CdTe PV system mounted on a slanted roof.

5.5.5 Comparison of the three system sizes integrated, mounted and open ground

The comparison of the environmental impacts per life cycle stage of the three different systems shows that the residential scale systems do only differ slightly while the large scale system causes substantially higher impacts per kWh produced (see Fig. 5.20). The smallest impacts are caused by the PV system integrated in a residential roof. Slightly higher impacts are caused by the mounted system whereas the open ground system causes the highest impacts. Overall the impacts of the open ground system are about 150 % higher compared to the mounted residential system. The product stage, the use stage, end-of-life stage as well as the potential benefits for recycling are similarly important in all three systems. The main difference is caused by the construction stage and, more specifically, by the land use of the different PV systems. The land use impacts of the open ground CdTe PV system are much higher than of the systems installed on a slanted roof. This is due to the fact that the open ground system directly occupies a certain land area while the land use of the residential PV systems is attributed to the building.

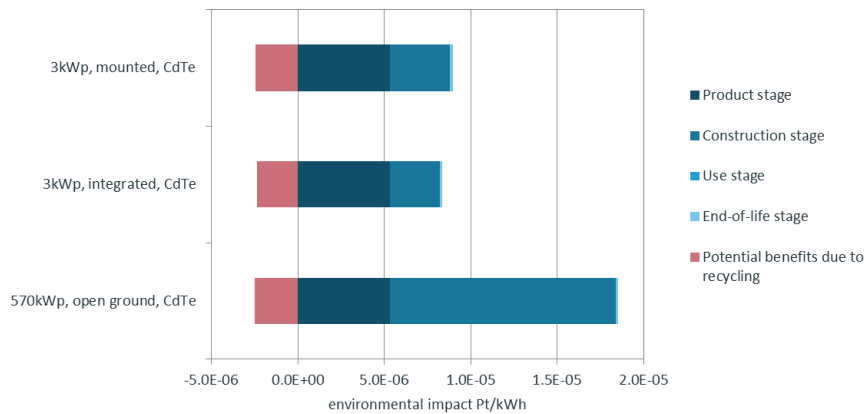


Fig. 5.20 Environmental impact results (weighted) of 1 kWh of DC electricity produced with a residential scale (3 kWp) PV system with CdTe PV panels mounted on and integrated in a slanted roof as well as produced with a large scale open ground system. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.

5.6 Results CIS PV technology

5.6.1 Overview

In this subchapter the results of electricity production with a residential scale CIS PV system, mounted on a slanted roof, prior to normalization (section 5.6.2) as well as the normalized (section 5.6.3) and the weighted results (section 5.6.4) are described.

5.6.2 Results prior to normalization

The results prior to normalization of the production of 1 kWh of DC electricity with CIS PV panels mounted on a slanted roof in a 3 kWp installation system are summarized in Tab. 5.8. The life cycle stages analyzed are the product stage, the construction stage, the operation stage and the end-of-life stage. Net benefits caused by end-of-life recycling are allocated to the production of electricity with a share of 50 %. The other half is allocated to the life cycle of products potentially using the secondary products (see also subchapter 3.5).

Tab. 5.8 Environmental impact results (characterized) of 1 kWh of DC electricity produced with a 3 kWp power system with CIS PV panels mounted on a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, mounted with CIS PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change	kg CO ₂ eq	2.63E-02	1.27E-02	4.46E-06	2.55E-03	4.16E-02	-4.98E-03	3.66E-02
Ozone depletion	kg CFC-11 eq	1.36E-09	3.61E-10	2.75E-13	8.65E-11	1.81E-09	-1.08E-10	1.70E-09
Human toxicity, cancer effects	CTUh, c	3.14E-10	9.26E-10	6.80E-13	5.97E-11	1.30E-09	-2.97E-10	1.00E-09
Human toxicity, non-cancer effects	CTUh, n-c	4.07E-09	4.91E-09	1.06E-11	1.79E-10	9.17E-09	-2.67E-11	9.14E-09
Particulate matter	kg PM _{2.5} eq	9.44E-06	8.12E-06	2.45E-09	7.74E-07	1.83E-05	-3.13E-06	1.52E-05
Ionizing radiation HH	kBq U235 eq	2.10E-03	1.15E-03	8.28E-07	2.83E-04	3.53E-03	-4.15E-04	3.12E-03
Photochemical ozone formation	kg NMVOC eq	7.90E-05	3.33E-05	1.61E-08	9.26E-06	1.22E-04	-1.03E-05	1.11E-04
Acidification	mol H ⁺ eq	1.40E-04	8.38E-05	3.36E-08	1.14E-05	2.35E-04	-2.97E-05	2.05E-04
Terrestrial eutrophication	mol N eq	2.99E-04	1.13E-04	8.79E-08	3.34E-05	4.46E-04	-3.86E-05	4.07E-04
Freshwater eutrophication	kg P eq	2.46E-06	4.07E-06	5.92E-09	1.39E-07	6.67E-06	-1.04E-06	5.62E-06
Marine eutrophication	kg N eq	2.55E-05	9.59E-06	1.30E-07	3.30E-06	3.85E-05	-3.06E-06	3.55E-05
Freshwater ecotoxicity	CTUe	1.19E-02	7.87E-02	2.63E-05	1.98E-03	9.27E-02	-4.29E-03	8.84E-02
Land use	kg C deficit	4.21E-02	1.26E-02	1.84E-05	2.34E-03	5.71E-02	-3.92E-03	5.31E-02
Water resource depletion	m ³ water eq	2.14E-05	2.07E-05	1.18E-07	1.46E-06	4.37E-05	-9.24E-06	3.44E-05
Mineral, fossil & ren resource depletion	kg Sb eq	5.12E-04	5.40E-07	-4.69E-10	1.04E-08	5.12E-04	-8.99E-09	5.12E-04
Cumulative energy demand non renewable	MJ oil eq	4.05E-01	1.82E-01	6.84E-05	3.62E-02	6.23E-01	-6.21E-02	5.61E-01
Cumulative energy demand renewable	MJ oil eq	4.15E-02	3.66E-02	3.60E+00	2.36E-03	3.68E+00	-1.53E-02	3.67E+00
Nuclear waste	m ³ HAA eq	5.18E-11	2.53E-11	1.54E-14	5.91E-12	8.30E-11	-9.28E-12	7.37E-11

The following Fig. 5.21 illustrates the shares of environmental impacts per life cycle stage relative to the overall results per impact category, from production to end-of-life stage. The potential benefits due to recycling are also expressed relative to the overall results. The product stage and the construction stage are most important for most of the impact indicators. The end-of-life stage is much less important (9 % maximum) and the operation stage is insignificant with the exception of the renewable cumulative energy demand.

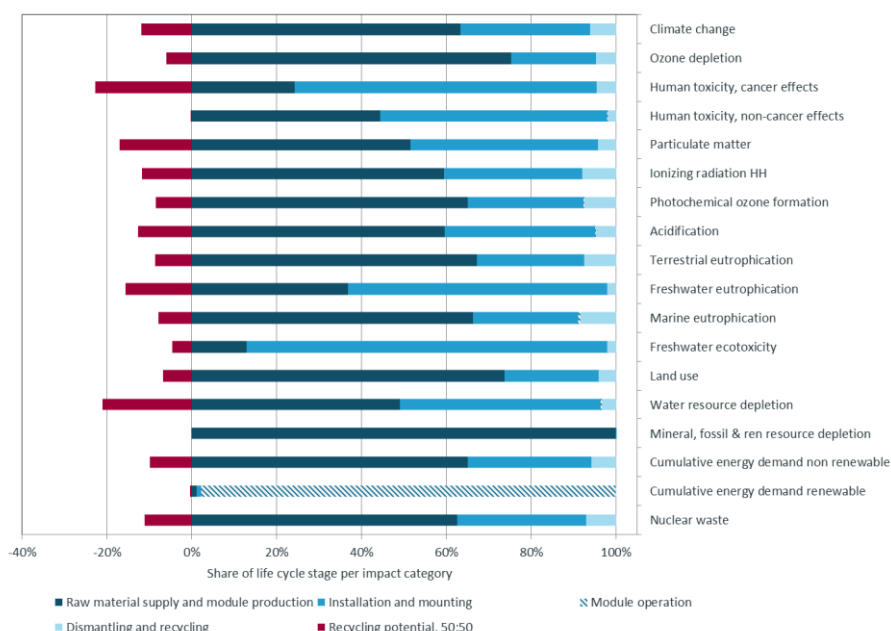


Fig. 5.21 Environmental impact results (characterized, indexed to 100 %) of 1 kWh of DC electricity produced with a 3 kWp power system with CIS PV panels mounted on a slanted roof. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.

The impact category **climate change** is mainly influenced by the production and construction stages, contributing to 64 % and 31 % to the overall greenhouse gas emissions, respectively. The operation

stage has a negligible climate change impact (<1 %). The end-of-life stage causes minor climate change impacts (6 %). Within the product stage the panel production (electricity consumption) and the production of the raw materials (supply chain of flat and solar glass as well as of liquid nitrogen as protection gas) cause the major share of climate change impacts. The construction stage is mainly influenced by the supply chain of the mounting system (electricity used for aluminium production). Potential benefits due to recycling at the end-of-life sum up to about -12 % of the overall greenhouse gas emissions and are mainly owed to the recycling of the aluminium contained in the mounting structure.

The **ozone depletion potential** has its largest impact in the product stage, mainly in the electricity consumption and its supply chain. The product stage causes about 75 % of the total ozone depletion potential. The electricity consumption in the CIS panel production, the consumption of silicon products, solar and flat glass as well as liquid nitrogen (and their supply chains) dominate the impacts in the product stage. The construction stage causes the second highest impact (20 %) and is dominated by the aluminium consumption within the slanted roof mounting system. The operation stage is negligible and causes an impact of less than 1 %. The end-of-life stage contributes 5 % to the total impacts with the electricity consumption during panel recycling being most important. However the impact of recycling efforts itself is small compared to the production impacts. The potential benefits due to recycling at the end-of-life sum up to about -6 %. They are mainly due to the savings of aluminium from the mounting structure as well as savings in fossil fuels (natural gas) when producing flat glass from recycled glass cullets instead of primary mineral raw materials.

About 24 % of the **human toxicity, cancer effects** are caused by the product stage, mainly due to the aluminium needed for the panel framing and the electricity consumption in the laminate production. The construction stage causes 71 % of the impacts. Therein the slanted roof mounting system and its aluminium and steel consumption cause the largest impact. The electric installation (copper) is of minor importance. The contribution of the use stage to the total impacts is negligible (<1 %). At the end-of-life impacts to human toxicity cancer effects (5 %) are mainly caused by the hydrogen peroxide production. The potential benefits of recycling sum up to -23 % and are primarily owed to the recycling of aluminium from the mounting structure.

The **human toxicity non-cancer effects** are dominated by the production of the CIS PV panel, which contributes about 44 % to the total impact. Indium production in the supply chain of the laminate production causes the main impact. Furthermore aluminium for framing and the electricity consumption in the production of laminates are important. The construction stage contributes to 54 % to the overall impact with the beneficiation of copper (electric installation) and the disposal of redmud from bauxite digestion in the supply chain of aluminium (mounting structure) being most important. The contribution of the use stage to the total impacts is negligible (<1 %). The end-of-life stage contributes with 2 % to the overall impact. The potential benefits of recycling are very small (<1 %). Benefits for substituting primary raw materials in flat glass production and recycling metals are compensated by the reprocessing of scrap aluminium to secondary aluminium.

The product stage and the construction stage are the most important contributors to the **particulate matter** impacts (51 % and 44 %, respectively). Within the product stage, the production and supply chain of solar and flat glass as well as the electricity consumption are most important. The construction stage is dominated by the slanted roof mounting system (aluminium consumption) and the electric installation (copper consumption). The operation stage causes a negligible impact of less than 1 %. The end-of-life stage causes a share of 4 % on the overall PM impacts, mainly caused by the transportation of the panels with lorries, but also due to the electricity consumption required for

panel recycling. The potential benefits from recycling sum up to -17 % of the overall PM impacts and are associated with the savings when recycling aluminium and copper.

The impact category **ionizing radiation HH** is mainly influenced by the product stage (59 %). It is dominated by the supply chains of the laminate production (electricity consumption, liquid nitrogen, solar and flat glass). The construction stage accounts for 33 %. Within the installation the slanted roof mounting system causes the largest impact (aluminium consumption). The contribution of the use stage to the total impacts is negligible (<1 %). The end-of-life accounts for 8 % of the overall impact with the electricity consumption for the recycling process being the driving force. Potential benefits due to recycling contribute to -12 % relative to the overall impact and are due to the recycling of aluminium.

The product stage is the most dominant life cycle stage regarding the **photochemical ozone formation potential** (65 %). On the one hand, the solar glass and flat glass production causes a large impact. On the other hand, the electricity consumption (and therein the use of fossil fuels) when producing CIS laminates is important. The construction stage causes the second largest impact (27 %) and is dominated by the slanted roof mounting system (aluminium production and electricity consumption in its supplied chain). The contribution of the use stage to the total impacts is negligible (<1 %). The end-of-life accounts for 8 % of the overall impact with the transportation by lorry being the most important process. The potential benefits due to recycling amount to -8 % and are owed to the recycling of aluminium and copper.

The **acidification potential** is likewise dominated by the product stage (60 %). In the supply chain of CIS PV panels, the production of flat glass, transoceanic transports and electricity generated in fossil power plants are responsible most of the acidification impacts. The construction stage causes 36 % of the overall acidification impacts, which is due to the slanted roof mounting system as well as the electric installation that contribute with the use of aluminium and copper, respectively. The contribution of the use stage to the total impacts is negligible (<1 %). The end-of-life stage accounts for 5 % of the overall impact where the consumption of fossil fuels in the transport supply chain and the electricity supply chain are most relevant. The benefits for recycling contribute with -13 % to the acidification potential. The savings in avoiding primary aluminium and primary copper production are the main drivers.

The impact category **terrestrial eutrophication potential** is dominated by the product stage (67 %). The emissions of nitrogen oxides into the air cause the main impact. The most relevant processes with regard to the terrestrial eutrophication potential are transports by transoceanic freight ship, the production of flat and solar glass, the supply chain of electricity production. The construction stage contributes 25 % to the overall terrestrial eutrophication potential. The slanted roof mounting system and the electric installation (transport and electricity consumption within the aluminium supply chain and blasting in the copper supply chain) cause the main impact. The contribution of the use stage to the total impacts is negligible (<1 %). The end-of-life stage causes a share of 7 % and is dominated by the nitrogen oxide emissions into air from road transport and electricity production. The potential benefits from recycling sum up to -9 % and are due to the savings in avoiding primary aluminium and copper production as well as in avoiding soda powder production replaced by recycled glass cullets.

A share of 61 % of the **freshwater eutrophication potential** impact comes from the construction stage, especially from the electric installation. Phosphate emitted into groundwater bodies is the most relevant substance in terms of freshwater eutrophication. The electric installation (disposal of sulfidic tailings from copper-beneficiation) and, to a lesser extent, also the supply chain of the slanted roof

mounting system (aluminium consumption) cause the main emissions of phosphate into water bodies. The product stage contributes a share of 37 % which comes from consumption of fossil energy carriers (lignite) in the supply chain of electricity in the laminate production process. The contribution of the use stage to the total impacts is negligible (<1 %). The impacts due to takeback and recycling at the end-of-life (2 %) are caused by the consumption of fossil fuels (lignite and hard coal) in the supply chain of electricity production. Potential benefits from recycling (-16 %) are owed to avoiding primary copper and aluminium production and the savings in emissions in the copper and aluminium supply chains.

The **marine eutrophication potential** is primarily determined by the emissions of nitrogen oxides into air. The same explanations apply as for the terrestrial eutrophication potential. The product stage is most important with a share of 66 % in the total impacts, followed by the construction stage (25 %) and the end-of-life stage (9 %). The share of the use stage is lower than 1 %. The potential benefits due to recycling amount to -8 %.

The product stage has a share of 13 % in the impact category **freshwater ecotoxicity potential**. Emissions of heavy metals water are the main drivers for the freshwater ecotoxicity impacts. Indium rich leaching residues in the laminate supply chain lead to chlorobenzene, antimony, and chromium IV emissions into water as well as to zinc emissions into air. In addition, the production of polyphenylene sulphide and the supply chain of electricity contribute to the freshwater ecotoxicity potential in the product stage. The construction stage contributes 85 % to the impacts and is dominated by the emissions of antimony into water when disposing of industrial electronics (from the production of the electric installation) in the municipal waste incineration plant. The contribution of the use stage to the total impacts is negligible (<1 %). The end-of-life stage accounts for 2 % of the overall impacts. The largest potential benefits due to recycling (-5 %) are gained in the savings of primary aluminium production (savings in redmud disposal) and primary copper production (disposal of sulphidic tailings).

Land use impacts are mainly caused by the production and the construction stages (74 % and 22 %, respectively). The production facilities (laminate factory), the wells for exploration and production of natural gas and oil as well as the supply chain of metals (bauxite mine) contribute most to the total land use impacts. The contribution of the use stage to the total impacts is negligible (<1 %). The land use of the end-of-life stage (4 %) is dominated by the land use for electricity production. Potential benefits from recycling amount to around -7 % and are owed to avoiding primary aluminium production.

The impact category **water resource depletion** is mainly caused by the product stage (49 %) and by the construction stage (47 %). The end-of-life stage causes 3 % of the impacts and the use stage has a share of less than 1 %. Within the three relevant life cycle stages, the supply chain of electricity consumed in the production and processing metals (and therein the water evaporated from hydroelectric power systems) is the main source. Potential benefits due to recycling (-21 %) are mainly owed to the savings in replacing primary aluminium with secondary aluminium.

Mineral, fossil and renewable resource depletion is fully dominated by the product stage, namely by the consumption of indium and its supply chain. All other life cycle stages as well as the potential benefits for recycling are negligible (<1 %).

The **cumulative energy demand non-renewable** (CED nr) is on one hand caused by the product stage (65 %, electricity consumption based on fossil fuels when manufacturing the CIS PV panels)

and on the other hand by the slanted roof mounting system (29 %, electricity consumption in aluminium supply chain). The same explanation about electricity applies for the end-of-life stage, which accounts for 6 % to the overall CED nr. The contribution of the use stage to the total impacts is negligible (<1 %). The potential benefits of recycling (-10 %) are credited for the savings in electricity consumption when replacing primary aluminium with secondary aluminium.

The indicator **cumulative energy demand renewable** (CED r) is the only indicator for which the major contribution is in the operation stage (98 %). It mainly reflects the harvested amount of solar energy needed to produce electricity with the PV panel. The product and the construction stage contribute approximately 1 % each, whereas the end-of-life stage and the potential benefits for recycling are negligible.

The impact category **nuclear waste** is mainly influenced by the product stage and within by the direct electricity consumption and the electricity consumption in the supply chain of liquid nitrogen, sourcing from nuclear power. This stage accounts for 62 %. The construction stage contributes 30 % to the overall impact. It is dominated by the mounting structure (electricity consumption in the supply chain of aluminium production). The end-of-life stage accounts for 7 % of the overall impact. The electricity supply chain is the driving process. Potential benefits due to recycling contribute -11 % to the total impact and are due to the savings in electricity consumption in the aluminium production when saving primary aluminium.

5.6.3 *Results normalized*

The normalized environmental impact results of the electricity production are listed in Tab. 5.9. The three additional impact categories cumulative energy demand non-renewable, cumulative energy demand renewable and nuclear waste are not subject to normalization.

The production of 1 kWh of DC electricity contributes the largest share to the mineral, fossil and renewable resource depletion potential of an average European citizen. Producing 1 kWh electricity generated with solar power from CIS PV panels causes 0.5 % of the resource depletion potential per person in Europe. An average European person consumes around 5'000 kWh of electricity. Assuming it would all be produced with a residential CIS PV system, the electricity production would exceed the present personal annual budget of resource depletion by 2'500 %.

The second and third largest contributions to the per capita impacts are observed in the impact categories human toxicity, cancer effects, and human toxicity, non-cancer effects.

Tab. 5.9 Environmental impact results (normalized) of 1 kWh of DC electricity produced with a residential scale CIS PV system mounted on a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, mounted with CIS PV technology, normalized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change		2.86E-06	1.38E-06	4.84E-10	2.76E-07	4.51E-06	-5.40E-07	3.97E-06
Ozone depletion		6.29E-08	1.67E-08	1.27E-11	4.00E-09	8.37E-08	-4.99E-09	7.87E-08
Human toxicity, cancer effects		8.52E-06	2.51E-05	1.84E-08	1.62E-06	3.53E-05	-8.06E-06	2.72E-05
Human toxicity, non-cancer effects		7.63E-06	9.22E-06	1.98E-08	3.36E-07	1.72E-05	-5.00E-08	1.72E-05
Particulate matter		2.49E-06	2.14E-06	6.44E-10	2.04E-07	4.83E-06	-8.24E-07	4.00E-06
Ionizing radiation HH		1.86E-06	1.02E-06	7.33E-10	2.51E-07	3.13E-06	-3.67E-07	2.76E-06
Photochemical ozone formation		2.49E-06	1.05E-06	5.09E-10	2.92E-07	3.84E-06	-3.24E-07	3.51E-06
Acidification		2.96E-06	1.77E-06	7.10E-10	2.40E-07	4.97E-06	-6.28E-07	4.34E-06
Terrestrial eutrophication		1.70E-06	6.43E-07	4.99E-10	1.90E-07	2.53E-06	-2.20E-07	2.31E-06
Freshwater eutrophication		1.66E-06	2.75E-06	4.00E-09	9.42E-08	4.51E-06	-7.05E-07	3.80E-06
Marine eutrophication		1.51E-06	5.68E-07	7.69E-09	1.95E-07	2.28E-06	-1.81E-07	2.10E-06
Freshwater ecotoxicity		1.37E-06	9.01E-06	3.01E-09	2.27E-07	1.06E-05	-4.90E-07	1.01E-05
Land use		5.62E-07	1.69E-07	2.46E-10	3.13E-08	7.63E-07	-5.24E-08	7.10E-07
Water resource depletion		2.63E-07	2.54E-07	1.46E-09	1.80E-08	5.37E-07	-1.14E-07	4.23E-07
Mineral, fossil & ren resource depletion		5.06E-03	5.34E-06	-4.63E-09	1.03E-07	5.07E-03	-8.90E-08	5.07E-03

5.6.4 Results weighted

Tab. 5.10 shows the weighted environmental impact results (equal weighting) of the electricity production with CIS PV panels. The blue bars within column “Total” illustrate the importance of each impact category in the total of impacts. Fig. 5.22 illustrates the weighted impacts. The three additional impact categories cumulative energy demand non-renewable, cumulative energy demand renewable and nuclear waste are not included in the weighted results.

The environmental impacts of the production of 1 kWh electricity with a residential scale CIS PV system mounted on a slanted roof, are fully driven by mineral, fossil and renewable resource depletion (98 %). All other impact categories have minor shares of less than 1 %. This high impact is because of the resource consumption of indium in the panel production (high characterization factor for indium and high normalization factor for this category).

Tab. 5.10 Environmental impact results (weighted) of 1 kWh of DC electricity produced with a 3 kWp power system with CIS PV panels mounted on a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, mounted with CIS PV technology, weighted		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Total	Pt	3.40E-04	4.03E-06	3.57E-09	2.72E-07	3.44E-04	-8.43E-07	3.43E-04
Climate change	Pt	1.90E-07	9.19E-08	3.23E-11	1.84E-08	3.01E-07	-3.60E-08	2.65E-07
Ozone depletion	Pt	4.20E-09	1.12E-09	8.50E-13	2.67E-10	5.58E-09	-3.33E-10	5.25E-09
Human toxicity, cancer effects	Pt	5.68E-07	1.67E-06	1.23E-09	1.08E-07	2.35E-06	-5.37E-07	1.81E-06
Human toxicity, non-cancer effects	Pt	5.09E-07	6.15E-07	1.32E-09	2.24E-08	1.15E-06	-3.33E-09	1.14E-06
Particulate matter	Pt	1.66E-07	1.42E-07	4.30E-11	1.36E-08	3.22E-07	-5.50E-08	2.67E-07
Ionizing radiation HH	Pt	1.24E-07	6.77E-08	4.88E-11	1.67E-08	2.08E-07	-2.45E-08	1.84E-07
Photochemical ozone formation	Pt	1.66E-07	7.00E-08	3.40E-11	1.95E-08	2.56E-07	-2.16E-08	2.34E-07
Acidification	Pt	1.97E-07	1.18E-07	4.74E-11	1.60E-08	3.31E-07	-4.19E-08	2.89E-07
Terrestrial eutrophication	Pt	1.13E-07	4.29E-08	3.33E-11	1.26E-08	1.69E-07	-1.46E-08	1.54E-07
Freshwater eutrophication	Pt	1.11E-07	1.83E-07	2.67E-10	6.28E-09	3.00E-07	-4.70E-08	2.53E-07
Marine eutrophication	Pt	1.01E-07	3.78E-08	5.13E-10	1.30E-08	1.52E-07	-1.21E-08	1.40E-07
Freshwater ecotoxicity	Pt	9.11E-08	6.00E-07	2.01E-10	1.51E-08	7.07E-07	-3.27E-08	6.74E-07
Land use	Pt	3.75E-08	1.13E-08	1.64E-11	2.09E-09	5.09E-08	-3.50E-09	4.74E-08
Water resource depletion	Pt	1.75E-08	1.70E-08	9.71E-11	1.20E-09	3.58E-08	-7.57E-09	2.82E-08
Mineral, fossil & ren resource depletion	Pt	3.38E-04	3.56E-07	-3.15E-10	6.88E-09	3.38E-04	-5.94E-09	3.38E-04

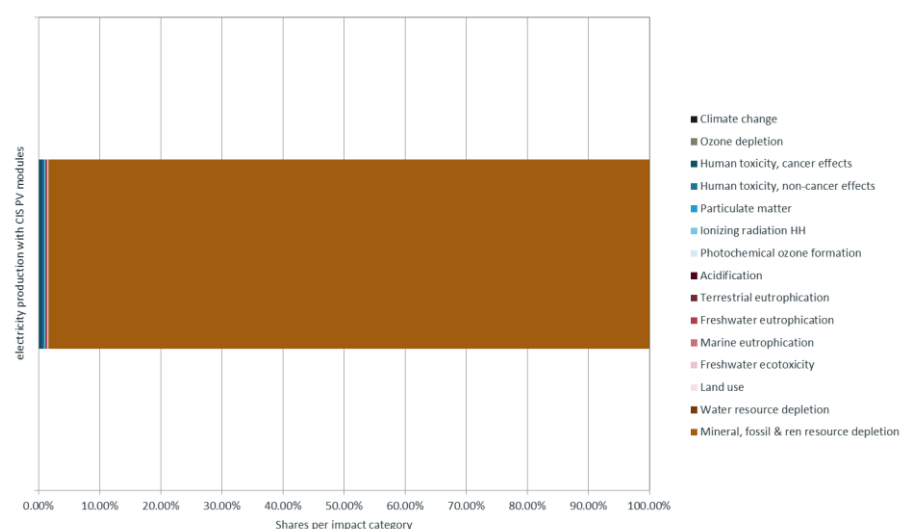


Fig. 5.22 Share per impact category on the weighted result for the production of 1 kWh of DC electricity produced with a residential scale CIS PV system mounted on a slanted roof.

5.6.5 Comparison of the three system sizes integrated, mounted and open ground

The comparison of the environmental impacts of the different systems shows only small differences (see Fig. 5.23). The environmental impacts of the open ground system are 4 % higher compared to the residential systems. The product stage is dominating the results. The main impact in the product stage is caused by the indium resource consumption. Differences occurring in the construction stage are hardly noticeable. The use stage, the end-of-life stage as well as potential benefits for recycling contribute negligible shares to the overall impacts. The construction stage is only relevant for the open ground CIS PV system.

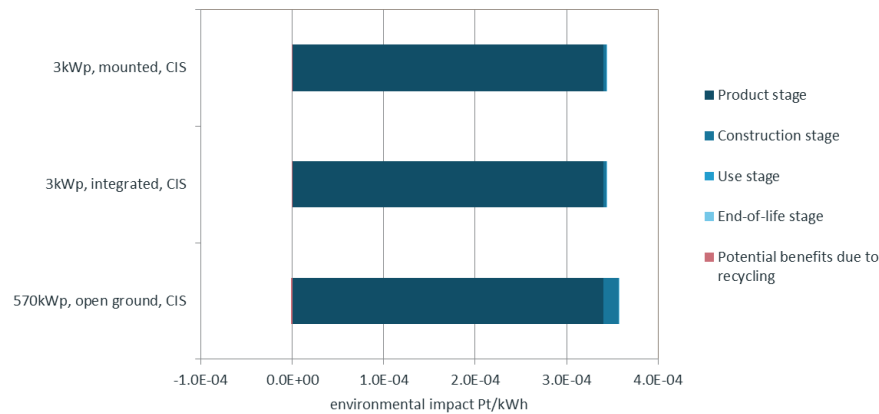


Fig. 5.23 Environmental impact results (weighted) of 1 kWh of DC electricity produced with a residential scale (3 kWp) PV system with CIS PV panels mounted on and integrated in a slanted roof as well as produced with a large scale open ground system. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.

5.7 Results micromorphous-Si PV technology

5.7.1 Overview

In this subchapter the results of electricity production with a residential scale micromorphous-Si PV system, mounted on a slanted roof, prior to normalization (section 5.7.2) as well as the normalized results (section 5.7.3) and the weighted results (section 5.7.4) are described.

5.7.2 Results prior to normalization

The results prior to normalization of the production of 1 kWh of electricity with micromorphous-Si PV panels mounted on a slanted roof in a 3 kWp installation system are summarized in Tab. 5.11. The life cycle stages analyzed are the product stage, the construction stage, the operation stage and the end-of-life stage. Net benefits caused by end-of-life recycling are allocated to the production of electricity with a share of 50 %. The other half is allocated to the life cycle of products potentially using the secondary products (see also subchapter 3.5).

Tab. 5.11 Environmental impact results (characterized) of 1 kWh of DC electricity produced with a residential scale micro-Si PV system mounted on a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, mounted with micro-Si PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change	kg CO ₂ eq	4.09E-02	1.36E-02	4.81E-06	2.56E-03	5.71E-02	-5.59E-03	5.15E-02
Ozone depletion	kg CFC-11 eq	7.50E-10	3.89E-10	2.98E-13	8.70E-11	1.23E-09	-1.21E-10	1.11E-09
Human toxicity, cancer effects	CTUh, c	4.00E-10	9.96E-10	7.34E-13	6.01E-11	1.46E-09	-3.34E-10	1.12E-09
Human toxicity, non-cancer effects	CTUh, n-c	4.63E-09	5.08E-09	1.14E-11	1.80E-10	9.90E-09	1.79E-11	9.91E-09
Particulate matter	kg PM _{2.5} eq	5.00E-05	8.59E-06	2.61E-09	7.79E-07	5.93E-05	-3.47E-06	5.59E-05
Ionizing radiation HH	kBq U235 eq	1.20E-03	1.24E-03	8.95E-07	2.85E-04	2.72E-03	-4.68E-04	2.25E-03
Photochemical ozone formation	kg NMVOC eq	1.67E-04	3.53E-05	1.74E-08	9.32E-06	2.11E-04	-1.14E-05	2.00E-04
Acidification	mol H ⁺ eq	3.62E-04	8.81E-05	3.61E-08	1.14E-05	4.61E-04	-3.26E-05	4.29E-04
Terrestrial eutrophication	mol N eq	6.15E-04	1.20E-04	9.46E-08	3.36E-05	7.69E-04	-4.28E-05	7.26E-04
Freshwater eutrophication	kg P eq	1.84E-06	4.11E-06	6.39E-09	1.40E-07	6.10E-06	-1.06E-06	5.04E-06
Marine eutrophication	kg N eq	5.48E-05	1.02E-05	1.40E-07	3.32E-06	6.84E-05	-3.40E-06	6.50E-05
Freshwater ecotoxicity	CTUe	1.39E-02	7.97E-02	2.84E-05	1.99E-03	9.56E-02	-4.71E-03	9.09E-02
Land use	kg C deficit	3.51E-02	1.35E-02	1.98E-05	2.36E-03	5.10E-02	-4.36E-03	4.67E-02
Water resource depletion	m ³ water eq	6.33E-05	2.23E-05	1.28E-07	1.47E-06	8.73E-05	-1.04E-05	7.68E-05
Mineral, fossil & ren resource depletion	kg Sb eq	9.53E-07	5.50E-07	4.77E-11	1.05E-08	1.51E-06	2.33E-09	1.52E-06
Cumulative energy demand non renewable	MJ oil eq	4.31E-01	1.95E-01	7.39E-05	3.64E-02	6.62E-01	-6.97E-02	5.92E-01
Cumulative energy demand renewable	MJ oil eq	3.65E-02	3.95E-02	3.60E+00	2.38E-03	3.68E+00	-1.73E-02	3.66E+00
Nuclear waste	m ³ HAA eq	2.67E-11	2.72E-11	1.66E-14	5.95E-12	5.99E-11	-1.05E-11	4.94E-11

The following Fig. 5.24 illustrates the shares of environmental impacts per life cycle stage relative to the overall results per impact category, from production to end-of-life stage. The potential benefits due to recycling are also expressed relative to the overall results. It becomes evident, that the product stage and the construction stage are most important for most of the impact indicators. With a few exceptions the operation stage and the end-of-life stage are much less important.

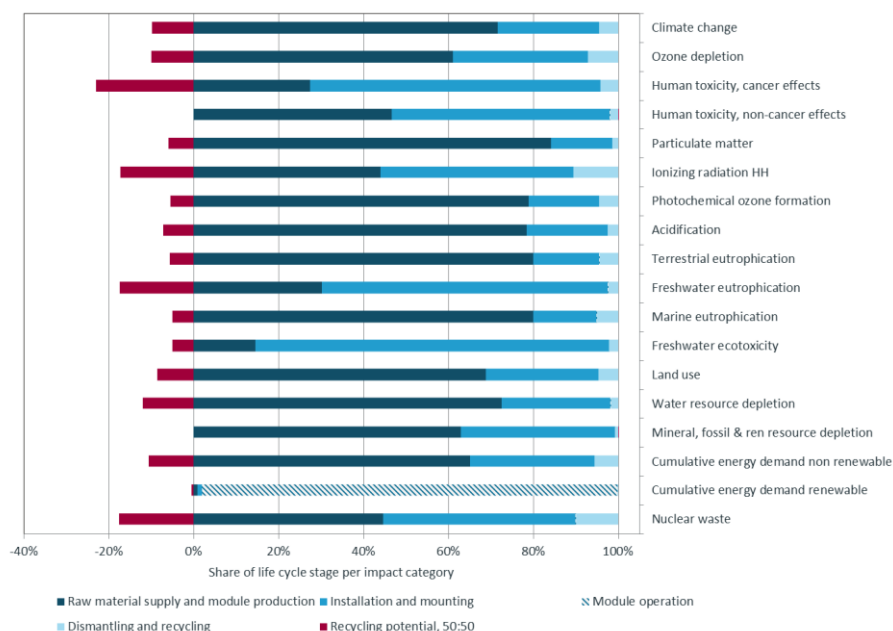


Fig. 5.24 Environmental impact results (characterized, indexed to 100 %) of 1 kWh of DC electricity produced with a residential scale micro-Si PV system mounted on a slanted roof. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.

Climate change of the electricity production with micromorphous-Si PV panels is mainly influenced by the product stage. It contributes 72 % to the overall impact. The construction stage is of lower importance but contributes still with 24 %. Module operation has a negligible effect on climate

change, whereas the end-of-life has a share of 4 % to the impacts. Within the product stage the production of the panel and the electricity consumption therein causes the main impact (fossil fuels, i.e. hard coal consumption within the supply chain of electricity). The construction stage is dominated by the slanted-roof construction and therein by the electricity production from fossil fuels in the supply chain of aluminium and by direct emissions in the aluminium electrolysis. The end-of-life stage is likewise dominated by the electricity supply chain and the consumption of fossil fuels for electricity production. In the same way the benefits for recycling (-10 %) are based on savings in fossil fuel consumption for electricity in the supply chain of aluminium.

The impact category **ozone depletion potential** has more than half of its impact in the product stage (61 %). Its largest impact stems from the production of the PV panel and the supply chains of solar glass and aluminium (for framing) and namely from halone emissions (bromochlorodifluoromethane, halone 1211) in the long distance pipeline transport of natural gas for electricity production in the supply chain of glass, aluminium and the direct electricity consumption. The construction stage accounts for 32 % of the impacts and is dominated likewise by the electricity consumption in the supply chain of aluminium. The use stage has a negligible share in the total impacts (<1 %). The end-of-life stage has a share of 7 % on the overall impacts. Its main impact comes from the electricity consumption (and therein the emissions of pipeline transport of natural gas) for the recycling process and for the production of steam in the supply chain of hydrogen peroxide. The benefits for recycling (-10 %) are likewise gained by the savings in electricity consumption when avoiding primary aluminium production.

The product stage has an impact of 27 % to **human toxicity cancer effects potential**. Within the panel production the aluminium consumption and aluminium supply chain (chromium IV emissions into water from redmud disposal in residual material landfills from bauxite digestion) cause the impact. The construction stage has the main impact (68 %). This stage is dominated by the supply chains of aluminium and steel. For steel the disposal of slag to residual material landfills leads to chromium IV emissions to water. The use stage causes a negligible share in the total impacts (<1 %). The end-of-life stage has only a minor impact (4 %) and is dominated by the hydrogen peroxide supply chain (chromium IV emissions from anthraquinone production). The benefits for recycling are owed to the savings in aluminium and the emissions in its supply chain (emissions from redmud disposal) and sum up to -23 %.

The share of the product stage on the impact category **human toxicity non cancer effects** is 47 %. The product stage is dominated by the supply chain electricity demand for production, in which heavy metals (arsenic, zinc, mercury and others) are released into water bodies and air. The construction stage has a share of 51 % in the total impacts. Its main impact comes from the electric installation (heavy metal release in the supply chain of copper) and the mounting structure (heavy metal emissions in the supply chain of aluminium). With 2 % the end-of-life stage has only a minor impact and the share of the operation stage is negligible. The potential benefits for recycling are slightly positive (<1 %), which is due to the fact that the impacts for producing secondary aluminium from aluminium scrap cannot outweigh the benefits for savings in primary aluminium production. More specifically, the zinc emissions in the supply chain of secondary aluminium production from scrap aluminium (from frame and mounting structure) compensate for the emissions saved by recycling material.

The impacts of **particulate matter formation** are dominated by the product stage and amount to 84 %. The main impact comes from the supply chain of the electricity consumption (direct electricity use as well as electricity consumption in the supply chains of solar and flat glass production and

aluminium production) in which the burning of hard coal causes PM2.5 emissions. The construction stage causes 14 % of the impacts and is likewise brought forth by micro fine dust emissions in the supply chain of aluminium production (direct emissions during electrolysis and within the supply chain of electricity). The use stage has a negligible share in the total impacts (<1 %). The end-of-life stage causes 1 % of the impacts and likewise the electricity consumption is the dominating factor. The benefits for recycling sum up to -6 % and are owed to the savings primary aluminium production and its electricity consumption.

The impact category **ionizing radiation HH** is dominated by the product stage and the construction stage (contributing 44 % and 45 %, respectively). In the end-of-life stage, 10 % of the ionizing radiation HH impacts are caused. The main contributor is the electricity consumption, where carbon-14 is released into air when spent nuclear fuel is reprocessed. The use stage has a negligible share in the total impacts (<1 %). Potential benefits for recycling are -17 % owed to the savings in electricity consumption when avoiding primary aluminium production (savings in carbon-14 emissions when reprocessing spent nuclear fuel).

Within the impact category **photochemical ozone formation** the product stage is the most important one (79 %). The impact is caused by nitrogen oxides released into air when burning hard coal for electricity production. The same applies for the construction stage (17 %), where besides the direct electricity used for production, electricity is consumed in the supply chains of aluminium and copper production. Furthermore the transport (lorry and freight ship within the supply chains of the materials used) causes emissions of nitrogen oxides. The end-of-life stage (4 %) is likewise dominated by the emissions in the supply chain of lorry transport and electricity. The use stage has a negligible share in the total impacts (<1 %). The potential benefits for recycling are -5 % and are owed to avoided primary aluminium production.

The **acidification potential** has its main impact from the product stage (78 %), in which sulfur dioxide emissions are released to air when burning hard coal for electricity production. The same explanation applies for the construction stage (19 %) where electricity is used in the supply chain of aluminium. Sulfur dioxide is further released directly in the copper refining process. Furthermore transports (transoceanic freight ships and lorries) cause sulfur dioxide emissions as well. The use stage has a negligible share in the total impacts (<1 %). The end-of-life stage contributes 2 % to the impacts. The transport as well as the electricity consumption and its supply chain are the driving forces. The potential benefits of recycling amount to -7 % and are due to savings in electricity consumption from hard coal and lignite when avoiding primary aluminium production.

Within the impact category **terrestrial eutrophication potential** the emission of nitrogen oxides into air is the driving force. The product stage (80 %) induces nitrogen oxides in the process of burning hard coal for electricity production. Also the production of flat glass and transports by transoceanic freight ships and lorries contribute significant shares to the terrestrial eutrophication potential. Within the construction stage (16 %) the supply chain of aluminium and copper mining (blasting) are most relevant. The use stage has a negligible share in the total impacts (<1 %). The end-of-life stage contributes to 4 % and is dominated by the lorry transport, where likewise nitrogen oxides are emitted in the operation. Potential benefits of recycling (-6 %) are gained from savings in electricity consumption and from reduced transport when avoiding primary aluminium production.

The **freshwater eutrophication potential** is dominated by the emissions of phosphate into water bodies from off-site disposal of sulfidic tailings when mining copper. Within the product stage copper is used in the cables in the panel. The share of the product stage on the total impact is 30 %. The

construction stage (67 %) is likewise dominated by the supply chain of copper from the electric installation. The use stage has a negligible share in the total impacts (<1 %). The end-of-life stage causes 2 % of the impacts and its main impact comes from the disposal of spoil from lignite mining within the supply chain of electricity production. The benefits for recycling have an impact of -17 % and are due to the savings in primary copper production and its mining emissions.

The **marine eutrophication potential** is primarily determined by the emissions of nitrogen oxides into air. The same explanations apply as for the terrestrial eutrophication potential. The product stage dominates this category with a share of 80 % in the total marine eutrophication impacts. The construction stage contributes 15 %, whereas the use and the end-of-life stage are much less important (<1 % and 5 %, respectively). The potential benefits for recycling amount to -5 %.

Emissions of heavy metals into water and air are the main drivers for the **freshwater ecotoxicity** impacts. These are emitted with a share of 15 % in the product stage and to the major share in the construction stage (83 %). Antimony, chromium VI and arsenic are the most relevant heavy metals emitted. Within the product stage, the electricity production in hard coal power plants and the aluminium frame (disposal of redmud from bauxite digestion) are the driving forces. The construction stage has its main impact when disposing of plastics from electric equipment and from the supply chain of the electric installation in municipal incineration plants. The impacts from the aluminium supply chain, used within the slanted roof mounting system, are noteworthy but not as important as the copper refining process. The use stage has a negligible share in the total impacts (<1 %). The end-of-life stage accounts for 2 % of the overall impacts. The impacts are caused by the disposal of plastics from electric equipment (cabling) and by the supply chain of chemical solutions used to dissolve the PV laminates. The largest potential benefits due to recycling (-5 %) are gained in avoiding primary aluminium production.

Land use is dominated by the mines and power plants used in the supply chains of non-renewable and renewable energy carriers. This applies for all life cycle stages. The product and construction stages contribute 69 % and 26 % to the total land use impacts, respectively. The share of the end-of-life stage (5 %) is less important and the share of the operation stage is negligible (<1 %). Potential benefits from recycling amount to -9 %.

The impact category **water resource depletion** has its main impact from the product stage (73 %). The electricity generation with hydro power is the driving force within the supply chain of aluminium but also for the direct electricity consumption. Water evaporating from the hydro power water reservoir causes the impacts on water resource depletion. The construction stage (26 %) and the end-of-life stage (2 %) are likewise dominated by the consumption of electricity and therein the share of hydropower. The use stage has a negligible share in the total impacts (<1 %). Benefits for recycling are owed to savings in electricity in the supply chain of avoided primary aluminium production (-12 %).

The product stage has a share of 63 % on the impact category **mineral, fossil and renewable resource depletion**. The production of silver and fluorspar used for nitrogen trifluoride production has the main impact. Within the construction stage (36 %) the consumption of copper from the electric installation causes the main impact. The use and the end-of-life stage have negligible shares of less than 1 % on the overall impacts. Potential benefits for recycling are slightly positive. Impacts for producing secondary aluminium from aluminium scrap compensate the benefits for savings in primary aluminium production. This is mainly due to the fact that primary zinc is added as an alloying element in secondary aluminium production (from old scrap).

The **cumulative energy demand non-renewable** (CED nr) is dominated by the electricity (electricity production from fossil and nuclear fuels) and the transport supply chains (diesel and heavy fuel oil production). The product stage has a share of 65 % whereas the construction stage causes 29 % of the impacts. The end-of-life stage has a share of 6 % while potential benefits for recycling (-11 %) are gained from savings in primary aluminium production and its electricity consumption.

The indicator **cumulative energy demand renewable** (CED r) is the only indicator for which the major contribution (98 %) is in the operation stage. It reflects the harvested amount of solar energy needed to produce electricity with the PV panel. The product and the construction stage contribute approximately 1 % each, whereas the end-of-life stage and the potential benefits for recycling are negligible.

The impact category **nuclear waste** is dominated by the electricity supply chain and the electricity produced from nuclear power, which influences all life cycle stages. The product and the construction stage both contribute 45 % to the total impacts and the end-of-life stage causes 10 %. The nuclear waste generated by the use stage is negligible compared to the total of all life cycle stages. The potential benefits for recycling have a share of 17 %.

5.7.3 *Results normalized*

Tab. 5.12 shows the normalized results of the environmental impact of the production of 1 kWh electricity with micromorphous-Si PV panels on a slanted roof, 3 kWp residential installation. The three additional impact categories cumulative energy demand non-renewable, cumulative energy demand renewable and nuclear waste are not subject to normalization.

The impact category human toxicity, cancer effects has the largest impact per European person. Producing 1 kWh electricity generated with solar power from micromorphous-Si PV panels causes 0.03 % of the human toxicity potential per person in Europe. An average European person consumes around 5'000 kWh of electricity. Assuming it would all be produced with such a residential scale micromorphous-Si PV system, the electricity production alone would cause 15 % of the present budget of the human toxicity potential. The second largest impact per person is caused by the impact category human toxicity, non-cancer effects, followed by the depletion of mineral, fossil and renewable resources.

Tab. 5.12 Environmental impact results (normalized) of 1 kWh of DC electricity produced with a 3 kWp power system with micro-Si PV panels mounted on a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, mounted with micro-Si PV technology, normalized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change		4.44E-06	1.48E-06	5.21E-10	2.78E-07	6.19E-06	-6.06E-07	5.59E-06
Ozone depletion		3.47E-08	1.80E-08	1.38E-11	4.03E-09	5.68E-08	-5.60E-09	5.12E-08
Human toxicity, cancer effects		1.08E-05	2.70E-05	1.99E-08	1.63E-06	3.95E-05	-9.06E-06	3.04E-05
Human toxicity, non-cancer effects		8.68E-06	9.53E-06	2.14E-08	3.38E-07	1.86E-05	3.36E-08	1.86E-05
Particulate matter		1.31E-05	2.26E-06	6.85E-10	2.05E-07	1.56E-05	-9.12E-07	1.47E-05
Ionizing radiation HH		1.06E-06	1.09E-06	7.92E-10	2.52E-07	2.41E-06	-4.14E-07	1.99E-06
Photochemical ozone formation		5.26E-06	1.11E-06	5.48E-10	2.94E-07	6.67E-06	-3.59E-07	6.31E-06
Acidification		7.65E-06	1.86E-06	7.63E-10	2.42E-07	9.75E-06	-6.89E-07	9.06E-06
Terrestrial eutrophication		3.49E-06	6.81E-07	5.38E-10	1.91E-07	4.37E-06	-2.43E-07	4.12E-06
Freshwater eutrophication		1.25E-06	2.77E-06	4.32E-09	9.49E-08	4.12E-06	-7.15E-07	3.40E-06
Marine eutrophication		3.24E-06	6.03E-07	8.31E-09	1.96E-07	4.05E-06	-2.01E-07	3.85E-06
Freshwater ecotoxicity		1.59E-06	9.12E-06	3.25E-09	2.28E-07	1.09E-05	-5.39E-07	1.04E-05
Land use		4.70E-07	1.80E-07	2.65E-10	3.15E-08	6.82E-07	-5.83E-08	6.24E-07
Water resource depletion		7.78E-07	2.74E-07	1.57E-09	1.81E-08	1.07E-06	-1.28E-07	9.44E-07
Mineral, fossil & ren resource depletion		9.44E-06	5.44E-06	4.73E-10	1.04E-07	1.50E-05	2.30E-08	1.50E-05

5.7.4 Results weighted

In Tab. 5.13 the weighted environmental impact results (equal weighting) of the electricity production with micromorphous-Si PV panels are listed. The blue bars within column “Total” illustrate the importance of each impact category in the total of impacts. Fig. 5.25 illustrates the weighted shares per impact category in more detail. The three additional impact categories cumulative energy demand non-renewable, cumulative energy demand renewable and nuclear waste are not included in the weighted results.

The weighted environmental impacts of the production of 1 kWh electricity with a 3 kWp power system with micromorphous-Si PV panels mounted on a slanted roof are mainly driven by the human toxicity, cancer effects potential (24 %), human toxicity, non cancer effects (15 %), mineral, fossil and renewable resource depletion potential (12 %) and particulate matter (12 %).

Tab. 5.13 Environmental impact results (weighted) of 1 kWh of DC electricity produced with a residential scale micro-Si PV system mounted on a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, mounted with micro-Si PV technology, weighted		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Total	Pt	4.76E-06	4.23E-06	4.22E-09	2.74E-07	9.26E-06	-9.25E-07	8.34E-06
Climate change	Pt	2.96E-07	9.84E-08	3.48E-11	1.85E-08	4.13E-07	-4.04E-08	3.72E-07
Ozone depletion	Pt	2.32E-09	1.20E-09	9.20E-13	2.69E-10	3.79E-09	-3.73E-10	3.41E-09
Human toxicity, cancer effects	Pt	7.22E-07	1.80E-06	1.33E-09	1.09E-07	2.63E-06	-6.04E-07	2.03E-06
Human toxicity, non-cancer effects	Pt	5.79E-07	6.35E-07	1.43E-09	2.25E-08	1.24E-06	2.24E-09	1.24E-06
Particulate matter	Pt	8.77E-07	1.51E-07	4.57E-11	1.37E-08	1.04E-06	-6.08E-08	9.80E-07
Ionizing radiation HH	Pt	7.07E-08	7.29E-08	5.28E-11	1.68E-08	1.60E-07	-2.76E-08	1.33E-07
Photochemical ozone formation	Pt	3.51E-07	7.41E-08	3.65E-11	1.96E-08	4.45E-07	-2.40E-08	4.21E-07
Acidification	Pt	5.10E-07	1.24E-07	5.09E-11	1.61E-08	6.50E-07	-4.59E-08	6.04E-07
Terrestrial eutrophication	Pt	2.33E-07	4.54E-08	3.58E-11	1.27E-08	2.91E-07	-1.62E-08	2.75E-07
Freshwater eutrophication	Pt	8.30E-08	1.85E-07	2.88E-10	6.32E-09	2.75E-07	-4.77E-08	2.27E-07
Marine eutrophication	Pt	2.16E-07	4.02E-08	5.54E-10	1.31E-08	2.70E-07	-1.34E-08	2.57E-07
Freshwater ecotoxicity	Pt	1.06E-07	6.08E-07	2.17E-10	1.52E-08	7.29E-07	-3.59E-08	6.93E-07
Land use	Pt	3.13E-08	1.20E-08	1.77E-11	2.10E-09	4.55E-08	-3.89E-09	4.16E-08
Water resource depletion	Pt	5.19E-08	1.83E-08	1.05E-10	1.21E-09	7.15E-08	-8.52E-09	6.29E-08
Mineral, fossil & ren resource depletion	Pt	6.29E-07	3.63E-07	3.15E-11	6.92E-09	9.99E-07	1.54E-09	1.00E-06

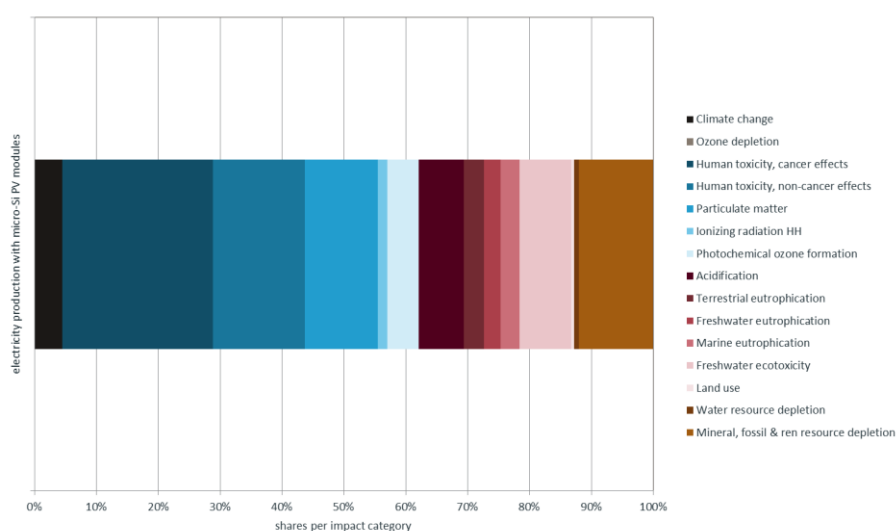


Fig. 5.25 Share per impact category on the weighted result for the production of 1 kWh of DC electricity produced with a residential scale micro-Si PV system mounted on a slanted roof.

5.7.5 Comparison of the three system sizes integrated, mounted and open ground

The comparison of the environmental impacts per life cycle stage of the three different systems shows that the residential scale systems do only differ slightly while the large scale system causes substantially higher impacts per kWh produced (see Fig. 5.26). The smallest impacts are caused by the PV system integrated in a residential roof. Slightly higher impacts are caused by the mounted system whereas the open ground system causes the highest impacts. Overall the impacts of the open ground system are about 165 % higher compared to the mounted residential system. The product stage, the use stage, end-of-life stage as well as the potential benefits for recycling are similarly important in all three systems. The main difference is caused by the construction stage and, more specifically, by the land use of the different PV systems. The land use impacts of the open ground micromorphous-Si PV system are much higher than of the systems installed on a slanted roof. This is

due to the fact that the open ground system directly occupies a certain land area while the land use of the residential PV systems is attributed to the building.

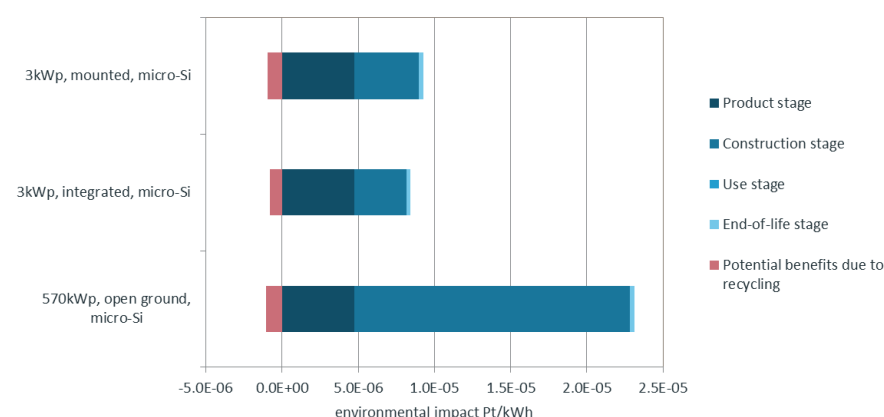


Fig. 5.26 Environmental impact results (weighted) of 1 kWh of DC electricity produced with a residential scale (3 kWp) PV system with micro-Si PV panels mounted on and integrated in a slanted roof as well as produced with a large scale open ground system. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.

5.8 Results multicrystalline-Si PV technology

5.8.1 Overview

In this subchapter the results of electricity production with multicrystalline PV panels are explained in detail. Section 5.8.2 shows the results prior to normalization while in section 5.8.3 the normalized results are presented. Section 5.8.4 contains the details of the weighted results.

5.8.2 Results prior to normalization

Tab. 5.14 shows the environmental impacts (prior to normalization) of the production of 1 kWh of DC electricity with a 3 kWp power system with multicrystalline PV panels, mounted on a slanted roof. The life cycle stages analyzed are the product stage, the construction stage, the operation stage and the end-of-life stage. Net benefits caused by end-of-life recycling are allocated to the production of electricity with a share of 50 %. The other half is allocated to the life cycle of products potentially using the secondary products (see also subchapter 3.5).

Tab. 5.14 Environmental impact results (characterized) of 1 kWh of DC electricity produced with a residential scale multi-Si PV system mounted on a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, mounted with multi-Si PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change	kg CO ₂ eq	4.52E-02	9.69E-03	3.25E-06	1.41E-03	5.63E-02	-3.43E-03	5.28E-02
Ozone depletion	kg CFC-11 eq	1.48E-09	2.70E-10	2.02E-13	4.78E-11	1.79E-09	-7.29E-11	1.72E-09
Human toxicity, cancer effects	CTUh, c	5.38E-10	6.93E-10	4.99E-13	3.30E-11	1.27E-09	-2.18E-10	1.05E-09
Human toxicity, non-cancer effects	CTUh, n-c	6.12E-09	4.37E-09	7.75E-12	9.89E-11	1.06E-08	-1.09E-10	1.05E-08
Particulate matter	kg PM _{2.5} eq	6.14E-05	6.53E-06	1.74E-09	4.28E-07	6.84E-05	-2.30E-06	6.61E-05
Ionizing radiation HH	kBq U235 eq	1.91E-03	8.53E-04	6.08E-07	1.57E-04	2.92E-03	-2.99E-04	2.62E-03
Photochemical ozone formation	kg NMVOC eq	1.66E-04	2.64E-05	1.17E-08	5.12E-06	1.97E-04	-7.56E-06	1.90E-04
Acidification	mol H ⁺ eq	4.05E-04	6.93E-05	2.44E-08	6.29E-06	4.81E-04	-2.26E-05	4.58E-04
Terrestrial eutrophication	mol N eq	5.97E-04	8.97E-05	6.42E-08	1.85E-05	7.05E-04	-2.78E-05	6.77E-04
Freshwater eutrophication	kg P eq	3.11E-06	3.94E-06	4.35E-09	7.71E-08	7.13E-06	-1.01E-06	6.12E-06
Marine eutrophication	kg N eq	5.51E-05	7.50E-06	9.55E-08	1.82E-06	6.46E-05	-2.23E-06	6.23E-05
Freshwater ecotoxicity	CTUe	2.42E-02	7.55E-02	1.93E-05	1.09E-03	1.01E-01	-3.30E-03	9.75E-02
Land use	kg C deficit	5.63E-02	9.70E-03	1.35E-05	1.29E-03	6.73E-02	-2.74E-03	6.46E-02
Water resource depletion	m ³ water eq	4.71E-05	1.53E-05	8.70E-08	8.09E-07	6.33E-05	-6.69E-06	5.66E-05
Mineral, fossil & ren resource depletion	kg Sb eq	1.88E-05	5.15E-07	1.49E-11	5.76E-09	1.93E-05	-3.22E-08	1.93E-05
Cumulative energy demand non renewable	MJ oil eq	5.46E-01	1.39E-01	5.00E-05	2.00E-02	7.05E-01	-4.35E-02	6.62E-01
Cumulative energy demand renewable	MJ oil eq	7.20E-02	2.72E-02	3.60E+00	1.31E-03	3.70E+00	-1.11E-02	3.69E+00
Nuclear waste	m ³ HAA eq	4.42E-11	1.88E-11	1.13E-14	3.27E-12	6.63E-11	-6.70E-12	5.96E-11

Fig. 5.27 shows the shares per environmental impact over all life cycle stages. The sum of the impacts from product stage to end-of-life stage is set to 100 %. The potential benefits due to recycling are expressed relative to this sum. The product stage and the construction stage are most important for most of the impact indicators. With a few exceptions the operation stage and the end-of-life stage are much less important.

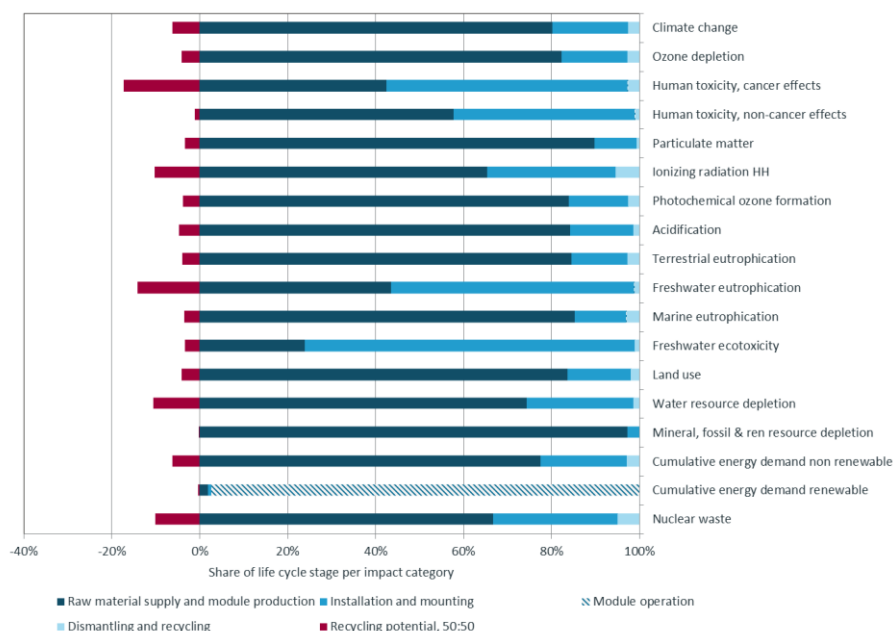


Fig. 5.27 Environmental impact results (characterized, indexed to 100 %) of 1 kWh of DC electricity produced with a residential scale multi-Si PV system mounted on a slanted roof. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.

The impact category **climate change** is mainly influenced by the product stage. It contributes to 80 % to the overall greenhouse gas emissions. The construction stage causes a share of 17 % of the overall impacts. The operation stage and the end-of-life stages cause minor climate change impacts (<1 %

and 3 %, respectively). Within the product stage the panel production, especially the silicon production (electricity consumption from hard coal), causes the major share of climate change impacts. The construction stage is dominated by the manufacture of the mounting system (electricity from fossil fuels used in the supply chain of aluminium and direct emissions during the electrolysis process). The potential benefits due to recycling at the end-of-life sum up to about -6 % of the overall greenhouse gas emissions and are mainly owed to the recycling of the aluminium and hence avoiding primary aluminium production.

The impact category **ozone depletion** has its main contribution from the product stage (82 %), driven by the supply chain of electricity production. Bromochlorodifluoromethane (halone 1211) is released when transporting natural gas in long distance pipelines. The most relevant processes in the product stage with regard to ozone depletion impacts is the production of silicon. In the construction stage the aluminium supply chain (likewise natural gas pipeline transport for electricity production) causes the dominant impact (15 %). The end-of-life stage causes 3 % of the impacts and is also driven by the electricity consumption and its supply chain. Benefits for recycling are as well given for savings of electricity consumption when avoiding primary aluminium production (-4 %).

The product stage contributes 43 % to the **human toxicity, cancer effects potential**. Chromium VI to water and chromium to air are the most important substances for this impact category. Within the product stage the emissions stem from the electricity supply chain (disposal of hard coal ash in residual material landfill). Electricity is mainly used in the silicon and wafer production. The construction stage causes 55 % of the impacts. The supply chains of aluminium and steel are the driving forces. Chromium IV is emitted to water by redmud disposal from bauxite digestion and by the disposal of steel slag, in residual material landfills. The use stage causes a negligible contribution to the total impacts (<1 %). The end-of-life stage has only a minor impact (3 %) and is dominated by the hydrogen peroxide supply chain (chromium IV emissions from anthraquinone production). The benefits for recycling are owed to the savings in aluminium and the emissions in its supply chain (emissions from redmud disposal) and sum up to -17 %.

Human toxicity, non cancer effects are dominated by the emission of arsenic to water and heavy metals (zinc, mercury, lead and others) to air. The product stage has a share of 58 % in the overall impacts. The heavy metals are released in the supply chain of electricity, which is used for the silicon and the wafer production. In the production of electricity by hard coal heavy metals are emitted directly, but the bigger part is caused when hard coal ash is disposed of in the residual landfill. The construction stage has a share of 41 % in the overall impacts where direct emissions are released during copper refining (electric installation) and in the disposal of redmud from bauxite digestion (mounting structure). The contributions of the operation and end-of-life stage as well as of the benefits for recycling are negligible (<1 %, 1 % and -1 %, respectively).

Ultrafine dust (PM_{2.5}) and sulfur dioxides emitted to air are the most important substances for the impact category **particulate matter formation potential**. These are released in the supply chain of electricity production (burning of hard coal). The product stage (90 %) and the construction stage (10 %) are concerned in the supply chain of silicon production, aluminium and copper processing. The contribution of the use stage is negligible (<1 %) and the end-of-life stage has a share of 1 % of the overall impact. Benefits for recycling (-3 %) are owed to savings in aluminium production.

Ionizing radiation HH is caused by the emission of carbon-14 in the reprocessing spent nuclear fuel in the supply chain of nuclear electricity production. Electricity is used in the supply chains of all product stages. The share of the product stage is 65 %, whereas the construction stage causes 29 %

and the end-of-life stage 5 % of the impacts. The use stage causes a negligible contribution to the total impacts (<1 %). Potential benefits for recycling (-10 %) are likewise gained from savings in electricity consumption.

Photochemical ozone formation and **acidification potential** are caused by the emission of sulfur dioxide and nitrogen oxides in air. The product stage has a share of 84 % in the overall impacts of both indicators. The airborne substances are emitted when burning hard coal in the supply chain of electricity production. In the construction stage (13 % and 14 % for photochemical ozone formation and acidification potential, respectively) impacts are caused in the supply chain of aluminium production (use of electricity produced from hard coal) and in transport processes from aluminium hydroxide in the aluminium supply chain. Blasting in the supply chain of copper used in the electric installation has a perceivable impact as well. The use stage causes a negligible contribution to the total impacts (<1 %). The end-of-life stage has a share of 3 % and 1 % (photochemical ozone formation and acidification potential, respectively) on the overall impact. The impact is caused by the lorry operation for transport processes. Potential benefits for recycling (-4 % for photochemical ozone formation and -5 % for the acidification potential) are gained from replacing primary aluminium by secondary aluminium.

Nitrogen oxides are the most important substance within the impact category **terrestrial eutrophication potential**. In the product stage nitrogen oxides are emitted in the supply chain of silicon production, namely by burning of hard coal for electricity production. The product stage causes a share of 85 % of the overall impacts whereas the construction stage causes 13 % of the impacts. In the construction stage nitrogen oxides are emitted also in the supply chain of electricity production used in the aluminium production. Transports by transoceanic freight ship and by lorry have a noteworthy impact as well. The use stage causes a negligible contribution to the total impacts (<1 %). The end-of-life stage contributes to 3 % and is dominated by lorry transports, where likewise nitrogen oxides are emitted in the operation of the lorries. Potential benefits for recycling (-4 %) are gained from avoided primary aluminium production.

Freshwater eutrophication has almost all of its impacts from the product stage (44 %) and the construction stage (55 %). Phosphate released into water causes the main impacts. In the product stage impacts stem from the supply chain of electricity production (spoil disposal from hard coal mining) and the silver mining (off-site disposal of sulfidic tailings). Within the construction stage the electric installation is the dominant aspect, where phosphate is emitted when sulfidic tailings are disposed of in the copper mining. 1 % of the impacts is caused by the end-of-life stage and therein by the lignite mining for electricity production. Recycling benefits are owed to avoiding primary copper production (-14 %) and hence to saved emissions from copper mining. Another potential benefit from recycling is gained by the reduced electricity consumption of secondary aluminium production compared to primary aluminium.

The **marine eutrophication potential** is primarily determined by the emissions of nitrogen oxides into air. The same explanations apply as for the terrestrial eutrophication potential. The production stage is the most important contributor to the total marine eutrophication potential and has a share of 85 %. The construction, use and end-of-life stage cause 12 %, <1 % and 3 % of the total impacts, respectively. The potential benefits due to recycling amount to -3 %.

The product stage causes 24 % of the **freshwater ecotoxicity potential**. On one hand the supply chain of silver used in the panel causes zinc, copper and arsenic emissions into water. On the other hand in the supply chain of electricity production from fossil fuels heavy metals are released into water (for

instance when burning hard coal). The construction stage causes with 75 % the main share of the impacts. Antimony emitted into water by the disposal of plastic components of industrial electronics (from the production of the electric installation) in municipal waste incineration plants causes the main impacts. Emissions from copper and aluminium mining have a small effect compared to the disposal of plastics from electronics. The use stage has a negligible contribution to the total impacts (<1 %). The end-of-life stage causes 1 % of the impacts. Benefits for recycling (-3 %) are mainly owed to savings in copper and bauxite mining.

Land use impacts are mainly caused by the production and the construction stages (84 % and 14 %, respectively). They are mainly caused by the production facilities (panel factory), the mines of metals (bauxite mine) and in the supply chain of fossil energy carriers. Within the electricity supply chain the land used by coal mines and wells for exploration and extraction of oil and gas are most important. The use stage has a negligible contribution to the total impacts (<1 %). The land use of the end-of-life stage (2 %) is dominated by the production of fossil fuels used in transportation and the land use for electricity production. Potential benefits from recycling amount to around -4 % and are owed to the savings for recycling aluminium and therein the savings of fossil fuels, raw materials and its processing.

Water resource depletion is dominated by the product stage (74 %). Electricity production from hydro power leads to water evaporation from the reservoirs. Within the silicon production (electronic and solar grade) electricity is the main energy carrier used. The impacts in the construction stage (24 %) are caused by the electricity from hydro power used in the supply chain of aluminium. In the end-of-life stage the electricity consumption is the driving force as well (1 %), however the impact is of minor importance. The use stage has a negligible contribution to the total impacts (<1 %). Potential benefits for recycling (-11 %) are gained by avoiding primary aluminium production and hence by savings in electricity consumption from hydro power.

In the impact category **mineral, fossil and renewable resource depletion potential** the product stage has a share of 97 % of all impacts. Within the panel production the silver used in the metallization paste causes the major impact. Silver is the crucial resource in this impact category. The construction stage has a share of 3 %, which comes from the mining of copper for the electric installation. The impacts of the use and the end-of-life stage are negligible (both < 1 %). The potential benefits for recycling have likewise a negligible effect (-0.2 %). Benefits for avoiding primary aluminium and copper production are almost outweighed by the additional impacts of zinc used as an alloying element for producing secondary aluminium from aluminium scrap.

The product stage is most important for the impact category **cumulative energy demand non-renewable** (CED nr; 77 %). Within the panel production the electricity from fossil fuels (hard coal) for the production of solar grade silicon causes the main impact. The construction stage (20 %) is dominated by the electricity generated by nuclear and fossil fuels for the production of aluminium. The use stage has a negligible contribution to the total impacts (<1 %). The end-of-life stage (3 %) causes its impact from fossil fuels used in the supply chain of electricity generation and the lorry transport. Potential benefits for recycling (-6 %) are owed to the savings in uranium and hard coal when reducing the primary production of aluminium.

The indicator **cumulative energy demand renewable** (CED r) is the only indicator for which the major contribution is in the operation stage (97 %). It reflects the harvested amount of solar energy needed to produce electricity with the PV panel. The product and the construction stage contribute

2 % and 1 % to the total impacts, respectively. The shares of the end-of-life stage and of the potential benefits for recycling are negligible.

The impact category **nuclear waste** has its major share from the product stage (67 %). The impact results from the electricity consumption within the supply chain of aluminium in the production of the panel. Electricity produced from nuclear power creates radioactive waste ending up in a final repository. In the construction stage (28 %) the impacts are likewise generated by the electricity consumption from nuclear power and its waste. The aluminium consumed and the electricity consumption in its supply chain is the driving force. The use stage has a negligible contribution to the total impacts (<1 %). The end-of-life has an impact of 5 %, which are as well caused by the supply chain of electricity from nuclear power. Benefits for recycling are granted for savings in electricity consumption when saving the production of primary aluminium (-10 %).

5.8.3 Results normalized

The following Tab. 5.15 shows the environmental impacts (normalized) of the production of 1 kWh of DC electricity with a residential scale multicrystalline-Si PV system mounted on a slanted roof. The three additional impact categories cumulative energy demand non-renewable, cumulative energy demand renewable and nuclear waste are not subject to normalization.

The impact category mineral, fossil and renewable resource depletion is the most important one. The generation of 1 kWh electricity causes 0.2 % of the resource depletion potential per person in Europe. An average European person consumes around 5'000 kWh of electricity. Assuming it would all be produced with such 3 kWp multicrystalline-Si PV systems, the electricity production would consume the present budget of resource depletion to 95 %.

The second largest contribution to the per capita impacts is observed in the impact category human toxicity cancer effects, followed by the human toxicity non-cancer effects.

Tab. 5.15 Environmental impact results (normalized) of 1 kWh of DC electricity produced with a residential scale multi-Si PV system mounted on a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, mounted with multi-Si PV technology, normalized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change	-	4.90E-06	1.05E-06	3.53E-10	1.53E-07	6.10E-06	-3.72E-07	5.73E-06
Ozone depletion	-	6.83E-08	1.25E-08	9.34E-12	2.21E-09	8.30E-08	-3.37E-09	7.96E-08
Human toxicity, cancer effects	-	1.46E-05	1.88E-05	1.35E-08	8.94E-07	3.43E-05	-5.92E-06	2.84E-05
Human toxicity, non-cancer effects	-	1.15E-05	8.19E-06	1.45E-08	1.86E-07	1.99E-05	-2.05E-07	1.97E-05
Particulate matter	-	1.62E-05	1.72E-06	4.58E-10	1.13E-07	1.80E-05	-6.04E-07	1.74E-05
Ionizing radiation HH	-	1.69E-06	7.55E-07	5.38E-10	1.39E-07	2.58E-06	-2.65E-07	2.32E-06
Photochemical ozone formation	-	5.23E-06	8.32E-07	3.71E-10	1.62E-07	6.22E-06	-2.38E-07	5.98E-06
Acidification	-	8.57E-06	1.47E-06	5.15E-10	1.33E-07	1.02E-05	-4.78E-07	9.69E-06
Terrestrial eutrophication	-	3.39E-06	5.10E-07	3.65E-10	1.05E-07	4.00E-06	-1.58E-07	3.85E-06
Freshwater eutrophication	-	2.10E-06	2.66E-06	2.94E-09	5.21E-08	4.82E-06	-6.80E-07	4.14E-06
Marine eutrophication	-	3.26E-06	4.44E-07	5.65E-09	1.08E-07	3.82E-06	-1.32E-07	3.69E-06
Freshwater ecotoxicity	-	2.77E-06	8.63E-06	2.21E-09	1.25E-07	1.15E-05	-3.78E-07	1.12E-05
Land use	-	7.53E-07	1.30E-07	1.80E-10	1.73E-08	9.00E-07	-3.67E-08	8.63E-07
Water resource depletion	-	5.78E-07	1.88E-07	1.07E-09	9.93E-09	7.78E-07	-8.21E-08	6.96E-07
Mineral, fossil & ren resource depletion	-	1.86E-04	5.09E-06	1.34E-10	5.70E-08	1.91E-04	-3.19E-07	1.91E-04

5.8.4 Results weighted

In Tab. 5.16 the weighted environmental impact results (equal weighting) of the electricity production with multicrystalline-Si PV panels are listed. The blue bars within column “Total” represent the

importance of each impact category in the total of impacts. Fig. 5.28 illustrates the shares in more detail. The three additional impact categories cumulative energy demand non-renewable, cumulative energy demand renewable and nuclear waste are not included in the weighted results.

The weighted environmental impacts of the production of 1 kWh electricity with a residential scale multicrystalline-Si PV system mounted on a slanted roof are mainly driven by the mineral, fossil and renewable resource depletion potential (63 %), human toxicity cancer effects (9 %), human toxicity non-cancer effects and particulate matter (both 6 %).

Tab. 5.16 Environmental impact results (weighted) of 1 kWh of DC electricity produced with a residential scale multi-Si PV system mounted on a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, mounted with multi-Si PV technology, weighted		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Total	Pt	1.74E-05	3.37E-06	2.86E-09	1.50E-07	2.10E-05	-6.58E-07	2.03E-05
Climate change	Pt	3.26E-07	7.01E-08	2.35E-11	1.02E-08	4.07E-07	-2.48E-08	3.82E-07
Ozone depletion	Pt	4.55E-09	8.32E-10	6.23E-13	1.48E-10	5.53E-09	-2.25E-10	5.31E-09
Human toxicity, cancer effects	Pt	9.72E-07	1.25E-06	9.02E-10	5.96E-08	2.29E-06	-3.94E-07	1.89E-06
Human toxicity, non-cancer effects	Pt	7.65E-07	5.46E-07	9.70E-10	1.24E-08	1.32E-06	-1.36E-08	1.31E-06
Particulate matter	Pt	1.08E-06	1.15E-07	3.06E-11	7.51E-09	1.20E-06	-4.03E-08	1.16E-06
Ionizing radiation HH	Pt	1.13E-07	5.03E-08	3.59E-11	9.24E-09	1.72E-07	-1.76E-08	1.55E-07
Photochemical ozone formation	Pt	3.48E-07	5.55E-08	2.47E-11	1.08E-08	4.15E-07	-1.59E-08	3.99E-07
Acidification	Pt	5.71E-07	9.77E-08	3.44E-11	8.86E-09	6.78E-07	-3.18E-08	6.46E-07
Terrestrial eutrophication	Pt	2.26E-07	3.40E-08	2.43E-11	6.99E-09	2.67E-07	-1.05E-08	2.56E-07
Freshwater eutrophication	Pt	1.40E-07	1.77E-07	1.96E-10	3.47E-09	3.21E-07	-4.53E-08	2.76E-07
Marine eutrophication	Pt	2.18E-07	2.96E-08	3.77E-10	7.19E-09	2.55E-07	-8.79E-09	2.46E-07
Freshwater ecotoxicity	Pt	1.84E-07	5.76E-07	1.47E-10	8.35E-09	7.69E-07	-2.52E-08	7.43E-07
Land use	Pt	5.02E-08	8.64E-09	1.20E-11	1.15E-09	6.00E-08	-2.45E-09	5.75E-08
Water resource depletion	Pt	3.86E-08	1.26E-08	7.13E-11	6.62E-10	5.18E-08	-5.48E-09	4.64E-08
Mineral, fossil & ren resource depletion	Pt	1.24E-05	3.40E-07	8.91E-12	3.80E-09	1.17E-05	-2.13E-08	1.17E-05

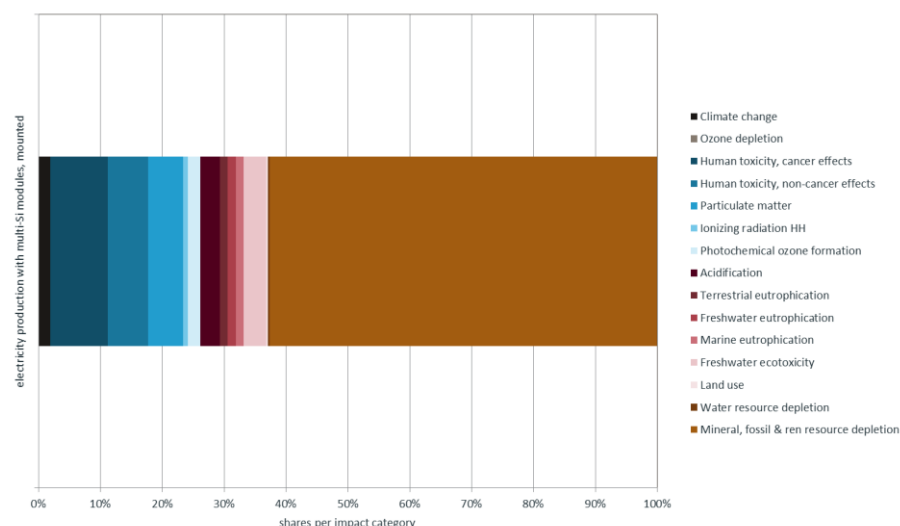


Fig. 5.28 Share per impact category on the weighted result for the production of 1 kWh of DC electricity produced with a residential scale multi-Si PV system mounted on a slanted roof.

5.8.5 Comparison of the three system sizes integrated, mounted and open ground

The comparison of the environmental impacts per life cycle stage of the three different systems shows that the residential scale systems do only differ slightly while the large scale system causes substantially higher impacts per kWh produced (see Fig. 5.29). The smallest impacts are caused by

the PV system integrated in a residential roof. Slightly higher impacts are caused by the mounted system whereas the open ground system causes the highest impacts. Overall the impacts of the open ground system are about 45 % higher compared to the mounted residential system. The product stage, the use stage, end-of-life stage as well as the potential benefits for recycling are similarly important in all three systems. The main difference is caused by the construction stage and, more specifically, by the land use of the different PV systems. The land use impacts of the open ground multicrystalline-Si PV system are much higher than of the systems installed on a slanted roof. This is due to the fact that the open ground system directly occupies a certain land area while the land use of the residential PV systems is attributed to the building.

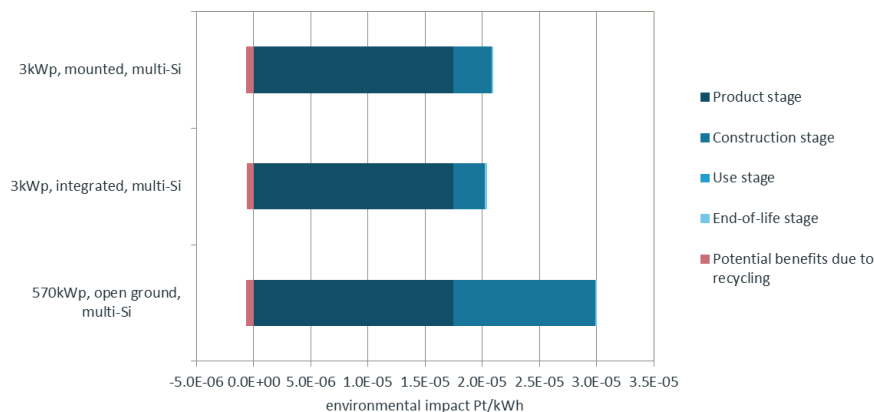


Fig. 5.29 Environmental impact results (weighted) of 1 kWh of DC electricity produced with a residential scale (3 kWp) PV system with multi-Si PV panels mounted on and integrated in a slanted roof as well as produced with a large scale open ground system. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.

5.9 Results monocrystalline-Si PV technology

5.9.1 Overview

In the following sections the results of electricity production with monocrystalline PV panels are explained in detail. Section 5.9.2 shows the results prior to normalization while in section 5.9.3 the normalized results are presented. Section 5.9.4 contains the details of the weighted results.

5.9.2 Results prior to normalization

Tab. 5.17 shows the environmental impacts (prior to normalization) of the production of 1 kWh of DC electricity with a residential scale monocrystalline PV system, mounted on a slanted roof. The life cycle stages analyzed are the product stage, the construction stage, the operation stage and the end-of-life stage. Net benefits caused by end-of-life recycling are allocated to the production of electricity with a share of 50 %. The other half is allocated to the life cycle of products potentially using the secondary products (see also subchapter 3.5).

Tab. 5.17 Environmental impact results (characterized) of 1 kWh of DC electricity produced with a residential scale mono-Si PV system mounted on a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, mounted with mono-Si PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change	kg CO ₂ eq	8.00E-02	9.47E-03	3.13E-06	1.43E-03	9.09E-02	-3.33E-03	8.75E-02
Ozone depletion	kg CFC-11 eq	2.12E-09	2.63E-10	1.96E-13	4.87E-11	2.43E-09	-7.10E-11	2.36E-09
Human toxicity, cancer effects	CTUh, c	7.09E-10	6.76E-10	4.86E-13	3.36E-11	1.42E-09	-2.13E-10	1.21E-09
Human toxicity, non-cancer effects	CTUh, n-c	9.12E-09	4.33E-09	7.54E-12	1.01E-10	1.36E-08	-1.16E-10	1.34E-08
Particulate matter	kg PM _{2.5} eq	1.26E-04	6.42E-06	1.63E-09	4.36E-07	1.33E-04	-2.25E-06	1.30E-04
Ionizing radiation HH	kBq U235 eq	2.82E-03	8.32E-04	5.91E-07	1.59E-04	3.82E-03	-2.91E-04	3.53E-03
Photochemical ozone formation	kg NMVOC eq	2.96E-04	2.59E-05	1.13E-08	5.21E-06	3.27E-04	-7.39E-06	3.20E-04
Acidification	mol H ⁺ eq	7.42E-04	6.83E-05	2.34E-08	6.40E-06	8.17E-04	-2.22E-05	7.94E-04
Terrestrial eutrophication	mol N eq	1.05E-03	8.82E-05	6.20E-08	1.88E-05	1.15E-03	-2.71E-05	1.13E-03
Freshwater eutrophication	kg P eq	3.77E-06	3.93E-06	4.23E-09	7.85E-08	7.78E-06	-1.00E-06	6.77E-06
Marine eutrophication	kg N eq	1.05E-04	7.37E-06	9.29E-08	1.86E-06	1.15E-04	-2.18E-06	1.12E-04
Freshwater ecotoxicity	CTUe	2.90E-02	7.52E-02	1.88E-05	1.11E-03	1.05E-01	-3.24E-03	1.02E-01
Land use	kg C deficit	7.39E-02	9.50E-03	1.31E-05	1.32E-03	8.47E-02	-2.68E-03	8.21E-02
Water resource depletion	m ³ water eq	9.15E-05	1.49E-05	8.47E-08	8.23E-07	1.07E-04	-6.51E-06	1.01E-04
Mineral, fossil & ren resource depletion	kg Sb eq	1.21E-05	5.13E-07	2.01E-11	5.87E-09	1.26E-05	-3.40E-08	1.25E-05
Cumulative energy demand non renewable	MJ oil eq	9.21E-01	1.36E-01	4.82E-05	2.04E-02	1.08E+00	-4.24E-02	1.03E+00
Cumulative energy demand renewable	MJ oil eq	1.13E-01	2.66E-02	3.60E+00	1.33E-03	3.74E+00	-1.08E-02	3.73E+00
Nuclear waste	m ³ HAA eq	6.70E-11	1.83E-11	1.10E-14	3.33E-12	8.86E-11	-6.52E-12	8.21E-11

Fig. 5.30 shows the shares per environmental impact over all life cycle stages. The impacts from product stage to end-of-life stage are indexed to a total of 100 %. The potential benefits due to recycling are expressed relative to this sum. The product stage and the construction stage are most important for most of the impact indicators. With a few exceptions the operation stage and the end-of-life stage are much less important.

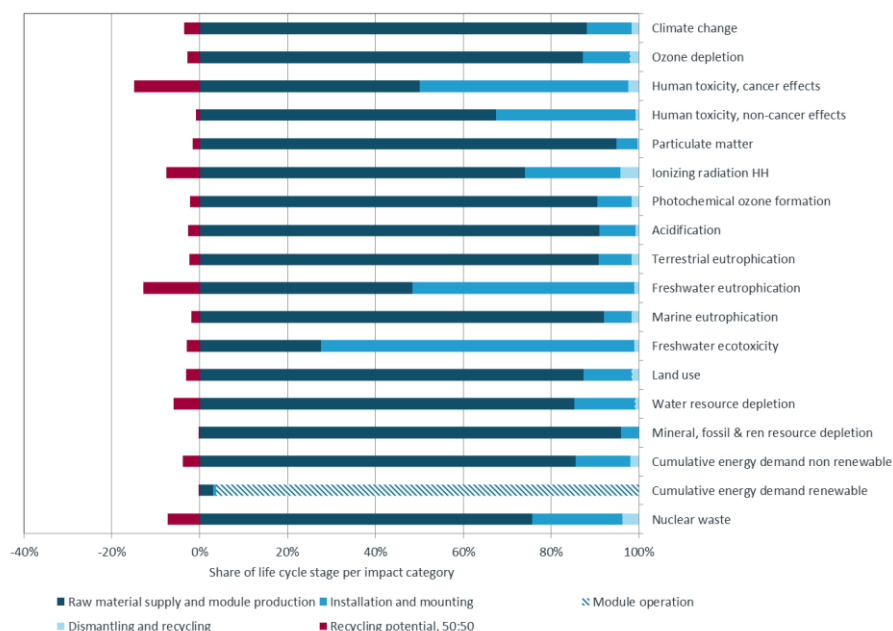


Fig. 5.30 Environmental impact results (characterized, indexed to 100 %) of 1 kWh of DC electricity produced with a residential scale mono-Si PV system mounted on a slanted roof. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.

The impact category **climate change** is mainly influenced by the consumption of fossil fuels for electricity production. The product stage contributes to 88 % to the overall greenhouse gas emissions. The construction stage causes a share of 10 % on the overall impacts. The operation stage and the

end-of-life stages cause minor climate change impacts (<1 % and 2 %, respectively). Within the product stage the wafer production and the CZ monocrystalline silicon production (electricity consumption from hard coal) cause the major share of climate change impacts. The construction stage is dominated by the production of the mounting system (electricity from fossil fuels used in the supply chain of aluminium). The potential benefits due to recycling at the end-of-life sum up to about -4 % of the overall greenhouse gas emissions and are mainly owed to the recycling of the aluminium and hence to avoiding primary aluminium production.

The impact category **ozone depletion** has its main impact from the product stage (87 %), driven by the supply chain of electricity production. Bromochlorodifluoromethane (halone 1211) is released when transporting natural gas in long distance pipelines. The most relevant processes in the product stage with regard to ozone depletion impacts is the production of silicon. Furthermore tetrachloromethane (CFC11) is emitted to air from the chlorine production in the supply chain of solar grade silicon. In the construction stage the aluminium supply chain (likewise natural gas pipeline transport for electricity production) causes the dominant impact (11 %). The use stage has a negligible share in the total impacts (<1 %). The end-of-life stage causes 2 % of the impacts and is also driven by the electricity consumption and its supply chain. Potential benefits for recycling are given for savings of electricity consumption when avoiding primary aluminium production as well (-3 %).

The product stage has an impact of 50 % to **human toxicity, cancer effects potential**. Chromium VI to water and chromium to air are the most important substances for this impact category. Within the product stage the emissions stem from the electricity supply chain (disposal of hard coal ash in residual material landfill). Electricity is mainly used in the solar grade silicon and CZ monocrystalline silicon production. The construction stage causes 48 % of the impacts. The supply chains of aluminium and steel are the driving forces. Chromium IV emissions are caused by redmud disposal from bauxite digestion and from disposal of steel slag in residual material landfills. The use stage has a negligible share in the total impacts (<1 %). The end-of-life stage has only a minor impact (2 %) and is dominated by the hydrogen peroxide supply chain (chromium IV emissions from anthraquinone production). The potential benefits for recycling are owed to avoiding primary aluminium production and the emissions in its supply chain (emissions from redmud disposal) and sum up to -15 %.

Human toxicity, non cancer effects are dominated by the emission of arsenic to water and heavy metals (mercury, lead, zinc and others) to air. The product stage has a share of 67 % in the overall impacts. The heavy metals are released in the supply chain of electricity, which is used for the solar grade silicon and the CZ monocrystalline silicon production. In the production of electricity from hard coal heavy metals are emitted directly, but also indirect emissions occur when hard coal ash is disposed of in the residual landfill. The construction stage has a share of 32 % in the overall impacts where direct emissions are released during copper refining (electric installation) and in the disposal of redmud from bauxite digestion (mounting structure). The contributions of the use and the end-of-life stage as well as the potential benefits for recycling are negligible, contributing <1 %, 1 % and -1 % to the total human toxicity non cancer effects, respectively.

Ultrafine dust (PM_{2.5}) and sulfur dioxides emitted to air are the most important substances for the impact category **particulate matter formation potential**. These are released in the supply chain of electricity production (burning of hard coal). The product stage (95 %) generates its impacts by the consumption of electricity for the production of silicon and CZ monocrystalline silicon. The construction stage (5 %) is concerned in the supply chain of aluminium processing. The use and the end-of-life stage have a share of <1 % on the overall impact. Potential benefits for recycling (-2 %) are owed to savings in primary aluminium production.

Ionizing radiation HH is caused by the emission of carbon-14 in the reprocessing spent nuclear fuel in the supply chain of electricity production from nuclear fuels. Electricity is used in the supply chains of all product stages. The share of the product stage is 74 %, whereas the construction stage causes 22 % and the end-of-life stage 4 % of the impacts. The use stage has a negligible share in the total impacts (<1 %). Potential benefits for recycling (-8 %) are likewise gained from savings in electricity consumption.

Photochemical ozone formation and **acidification potential** are caused by the emission of sulfur dioxide and nitrogen oxides in air. The product stage has a share of 90 % on photochemical ozone formation and of 91 % on the total acidification potential. The airborne substances are emitted when burning hard coal in the supply chain of electricity production. In the construction stage (8 % for both indicators) impacts on the acidification potential are caused in the supply chain of aluminium production (use of electricity produced from hard coal). Blasting of copper in the supply chain of the electric installation has a noteworthy impact as well. The use stage has a negligible share in the total impacts (<1 %). The end-of-life stage has a share of 2 % and 1 %, respectively, on the overall impact. Potential benefits for recycling (-2 % and -3 % for photochemical ozone formation and acidification potential, respectively) are gained from avoided primary aluminium production.

Nitrogen oxides are the most important substance within the impact category **terrestrial eutrophication potential**. In the product stage, nitrogen oxides are emitted in the supply chain of silicon production, namely when burning hard coal for electricity production. The product stage causes a share of 91 % of the overall impacts whereas the construction stage causes 8 % of the impacts. In the construction stage nitrogen oxides are emitted also in the supply chain of electricity production used in the aluminium production. Transports by transoceanic freight ship have a noteworthy impact as well. The use stage has a negligible share in the total impacts (<1 %). The end-of-life stage contributes to 2 % and is dominated by the lorry transport, where likewise nitrogen oxides are emitted. Potential benefits for recycling (-2 %) are gained from avoided primary aluminium production.

Freshwater eutrophication has almost all of its impacts from the product stage and the construction stage (48 % and 51 %, respectively). Phosphate released into water causes the main impacts. In the product stage, the freshwater eutrophication impacts stem from the supply chain of electricity production (spoil disposal from hard coal and lignite mining) and the silver mining (off-site disposal of sulfidic tailings). Within the construction stage the electric installation is the dominant aspect, where phosphate is emitted when sulfidic tailings are disposed of in the copper mining. The use stage has a negligible share in the total impacts (<1 %). 1 % of the impacts are caused by the end-of-life stage and therein by the lignite mining for electricity production. Potential recycling benefits are owed to avoiding primary copper production (-13 %) and hence to saved emissions from copper mining.

The **marine eutrophication potential** is primarily determined by the emissions of nitrogen oxides into air. The same explanations apply as for the terrestrial eutrophication potential. The product stage causes 92 % of the total impacts, whereas the construction, the use and the end-of-life stages have shares of 6 %, <1 % and 2 %, respectively. The potential benefits due to recycling are -2 %.

The product stage causes 28 % of the impacts in the category **freshwater ecotoxicity potential**. On the one hand in the supply chain of electricity production from fossil fuels heavy metals are released into air (for instance when burning hard coal). On the other hand the supply chain of silver used in the panel causes zinc, copper and arsenic emissions into water. The construction stage causes the main share of the freshwater ecotoxicity impacts (71 %). Antimony emitted into water by the disposal of

plastic components of the electric installation in municipal waste incineration plants causes the main impacts. Emissions from copper and aluminium mining have a small effect compared to the disposal of plastics from electronics. The use stage has a negligible share in the total impacts (<1 %). The end-of-life stage causes 1 % of the impacts. Potential benefits for recycling (-3 %) are mainly owed to savings in bauxite mining in the supply chain of avoided primary aluminium production.

Land use impacts are mainly caused by the production and the construction stages (87 % and 11 %, respectively). They are mainly caused by the production facilities (panel factory), the mines of metals (bauxite mine) and in the supply chain of fossil energy carriers. Within the electricity supply chain the land used by coal mines and wells for exploration and extraction of oil and gas are most important. The land use of the end-of-life stage (2 %) is dominated by the production of fossil fuels used in transportation and the land use for electricity production. Potential benefits from recycling amount to around -3 % and are owed to avoiding primary aluminium production and therein the savings of fossil fuels, raw materials and its processing.

Water resource depletion is dominated by the product stage (85 %). Electricity production from hydro power leads to water evaporation from their reservoirs. Within the silicon production (electronic and solar grade) and the CZ monocrystalline silicon production, electricity is the main energy carrier used. The construction stage (14 %) has its impacts caused by the electricity from hydro power used in the supply chain of aluminium. The use stage has a negligible share in the total impacts (<1 %). In the end-of-life stage the electricity consumption is the driving force (1 %), however the impact is only minor. Potential benefits for recycling (-6 %) are gained by avoiding primary aluminium production and hence savings in electricity consumption from hydro power.

In the impact category **mineral, fossil and renewable resource depletion potential** the product stage has a share of 96 % of all impacts. Within the panel production the silver used in the metallization paste causes the major impact. Silver is the crucial resource within this impact category. The construction stage has a share of 4 % and comes from the mining of copper for the electric installation. The use and the end-of-life stage have no noteworthy impact (<1 %). The potential benefits for recycling have likewise a negligible effect. Benefits for avoiding primary aluminium and copper production are almost outweighed by the impacts of zinc used for producing secondary aluminium from aluminium scrap.

The product stage is most important for the impact category **cumulative energy demand non-renewable** (CED nr; 85 %). Within the panel production the electricity from fossil fuels (hard coal) for the production of CZ monocrystalline silicon and the wafer production cause the main impacts. The construction stage (13 %) is dominated by the consumption of nuclear and fossil fuels used in the production of aluminium. The use stage has a negligible share in the total impacts (<1 %). The end-of-life stage (2 %) causes its impact from fossil fuels used in the supply chain of electricity generation and lorry transports. Potential benefits for recycling (-4 %) are owed to the savings in uranium and hard coal when avoiding primary aluminium production.

The indicator **cumulative energy demand renewable** (CED r) is the only indicator for which the major contribution is in the operation stage (96 %). It reflects the harvested amount of solar energy needed to produce electricity with the PV panel. The product and the construction stage contribute 3 % and 1 % to the total impacts, respectively. The shares of the end-of-life stage and of the potential benefits for recycling are negligible.

The impact category **nuclear waste** has its major share from the product stage (76 %). The impact results from the electricity consumption within the supply chain of CZ monocrystalline silicon production and of wafer production. Electricity produced from nuclear power creates radioactive waste ending up in a final repository. In the construction stage (21 %) the impacts are likewise generated by the electricity consumption from nuclear power and its waste. The aluminium consumed and the electricity consumption in its supply chain is the driving force. The use stage has a negligible share in the total impacts (<1 %). The end-of-life has an impact of 4 %, which are as well caused by the supply chain of electricity from nuclear power. Potential benefits for recycling are granted for savings in electricity consumption when avoiding the production of primary aluminium (-7 %).

5.9.3 Results normalized

The following Tab. 5.18 shows the environmental impacts (normalized) of the production of 1 kWh of DC electricity with a residential scale monocrystalline-Si PV system mounted on a slanted roof. The three additional impact categories cumulative energy demand non-renewable, cumulative energy demand renewable and nuclear waste are not subject to normalization.

The impact category mineral, fossil and renewable resource depletion is the most important one. The generation of 1 kWh electricity causes a bit more than 0.1 ‰ of the resource depletion potential per person in Europe. An average European person consumes around 5'000 kWh of electricity. Assuming it would all be produced with such 3 kWp multicrystalline-Si PV systems, the electricity production would consume the present budget of resource depletion to 62 %.

The second largest contribution to the per capita impacts is observed in the impact category particulate matter followed by the human toxicity, cancer effects.

Tab. 5.18 Environmental impact results (normalized) of 1 kWh of DC electricity produced with a 3 kWp power system with mono-Si PV panels mounted on a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, mounted with mono-Si PV technology, normalized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change		8.67E-06	1.03E-06	3.39E-10	1.55E-07	9.86E-06	-3.62E-07	9.50E-06
Ozone depletion		9.81E-08	1.22E-08	9.06E-12	2.25E-09	1.13E-07	-3.29E-09	1.09E-07
Human toxicity, cancer effects		1.92E-05	1.83E-05	1.32E-08	9.11E-07	3.85E-05	-5.77E-06	3.27E-05
Human toxicity, non-cancer effects		1.71E-05	8.12E-06	1.42E-08	1.89E-07	2.54E-05	-2.17E-07	2.52E-05
Particulate matter		3.31E-05	1.69E-06	4.30E-10	1.15E-07	3.49E-05	-5.91E-07	3.43E-05
Ionizing radiation HH		2.50E-06	7.36E-07	5.23E-10	1.41E-07	3.38E-06	-2.58E-07	3.12E-06
Photochemical ozone formation		9.35E-06	8.18E-07	3.57E-10	1.64E-07	1.03E-05	-2.33E-07	1.01E-05
Acidification		1.57E-05	1.44E-06	4.95E-10	1.35E-07	1.73E-05	-4.69E-07	1.68E-05
Terrestrial eutrophication		5.95E-06	5.01E-07	3.52E-10	1.07E-07	6.56E-06	-1.54E-07	6.41E-06
Freshwater eutrophication		2.54E-06	2.65E-06	2.86E-09	5.30E-08	5.25E-06	-6.79E-07	4.58E-06
Marine eutrophication		6.23E-06	4.36E-07	5.50E-09	1.10E-07	6.78E-06	-1.29E-07	6.65E-06
Freshwater ecotoxicity		3.32E-06	8.61E-06	2.15E-09	1.27E-07	1.21E-05	-3.71E-07	1.17E-05
Land use		9.88E-07	1.27E-07	1.75E-10	1.76E-08	1.13E-06	-3.58E-08	1.10E-06
Water resource depletion		1.12E-06	1.84E-07	1.04E-09	1.01E-08	1.32E-06	-8.00E-08	1.24E-06
Mineral, fossil & ren resource depletion		1.19E-04	5.08E-06	1.99E-10	5.81E-08	1.25E-04	-3.36E-07	1.24E-04

5.9.4 Results weighted

In Tab. 5.19 the weighted environmental impact results (equal weighting) of the electricity production with monocrystalline-Si PV panels are listed. The blue bars within column “Total” represent the importance of each impact category in the total of impacts. Fig. 5.31 illustrates the shares in more detail. The three additional impact categories cumulative energy demand non-renewable, cumulative energy demand renewable and nuclear waste are not included in the weighted results.

The weighted environmental impacts of the production of 1 kWh of DC electricity with a residential scale monocrystalline-Si PV system mounted on a slanted roof are mainly driven by the mineral, fossil and renewable resource depletion potential (43 %), particulate matter (12 %) and human toxicity, cancer effects (11 %).

Tab. 5.19 Environmental impact results (weighted) of 1 kWh of DC electricity produced with a residential scale mono-Si PV system mounted on a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, mounted with mono-Si PV technology, weighted		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Total	Pt	1.64E-05	3.32E-06	2.78E-09	1.53E-07	1.98E-05	-6.46E-07	1.92E-05
Climate change	Pt	5.78E-07	6.85E-08	2.26E-11	1.04E-08	6.57E-07	-2.41E-08	6.33E-07
Ozone depletion	Pt	6.54E-09	8.12E-10	6.04E-13	1.50E-10	7.50E-09	-2.19E-10	7.28E-09
Human toxicity, cancer effects	Pt	1.28E-06	1.22E-06	8.78E-10	6.07E-08	2.56E-06	-3.84E-07	2.18E-06
Human toxicity, non-cancer effects	Pt	1.14E-06	5.41E-07	9.44E-10	1.26E-08	1.70E-06	-1.45E-08	1.68E-06
Particulate matter	Pt	2.21E-06	1.13E-07	2.86E-11	7.64E-09	2.33E-06	-3.94E-08	2.29E-06
Ionizing radiation HH	Pt	1.67E-07	4.91E-08	3.49E-11	9.41E-09	2.25E-07	-1.72E-08	2.08E-07
Photochemical ozone formation	Pt	6.23E-07	5.45E-08	2.38E-11	1.10E-08	6.89E-07	-1.55E-08	6.73E-07
Acidification	Pt	1.05E-06	9.63E-08	3.30E-11	9.02E-09	1.15E-06	-3.12E-08	1.12E-06
Terrestrial eutrophication	Pt	3.97E-07	3.34E-08	2.35E-11	7.12E-09	4.37E-07	-1.03E-08	4.27E-07
Freshwater eutrophication	Pt	1.70E-07	1.77E-07	1.91E-10	3.54E-09	3.50E-07	-4.52E-08	3.05E-07
Marine eutrophication	Pt	4.15E-07	2.91E-08	3.67E-10	7.32E-09	4.52E-07	-8.59E-09	4.44E-07
Freshwater ecotoxicity	Pt	2.21E-07	5.74E-07	1.43E-10	8.50E-09	8.04E-07	-2.47E-08	7.79E-07
Land use	Pt	6.59E-08	8.47E-09	1.17E-11	1.17E-09	7.55E-08	-2.39E-09	7.31E-08
Water resource depletion	Pt	7.49E-08	1.22E-08	6.94E-11	6.74E-10	8.79E-08	-5.33E-09	8.26E-08
Mineral, fossil & ren resource depletion	Pt	7.96E-06	3.39E-07	1.32E-11	3.87E-09	8.30E-06	-2.24E-08	8.28E-06

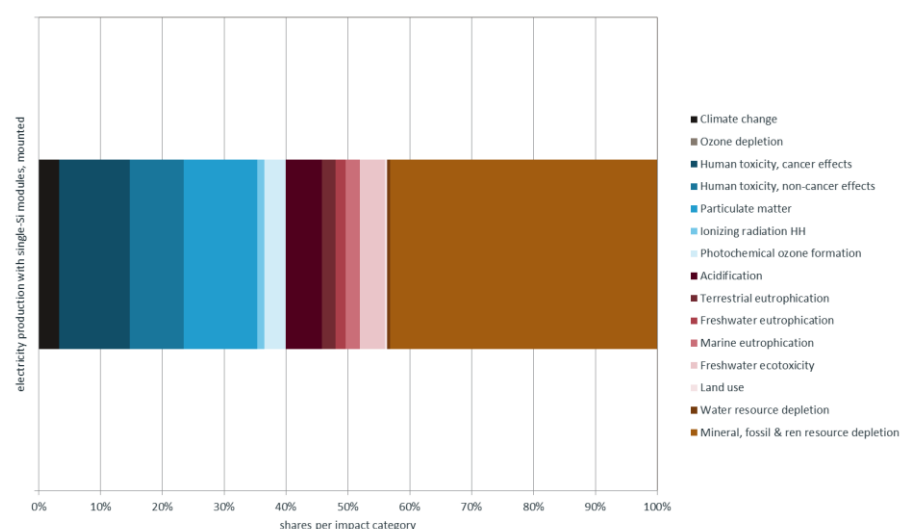


Fig. 5.31 Share per impact category on the weighted result for the production of 1 kWh of DC electricity produced with a residential scale mono-Si PV system mounted on a slanted roof.

5.9.5 Comparison of the three system sizes integrated, mounted and open ground

The comparison of the environmental impacts per life cycle stage of the three different systems shows that the residential scale systems do only differ slightly while the large scale system causes substantially higher impacts per kWh produced. The smallest impacts are caused by the PV system integrated in a residential roof. Slightly higher impacts are caused by the mounted system whereas the open ground system causes the highest impacts (see Fig. 5.32). Overall the impacts of the open ground system are about 46 % higher compared to the mounted residential system. The product stage, the use stage, end-of-life stage as well as the potential benefits for recycling are similarly important in

all three systems. The main difference is caused by the construction stage and, more specifically, by the land use of the different PV systems. The land use impacts of the open ground monocrystalline-Si PV system are much higher than of the systems installed on a slanted roof. This is due to the fact that the open ground system directly occupies a certain land area while the land use of the residential PV systems is attributed to the building.

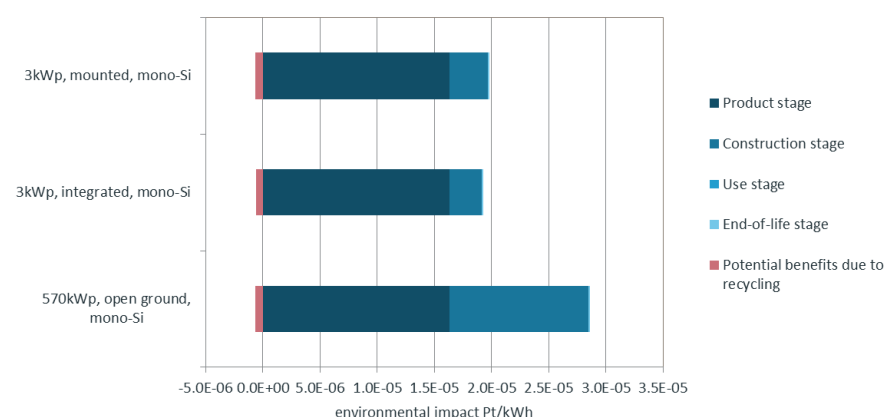


Fig. 5.32 Environmental impact results (weighted) of 1 kWh of DC electricity produced with a residential scale (3 kWp) PV system with mono-Si PV panels mounted on and integrated in a slanted roof as well as produced with a large scale open ground system. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.

5.10 Electricity production with average PV module

Tab. 5.20 shows the technology mix of global PV production in the year 2012 based on FHI-ISE (2013)¹⁰. More than 85 % of the PV modules produced in 2012 were crystalline silicon based of which 45.2 % were multi-crystalline and 40.5 % mono-crystalline. 14.4 % of the production were thin film modules with CdTe having the largest share, followed by micromorphous modules. A-ribbon PV modules are produced in small amounts and thus are not included in the analyses.

Tab. 5.20 Global PV module production in Gigawatt peak (GWp, the maximum power output under standard conditions of the modules) by technology, reference year 2012, based on FHI-ISE (2013)¹⁰

PV technology	share
CdTe	6.3%
CIS	3.5%
micromorphous-Si	4.5%
multicrystalline-Si	45.2%
monocrystalline-Si	40.5%

Based on the market mix of PV panels in the year 2012 the environmental performance of the electricity production with the average PV module mounted on a slanted roof in Europe was assessed. The indexed characterized results per life cycle stage are illustrated in Fig. 5.33 and listed in detail in Tab. 5.21. All life cycle stages, from production to end-of-life, including 50 % of the potential net benefits of recycling are included. The environmental performance of a kWh of DC electricity produced with the average PV panel mix in Europe is mainly influenced by the production of the panels with exception of human toxicity cancer effects, freshwater ecotoxicity and eutrophication as well as CED renewable. The use phase is of no noteworthy importance except for the CED renewable. The end-of-life stage has impacts from 0 % to 5 % while potential benefits for recycling can add up to -17 %.

¹⁰ FHI-ISE (2013) cites the data originally published by Navigant consult 2012.

Tab. 5.21 Environmental impact results (characterized) of 1 kWh of DC electricity produced with a residential scale (3 kWp) PV system with average PV panels mounted on a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, mounted with average PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change	kg CO ₂ eq	5.64E-02	9.91E-03	3.33E-06	1.52E-03	6.78E-02	-3.54E-03	6.43E-02
Ozone depletion	kg CFC-11 eq	1.65E-09	2.76E-10	2.07E-13	5.16E-11	1.98E-09	-7.57E-11	1.90E-09
Human toxicity, cancer effects	CTUh, c	5.64E-10	7.10E-10	5.12E-13	3.55E-11	1.31E-09	-2.22E-10	1.09E-09
Human toxicity, non-cancer effects	CTUh, n-c	6.86E-09	4.41E-09	7.96E-12	1.06E-10	1.14E-08	-1.10E-10	1.13E-08
Particulate matter	kg PM _{2.5} eq	8.16E-05	6.65E-06	1.77E-09	4.55E-07	8.87E-05	-2.36E-06	8.63E-05
Ionizing radiation HH	kBq U235 eq	2.15E-03	8.75E-04	6.24E-07	1.64E-04	3.19E-03	-3.05E-04	2.89E-03
Photochemical ozone formation	kg NMVOC eq	2.09E-04	2.69E-05	1.20E-08	5.49E-06	2.41E-04	-7.74E-06	2.34E-04
Acidification	mol H+ eq	5.12E-04	7.04E-05	2.49E-08	6.69E-06	5.89E-04	-2.31E-05	5.66E-04
Terrestrial eutrophication	mol N eq	7.47E-04	9.16E-05	6.57E-08	1.98E-05	8.58E-04	-2.86E-05	8.30E-04
Freshwater eutrophication	kg P eq	3.13E-06	3.95E-06	4.46E-09	8.56E-08	7.17E-06	-1.01E-06	6.16E-06
Marine eutrophication	kg N eq	7.23E-05	7.67E-06	9.80E-08	1.96E-06	8.20E-05	-2.28E-06	7.97E-05
Freshwater ecotoxicity	CTUe	2.40E-02	7.57E-02	1.98E-05	1.18E-03	1.01E-01	-3.37E-03	9.75E-02
Land use	kg C deficit	5.95E-02	9.92E-03	1.38E-05	1.39E-03	7.08E-02	-2.84E-03	6.80E-02
Water resource depletion	m ³ water eq	6.22E-05	1.57E-05	8.93E-08	8.85E-07	7.89E-05	-6.81E-06	7.21E-05
Mineral, fossil & ren resource depletion	kg Sb eq	3.18E-05	5.17E-07	2.29E-12	6.20E-09	3.23E-05	-2.08E-07	3.21E-05
Cumulative energy demand non renewable	MJ oil eq	6.65E-01	1.42E-01	5.12E-05	2.16E-02	8.29E-01	-4.47E-02	7.84E-01
Cumulative energy demand renewable	MJ oil eq	8.19E-02	2.79E-02	3.60E+00	1.39E-03	3.71E+00	-1.13E-02	3.70E+00
Nuclear waste	m ³ HAA eq	5.06E-11	1.92E-11	1.16E-14	3.45E-12	7.33E-11	-6.82E-12	6.64E-11

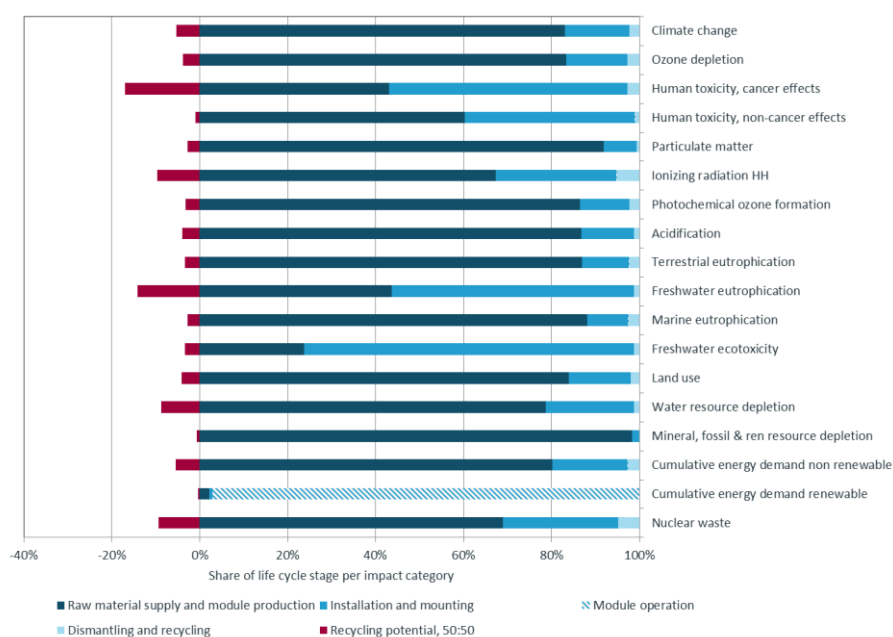


Fig. 5.33 Environmental impact results (characterized, indexed to 100 %) of 1 kWh of DC electricity produced with a residential scale (3 kWp) PV system with average PV panels mounted on a slanted roof. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.

The production of 1 kWh DC electricity with an average residential scale PV system mounted on a rooftop causes on average 64.3 grams of CO₂-eq and requires 0.784 MJ of non-renewable primary energy. The particulate matter emissions and the acidification impacts of 1 kWh DC electricity amount to 86.3 mg and 0.566 mmol H+eq, respectively. 1 kWh of DC electricity produced with an average PV module requires 32.1 mg Sb-eq of abiotic resources and consumes 72.1 g water-eq of water. The human toxicity impacts are $1.09 \cdot 10^{-9}$ CTUh_c (cancer effects) and $1.13 \cdot 10^{-8}$ CTUh_{n-c} (non cancer effects) per kWh of PV electricity and the freshwater ecotoxicity impacts amount to $9.75 \cdot 10^{-2}$ CTUe/kWh. The corresponding table with the detailed data is listed in Tab. A. 61 in ANNEX II.

The following Fig. 5.34 shows the shares per technology of the average PV panel per impact category (all life cycle stages are included). The technologies multicrystalline-Si and monocrystalline-Si cause the higher share of environmental impacts per impact category as their share in the average PV panel is clearly higher than for the technologies CdTe, CIS or micromorphous-Si.

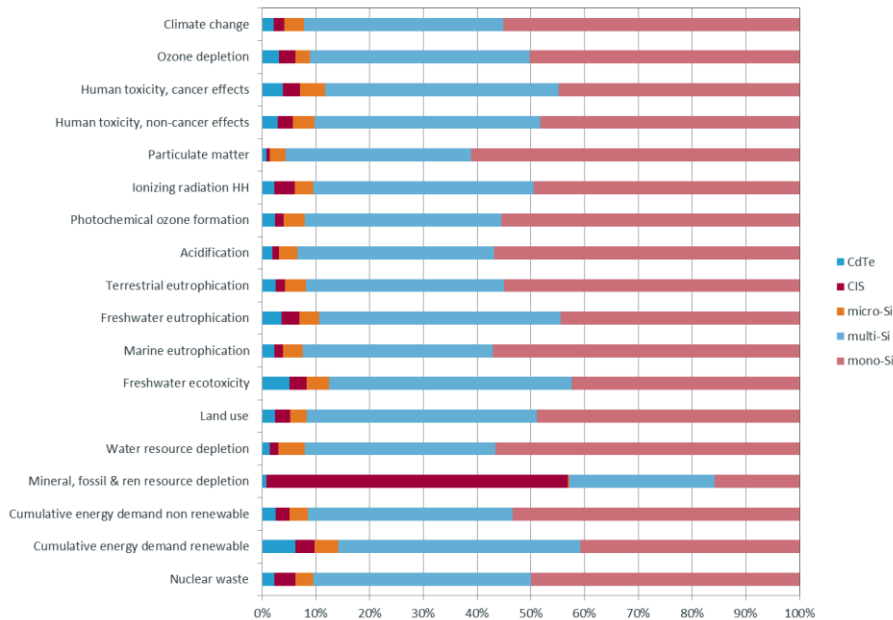


Fig. 5.34 Environmental impact (indexed to 100 %) of 1 kWh of DC electricity production (all life cycle stages included) with the average PV module, mounted on a slanted residential roof in Europe.

CdTe PV panels have a market share of 6.3 %. However the share of environmental impacts of the electricity production with CdTe PV panels does not exceed 6.2 %. Hence, the share of environmental impacts is (at least) slightly lower than its market share. The share of environmental impacts from the electricity production with CIS PV modules exceeds its market share substantially in the impact category mineral, fossil and renewable resource consumption due to the high scores of indium consumption. The environmental impacts caused by electricity production with micromorphous-Si PV and multicrystalline-Si panels are close to their respective market shares. The environmental impacts caused by the electricity production with monocrystalline-Si PV panels are higher than their market share. In summary CdTe, CIS, micro-Si and multi-Si PV systems contribute underproportionally to most of the environmental impacts of the average kWh, while monocrystalline silicon PV systems rather contribute overproportionally.

5.11 Sensitivity analyses

5.11.1 Overview

Three sensitivity analyses were conducted to test the results in terms of changes to the boundary conditions:

- Three different allocation approaches were applied for the allocation of benefits for recycling at the end-of-life stage
- The inverter is also considered within the mounting system
- Variation of life time and annual degradation rate

In the following sections 5.11.2 to 5.11.4 the results of sensitivity of these three additional analyses are shown.

5.11.2 Sensitivity of allocation approaches at the end-of-life stage

It is assumed that the PV modules are recycled at the end-of-life. The recycling processes modelled in the screening LCA and in the PEF supporting studies comply with the WEEE directive on the disposal of waste electrical and electronic equipment (EU Parliament 2012). Potential environmental benefits which result from recycling are allocated according to the formula provided in the recommendation of the European Commission (2013). Two different additional allocation approaches were applied:

- recycled content approach (100 % / 0 %)
- the avoided burden approach (0 % / 100 %)

Both alternative formulas are provided in the PEF guidance document (European Commission 2015). The sensitivity analysis is conducted for one PV system, namely electricity production with a residential scale CdTe PV system mounted on a slanted roof top. This system is chosen, because primary data on the end-of-life treatment of panels are available from First Solar. The end-of-life stage and the potential benefits for recycling of the other PV systems were derived from the former. Hence data quality is considered to be most reliable for the CdTe PV system.

Fig. 5.35 shows the relative results (characterized) of the sensitivity analyses of producing 1 kWh of DC electricity with a residential scale CdTe PV system mounted on a slanted roof. The results of the base case (allocation 50:50) are considered as 100 %. The results of the other allocation approaches are shown in relation to the base case. Naturally the recycled content approach leads to the highest impacts, as no benefits are allocated to the electricity produced. The avoided burden approach causes lower impacts, because all benefits are allocated to the electricity produced.

The change in total impacts varies from <1 % up to ± 67 % between the standard case and the sensitivity analyses, depending on the impact category. The impact category mineral, fossil and renewable resource consumption shows the highest sensitivity to the allocation approach chosen. Here the results of the two new allocation approaches differ with ± 67 % from the impacts of the standard case. In this impact category the resource consumption of cadmium and telluride from the CdTe panel, copper from the electrical installation as well as aluminium from the mounting structure are responsible for the high variability.

The impact category water resource depletion is likewise highly sensitive to the allocation approach and shows deviations of ± 36 % compared to the base case. This high sensitivity results from the supply chain of aluminium production and therein the electricity production with hydropower. The recycling of aluminium and hence the avoidance of producing primary aluminium results in high benefits.

Furthermore the impact category freshwater eutrophication features high sensitivity to the allocation approach applied. The results of the alternative allocation approaches deviate with ± 29 % from the standard case and is based on the fact that recycling copper (from the electrical installation) results in savings of phosphate emissions into water from off-site disposal of sulphidic tailings in the avoided supply chain of copper.

Impacts on the human toxicity cancer effects react rather sensitive to the allocation approach as well and show variances of ± 28 % compared to the standard case. The potential benefits for recycling, which are allocated to the electricity or the secondary product, come from recycling of aluminium. Within the supply chain of primary aluminium production the benefits result from savings of chromium VI released into water when disposing redmud from bauxite digestion to residual landfills.

The impact categories ionizing radiation HH, nuclear waste, particulate matter, climate change, acidification, CED non-renewable and land use show deviations from the standard case between $\pm 10\%$ and $\pm 25\%$ (categories ranked in descending order). For all of them the savings in emissions in the avoided supply chains of primary production when recycling aluminium and copper are responsible for the benefits allocated to the electricity or the secondary product.

The impact categories terrestrial eutrophication, photochemical ozone formation, ozone depletion, marine eutrophication, human toxicity non-cancer effects, freshwater ecotoxicity and CED renewable react only marginally sensitive to the allocation approach applied (categories ranked in descending order). The sensitivities range from 0% to $\pm 9\%$.

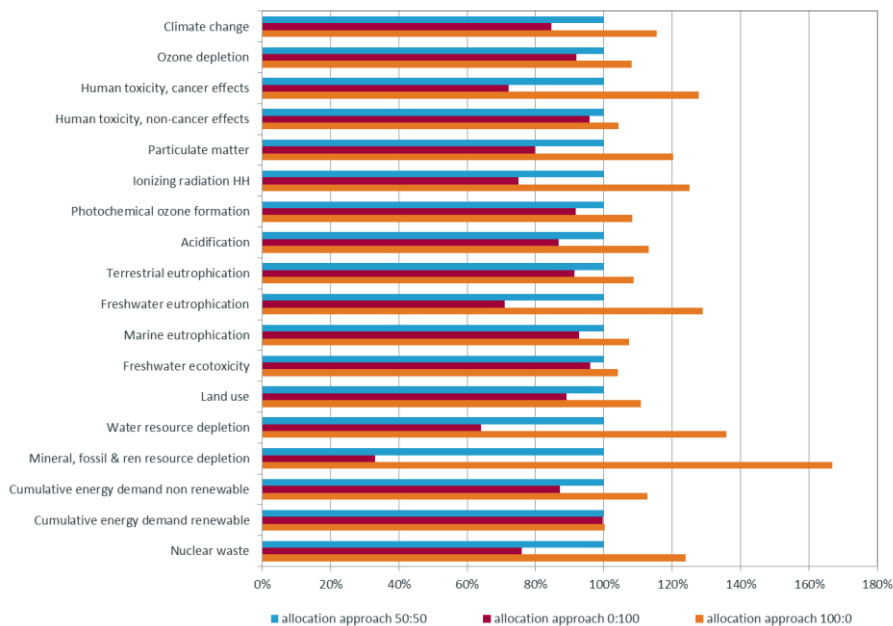


Fig. 5.35 Relative results (characterized) of the sensitivity analyses of allocation of benefits for recycling at the end-of-life stage when producing 1 kWh DC electricity with a residential scale CdTe PV system mounted on a slanted roof top. The allocation 50:50 is the base case and its impacts are indexed to 100 %. The results of the other allocation approaches are shown in relation to the base case.

It is assumed that the sensitivities for the electricity production with CIS, micromorphous-Si, multicrystalline-Si and monocrystalline-Si PV panels are comparable to the results shown here or even react less sensitive to the change in allocation applied. The smaller sensitivity is justified with the less detailed modelling of the potential benefits for recycling for these PV systems, as no data was available on the recovery of semiconductor or wafer material. The modelling is derived from the CdTe PV system. The CIS, micromorphous-Si, multicrystalline-Si and monocrystalline-Si PV panels are all equipped with aluminium frames and hence a larger amount of aluminium can be recycled from these systems compared to the CdTe PV system. The impact categories being dominated by the aluminium consumption and its supply chain react most sensitive (freshwater eutrophication, human toxicity cancer effects, ionizing radiation potential HH, nuclear waste and water resource depletion).

For CdTe PV panels the semiconductor materials cadmium and telluride are recycled. Hence the impact category mineral, fossil and renewable resource depletion reacts most sensitive. Because no semiconductor materials are gained back from CIS, micromorphous-Si, multicrystalline-Si and monocrystalline-Si panels, the sensitivity of their impact on mineral, fossil and renewable resource depletion is only marginal.

5.11.3 *Sensitivity of inclusion of the inverter to the mounting system*

The inverters transform electricity from direct current into alternating current. The frequency in Europe is 50 cycles per second. The electricity can be fed into the grid after transforming it to low-voltage level. In the standard case the inverter is not part of the PV system.

In this sensitivity analysis the manufacture of the inverter is included in order to illustrate its environmental impact and its influence on the overall impact of PV electricity. As the inverters are identical for all systems, the sensitivity analysis is only conducted for the residential scale multicrystalline-Si PV system mounted on a slanted roof.

The service life of an average inverter is assumed to last for 15 years (Frischknecht et al. 2015b). Hence, two inverters are required during the life time of the PV system considered in this study. The demand of the inverter with a capacity of 2.5 kWp is scaled to the maximum power output of the residential scale PV system.

The following Fig. 5.36 shows the relative results of the electricity production with and without the inclusion of the manufacture of the inverter. Depending on the impact category the inclusion of the inverter causes an additional impact of between 0 % and 83 %. The impact category freshwater eutrophication reacts most sensitive to the inclusion of the inverter to the PV system. The electricity production causes impacts which are 83 % higher than in the standard case. This additional impact is caused by the supply chain of copper (phosphate emissions when disposing sulphidic tailings off-site) used in the inverter. The impact category mineral, fossil and renewable resource depletion is sensitive to the inclusion of the inverter as well and shows a 56 % higher score. It is caused by the supply chains of silver and tantalum from the inverter. Human toxicity cancer effects and human toxicity non-cancer effects show sensitivities of 26 % and 37 %, respectively, which are caused by the supply chains of copper, steel and aluminium.

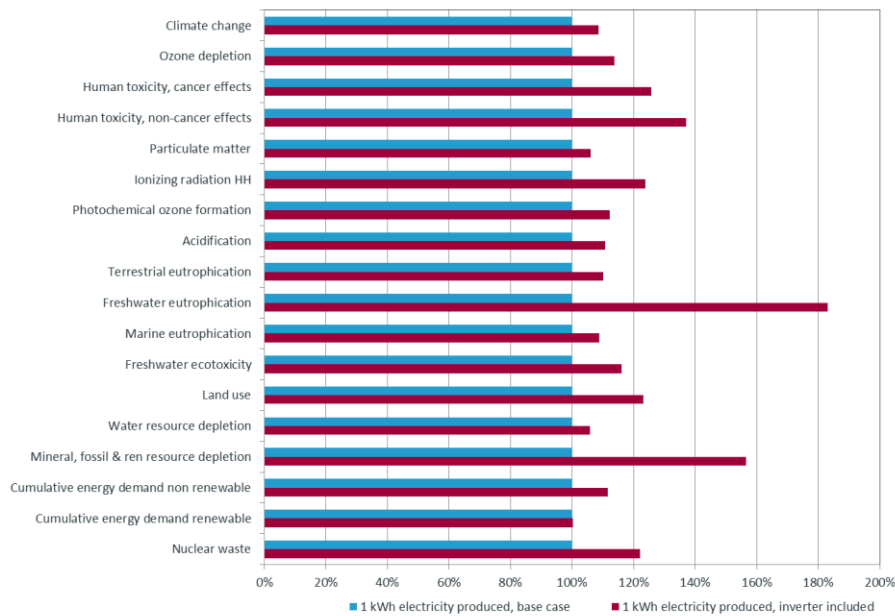


Fig. 5.36 Relative results (characterized) of the sensitivity analyses when producing 1 kWh of DC electricity including and excluding the inverter with a residential scale multi-Si PV system mounted on a slanted roof top. The base case represents the situation without inverter. Results are indexed to the base case (100 %).

5.11.4 Sensitivity of life time and degradation rate

In this third sensitivity analysis the life time and annual degradation rate are varied. A degradation rate of 0.5 % per year, including 2 % in the first year and a life span of 25 years are assumed according to Frischknecht and Itten (2014). This degradation results in a loss in yield of 14 % in the 25th year of operation of the PV system. Hence, the weighted average yield of a PV module operated during 25 years is 7 % below the average yield of 1090 kWh per kWp and amounts to 1013 kWh/kWp. The sensitivity analysis is calculated on the 3 kWp PV installation with micromorphous-Si PV panels mounted on a slanted roof top.

The environmental impacts of electricity production with a degradation rate of 0.5 % and a service life time of 25 years is shown in Fig. 5.37. The results are given in relation to the electricity produced with an average annual yield of 975 kWp/kWh, a degradation rate of 0.7 % and a life span of 30 years (base case). Included are all life cycle stages from product stage, construction stage, operation stage to the end-of-life stage and potential benefits for recycling. An enhanced yield, based on a reduction in the degradation rate would lead to diminished environmental impacts. However any environmental benefits are overcompensated by the reduction in service life time and as a result the environmental impacts are larger compared to the base case.

The environmental impacts of all life cycle stages of electricity produced with this adapted PV system are approximately 15 % higher compared to the base case. The environmental performance is directly related to the system performance. Hence the change of the environmental impacts is equal for all impact categories. The only exception is the impact category CED renewable, where naturally no significant differences occur. This is due to the dominance of the solar energy harvested by the PV panel to produce electricity.

These tendencies are independent of the PV technology and very similar results would arise when assessing the environmental impacts of electricity production with a reduced degradation rate and a reduced service life time.

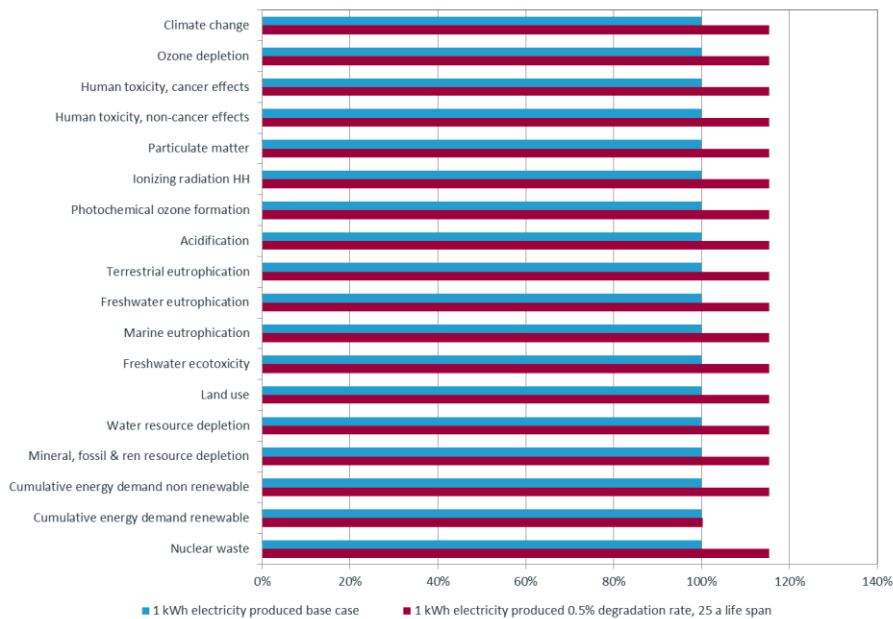


Fig. 5.37 Relative results (characterized) of the sensitivity analyses when producing 1 kWh of DC electricity applying a degradation rate of 0.5 % (2 % in the first year) and a life span of 25 years with a residential scale multi-Si PV system mounted on a slanted roof top. The base case represents the situation with a degradation rate of 0.7 % and a life span of 30 years. All results are indexed to the base case (100 %).

6. Interpretation

6.1 Overview

The hotspots of PV electricity generation with the different technologies are identified with regard to the most relevant impact categories, life cycle stages, processes and elementary flows. This analysis is done for the 3 kWp residential scale PV system mounted on a slanted roof. The most relevant impact categories are presented in subchapter 6.2. In subchapter 6.3 the environmental hotspots per PV technology are illustrated. Subchapter 6.4 lists the most relevant life cycle stages, subchapter 6.5 presents the most relevant processes subchapter 6.6 shows the most relevant elementary flows. In addition, the matching of the ILCD method with the elementary flows is assessed in subchapter 6.7.

6.2 Most relevant impact categories

The most relevant impact categories are similar for all five PV technologies but in changing orders:

- mineral, fossil and renewable resource depletion
- human toxicity (cancer and non-cancer effects)
- freshwater ecotoxicity
- particulate matter potential
- acidification potential

The significance of each impact category is given based on the weighed results of the production of 1 kWh DC electricity with a 3 kWp PV installation mounted on a slanted roof top.

Regarding electricity produced with CdTe PV panels on residential scale plants mounted on roof tops, the most important impact categories are:

1. mineral, fossil and renewable resource depletion (43 % of weighted results)
2. human toxicity cancer effects (18 % of weighted results)
3. human toxicity non cancer effects (10 % of weighted results)
4. freshwater ecotoxicity (9 % of weighted results)

5. acidification potential (4 % of weighted results)

Electricity production with a 3 kWp PV system with CIS PV panels causes most environmental impacts in the following impact categories:

1. mineral, fossil and renewable resource depletion (98 % of weighted results)
2. human toxicity cancer effects (0.5 % of weighted results)
3. human toxicity non cancer effects (0.3 % of weighted results)
4. freshwater ecotoxicity (0.2 % of weighted results)
5. acidification potential (0.1 % of weighted results)

The most important environmental impact categories when producing electricity with a 3 kWp PV system with micromorphous-Si PV panels are:

1. human toxicity cancer effects (25 % of weighted results)
2. human toxicity non cancer effects (15 % of weighted results)
3. mineral, fossil and renewable resource depletion (12 % of weighted results)
4. particulate matter potential (12 % of weighted results)
5. freshwater ecotoxicity (8 % of weighted results)

Regarding electricity produced with multicrystalline-Si PV panels on residential scale plants mounted on roof tops, the most important impact categories are:

1. mineral, fossil and renewable resource depletion (62 % of weighted results)
2. human toxicity cancer effects (9 % of weighted results)
3. human toxicity non cancer effects (6 % of weighted results)
4. particulate matter potential (6 % of weighted results)
5. freshwater ecotoxicity (4 % of weighted results)

The most relevant impact categories regarding the electricity production with monocrystalline-Si PV panels on a residential scale PV system are:

1. mineral, fossil and renewable resource depletion (43 % of weighted results)
2. particulate matter (12 % of weighted results)
3. human toxicity cancer effects (11 % of weighted results)
4. human toxicity non cancer effects (9 % of weighted results)
5. acidification potential (6 % of weighted results)

6.3 Environmental hotspots

The environmental hotspots are identified as the two most relevant life cycle stages, processes and elementary flows based on the characterized results for each impact category (i.e., according to option B of the Product Environmental Footprint Pilot Guidance (version 5.2, European Commission 2015)). The environmental hotspots for the five PV technologies analysed are listed in Tab. 6.1 to Tab. 6.5.

In the following paragraphs, the environmental hotspots are shortly highlighted and explained. The focus of this analysis is on the most important impact categories listed in Subchapter 6.2. Detailed information on the most relevant life cycle stages, processes and elementary flows for each impact category can be found in the Subchapters 6.4, 6.5 and 6.6, respectively.

The two **most relevant life cycle stages** are the product and the construction stage for all PV technologies analysed and for most of the impact categories. The use stage is the most important life cycle stage when considering the renewable cumulative energy demand. The end-of-life stage is

among the two most important life cycle stages in a few impact categories (human toxicity, cancer effects, mineral, fossil and renewable resource depletion, freshwater ecotoxicity, water resource depletion) only in the case of CdTe PV modules mounted on a slanted roof.

For electricity generated with a residential scale **CdTe PV system** mounted on a slanted roof, the most relevant impact category is mineral, fossil and renewable resource depletion. The environmental impacts in this category are mainly caused by the mining and production of cadmium, silver and tellurium. The silver resource, which is identified as the second elementary flow hotspot, is used in the production of copper telluride. The impact category human toxicity cancer effects is primarily influenced by the supply chains of aluminium (redmud disposal from bauxite mining) and steel (slag disposal). Chromium emissions to water and air are identified as the two most relevant elementary flows. These substances are emitted during the production of steel. The impact category human toxicity non cancer effects is dominated by heavy metal emissions to air and water caused by the production of copper. Arsenic emissions to water, which are mainly caused by natural gas extraction, and lead emissions to air caused by sinter iron production are the two most relevant elementary flows in the impact category human toxicity non cancer effects. The impacts in the category freshwater ecotoxicity are dominated by the treatment of plastics from the electric installation in municipal incineration plants. The second process hotspot is the disposal of redmud from bauxite digestion. Emissions of antimony to water are identified as the most relevant elementary flow and originate from natural gas extraction. Chromium VI, the second most important substance, is emitted to water during the beneficiation of iron ore. The acidification impacts are mainly caused by ship transports and by flat glass production. The hotspot substances sulfur dioxide (air) and nitrogen oxides (air) are mainly emitted by flat glass production. The aluminium supply chain is another important source of sulfur dioxide emissions to air.

The impact category mineral, fossil and renewable resource depletion strongly dominates the weighted environmental impacts of electricity produced with a residential scale **CIS PV system** mounted on a slanted roof. The extraction of indium is responsible for the major share of the environmental impacts in this category. Similar to the CdTe PV system, the supply chains of aluminium and steel are most relevant when considering the impact category human toxicity cancer effects. The chromium emissions to water mainly occur during the extraction of iron ore and the chromium emissions to air are primarily caused by steel production. The two most relevant processes in the impact category human toxicity non cancer effects are the production of primary zinc and of indium-rich leaching residues. The hotspot substances zinc and mercury are emitted to air in the production of steel. The freshwater ecotoxicity impacts are mainly caused by the incineration of plastic components of the electric installation and by the disposal of redmud from bauxite digestion. The most relevant elementary flows in this impact category, namely antimony and chromium VI emissions to water, are mainly caused by the extraction of natural gas and the processing of iron ore, respectively. The fifth most important impact category is the acidification potential, which is primarily caused by transoceanic freight ships and the production of flat glass. The substances sulfur dioxide and nitrogen oxides are mostly emitted by the latter process.

The most relevant impact category for the residential scale **micromorphous-Si PV system** mounted on a slanted roof is human toxicity cancer effects. The two most relevant processes are the disposal of redmud from bauxite digestion and the disposal of slag from steel production. The most relevant elementary flows in this impact category are the same as for the CdTe and CIS PV technologies, namely chromium VI emissions to water and chromium emissions to air. Both of them are mainly associated with the supply chain of steel production. The process hotspots in the impact category human toxicity non cancer effects are primary zinc and primary copper production. The most

important elementary flow is zinc emitted to air by steel production and by lorry transports. Arsenic emissions to water are mainly caused by the beneficiation of iron ore. The third most important impact category is mineral, fossil and renewable resource depletion with zinc concentrate at beneficiation being the most important process. The raw elementary flow silver is mainly used in parkes process crust from desilvering of lead. The burning of hard coal in Chinese power plants dominates the particulate matter potential. The emissions of PM_{2.5} and sulfur dioxide, the most relevant elementary flows in this impact category, are caused by the production of aluminium and flat glass. The incineration of plastics and the disposal of redmud from bauxite digestion are identified as the two most relevant processes in the impact category freshwater ecotoxicity. These processes and also the elementary flow hotspots are in perfect agreement with the results observed for CdTe and CIS PV systems.

For electricity produced with a residential scale **multicrystalline-Si PV** system mounted on a slanted roof, the impact category mineral, fossil and renewable resource depletion is most important. The first process hotspot in this impact category is the resource correction for silver used in the production of parkes process crust from desilvering of lead. Also the second most relevant process is associated with the silver supply chain, namely the production of silver and tellurium containing anode slime. Silver is then used in the metallization paste for the production of multicrystalline-Si PV cells. The most relevant elementary flows with regard to mineral, fossil and renewable resource depletion are silver and tellurium. The impact category human toxicity, cancer effects is dominated by the disposal of redmud from bauxite digestion and the disposal of slag from the production of unalloyed electric steel. Emissions of chromium VI to water and of chromium to air are identified as substance hotspots. The impact category human toxicity non cancer effects are mainly caused by the production of primary aluminium and by the disposal of hard coal ash. The most relevant elementary flow, arsenic emitted to water, originates from the beneficiation of iron ore and from natural gas extraction. The mercury emissions to air, which is the second substance hotspot in the impact category human toxicity non cancer effects, are released by the production of unalloyed electric steel. The particulate matter impacts are dominated by the burning of hard coal in Chinese power plants. The emissions of PM_{2.5} are associated with the electrolysis of aluminium oxide and silicon production. Sulfur dioxide is emitted to air during the production of flat glass, primary aluminium and silicon. Freshwater ecotoxicity, the fifth most important impact category, has the same environmental hotspots as for the PV technologies described previously.

The depletion of mineral, fossil and renewable resources is the most important impact category for electricity produced with a 3 kWp **monocrystalline-Si PV system** mounted on a slanted roof. The process and substance hotspots in this impact category are identical to those of the multicrystalline-Si PV system. The particulate matter potential is dominated by the production of electricity in Chinese hard coal power plants. The major part of PM_{2.5} emissions are emitted in the aluminium electrolysis process. The emissions of sulfur dioxide are primarily caused by the production of flat glass, liquid primary aluminium and MG-silicon. The third most relevant impact category is human toxicity cancer effects, which has the same process and substance hotspots as the ones identified for the other PV technologies. The human toxicity non cancer effects are mainly caused by the disposal of hard coal ash and by primary copper production. The first substance hotspot, arsenic emitted to water, is associated with the extraction of natural gas in North America and with the beneficiation of iron ore. Mercury emissions to air occur during the production of electric steel. The process hotspots with regard to the acidification potential are the electricity generation by hard coal power plants as well as the mining of hard coal.

Tab. 6.1 Identification of the two most relevant life cycle stages, processes and elementary flows for each impact category for 1 kWh of DC electricity produced with a residential scale CdTe PV system mounted on a slanted roof.

Rank	Impact category	Unit per kWh electricity	Most relevant life cycle stages	Most relevant processes	Most relevant elementary flows
	electricity, 3 kWp installation, mounted with CdTe PV technology, characterized				
1	Climate change	kg CO2 eq	Product stage	Aluminium, primary, liquid, at plant/RER U	Carbon dioxide, fossil / Air
2	Climate change	kg CO2 eq	Construction stage	Flat glass, uncoated, at plant/RER U	Methane, tetrafluoro-, CFC-14 / Air
1	Ozone depletion	kg CFC-11 eq	Product stage	transport, natural gas, pipeline, long distance/tkm/RU U	Methane, bromochlorodifluoro-, Halon 1211 / Air
2	Ozone depletion	kg CFC-11 eq	Construction stage	Transport, natural gas, onshore pipeline, long distance/tkm/NAC U	Methane, bromotrifluoro-, Halon 1301 / Air
1	Human toxicity, cancer effects	CTUh, c	Construction stage	Disposal, redmud from bauxite digestion, 0% water, to residual material landfill/CH U	Chromium VI / Water
2	Human toxicity, cancer effects	CTUh, c	End-of-life stage	Disposal, slag, unalloyed electr. steel, 0% water, to residual material landfill/CH U	Chromium / Air
1	Human toxicity, non-cancer effects	CTUh, n-c	Construction stage	Copper, primary, at refinery/RLA U	Arsenic / Water
2	Human toxicity, non-cancer effects	CTUh, n-c	Product stage	Disposal, redmud from bauxite digestion, 0% water, to residual material landfill/CH U	Lead / Air
1	Particulate matter	kg PM2.5 eq	Construction stage	Aluminium, primary, liquid, at plant/RER U	Particulates, < 2.5 um / Air
2	Particulate matter	kgBq U235 eq	Product stage	Flat glass, uncoated, at plant/RER U	Sulfur dioxide / Air
1	Ionizing radiation HH	kg PM2.5 eq	Construction stage	Nuclear spent fuel, in reprocessing, at plant/RER U	Carbon-14 / Air
2	Ionizing radiation HH	kg PM2.5 eq	Product stage	Uranium natural, at underground mine/RNA U	Radon-222 / Air
1	Photochemical ozone formation	kg NMVOC eq	Product stage	Operation, transoceanic freight ship/OCE U	Nitrogen oxides / Air
2	Photochemical ozone formation	kg NMVOC eq	Construction stage	Flat glass, uncoated, at plant/RER U	Sulfur dioxide / Air
1	Acidification	mol H+ eq	Product stage	Operation, transoceanic freight ship/OCE U	Sulfur dioxide / Air
2	Acidification	mol H+ eq	Construction stage	Flat glass, uncoated, at plant/RER U	Nitrogen oxides / Air
1	Terrestrial eutrophication	mol N eq	Product stage	Operation, transoceanic freight ship/OCE U	Nitrogen oxides / Air
2	Terrestrial eutrophication	mol N eq	Construction stage	Flat glass, uncoated, at plant/RER U	Ammonia / Air
1	Freshwater eutrophication	kg P eq	Construction stage	Disposal, sulfidic tailings, off-site/GLO U	Phosphate / Water
2	Freshwater eutrophication	kg P eq	End-of-life stage	Disposal, spoil from coal mining, in surface landfill/GLO U	Phosphorus / Water
1	Marine eutrophication	kg N eq	Product stage	Operation, transoceanic freight ship/OCE U	Nitrogen oxides / Air
2	Marine eutrophication	kg N eq	Construction stage	Flat glass, uncoated, at plant/RER U	Nitrate / Water
1	Freshwater ecotoxicity	CTUe	Construction stage	disposal, plastic, industr. electronics, 15.3% water, to municipal incineration/kg/CH U	Antimony / Water
2	Freshwater ecotoxicity	CTUe	Product stage	Disposal, redmud from bauxite digestion, 0% water, to residual material landfill/CH U	Chromium VI / Water
1	Land use	kg C deficit	Product stage	Well for exploration and production, onshore/GLO/I U	Transformation, to mineral extraction site / Raw
2	Land use	kg C deficit	Construction stage	Well for exploration and production, offshore/OCE/I U	Transformation, to arable, non-irrigated / Raw
1	Water resource depletion	m3 water eq	Construction stage	electricity, hydropower, at run-of-river power plant without reservoir/kWh/RER U	Water, OECD / Air
2	Water resource depletion	m3 water eq	End-of-life stage	tap water, at user/kg/RER U	Water, cooling, unspecified natural origin/m3 / Raw
1	Mineral, fossil & ren resource depletion	kg Sb eq	Product stage	Resource correction, PbZn, cadmium, positive/GLO U	Cadmium / Raw
2	Mineral, fossil & ren resource depletion	kg Sb eq	End-of-life stage	Zinc concentrate, at beneficiation/GLO U	Silver / Raw
1	Cumulative energy demand non renewable	MJ oil eq	Product stage	natural gas, at production onshore/m3/NAC U	Gas, natural/m3 / Raw
2	Cumulative energy demand non renewable	MJ oil eq	Construction stage	natural gas, at production offshore/m3/NO U	Oil, crude / Raw
1	Cumulative energy demand renewable	MJ oil eq	Use stage	hotspot electricity, PV, at 3kWp slanted-roof, CdTe, panel, mounted, incl. EoL absolute/RER	Energy, solar, converted / Raw
2	Cumulative energy demand renewable	MJ oil eq	Construction stage	electricity, hydropower, at run-of-river power plant without reservoir/kWh/RER U	Energy, potential (in hydropower reservoir), converted / Raw
1	Nuclear waste	m3 HAA eq	Construction stage	Radioactive waste, in final repository for nuclear waste SF, HLW, and ILW/CH U	Volume occupied, final repository for radioactive waste / Raw
2	Nuclear waste	m3 HAA eq	Product stage	Titanium dioxide, chloride process, at plant/RER S	Volume occupied, final repository for low-active radioactive waste / Raw

Tab. 6.2 Identification of the two most relevant life cycle stages, processes and elementary flows for each impact category for 1 kWh of DC electricity produced with a residential scale CIS PV system mounted on a slanted roof.

Rank	Impact category	Unit per kWh electricity	Most relevant life cycle stages	Most relevant processes	Most relevant elementary flows
	electricity, 3 kWp installation, mounted with CIS PV technology, characterized				
1	Climate change	kg CO2 eq	Product stage	Lignite, burned in power plant/DE U	Carbon dioxide, fossil / Air
2	Climate change	kg CO2 eq	Construction stage	Aluminium, primary, liquid, at plant/RER U	Methane, tetrafluoro-, CFC-14 / Air
1	Ozone depletion	kg CFC-11 eq	Product stage	transport, natural gas, pipeline, long distance/tkm/RU U	Methane, bromochlorodifluoro-, Halon 1211 / Air
2	Ozone depletion	kg CFC-11 eq	Construction stage	Methylchloride, at plant/WEU U	Ethane, 1,2-dichloro-1,1,2,2-tetrafluoro-, CFC-114 / Air
1	Human toxicity, cancer effects	CTUh, c	Construction stage	Disposal, redmud from bauxite digestion, 0% water, to residual material landfill/CH U	Chromium VI / Water
2	Human toxicity, cancer effects	CTUh, c	Product stage	Disposal, slag, unalloyed electr. steel, 0% water, to residual material landfill/CH U	Chromium / Air
1	Human toxicity, non-cancer effects	CTUh, n-c	Construction stage	Zinc, primary, at regional storage/RER U	Zinc / Air
2	Human toxicity, non-cancer effects	CTUh, n-c	Product stage	Leaching residues, indium rich, from zinc circuit, at smelter/GLO U	Mercury / Air
1	Particulate matter	kg PM2.5 eq	Product stage	Aluminium, primary, liquid, at plant/RER U	Particulates, < 2.5 um / Air
2	Particulate matter	kg PM2.5 eq	Construction stage	Flat glass, uncoated, at plant/RER U	Sulfur dioxide / Air
1	Ionizing radiation HH	kg PM2.5 eq	Product stage	Nuclear spent fuel, in reprocessing, at plant/RER U	Carbon-14 / Air
2	Ionizing radiation HH	kg PM2.5 eq	Construction stage	Uranium natural, at underground mine/RNA U	Radon-222 / Air
1	Photochemical ozone formation	kg NMVOC eq	Product stage	Operation, transoceanic freight ship/OCE U	Nitrogen oxides / Air
2	Photochemical ozone formation	kg NMVOC eq	Construction stage	Flat glass, uncoated, at plant/RER U	NMVOC, non-methane volatile organic compounds, unspecified origin / Air
1	Acidification	mol H+ eq	Product stage	Operation, transoceanic freight ship/OCE U	Sulfur dioxide / Air
2	Acidification	mol H+ eq	Construction stage	Flat glass, uncoated, at plant/RER U	Nitrogen oxides / Air
1	Terrestrial eutrophication	mol N eq	Product stage	Operation, transoceanic freight ship/OCE U	Nitrogen oxides / Air
2	Terrestrial eutrophication	mol N eq	Construction stage	Flat glass, uncoated, at plant/RER U	Ammonia / Air
1	Freshwater eutrophication	kg P eq	Construction stage	Disposal, sulfidic tailings, off-site/GLO U	Phosphate / Water
2	Freshwater eutrophication	kg P eq	Product stage	Disposal, spoil from lignite mining, in surface landfill/GLO U	Phosphorus / Water
1	Marine eutrophication	kg N eq	Product stage	Operation, transoceanic freight ship/OCE U	Nitrogen oxides / Air
2	Marine eutrophication	kg N eq	Construction stage	Flat glass, uncoated, at plant/RER U	Nitrate / Water
1	Freshwater ecotoxicity	CTUe	Construction stage	disposal, plastic, industr. electronics, 15.3% water, to municipal incineration/kg/CH U	Antimony / Water
2	Freshwater ecotoxicity	CTUe	Product stage	Disposal, redmud from bauxite digestion, 0% water, to residual material landfill/CH U	Chromium VI / Water
1	Land use	kg C deficit	Product stage	Photovoltaic panel factory/GLO/I U	Transformation, to mineral extraction site / Raw
2	Land use	kg C deficit	Construction stage	Well for exploration and production, onshore/GLO/I U	Transformation, from unknown / Raw
1	Water resource depletion	m3 water eq	Product stage	electricity, hydropower, at reservoir power plant, non alpine regions/kWh/RER U	Water, OECD / Air
2	Water resource depletion	m3 water eq	Construction stage	electricity, hydropower, at pumped storage power plant/kWh/DE U	Water, cooling, unspecified natural origin/m3 / Raw
1	Mineral, fossil & ren resource depletion	kg Sb eq	Product stage	Resource correction, PbZn, indium, positive/GLO U	Indium / Raw
2	Mineral, fossil & ren resource depletion	kg Sb eq	Construction stage	Zinc concentrate, at beneficiation/GLO U	Tin / Raw
1	Cumulative energy demand non renewable	MJ oil eq	Product stage	Uranium natural, at underground mine/RNA U	Oil, crude / Raw
2	Cumulative energy demand non renewable	MJ oil eq	Construction stage	Lignite, at mine/RER U	Gas, natural/m3 / Raw
1	Cumulative energy demand renewable	MJ oil eq	Use stage	hotspot electricity, PV, at 3kWp slanted-roof, CIS, panel, mounted, incl. EoL absolute/RER U	Energy, solar, converted / Raw
2	Cumulative energy demand renewable	MJ oil eq	Product stage	electricity, hydropower, at run-of-river power plant without reservoir/kWh/RER U	Energy, potential (in hydropower reservoir), converted / Raw
1	Nuclear waste	m3 HAA eq	Product stage	Radioactive waste, in final repository for nuclear waste SF, HLW, and ILW/CH U	Volume occupied, final repository for radioactive waste / Raw
2	Nuclear waste	m3 HAA eq	Construction stage	Radioactive waste, in final repository for nuclear waste LLW/CH U	Volume occupied, final repository for low-active radioactive waste / Raw

Tab. 6.3 Identification of the two most relevant life cycle stages, processes and elementary flows for each impact category for 1 kWh of DC electricity produced with a residential scale micro-Si PV system mounted on a slanted roof.

Rank	Impact category	Unit per kWh electricity	Most relevant life cycle stages	Most relevant processes	Most relevant elementary flows
	electricity, 3 kWp installation, mounted with micro-Si PV technology, characterized				
1	Climate change	kg CO2 eq	Product stage	Hard coal, burned in power plant/CN U	Carbon dioxide, fossil / Air
2	Climate change	kg CO2 eq	Construction stage	Aluminium, primary, liquid, at plant/RER U	Nitrogen fluoride / Air
1	Ozone depletion	kg CFC-11 eq	Product stage	transport, natural gas, pipeline, long distance/tkm/RU U	Methane, bromochlorodifluoro-, Halon 1211 / Air
2	Ozone depletion	kg CFC-11 eq	Construction stage	Crude oil, at production/NG U	Methane, bromotrifluoro-, Halon 1301 / Air
1	Human toxicity, cancer effects	CTUh, c	Construction stage	Disposal, redmud from bauxite digestion, 0% water, to residual material landfill/CH U	Chromium VI / Water
2	Human toxicity, cancer effects	CTUh, c	Product stage	Disposal, slag, unalloyed electr. steel, 0% water, to residual material landfill/CH U	Chromium / Air
1	Human toxicity, non-cancer effects	CTUh, n-c	Construction stage	Zinc, primary, at regional storage/RER U	Zinc / Air
2	Human toxicity, non-cancer effects	CTUh, n-c	Product stage	Copper, primary, at refinery/RLA U	Arsenic / Water
1	Particulate matter	kg PM2.5 eq	Product stage	Hard coal, burned in power plant/CN U	Particulates, < 2.5 um / Air
2	Particulate matter	kBq U235 eq	Construction stage	Hard coal, burned in coal mine power plant/CN U	Sulfur dioxide / Air
1	Ionizing radiation HH	kg PM2.5 eq	Construction stage	Nuclear spent fuel, in reprocessing, at plant/RER U	Carbon-14 / Air
2	Ionizing radiation HH	kg PM2.5 eq	Product stage	Uranium natural, at underground mine/RNA U	Radon-222 / Air
1	Photochemical ozone formation	kg NMVOC eq	Product stage	Hard coal, burned in power plant/CN U	Nitrogen oxides / Air
2	Photochemical ozone formation	kg NMVOC eq	Construction stage	Operation, transoceanic freight ship/OCE U	Sulfur dioxide / Air
1	Acidification	mol H+ eq	Product stage	Hard coal, burned in power plant/CN U	Sulfur dioxide / Air
2	Acidification	mol H+ eq	Construction stage	Operation, transoceanic freight ship/OCE U	Nitrogen oxides / Air
1	Terrestrial eutrophication	mol N eq	Product stage	Hard coal, burned in power plant/CN U	Nitrogen oxides / Air
2	Terrestrial eutrophication	mol N eq	Construction stage	Operation, transoceanic freight ship/OCE U	Ammonia / Air
1	Freshwater eutrophication	kg P eq	Construction stage	Disposal, sulfidic tailings, off-site/GLO U	Phosphate / Water
2	Freshwater eutrophication	kg P eq	Product stage	Disposal, spoil from coal mining, in surface landfill/GLO U	Phosphorus / Water
1	Marine eutrophication	kg N eq	Product stage	Hard coal, burned in power plant/CN U	Nitrogen oxides / Air
2	Marine eutrophication	kg N eq	Construction stage	Operation, transoceanic freight ship/OCE U	Ammonia / Air
1	Freshwater ecotoxicity	CTUe	Construction stage	disposal, plastic, industr. electronics, 15.3% water, to municipal incineration/kg/CH U	Antimony / Water
2	Freshwater ecotoxicity	CTUe	Product stage	Disposal, redmud from bauxite digestion, 0% water, to residual material landfill/CH U	Chromium VI / Water
1	Land use	kg C deficit	Product stage	Well for exploration and production, onshore/GLO/I U	Transformation, to mineral extraction site / Raw
2	Land use	kg C deficit	Construction stage	Well for exploration and production, offshore/OCE/I U	Transformation, from unknown / Raw
1	Water resource depletion	m3 water eq	Product stage	electricity, hydropower, at reservoir power plant, non alpine regions/kWh/RER U	Water, OECD / Air
2	Water resource depletion	m3 water eq	Construction stage	photovoltaic panel, micro-Si, at plant/m2/CN/I U	Water, CN / Air
1	Mineral, fossil & ren resource depletion	kg Sb eq	Product stage	Zinc concentrate, at beneficiation/GLO U	Silver / Raw
2	Mineral, fossil & ren resource depletion	kg Sb eq	Construction stage	Resource correction, PbZn, indium, negative/GLO U	Copper / Raw
1	Cumulative energy demand non renewable	MJ oil eq	Product stage	Hard coal, at mine/CN U	Coal, hard / Raw
2	Cumulative energy demand non renewable	MJ oil eq	Construction stage	Uranium natural, at underground mine/RNA U	Oil, crude / Raw
1	Cumulative energy demand renewable	MJ oil eq	Use stage	hotspot electricity, PV, at 3kWp slanted-roof, micro-Si, panel, mounted, incl. EoL absolute/F	Energy, solar, converted / Raw
2	Cumulative energy demand renewable	MJ oil eq	Construction stage	electricity, hydropower, at reservoir power plant, non alpine regions/kWh/RER U	Energy, potential (in hydropower reservoir), converted / Raw
1	Nuclear waste	m3 HAA eq	Construction stage	Radioactive waste, in final repository for nuclear waste SF, HLW, and ILW/CH U	Volume occupied, final repository for radioactive waste / Raw
2	Nuclear waste	m3 HAA eq	Product stage	Zeolite, slurry, 50% in H2O, at plant/RER S	Volume occupied, final repository for low-active radioactive waste / Raw

Tab. 6.4 Identification of the two most relevant life cycle stages, processes and elementary flows for each impact category for 1 kWh of DC electricity produced with a residential scale multi-Si PV system mounted on a slanted roof.

Rank	Impact category	Unit per kWh electricity	Most relevant life cycle stages	Most relevant processes	Most relevant elementary flows
	electricity, 3 kWp installation, mounted with multi-Si PV technology, characterized				
1	Climate change	kg CO2 eq	Product stage	Hard coal, burned in power plant/CN U	Carbon dioxide, fossil / Air
2	Climate change	kg CO2 eq	Construction stage	Hard coal, at mine/CN U	Methane, fossil / Air
1	Ozone depletion	kg CFC-11 eq	Product stage	transport, natural gas, pipeline, long distance/tkm/RU U	Methane, bromochlorodifluoro-, Halon 1211 / Air
2	Ozone depletion	kg CFC-11 eq	Construction stage	Chlorine, gaseous, mercury cell, at plant/RER U	Methane, tetrachloro-, CFC-10 / Air
1	Human toxicity, cancer effects	CTUh, c	Construction stage	Disposal, redmud from bauxite digestion, 0% water, to residual material landfill/CH U	Chromium VI / Water
2	Human toxicity, cancer effects	CTUh, c	Product stage	Disposal, slag, unalloyed electr. steel, 0% water, to residual material landfill/CH U	Chromium / Air
1	Human toxicity, non-cancer effects	CTUh, n-c	Product stage	Copper, primary, at refinery/RLA U	Arsenic / Water
2	Human toxicity, non-cancer effects	CTUh, n-c	Construction stage	Disposal, hard coal ash, 0% water, to residual material landfill/PL U	Mercury / Air
1	Particulate matter	kg PM2.5 eq	Product stage	Hard coal, burned in power plant/CN U	Particulates, < 2.5 um / Air
2	Particulate matter	kg PM2.5 eq	Construction stage	Hard coal, burned in coal mine power plant/CN U	Sulfur dioxide / Air
1	Ionizing radiation HH	kg PM2.5 eq	Product stage	Nuclear spent fuel, in reprocessing, at plant/RER U	Carbon-14 / Air
2	Ionizing radiation HH	kg PM2.5 eq	Construction stage	Uranium natural, at underground mine/RNA U	Radon-222 / Air
1	Photochemical ozone formation	kg NMVOC eq	Product stage	Hard coal, burned in power plant/CN U	Nitrogen oxides / Air
2	Photochemical ozone formation	kg NMVOC eq	Construction stage	Operation, transoceanic freight ship/OCE U	Sulfur dioxide / Air
1	Acidification	mol H+ eq	Product stage	Hard coal, burned in power plant/CN U	Sulfur dioxide / Air
2	Acidification	mol H+ eq	Construction stage	Hard coal, at mine/CN U	Nitrogen oxides / Air
1	Terrestrial eutrophication	mol N eq	Product stage	Hard coal, burned in power plant/CN U	Nitrogen oxides / Air
2	Terrestrial eutrophication	mol N eq	Construction stage	Operation, transoceanic freight ship/OCE U	Ammonia / Air
1	Freshwater eutrophication	kg P eq	Construction stage	Disposal, sulfidic tailings, off-site/GLO U	Phosphate / Water
2	Freshwater eutrophication	kg P eq	Product stage	Disposal, spoil from coal mining, in surface landfill/GLO U	Phosphorus / Water
1	Marine eutrophication	kg N eq	Product stage	Hard coal, burned in power plant/CN U	Nitrogen oxides / Air
2	Marine eutrophication	kg N eq	Construction stage	Operation, transoceanic freight ship/OCE U	Nitrate / Water
1	Freshwater ecotoxicity	CTUe	Construction stage	disposal, plastic, industr. electronics, 15.3% water, to municipal incineration/kg/CH U	Antimony / Water
2	Freshwater ecotoxicity	CTUe	Product stage	Disposal, redmud from bauxite digestion, 0% water, to residual material landfill/CH U	Chromium VI / Water
1	Land use	kg C deficit	Product stage	Well for exploration and production, onshore/GLO/I U	Transformation, to mineral extraction site / Raw
2	Land use	kg C deficit	Construction stage	Photovoltaic panel factory/GLO/I U	Transformation, to arable, non-irrigated / Raw
1	Water resource depletion	m3 water eq	Product stage	electricity, hydropower, at reservoir power plant, non alpine regions/kWh/RER U	Water, OECD / Air
2	Water resource depletion	m3 water eq	Construction stage	silicon, electronic grade, at plant/kg/CN U	Water, cooling, unspecified natural origin/m3 / Raw
1	Mineral, fossil & ren resource depletion	kg Sb eq	Product stage	Resource correction, PbZn, silver, positive/GLO U	Silver / Raw
2	Mineral, fossil & ren resource depletion	kg Sb eq	Construction stage	Anode slime, silver and tellurium containing, primary copper production/GLO U	Tellurium / Raw
1	Cumulative energy demand non renewable	MJ oil eq	Product stage	Hard coal, at mine/CN U	Coal, hard / Raw
2	Cumulative energy demand non renewable	MJ oil eq	Construction stage	Uranium natural, at underground mine/RNA U	Oil, crude / Raw
1	Cumulative energy demand renewable	MJ oil eq	Use stage	hotspot electricity, PV, at 3kWp slanted-roof, multi-Si, panel, mounted, incl. EoL absolute/R	Energy, solar, converted / Raw
2	Cumulative energy demand renewable	MJ oil eq	Product stage	electricity, hydropower, at run-of-river power plant without reservoir/kWh/RER U	Energy, potential (in hydropower reservoir), converted / Raw
1	Nuclear waste	m3 HAA eq	Product stage	Radioactive waste, in final repository for nuclear waste SF, HLW, and ILW/CH U	Volume occupied, final repository for radioactive waste / Raw
2	Nuclear waste	m3 HAA eq	Construction stage	Titanium dioxide, chloride process, at plant/RER S	Volume occupied, final repository for low-active radioactive waste / Raw

Tab. 6.5 Identification of the two most relevant life cycle stages, processes and elementary flows for each impact category for 1 kWh of DC electricity produced with a residential scale mono-Si PV system mounted on a slanted roof.

Rank	Impact category	Unit per kWh electricity	Most relevant life cycle stages	Most relevant processes	Most relevant elementary flows
	electricity, 3 kWp installation, mounted with mono-Si PV technology, characterized				
1	Climate change	kg CO2 eq	Product stage	Hard coal, burned in power plant/CN U	Carbon dioxide, fossil / Air
2	Climate change	kg CO2 eq	Construction stage	Hard coal, at mine/CN U	Methane, fossil / Air
1	Ozone depletion	kg CFC-11 eq	Product stage	transport, natural gas, pipeline, long distance/tkm/RU U	Methane, bromochlorodifluoro-, Halon 1211 / Air
2	Ozone depletion	kg CFC-11 eq	Construction stage	Chlorine, gaseous, mercury cell, at plant/RER U	Methane, tetrachloro-, CFC-10 / Air
1	Human toxicity, cancer effects	CTUh, c	Product stage	Disposal, redmud from bauxite digestion, 0% water, to residual material landfill/CH U	Chromium VI / Water
2	Human toxicity, cancer effects	CTUh, c	Construction stage	Disposal, slag, unalloyed electr. steel, 0% water, to residual material landfill/CH U	Chromium / Air
1	Human toxicity, non-cancer effects	CTUh, n-c	Product stage	Disposal, hard coal ash, 0% water, to residual material landfill/PL U	Arsenic / Water
2	Human toxicity, non-cancer effects	CTUh, n-c	Construction stage	Copper, primary, at refinery/RLA U	Mercury / Air
1	Particulate matter	kg PM2.5 eq	Product stage	Hard coal, burned in power plant/CN U	Particulates, < 2.5 um / Air
2	Particulate matter	kg PM2.5 eq	Construction stage	Hard coal, burned in coal mine power plant/CN U	Sulfur dioxide / Air
1	Ionizing radiation HH	kg PM2.5 eq	Product stage	Nuclear spent fuel, in reprocessing, at plant/RER U	Carbon-14 / Air
2	Ionizing radiation HH	kg PM2.5 eq	Construction stage	Uranium natural, at underground mine/RNA U	Radon-222 / Air
1	Photochemical ozone formation	kg NMVOC eq	Product stage	Hard coal, burned in power plant/CN U	Nitrogen oxides / Air
2	Photochemical ozone formation	kg NMVOC eq	Construction stage	Operation, transoceanic freight ship/OCE U	Sulfur dioxide / Air
1	Acidification	mol H+ eq	Product stage	Hard coal, burned in power plant/CN U	Sulfur dioxide / Air
2	Acidification	mol H+ eq	Construction stage	Hard coal, at mine/CN U	Nitrogen oxides / Air
1	Terrestrial eutrophication	mol N eq	Product stage	Hard coal, burned in power plant/CN U	Nitrogen oxides / Air
2	Terrestrial eutrophication	mol N eq	Construction stage	Operation, transoceanic freight ship/OCE U	Ammonia / Air
1	Freshwater eutrophication	kg P eq	Construction stage	Disposal, sulfidic tailings, off-site/GLO U	Phosphate / Water
2	Freshwater eutrophication	kg P eq	Product stage	Disposal, spoil from coal mining, in surface landfill/GLO U	Phosphorus / Water
1	Marine eutrophication	kg N eq	Product stage	Hard coal, burned in power plant/CN U	Nitrogen oxides / Air
2	Marine eutrophication	kg N eq	Construction stage	CZ single crystalline silicon, photovoltaics, at plant/kg/CN U	Nitrate / Water
1	Freshwater ecotoxicity	CTUe	Construction stage	disposal, plastic, industr. electronics, 15.3% water, to municipal incineration/kg/CH U	Antimony / Water
2	Freshwater ecotoxicity	CTUe	Product stage	Disposal, redmud from bauxite digestion, 0% water, to residual material landfill/CH U	Chromium VI / Water
1	Land use	kg C deficit	Product stage	Well for exploration and production, onshore/GLO/I U	Transformation, to mineral extraction site / Raw
2	Land use	kg C deficit	Construction stage	Hard coal, at mine/CN U	Transformation, from unknown / Raw
1	Water resource depletion	m3 water eq	Product stage	electricity, hydropower, at reservoir power plant, non alpine regions/kWh/RER U	Water, cooling, unspecified natural origin/m3 / Raw
2	Water resource depletion	m3 water eq	Construction stage	CZ single crystalline silicon, photovoltaics, at plant/kg/CN U	Water, OECD / Air
1	Mineral, fossil & ren resource depletion	kg Sb eq	Product stage	Resource correction, PbZn, silver, positive/GLO U	Silver / Raw
2	Mineral, fossil & ren resource depletion	kg Sb eq	Construction stage	Anode slime, silver and tellurium containing, primary copper production/GLO U	Tellurium / Raw
1	Cumulative energy demand non renewable	MJ oil eq	Product stage	Hard coal, at mine/CN U	Coal, hard / Raw
2	Cumulative energy demand non renewable	MJ oil eq	Construction stage	Uranium natural, at underground mine/RNA U	Gas, natural/m3 / Raw
1	Cumulative energy demand renewable	MJ oil eq	Use stage	hotspot electricity, PV, at 3kWp slanted-roof, mono-Si, panel, mounted, incl. EoL absolute/f	Energy, solar, converted / Raw
2	Cumulative energy demand renewable	MJ oil eq	Product stage	electricity, hydropower, at run-of-river power plant without reservoir/kWh/RER U	Energy, potential (in hydropower reservoir), converted / Raw
1	Nuclear waste	m3 HAA eq	Product stage	Radioactive waste, in final repository for nuclear waste SF, HLW, and ILW/CH U	Volume occupied, final repository for radioactive waste / Raw
2	Nuclear waste	m3 HAA eq	Construction stage	Titanium dioxide, chloride process, at plant/RER S	Volume occupied, final repository for low-active radioactive waste / Raw

6.4 Most relevant life cycle stages

The most relevant life cycle stages for the five different PV technologies are compared on basis of the residential scale (3 kWp) PV systems mounted on a slanted roof. In this analysis, it is not possible to disaggregate the production of the main product and raw material acquisition and pre-processing. The unit process inventory data of module manufacturing include several process steps and are regionally differentiated for some of the technologies considered. A disaggregation of the production of PV modules into separate life cycle stages as recommended by the Product Environmental Footprint Pilot Guidance (version 5.2, European Commission 2015) would therefore be very complex. The environmental impacts of the end-of-life stage and the potential benefits due to recycling are aggregated for the identification of the most relevant life cycle stages. Only absolute contributions to the total impacts are considered.

For the production of electricity with a residential scale CdTe PV system mounted on a slanted roof the product stage and construction stage are the most important life cycle stages in many impact categories (Tab. 6.6). The use stage causes 99 % of the total impacts in the category renewable cumulative energy demand but contributes very little to the other impact categories. The end-of-life stage is the second most relevant life cycle stage in the impact categories human toxicity, cancer effects, freshwater eutrophication, water resource depletion as well as mineral, fossil and renewable resource depletion.

Tab. 6.6 Identification of the most relevant life cycle stages for each impact category for 1 kWh of DC electricity produced with a residential scale CdTe PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant life cycle stages to the characterized environmental impacts is reached.

Impact category	Unit per kWh electricity	1st most relevant LC stage	Share of 1st most relevant LC stage	2nd most relevant LC stage	Share of 2nd most relevant LC stage	Share of 2 most relevant LC stages
electricity, 3 kWp installation, mounted with CdTe PV technology, characterized			Absolute contribution		Absolute contribution	Checksum
Climate change	kg CO ₂ eq	Product stage	53%	Construction stage	40%	93%
Ozone depletion	kg CFC-11 eq	Product stage	69%	Construction stage	29%	98%
Human toxicity, cancer effects	CTUh, c	Construction stage	76%	End-of-life stage	16%	91%
Human toxicity, non-cancer effects	CTUh, n-c	Construction stage	82%	Product stage	15%	98%
Particulate matter	kg PM _{2.5} eq	Construction stage	45%	Product stage	42%	87%
Ionizing radiation HH	kBq U235 eq	Construction stage	67%	Product stage	21%	88%
Photochemical ozone formation	kg NMVOC eq	Product stage	67%	Construction stage	30%	98%
Acidification	mol H ⁺ eq	Product stage	56%	Construction stage	35%	92%
Terrestrial eutrophication	mol N eq	Product stage	70%	Construction stage	27%	97%
Freshwater eutrophication	kg P eq	Construction stage	73%	End-of-life stage	17%	90%
Marine eutrophication	kg N eq	Product stage	72%	Construction stage	26%	99%
Freshwater ecotoxicity	CTUe	Construction stage	93%	Product stage	5%	98%
Land use	kg C deficit	Product stage	60%	Construction stage	35%	95%
Water resource depletion	m ³ water eq	Construction stage	64%	End-of-life stage	18%	83%
Mineral, fossil & ren resource depletion	kg Sb eq	Product stage	66%	End-of-life stage	29%	95%
Cumulative energy demand non renewable	MJ oil eq	Product stage	54%	Construction stage	41%	95%
Cumulative energy demand renewable	MJ oil eq	Use stage	99%	Construction stage	1%	100%
Nuclear waste	m ³ HAA eq	Construction stage	65%	Product stage	24%	89%

When electricity with a residential scale CIS PV system mounted on a slanted roof top is produced, the product stage is among the two most important life cycle stages in any impact category (Tab. 6.7). The production of the CIS PV module contributes almost exclusively to the impact category mineral, fossil and renewable resource depletion. The construction stage is another very important life cycle stage and contributes the most to the impact categories human toxicity, cancer and non-cancer effects, freshwater eutrophication and freshwater ecotoxicity. The use stage contributes 98 % to the total

impacts in the category renewable cumulative energy demand but is not among the two most important life cycle stages when considering other impact categories. The end-of-life stage is of minor importance in the case of 3 kWp CIS PV systems mounted on a slanted roof.

Tab. 6.7 Identification of the most relevant life cycle stages for each impact category for 1 kWh of DC electricity produced with a residential scale CIS PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant life cycle stages to the characterized environmental impacts is reached.

Impact category	Unit per kWh electricity	1st most relevant LC stage	Share of 1st most relevant LC stage	2nd most relevant LC stage	Share of 2nd most relevant LC stage	Share of 2 most relevant LC stages
electricity, 3 kWp installation, mounted with CIS PV technology, characterized			Absolute contribution		Absolute contribution	Checksum
Climate change	kg CO ₂ eq	Product stage	63%	Construction stage	31%	94%
Ozone depletion	kg CFC-11 eq	Product stage	78%	Construction stage	21%	99%
Human toxicity, cancer effects	CTUh, c	Construction stage	63%	Product stage	21%	84%
Human toxicity, non-cancer effects	CTUh, n-c	Construction stage	54%	Product stage	44%	98%
Particulate matter	kg PM _{2.5} eq	Product stage	47%	Construction stage	41%	88%
Ionizing radiation HH	kBq U ₂₃₅ eq	Product stage	62%	Construction stage	34%	96%
Photochemical ozone formation	kg NMVOC eq	Product stage	70%	Construction stage	29%	99%
Acidification	mol H ⁺ eq	Product stage	58%	Construction stage	35%	92%
Terrestrial eutrophication	mol N eq	Product stage	72%	Construction stage	27%	99%
Freshwater eutrophication	kg P eq	Construction stage	55%	Product stage	33%	88%
Marine eutrophication	kg N eq	Product stage	72%	Construction stage	27%	99%
Freshwater ecotoxicity	CTUe	Construction stage	85%	Product stage	13%	97%
Land use	kg C deficit	Product stage	75%	Construction stage	22%	97%
Water resource depletion	m ³ water eq	Product stage	43%	Construction stage	41%	84%
Mineral, fossil & ren resource depletion	kg Sb eq	Product stage	100%	Construction stage	0%	100%
Cumulative energy demand non renewable	MJ oil eq	Product stage	66%	Construction stage	30%	96%
Cumulative energy demand renewable	MJ oil eq	Use stage	98%	Product stage	1%	99%
Nuclear waste	m ³ HAA eq	Product stage	64%	Construction stage	31%	96%

The most important life cycle stages in the production of electricity with a residential scale micromorphous-Si PV system mounted on a slanted roof are the product stage and the construction stage (Tab. 6.8). Together they contribute at least 84 % to each impact category with the exception of the renewable cumulative energy demand. The use stage has a share of 98 % in the total impacts in the category renewable cumulative energy demand. The end-of-life stage is of minor importance in the case of 3 kWp micromorphous-Si PV systems mounted on a slanted roof.

Tab. 6.8 Identification of the most relevant life cycle stages for each impact category for 1 kWh of DC electricity produced with a residential scale micro-Si PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant life cycle stages to the characterized environmental impacts is reached.

Impact category	Unit per kWh electricity	1st most relevant LC stage	Share of 1st most relevant LC stage	2nd most relevant LC stage	Share of 2nd most relevant LC stage	Share of 2 most relevant LC stages
electricity, 3 kWp installation, mounted with micro-Si PV technology, characterized			Absolute contribution		Absolute contribution	Checksum
Climate change	kg CO ₂ eq	Product stage	71%	Construction stage	24%	95%
Ozone depletion	kg CFC-11 eq	Product stage	64%	Construction stage	33%	97%
Human toxicity, cancer effects	CTUh, c	Construction stage	60%	Product stage	24%	84%
Human toxicity, non-cancer effects	CTUh, n-c	Construction stage	51%	Product stage	47%	98%
Particulate matter	kg PM _{2.5} eq	Product stage	82%	Construction stage	14%	96%
Ionizing radiation HH	kBq U ₂₃₅ eq	Construction stage	47%	Product stage	46%	93%
Photochemical ozone formation	kg NMVOC eq	Product stage	82%	Construction stage	17%	99%
Acidification	mol H ⁺ eq	Product stage	77%	Construction stage	19%	96%
Terrestrial eutrophication	mol N eq	Product stage	83%	Construction stage	16%	99%
Freshwater eutrophication	kg P eq	Construction stage	60%	Product stage	27%	87%
Marine eutrophication	kg N eq	Product stage	84%	Construction stage	16%	100%
Freshwater ecotoxicity	CTUe	Construction stage	83%	Product stage	14%	97%
Land use	kg C deficit	Product stage	69%	Construction stage	27%	96%
Water resource depletion	m ³ water eq	Product stage	67%	Construction stage	24%	90%
Mineral, fossil & ren resource depletion	kg Sb eq	Product stage	63%	Construction stage	36%	99%
Cumulative energy demand non renewable	MJ oil eq	Product stage	65%	Construction stage	30%	95%
Cumulative energy demand renewable	MJ oil eq	Use stage	98%	Construction stage	1%	99%
Nuclear waste	m ³ HAA eq	Construction stage	47%	Product stage	46%	92%

The product stage is the most important life cycle stage according to 14 impact categories when considering the production of electricity with a residential scale multicrystalline-Si PV system mounted on a slanted roof (Tab. 6.9). In the remaining impact categories analysed it causes the second highest contribution. The construction stage is another very important life cycle stage and causes the highest share in the total impacts in the categories human toxicity, cancer effects, freshwater eutrophication and freshwater ecotoxicity. The product and the construction stage together contribute at least 87 % to each impact category with the exception of the renewable cumulative energy demand. The use stage is most important in this impact category (97 %). The end-of-life stage is of minor importance in the case of 3 kWp multicrystalline-Si PV systems mounted on a slanted roof.

Tab. 6.9 Identification of the most relevant life cycle stages for each impact category for 1 kWh of DC electricity produced with a residential scale multi-Si PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant life cycle stages to the characterized environmental impacts is reached.

Impact category	Unit per kWh electricity	1st most relevant LC stage	Share of 1st most relevant LC stage	2nd most relevant LC stage	Share of 2nd most relevant LC stage	Share of 2 most relevant LC stages
electricity, 3 kWp installation, mounted with multi-Si PV technology, characterized			Absolute contribution		Absolute contribution	Checksum
Climate change	kg CO ₂ eq	Product stage	79%	Construction stage	17%	96%
Ozone depletion	kg CFC-11 eq	Product stage	83%	Construction stage	15%	99%
Human toxicity, cancer effects	CTUh, c	Construction stage	49%	Product stage	38%	87%
Human toxicity, non-cancer effects	CTUh, n-c	Product stage	58%	Construction stage	42%	100%
Particulate matter	kg PM _{2.5} eq	Product stage	88%	Construction stage	9%	97%
Ionizing radiation HH	kBq U235 eq	Product stage	66%	Construction stage	29%	95%
Photochemical ozone formation	kg NMVOC eq	Product stage	85%	Construction stage	14%	99%
Acidification	mol H ⁺ eq	Product stage	83%	Construction stage	14%	97%
Terrestrial eutrophication	mol N eq	Product stage	86%	Construction stage	13%	99%
Freshwater eutrophication	kg P eq	Construction stage	49%	Product stage	39%	88%
Marine eutrophication	kg N eq	Product stage	87%	Construction stage	12%	99%
Freshwater ecotoxicity	CTUe	Construction stage	74%	Product stage	24%	98%
Land use	kg C deficit	Product stage	83%	Construction stage	14%	98%
Water resource depletion	m ³ water eq	Product stage	69%	Construction stage	22%	91%
Mineral, fossil & ren resource depletion	kg Sb eq	Product stage	97%	Construction stage	3%	100%
Cumulative energy demand non renewable	MJ oil eq	Product stage	77%	Construction stage	20%	97%
Cumulative energy demand renewable	MJ oil eq	Use stage	97%	Product stage	2%	99%
Nuclear waste	m ³ HAA eq	Product stage	67%	Construction stage	28%	95%

The product stage is the most important life cycle stage according to 15 impact categories when considering the production of electricity with a residential scale monocrystalline-Si PV system mounted on a slanted roof (Tab. 6.10). In the remaining impact categories analysed it causes the second highest contribution. The construction stage is another very important life cycle stage and causes the highest share in the total impacts in the categories freshwater eutrophication and freshwater ecotoxicity. The product and the construction stage together contribute at least 89 % to each impact category with the exception of the renewable cumulative energy demand. The use stage is most important in this impact category (96 %). The end-of-life stage is of minor importance in the case of 3 kWp monocrystalline-Si PV systems mounted on a slanted roof.

Tab. 6.10 Identification of the most relevant life cycle stages for each impact category for 1 kWh of DC electricity produced with a residential scale mono-Si PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant life cycle stages to the characterized environmental impacts is reached.

Impact category	Unit per kWh electricity	1st most relevant LC stage	Share of 1st most relevant LC stage	2nd most relevant LC stage	Share of 2nd most relevant LC stage	Share of 2 most relevant LC stages
electricity, 3 kWp installation, mounted with mono-Si PV technology, characterized			Absolute contribution		Absolute contribution	Checksum
Climate change	kg CO ₂ eq	Product stage	88%	Construction stage	10%	98%
Ozone depletion	kg CFC-11 eq	Product stage	88%	Construction stage	11%	99%
Human toxicity, cancer effects	CTUh, c	Product stage	45%	Construction stage	43%	89%
Human toxicity, non-cancer effects	CTUh, n-c	Product stage	68%	Construction stage	32%	100%
Particulate matter	kg PM _{2.5} eq	Product stage	94%	Construction stage	5%	99%
Ionizing radiation HH	kBq U235 eq	Product stage	75%	Construction stage	22%	97%
Photochemical ozone formation	kg NMVOC eq	Product stage	91%	Construction stage	8%	99%
Acidification	mol H ⁺ eq	Product stage	90%	Construction stage	8%	98%
Terrestrial eutrophication	mol N eq	Product stage	92%	Construction stage	8%	99%
Freshwater eutrophication	kg P eq	Construction stage	46%	Product stage	44%	89%
Marine eutrophication	kg N eq	Product stage	93%	Construction stage	7%	100%
Freshwater ecotoxicity	CTUe	Construction stage	71%	Product stage	27%	98%
Land use	kg C deficit	Product stage	87%	Construction stage	11%	98%
Water resource depletion	m ³ water eq	Product stage	82%	Construction stage	13%	95%
Mineral, fossil & ren resource depletion	kg Sb eq	Product stage	96%	Construction stage	4%	100%
Cumulative energy demand non renewable	MJ oil eq	Product stage	85%	Construction stage	13%	98%
Cumulative energy demand renewable	MJ oil eq	Use stage	96%	Product stage	3%	99%
Nuclear waste	m ³ HAA eq	Product stage	76%	Construction stage	21%	96%

6.5 Most relevant processes

The most relevant processes for the five different PV technologies are compared on basis of the residential scale (3 kWp) PV systems mounted on a slanted roof. The most relevant processes are defined as those contributing cumulatively at least 80 % to the total impacts in each impact category. The shares of the individual life cycle stages (product stage, construction stage, use stage, end-of-life stage) to the total contribution of each process hotspot are also calculated. The most relevant processes for the five PV technologies analysed in this study are listed for each impact category in Tab. 6.11 to Tab. 6.15.

Tab. 6.11

Identification of the most relevant processes for each impact category and each life cycle stage for 1 kWh of DC electricity produced with a residential scale CdTe PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant processes to the characterized environmental impacts is reached.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total	Most relevant processes	Cumulative share
electricity, 3 kWp installation, mounted with CdTe PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling incl. recycling potential, 50:50	Contribution		
Climate change	kg CO2 eq	1.35E-02	1.01E-02	3.47E-06	1.72E-03	2.53E-02		
Climate change	kg CO2 eq	1.71E-05	2.35E-03	3.61E-09	8.80E-04	3.25E-03	Aluminium, primary, liquid, at plant/RER U	13%
Climate change	kg CO2 eq	3.00E-03	6.82E-07	2.44E-10	2.94E-08	3.00E-03	Flat glass, uncoated, at plant/RER U	25%
Climate change	kg CO2 eq	2.18E-03	5.58E-04	2.58E-08	2.02E-04	2.94E-03	Natural gas, burned in power plant/UCTE U	36%
Climate change	kg CO2 eq	8.43E-04	6.10E-04	1.40E-07	-5.66E-06	1.45E-03	Hard coal, burned in power plant/DE U	42%
Climate change	kg CO2 eq	1.01E-03	1.87E-04	1.78E-08	6.47E-05	1.26E-03	Operation, transoceanic freight ship/OCE U	47%
Climate change	kg CO2 eq	6.18E-05	7.61E-04	1.25E-08	3.06E-04	1.13E-03	Light fuel oil, burned in industrial furnace 1MW, non-modulating/CH U	51%
Climate change	kg CO2 eq	3.33E-04	2.42E-04	5.66E-08	7.29E-05	6.48E-04	Hard coal, burned in power plant/ES U	54%
Climate change	kg CO2 eq	3.63E-07	5.82E-04	8.33E-11	2.62E-08	5.82E-04	disposal, plastic, industr. electronics, 15.3% water, to municipal incineration/kg/CH U	56%
Climate change	kg CO2 eq	2.98E-04	1.37E-04	4.01E-08	9.71E-05	5.32E-04	Hard coal, burned in industrial furnace 1-10MW/RER U	58%
Climate change	kg CO2 eq	6.04E-05	2.39E-04	3.45E-08	1.35E-04	4.35E-04	Natural gas, burned in industrial furnace >100kW/RER U	60%
Climate change	kg CO2 eq	6.15E-05	2.52E-04	1.19E-07	9.79E-05	4.11E-04	Pig iron, at plant/GLO U	62%
Climate change	kg CO2 eq	5.35E-04	9.05E-05	7.33E-09	-2.59E-04	3.67E-04	Operation, lorry>16t, fleet average/RER U	63%
Climate change	kg CO2 eq	0.00E+00	0.00E+00	0.00E+00	3.66E-04	3.66E-04	Avoided burden from recycling, CdTe PV module, mounted construction/m2/RER U	65%
Climate change	kg CO2 eq	6.01E-06	3.18E-04	1.21E-07	-1.97E-06	3.22E-04	Polyethylene, HDPE, granulate, at plant/RER U	66%
Climate change	kg CO2 eq	1.21E-04	3.95E-04	1.46E-07	-1.95E-04	3.21E-04	Lignite, burned in power plant/DE U	67%
Climate change	kg CO2 eq	1.64E-04	1.18E-04	2.66E-08	3.53E-05	3.17E-04	Hard coal, burned in power plant/FR U	68%
Climate change	kg CO2 eq	1.68E-04	1.46E-04	1.18E-08	-1.13E-07	3.14E-04	Natural gas, burned in industrial furnace low-NOx >100kW/RER U	70%
Climate change	kg CO2 eq	1.59E-04	1.20E-04	3.98E-08	3.29E-05	3.12E-04	Hard coal, burned in power plant/IT U	71%
Climate change	kg CO2 eq	1.40E-04	1.02E-04	2.54E-08	2.99E-05	2.72E-04	Hard coal, burned in power plant/NL U	72%
Climate change	kg CO2 eq	8.39E-05	1.29E-04	2.03E-08	4.19E-05	2.55E-04	Heavy fuel oil, burned in power plant/IT U	73%
Climate change	kg CO2 eq	1.21E-04	9.88E-05	3.80E-08	1.14E-05	2.31E-04	Hard coal, at mine/WEU U	74%
Climate change	kg CO2 eq	2.17E-04	3.83E-08	3.52E-11	-5.58E-07	2.17E-04	Hard coal, burned in power plant/RFC U	75%
Climate change	kg CO2 eq	2.05E-04	2.49E-06	3.22E-09	-6.56E-07	2.06E-04	Hard coal, burned in power plant/SERC U	76%
Climate change	kg CO2 eq	1.95E-04	3.06E-07	3.03E-10	-1.75E-07	1.95E-04	Nylon 66, glass-filled, at plant/RER U	76%
Climate change	kg CO2 eq	2.06E-05	7.14E-05	7.55E-09	1.00E-04	1.92E-04	Heavy fuel oil, burned in industrial furnace 1MW, non-modulating/RER U	77%
Climate change	kg CO2 eq	9.41E-05	8.09E-05	1.16E-08	1.28E-05	1.88E-04	Refinery gas, burned in furnace/MJ/RER U	78%
Climate change	kg CO2 eq	8.35E-05	6.00E-05	1.30E-08	1.84E-05	1.62E-04	Hard coal, burned in power plant/PT U	79%
Climate change	kg CO2 eq	1.57E-04	1.01E-06	1.48E-09	8.65E-08	1.58E-04	Ethylene, average, at plant/RER U	79%
Climate change	kg CO2 eq	2.69E-05	9.48E-05	3.25E-08	2.56E-05	1.47E-04	Lignite, burned in power plant/GR U	80%
Climate change	kg CO2 eq	7.46E-05	5.24E-05	8.54E-09	1.69E-05	1.44E-04	Hard coal, burned in power plant/BE U	80%
Ozone depletion	kg CFC-11 eq	6.73E-10	2.83E-10	2.14E-13	2.28E-11	9.79E-10		
Ozone depletion	kg CFC-11 eq	2.61E-10	8.17E-11	2.26E-14	7.38E-12	3.50E-10	transport, natural gas, pipeline, long distance/tkm/RU U	36%
Ozone depletion	kg CFC-11 eq	9.38E-11	3.30E-11	1.36E-14	5.12E-12	1.32E-10	Transport, natural gas, onshore pipeline, long distance/tkm/NAC U	49%
Ozone depletion	kg CFC-11 eq	9.23E-11	3.11E-11	1.14E-14	2.47E-12	1.26E-10	transport, natural gas, pipeline, long distance/tkm/RER U	62%
Ozone depletion	kg CFC-11 eq	3.51E-11	4.65E-11	1.60E-14	9.81E-12	9.15E-11	Crude oil, at production/NG U	71%
Ozone depletion	kg CFC-11 eq	4.69E-11	1.47E-11	4.03E-15	1.61E-12	6.32E-11	Transport, natural gas, onshore pipeline, long distance/NO U	78%
Ozone depletion	kg CFC-11 eq	3.62E-11	1.14E-11	3.16E-15	1.25E-12	4.88E-11	Transport, natural gas, offshore pipeline, long distance/NO U	83%
Human toxicity, cancer effects	CTUh, c	8.10E-11	7.26E-10	5.25E-13	1.49E-10	9.56E-10		
Human toxicity, cancer effects	CTUh, c	3.24E-12	4.32E-10	3.37E-15	1.61E-10	5.97E-10	Disposal, redmud from bauxite digestion, 0% water, to residual material landfill/CH U	62%
Human toxicity, cancer effects	CTUh, c	2.59E-11	1.01E-10	7.07E-14	-4.91E-13	1.26E-10	Disposal, slag, unalloyed electr. steel, 0% water, to residual material landfill/CH U	76%
Human toxicity, cancer effects	CTUh, c	7.94E-12	7.07E-11	4.28E-14	3.77E-12	8.25E-11	Ferrochromium, high-carbon, 68% Cr, at plant/GLO U	84%
Human toxicity, non-cancer effects	CTUh, n-c	8.36E-10	4.44E-09	8.15E-12	1.15E-10	5.40E-09		
Human toxicity, non-cancer effects	CTUh, n-c	6.07E-11	1.74E-09	2.42E-14	4.63E-10	2.26E-09	Copper, primary, at refinery/RLA U	42%
Human toxicity, non-cancer effects	CTUh, n-c	5.26E-12	7.01E-10	5.46E-15	2.62E-10	9.68E-10	Disposal, redmud from bauxite digestion, 0% water, to residual material landfill/CH U	49%
Human toxicity, non-cancer effects	CTUh, n-c	8.18E-11	3.49E-10	1.28E-13	-2.68E-12	4.28E-10	Steel, electric, un- and low-alloyed, at plant/RER U	68%
Human toxicity, non-cancer effects	CTUh, n-c	1.37E-10	2.98E-11	1.80E-14	-5.88E-10	-4.20E-10	Zinc, primary, at regional storage/RER U	60%
Human toxicity, non-cancer effects	CTUh, n-c	8.75E-12	2.51E-10	3.49E-15	6.68E-11	3.26E-10	Copper, primary, at refinery/ID U	66%
Human toxicity, non-cancer effects	CTUh, n-c	7.06E-12	2.02E-10	2.81E-15	5.38E-11	2.63E-10	Copper, primary, at refinery/RER U	71%
Human toxicity, non-cancer effects	CTUh, n-c	9.20E-12	1.13E-10	1.86E-15	4.55E-11	1.68E-10	Light fuel oil, burned in industrial furnace 1MW, non-modulating/CH U	74%
Human toxicity, non-cancer effects	CTUh, n-c	3.85E-12	1.10E-10	1.53E-15	2.94E-11	1.43E-10	Copper, from imported concentrates, at refinery/DE U	77%
Human toxicity, non-cancer effects	CTUh, n-c	9.96E-11	1.24E-14	1.37E-17	4.35E-11	1.43E-10	Copper, primary, at refinery/GLO U	79%
Human toxicity, non-cancer effects	CTUh, n-c	8.63E-14	1.38E-10	1.98E-17	6.21E-15	1.38E-10	disposal, plastic, industr. electronics, 15.3% water, to municipal incineration/kg/CH U	82%
Particulate matter	kg PM2.5 eq	6.25E-06	6.76E-06	1.90E-09	1.88E-06	1.49E-05		
Particulate matter	kg PM2.5 eq	1.47E-08	2.02E-06	3.10E-12	7.57E-07	2.79E-06	Aluminium, primary, liquid, at plant/RER U	19%
Particulate matter	kg U235 eq	1.97E-06	4.47E-10	1.60E-13	1.93E-11	1.97E-06	Flat glass, uncoated, at plant/RER U	32%
Particulate matter	kg PM2.5 eq	1.24E-06	2.29E-07	2.17E-11	7.92E-08	1.54E-06	Operation, transoceanic freight ship/OCE U	42%
Particulate matter	kg PM2.5 eq	2.09E-08	5.98E-07	8.32E-12	1.59E-07	7.79E-07	Copper, primary, at refinery/RLA U	48%
Particulate matter	kg PM2.5 eq	3.29E-07	1.51E-07	4.43E-11	1.07E-07	5.87E-07	Hard coal, burned in industrial furnace 1-10MW/RER U	52%
Particulate matter	kg PM2.5 eq	2.22E-07	1.61E-07	3.78E-11	4.87E-08	4.32E-07	Hard coal, burned in power plant/ES U	54%
Particulate matter	kg PM2.5 eq	1.05E-08	3.02E-07	4.20E-12	8.05E-08	3.93E-07	Copper concentrate, at beneficiation/RLA U	57%
Particulate matter	kg PM2.5 eq	9.82E-09	2.81E-07	3.91E-12	7.50E-08	3.66E-07	Copper, primary, at refinery/ID U	60%
Particulate matter	kg PM2.5 eq	3.77E-08	1.31E-07	1.39E-11	1.84E-07	3.53E-07	Heavy fuel oil, burned in industrial furnace 1MW, non-modulating/RER U	62%
Particulate matter	kg PM2.5 eq	7.46E-09	2.13E-07	2.97E-12	3.53E-08	2.56E-07	Copper concentrate, at beneficiation/RER U	64%
Particulate matter	kg PM2.5 eq	1.29E-07	9.97E-08	1.28E-11	2.05E-08	2.49E-07	Natural gas, sour, burned in production flare/MJ/GLO U	65%
Particulate matter	kg PM2.5 eq	7.54E-08	1.16E-07	1.82E-11	3.77E-08	2.29E-07	Heavy fuel oil, burned in power plant/IT U	67%
Particulate matter	kg PM2.5 eq	1.16E-09	1.47E-07	1.23E-12	5.50E-08	2.04E-07	Bauxite, at mine/GLO U	68%
Particulate matter	kg PM2.5 eq	9.91E-08	8.52E-08	1.22E-11	1.35E-08	1.98E-07	Heavy fuel oil, burned in refinery furnace/MJ/RER U	70%
Particulate matter	kg PM2.5 eq	3.03E-08	1.25E-07	7.86E-12	2.84E-08	1.84E-07	Blasting/RER U	71%
Particulate matter	kg PM2.5 eq	1.01E-07	7.57E-08	1.11E-10	2.46E-09	1.79E-07	Diesel, burned in building machine/GLO U	72%
Particulate matter	kg PM2.5 eq	6.98E-08	1.10E-07	7.12E-11	-6.38E-09	1.73E-07	Electricity, at cogen 6400kWth, wood, allocation exergy/CH U	73%
Particulate matter	kg PM2.5 eq	3.43E-08	1.02E-07	4.36E-11	2.51E-08	1.61E-07	Lignite, burned in power plant/CS U	74%
Particulate matter	kg PM2.5 eq	2.25E-08	9.23E-08	4.37E-11	3.59E-08	1.51E-07	Sinter, iron, at plant/GLO U	75%
Particulate matter	kg PM2.5 eq	2.64E-09	1.40E-07	5.31E-11	-8.64E-10	1.41E-07	Polyethylene, HDPE, granulate, at plant/RER U	76%
Particulate matter	kg PM2.5 eq	2.01E-07	3.40E-08	2.75E-12	-9.71E-08	1.38E-07	Operation, lorry>16t, fleet average/RER U	77%
Particulate matter	kg PM2.5 eq	6.49E-08	4.68E-08	1.06E-11	1.40E-08	1.26E-07	Hard coal, burned in power plant/FR U	78%
Particulate matter	kg PM2.5 eq	3.23E-09	9.23E-08	1.29E-12	2.46E-08	1.20E-07	Copper, primary, at refinery/RER U	79%
Particulate matter	kg PM2.5 eq	1.12E-07	1.76E-10	1.74E-13	-1.00E-10	1.12E-07	Nylon 66, glass-filled, at plant/RER U	80%
Particulate matter	kg PM2.5 eq	5.70E-08	4.31E-08	1.43E-11	1.18E-08	1.12E-07	Hard coal, burned in power plant/IT U	80%
Ionizing radiation HH	kg PM2.5 eq	2.84E-04	8.95E-04	6.41E-07	1.61E-04	3.34E-03		
Ionizing radiation HH	kg PM2.5 eq	2.42E-04	8.26E-04	4.82E-07	1.54E-04	1.22E-03	Nuclear spent fuel, in reprocessing, at plant/RER U	91%
Ionizing radiation HH	kg PM2.5 eq	1.31E-05	3.30E-05	1.43E-08	5.03E-06	5.11E-05	Uranium natural, at underground mine/RNA U	95%

Tab. 6.11 Identification of the most relevant processes for each impact category and each life cycle stage for 1 kWh of DC electricity produced with a residential scale CdTe PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant processes to the characterized environmental impacts is reached. (Continued)

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total	Most relevant processes	Cumulative share
electricity, 3 kWp installation, mounted with CdTe PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling incl. recycling potential, 50:50	Contribution		
Photochemical ozone formation	kg NMVOC eq	6.14E-05	2.75E-05	1.25E-08	2.11E-06	9.10E-05		
Photochemical ozone formation	kg NMVOC eq	1.95E-05	3.62E-06	3.45E-10	1.25E-06	2.45E-05	Operation, transoceanic freight ship/OCE U	27%
Photochemical ozone formation	kg NMVOC eq	1.58E-05	3.59E-09	1.29E-12	1.55E-10	1.58E-05	Flat glass, uncoated, at plant/RER U	44%
Photochemical ozone formation	kg NMVOC eq	1.01E-06	4.17E-06	2.62E-10	9.45E-07	6.13E-06	Blasting/RER U	51%
Photochemical ozone formation	kg NMVOC eq	5.64E-06	9.54E-07	7.73E-11	-2.73E-06	3.86E-06	Operation, lorry >16t, fleet average/RER U	55%
Photochemical ozone formation	kg NMVOC eq	1.46E-06	1.06E-06	2.48E-10	3.19E-07	2.84E-06	Hard coal, burned in power plant/ES U	58%
Photochemical ozone formation	kg NMVOC eq	2.03E-06	5.21E-07	2.41E-11	1.88E-07	2.74E-06	Natural gas, burned in power plant/UCTE U	61%
Photochemical ozone formation	kg NMVOC eq	1.13E-06	8.45E-07	1.24E-09	2.75E-08	2.00E-06	Diesel, burned in building machine/GLO U	64%
Photochemical ozone formation	kg NMVOC eq	7.86E-07	8.20E-07	2.16E-10	1.71E-07	1.78E-06	Natural gas, vented/GLO U	66%
Photochemical ozone formation	kg NMVOC eq	8.09E-07	3.71E-07	1.09E-10	2.63E-07	1.44E-06	Hard coal, burned in industrial furnace 1-10MW/RER U	67%
Photochemical ozone formation	kg NMVOC eq	2.52E-08	1.33E-06	5.07E-10	-8.25E-09	1.35E-06	Polyethylene, HDPE, granulate, at plant/RER U	69%
Photochemical ozone formation	kg NMVOC eq	6.36E-07	4.73E-07	1.00E-10	9.02E-08	1.20E-06	Diesel, burned in diesel-electric generating set/GLO U	70%
Photochemical ozone formation	kg NMVOC eq	6.12E-07	4.43E-07	1.02E-10	-4.11E-09	1.05E-06	Hard coal, burned in power plant/DE U	71%
Photochemical ozone formation	kg NMVOC eq	2.77E-08	7.95E-07	1.10E-11	2.12E-07	1.03E-06	Copper, primary, at refinery/RLA U	72%
Photochemical ozone formation	kg NMVOC eq	1.42E-08	9.67E-07	8.84E-13	4.13E-10	9.82E-07	Operation, lorry 3.5-16t, fleet average/RER U	73%
Photochemical ozone formation	kg NMVOC eq	3.11E-07	4.79E-07	7.51E-11	1.55E-07	9.45E-07	Heavy fuel oil, burned in power plant/IT U	74%
Photochemical ozone formation	kg NMVOC eq	4.58E-08	5.64E-07	9.24E-12	2.26E-07	8.36E-07	Light fuel oil, burned in industrial furnace 1MW, non-modulating/CH U	75%
Photochemical ozone formation	kg NMVOC eq	4.01E-07	2.90E-07	6.53E-11	8.65E-08	7.77E-07	Hard coal, burned in power plant/FR U	76%
Photochemical ozone formation	kg NMVOC eq	1.91E-09	7.66E-07	2.45E-14	6.64E-11	7.68E-07	Operation, van < 3.5t/RER U	77%
Photochemical ozone formation	kg NMVOC eq	3.83E-07	2.90E-07	9.60E-11	7.94E-08	7.53E-07	Hard coal, burned in power plant/IT U	78%
Photochemical ozone formation	kg NMVOC eq	3.60E-09	4.96E-07	7.61E-13	1.86E-07	6.85E-07	Aluminium, primary, liquid, at plant/RER U	79%
Photochemical ozone formation	kg NMVOC eq	3.13E-07	2.42E-07	3.11E-11	4.99E-08	6.05E-07	Natural gas, sour, burned in production flare/Mi/GLO U	79%
Photochemical ozone formation	kg NMVOC eq	4.50E-07	1.42E-07	4.09E-11	6.77E-09	5.98E-07	Sweetening, natural gas/DE U	80%
Photochemical ozone formation	kg NMVOC eq	3.37E-07	2.24E-07	8.53E-11	2.08E-08	5.82E-07	Operation, freight train/RER U	80%
Acidification	mol H+ eq	1.14E-04	7.15E-05	2.61E-08	1.69E-05	2.02E-04		
Acidification	mol H+ eq	3.36E-05	6.20E-06	5.90E-10	2.15E-06	4.19E-05	Operation, transoceanic freight ship/OCE U	21%
Acidification	mol H+ eq	3.34E-05	7.59E-09	2.72E-12	3.27E-10	3.34E-05	Flat glass, uncoated, at plant/RER U	37%
Acidification	mol H+ eq	4.48E-07	1.28E-05	1.78E-10	3.42E-06	1.67E-05	Copper, primary, at refinery/RLA U	45%
Acidification	mol H+ eq	5.36E-08	7.38E-06	1.13E-11	2.76E-06	1.02E-05	Aluminium, primary, liquid, at plant/RER U	51%
Acidification	mol H+ eq	4.24E-06	3.07E-06	7.20E-10	9.28E-07	8.25E-06	Hard coal, burned in power plant/ES U	55%
Acidification	mol H+ eq	2.11E-07	6.03E-06	8.39E-11	1.61E-06	7.85E-06	Copper, primary, at refinery/ID U	58%
Acidification	mol H+ eq	1.09E-06	4.52E-06	2.83E-10	1.02E-06	6.63E-06	Blasting/RER U	62%
Acidification	mol H+ eq	2.83E-06	2.19E-06	2.81E-10	4.51E-07	5.47E-06	Natural gas, sour, burned in production flare/Mi/GLO U	64%
Acidification	mol H+ eq	2.60E-06	1.19E-06	3.50E-10	8.47E-07	4.64E-06	Hard coal, burned in industrial furnace 1-10MW/RER U	67%
Acidification	mol H+ eq	1.01E-06	1.56E-06	2.45E-10	5.07E-07	3.08E-06	Heavy fuel oil, burned in power plant/IT U	68%
Acidification	mol H+ eq	1.99E-06	4.60E-08	2.30E-12	8.10E-07	2.85E-06	Soda, powder, at plant/RER U	70%
Acidification	mol H+ eq	4.10E-06	6.94E-07	5.62E-11	-1.98E-06	2.81E-06	Operation, lorry >16t, fleet average/RER U	71%
Acidification	mol H+ eq	6.93E-08	1.98E-06	2.76E-11	5.27E-07	2.58E-06	Copper, primary, at refinery/RER U	72%
Acidification	mol H+ eq	1.27E-06	9.19E-07	2.07E-10	2.74E-07	2.47E-06	Hard coal, burned in power plant/FR U	74%
Acidification	mol H+ eq	1.11E-06	8.36E-07	2.77E-10	2.29E-07	2.17E-06	Hard coal, burned in power plant/IT U	75%
Acidification	mol H+ eq	2.12E-06	3.73E-10	3.43E-13	-5.43E-09	2.11E-06	Hard coal, burned in power plant/RFC U	76%
Acidification	mol H+ eq	1.19E-06	8.61E-07	1.98E-10	-7.99E-09	2.04E-06	Hard coal, burned in power plant/DE U	77%
Acidification	mol H+ eq	4.31E-07	1.28E-06	5.48E-10	3.15E-07	2.03E-06	Lignite, burned in power plant/CS U	78%
Acidification	mol H+ eq	1.04E-07	1.40E-06	2.43E-12	5.18E-07	2.02E-06	Lignite, burned in power plant/ES U	79%
Acidification	mol H+ eq	1.46E-06	3.73E-07	1.73E-11	1.35E-07	1.96E-06	Natural gas, burned in power plant/UCTE U	80%
Acidification	mol H+ eq	9.19E-07	7.33E-07	1.21E-10	1.42E-07	1.79E-06	Operation, transoceanic tanker/OCE U	81%
Terrestrial eutrophication	mol N eq	2.41E-04	9.35E-05	6.81E-08	9.87E-06	3.44E-04		
Terrestrial eutrophication	mol N eq	7.63E-05	1.41E-05	1.34E-09	4.88E-06	9.53E-05	Operation, transoceanic freight ship/OCE U	28%
Terrestrial eutrophication	mol N eq	6.04E-05	1.37E-08	4.91E-12	5.91E-10	6.04E-05	Flat glass, uncoated, at plant/RER U	45%
Terrestrial eutrophication	mol N eq	5.69E-06	2.35E-05	1.48E-09	5.34E-06	3.46E-05	Blasting/RER U	55%
Terrestrial eutrophication	mol N eq	2.35E-05	3.97E-06	3.22E-10	-1.14E-05	1.61E-05	Operation, lorry >16t, fleet average/RER U	60%
Terrestrial eutrophication	mol N eq	8.91E-06	2.06E-07	1.03E-11	3.62E-06	1.27E-05	Soda, powder, at plant/RER U	64%
Terrestrial eutrophication	mol N eq	8.24E-06	2.11E-06	9.76E-11	7.62E-07	1.11E-05	Natural gas, burned in power plant/UCTE U	67%
Terrestrial eutrophication	mol N eq	5.34E-06	3.87E-06	9.06E-10	1.17E-06	1.04E-05	Hard coal, burned in power plant/ES U	70%
Terrestrial eutrophication	mol N eq	4.30E-06	3.22E-06	4.73E-09	1.05E-07	7.63E-06	Diesel, burned in building machine/GLO U	72%
Terrestrial eutrophication	mol N eq	6.96E-06	1.71E-07	1.13E-10	6.09E-09	7.14E-06	Zinc coating, coils/RER U	74%
Terrestrial eutrophication	mol N eq	2.76E-06	1.27E-06	3.72E-10	8.99E-07	4.93E-06	Hard coal, burned in industrial furnace 1-10MW/RER U	76%
Terrestrial eutrophication	mol N eq	2.53E-06	1.88E-06	3.99E-10	3.59E-07	4.78E-06	Diesel, burned in diesel-electric generating set/GLO U	77%
Terrestrial eutrophication	mol N eq	2.39E-06	1.73E-06	3.97E-10	-1.60E-08	4.10E-06	Hard coal, burned in power plant/DE U	78%
Terrestrial eutrophication	mol N eq	5.63E-08	3.85E-06	3.52E-12	1.64E-09	3.91E-06	Operation, lorry 3.5-16t, fleet average/RER U	79%
Terrestrial eutrophication	mol N eq	1.10E-06	1.70E-06	2.66E-10	5.50E-07	3.35E-06	Heavy fuel oil, burned in power plant/IT U	80%
Freshwater eutrophication	kg P eq	5.32E-07	3.96E-06	4.57E-09	9.15E-07	5.41E-06		
Freshwater eutrophication	kg P eq	2.48E-07	3.63E-06	5.35E-11	9.22E-07	4.80E-06	Disposal, sulfidic tailings, off-site/GLO U	89%
Freshwater eutrophication	kg P eq	1.79E-07	1.17E-07	3.77E-11	3.01E-08	3.25E-07	Disposal, spoil from coal mining, in surface landfill/GLO U	95%
Marine eutrophication	kg N eq	2.14E-05	7.84E-06	1.00E-07	3.06E-07	2.97E-05		
Marine eutrophication	kg N eq	6.82E-06	1.26E-06	1.20E-10	4.37E-07	8.52E-06	Operation, transoceanic freight ship/OCE U	29%
Marine eutrophication	kg N eq	5.51E-06	1.25E-09	4.49E-13	5.39E-11	5.51E-06	Flat glass, uncoated, at plant/RER U	47%
Marine eutrophication	kg N eq	3.46E-07	1.43E-06	8.98E-11	3.24E-07	2.10E-06	Blasting/RER U	54%
Marine eutrophication	kg N eq	2.14E-06	3.62E-07	2.93E-11	-1.03E-06	1.47E-06	Operation, lorry >16t, fleet average/RER U	59%
Marine eutrophication	kg N eq	7.52E-07	1.93E-07	8.91E-12	6.96E-08	1.01E-06	Natural gas, burned in power plant/UCTE U	63%
Marine eutrophication	kg N eq	4.87E-07	3.53E-07	8.28E-11	1.07E-07	9.47E-07	Hard coal, burned in power plant/ES U	66%
Marine eutrophication	kg N eq	7.07E-07	5.99E-14	8.33E-17	-6.68E-14	7.07E-07	photovoltaic laminate, CdTe, at plant/m2/MY/I U	68%
Marine eutrophication	kg N eq	3.92E-07	2.94E-07	4.31E-10	9.55E-09	6.95E-07	Diesel, burned in building machine/GLO U	71%
Marine eutrophication	kg N eq	2.52E-07	1.16E-07	3.39E-11	8.21E-08	4.50E-07	Hard coal, burned in industrial furnace 1-10MW/RER U	72%
Marine eutrophication	kg N eq	2.31E-07	1.72E-07	3.64E-11	3.28E-08	4.36E-07	Diesel, burned in diesel-electric generating set/GLO U	74%
Marine eutrophication	kg N eq	2.18E-07	1.58E-07	3.63E-11	-1.46E-09	3.74E-07	Hard coal, burned in power plant/DE U	75%
Marine eutrophication	kg N eq	5.13E-09	3.50E-07	3.20E-13	1.49E-10	3.56E-07	Operation, lorry 3.5-16t, fleet average/RER U	76%
Marine eutrophication	kg N eq	1.01E-07	1.55E-07	2.43E-11	5.02E-08	3.05E-07	Heavy fuel oil, burned in power plant/IT U	77%
Marine eutrophication	kg N eq	1.62E-08	2.00E-07	3.27E-12	8.01E-08	2.96E-07	Light fuel oil, burned in industrial furnace 1MW, non-modulating/CH U	78%
Marine eutrophication	kg N eq	2.84E-07	4.45E-10	4.40E-13	-2.54E-10	2.84E-07	Nylon 66, glass-filled, at plant/RER U	79%
Marine eutrophication	kg N eq	1.31E-07	9.47E-08	2.13E-11	2.83E-08	2.54E-07	Hard coal, burned in power plant/FR U	80%
Marine eutrophication	kg N eq	1.28E-07	9.68E-08	3.21E-11	2.65E-08	2.51E-07	Hard coal, burned in power plant/IT U	81%
Freshwater ecotoxicity	CTue	3.80E-03	7.59E-02	2.04E-05	2.02E-03	8.18E-02		
Freshwater ecotoxicity	CTue	3.80E-05	6.09E-02	8.72E-09	2.74E-06	6.09E-02	disposal, plastic, industr. electronics, 15.3% water, to municipal incineration/kg/CH U	75%
Freshwater ecotoxicity	CTue	4.04E-05	5.38E-03	4.19E-08	2.01E-03	7.43E-03	Disposal, redmud from bauxite digestion, 0% water, to residual material landfill/CH U	84%

Tab. 6.11 Identification of the most relevant processes for each impact category and each life cycle stage for 1 kWh of DC electricity produced with a residential scale CdTe PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant processes to the characterized environmental impacts is reached. (Continued)

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total	Most relevant processes	Cumulative share
electricity, 3 kWp installation, mounted with CdTe PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling incl. recycling potential, 50:50	Contribution		
Land use	kg C deficit	1.75E-02	1.01E-02	1.42E-05	1.53E-03	2.92E-02		
Land use	kg C deficit	4.94E-03	3.62E-03	6.28E-07	7.15E-04	9.28E-03	Well for exploration and production, onshore/GLO/I U	32%
Land use	kg C deficit	3.84E-03	1.92E-03	3.80E-07	3.30E-04	6.09E-03	Well for exploration and production, offshore/OCE/I U	53%
Land use	kg C deficit	2.47E-05	3.13E-03	2.62E-08	1.17E-03	4.32E-03	Bauxite, at mine/GLO U	67%
Land use	kg C deficit	-4.58E-05	-3.11E-03	-7.31E-08	-1.16E-03	-4.31E-03	Recultivation, bauxite mine/GLO U	53%
Land use	kg C deficit	1.92E-03	1.63E-10	2.26E-13	-1.81E-10	1.92E-03	photovoltaic panel factory CdTe/p/US/I U	59%
Land use	kg C deficit	-2.69E-05	-1.28E-03	-1.95E-08	-4.66E-04	-1.77E-03	electricity, hydropower, at reservoir power plant, non alpine regions/kWh/RER U	53%
Land use	kg C deficit	8.07E-04	6.34E-04	8.53E-08	1.30E-04	1.57E-03	production plant crude oil, onshore/p/GLO/I U	59%
Land use	kg C deficit	9.56E-04	3.17E-07	8.86E-10	1.94E-08	9.57E-04	Flat glass plant/RER/I U	62%
Land use	kg C deficit	4.58E-05	6.69E-04	9.85E-09	1.70E-04	8.84E-04	Disposal, sulfidic tailings, off-site/GLO U	65%
Land use	kg C deficit	3.75E-04	2.96E-04	1.18E-07	6.10E-05	7.32E-04	Underground mine, hard coal/GLO/I U	67%
Land use	kg C deficit	5.49E-04	1.16E-04	1.61E-09	6.27E-05	7.27E-04	Softwood, Scandinavian, standing, under bark, in forest/NORDEL U	70%
Land use	kg C deficit	3.95E-04	2.49E-04	3.20E-08	2.21E-05	6.66E-04	Softwood, standing, under bark, in forest/RER U	72%
Land use	kg C deficit	4.38E-04	2.33E-05	1.14E-07	1.61E-04	6.23E-04	Sand, at mine/CH U	74%
Land use	kg C deficit	1.24E-05	4.37E-04	3.17E-08	1.11E-04	5.61E-04	Aluminium electrolysis, plant/RER/I U	76%
Land use	kg C deficit	-3.50E-04	-6.95E-05	-5.14E-07	-1.13E-04	-5.33E-04	Recultivation, limestone mine/CH U	74%
Land use	kg C deficit	3.39E-05	3.14E-04	4.78E-08	8.77E-05	4.36E-04	Residual material landfill facility/CH/I U	76%
Land use	kg C deficit	1.27E-05	3.04E-04	3.61E-09	8.65E-05	4.03E-04	Aluminium hydroxide, plant/RER/I U	77%
Land use	kg C deficit	2.76E-04	6.04E-05	1.85E-08	1.69E-05	3.54E-04	Hard coal, at mine/RNA U	78%
Land use	kg C deficit	3.33E-06	3.40E-04	1.02E-09	1.42E-07	3.44E-04	Aluminium casting, plant/RER/I U	80%
Land use	kg C deficit	5.51E-05	1.92E-04	9.05E-08	7.19E-05	3.19E-04	Bentonite, at mine/DE U	81%
Water resource depletion	m3 water eq	4.18E-06	1.61E-05	9.14E-08	4.58E-06	2.49E-05		
Water resource depletion	m3 water eq	2.75E-07	1.31E-05	2.00E-10	4.77E-06	1.81E-05	electricity, hydropower, at reservoir power plant, non alpine regions/kWh/RER U	73%
Water resource depletion	m3 water eq	9.14E-07	1.08E-08	8.97E-08	-3.06E-08	9.84E-07	tap water, at user/kg/RER U	77%
Water resource depletion	m3 water eq	4.28E-07	9.87E-09	4.94E-13	1.74E-07	6.11E-07	Soda, powder, at plant/RER U	79%
Water resource depletion	m3 water eq	2.85E-07	2.35E-07	1.49E-10	1.82E-08	5.38E-07	Water, decarbonised, at plant/RER U	81%
Mineral, fossil & ren resource depletion	kg Sb eq	6.57E-06	5.19E-07	3.52E-11	2.84E-06	9.92E-06		
Mineral, fossil & ren resource depletion	kg Sb eq	4.04E-06	1.27E-12	1.38E-15	1.73E-06	5.77E-06	Resource correction, PbZn, cadmium, positive/GLO U	58%
Mineral, fossil & ren resource depletion	kg Sb eq	1.50E-06	3.26E-07	1.97E-10	-6.41E-06	-4.59E-06	Zinc concentrate, at beneficiation/GLO U	12%
Mineral, fossil & ren resource depletion	kg Sb eq	-1.32E-06	-2.88E-07	-1.74E-10	5.66E-06	4.05E-06	Resource correction, PbZn, indium, negative/GLO U	53%
Mineral, fossil & ren resource depletion	kg Sb eq	2.44E-06	1.54E-10	1.74E-13	1.07E-06	3.51E-06	Anode slime, silver and tellurium containing, primary copper production/GLO U	88%
Cumulative energy demand non renewable	MJ oil eq	1.92E-01	1.45E-01	5.32E-05	1.92E-02	3.57E-01		
Cumulative energy demand non renewable	MJ oil eq	2.08E-02	7.01E-03	2.60E-06	1.13E-03	2.90E-02	natural gas, at production onshore/m3/NAC U	8%
Cumulative energy demand non renewable	MJ oil eq	1.96E-02	6.07E-03	1.59E-06	6.48E-04	2.63E-02	natural gas, at production offshore/m3/NO U	16%
Cumulative energy demand non renewable	MJ oil eq	6.69E-03	1.69E-02	7.29E-06	2.57E-03	2.61E-02	Uranium natural, at underground mine/RNA U	23%
Cumulative energy demand non renewable	MJ oil eq	1.26E-02	9.98E-03	1.46E-06	1.96E-03	2.46E-02	Crude oil, at production onshore/RME U	30%
Cumulative energy demand non renewable	MJ oil eq	1.83E-02	5.72E-03	1.59E-06	5.21E-04	2.45E-02	natural gas, at production onshore/m3/RU U	37%
Cumulative energy demand non renewable	MJ oil eq	1.07E-02	8.78E-03	3.38E-06	1.01E-03	2.05E-02	Hard coal, at mine/WEU U	42%
Cumulative energy demand non renewable	MJ oil eq	1.04E-02	8.10E-03	1.04E-06	1.67E-03	2.02E-02	crude oil, at production offshore/kg/NO U	48%
Cumulative energy demand non renewable	MJ oil eq	4.46E-03	1.12E-02	4.86E-06	1.72E-03	1.74E-02	Uranium natural, at open pit mine/RNA U	53%
Cumulative energy demand non renewable	MJ oil eq	8.71E-03	6.76E-03	8.65E-07	1.39E-03	1.69E-02	crude oil, at production onshore/kg/RU U	58%
Cumulative energy demand non renewable	MJ oil eq	8.67E-03	6.73E-03	8.61E-07	1.39E-03	1.68E-02	crude oil, at production offshore/kg/GB U	62%
Cumulative energy demand non renewable	MJ oil eq	6.08E-03	6.58E-03	1.81E-06	1.38E-03	1.40E-02	Crude oil, at production onshore/RAF U	66%
Cumulative energy demand non renewable	MJ oil eq	2.40E-04	1.27E-02	4.83E-06	-7.86E-05	1.29E-02	Polyethylene, HDPE, granulate, at plant/RER U	70%
Cumulative energy demand non renewable	MJ oil eq	9.41E-03	2.06E-03	6.32E-07	5.75E-04	1.20E-02	Hard coal, at mine/RNA U	73%
Cumulative energy demand non renewable	MJ oil eq	4.51E-03	5.40E-03	2.70E-06	1.46E-03	1.14E-02	Hard coal, at mine/EEU U	76%
Cumulative energy demand non renewable	MJ oil eq	3.06E-03	8.41E-03	3.88E-06	-1.22E-03	1.03E-02	Lignite, at mine/RER U	79%
Cumulative energy demand non renewable	MJ oil eq	7.49E-03	2.38E-03	7.05E-07	2.65E-04	1.01E-02	Natural gas, at production onshore/NL U	82%
Cumulative energy demand renewable	MJ oil eq	8.05E-03	2.85E-02	3.60E+00	8.39E-03	3.64E+00		
Cumulative energy demand renewable	MJ oil eq	0.00E+00	0.00E+00	3.60E+00	0.00E+00	3.60E+00	hotspot electricity, PV, at 3kWp slanted-roof, CdTe, panel, mounted, incl. EoL absolute/RER	99%
Cumulative energy demand renewable	MJ oil eq	1.05E-03	1.22E-02	1.32E-06	3.92E-03	1.72E-02	electricity, hydropower, at run-of-river power plant without reservoir/kWh/RER U	99%
Nuclear waste	m3 HAA eq	7.25E-12	1.97E-11	1.19E-14	3.20E-12	3.01E-11		
Nuclear waste	m3 HAA eq	7.24E-12	1.97E-11	1.19E-14	3.21E-12	3.01E-11	Radioactive waste, in final repository for nuclear waste SF, HLW, and ILW/CH U	100%
Nuclear waste	m3 HAA eq	5.09E-15	2.70E-15	7.22E-19	8.48E-17	7.88E-15	Titanium dioxide, chloride process, at plant/RER S	100%

Tab. 6.12

Identification of the most relevant processes for each impact category and each life cycle stage for 1 kWh of DC electricity produced with a residential scale CIS PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant processes to the characterized environmental impacts is reached.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total	Most relevant processes	Cumulative share
electricity, 3 kWp installation, mounted with CIS PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling incl. recycling potential, 50:50	Contribution		
Climate change	kg CO2 eq	2.63E-02	1.27E-02	4.50E-06	2.43E-03	4.15E-02		
Climate change	kg CO2 eq	4.56E-03	5.07E-04	1.90E-07	7.25E-05	5.14E-03	Lignite, burned in power plant/DE U	12%
Climate change	kg CO2 eq	5.82E-04	3.05E-03	4.68E-09	1.47E-03	5.09E-03	Aluminium, primary, liquid, at plant/RER U	25%
Climate change	kg CO2 eq	3.16E-03	7.86E-04	1.82E-07	2.13E-04	4.16E-03	Hard coal, burned in power plant/DE U	35%
Climate change	kg CO2 eq	3.14E-03	7.85E-07	3.17E-10	-2.96E-08	3.15E-03	Flat glass, uncoated, at plant/RER U	42%
Climate change	kg CO2 eq	2.50E-04	9.86E-04	1.62E-08	4.96E-04	1.73E-03	Light fuel oil, burned in industrial furnace 1MW, non-modulating/CH U	46%
Climate change	kg CO2 eq	1.45E-03	9.20E-09	1.05E-11	-4.57E-09	1.45E-03	Magnesium, at plant/RER U	50%
Climate change	kg CO2 eq	2.49E-04	7.23E-04	3.34E-08	3.26E-04	1.30E-03	Natural gas, burned in power plant/UCTE U	53%
Climate change	kg CO2 eq	8.55E-04	2.38E-04	2.30E-08	1.01E-04	1.19E-03	Operation, transoceanic freight ship/OCE U	56%
Climate change	kg CO2 eq	9.80E-04	3.40E-05	4.16E-08	-2.09E-05	9.93E-04	Natural gas, burned in power plant/DE U	58%
Climate change	kg CO2 eq	3.33E-04	3.02E-04	4.48E-08	1.32E-04	7.68E-04	Natural gas, burned in industrial furnace >100kW/RER U	60%
Climate change	kg CO2 eq	4.17E-04	1.67E-04	5.03E-09	1.04E-04	6.88E-04	Hard coal, burned in industrial furnace 1-10MW/RER U	62%
Climate change	kg CO2 eq	2.35E-04	3.11E-04	7.34E-08	8.32E-05	6.29E-04	Hard coal, burned in power plant/ES U	63%
Climate change	kg CO2 eq	7.55E-07	5.82E-04	1.08E-10	6.58E-09	5.83E-04	disposal, plastic, industr. electronics, 15.3% water, to municipal incineration/kg/CH U	65%
Climate change	kg CO2 eq	1.07E-04	3.24E-04	1.54E-07	1.25E-04	5.56E-04	Pig iron, at plant/GLO U	66%
Climate change	kg CO2 eq	3.79E-04	1.27E-04	4.92E-08	3.55E-05	5.42E-04	Hard coal, at mine/WEU U	67%
Climate change	kg CO2 eq	3.16E-04	1.86E-04	1.53E-08	-5.62E-07	5.01E-04	Natural gas, burned in industrial furnace low-NOx >100kW/RER U	69%
Climate change	kg CO2 eq	4.51E-04	2.42E-05	5.03E-09	-6.05E-07	4.75E-04	Light fuel oil, burned in industrial furnace 1MW, non-modulating/RER U	70%
Climate change	kg CO2 eq	7.87E-04	1.16E-04	9.50E-09	-4.50E-04	4.53E-04	Operation, lorry >16t, fleet average/RER U	71%
Climate change	kg CO2 eq	4.41E-04	1.20E-06	1.92E-09	5.25E-08	4.43E-04	Ethylene, average, at plant/RER U	72%
Climate change	kg CO2 eq	3.67E-04	4.31E-05	5.28E-08	-2.66E-05	3.84E-04	Blast furnace gas, burned in power plant/RER U	73%
Climate change	kg CO2 eq	0.00E+00	0.00E+00	0.00E+00	3.80E-04	3.80E-04	Avoided burden from recycling, CIS PV module, mounted construction/m2/RER U	74%
Climate change	kg CO2 eq	4.17E-05	3.19E-04	1.57E-07	-3.52E-06	3.57E-04	Polyethylene, HDPE, granulate, at plant/RER U	75%
Climate change	kg CO2 eq	1.47E-04	1.55E-04	5.16E-08	2.77E-05	3.29E-04	Hard coal, burned in power plant/IT U	75%
Climate change	kg CO2 eq	9.98E-05	1.67E-04	2.63E-08	5.62E-05	3.23E-04	Heavy fuel oil, burned in power plant/IT U	76%
Climate change	kg CO2 eq	1.22E-04	1.52E-04	3.45E-08	4.19E-05	3.16E-04	Hard coal, burned in power plant/FR U	77%
Climate change	kg CO2 eq	2.72E-04	5.89E-05	7.27E-08	-3.67E-05	2.95E-04	Lignite, burned in power plant/CZ U	78%
Climate change	kg CO2 eq	1.52E-04	1.04E-04	1.50E-08	1.67E-05	2.72E-04	Refinery gas, burned in furnace/MJ/RER U	78%
Climate change	kg CO2 eq	1.08E-04	1.31E-04	3.30E-08	3.31E-05	2.72E-04	Hard coal, burned in power plant/NL U	79%
Climate change	kg CO2 eq	2.32E-04	8.47E-05	1.05E-07	-5.29E-05	2.64E-04	Natural gas, burned in power plant/IT U	80%
Climate change	kg CO2 eq	1.19E-04	1.22E-04	4.21E-08	2.06E-05	2.61E-04	Lignite, burned in power plant/GR U	80%
Ozone depletion	kg CFC-11 eq	1.36E-09	3.62E-10	2.77E-13	2.13E-11	1.74E-09		
Ozone depletion	kg CFC-11 eq	3.11E-10	1.04E-10	2.93E-14	5.88E-12	4.21E-10	transport, natural gas, pipeline, long distance/tkm/RU U	24%
Ozone depletion	kg CFC-11 eq	2.32E-10	2.93E-15	8.19E-18	4.26E-16	2.32E-10	Methylchloride, at plant/WEU U	37%
Ozone depletion	kg CFC-11 eq	1.99E-10	1.70E-11	9.65E-15	-5.14E-13	2.15E-10	Uranium, enriched 4.0%, at USEC enrichment plant/US U	50%
Ozone depletion	kg CFC-11 eq	1.19E-10	3.97E-11	1.47E-14	-2.99E-13	1.58E-10	transport, natural gas, pipeline, long distance/tkm/RER U	59%
Ozone depletion	kg CFC-11 eq	6.08E-11	5.98E-11	2.07E-14	1.43E-11	1.35E-10	Crude oil, at production/NG U	67%
Ozone depletion	kg CFC-11 eq	9.00E-11	4.21E-11	1.77E-14	-1.84E-12	1.30E-10	Transport, natural gas, onshore pipeline, long distance/tkm/NAC U	74%
Ozone depletion	kg CFC-11 eq	5.20E-11	1.88E-11	5.22E-15	1.08E-12	7.19E-11	Transport, natural gas, onshore pipeline, long distance/NO U	78%
Ozone depletion	kg CFC-11 eq	1.97E-11	3.17E-11	4.42E-15	1.12E-11	6.25E-11	Uranium, enriched 3.9%, at USEC enrichment plant/US U	82%
Human toxicity, cancer effects	CTUh, c	3.14E-10	9.26E-10	6.81E-13	2.38E-10	1.48E-09		
Human toxicity, cancer effects	CTUh, c	1.07E-10	5.60E-10	4.36E-15	2.69E-10	9.36E-10	Disposal, redmud from bauxite digestion, 0% water, to residual material landfill/CH U	63%
Human toxicity, cancer effects	CTUh, c	4.61E-11	1.30E-10	9.16E-14	-1.90E-12	1.74E-10	Disposal, slag, unalloyed electr. steel, 0% water, to residual material landfill/CH U	75%
Human toxicity, cancer effects	CTUh, c	2.39E-11	8.75E-11	5.55E-14	3.53E-12	1.15E-10	Ferrochromium, high-carbon, 68% Cr, at plant/GLO U	83%
Human toxicity, non-cancer effects	CTUh, n-c	4.07E-09	4.91E-09	1.06E-11	1.52E-10	9.14E-09		
Human toxicity, non-cancer effects	CTUh, n-c	8.89E-10	3.23E-11	2.34E-14	9.82E-10	1.90E-09	Zinc, primary, at regional storage/RER U	21%
Human toxicity, non-cancer effects	CTUh, n-c	1.44E-09	2.45E-16	3.24E-19	1.49E-16	1.44E-09	Leaching residues, indium rich, from zinc circuit, at smelter/GLO U	37%
Human toxicity, non-cancer effects	CTUh, n-c	1.38E-10	1.75E-09	3.13E-14	-4.69E-10	1.41E-09	Copper, primary, at refinery/RLA U	52%
Human toxicity, non-cancer effects	CTUh, n-c	1.74E-10	9.08E-10	7.07E-15	-4.36E-10	6.45E-10	Disposal, redmud from bauxite digestion, 0% water, to residual material landfill/CH U	59%
Human toxicity, non-cancer effects	CTUh, n-c	1.35E-10	4.48E-10	1.66E-13	7.08E-12	5.91E-10	Steel, electric, un- and low-alloyed, at plant/RER U	66%
Human toxicity, non-cancer effects	CTUh, n-c	2.17E-11	2.75E-10	4.94E-15	2.66E-10	5.63E-10	Copper, secondary, at refinery/RER U	72%
Human toxicity, non-cancer effects	CTUh, n-c	1.99E-11	2.52E-10	4.52E-15	-6.76E-11	2.04E-10	Copper, primary, at refinery/ID U	74%
Human toxicity, non-cancer effects	CTUh, n-c	1.15E-10	1.65E-12	1.20E-15	5.03E-11	1.67E-10	Zinc concentrate, at beneficiation/GLO U	76%
Human toxicity, non-cancer effects	CTUh, n-c	1.66E-11	2.02E-10	3.64E-15	-5.44E-11	1.65E-10	Copper, primary, at refinery/RER U	78%
Human toxicity, non-cancer effects	CTUh, n-c	1.23E-10	3.07E-11	7.10E-15	-8.33E-12	1.46E-10	Hard coal, burned in power plant/DE U	79%
Human toxicity, non-cancer effects	CTUh, n-c	1.79E-13	1.38E-10	2.57E-17	-1.56E-15	1.38E-10	disposal, plastic, industr. electronics, 15.3% water, to municipal incineration/kg/CH U	81%
Particulate matter	kg PM2.5 eq	9.44E-06	8.12E-06	2.47E-09	2.36E-06	1.99E-05		
Particulate matter	kg PM2.5 eq	5.00E-07	2.62E-06	4.02E-12	1.26E-06	4.38E-06	Aluminium, primary, liquid, at plant/RER U	22%
Particulate matter	kg U235 eq	2.06E-06	5.15E-10	2.08E-13	-1.94E-11	2.06E-06	Flat glass, uncoated, at plant/RER U	32%
Particulate matter	kg PM2.5 eq	1.05E-06	2.91E-07	2.82E-11	1.23E-07	1.46E-06	Operation, transoceanic freight ship/OCE U	40%
Particulate matter	kg PM2.5 eq	4.75E-08	6.01E-07	1.08E-11	1.61E-07	8.10E-07	Copper, primary, at refinery/RLA U	44%
Particulate matter	kg PM2.5 eq	4.60E-07	1.84E-07	5.74E-11	1.15E-07	7.59E-07	Hard coal, burned in industrial furnace 1-10MW/RER U	48%
Particulate matter	kg PM2.5 eq	6.18E-07	1.40E-07	9.22E-11	-1.56E-08	7.43E-07	Electricity, at cogen 6400kWth, wood, allocation exergy/CH U	51%
Particulate matter	kg PM2.5 eq	1.73E-07	1.40E-07	1.80E-11	1.57E-07	4.70E-07	Heavy fuel oil, burned in industrial furnace 1MW, non-modulating/RER U	54%
Particulate matter	kg PM2.5 eq	1.57E-07	2.08E-07	4.90E-11	5.55E-08	4.20E-07	Hard coal, burned in power plant/ES U	56%
Particulate matter	kg PM2.5 eq	2.40E-08	3.03E-07	5.44E-12	8.14E-08	4.08E-07	Copper concentrate, at beneficiation/RLA U	58%
Particulate matter	kg PM2.5 eq	2.23E-08	2.82E-07	5.07E-12	7.59E-08	3.81E-07	Copper, primary, at refinery/ID U	60%
Particulate matter	kg PM2.5 eq	1.97E-07	1.28E-07	1.66E-11	2.80E-08	3.53E-07	Natural gas, sour, burned in production flare/MJ/GLO U	61%
Particulate matter	kg PM2.5 eq	3.66E-08	1.91E-07	1.60E-12	9.18E-08	3.20E-07	Bauxite, at mine/GLO U	63%
Particulate matter	kg PM2.5 eq	2.06E-07	9.54E-08	1.44E-10	-1.75E-09	3.00E-07	Diesel, burned in building machine/GLO U	65%
Particulate matter	kg PM2.5 eq	1.50E-07	1.31E-07	5.65E-11	1.19E-08	2.93E-07	Lignite, burned in power plant/CS U	66%
Particulate matter	kg PM2.5 eq	8.97E-08	1.50E-07	2.36E-11	5.05E-08	2.90E-07	Heavy fuel oil, burned in power plant/IT U	67%
Particulate matter	kg PM2.5 eq	1.60E-07	1.09E-07	1.58E-11	1.76E-08	2.87E-07	Heavy fuel oil, burned in refinery furnace/MJ/RER U	69%
Particulate matter	kg PM2.5 eq	2.07E-07	5.14E-08	1.19E-11	1.40E-08	2.72E-07	Hard coal, burned in power plant/DE U	70%
Particulate matter	kg PM2.5 eq	1.74E-08	2.14E-07	3.85E-12	3.55E-08	2.67E-07	Copper concentrate, at beneficiation/RER U	72%
Particulate matter	kg PM2.5 eq	2.28E-07	2.53E-08	9.48E-12	3.62E-09	2.57E-07	Lignite, burned in power plant/DE U	73%
Particulate matter	kg PM2.5 eq	7.21E-08	1.33E-07	1.02E-11	2.80E-08	2.33E-07	Blasting/RER U	74%
Particulate matter	kg PM2.5 eq	3.92E-08	1.19E-07	5.66E-11	4.58E-08	2.04E-07	Sinter, iron, at plant/GLO U	75%
Particulate matter	kg PM2.5 eq	2.95E-07	4.36E-08	3.57E-12	-1.69E-07	1.70E-07	Operation, lorry >16t, fleet average/RER U	76%
Particulate matter	kg PM2.5 eq	1.64E-07	4.46E-10	7.11E-13	1.95E-11	1.64E-07	Ethylene, average, at plant/RER U	77%
Particulate matter	kg PM2.5 eq	1.83E-08	1.40E-07	6.89E-11	-1.54E-09	1.57E-07	Polyethylene, HDPE, granulate, at plant/RER U	78%
Particulate matter	kg PM2.5 eq	2.56E-08	8.78E-08	1.52E-13	4.17E-08	1.55E-07	Lignite, burned in power plant/ES U	78%
Particulate matter	kg PM2.5 eq	3.04E-08	1.11E-07	7.06E-11	4.48E-09	1.46E-07	Ferrochromium, high-carbon, 68% Cr, at plant/GLO U	79%
Particulate matter	kg PM2.5 eq	6.52E-08	6.69E-08	2.31E-11	1.13E-08	1.43E-07	Lignite, burned in power plant/GR U	80%
Particulate matter	kg PM2.5 eq	4.82E-08	6.04E-08	1.37E-11	1.66E-08	1.25E-07	Hard coal, burned in power plant/FR U	80%
Ionizing radiation HH	kg PM2.5 eq	2.10E-03	1.15E-03	8.31E-07	1.32E-04	3.38E-03		
Ionizing radiation HH	kg PM2.5 eq	1.90E-03	1.06E-03	6.25E-07	1.28E-04	3.09E-03	Nuclear spent fuel, in reprocessing, at plant/RER U	91%
Ionizing radiation HH	kg PM2.5 eq	9.40E-05	4.24E-05	1.85E-08	6.58E-06	1.43E-04	Uranium natural, at underground mine/RNA U	96%

Tab. 6.12 Identification of the most relevant processes for each impact category and each life cycle stage for 1 kWh of DC electricity produced with a residential scale CIS PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant processes to the characterized environmental impacts is reached. (Continued)

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total	Most relevant processes	Cumulative share
electricity, 3 kWp installation, mounted with CIS PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling incl. recycling potential, 50:50	Contribution		
Photochemical ozone formation	kg NMVOC eq	7.90E-05	3.33E-05	1.63E-08	1.01E-06	1.13E-04	Operation, transoceanic freight ship/OCE U	20%
Photochemical ozone formation	kg NMVOC eq	1.66E-05	4.61E-06	4.47E-10	1.95E-06	2.31E-05	Flat glass, uncoated, at plant/RER U	35%
Photochemical ozone formation	kg NMVOC eq	2.40E-06	4.41E-06	3.39E-10	9.30E-07	7.74E-06	Blasting/RER U	42%
Photochemical ozone formation	kg NMVOC eq	8.29E-06	1.22E-06	1.00E-10	-4.75E-06	4.77E-06	Operation, lorry >16t, fleet average/RER U	46%
Photochemical ozone formation	kg NMVOC eq	3.19E-06	3.55E-07	1.33E-10	5.08E-08	3.60E-06	Lignite, burned in power plant/DE U	49%
Photochemical ozone formation	kg NMVOC eq	2.30E-06	1.06E-06	1.61E-09	-1.95E-08	3.35E-06	Diesel, burned in building machine/GLO U	52%
Photochemical ozone formation	kg NMVOC eq	2.30E-06	5.71E-07	1.32E-10	1.55E-07	3.02E-06	Hard coal, burned in power plant/DE U	55%
Photochemical ozone formation	kg NMVOC eq	1.03E-06	1.36E-06	3.21E-10	3.64E-07	2.76E-06	Hard coal, burned in power plant/ES U	57%
Photochemical ozone formation	kg NMVOC eq	1.28E-06	1.05E-06	2.80E-10	2.43E-07	2.58E-06	Natural gas, vented/GLO U	60%
Photochemical ozone formation	kg NMVOC eq	1.13E-06	4.52E-07	1.41E-10	2.81E-07	1.86E-06	Hard coal, burned in industrial furnace 1-10MW/RER U	61%
Photochemical ozone formation	kg NMVOC eq	9.92E-07	6.07E-07	1.30E-10	1.18E-07	1.72E-06	Diesel, burned in diesel-electric generating set/GLO U	63%
Photochemical ozone formation	kg NMVOC eq	1.34E-06	2.83E-07	1.11E-10	2.40E-08	1.65E-06	Operation, freight train/RER U	64%
Photochemical ozone formation	kg NMVOC eq	1.75E-07	1.34E-06	6.58E-10	-1.48E-08	1.50E-06	Polyethylene, HDPE, granulate, at plant/RER U	65%
Photochemical ozone formation	kg NMVOC eq	1.49E-06	4.06E-09	6.48E-12	1.78E-10	1.50E-06	Ethylene, average, at plant/RER U	67%
Photochemical ozone formation	kg NMVOC eq	2.26E-08	1.27E-06	1.15E-12	3.90E-10	1.29E-06	Operation, lorry 3.5-16t, fleet average/RER U	68%
Photochemical ozone formation	kg NMVOC eq	1.85E-07	7.31E-07	1.20E-11	3.67E-07	1.28E-06	Light fuel oil, burned in industrial furnace 1MW, non-modulating/CH U	69%
Photochemical ozone formation	kg NMVOC eq	2.33E-07	6.75E-07	3.12E-11	3.05E-07	1.21E-06	Natural gas, burned in power plant/UCTE U	70%
Photochemical ozone formation	kg NMVOC eq	3.70E-07	6.18E-07	9.74E-11	2.08E-07	1.20E-06	Heavy fuel oil, burned in power plant/IT U	71%
Photochemical ozone formation	kg NMVOC eq	6.31E-08	7.98E-07	1.43E-11	2.14E-07	1.07E-06	Copper, primary, at refinery/RLA U	72%
Photochemical ozone formation	kg NMVOC eq	1.23E-07	6.42E-07	9.87E-13	3.10E-07	1.07E-06	Aluminium, primary, liquid, at plant/RER U	73%
Photochemical ozone formation	kg NMVOC eq	4.46E-10	9.94E-07	3.18E-14	8.86E-11	9.94E-07	Operation, van < 3.5t/RER U	74%
Photochemical ozone formation	kg NMVOC eq	4.78E-07	3.11E-07	4.03E-11	6.80E-08	8.57E-07	Natural gas, sour, burned in production flare/Mi/GLO U	75%
Photochemical ozone formation	kg NMVOC eq	6.24E-07	1.81E-07	5.31E-11	8.81E-09	8.14E-07	Sweetening, natural gas/DE U	75%
Photochemical ozone formation	kg NMVOC eq	3.54E-07	3.73E-07	1.24E-10	6.69E-08	7.93E-07	Hard coal, burned in power plant/IT U	76%
Photochemical ozone formation	kg NMVOC eq	2.98E-07	3.73E-07	8.46E-11	1.03E-07	7.74E-07	Hard coal, burned in power plant/FR U	77%
Photochemical ozone formation	kg NMVOC eq	4.13E-07	2.71E-07	4.54E-11	5.53E-08	7.39E-07	Operation, transoceanic tanker/OCE U	77%
Photochemical ozone formation	kg NMVOC eq	3.55E-07	3.06E-07	8.80E-11	7.09E-08	7.33E-07	Natural gas, sweet, burned in production flare/Mi/GLO U	78%
Photochemical ozone formation	kg NMVOC eq	6.68E-07	2.31E-08	2.83E-11	-1.42E-08	6.77E-07	Natural gas, burned in power plant/DE U	79%
Photochemical ozone formation	kg NMVOC eq	4.41E-07	2.24E-07	1.04E-09	7.44E-09	6.73E-07	Operation, lorry 20-28t, fleet average/CH U	79%
Photochemical ozone formation	kg NMVOC eq	3.28E-07	2.13E-07	2.76E-11	4.67E-08	5.88E-07	crude oil, at production offshore/kg/GB U	80%
Photochemical ozone formation	kg NMVOC eq	3.00E-07	2.62E-07	1.13E-10	2.39E-08	5.87E-07	Lignite, burned in power plant/CS U	80%
Acidification	mol H+ eq	1.40E-04	8.38E-05	3.38E-08	1.83E-05	2.42E-04		
Acidification	mol H+ eq	2.84E-05	7.90E-06	7.65E-10	3.34E-06	3.96E-05	Operation, transoceanic freight ship/OCE U	16%
Acidification	mol H+ eq	3.50E-05	8.74E-09	3.53E-12	-3.30E-10	3.50E-05	Flat glass, uncoated, at plant/RER U	31%
Acidification	mol H+ eq	1.02E-06	1.29E-05	2.31E-10	3.46E-06	1.74E-05	Copper, primary, at refinery/RLA U	38%
Acidification	mol H+ eq	1.83E-06	9.56E-06	1.47E-11	4.61E-06	1.60E-05	Aluminium, primary, liquid, at plant/RER U	45%
Acidification	mol H+ eq	2.59E-06	4.77E-06	3.67E-10	1.01E-06	8.37E-06	Blasting/RER U	48%
Acidification	mol H+ eq	4.79E-07	6.05E-06	1.09E-10	1.63E-06	8.16E-06	Copper, primary, at refinery/ID U	51%
Acidification	mol H+ eq	2.99E-06	3.96E-06	9.34E-10	1.06E-06	8.01E-06	Hard coal, burned in power plant/ES U	55%
Acidification	mol H+ eq	4.32E-06	2.81E-06	3.64E-10	6.14E-07	7.75E-06	Natural gas, sour, burned in production flare/Mi/GLO U	58%
Acidification	mol H+ eq	3.64E-06	1.45E-06	4.54E-10	9.06E-07	6.00E-06	Hard coal, burned in industrial furnace 1-10MW/RER U	60%
Acidification	mol H+ eq	4.46E-06	1.11E-06	2.57E-10	3.01E-07	5.87E-06	Hard coal, burned in power plant/DE U	63%
Acidification	mol H+ eq	5.11E-06	5.69E-07	2.13E-10	8.13E-08	5.76E-06	Lignite, burned in power plant/DE U	65%
Acidification	mol H+ eq	1.21E-06	2.02E-06	3.17E-10	6.80E-07	3.90E-06	Heavy fuel oil, burned in power plant/IT U	67%
Acidification	mol H+ eq	1.88E-06	1.64E-06	7.10E-10	1.50E-07	3.68E-06	Lignite, burned in power plant/CS U	68%
Acidification	mol H+ eq	6.03E-06	8.91E-07	7.29E-11	-3.45E-06	3.47E-06	Operation, lorry >16t, fleet average/RER U	70%
Acidification	mol H+ eq	5.29E-07	1.82E-06	3.15E-12	8.63E-07	3.21E-06	Lignite, burned in power plant/ES U	71%
Acidification	mol H+ eq	2.10E-06	5.94E-08	2.99E-12	8.41E-07	3.00E-06	Soda, powder, at plant/RER U	72%
Acidification	mol H+ eq	1.63E-07	1.99E-06	3.57E-11	5.34E-07	2.68E-06	Copper, primary, at refinery/RER U	73%
Acidification	mol H+ eq	1.43E-06	9.39E-07	1.57E-10	1.92E-07	2.56E-06	Operation, transoceanic tanker/OCE U	75%
Acidification	mol H+ eq	9.46E-07	1.18E-06	2.68E-10	3.26E-07	2.46E-06	Hard coal, burned in power plant/FR U	76%
Acidification	mol H+ eq	1.59E-06	7.34E-07	1.11E-09	-1.34E-08	2.31E-06	Diesel, burned in building machine/GLO U	76%
Acidification	mol H+ eq	3.33E-07	1.31E-06	2.15E-11	6.60E-07	2.31E-06	Light fuel oil, burned in industrial furnace 1MW, non-modulating/CH U	77%
Acidification	mol H+ eq	1.02E-06	1.08E-06	3.59E-10	1.93E-07	2.29E-06	Hard coal, burned in power plant/IT U	78%
Acidification	mol H+ eq	2.07E-06	4.68E-08	3.28E-11	-1.82E-09	2.12E-06	Zinc coating, coils/RER U	79%
Acidification	mol H+ eq	1.09E-06	7.45E-07	1.08E-10	1.20E-07	1.95E-06	Heavy fuel oil, burned in refinery furnace/Mi/RER U	80%
Terrestrial eutrophication	mol N eq	2.99E-04	1.13E-04	8.83E-08	5.26E-06	4.18E-04		
Terrestrial eutrophication	mol N eq	6.45E-05	1.80E-05	1.74E-09	7.59E-06	9.01E-05	Operation, transoceanic freight ship/OCE U	22%
Terrestrial eutrophication	mol N eq	6.32E-05	1.58E-08	6.37E-12	-5.96E-10	6.32E-05	Flat glass, uncoated, at plant/RER U	37%
Terrestrial eutrophication	mol N eq	1.35E-05	2.49E-05	1.91E-09	5.25E-06	4.37E-05	Blasting/RER U	47%
Terrestrial eutrophication	mol N eq	3.45E-05	5.10E-06	4.17E-10	-1.98E-05	1.99E-05	Operation, lorry >16t, fleet average/RER U	52%
Terrestrial eutrophication	mol N eq	1.28E-05	1.42E-06	5.31E-10	2.03E-07	1.44E-05	Lignite, burned in power plant/DE U	55%
Terrestrial eutrophication	mol N eq	9.38E-06	2.65E-07	1.34E-11	3.76E-06	1.34E-05	Soda, powder, at plant/RER U	59%
Terrestrial eutrophication	mol N eq	8.77E-06	4.06E-06	6.13E-09	-7.44E-08	1.28E-05	Diesel, burned in building machine/GLO U	62%
Terrestrial eutrophication	mol N eq	8.95E-06	2.23E-06	5.15E-10	6.04E-07	1.18E-05	Hard coal, burned in power plant/DE U	64%
Terrestrial eutrophication	mol N eq	3.76E-06	4.98E-06	1.17E-09	1.33E-06	1.01E-05	Hard coal, burned in power plant/ES U	67%
Terrestrial eutrophication	mol N eq	9.27E-06	2.09E-07	1.47E-10	-8.12E-09	9.47E-06	Zinc coating, coils/RER U	69%
Terrestrial eutrophication	mol N eq	3.95E-06	2.42E-06	5.17E-10	4.69E-07	6.83E-06	Diesel, burned in diesel-electric generating set/GLO U	71%
Terrestrial eutrophication	mol N eq	5.24E-06	1.10E-06	4.31E-10	9.34E-08	6.43E-06	Operation, freight train/RER U	72%
Terrestrial eutrophication	mol N eq	3.86E-06	1.54E-06	4.82E-10	9.61E-07	6.37E-06	Hard coal, burned in industrial furnace 1-10MW/RER U	74%
Terrestrial eutrophication	mol N eq	9.00E-08	5.05E-06	4.56E-12	1.55E-09	5.14E-06	Operation, lorry 3.5-16t, fleet average/RER U	75%
Terrestrial eutrophication	mol N eq	7.25E-07	2.86E-06	4.68E-11	1.44E-06	5.02E-06	Light fuel oil, burned in industrial furnace 1MW, non-modulating/CH U	76%
Terrestrial eutrophication	mol N eq	9.43E-07	2.73E-06	1.26E-10	1.23E-06	4.91E-06	Natural gas, burned in power plant/UCTE U	77%
Terrestrial eutrophication	mol N eq	1.31E-06	2.19E-06	3.45E-10	7.39E-07	4.24E-06	Heavy fuel oil, burned in power plant/IT U	78%
Terrestrial eutrophication	mol N eq	1.60E-09	3.56E-06	1.14E-13	3.17E-10	3.56E-06	Operation, van < 3.5t/RER U	79%
Terrestrial eutrophication	mol N eq	1.51E-06	1.30E-06	3.74E-10	3.01E-07	3.11E-06	Natural gas, sweet, burned in production flare/Mi/GLO U	80%
Freshwater eutrophication	kg P eq	2.46E-06	4.07E-06	5.92E-09	9.05E-07	7.43E-06		
Freshwater eutrophication	kg P eq	8.03E-07	3.65E-06	6.93E-11	8.66E-07	5.31E-06	Disposal, sulfidic tailings, off-site/GLO U	72%
Freshwater eutrophication	kg P eq	1.37E-06	2.63E-07	1.22E-10	1.58E-08	1.65E-06	Disposal, spoil from lignite mining, in surface landfill/GLO U	94%

Tab. 6.12 Identification of the most relevant processes for each impact category and each life cycle stage for 1 kWh of DC electricity produced with a residential scale CIS PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant processes to the characterized environmental impacts is reached. (Continued)

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total	Most relevant processes	Cumulative share
electricity, 3 kWp installation, mounted with CIS PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling incl. recycling potential, 50:50	Contribution		
Marine eutrophication	kg N eq	2.55E-05	9.99E-06	1.30E-07	2.40E-07	3.55E-05	Operation, transoceanic freight ship/OCE U	19%
Marine eutrophication	kg N eq	5.77E-06	1.61E-06	1.55E-10	-6.79E-07	6.69E-06	Flat glass, uncoated, at plant/RER U	35%
Marine eutrophication	kg N eq	5.77E-06	1.44E-09	5.81E-13	5.44E-11	5.77E-06	Operation, lorry >16t, fleet average/RER U	50%
Marine eutrophication	kg N eq	3.15E-06	4.65E-07	3.80E-11	1.80E-06	5.41E-06	blasting/RER U	56%
Marine eutrophication	kg N eq	8.23E-07	1.51E-06	1.16E-10	-3.19E-07	2.02E-06	Uignite, burned in power plant/DE U	60%
Marine eutrophication	kg N eq	1.17E-06	1.30E-07	4.85E-11	-1.85E-08	1.28E-06	Diesel, burned in building machine/GLO U	63%
Marine eutrophication	kg N eq	8.00E-07	3.70E-07	5.59E-10	6.78E-09	1.18E-06	Hard coal, burned in power plant/DE U	66%
Marine eutrophication	kg N eq	8.17E-07	2.03E-07	4.70E-11	-5.51E-08	9.66E-07	Hard coal, burned in power plant/ES U	68%
Marine eutrophication	kg N eq	4.78E-07	1.01E-07	3.93E-11	-8.52E-09	5.70E-07	Operation, freight train/RER U	69%
Marine eutrophication	kg N eq	3.61E-07	2.21E-07	4.72E-11	-4.29E-08	5.38E-07	Diesel, burned in diesel-electric generating set/GLO U	71%
Marine eutrophication	kg N eq	8.19E-09	4.59E-07	4.15E-13	-1.41E-10	4.67E-07	Operation, lorry 3.5-16t, fleet average/RER U	72%
Marine eutrophication	kg N eq	3.53E-07	1.41E-07	4.40E-11	-8.78E-08	4.06E-07	Hard coal, burned in industrial furnace 1-10MW/RER U	73%
Marine eutrophication	kg N eq	1.44E-10	3.21E-07	1.03E-14	-2.86E-11	3.21E-07	Operation, van < 3.5t/RER U	74%
Marine eutrophication	kg N eq	2.82E-07	7.66E-10	1.22E-12	-3.35E-11	2.82E-07	Ethylene, average, at plant/RER U	75%
Marine eutrophication	kg N eq	1.40E-07	2.26E-11	1.27E-07	-3.63E-12	2.67E-07	Treatment, sewage, from residence, to wastewater treatment, class 2/CH U	76%
Marine eutrophication	kg N eq	4.26E-10	1.92E-11	2.52E-14	2.60E-07	2.60E-07	Treatment, PV cell production effluent, to wastewater treatment, class 3/CH U	76%
Marine eutrophication	kg N eq	2.42E-07	8.40E-09	1.03E-11	5.17E-09	2.56E-07	Natural gas, burned in power plant/DE U	77%
Marine eutrophication	kg N eq	1.20E-07	2.00E-07	3.15E-11	-6.74E-08	2.52E-07	Heavy fuel oil, burned in power plant/IT U	78%
Marine eutrophication	kg N eq	1.63E-07	8.26E-08	3.85E-10	-2.75E-09	2.43E-07	Operation, lorry 20-28t, fleet average/CH U	79%
Marine eutrophication	kg N eq	2.75E-08	2.10E-07	1.03E-10	2.32E-09	2.40E-07	Polyethylene, HDPE, granulate, at plant/RER U	79%
Marine eutrophication	kg N eq	1.38E-07	1.19E-07	3.41E-11	-2.75E-08	2.29E-07	Natural gas, sweet, burned in production flare/MJ/GLO U	80%
Marine eutrophication	kg N eq	1.66E-07	3.60E-08	4.44E-11	2.24E-08	2.25E-07	Uignite, burned in power plant/CZ U	80%
Freshwater ecotoxicity	CTUe	1.19E-02	7.87E-02	2.64E-05	2.31E-03	9.30E-02		
Freshwater ecotoxicity	CTUe	7.90E-05	6.09E-02	1.13E-08	6.89E-07	6.10E-02	disposal, plastic, industr. electronics, 15.3% water, to municipal incineration/kg/CH U	66%
Freshwater ecotoxicity	CTUe	1.33E-03	6.98E-03	5.44E-08	3.35E-03	1.17E-02	Disposal, redmud from bauxite digestion, 0% water, to residual material landfill/CH U	78%
Freshwater ecotoxicity	CTUe	2.32E-04	2.93E-03	5.26E-08	7.87E-04	3.95E-03	Copper, primary, at refinery/RLA U	82%
Land use	kg C deficit	4.21E-02	1.26E-02	1.84E-05	1.58E-03	5.63E-02		
Land use	kg C deficit	1.31E-02	6.69E-08	8.78E-11	-4.11E-08	1.31E-02	Photovoltaic panel factory/GLO/I U	23%
Land use	kg C deficit	7.32E-03	4.64E-03	8.14E-07	9.51E-04	1.29E-02	Well for exploration and production, onshore/GLO/I U	46%
Land use	kg C deficit	4.85E-03	2.46E-03	4.92E-07	3.63E-04	7.67E-03	Well for exploration and production, offshore/OCE/I U	60%
Land use	kg C deficit	7.77E-04	4.06E-03	3.40E-08	1.95E-03	6.78E-03	Bauxite, at mine/GLO U	72%
Land use	kg C deficit	-8.04E-04	-4.03E-03	-9.47E-08	-1.93E-03	-6.76E-03	Recultivation, bauxite mine/GLO U	60%
Land use	kg C deficit	3.22E-03	3.15E-04	4.15E-08	1.36E-05	3.54E-03	Softwood, standing, under bark, in forest/RER U	66%
Land use	kg C deficit	3.44E-03	3.14E-06	2.07E-09	1.53E-07	3.44E-03	Tin, at regional storage/RER U	72%
Land use	kg C deficit	-5.51E-04	-1.66E-03	-2.53E-08	-7.73E-04	-2.98E-03	electricity, hydropower, at reservoir power plant, non alpine regions/kWh/RER U	67%
Land use	kg C deficit	1.24E-03	8.13E-04	1.11E-07	1.78E-04	2.23E-03	production plant crude oil, onshore/p/GLO/I U	71%
Land use	kg C deficit	8.21E-04	3.80E-04	1.52E-07	9.82E-05	1.30E-03	Underground mine, hard coal/GLO/I U	73%
Land use	kg C deficit	1.00E-03	3.61E-07	1.15E-09	-1.61E-08	1.00E-03	Flat glass plant/RER/I U	75%
Land use	kg C deficit	1.48E-04	6.72E-04	1.28E-08	1.60E-04	9.80E-04	Disposal, sulfidic tailings, off-site/GLO U	77%
Land use	kg C deficit	1.08E-04	5.41E-04	4.11E-08	1.86E-04	8.35E-04	Aluminium electrolysis, plant/RER/I U	78%
Land use	kg C deficit	3.47E-04	4.41E-04	1.32E-09	5.15E-08	7.88E-04	Aluminium casting, plant/RER/I U	80%
Land use	kg C deficit	6.88E-04	6.74E-05	8.88E-09	2.91E-06	7.58E-04	Softwood, stand establishment / tending / site development, under bark/RER U	81%
Water resource depletion	m3 water eq	2.14E-05	2.07E-05	1.19E-07	7.78E-06	5.00E-05		
Water resource depletion	m3 water eq	5.63E-06	1.70E-05	2.59E-10	7.91E-06	3.05E-05	electricity, hydropower, at reservoir power plant, non alpine regions/kWh/RER U	61%
Water resource depletion	m3 water eq	6.06E-06	2.10E-07	2.57E-10	-1.29E-07	6.15E-06	electricity, hydropower, at pumped storage power plant/kWh/DE U	73%
Water resource depletion	m3 water eq	1.19E-06	1.32E-07	4.95E-11	1.89E-08	1.34E-06	Lignite, burned in power plant/DE U	76%
Water resource depletion	m3 water eq	9.59E-07	2.39E-07	5.52E-11	6.47E-08	1.26E-06	Hard coal, burned in power plant/DE U	78%
Water resource depletion	m3 water eq	7.55E-07	3.00E-07	1.93E-10	2.67E-08	1.08E-06	Water, decarbonised, at plant/RER U	81%
Mineral, fossil & ren resource depletion	kg Sb eq	5.12E-04	5.42E-07	4.56E-11	1.43E-09	5.12E-04		
Mineral, fossil & ren resource depletion	kg Sb eq	4.97E-04	9.31E-11	1.21E-13	5.34E-11	4.97E-04	Resource correction, PbZn, indium, positive/GLO U	97%
Mineral, fossil & ren resource depletion	kg Sb eq	2.46E-05	3.52E-07	2.55E-10	1.07E-05	3.57E-05	Zinc concentrate, at beneficiation/GLO U	104%
Mineral, fossil & ren resource depletion	kg Sb eq	-8.57E-06	-3.11E-07	-2.25E-10	-9.47E-06	-1.84E-05	Resource correction, PbZn, indium, negative/GLO U	100%
Mineral, fossil & ren resource depletion	kg Sb eq	-2.70E-06	-3.74E-08	-2.71E-11	-1.14E-06	-3.87E-06	Resource correction, PbZn, cadmium, negative/GLO U	100%
Cumulative energy demand non renewable	MJ oil eq	4.05E-01	1.82E-01	6.90E-05	2.59E-02	6.13E-01		
Cumulative energy demand non renewable	MJ oil eq	4.80E-02	2.17E-02	9.45E-06	3.37E-03	7.31E-02	Uranium natural, at underground mine/RNA U	12%
Cumulative energy demand non renewable	MJ oil eq	5.63E-02	1.08E-02	5.02E-06	6.50E-04	6.77E-02	Lignite, at mine/RER U	23%
Cumulative energy demand non renewable	MJ oil eq	3.20E-02	1.44E-02	6.30E-06	2.24E-03	4.87E-02	Uranium natural, at open pit mine/RNA U	31%
Cumulative energy demand non renewable	MJ oil eq	3.37E-02	1.13E-02	4.38E-06	3.16E-03	4.81E-02	Hard coal, at mine/WEU U	39%
Cumulative energy demand non renewable	MJ oil eq	1.94E-02	1.28E-02	1.89E-06	2.65E-03	3.48E-02	Crude oil, at production onshore/RME U	44%
Cumulative energy demand non renewable	MJ oil eq	2.19E-02	7.76E-03	2.06E-06	5.21E-04	3.01E-02	natural gas, at production offshore/m3/NO U	49%
Cumulative energy demand non renewable	MJ oil eq	2.17E-02	7.31E-03	2.06E-06	4.10E-04	2.94E-02	natural gas, at production onshore/m3/RU U	54%
Cumulative energy demand non renewable	MJ oil eq	1.99E-02	8.94E-03	3.37E-06	-1.08E-04	2.88E-02	natural gas, at production onshore/m3/NAC U	59%
Cumulative energy demand non renewable	MJ oil eq	1.60E-02	1.04E-02	1.34E-06	2.28E-03	2.86E-02	crude oil, at production offshore/kg/NO U	64%
Cumulative energy demand non renewable	MJ oil eq	1.33E-02	8.66E-03	1.12E-06	1.90E-03	2.39E-02	crude oil, at production offshore/kg/RU U	67%
Cumulative energy demand non renewable	MJ oil eq	1.33E-02	8.63E-03	1.12E-06	1.89E-03	2.38E-02	crude oil, at production offshore/kg/GB U	71%
Cumulative energy demand non renewable	MJ oil eq	2.12E-02	5.77E-05	9.21E-08	2.52E-06	2.13E-02	Ethylene, average, at plant/RER U	75%
Cumulative energy demand non renewable	MJ oil eq	9.99E-03	8.46E-03	2.35E-06	1.97E-03	2.04E-02	Crude oil, at production onshore/RAF U	78%
Cumulative energy demand non renewable	MJ oil eq	1.20E-02	6.88E-03	3.50E-06	1.41E-03	2.03E-02	Hard coal, at mine/EEU U	81%
Cumulative energy demand renewable	MJ oil eq	4.15E-02	3.66E-02	3.60E+00	1.30E-02	3.69E+00		
Cumulative energy demand renewable	MJ oil eq	0.00E+00	0.00E+00	3.60E+00	0.00E+00	3.60E+00	hotspot electricity, PV, at 3kWp slanted-roof, CIS, panel, mounted, incl. EoL absolute/RER U	98%
Cumulative energy demand renewable	MJ oil eq	9.35E-03	1.57E-02	1.71E-06	6.10E-03	3.11E-02	electricity, hydropower, at run-of-river power plant without reservoir/kWh/RER U	98%
Nuclear waste	m3 HAA eq	5.18E-11	2.53E-11	1.54E-14	3.38E-12	8.05E-11		
Nuclear waste	m3 HAA eq	5.18E-11	2.52E-11	1.54E-14	3.38E-12	8.04E-11	Radioactive waste, in final repository for nuclear waste SF, HLW, and ILW/CH U	100%
Nuclear waste	m3 HAA eq	1.38E-14	6.36E-15	3.99E-18	9.52E-16	2.11E-14	Radioactive waste, in final repository for nuclear waste LLW/CH U	100%

Tab. 6.13

Identification of the most relevant processes for each impact category and each life cycle stage for 1 kWh of DC electricity produced with a residential scale micro-Si PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant processes to the characterized environmental impacts is reached.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total	Most relevant processes	Cumulative share
electricity, 3 kWp installation, mounted with micro-Si PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling incl. recycling potential, 50:50	Contribution		
Climate change	kg CO ₂ eq	4.09E-02	1.36E-02	4.86E-06	3.03E-03	5.76E-02		
Climate change	kg CO ₂ eq	1.17E-02	1.37E-07	1.84E-10	-7.67E-08	1.17E-02	Hard coal, burned in power plant/CN U	20%
Climate change	kg CO ₂ eq	7.00E-04	3.29E-03	5.05E-09	1.66E-03	5.64E-03	Aluminium, primary, liquid, at plant/RER U	30%
Climate change	kg CO ₂ eq	4.47E-03	5.25E-08	7.06E-11	-2.94E-08	4.47E-03	Hard coal, at mine/CN U	38%
Climate change	kg CO ₂ eq	4.30E-03	0.00E+00	0.00E+00	0.00E+00	4.30E-03	nitrogen trifluoride, at plant/RER U	45%
Climate change	kg CO ₂ eq	3.66E-03	8.21E-07	3.42E-10	-2.12E-08	3.67E-03	Flat glass, uncoated, at plant/RER U	52%
Climate change	kg CO ₂ eq	2.24E-03	1.20E-12	6.23E-18	3.76E-14	2.24E-03	Operation, lorry 16-32t, EURO3/RER U	56%
Climate change	kg CO ₂ eq	2.96E-04	1.06E-03	1.75E-08	5.59E-04	1.92E-03	Light fuel oil, burned in industrial furnace 1MW, non-modulating/CH U	59%
Climate change	kg CO ₂ eq	1.89E-03	9.87E-09	1.13E-11	-4.44E-09	1.89E-03	Magnesium, at plant/RER U	62%
Climate change	kg CO ₂ eq	5.36E-04	8.48E-04	1.96E-07	2.54E-04	1.64E-03	Hard coal, burned in power plant/DE U	65%
Climate change	kg CO ₂ eq	1.26E-03	2.56E-04	2.49E-08	1.13E-04	1.63E-03	Operation, transoceanic freight ship/OCE U	68%
Climate change	kg CO ₂ eq	2.81E-04	7.80E-04	3.61E-08	3.69E-04	1.43E-03	Natural gas, burned in power plant/UCTE U	70%
Climate change	kg CO ₂ eq	4.53E-04	5.47E-04	2.05E-07	9.56E-05	1.10E-03	Lignite, burned in power plant/DE U	72%
Climate change	kg CO ₂ eq	4.72E-04	1.77E-04	5.62E-08	1.19E-04	7.68E-04	Hard coal, burned in industrial furnace 1-10MW/RER U	74%
Climate change	kg CO ₂ eq	2.18E-04	3.24E-04	4.83E-08	1.58E-04	7.01E-04	Natural gas, burned in industrial furnace >100kW/RER U	75%
Climate change	kg CO ₂ eq	2.14E-04	3.36E-04	7.92E-08	9.91E-05	6.49E-04	Hard coal, burned in power plant/ES U	76%
Climate change	kg CO ₂ eq	1.53E-04	3.48E-04	1.67E-07	1.36E-04	6.38E-04	Pig iron, at plant/GLO U	77%
Climate change	kg CO ₂ eq	2.01E-05	4.44E-06	2.53E-07	-6.19E-04	-5.94E-04	disposal, plastics, mixture, 15.3% water, to municipal incineration/kg/CH U	76%
Climate change	kg CO ₂ eq	6.03E-07	5.82E-04	1.17E-10	1.65E-08	5.83E-04	disposal, plastic, industr. electronics, 15.3% water, to municipal incineration/kg/CH U	77%
Climate change	kg CO ₂ eq	3.75E-04	2.00E-04	1.65E-08	-7.99E-08	5.75E-04	Natural gas, burned in industrial furnace low-NOx >100kW/RER U	78%
Climate change	kg CO ₂ eq	7.97E-04	1.25E-04	1.03E-08	-4.47E-04	4.73E-04	Operation, lorry >16t, fleet average/RER U	79%
Climate change	kg CO ₂ eq	1.11E-04	3.19E-04	1.69E-07	-3.45E-06	4.27E-04	Polyethylene, HDPE, granulate, at plant/RER U	80%
Climate change	kg CO ₂ eq	0.00E+00	0.00E+00	0.00E+00	4.24E-04	4.24E-04	Avoided burden from recycling, micro-Si PV module, mounted construction/m2/RER U	80%
Ozone depletion	kg CFC-11 eq	7.50E-10	3.89E-10	2.99E-13	3.39E-11	1.17E-09		
Ozone depletion	kg CFC-11 eq	2.48E-10	1.12E-10	3.17E-14	8.96E-12	3.69E-10	transport, natural gas, pipeline, long distance/tkm/RU U	31%
Ozone depletion	kg CFC-11 eq	1.08E-10	6.44E-11	2.23E-14	1.75E-11	1.90E-10	Crude oil, at production/NG U	48%
Ozone depletion	kg CFC-11 eq	8.78E-11	4.53E-11	1.91E-14	-7.10E-13	1.32E-10	Transport, natural gas, onshore pipeline, long distance/tkm/NAC U	59%
Ozone depletion	kg CFC-11 eq	8.88E-11	4.27E-11	1.59E-14	8.02E-13	1.32E-10	transport, natural gas, pipeline, long distance/tkm/RER U	70%
Ozone depletion	kg CFC-11 eq	1.80E-11	3.42E-11	4.78E-15	1.29E-11	6.51E-11	Uranium, enriched 3.9%, at USEC enrichment plant/US U	76%
Ozone depletion	kg CFC-11 eq	4.02E-11	2.02E-11	5.64E-15	1.63E-12	6.20E-11	Transport, natural gas, onshore pipeline, long distance/NO U	81%
Human toxicity, cancer effects	CTUh, c	4.00E-10	9.96E-10	7.36E-13	2.74E-10	1.67E-09		
Human toxicity, cancer effects	CTUh, c	1.29E-10	6.05E-10	4.71E-15	3.04E-10	1.04E-09	Disposal, redmud from bauxite digestion, 0% water, to residual material landfill/CH U	62%
Human toxicity, cancer effects	CTUh, c	6.38E-11	1.40E-10	9.90E-14	-1.33E-12	2.02E-10	Disposal, slag, unalloyed electr. steel, 0% water, to residual material landfill/CH U	74%
Human toxicity, cancer effects	CTUh, c	2.38E-11	9.33E-11	5.99E-14	3.78E-12	1.21E-10	Ferrochromium, high-carbon, 68% Cr, at plant/GLO U	81%
Human toxicity, non-cancer effects	CTUh, n-c	4.63E-09	5.08E-09	1.14E-11	1.98E-10	9.91E-09		
Human toxicity, non-cancer effects	CTUh, n-c	8.66E-10	3.31E-11	2.53E-14	1.11E-09	2.01E-09	Zinc, primary, at regional storage/RER U	20%
Human toxicity, non-cancer effects	CTUh, n-c	5.08E-10	1.75E-09	3.38E-14	-4.69E-10	1.79E-09	Copper, primary, at refinery/RLA U	38%
Human toxicity, non-cancer effects	CTUh, n-c	1.24E-09	1.38E-11	1.09E-14	-5.80E-13	1.25E-09	Disposal, hard coal ash, 0% water, to residual material landfill/PL U	51%
Human toxicity, non-cancer effects	CTUh, n-c	2.09E-10	9.81E-10	7.64E-15	-4.92E-10	6.97E-10	Disposal, redmud from bauxite digestion, 0% water, to residual material landfill/CH U	58%
Human toxicity, non-cancer effects	CTUh, n-c	1.76E-10	4.83E-10	1.79E-13	5.65E-12	6.65E-10	electric, un- and low-alloyed, at plant/RER U	65%
Human toxicity, non-cancer effects	CTUh, n-c	8.01E-11	2.75E-10	5.33E-15	2.66E-10	6.22E-10	Copper, secondary, at refinery/RER U	71%
Human toxicity, non-cancer effects	CTUh, n-c	5.69E-10	6.68E-15	8.98E-18	3.74E-15	5.69E-10	Hard coal, burned in power plant/CN U	77%
Human toxicity, non-cancer effects	CTUh, n-c	7.33E-11	2.52E-10	4.88E-15	-6.76E-11	2.58E-10	Copper, primary, at refinery/ID U	79%
Human toxicity, non-cancer effects	CTUh, n-c	5.90E-11	2.03E-10	3.93E-15	-5.44E-11	2.07E-10	Copper, primary, at refinery/RER U	81%
Particulate matter	kg PM2.5 eq	5.00E-05	8.59E-06	2.66E-09	2.69E-06	6.12E-05		
Particulate matter	kg PM2.5 eq	2.39E-05	2.81E-10	3.78E-13	-1.58E-10	2.39E-05	Hard coal, burned in power plant/CN U	39%
Particulate matter	kg U235 eq	1.41E-05	1.66E-10	2.23E-13	-9.28E-11	1.41E-05	Hard coal, burned in coal mine power plant/CN U	62%
Particulate matter	kg PM2.5 eq	6.02E-07	2.83E-06	4.34E-12	1.42E-06	4.85E-06	Aluminium, primary, liquid, at plant/RER U	70%
Particulate matter	kg PM2.5 eq	2.40E-06	5.38E-10	2.24E-13	-1.39E-11	2.40E-06	Flat glass, uncoated, at plant/RER U	74%
Particulate matter	kg PM2.5 eq	1.54E-06	3.13E-07	3.04E-11	1.39E-07	1.99E-06	Operation, transoceanic freight ship/OCE U	77%
Particulate matter	kg PM2.5 eq	1.09E-06	1.28E-11	1.72E-14	-7.16E-12	1.09E-06	Hard coal, at mine/CN U	79%
Particulate matter	kg PM2.5 eq	1.75E-07	6.01E-07	1.16E-11	1.61E-07	9.38E-07	Copper, primary, at refinery/RLA U	81%
Ionizing radiation HH	kg PM2.5 eq	1.20E-03	1.24E-03	8.98E-07	1.83E-04	2.62E-03		
Ionizing radiation HH	kg PM2.5 eq	1.06E-03	1.14E-03	6.75E-07	1.75E-04	2.37E-03	Nuclear spent fuel, in reprocessing, at plant/RER U	91%
Ionizing radiation HH	kg PM2.5 eq	4.50E-05	4.57E-05	2.00E-08	8.54E-06	9.92E-05	Uranium natural, at underground mine/RNA U	94%
Photochemical ozone formation	kg NMVOC eq	1.67E-04	3.53E-05	1.76E-08	2.07E-06	2.04E-04		
Photochemical ozone formation	kg NMVOC eq	5.77E-05	6.78E-10	9.11E-13	-3.80E-10	5.77E-05	Hard coal, burned in power plant/CN U	28%
Photochemical ozone formation	kg NMVOC eq	2.44E-05	4.96E-06	4.82E-10	2.20E-06	3.16E-05	Operation, transoceanic freight ship/OCE U	44%
Photochemical ozone formation	kg NMVOC eq	1.93E-05	4.32E-09	1.80E-12	-1.11E-10	1.93E-05	Flat glass, uncoated, at plant/RER U	53%
Photochemical ozone formation	kg NMVOC eq	1.92E-05	1.03E-14	5.36E-20	3.24E-16	1.92E-05	Operation, lorry 16-32t, EURO3/RER U	63%
Photochemical ozone formation	kg NMVOC eq	2.29E-06	4.49E-06	3.66E-10	9.50E-07	7.73E-06	Blasting/RER U	66%
Photochemical ozone formation	kg NMVOC eq	8.40E-06	1.32E-06	1.08E-10	-4.73E-06	4.98E-06	Operation, lorry >16t, fleet average/RER U	69%
Photochemical ozone formation	kg NMVOC eq	2.97E-06	1.14E-06	1.74E-09	1.79E-08	4.13E-06	Diesel, burned in building machine/GLO U	71%
Photochemical ozone formation	kg NMVOC eq	2.12E-06	1.13E-06	3.02E-10	2.98E-07	3.55E-06	Natural gas, vented/GLO U	73%
Photochemical ozone formation	kg NMVOC eq	9.38E-07	1.47E-06	3.47E-10	4.34E-07	2.84E-06	Hard coal, burned in power plant/ES U	74%
Photochemical ozone formation	kg NMVOC eq	2.46E-06	2.89E-11	3.88E-14	-1.62E-11	2.46E-06	Hard coal, at mine/CN U	75%
Photochemical ozone formation	kg NMVOC eq	1.33E-06	6.53E-07	1.40E-10	1.48E-07	2.13E-06	Diesel, burned in diesel-electric generating set/GLO U	76%
Photochemical ozone formation	kg NMVOC eq	2.08E-06	2.45E-11	3.29E-14	-1.37E-11	2.08E-06	Hard coal, burned in coal mine power plant/CN U	77%
Photochemical ozone formation	kg NMVOC eq	1.28E-06	4.80E-07	1.52E-10	3.22E-07	2.08E-06	Hard coal, burned in industrial furnace 1-10MW/RER U	78%
Photochemical ozone formation	kg NMVOC eq	1.75E-06	2.37E-07	1.13E-09	1.54E-08	2.01E-06	Operation, lorry 20-28t, fleet average/CH U	79%
Photochemical ozone formation	kg NMVOC eq	4.64E-07	1.34E-06	7.10E-10	-1.44E-08	1.79E-06	Polyethylene, HDPE, granulate, at plant/RER U	80%
Acidification	mol H+ eq	3.62E-04	8.81E-05	3.65E-08	2.11E-05	4.71E-04		
Acidification	mol H+ eq	1.61E-04	1.89E-09	2.54E-12	-1.06E-09	1.61E-04	Hard coal, burned in power plant/CN U	34%
Acidification	mol H+ eq	4.19E-05	8.50E-06	8.26E-10	3.77E-06	5.41E-05	Operation, transoceanic freight ship/OCE U	46%
Acidification	mol H+ eq	4.08E-05	9.13E-09	3.81E-12	-2.36E-10	4.08E-05	Flat glass, uncoated, at plant/RER U	54%
Acidification	mol H+ eq	2.33E-05	2.74E-10	3.69E-13	-1.54E-10	2.33E-05	Hard coal, at mine/CN U	59%
Acidification	mol H+ eq	3.75E-06	1.29E-05	2.50E-10	3.46E-06	2.01E-05	Copper, primary, at refinery/RLA U	64%
Acidification	mol H+ eq	2.20E-06	1.03E-05	1.59E-11	5.20E-06	1.77E-05	Aluminium, primary, liquid, at plant/RER U	67%
Acidification	mol H+ eq	1.42E-05	7.60E-15	3.95E-20	2.39E-16	1.42E-05	Operation, lorry 16-32t, EURO3/RER U	70%
Acidification	mol H+ eq	6.61E-06	3.02E-06	3.93E-10	7.61E-07	1.04E-05	Natural gas, sour, burned in production flare/MJ/GLO U	73%
Acidification	mol H+ eq	1.76E-06	6.06E-06	1.17E-10	1.63E-06	9.45E-06	Copper, primary, at refinery/ID U	75%
Acidification	mol H+ eq	2.47E-06	4.86E-06	3.96E-10	1.03E-06	8.36E-06	Blasting/RER U	76%
Acidification	mol H+ eq	2.73E-06	4.27E-06	1.01E-09	1.26E-06	8.26E-06	Hard coal, burned in power plant/ES U	78%
Acidification	mol H+ eq	7.92E-06	9.31E-11	1.25E-13	-5.22E-11	7.92E-06	Hard coal, burned in coal mine power plant/CN U	80%
Acidification	mol H+ eq	4.12E-06	1.55E-06	4.90E-10	1.04E-06	6.70E-06	Hard coal, burned in industrial furnace 1-10MW/RER U	81%

Tab. 6.13 Identification of the most relevant processes for each impact category and each life cycle stage for 1 kWh of DC electricity produced with a residential scale micro-Si PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant processes to the characterized environmental impacts is reached. (Continued)

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total	Most relevant processes	Cumulative share
electricity, 3 kWp installation, mounted with micro-Si PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling incl. recycling potential, 50:50	Contribution		
Terrestrial eutrophication	mol N eq	6.15E-04	1.20E-04	9.54E-08	9.17E-06	7.44E-04		
Terrestrial eutrophication	mol N eq	2.13E-04	2.50E-09	3.36E-12	-1.40E-09	2.13E-04	Hard coal, burned in power plant/CN U	29%
Terrestrial eutrophication	mol N eq	9.52E-05	1.93E-05	1.88E-09	8.57E-06	1.23E-04	Operation, transoceanic freight ship/OCE U	45%
Terrestrial eutrophication	mol N eq	8.11E-05	4.34E-14	2.26E-19	1.36E-15	8.11E-05	Operation, lorry 16-32t, EURO3/RER U	56%
Terrestrial eutrophication	mol N eq	7.37E-05	1.65E-08	6.88E-12	-4.26E-10	7.37E-05	Flat glass, uncoated, at plant/RER U	66%
Terrestrial eutrophication	mol N eq	1.29E-05	2.54E-05	2.07E-09	5.36E-06	4.36E-05	Blasting/RER U	72%
Terrestrial eutrophication	mol N eq	3.50E-05	5.49E-06	4.50E-10	-1.97E-05	2.07E-05	Operation, lorry >16t, fleet average/RER U	75%
Terrestrial eutrophication	mol N eq	1.13E-05	4.34E-06	6.62E-09	6.82E-08	1.57E-05	Diesel, burned in building machine/GLO U	77%
Terrestrial eutrophication	mol N eq	1.09E-05	2.86E-07	1.44E-11	4.20E-06	1.54E-05	Soda, powder, at plant/RER U	79%
Terrestrial eutrophication	mol N eq	3.43E-06	5.37E-06	1.27E-09	1.59E-06	1.04E-05	Hard coal, burned in power plant/ES U	80%
Freshwater eutrophication	kg P eq	1.84E-06	4.11E-06	6.40E-09	9.18E-07	6.87E-06		
Freshwater eutrophication	kg P eq	1.14E-06	3.65E-06	7.48E-11	8.60E-07	5.65E-06	Disposal, sulfidic tailings, off-site/GLO U	82%
Freshwater eutrophication	kg P eq	3.77E-07	1.61E-07	5.27E-11	4.89E-08	5.87E-07	Disposal, spoil from coal mining, in surface landfill/GLO U	91%
Marine eutrophication	kg N eq	5.48E-05	1.02E-05	1.40E-07	7.64E-08	6.52E-05		
Marine eutrophication	kg N eq	1.94E-05	2.28E-10	3.07E-13	-1.28E-10	1.94E-05	Hard coal, burned in power plant/CN U	30%
Marine eutrophication	kg N eq	8.51E-06	1.73E-06	1.68E-10	7.66E-07	1.10E-05	Operation, transoceanic freight ship/OCE U	47%
Marine eutrophication	kg N eq	7.39E-06	3.96E-15	2.06E-20	1.24E-16	7.39E-06	Operation, lorry 16-32t, EURO3/RER U	58%
Marine eutrophication	kg N eq	6.73E-06	1.51E-09	6.28E-13	-3.89E-11	6.73E-06	Flat glass, uncoated, at plant/RER U	68%
Marine eutrophication	kg N eq	7.84E-07	1.54E-06	1.26E-10	3.26E-07	2.65E-06	Blasting/RER U	72%
Marine eutrophication	kg N eq	3.19E-06	5.01E-07	4.11E-11	-1.80E-06	1.89E-06	Operation, lorry >16t, fleet average/RER U	75%
Marine eutrophication	kg N eq	1.03E-06	3.96E-07	6.03E-10	6.22E-09	1.43E-06	Diesel, burned in building machine/GLO U	77%
Marine eutrophication	kg N eq	3.13E-07	4.91E-07	1.16E-10	1.45E-07	9.49E-07	Hard coal, burned in power plant/ES U	79%
Marine eutrophication	kg N eq	4.83E-07	2.37E-07	5.10E-11	5.38E-08	7.74E-07	Diesel, burned in diesel-electric generating set/GLO U	80%
Freshwater ecotoxicity	CTUe	1.39E-02	7.97E-02	2.85E-05	2.72E-03	9.63E-02		
Freshwater ecotoxicity	CTUe	6.31E-05	6.09E-02	1.22E-08	1.72E-06	6.10E-02	disposal, plastic, industr. electronics, 15.3% water, to municipal incineration/kg/CH U	63%
Freshwater ecotoxicity	CTUe	1.60E-03	7.53E-03	5.87E-08	3.78E-03	1.29E-02	Disposal, redmud from bauxite digestion, 0% water, to residual material landfill/CH U	77%
Freshwater ecotoxicity	CTUe	8.53E-04	2.93E-03	5.68E-08	7.87E-04	4.57E-03	Copper, primary, at refinery/RLA U	81%
Land use	kg C deficit	3.51E-02	1.35E-02	1.99E-05	2.00E-03	5.07E-02		
Land use	kg C deficit	1.06E-02	5.00E-03	8.79E-07	1.19E-03	1.67E-02	Well for exploration and production, onshore/GLO/I U	33%
Land use	kg C deficit	5.59E-03	2.65E-03	5.32E-07	4.69E-04	8.71E-03	Well for exploration and production, offshore/OCE/I U	50%
Land use	kg C deficit	9.33E-04	4.38E-03	3.67E-08	2.20E-03	7.51E-03	Bauxite, at mine/GLO U	65%
Land use	kg C deficit	-9.45E-04	-4.35E-03	-1.02E-07	-2.18E-03	-7.47E-03	Recultivation, bauxite mine/GLO U	50%
Land use	kg C deficit	-1.46E-03	-1.79E-03	-2.73E-08	-8.73E-04	-4.13E-03	electricity, hydropower, at reservoir power plant, non alpine regions/kWh/RER U	42%
Land use	kg C deficit	3.92E-03	4.60E-08	6.19E-11	-2.58E-08	3.92E-03	Hard coal, at mine/CN U	50%
Land use	kg C deficit	1.90E-03	8.74E-04	1.19E-07	2.21E-04	3.00E-03	production plant crude oil, onshore/p/GLO/I U	56%
Land use	kg C deficit	2.62E-03	3.08E-08	4.14E-11	-1.73E-08	2.62E-03	Underground mine, hard coal/CN/I U	61%
Land use	kg C deficit	1.31E-03	1.84E-04	3.14E-07	3.87E-05	1.54E-03	Hardwood, standing, under bark, in forest/RER U	64%
Land use	kg C deficit	1.17E-03	3.76E-07	1.24E-09	-1.24E-08	1.17E-03	Flat glass plant/RER/I U	66%
Land use	kg C deficit	2.09E-04	6.73E-04	1.38E-08	1.58E-04	1.04E-03	Disposal, sulfidic tailings, off-site/GLO U	68%
Land use	kg C deficit	3.47E-04	4.26E-04	6.69E-08	1.56E-04	9.30E-04	Residual material landfill facility/CH/I U	70%
Land use	kg C deficit	1.34E-04	5.78E-04	4.44E-08	2.11E-04	9.22E-04	Aluminium electrolysis, plant/RER/I U	72%
Land use	kg C deficit	4.42E-04	4.76E-04	1.42E-09	1.08E-07	9.18E-04	Aluminium casting, plant/RER/I U	74%
Land use	kg C deficit	7.21E-04	2.46E-04	2.60E-08	-1.00E-04	8.67E-04	Operation, maintenance, road/CH/I U	76%
Land use	kg C deficit	3.41E-04	4.09E-04	1.65E-07	1.16E-04	8.65E-04	Underground mine, hard coal/GLO/I U	77%
Land use	kg C deficit	5.63E-04	2.70E-05	1.59E-07	1.78E-04	7.68E-04	Sand, at mine/CH U	79%
Land use	kg C deficit	-5.40E-04	-9.22E-05	-7.20E-07	-1.30E-04	-7.63E-04	Recultivation, limestone mine/CH U	77%
Land use	kg C deficit	6.71E-04	0.00E+00	0.00E+00	0.00E+00	6.71E-04	Solar collector factory/RER/I U	79%
Land use	kg C deficit	6.59E-04	1.03E-04	9.05E-08	-1.01E-04	6.60E-04	Gravel, crushed, at mine/CH U	80%
Land use	kg C deficit	3.03E-04	3.38E-04	4.48E-08	1.66E-05	6.58E-04	Softwood, standing, under bark, in forest/RER U	81%
Water resource depletion	m3 water eq	6.33E-05	2.23E-05	1.28E-07	8.94E-06	9.47E-05		
Water resource depletion	m3 water eq	1.50E-05	1.83E-05	2.80E-10	8.93E-06	4.22E-05	electricity, hydropower, at reservoir power plant, non alpine regions/kWh/RER U	45%
Water resource depletion	m3 water eq	3.78E-05	0.00E+00	0.00E+00	0.00E+00	3.78E-05	photovoltaic panel, micro-Si, at plant/m2/CN/I U	85%
Mineral, fossil & ren resource depletion	kg Sb eq	9.53E-07	5.50E-07	4.93E-11	1.28E-08	1.52E-06		
Mineral, fossil & ren resource depletion	kg Sb eq	9.45E-06	3.62E-07	2.76E-10	1.21E-05	2.19E-05	Zinc concentrate, at beneficiation/GLO U	1444%
Mineral, fossil & ren resource depletion	kg Sb eq	-8.35E-06	-3.19E-07	-2.44E-10	-1.07E-05	-1.93E-05	Resource correction, PbZn, indium, negative/GLO U	168%
Mineral, fossil & ren resource depletion	kg Sb eq	-1.00E-06	-3.83E-08	-2.92E-11	-1.28E-06	-2.32E-06	Resource correction, PbZn, cadmium, negative/GLO U	15%
Mineral, fossil & ren resource depletion	kg Sb eq	6.54E-08	2.25E-07	4.36E-12	-3.73E-08	2.53E-07	Copper concentrate, at beneficiation/RER U	32%
Mineral, fossil & ren resource depletion	kg Sb eq	2.30E-07	7.88E-09	6.04E-13	-3.24E-09	2.35E-07	Fluorspar, 97%, at plant/GLO U	47%
Mineral, fossil & ren resource depletion	kg Sb eq	1.71E-07	2.43E-10	2.91E-13	1.22E-10	1.72E-07	Resource correction, PbZn, silver, positive/GLO U	59%
Mineral, fossil & ren resource depletion	kg Sb eq	4.44E-08	1.52E-07	2.95E-12	-4.09E-08	1.56E-07	Copper concentrate, at beneficiation/RLA U	69%
Mineral, fossil & ren resource depletion	kg Sb eq	1.43E-07	2.04E-10	2.44E-13	1.03E-10	1.44E-07	Anode slime, silver and tellurium containing, primary copper production/GLO U	78%
Mineral, fossil & ren resource depletion	kg Sb eq	4.23E-08	5.66E-10	3.77E-13	-5.07E-11	4.28E-08	Tin, at regional storage/RER U	81%
Cumulative energy demand non renewable	MJ oil eq	4.31E-01	1.95E-01	7.45E-05	3.33E-02	6.59E-01		
Cumulative energy demand non renewable	MJ oil eq	1.40E-01	1.65E-06	2.22E-09	-9.23E-07	1.40E-01	Hard coal, at mine/CN U	21%
Cumulative energy demand non renewable	MJ oil eq	2.30E-02	2.33E-02	1.02E-05	4.36E-03	5.07E-02	Uranium natural, at underground mine/RNA U	29%
Cumulative energy demand non renewable	MJ oil eq	3.02E-02	1.38E-02	2.04E-06	3.31E-03	4.73E-02	Crude oil, at production onshore/RME U	36%
Cumulative energy demand non renewable	MJ oil eq	2.45E-02	1.12E-02	1.45E-06	2.82E-03	3.84E-02	crude oil, at production offshore/kg/NO U	42%
Cumulative energy demand non renewable	MJ oil eq	1.53E-02	1.56E-02	6.81E-06	2.91E-03	3.38E-02	Uranium natural, at open pit mine/RNA U	47%
Cumulative energy demand non renewable	MJ oil eq	2.04E-02	9.32E-03	1.21E-06	2.35E-03	3.21E-02	crude oil, at production onshore/kg/RU U	52%
Cumulative energy demand non renewable	MJ oil eq	2.03E-02	9.28E-03	1.21E-06	2.34E-03	3.19E-02	crude oil, at production offshore/kg/GB U	57%
Cumulative energy demand non renewable	MJ oil eq	1.89E-02	9.61E-03	3.64E-06	1.39E-04	2.86E-02	natural gas, at production onshore/m3/NAC U	61%
Cumulative energy demand non renewable	MJ oil eq	1.67E-02	9.11E-03	2.54E-06	2.42E-03	2.82E-02	Crude oil, at production onshore/RAF U	65%
Cumulative energy demand non renewable	MJ oil eq	1.73E-02	7.87E-03	2.22E-06	6.25E-04	2.58E-02	natural gas, at production onshore/m3/RU U	69%
Cumulative energy demand non renewable	MJ oil eq	1.67E-02	8.35E-03	2.22E-06	7.53E-04	2.58E-02	natural gas, at production offshore/m3/NO U	73%
Cumulative energy demand non renewable	MJ oil eq	9.10E-03	1.22E-02	4.73E-06	3.71E-03	2.50E-02	Hard coal, at mine/WEU U	77%
Cumulative energy demand non renewable	MJ oil eq	1.11E-02	1.16E-02	5.43E-06	1.10E-03	2.38E-02	Lignite, at mine/RER U	81%
Cumulative energy demand renewable	MJ oil eq	3.65E-02	3.95E-02	3.60E+00	1.49E-02	3.69E+00		
Cumulative energy demand renewable	MJ oil eq	0.00E+00	0.00E+00	3.60E+00	0.00E+00	3.60E+00	hotspot electricity, PV, at 3kWp slanted-roof, micro-Si, panel, mounted, incl. EoL absolute/F	98%
Cumulative energy demand renewable	MJ oil eq	1.40E-02	1.71E-02	2.62E-07	8.36E-03	3.95E-02	electricity, hydropower, at reservoir power plant, non alpine regions/kWh/RER U	99%
Nuclear waste	m3 HAA eq	2.67E-11	2.72E-11	1.67E-14	4.52E-12	5.84E-11		
Nuclear waste	m3 HAA eq	2.66E-11	2.72E-11	1.67E-14	4.52E-12	5.84E-11	Radioactive waste, in final repository for nuclear waste SF, HLW, and ILW/CH U	100%
Nuclear waste	m3 HAA eq	5.76E-14	2.21E-19	2.58E-22	-1.08E-19	5.76E-14	Zeolite, slurry, 50% in H2O, at plant/RER S	100%

Tab. 6.14

Identification of the most relevant processes for each impact category and each life cycle stage for 1 kWh of DC electricity produced with a residential scale multi-Si PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant processes to the characterized environmental impacts is reached.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total	Most relevant processes	Cumulative share
electricity, 3 kWp installation, mounted with multi-Si PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling incl. recycling potential, 50:50	Contribution		
Climate change	kg CO2 eq	4.52E-02	9.69E-03	3.31E-06	2.02E-03	5.69E-02		
Climate change	kg CO2 eq	1.49E-02	9.73E-08	1.25E-10	-4.12E-08	1.49E-02	Hard coal, burned in power plant/CN U	26%
Climate change	kg CO2 eq	5.72E-03	3.73E-08	4.80E-11	-1.58E-08	5.72E-03	Hard coal, at mine/CN U	36%
Climate change	kg CO2 eq	4.43E-04	2.24E-03	3.44E-09	1.07E-03	3.75E-03	Aluminium, primary, liquid, at plant/RER U	43%
Climate change	kg CO2 eq	1.65E-03	6.63E-07	2.33E-10	1.44E-08	1.65E-03	Flat glass, uncoated, at plant/RER U	46%
Climate change	kg CO2 eq	1.58E-03	5.00E-05	8.13E-09	1.75E-05	1.65E-03	Hard coal, burned in power plant/BE U	49%
Climate change	kg CO2 eq	5.67E-04	5.82E-04	1.34E-07	1.74E-04	1.32E-03	Hard coal, burned in power plant/DE U	51%
Climate change	kg CO2 eq	1.92E-04	7.25E-04	1.19E-08	3.57E-04	1.27E-03	Light fuel oil, burned in industrial furnace 1MW, non-modulating/CH U	53%
Climate change	kg CO2 eq	1.03E-03	6.99E-09	7.72E-12	-2.41E-09	1.03E-03	Magnesium, at plant/RER U	55%
Climate change	kg CO2 eq	5.52E-04	3.77E-04	1.39E-07	7.22E-05	1.00E-03	Lignite, burned in power plant/DE U	57%
Climate change	kg CO2 eq	1.96E-04	5.32E-04	2.46E-08	2.40E-04	9.67E-04	Natural gas, burned in power plant/UCTE U	59%
Climate change	kg CO2 eq	9.18E-04	6.97E-09	8.66E-12	-3.04E-09	9.18E-04	Electricity, at cogen 1MW lean burn, allocation exergy/RER U	60%
Climate change	kg CO2 eq	6.21E-04	1.78E-04	1.69E-08	7.52E-05	8.75E-04	Operation, transoceanic freight ship/OCE U	62%
Climate change	kg CO2 eq	4.91E-04	2.29E-04	3.29E-08	8.21E-05	8.02E-04	Natural gas, burned in industrial furnace >100kW/RER U	63%
Climate change	kg CO2 eq	6.71E-04	3.98E-09	5.03E-12	-1.70E-09	6.71E-04	Heat, at cogen 1MW lean burn, allocation exergy/RER U	64%
Climate change	kg CO2 eq	5.91E-04	3.28E-05	3.88E-08	-1.41E-05	6.10E-04	Blast furnace gas, burned in power plant/RER U	65%
Climate change	kg CO2 eq	9.44E-04	3.12E-06	1.72E-07	-3.40E-04	6.08E-04	disposal, plastics, mixture, 15.3% water, to municipal incineration/kg/CH U	66%
Climate change	kg CO2 eq	2.56E-04	2.40E-04	1.13E-07	9.56E-05	5.92E-04	Pig iron, at plant/GLO U	67%
Climate change	kg CO2 eq	1.19E-06	5.82E-04	7.94E-11	-4.99E-10	5.83E-04	disposal, plastic, industr. electronics, 15.3% water, to municipal incineration/kg/CH U	68%
Climate change	kg CO2 eq	3.85E-04	1.32E-04	3.82E-08	4.40E-05	5.61E-04	Hard coal, burned in industrial furnace 1-10MW/RER U	69%
Climate change	kg CO2 eq	2.27E-04	2.30E-04	5.39E-08	6.81E-05	5.26E-04	Hard coal, burned in power plant/ES U	70%
Climate change	kg CO2 eq	6.81E-04	8.63E-05	6.98E-09	-2.43E-04	5.24E-04	Operation, lorry>16t, fleet average/RER U	71%
Climate change	kg CO2 eq	5.20E-04	2.93E-07	2.88E-10	-7.98E-08	5.20E-04	Nylon 66, glass-filled, at plant/RER U	72%
Climate change	kg CO2 eq	4.98E-04	3.25E-09	4.18E-12	-1.38E-09	4.98E-04	Hard coal, burned in coal mine power plant/CN U	73%
Climate change	kg CO2 eq	4.39E-04	4.73E-05	6.67E-07	-1.69E-06	4.85E-04	Clinker, at plant/CH U	74%
Climate change	kg CO2 eq	4.27E-04	2.48E-09	3.14E-12	-1.05E-09	4.27E-04	Natural gas, burned in power plant/IP U	75%
Climate change	kg CO2 eq	2.73E-04	1.40E-04	1.12E-08	4.09E-07	4.13E-04	Natural gas, burned in industrial furnace low-NOx >100kW/RER U	75%
Climate change	kg CO2 eq	3.89E-04	9.82E-07	1.41E-09	9.31E-08	3.90E-04	Ethylene, average, at plant/RER U	76%
Climate change	kg CO2 eq	1.96E-04	1.23E-04	1.93E-08	4.37E-05	3.63E-04	Heavy fuel oil, burned in power plant/IT U	77%
Climate change	kg CO2 eq	2.35E-05	3.18E-04	1.15E-07	-1.83E-06	3.39E-04	Polyethylene, HDPE, granulate, at plant/RER U	77%
Climate change	kg CO2 eq	2.51E-04	6.45E-05	7.68E-08	-2.79E-05	2.87E-04	Natural gas, burned in power plant/IT U	78%
Climate change	kg CO2 eq	1.47E-04	1.15E-04	3.79E-08	2.55E-05	2.87E-04	Hard coal, burned in power plant/IT U	78%
Climate change	kg CO2 eq	2.79E-04	1.58E-09	2.00E-12	-6.72E-10	2.79E-04	MG-Silicon, at plant/kg/CN U	79%
Climate change	kg CO2 eq	2.74E-04	3.14E-08	1.89E-12	-6.64E-10	2.74E-04	Silicon carbide, at plant/RER U	79%
Climate change	kg CO2 eq	1.77E-04	7.67E-05	1.10E-08	1.57E-05	2.70E-04	Refinery gas, burned in furnace/MJ/RER U	80%
Climate change	kg CO2 eq	2.64E-04	3.72E-08	3.36E-11	-6.75E-07	2.64E-04	Hard coal, burned in power plant/RFC U	80%
Ozone depletion	kg CFC-11 eq	1.48E-09	2.70E-10	2.04E-13	2.51E-11	1.77E-09		
Ozone depletion	kg CFC-11 eq	3.08E-10	7.81E-11	2.16E-14	5.71E-12	3.91E-10	transport, natural gas, pipeline, long distance/tkm/RU U	22%
Ozone depletion	kg CFC-11 eq	1.95E-10	1.76E-12	6.19E-14	-5.08E-13	1.97E-10	Chlorine, gaseous, mercury cell, at plant/RER U	33%
Ozone depletion	kg CFC-11 eq	1.14E-10	3.16E-11	1.30E-14	-7.55E-14	1.45E-10	Transport, natural gas, onshore pipeline, long distance/tkm/NAC U	41%
Ozone depletion	kg CFC-11 eq	1.10E-10	2.97E-11	1.08E-14	7.38E-13	1.40E-10	transport, natural gas, pipeline, long distance/tkm/RER U	49%
Ozone depletion	kg CFC-11 eq	6.63E-11	4.41E-11	1.52E-14	1.21E-11	1.22E-10	Crude oil, at production/NG U	56%
Ozone depletion	kg CFC-11 eq	8.18E-11	7.35E-13	8.32E-16	-1.25E-13	8.24E-11	Chlorine, gaseous, diaphragm cell, at plant/RER U	61%
Ozone depletion	kg CFC-11 eq	7.52E-11	6.78E-13	1.37E-14	-1.60E-13	7.57E-11	Chlorine, gaseous, membrane cell, at plant/RER U	65%
Ozone depletion	kg CFC-11 eq	4.09E-11	2.34E-11	3.25E-15	8.59E-12	7.29E-11	Uranium, enriched 3.9%, at USEC enrichment plant/US U	69%
Ozone depletion	kg CFC-11 eq	5.02E-11	1.40E-11	3.84E-15	1.04E-12	6.53E-11	Transport, natural gas, onshore pipeline, long distance/NO U	73%
Ozone depletion	kg CFC-11 eq	6.24E-11	3.03E-16	3.82E-19	-1.29E-16	6.24E-11	Refrigerant R134a, at plant/RER U	77%
Ozone depletion	kg CFC-11 eq	5.85E-11	3.27E-14	9.22E-17	-1.96E-12	5.66E-11	Tetrachloroethylene, at plant/WEU U	80%
Ozone depletion	kg CFC-11 eq	5.15E-11	2.18E-15	6.02E-18	4.15E-16	5.15E-11	Methylchloride, at plant/WEU U	83%
Human toxicity, cancer effects	CTUh, c	5.38E-10	6.93E-10	5.00E-13	1.85E-10	1.42E-09		
Human toxicity, cancer effects	CTUh, c	8.16E-11	4.12E-10	3.21E-15	1.97E-10	6.90E-10	Disposal, redmud from bauxite digestion, 0% water, to residual material landfill/CH U	49%
Human toxicity, cancer effects	CTUh, c	1.09E-10	9.63E-11	6.73E-14	-1.55E-13	2.05E-10	Disposal, slag, unalloyed electr. steel, 0% water, to residual material landfill/CH U	63%
Human toxicity, cancer effects	CTUh, c	6.23E-11	6.80E-11	4.08E-14	3.40E-12	1.34E-10	Ferrochromium, high-carbon, 68% Cr, at plant/GLO U	73%
Human toxicity, cancer effects	CTUh, c	8.40E-11	5.35E-13	3.93E-16	-2.42E-14	8.45E-11	Disposal, hard coal ash, 0% water, to residual material landfill/PL U	79%
Human toxicity, cancer effects	CTUh, c	2.96E-11	2.80E-11	5.69E-15	-4.38E-14	5.76E-11	Disposal, sludge from steel rolling, 20% water, to residual material landfill/CH U	83%
Human toxicity, non-cancer effects	CTUh, n-c	6.12E-09	4.37E-09	7.76E-12	1.02E-11	1.05E-08		
Human toxicity, non-cancer effects	CTUh, n-c	3.78E-10	1.74E-09	2.30E-14	4.71E-10	2.59E-09	Copper, primary, at refinery/RLA U	25%
Human toxicity, non-cancer effects	CTUh, n-c	1.58E-09	1.01E-11	7.40E-15	-4.56E-13	1.59E-09	Disposal, hard coal ash, 0% water, to residual material landfill/PL U	40%
Human toxicity, non-cancer effects	CTUh, n-c	1.32E-10	6.67E-10	5.20E-15	3.19E-10	1.12E-09	Disposal, redmud from bauxite digestion, 0% water, to residual material landfill/CH U	50%
Human toxicity, non-cancer effects	CTUh, n-c	7.28E-10	4.75E-15	6.11E-18	-2.01E-15	7.28E-10	Hard coal, burned in power plant/CN U	57%
Human toxicity, non-cancer effects	CTUh, n-c	3.56E-10	3.32E-10	1.22E-13	-1.99E-12	6.86E-10	Steel, electric, un- and low-alloyed, at plant/RER U	64%
Human toxicity, non-cancer effects	CTUh, n-c	5.45E-11	2.51E-10	3.32E-15	6.79E-11	3.73E-10	Copper, primary, at refinery/ID U	67%
Human toxicity, non-cancer effects	CTUh, n-c	4.47E-11	2.02E-10	2.68E-15	5.46E-11	3.01E-10	Copper, primary, at refinery/RER U	70%
Human toxicity, non-cancer effects	CTUh, n-c	2.53E-10	1.16E-14	1.31E-17	-5.63E-15	2.53E-10	Copper, primary, at refinery/GLO U	73%
Human toxicity, non-cancer effects	CTUh, n-c	2.26E-10	1.08E-12	7.92E-16	-3.04E-13	2.27E-10	Disposal, lead smelter slag, 0% water, to residual material landfill/GLO U	75%
Human toxicity, non-cancer effects	CTUh, n-c	2.86E-11	1.08E-10	1.77E-15	5.32E-11	1.90E-10	Light fuel oil, burned in industrial furnace 1MW, non-modulating/CH U	77%
Human toxicity, non-cancer effects	CTUh, n-c	2.40E-11	1.10E-10	1.46E-15	2.99E-11	1.64E-10	Copper, from imported concentrates, at refinery/DE U	78%
Human toxicity, non-cancer effects	CTUh, n-c	2.82E-13	1.38E-10	1.89E-17	-1.18E-16	1.38E-10	disposal, plastic, industr. electronics, 15.3% water, to municipal incineration/kg/CH U	80%
Human toxicity, non-cancer effects	CTUh, n-c	6.02E-11	6.70E-11	3.82E-14	2.84E-12	1.30E-10	Ferronickel, 25% Ni, at plant/GLO U	81%
Particulate matter	kg PM2.5 eq	6.14E-05	6.53E-06	1.81E-09	1.87E-06	6.98E-05		
Particulate matter	kg PM2.5 eq	3.07E-05	2.00E-10	2.57E-13	-8.47E-11	3.07E-05	Hard coal, burned in power plant/CN U	44%
Particulate matter	kg U235 eq	1.80E-05	1.18E-10	1.52E-13	-4.98E-11	1.80E-05	Hard coal, burned in coal mine power plant/CN U	70%
Particulate matter	kg PM2.5 eq	3.81E-07	1.92E-06	2.96E-12	9.21E-07	3.23E-06	Aluminium, primary, liquid, at plant/RER U	74%
Particulate matter	kg PM2.5 eq	1.39E-06	9.09E-12	1.17E-14	-3.85E-12	1.39E-06	Hard coal, at mine/CN U	76%
Particulate matter	kg PM2.5 eq	1.08E-06	4.35E-10	1.53E-13	9.41E-12	1.08E-06	Flat glass, uncoated, at plant/RER U	78%
Particulate matter	kg PM2.5 eq	7.60E-07	2.18E-07	2.07E-11	9.20E-08	1.07E-06	Operation, transoceanic freight ship/OCE U	79%
Particulate matter	kg PM2.5 eq	1.30E-07	5.98E-07	7.92E-12	1.62E-07	8.90E-07	Copper, primary, at refinery/RLA U	81%
Particulate matter	kg PM2.5 eq	2.56E-07	3.24E-08	2.62E-12	-9.13E-08	1.97E-07	Operation, lorry>16t, fleet average/RER U	92%
Ionizing radiation HH	kg PM2.5 eq	1.91E-03	8.53E-04	6.11E-07	1.43E-04	2.90E-03		
Ionizing radiation HH	kg PM2.5 eq	1.73E-03	7.88E-04	4.59E-07	1.36E-04	2.65E-03	Nuclear spent fuel, in reprocessing, at plant/RER U	91%
Ionizing radiation HH	kg PM2.5 eq	7.72E-05	3.15E-05	1.36E-08	6.34E-06	1.15E-04	Uranium natural, at underground mine/RNA U	95%

Tab. 6.14 Identification of the most relevant processes for each impact category and each life cycle stage for 1 kWh of DC electricity produced with a residential scale multi-Si PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant processes to the characterized environmental impacts is reached. (Continued)

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total	Most relevant processes	Cumulative share
electricity, 3 kWp installation, mounted with multi-Si PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling incl. recycling potential, 50:50	Contribution		
Photochemical ozone formation	kg NMVOC eq	1.66E-04	2.64E-05	1.19E-08	2.44E-06	1.94E-04	Hard coal, burned in power plant/CN U	38%
Photochemical ozone formation	kg NMVOC eq	7.38E-05	4.82E-10	6.20E-13	-2.04E-10	7.38E-05	Operation, transoceanic freight ship/OCE U	47%
Photochemical ozone formation	kg NMVOC eq	1.20E-05	3.46E-06	3.28E-10	1.46E-06	1.70E-05	Blasting/RER U	52%
Photochemical ozone formation	kg NMVOC eq	4.40E-06	4.13E-06	2.49E-10	9.04E-07	8.69E-06	Flat glass, uncoated, at plant/RER U	56%
Photochemical ozone formation	kg NMVOC eq	8.68E-06	3.49E-09	1.23E-12	7.55E-11	5.52E-06	Operation, lorry >16t, fleet average/RER U	59%
Photochemical ozone formation	kg NMVOC eq	7.18E-06	9.10E-07	7.36E-11	-2.56E-06	3.89E-06	Diesel, burned in building machine/GLO U	61%
Photochemical ozone formation	kg NMVOC eq	3.05E-06	8.02E-07	1.18E-09	4.00E-08	3.21E-06	Hard coal, burned in power plant/BE U	62%
Photochemical ozone formation	kg NMVOC eq	3.08E-06	9.74E-08	1.59E-11	3.41E-08	3.15E-06	Hard coal, at mine/CN U	64%
Photochemical ozone formation	kg NMVOC eq	3.15E-06	2.05E-11	2.64E-14	-8.69E-12	2.93E-06	photovoltaic cell, multi-Si, at plant/m2/CN U	66%
Photochemical ozone formation	kg NMVOC eq	2.93E-06	9.42E-12	1.18E-14	-4.02E-12	2.67E-06	Hard coal, burned in coal mine power plant/CN U	67%
Photochemical ozone formation	kg NMVOC eq	2.67E-06	1.74E-11	2.24E-14	-7.37E-12	2.35E-06	Operation, freight train/RER U	68%
Photochemical ozone formation	kg NMVOC eq	2.11E-06	2.15E-07	8.12E-11	2.17E-08	2.32E-06	Natural gas, vented/GLO U	69%
Photochemical ozone formation	kg NMVOC eq	1.34E-06	7.78E-07	2.06E-10	2.05E-07	2.30E-06	Hard coal, burned in power plant/ES U	71%
Photochemical ozone formation	kg NMVOC eq	9.95E-07	1.01E-06	2.36E-10	2.98E-07	1.80E-06	Diesel, burned in diesel-electric generating set/GLO U	72%
Photochemical ozone formation	kg NMVOC eq	1.25E-06	4.49E-07	9.55E-11	1.02E-07	1.71E-06	Electricity, at cogen 1MW lean burn, allocation exergy/RER U	72%
Photochemical ozone formation	kg NMVOC eq	1.71E-06	1.30E-11	1.61E-14	-5.67E-12	1.54E-06	photovoltaic panel, multi-Si, at plant/m2/CN U	73%
Photochemical ozone formation	kg NMVOC eq	1.54E-06	4.92E-12	6.16E-15	-2.10E-12	1.52E-06	Hard coal, burned in industrial furnace 1-10MW/RER U	74%
Photochemical ozone formation	kg NMVOC eq	1.04E-06	3.58E-07	1.04E-10	1.19E-07	1.42E-06	Polyethylene, HDPE, granulate, at plant/RER U	75%
Photochemical ozone formation	kg NMVOC eq	9.86E-08	1.33E-06	4.83E-10	-7.65E-09	1.35E-06	Heavy fuel oil, burned in power plant/IT U	75%
Photochemical ozone formation	kg NMVOC eq	7.27E-07	4.56E-07	7.15E-11	1.62E-07	1.32E-06	Ethylene, average, at plant/RER U	76%
Photochemical ozone formation	kg NMVOC eq	1.32E-06	3.32E-09	4.76E-12	3.15E-10	1.25E-06	Heat, at cogen 1MW lean burn, allocation exergy/RER U	77%
Photochemical ozone formation	kg NMVOC eq	1.25E-06	7.43E-12	9.39E-15	-3.17E-12	1.21E-06	Nylon 66, glass-filled, at plant/RER U	77%
Photochemical ozone formation	kg NMVOC eq	1.21E-06	6.83E-10	6.72E-13	-1.86E-10	1.18E-06	Copper, primary, at refinery/RLA U	78%
Photochemical ozone formation	kg NMVOC eq	1.73E-07	7.94E-07	1.05E-11	2.15E-07	1.18E-06	Operation, coal freight train, diesel/CN U	79%
Photochemical ozone formation	kg NMVOC eq	1.18E-06	7.74E-12	1.02E-14	-3.27E-12	9.60E-07	Hard coal, burned in power plant/DE U	79%
Photochemical ozone formation	kg NMVOC eq	4.12E-07	4.22E-07	9.70E-11	1.26E-07	9.44E-07	Light fuel oil, burned in industrial furnace 1MW, non-modulating/CH U	80%
Photochemical ozone formation	kg NMVOC eq	1.42E-07	5.37E-07	8.80E-12	2.64E-07	9.03E-07	Natural gas, burned in power plant/UCTE U	80%
Photochemical ozone formation	kg NMVOC eq	1.83E-07	4.96E-07	2.29E-11	2.24E-07	8.43E-07	MG-silicon, at plant/kg/CN U	80%
Photochemical ozone formation	kg NMVOC eq	8.43E-07	4.78E-12	6.05E-15	-2.03E-12	4.91E-04		
Acidification	mol H+ eq	4.05E-04	6.93E-05	2.49E-08	1.63E-05	2.06E-04	Hard coal, burned in power plant/CN U	42%
Acidification	mol H+ eq	2.06E-04	1.34E-09	1.73E-12	-5.69E-10	2.99E-05	Hard coal, at mine/CN U	48%
Acidification	mol H+ eq	2.99E-05	1.95E-10	2.51E-13	-8.25E-11	2.90E-05	Operation, transoceanic freight ship/OCE U	54%
Acidification	mol H+ eq	2.06E-05	5.93E-06	5.62E-10	2.50E-06	1.91E-05	Copper, primary, at refinery/RLA U	58%
Acidification	mol H+ eq	2.79E-06	1.28E-05	1.70E-10	3.48E-06	1.84E-05	Flat glass, uncoated, at plant/RER U	62%
Acidification	mol H+ eq	1.84E-05	7.37E-09	2.59E-12	1.60E-10	1.18E-05	Aluminium, primary, liquid, at plant/RER U	64%
Acidification	mol H+ eq	1.39E-06	7.03E-06	1.08E-11	3.36E-06	1.02E-05	Blasting/RER U	66%
Acidification	mol H+ eq	4.76E-06	4.47E-06	2.70E-10	9.79E-07	1.01E-05	Hard coal, burned in coal mine power plant/CN U	68%
Acidification	mol H+ eq	1.01E-05	6.62E-11	8.52E-14	-2.80E-11	9.88E-06	Hard coal, burned in power plant/BE U	70%
Acidification	mol H+ eq	9.47E-06	3.00E-07	4.88E-11	1.05E-07	8.97E-06	Copper, primary, at refinery/ID U	72%
Acidification	mol H+ eq	1.31E-06	6.03E-06	7.99E-11	1.63E-06	6.90E-06	Natural gas, sour, burned in production flare/MJ/GLO U	73%
Acidification	mol H+ eq	4.31E-06	2.08E-06	2.68E-10	5.20E-07	6.69E-06	Hard coal, burned in power plant/ES U	75%
Acidification	mol H+ eq	2.89E-06	2.93E-06	6.86E-10	8.67E-07	6.12E-06	Secondary sulphur, at refinery/RER U	76%
Acidification	mol H+ eq	6.18E-06	1.00E-07	2.18E-10	-1.61E-07	5.61E-06	Sulphuric acid, liquid, at plant/RER U	77%
Acidification	mol H+ eq	5.66E-06	9.46E-08	2.07E-10	-1.53E-07	4.89E-06	Hard coal, burned in industrial furnace 1-10MW/RER U	78%
Acidification	mol H+ eq	3.35E-06	1.15E-06	3.33E-10	3.84E-07	4.39E-06	Heavy fuel oil, burned in power plant/IT U	79%
Acidification	mol H+ eq	2.37E-06	1.49E-06	2.33E-10	5.28E-07	4.02E-06	Operation, lorry >16t, fleet average/RER U	80%
Acidification	mol H+ eq	5.22E-06	6.62E-07	5.35E-11	-1.87E-06	3.40E-06	Lignite, burned in power plant/CS U	81%
Acidification	mol H+ eq	1.99E-06	1.22E-06	5.22E-10	1.82E-07	6.96E-04		
Terrestrial eutrophication	mol N eq	5.97E-04	8.98E-05	6.49E-08	9.30E-06	2.72E-04	Hard coal, burned in power plant/CN U	39%
Terrestrial eutrophication	mol N eq	2.72E-04	1.78E-09	2.29E-12	-7.52E-10	6.60E-05	Operation, transoceanic freight ship/OCE U	49%
Terrestrial eutrophication	mol N eq	4.69E-05	1.35E-05	1.28E-09	5.68E-06	5.32E-05	Blasting/RER U	56%
Terrestrial eutrophication	mol N eq	2.48E-05	2.33E-05	1.41E-09	5.10E-06	3.32E-05	Flat glass, uncoated, at plant/RER U	61%
Terrestrial eutrophication	mol N eq	3.32E-05	1.33E-08	4.68E-12	2.88E-10	2.30E-05	Operation, lorry >16t, fleet average/RER U	64%
Terrestrial eutrophication	mol N eq	2.99E-05	3.79E-06	3.06E-10	-1.07E-05	1.48E-05	Diesel, burned in building machine/GLO U	67%
Terrestrial eutrophication	mol N eq	1.16E-05	3.06E-06	4.50E-09	1.53E-07	1.34E-05	photovoltaic cell, multi-Si, at plant/m2/CN U	68%
Terrestrial eutrophication	mol N eq	1.34E-05	4.33E-11	5.43E-14	-1.85E-11	1.16E-05	Hard coal, burned in power plant/BE U	70%
Terrestrial eutrophication	mol N eq	1.11E-05	3.51E-07	5.71E-11	1.23E-07	9.14E-06	Operation, freight train/RER U	71%
Terrestrial eutrophication	mol N eq	8.22E-06	8.37E-07	3.16E-10	8.44E-08	9.09E-06	Hard coal, burned in coal mine power plant/CN U	73%
Terrestrial eutrophication	mol N eq	9.09E-06	5.93E-11	7.63E-14	-2.51E-11	8.42E-06	Hard coal, burned in power plant/ES U	74%
Terrestrial eutrophication	mol N eq	3.64E-06	3.69E-06	8.63E-10	1.09E-06	8.22E-06	Zinc coating, coils/RER U	75%
Terrestrial eutrophication	mol N eq	8.05E-06	1.65E-07	1.08E-10	-2.02E-09	7.17E-06	Diesel, burned in diesel-electric generating set/GLO U	76%
Terrestrial eutrophication	mol N eq	4.97E-06	1.79E-06	3.80E-10	4.06E-07	7.05E-06	Soda, powder, at plant/RER U	77%
Terrestrial eutrophication	mol N eq	4.98E-06	1.96E-07	9.81E-12	1.88E-06	6.54E-06	Electricity, at cogen 1MW lean burn, allocation exergy/RER U	78%
Terrestrial eutrophication	mol N eq	6.54E-06	4.96E-11	6.16E-14	-2.16E-11	5.19E-06	Hard coal, burned in industrial furnace 1-10MW/RER U	79%
Terrestrial eutrophication	mol N eq	3.56E-06	1.22E-06	3.54E-10	4.07E-07	4.78E-06	Heat, at cogen 1MW lean burn, allocation exergy/RER U	80%
Terrestrial eutrophication	mol N eq	4.78E-06	2.83E-11	3.58E-14	-1.21E-11	4.77E-06	Heavy fuel oil, burned in power plant/IT U	80%
Terrestrial eutrophication	mol N eq	2.58E-06	1.62E-06	2.53E-10	5.74E-07	7.98E-06		
Freshwater eutrophication	kg P eq	3.11E-06	3.94E-06	4.35E-09	9.29E-07	6.21E-06	Disposal, sulfidic tailings, off-site/GLO U	78%
Freshwater eutrophication	kg P eq	1.70E-06	3.62E-06	5.09E-11	8.83E-07	8.58E-07	Disposal, spoil from coal mining, in surface landfill/GLO U	89%
Freshwater eutrophication	kg P eq	7.15E-07	1.11E-07	3.59E-11	3.25E-08	6.31E-05		
Marine eutrophication	kg N eq	5.51E-05	7.51E-06	9.56E-08	4.05E-07	2.49E-05	Hard coal, burned in power plant/CN U	39%
Marine eutrophication	kg N eq	2.49E-05	1.62E-10	2.09E-13	-6.87E-11	5.90E-06	Operation, transoceanic freight ship/OCE U	49%
Marine eutrophication	kg N eq	4.19E-06	1.20E-06	1.14E-10	5.07E-07	3.24E-06	Blasting/RER U	54%
Marine eutrophication	kg N eq	1.51E-06	1.42E-06	8.55E-11	3.10E-07	3.03E-06	Flat glass, uncoated, at plant/RER U	59%
Marine eutrophication	kg N eq	3.03E-06	1.22E-09	4.27E-13	2.63E-11	2.11E-06	Treatment, PV cell production effluent, to wastewater treatment, class 3/CH U	62%
Marine eutrophication	kg N eq	2.25E-06	1.46E-11	1.85E-14	-1.44E-07	2.10E-06	Operation, lorry >16t, fleet average/RER U	65%
Marine eutrophication	kg N eq	2.72E-06	3.45E-07	2.79E-11	-9.73E-07	1.35E-06	Diesel, burned in building machine/GLO U	67%
Marine eutrophication	kg N eq	1.06E-06	2.79E-07	4.10E-10	1.39E-08	1.12E-06	photovoltaic cell, multi-Si, at plant/m2/CN U	69%
Marine eutrophication	kg N eq	1.12E-06	3.61E-12	4.53E-15	-1.54E-12	1.06E-06	Hard coal, burned in power plant/BE U	71%
Marine eutrophication	kg N eq	1.01E-06	3.21E-08	5.22E-12	1.12E-08	8.34E-07	Operation, freight train/RER U	72%
Marine eutrophication	kg N eq	7.50E-07	7.63E-08	2.89E-11	7.70E-09	8.30E-07	Hard coal, burned in coal mine power plant/CN U	74%
Marine eutrophication	kg N eq	8.30E-07	5.41E-12	6.97E-15	-2.29E-12	7.69E-07	Hard coal, burned in power plant/ES U	75%
Marine eutrophication	kg N eq	3.32E-07	3.37E-07	7.88E-11	9.96E-08	7.55E-07	Nylon 66, glass-filled, at plant/RER U	76%
Marine eutrophication	kg N eq	7.55E-07	4.26E-10	4.19E-13	-1.16E-10	6.54E-07	Diesel, burned in diesel-electric generating set/GLO U	77%
Marine eutrophication	kg N eq	4.54E-07	1.63E-07	3.47E-11	3.70E-08	5.97E-07	Electricity, at cogen 1MW lean burn, allocation exergy/RER U	78%
Marine eutrophication	kg N eq	5.97E-07	4.53E-12	5.63E-15	-1.97E-12	4.74E-07	Hard coal, burned in industrial furnace 1-10MW/RER U	79%
Marine eutrophication	kg N eq	3.25E-07	1.12E-07	3.23E-11	3.72E-08	4.36E-07	Heat, at cogen 1MW lean burn, allocation exergy/RER U	79%
Marine eutrophication	kg N eq	4.36E-07	2.59E-12	3.27E-15	-1.10E-12	4.35E-07	Heavy fuel oil, burned in power plant/IT U	80%
Marine eutrophication	kg N eq	2.35E-07	1.47E-07	2.31E-11	5.23E-08	1.02E-01		
Freshwater ecotoxicity	CTUe	2.42E-02	7.55E-02	1.94E-05	2.21E-03	6.10E-02	disposal, plastic, industr. electronics, 15.3% water, to municipal incineration/kg/CH U	60%
Freshwater ecotoxicity	CTUe	1.24E-04	6.09E-02	8.31E-09	-5.22E-08	8.59E-03	Disposal, redmud from bauxite digestion, 0% water, to residual material landfill/CH U	68%
Freshwater ecotoxicity	CTUe	1.02E-03	5.13E-03	3.99E-08	2.45E-03	4.44E-03	Silver, from combined gold-silver production, at refinery/PE U	73%
Freshwater ecotoxicity	CTUe	4.44E-03	1.04E-07	1.19E-10	-4.98E-08	4.34E-03	Copper, primary, at refinery/RLA U	77%
Freshwater ecotoxicity	CTUe	6.34E-04	2.92E-03	3.87E-08	7.91E-04	3.55E-03	Disposal, hard coal ash, 0% water, to residual material landfill/PL U	80%

Tab. 6.14 Identification of the most relevant processes for each impact category and each life cycle stage for 1 kWh of DC electricity produced with a residential scale multi-Si PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant processes to the characterized environmental impacts is reached. (Continued)

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total	Most relevant processes	Cumulative share
electricity, 3 kWp installation, mounted with multi-Si PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling incl. recycling potential, 50:50	Contribution		
Land use	kg C deficit	5.63E-02	9.70E-03	1.35E-05	1.45E-03	6.75E-02		
Land use	kg C deficit	7.50E-03	3.44E-03	5.98E-07	8.12E-04	1.18E-02	Well for exploration and production, onshore/GLO/I U	17%
Land use	kg C deficit	9.66E-03	5.10E-08	6.45E-11	-2.17E-08	9.66E-03	Photovoltaic panel factory/GLO/I U	32%
Land use	kg C deficit	4.86E-03	1.83E-03	3.62E-07	3.20E-04	7.01E-03	Well for exploration and production, offshore/OCE/I U	42%
Land use	kg C deficit	5.01E-03	3.27E-08	4.21E-11	-1.38E-08	5.01E-03	Hard coal, at mine/CN U	50%
Land use	kg C deficit	-6.42E-04	-2.96E-03	-6.96E-08	-1.41E-03	-5.01E-03	Recultivation, bauxite mine/GLO U	42%
Land use	kg C deficit	5.94E-04	2.98E-03	2.49E-08	1.42E-03	5.00E-03	Bauxite, at mine/GLO U	50%
Land use	kg C deficit	3.45E-03	1.66E-08	2.10E-11	-7.10E-09	3.45E-03	Wafer factory/DE/I U	55%
Land use	kg C deficit	3.36E-03	2.19E-08	2.82E-11	-9.28E-09	3.36E-03	Underground mine, hard coal/CN/I U	60%
Land use	kg C deficit	-1.51E-03	-1.22E-03	-1.86E-08	-5.66E-04	-3.29E-03	electricity, hydropower, at reservoir power plant, non alpine regions/kWh/RER U	55%
Land use	kg C deficit	2.59E-03	2.38E-04	3.05E-08	6.63E-06	2.83E-03	Softwood, standing, under bark, in forest/RER U	59%
Land use	kg C deficit	2.60E-03	2.39E-06	1.52E-09	1.82E-07	2.61E-03	Tin, at regional storage/RER U	63%
Land use	kg C deficit	2.07E-03	1.27E-04	2.14E-07	2.63E-05	2.22E-03	Hardwood, standing, under bark, in forest/RER U	66%
Land use	kg C deficit	1.24E-03	6.01E-04	8.12E-08	1.51E-04	1.99E-03	production plant crude oil, onshore/p/GLO/I U	69%
Land use	kg C deficit	1.48E-03	7.23E-09	9.11E-12	-3.09E-09	1.48E-03	Photovoltaic cell factory/DE/I U	71%
Land use	kg C deficit	8.22E-04	3.00E-04	4.55E-08	1.03E-04	1.23E-03	Residual material landfill facility/CH/I U	73%
Land use	kg C deficit	3.14E-04	6.68E-04	9.38E-09	1.63E-04	1.14E-03	Disposal, sulfidic tailings, off-site/GLO U	75%
Land use	kg C deficit	6.76E-04	2.83E-04	1.12E-07	7.72E-05	1.04E-03	Underground mine, hard coal/GLO/I U	76%
Land use	kg C deficit	6.58E-04	1.11E-04	1.53E-09	3.36E-05	8.02E-04	Softwood, Scandinavian, standing, under bark, in forest/NORDEL U	77%
Land use	kg C deficit	6.93E-04	2.46E-07	2.73E-10	-1.20E-07	6.94E-04	Mine, gold and silver/CL/I U	78%
Land use	kg C deficit	1.26E-04	4.21E-04	3.02E-08	1.35E-04	6.82E-04	Aluminium electrolysis, plant/RER/I U	80%
Land use	kg C deficit	6.21E-04	8.11E-08	9.03E-11	-3.96E-08	6.21E-04	Mine, gold-silver-zinc-lead-copper/SE/I U	80%
Water resource depletion	m3 water eq	4.71E-05	1.53E-05	8.71E-08	5.88E-06	6.84E-05		
Water resource depletion	m3 water eq	1.54E-05	1.25E-05	1.90E-10	5.79E-06	3.37E-05	electricity, hydropower, at reservoir power plant, non alpine regions/kWh/RER U	49%
Water resource depletion	m3 water eq	5.13E-06	2.91E-11	3.68E-14	-1.24E-11	5.13E-06	silicon, electronic grade, at plant/kg/CN U	57%
Water resource depletion	m3 water eq	4.44E-06	2.90E-11	3.73E-14	-1.23E-11	4.44E-06	Hard coal, burned in power plant/CN U	63%
Water resource depletion	m3 water eq	2.71E-06	2.66E-11	3.22E-14	-1.19E-11	2.71E-06	silicon, electronic grade, at plant/kg/DE U	67%
Water resource depletion	m3 water eq	2.33E-06	1.32E-11	1.67E-14	-5.62E-12	2.33E-06	silicon, electronic grade, at plant/US U	71%
Water resource depletion	m3 water eq	2.04E-06	1.15E-11	1.46E-14	-4.91E-12	2.04E-06	silicon, electronic grade, at plant/APAC U	74%
Water resource depletion	m3 water eq	1.93E-06	1.03E-08	8.54E-08	-2.91E-08	1.99E-06	tap water, at user/kg/RER U	77%
Water resource depletion	m3 water eq	1.50E-06	4.83E-12	6.06E-15	-2.06E-12	1.50E-06	silicon, multi-Si, casted, at plant/kg/CN U	79%
Water resource depletion	m3 water eq	9.78E-07	2.25E-07	1.41E-10	3.31E-08	1.24E-06	Water, decarbonised, at plant/RER U	81%
Mineral, fossil & ren resource depletion	kg Sb eq	1.88E-05	5.15E-07	3.35E-11	2.65E-08	1.93E-05		
Mineral, fossil & ren resource depletion	kg Sb eq	7.41E-06	1.74E-10	1.98E-13	-8.31E-11	7.41E-06	Resource correction, PbZn, silver, positive/GLO U	38%
Mineral, fossil & ren resource depletion	kg Sb eq	6.20E-06	1.46E-10	1.66E-13	-6.96E-11	6.20E-06	Anode slime, silver and tellurium containing, primary copper production/GLO U	70%
Mineral, fossil & ren resource depletion	kg Sb eq	1.16E-06	2.72E-11	3.09E-14	-1.30E-11	1.16E-06	Silver, from combined gold-silver production, at refinery/CL U	76%
Mineral, fossil & ren resource depletion	kg Sb eq	1.14E-06	2.66E-11	3.03E-14	-1.27E-11	1.14E-06	Silver, from combined gold-silver production, at refinery/PE U	82%
Cumulative energy demand non renewable	MJ oil eq	5.46E-01	1.39E-01	5.07E-05	2.35E-02	7.09E-01		
Cumulative energy demand non renewable	MJ oil eq	1.80E-01	1.17E-06	1.51E-09	-4.96E-07	1.80E-01	Hard coal, at mine/CN U	25%
Cumulative energy demand non renewable	MJ oil eq	3.95E-02	1.61E-02	6.95E-06	3.24E-03	5.88E-02	Uranium natural, at underground mine/RNA U	34%
Cumulative energy demand non renewable	MJ oil eq	3.41E-02	6.69E-03	2.48E-06	1.44E-04	4.09E-02	natural gas, at production onshore/m3/NAC U	39%
Cumulative energy demand non renewable	MJ oil eq	2.63E-02	1.07E-02	4.63E-06	2.16E-03	3.92E-02	Uranium natural, at open pit mine/RNA U	45%
Cumulative energy demand non renewable	MJ oil eq	1.94E-02	9.46E-03	1.39E-06	2.28E-03	3.11E-02	Crude oil, at production onshore/RME U	49%
Cumulative energy demand non renewable	MJ oil eq	2.15E-02	5.46E-03	1.51E-06	3.99E-04	2.74E-02	natural gas, at production onshore/m3/RU U	53%
Cumulative energy demand non renewable	MJ oil eq	2.08E-02	5.80E-03	1.51E-06	4.73E-04	2.71E-02	natural gas, at production offshore/m3/NO U	57%
Cumulative energy demand non renewable	MJ oil eq	1.59E-02	7.68E-03	9.87E-07	1.93E-03	2.55E-02	crude oil, at production offshore/kg/NO U	61%
Cumulative energy demand non renewable	MJ oil eq	1.41E-02	8.03E-03	3.69E-06	9.85E-04	2.31E-02	Lignite, at mine/RER U	64%
Cumulative energy demand non renewable	MJ oil eq	1.21E-02	8.38E-03	3.22E-06	2.48E-03	2.29E-02	Hard coal, at mine/WEU U	67%
Cumulative energy demand non renewable	MJ oil eq	1.33E-02	6.41E-03	8.23E-07	1.61E-03	2.13E-02	crude oil, at production onshore/kg/RU U	70%
Cumulative energy demand non renewable	MJ oil eq	1.32E-02	6.38E-03	8.20E-07	1.60E-03	2.12E-02	crude oil, at production offshore/kg/GB U	73%
Cumulative energy demand non renewable	MJ oil eq	1.28E-02	5.16E-03	2.57E-06	1.12E-03	1.91E-02	Hard coal, at mine/EEU U	76%
Cumulative energy demand non renewable	MJ oil eq	1.87E-02	4.72E-05	6.77E-08	4.47E-06	1.87E-02	Ethylene, average, at plant/RER U	78%
Cumulative energy demand non renewable	MJ oil eq	1.05E-02	6.24E-03	1.73E-06	1.66E-03	1.84E-02	Crude oil, at production onshore/RAF U	81%
Cumulative energy demand renewable	MJ oil eq	7.20E-02	2.72E-02	3.60E+00	9.82E-03	3.71E+00		
Cumulative energy demand renewable	MJ oil eq	0.00E+00	0.00E+00	3.60E+00	0.00E+00	3.60E+00	hotspot electricity, PV, at 3kWp slanted-roof, multi-Si, panel, mounted, incl. EoL absolute/R	97%
Cumulative energy demand renewable	MJ oil eq	2.86E-02	1.17E-02	1.26E-06	4.61E-03	4.49E-02	electricity, hydropower, at run-of-river power plant without reservoir/kWh/RER U	98%
Nuclear waste	m3 HAA eq	4.42E-11	1.88E-11	1.13E-14	3.43E-12	6.64E-11		
Nuclear waste	m3 HAA eq	4.41E-11	1.88E-11	1.13E-14	3.44E-12	6.63E-11	Radioactive waste, in final repository for nuclear waste SF, HLW, and ILW/CH U	100%
Nuclear waste	m3 HAA eq	4.71E-14	2.65E-15	6.88E-19	1.92E-16	5.00E-14	Titanium dioxide, chloride process, at plant/RER S	100%

Tab. 6.15 Identification of the most relevant processes for each impact category and each life cycle stage for 1 kWh of DC electricity produced with a residential scale mono-Si PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant processes to the characterized environmental impacts is reached.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total	Most relevant processes	Cumulative share
electricity, 3 kWp installation, mounted with mono-Si PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling incl. recycling potential, 50:50	Contribution		
Climate change	kg CO2 eq	8.00E-02	9.47E-03	3.22E-06	1.90E-03	9.14E-02	Hard coal, burned in power plant/CN U	36%
Climate change	kg CO2 eq	3.33E-02	9.51E-08	1.22E-10	-4.25E-08	3.33E-02	Hard coal, at mine/CN U	50%
Climate change	kg CO2 eq	1.28E-02	3.65E-08	4.68E-11	-1.63E-08	1.28E-02	Aluminium, primary, liquid, at plant/RER U	54%
Climate change	kg CO2 eq	4.41E-04	2.18E-03	3.35E-09	1.04E-03	2.93E-03	Hard coal, burned in power plant/BE U	58%
Climate change	kg CO2 eq	2.86E-03	4.87E-05	7.92E-09	1.69E-05	2.93E-03	Natural gas, burned in industrial furnace low-NOx >100kW/RER U	60%
Climate change	kg CO2 eq	1.62E-03	1.37E-04	1.09E-08	2.82E-07	1.60E-03	Flat glass, uncoated, at plant/RER U	61%
Climate change	kg CO2 eq	1.60E-03	6.54E-07	2.27E-10	1.13E-08	1.52E-03	Electricity, at cogen 1MW lean burn, allocation exergy/RER U	63%
Climate change	kg CO2 eq	1.52E-03	6.82E-09	8.43E-12	-3.16E-09	1.47E-03	Hard coal, burned in power plant/DE U	65%
Climate change	kg CO2 eq	1.96E-04	7.06E-04	1.16E-08	3.47E-04	1.25E-03	Light fuel oil, burned in industrial furnace 1MW, non-modulating/CH U	66%
Climate change	kg CO2 eq	7.43E-04	3.67E-04	1.36E-07	6.74E-05	1.18E-03	Lignite, burned in power plant/DE U	67%
Climate change	kg CO2 eq	1.11E-03	3.20E-05	3.78E-08	-1.45E-05	1.13E-03	Blast furnace gas, burned in power plant/RER U	68%
Climate change	kg CO2 eq	1.12E-03	3.89E-09	4.90E-12	-1.75E-09	1.12E-03	Heat, at cogen 1MW lean burn, allocation exergy/RER U	70%
Climate change	kg CO2 eq	1.11E-03	3.17E-09	4.07E-12	-1.42E-09	1.11E-03	Hard coal, burned in coal mine power plant/CN U	71%
Climate change	kg CO2 eq	9.99E-04	6.84E-09	7.51E-12	-2.50E-09	9.99E-04	Magnesium, at plant/RER U	72%
Climate change	kg CO2 eq	2.12E-04	5.18E-04	2.39E-08	2.33E-04	9.63E-04	Natural gas, burned in power plant/UCTE U	73%
Climate change	kg CO2 eq	7.13E-04	1.74E-04	1.65E-08	7.31E-05	9.60E-04	Operation, transoceanic freight ship/OCE U	74%
Climate change	kg CO2 eq	4.93E-04	2.24E-04	3.20E-08	7.78E-05	7.95E-04	Natural gas, burned in industrial furnace >100kW/RER U	75%
Climate change	kg CO2 eq	7.53E-04	2.42E-09	3.06E-12	-1.09E-09	7.53E-04	Natural gas, burned in power plant/IP U	76%
Climate change	kg CO2 eq	4.98E-04	1.29E-04	3.72E-08	4.22E-05	6.69E-04	Hard coal, burned in industrial furnace 1-10MW/RER U	77%
Climate change	kg CO2 eq	2.97E-04	2.34E-04	1.10E-07	9.27E-05	6.24E-04	Pig iron, at plant/GLO U	77%
Climate change	kg CO2 eq	2.95E-04	2.24E-04	5.25E-08	6.52E-05	5.85E-04	Hard coal, burned in power plant/ES U	78%
Climate change	kg CO2 eq	1.15E-06	5.82E-04	7.73E-11	-2.51E-09	5.83E-04	disposal, plastic, industr. electronics, 15.3% water, to municipal incineration/kg/CH U	79%
Climate change	kg CO2 eq	9.21E-04	3.05E-06	1.68E-07	-3.46E-04	5.79E-04	disposal, plastics, mixture, 15.3% water, to municipal incineration/kg/CH U	79%
Climate change	kg CO2 eq	7.16E-04	8.41E-05	6.80E-09	-2.49E-04	5.52E-04	Operation, lorry >16t, fleet average/RER U	80%
Climate change	kg CO2 eq	4.68E-04	4.62E-05	6.49E-07	-1.98E-06	5.13E-04	Clinker, at plant/CH U	80%
Ozone depletion	kg CFC-11 eq	2.12E-09	2.63E-10	1.98E-13	2.23E-11	2.40E-09	transport, natural gas, pipeline, long distance/tkm/RU U	25%
Ozone depletion	kg CFC-11 eq	5.22E-10	7.61E-11	2.10E-14	5.06E-12	6.03E-10	Chlorine, gaseous, mercury cell, at plant/RER U	36%
Ozone depletion	kg CFC-11 eq	2.71E-10	1.73E-12	6.03E-14	-5.22E-13	2.73E-10	Transport, natural gas, onshore pipeline, long distance/tkm/NAC U	45%
Ozone depletion	kg CFC-11 eq	1.82E-10	3.08E-11	1.26E-14	-3.61E-13	2.12E-10	transport, natural gas, pipeline, long distance/tkm/RER U	54%
Ozone depletion	kg CFC-11 eq	1.79E-10	2.90E-11	1.05E-14	4.77E-13	2.09E-10	Crude oil, at production/NG U	59%
Ozone depletion	kg CFC-11 eq	7.51E-11	4.30E-11	1.48E-14	1.14E-11	1.30E-10	Chlorine, gaseous, diaphragm cell, at plant/RER U	64%
Ozone depletion	kg CFC-11 eq	1.13E-10	7.21E-13	8.10E-16	-1.28E-13	1.14E-10	Chlorine, gaseous, membrane cell, at plant/RER U	68%
Ozone depletion	kg CFC-11 eq	1.04E-10	6.65E-13	1.33E-14	-1.64E-13	1.05E-10	Transport, natural gas, onshore pipeline, long distance/NO U	72%
Ozone depletion	kg CFC-11 eq	8.25E-11	1.37E-11	3.73E-15	9.20E-13	9.71E-11	Uranium, enriched 3.9%, at USEC enrichment plant/US U	76%
Ozone depletion	kg CFC-11 eq	6.56E-11	2.28E-11	3.16E-15	8.29E-12	7.50E-11	Transport, natural gas, offshore pipeline, long distance/NO U	80%
Ozone depletion	kg CFC-11 eq	6.37E-11	1.06E-11	2.93E-15	6.86E-13	6.94E-11	Refrigerant R134a, at plant/RER U	82%
Human toxicity, cancer effects	CTUh, c	7.09E-10	6.76E-10	4.87E-13	1.79E-10	1.56E-09	Disposal, redmud from bauxite digestion, 0% water, to residual material landfill/CH U	43%
Human toxicity, cancer effects	CTUh, c	8.13E-11	4.01E-10	3.12E-15	1.91E-10	2.19E-10	Disposal, slag, unalloyed electr. steel, 0% water, to residual material landfill/CH U	57%
Human toxicity, cancer effects	CTUh, c	1.25E-10	9.39E-11	6.55E-14	-3.21E-13	1.86E-10	Disposal, hard coal ash, 0% water, to residual material landfill/PL U	59%
Human toxicity, cancer effects	CTUh, c	1.86E-10	5.24E-13	3.83E-16	-3.13E-14	1.35E-10	Ferrochromium, high-carbon, 68% Cr, at plant/GLO U	78%
Human toxicity, cancer effects	CTUh, c	6.46E-11	6.66E-11	3.97E-14	3.34E-12	5.91E-11	Disposal, sludge from steel rolling, 20% water, to residual material landfill/CH U	81%
Human toxicity, non-cancer effects	CTUh, n-c	9.12E-09	4.33E-09	7.56E-12	1.52E-11	1.35E-08	Disposal, hard coal ash, 0% water, to residual material landfill/PL U	26%
Human toxicity, non-cancer effects	CTUh, n-c	3.50E-09	9.86E-12	7.21E-15	-5.90E-13	2.60E-09	Copper, primary, at refinery/RLA U	45%
Human toxicity, non-cancer effects	CTUh, n-c	3.95E-10	1.74E-09	2.24E-14	4.71E-10	1.62E-09	Hard coal, burned in power plant/CN U	57%
Human toxicity, non-cancer effects	CTUh, n-c	1.62E-09	4.64E-15	5.95E-18	-2.07E-15	1.09E-09	Disposal, redmud from bauxite digestion, 0% water, to residual material landfill/CH U	66%
Human toxicity, non-cancer effects	CTUh, n-c	1.32E-10	6.50E-10	5.06E-15	3.10E-10	7.31E-10	Steel, electric, un- and low-alloyed, at plant/RER U	71%
Human toxicity, non-cancer effects	CTUh, n-c	4.10E-10	3.24E-10	1.19E-13	-2.43E-12	3.75E-10	Copper, primary, at refinery/ID U	74%
Human toxicity, non-cancer effects	CTUh, n-c	5.70E-11	2.51E-10	3.23E-15	6.79E-11	3.03E-10	Copper, primary, at refinery/RER U	76%
Human toxicity, non-cancer effects	CTUh, n-c	4.66E-11	2.02E-10	2.61E-15	5.46E-11	1.86E-10	Light fuel oil, burned in industrial furnace 1MW, non-modulating/CH U	77%
Human toxicity, non-cancer effects	CTUh, n-c	2.92E-11	1.05E-10	1.72E-15	5.17E-11	1.67E-10	Mercury, liquid, at plant/GLO U	79%
Human toxicity, non-cancer effects	CTUh, n-c	1.49E-10	1.66E-11	1.60E-14	1.40E-12	1.65E-10	Copper, from imported concentrates, at refinery/DE U	80%
Human toxicity, non-cancer effects	CTUh, n-c	2.50E-11	1.10E-10	1.42E-15	2.99E-11	1.62E-10	Copper, primary, at refinery/GLO U	81%
Particulate matter	kg PM2.5 eq	1.26E-04	6.42E-06	1.76E-09	1.81E-06	1.34E-04	Hard coal, burned in power plant/CN U	51%
Particulate matter	kg PM2.5 eq	6.84E-05	1.95E-10	2.51E-13	-8.73E-11	4.03E-05	Hard coal, burned in coal mine power plant/CN U	81%
Ionizing radiation HH	kg PM2.5 eq	2.82E-03	8.32E-04	5.94E-07	1.32E-04	3.79E-03	Nuclear spent fuel, in reprocessing, at plant/RER U	91%
Ionizing radiation HH	kg PM2.5 eq	2.56E-03	7.68E-04	4.47E-07	1.26E-04	1.56E-04	Uranium natural, at underground mine/RNA U	95%
Ionizing radiation HH	kg PM2.5 eq	1.20E-04	3.07E-05	1.32E-08	5.94E-06	3.24E-04	Hard coal, burned in power plant/CN U	51%
Photochemical ozone formation	kg NMVOC eq	2.96E-04	2.59E-05	1.16E-08	2.18E-06	1.65E-04	Operation, transoceanic freight ship/OCE U	60%
Photochemical ozone formation	kg NMVOC eq	1.65E-04	4.71E-10	6.04E-13	-2.10E-10	1.03E-05	Blasting/RER U	63%
Photochemical ozone formation	kg NMVOC eq	1.38E-05	3.38E-06	3.19E-10	1.42E-06	9.96E-06	CZ single crystalline silicon, photovoltaics, at plant/kg/CN U	65%
Photochemical ozone formation	kg NMVOC eq	5.32E-06	4.11E-06	2.43E-10	9.00E-07	8.43E-06	Flat glass, uncoated, at plant/RER U	68%
Photochemical ozone formation	kg NMVOC eq	9.96E-06	1.43E-11	1.82E-14	-6.38E-12	7.02E-06	Hard coal, at mine/CN U	69%
Photochemical ozone formation	kg NMVOC eq	8.43E-06	3.44E-09	1.19E-12	5.95E-11	5.95E-06	Hard coal, burned in coal mine power plant/CN U	71%
Photochemical ozone formation	kg NMVOC eq	7.02E-06	2.01E-11	2.57E-14	-8.96E-12	5.81E-06	Operation, lorry >16t, fleet average/RER U	73%
Photochemical ozone formation	kg NMVOC eq	5.95E-06	1.70E-11	2.18E-14	-7.59E-12	5.71E-06	Hard coal, burned in power plant/BE U	74%
Photochemical ozone formation	kg NMVOC eq	7.55E-06	8.87E-07	7.16E-11	-2.62E-05	4.53E-06	Diesel, burned in building machine/GLO U	75%
Photochemical ozone formation	kg NMVOC eq	5.58E-06	9.49E-08	1.54E-11	3.29E-08	3.12E-06	photovoltaic cell, single-Si, at plant/m2/CN U	76%
Photochemical ozone formation	kg NMVOC eq	3.71E-06	7.85E-07	1.15E-09	3.05E-08	2.83E-06	Electricity, at cogen 1MW lean burn, allocation exergy/RER U	77%
Photochemical ozone formation	kg NMVOC eq	3.12E-06	4.47E-12	5.71E-15	-2.00E-12	2.64E-06	Operation, coal freight train, diesel/CN U	78%
Photochemical ozone formation	kg NMVOC eq	2.83E-06	1.27E-11	1.57E-14	-5.89E-12	2.56E-06	Hard coal, burned in power plant/ES U	79%
Photochemical ozone formation	kg NMVOC eq	2.64E-06	7.56E-12	9.90E-15	-3.37E-12	2.49E-06	Natural gas, vented/GLO U	79%
Photochemical ozone formation	kg NMVOC eq	1.29E-06	9.83E-07	2.30E-10	2.86E-07	2.46E-06	Operation, freight train/RER U	80%
Photochemical ozone formation	kg NMVOC eq	1.54E-06	7.59E-07	2.00E-10	1.94E-07	1.97E-06	Heat, at cogen 1MW lean burn, allocation exergy/RER U	81%
Photochemical ozone formation	kg NMVOC eq	2.23E-06	2.10E-07	7.91E-11	2.06E-08	8.26E-04	Hard coal, burned in power plant/CN U	56%
Acidification	mol H+ eq	7.42E-04	6.83E-05	2.42E-08	1.58E-05	6.67E-05	Hard coal, at mine/CN U	64%
Acidification	mol H+ eq	4.60E-04	1.31E-09	1.68E-12	-5.86E-10	3.19E-05	Operation, transoceanic freight ship/OCE U	68%
Acidification	mol H+ eq	6.67E-05	1.90E-10	2.44E-13	-8.50E-11	2.26E-05	Hard coal, burned in coal mine power plant/CN U	70%
Acidification	mol H+ eq	2.37E-05	5.78E-06	5.47E-10	2.43E-06	1.92E-05	Copper, primary, at refinery/RLA U	73%
Acidification	mol H+ eq	2.26E-05	6.47E-11	8.29E-14	-2.89E-11	1.78E-05	Flat glass, uncoated, at plant/RER U	75%
Acidification	mol H+ eq	2.91E-06	1.28E-05	1.65E-10	3.48E-06	1.76E-05	Hard coal, burned in power plant/BE U	77%
Acidification	mol H+ eq	1.78E-05	7.28E-09	2.52E-12	1.26E-10	1.15E-05	Aluminium, primary, liquid, at plant/RER U	78%
Acidification	mol H+ eq	1.72E-05	2.92E-07	4.75E-11	1.01E-07	1.12E-05	Blasting/RER U	80%
Acidification	mol H+ eq	1.39E-06	6.84E-06	1.05E-11	3.28E-06	9.03E-06	Copper, primary, at refinery/ID U	81%
Acidification	mol H+ eq	5.76E-06	4.45E-06	2.62E-10	9.74E-07			
Acidification	mol H+ eq	1.37E-06	6.03E-06	7.78E-11	1.63E-06			

Tab. 6.15

Identification of the most relevant processes for each impact category and each life cycle stage for 1 kWh of DC electricity produced with a residential scale mono-Si PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant processes to the characterized environmental impacts is reached. (Continued)

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total	Most relevant processes	Cumulative share
electricity, 3 kWp installation, mounted with mono-Si PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling incl. recycling potential, 50:50	Contribution		
Terrestrial eutrophication	mol N eq	1.05E-03	8.82E-05	6.32E-08	8.36E-06	1.14E-03		
Terrestrial eutrophication	mol N eq	6.08E-04	1.74E-09	2.23E-12	-7.75E-10	6.08E-04	Hard coal, burned in power plant/CN U	53%
Terrestrial eutrophication	mol N eq	5.38E-05	1.31E-05	1.24E-09	5.52E-06	7.25E-05	Operation, transoceanic freight ship/OCE U	59%
Terrestrial eutrophication	mol N eq	3.00E-05	2.32E-05	1.37E-09	5.08E-06	5.83E-05	Blasting/RER U	65%
Terrestrial eutrophication	mol N eq	4.24E-05	6.08E-11	7.77E-14	-2.72E-11	4.24E-05	CZ single crystalline silicon, photovoltaics, at plant/kg/CN U	68%
Terrestrial eutrophication	mol N eq	3.22E-05	1.32E-08	4.55E-12	2.27E-10	3.22E-05	Flat glass, uncoated, at plant/RER U	71%
Terrestrial eutrophication	mol N eq	3.14E-05	3.69E-06	2.98E-10	-1.09E-05	2.42E-05	Operation, lorry >16t, fleet average/RER U	73%
Terrestrial eutrophication	mol N eq	2.01E-05	3.42E-07	5.56E-11	1.19E-07	2.06E-05	Hard coal, burned in power plant/BE U	75%
Terrestrial eutrophication	mol N eq	2.03E-05	5.79E-11	7.43E-14	-2.59E-11	2.03E-05	Hard coal, burned in coal mine power plant/CN U	77%
Terrestrial eutrophication	mol N eq	1.41E-05	2.99E-06	4.38E-09	1.16E-07	1.73E-05	Diesel, burned in building machine/GLO U	78%
Terrestrial eutrophication	mol N eq	1.08E-05	4.85E-11	6.00E-14	-2.25E-11	1.08E-05	Electricity, at cogen 1MWt lean burn, allocation exergy/RER U	79%
Terrestrial eutrophication	mol N eq	1.03E-05	2.95E-11	3.86E-14	-1.31E-11	1.03E-05	Operation, coal freight train, diesel/CN U	80%
Freshwater eutrophication	kg P eq	3.77E-06	3.93E-06	4.24E-09	9.26E-07	8.62E-06		
Freshwater eutrophication	kg P eq	1.43E-06	3.62E-06	4.96E-11	8.84E-07	5.94E-06	Disposal, sulfidic tailings, off-site/GLO U	69%
Freshwater eutrophication	kg P eq	1.34E-06	1.09E-07	3.49E-11	3.12E-08	1.48E-06	Disposal, spoil from coal mining, in surface landfill/GLO U	86%
Marine eutrophication	kg N eq	1.05E-04	7.37E-06	9.30E-08	3.21E-07	1.13E-04		
Marine eutrophication	kg N eq	5.55E-05	1.59E-10	2.03E-13	-7.08E-11	5.55E-05	Hard coal, burned in power plant/CN U	49%
Marine eutrophication	kg N eq	9.43E-06	1.35E-11	1.73E-14	-6.03E-12	9.43E-06	CZ single crystalline silicon, photovoltaics, at plant/kg/CN U	57%
Marine eutrophication	kg N eq	4.81E-06	1.18E-06	1.11E-10	4.93E-07	6.48E-06	Operation, transoceanic freight ship/OCE U	63%
Marine eutrophication	kg N eq	4.42E-06	1.43E-11	1.81E-14	-1.46E-07	4.27E-06	Treatment, PV cell production effluent, to wastewater treatment, class 3/CH U	67%
Marine eutrophication	kg N eq	1.83E-06	1.41E-06	8.32E-11	3.09E-07	3.55E-06	Blasting/RER U	70%
Marine eutrophication	kg N eq	2.94E-06	1.20E-09	4.16E-13	2.08E-11	2.94E-06	Flat glass, uncoated, at plant/RER U	73%
Marine eutrophication	kg N eq	2.87E-06	3.37E-07	2.72E-11	-9.95E-07	2.21E-06	Operation, lorry >16t, fleet average/RER U	75%
Marine eutrophication	kg N eq	1.84E-06	3.12E-08	5.08E-12	1.08E-08	1.88E-06	Hard coal, burned in power plant/BE U	76%
Marine eutrophication	kg N eq	1.85E-06	5.29E-12	6.79E-15	-2.36E-12	1.85E-06	Hard coal, burned in coal mine power plant/CN U	78%
Marine eutrophication	kg N eq	1.29E-06	2.73E-07	4.00E-10	1.06E-08	1.57E-06	Diesel, burned in building machine/GLO U	79%
Marine eutrophication	kg N eq	1.47E-06	2.11E-12	2.69E-15	-9.41E-13	1.47E-06	CZ single crystalline silicon, photovoltaics, at plant/kg/RER U	81%
Freshwater ecotoxicity	CTUe	2.90E-02	7.52E-02	1.89E-05	2.13E-03	1.06E-01		
Freshwater ecotoxicity	CTUe	1.21E-04	6.09E-02	8.09E-09	-2.63E-07	6.10E-02	disposal, plastic, industr. electronics, 15.3% water, to municipal incineration/kg/CH U	57%
Freshwater ecotoxicity	CTUe	1.01E-03	4.99E-03	3.89E-08	2.38E-03	8.39E-03	Disposal, redmud from bauxite digestion, 0% water, to residual material landfill/CH U	65%
Freshwater ecotoxicity	CTUe	7.80E-03	2.20E-05	1.61E-08	-1.32E-06	7.83E-03	Disposal, hard coal ash, 0% water, to residual material landfill/PL U	73%
Freshwater ecotoxicity	CTUe	6.63E-04	2.92E-03	3.76E-08	7.91E-04	4.37E-03	Copper, primary, at refinery/RLA U	77%
Freshwater ecotoxicity	CTUe	2.85E-03	1.02E-07	1.15E-10	-5.32E-08	2.85E-03	Silver, from combined gold-silver production, at refinery/PE U	79%
Freshwater ecotoxicity	CTUe	1.24E-03	9.34E-04	6.52E-07	-3.19E-06	2.18E-03	Disposal, slag, unalloyed electr. steel, 0% water, to residual material landfill/CH U	81%
Land use	kg C deficit	7.39E-02	9.50E-03	1.32E-05	1.36E-03	8.48E-02		
Land use	kg C deficit	9.34E-03	3.35E-03	5.82E-07	7.67E-04	1.35E-02	Well for exploration and production, onshore/GLO/I U	16%
Land use	kg C deficit	1.12E-02	3.19E-08	4.10E-11	-1.43E-08	1.12E-02	Hard coal, at mine/CN U	29%
Land use	kg C deficit	9.40E-03	4.98E-08	6.28E-11	-2.23E-08	9.40E-03	Photovoltaic panel factory/GLO/I U	40%
Land use	kg C deficit	6.87E-03	1.78E-03	3.52E-07	2.98E-04	8.95E-03	Well for exploration and production, offshore/OCE/I U	51%
Land use	kg C deficit	7.50E-03	2.14E-08	2.74E-11	-9.56E-09	7.50E-03	Underground mine, hard coal/CN/I U	60%
Land use	kg C deficit	-6.42E-04	-2.88E-03	-6.77E-08	-1.37E-03	-4.90E-03	Reclamation, bauxite mine/GLO U	54%
Land use	kg C deficit	5.92E-04	2.90E-03	2.43E-08	1.38E-03	4.88E-03	Bauxite, at mine/GLO U	60%
Land use	kg C deficit	-2.89E-03	-1.19E-03	-1.81E-08	-5.51E-04	-4.63E-03	electricity, hydropower, at reservoir power plant, non alpine regions/kWh/RER U	54%
Land use	kg C deficit	4.02E-03	1.24E-04	2.08E-07	2.49E-05	4.17E-03	Hardwood, standing, under bark, in forest/RER U	59%
Land use	kg C deficit	3.32E-03	1.63E-08	2.05E-11	-7.31E-09	3.32E-03	Wafer factory/DE/I U	63%
Land use	kg C deficit	2.63E-03	2.33E-04	2.97E-08	6.13E-06	2.86E-03	Softwood, standing, under bark, in forest/RER U	66%
Land use	kg C deficit	2.54E-03	2.34E-06	1.48E-09	1.49E-07	2.54E-03	Tin, at regional storage/RER U	69%
Land use	kg C deficit	1.44E-03	5.87E-04	7.91E-08	1.43E-04	2.17E-03	production plant crude oil, onshore/p/GLO/I U	72%
Land use	kg C deficit	1.17E-03	2.93E-04	4.43E-08	1.00E-04	1.57E-03	Residual material landfill facility/CH/I U	74%
Land use	kg C deficit	1.44E-03	7.06E-09	8.87E-12	-3.18E-09	1.44E-03	Photovoltaic cell factory/DE/I U	75%
Land use	kg C deficit	9.89E-04	2.76E-04	1.09E-07	7.37E-05	1.34E-03	Underground mine, hard coal/GLO/I U	77%
Land use	kg C deficit	1.13E-03	3.24E-09	4.16E-12	-1.45E-09	1.13E-03	Hard coal supply mix/CN U	78%
Land use	kg C deficit	2.64E-04	6.68E-04	9.14E-09	1.63E-04	1.10E-03	Disposal, sulfidic tailings, off-site/GLO U	80%
Land use	kg C deficit	7.15E-04	5.62E-05	1.72E-08	1.59E-05	7.87E-04	Hard coal, at mine/RNA U	81%
Water resource depletion	m3 water eq	9.15E-05	1.49E-05	8.48E-08	5.69E-06	1.12E-04		
Water resource depletion	m3 water eq	2.96E-05	1.21E-05	1.85E-10	5.63E-06	4.73E-05	electricity, hydropower, at reservoir power plant, non alpine regions/kWh/RER U	42%
Water resource depletion	m3 water eq	1.21E-05	1.74E-11	2.22E-14	-7.77E-12	1.21E-05	CZ single crystalline silicon, photovoltaics, at plant/kg/CN U	53%
Water resource depletion	m3 water eq	9.92E-06	2.83E-11	3.63E-14	-1.26E-11	9.92E-06	Hard coal, burned in power plant/CN U	62%
Water resource depletion	m3 water eq	8.56E-06	2.84E-11	3.58E-14	-1.27E-11	8.56E-06	silicon, electronic grade, at plant/kg/CN U	69%
Water resource depletion	m3 water eq	4.53E-06	2.61E-11	3.14E-14	-1.25E-11	4.53E-06	silicon, electronic grade, at plant/kg/DE U	73%
Water resource depletion	m3 water eq	3.89E-06	1.29E-11	1.63E-14	-5.79E-12	3.89E-06	silicon, electronic grade, at plant/US U	77%
Water resource depletion	m3 water eq	3.40E-06	1.13E-11	1.42E-14	-5.06E-12	3.40E-06	silicon, electronic grade, at plant/APAC U	80%
Water resource depletion	m3 water eq	1.77E-06	2.19E-07	1.38E-10	3.02E-08	2.02E-06	Water, decarbonised, at plant/RER U	82%
Mineral, fossil & ren resource depletion	kg Sb eq	1.21E-05	5.13E-07	3.26E-11	2.81E-08	1.26E-05		
Mineral, fossil & ren resource depletion	kg Sb eq	4.75E-06	1.71E-10	1.93E-13	-8.87E-11	4.75E-06	Resource correction, PbZn, silver, positive/GLO U	38%
Mineral, fossil & ren resource depletion	kg Sb eq	3.98E-06	1.43E-10	1.61E-13	-7.43E-11	3.98E-06	Anode slime, silver and tellurium containing, primary copper production/GLO U	69%
Mineral, fossil & ren resource depletion	kg Sb eq	7.43E-07	2.67E-11	3.01E-14	-1.39E-11	7.43E-07	Silver, from combined gold-silver production, at refinery/CL U	75%
Mineral, fossil & ren resource depletion	kg Sb eq	7.29E-07	2.62E-11	2.95E-14	-1.36E-11	7.29E-07	Silver, from combined gold-silver production, at refinery/PE U	81%
Cumulative energy demand non renewable	MJ oil eq	9.21E-01	1.36E-01	4.93E-05	2.20E-02	1.08E+00		
Cumulative energy demand non renewable	MJ oil eq	4.01E-01	1.14E-06	1.47E-09	-5.11E-07	4.01E-01	Hard coal, at mine/CN U	37%
Cumulative energy demand non renewable	MJ oil eq	6.12E-02	1.57E-02	6.76E-06	3.04E-03	7.99E-02	Uranium natural, at underground mine/RNA U	45%
Cumulative energy demand non renewable	MJ oil eq	5.59E-02	6.53E-03	2.41E-06	8.47E-05	6.25E-02	natural gas, at production onshore/m3/NAC U	50%
Cumulative energy demand non renewable	MJ oil eq	4.08E-02	1.05E-02	4.51E-06	2.03E-03	5.33E-02	Uranium natural, at open pit mine/RNA U	55%
Cumulative energy demand non renewable	MJ oil eq	3.65E-02	5.33E-03	1.47E-06	3.53E-04	4.22E-02	natural gas, at production onshore/m3/RU U	59%
Cumulative energy demand non renewable	MJ oil eq	3.43E-02	5.65E-03	1.47E-06	4.24E-04	4.04E-02	natural gas, at production offshore/m3/NO U	63%
Cumulative energy demand non renewable	MJ oil eq	2.26E-02	9.24E-03	1.35E-06	2.16E-03	3.40E-02	Crude oil, at production onshore/RME U	66%
Cumulative energy demand non renewable	MJ oil eq	1.86E-02	7.50E-03	9.61E-07	1.82E-03	2.79E-02	crude oil, at production offshore/kg/NO U	69%
Cumulative energy demand non renewable	MJ oil eq	1.90E-02	7.83E-03	3.59E-06	8.83E-04	2.77E-02	Lignite, at mine/RER U	71%
Cumulative energy demand non renewable	MJ oil eq	2.44E-02	1.91E-03	5.86E-07	5.41E-04	2.68E-02	Hard coal, at mine/RNA U	74%
Cumulative energy demand non renewable	MJ oil eq	1.61E-02	8.16E-03	3.13E-06	2.38E-03	2.67E-02	Hard coal, at mine/WEU U	76%
Cumulative energy demand non renewable	MJ oil eq	1.79E-02	5.03E-03	2.50E-06	1.06E-03	2.40E-02	Hard coal, at mine/EEU U	78%
Cumulative energy demand non renewable	MJ oil eq	1.55E-02	6.25E-03	8.02E-07	1.52E-03	2.33E-02	crude oil, at production onshore/kg/RU U	81%
Cumulative energy demand renewable	MJ oil eq	1.13E-01	2.66E-02	3.60E+00	9.51E-03	3.75E+00		
Cumulative energy demand renewable	MJ oil eq	0.00E+00	0.00E+00	3.60E+00	0.00E+00	3.60E+00	hotspot electricity, PV, at 3kWp slanted-roof, mono-Si, panel, mounted, incl. EoL absolute/f	96%
Cumulative energy demand renewable	MJ oil eq	4.62E-02	1.14E-02	1.22E-06	4.47E-03	6.21E-02	electricity, hydropower, at run-of-river power plant without reservoir/kWh/RER U	98%
Nuclear waste	m3 HAA eq	6.70E-11	1.83E-11	1.10E-14	3.20E-12	8.85E-11		
Nuclear waste	m3 HAA eq	6.69E-11	1.83E-11	1.10E-14	3.20E-12	8.84E-11	Radioactive waste, in final repository for nuclear waste SF, HLW, and ILW/CH U	100%
Nuclear waste	m3 HAA eq	4.74E-14	2.63E-15	6.70E-19	1.80E-16	5.02E-14	Titanium dioxide, chloride process, at plant/RER U	100%

6.6 Most relevant elementary flows

The analysis of the most relevant elementary flows is done for all five PV technologies for the 3 kWp installation mounted on a slanted roof top. The most relevant elementary flows are defined as those contributing cumulatively at least 80 % to the total impacts in each impact category or those having an individual share of more than 5 % in the total impacts. The elementary flow hotspots are identified at the whole life cycle level and only absolute contributions to the total impacts are considered (European Commission 2015). The most relevant elementary flows are not linked to the responsible most relevant processes as required by the Product Environmental Footprint Pilot Guidance since this analysis would be too complex and time-consuming. The most relevant elementary flows for the five PV technologies analysed in this study are listed for each impact category in Tab. 6.16 to Tab. 6.20.

Tab. 6.16 Identification of the most relevant elementary flows for each impact category for 1 kWh of DC electricity produced with a residential scale CdTe PV system mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant elementary flows to the characterized environmental impacts is reached.

Impact category	Unit per kWh electricity	Total	Most relevant elementary flows	Compartment	Cumulative share	Share
electricity, 3 kWp installation, mounted with CdTe PV technology, characterized		Contribution				
Climate change	kg CO2 eq	2.53E-02				
Climate change	kg CO2 eq	2.20E-02	Carbon dioxide, fossil	Air	87%	87%
Climate change	kg CO2 eq	1.63E-03	Methane, tetrafluoro-, CFC-14	Air	93%	6%
Ozone depletion	kg CFC-11 eq	9.79E-10				
Ozone depletion	kg CFC-11 eq	7.64E-10	Methane, bromochlorodifluoro-, Halon 1211	Air	78%	78%
Ozone depletion	kg CFC-11 eq	9.46E-11	Methane, bromotrifluoro-, Halon 1301	Air	88%	10%
Ozone depletion	kg CFC-11 eq	7.25E-11	Ethane, 1,2-dichloro-1,1,2,2-tetrafluoro-, CFC-114	Air	95%	7%
Human toxicity, cancer effects	CTUh, c	9.56E-10				
Human toxicity, cancer effects	CTUh, c	8.19E-10	Chromium VI	Water	86%	86%
Human toxicity, cancer effects	CTUh, c	8.22E-11	Chromium	Air	94%	9%
Human toxicity, non-cancer effects	CTUh, n-c	5.40E-09				
Human toxicity, non-cancer effects	CTUh, n-c	1.19E-09	Arsenic	Water	22%	22%
Human toxicity, non-cancer effects	CTUh, n-c	1.09E-09	Lead	Air	42%	20%
Human toxicity, non-cancer effects	CTUh, n-c	7.48E-10	Mercury	Air	56%	14%
Human toxicity, non-cancer effects	CTUh, n-c	6.99E-10	Arsenic	Air	69%	13%
Human toxicity, non-cancer effects	CTUh, n-c	6.90E-10	Zinc	Air	82%	13%
Human toxicity, non-cancer effects	CTUh, n-c	6.70E-10	Cadmium	Air	94%	12%
Particulate matter	kg PM2.5 eq	1.49E-05				
Particulate matter	kg PM2.5 eq	7.52E-06	Particulates, < 2.5 um	Air	50%	50%
Particulate matter	kg U235 eq	6.62E-06	Sulfur dioxide	Air	95%	44%
Ionizing radiation HH	kg PM2.5 eq	1.34E-03				
Ionizing radiation HH	kg PM2.5 eq	1.26E-03	Carbon-14	Air	94%	94%
Ionizing radiation HH	kg PM2.5 eq	6.85E-05	Radon-222	Air	99%	5%
Photochemical ozone formation	kg NMVOC eq	9.10E-05				
Photochemical ozone formation	kg NMVOC eq	7.22E-05	Nitrogen oxides	Air	79%	79%
Photochemical ozone formation	kg NMVOC eq	8.71E-06	Sulfur dioxide	Air	89%	10%
Photochemical ozone formation	kg NMVOC eq	8.50E-06	NMVOC, non-methane volatile organic compounds, unspecified origin	Air	98%	9%
Acidification	mol H+ eq	2.02E-04				
Acidification	mol H+ eq	1.41E-04	Sulfur dioxide	Air	70%	70%
Acidification	mol H+ eq	5.35E-05	Nitrogen oxides	Air	96%	26%
Terrestrial eutrophication	mol N eq	3.44E-04				
Terrestrial eutrophication	mol N eq	3.08E-04	Nitrogen oxides	Air	89%	89%
Terrestrial eutrophication	mol N eq	3.66E-05	Ammonia	Air	100%	11%
Freshwater eutrophication	kg P eq	5.41E-06				
Freshwater eutrophication	kg P eq	5.37E-06	Phosphate	Water	99%	99%
Freshwater eutrophication	kg P eq	3.11E-08	Phosphorus	Water	100%	1%
Marine eutrophication	kg N eq	2.97E-05				
Marine eutrophication	kg N eq	2.81E-05	Nitrogen oxides	Air	95%	95%
Marine eutrophication	kg N eq	1.18E-06	Nitrate	Water	99%	4%
Freshwater ecotoxicity	CTUe	8.18E-02				
Freshwater ecotoxicity	CTUe	6.15E-02	Antimony	Water	75%	75%
Freshwater ecotoxicity	CTUe	8.11E-03	Chromium VI	Water	85%	10%
Land use	kg C deficit	2.92E-02				
Land use	kg C deficit	1.85E-02	Transformation, to mineral extraction site	Raw	64%	64%
Land use	kg C deficit	1.21E-02	Transformation, to arable, non-irrigated	Raw	105%	41%
Land use	kg C deficit	-1.21E-02	Transformation, from arable, non-irrigated	Raw	64%	-41%
Land use	kg C deficit	6.07E-03	Transformation, to dump site, benthos	Raw	84%	21%
Land use	kg C deficit	-5.66E-03	Transformation, from mineral extraction site	Raw	65%	-19%
Land use	kg C deficit	-5.62E-03	Transformation, from unknown	Raw	46%	-19%
Land use	kg C deficit	3.83E-03	Transformation, to industrial area, built up	Raw	59%	13%
Land use	kg C deficit	2.26E-03	Transformation, to industrial area	Raw	67%	8%
Land use	kg C deficit	1.77E-03	Transformation, to traffic area, road network	Raw	73%	6%
Land use	kg C deficit	1.47E-03	Transformation, to dump site	Raw	78%	5%
Land use	kg C deficit	1.40E-03	Transformation, to industrial area, vegetation	Raw	82%	5%
Water resource depletion	m3 water eq	2.49E-05				
Water resource depletion	m3 water eq	1.86E-05	Water, OECD	Air	75%	75%
Water resource depletion	m3 water eq	2.65E-06	Water, cooling, unspecified natural origin/m3	Raw	85%	11%
Water resource depletion	m3 water eq	1.44E-06	Water, river	Raw	91%	6%
Mineral, fossil & ren resource depletion	kg Sb eq	9.92E-06				
Mineral, fossil & ren resource depletion	kg Sb eq	5.77E-06	Cadmium	Raw	58%	58%
Mineral, fossil & ren resource depletion	kg Sb eq	3.11E-06	Silver	Raw	89%	31%
Cumulative energy demand non renewable	MJ oil eq	3.57E-01				
Cumulative energy demand non renewable	MJ oil eq	1.21E-01	Gas, natural/m3	Raw	34%	34%
Cumulative energy demand non renewable	MJ oil eq	1.12E-01	Oil, crude	Raw	65%	31%
Cumulative energy demand non renewable	MJ oil eq	6.45E-02	Coal, hard	Raw	84%	18%
Cumulative energy demand non renewable	MJ oil eq	4.70E-02	Uranium	Raw	97%	13%
Cumulative energy demand renewable	MJ oil eq	3.64E+00				
Cumulative energy demand renewable	MJ oil eq	3.60E+00	Energy, solar, converted	Raw	99%	99%
Cumulative energy demand renewable	MJ oil eq	3.88E-02	Energy, potential (in hydropower reservoir), converted	Raw	100%	1%
Nuclear waste	m3 HAA eq	3.01E-11				
Nuclear waste	m3 HAA eq	3.01E-11	Volume occupied, final repository for radioactive waste	Raw	100%	100%
Nuclear waste	m3 HAA eq	7.56E-15	Volume occupied, final repository for low-active radioactive waste	Raw	100%	0%

Tab. 6.17 Identification of the most relevant elementary flows for each impact category for 1 kWh of DC electricity produced with a 3 kWp power system with CIS PV panels mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant elementary flows to the characterized environmental impacts is reached.

Impact category	Unit per kWh electricity	Total	Most relevant elementary flows	Compartment	Cumulative share	Share
electricity, 3 kWp installation, mounted with CIS PV technology, characterized		Contribution				
Climate change	kg CO2 eq	4.15E-02				
Climate change	kg CO2 eq	3.48E-02	Carbon dioxide, fossil	Air	84%	84%
Climate change	kg CO2 eq	2.56E-03	Methane, tetrafluoro-, CFC-14	Air	90%	6%
Ozone depletion	kg CFC-11 eq	1.74E-09				
Ozone depletion	kg CFC-11 eq	8.82E-10	Methane, bromochlorodifluoro-, Halon 1211	Air	51%	51%
Ozone depletion	kg CFC-11 eq	3.02E-10	Ethane, 1,2-dichloro-1,1,2,2-tetrafluoro-, CFC-114	Air	68%	17%
Ozone depletion	kg CFC-11 eq	2.37E-10	Methane, dichlorodifluoro-, CFC-12	Air	82%	14%
Ozone depletion	kg CFC-11 eq	1.39E-10	Methane, bromotrifluoro-, Halon 1301	Air	89%	8%
Ozone depletion	kg CFC-11 eq	8.87E-11	Methane, tetrachloro-, CFC-10	Air	95%	5%
Human toxicity, cancer effects	CTUh, c	1.48E-09				
Human toxicity, cancer effects	CTUh, c	1.29E-09	Chromium VI	Water	87%	87%
Human toxicity, cancer effects	CTUh, c	1.15E-10	Chromium	Air	95%	8%
Human toxicity, non-cancer effects	CTUh, n-c	9.14E-09				
Human toxicity, non-cancer effects	CTUh, n-c	4.46E-09	Zinc	Air	49%	49%
Human toxicity, non-cancer effects	CTUh, n-c	1.48E-09	Mercury	Air	65%	16%
Human toxicity, non-cancer effects	CTUh, n-c	1.09E-09	Arsenic	Water	77%	12%
Human toxicity, non-cancer effects	CTUh, n-c	8.92E-10	Lead	Air	87%	10%
Particulate matter	kg PM2.5 eq	1.99E-05				
Particulate matter	kg PM2.5 eq	1.11E-05	Particulates, < 2.5 um	Air	56%	56%
Particulate matter	kg U235 eq	7.90E-06	Sulfur dioxide	Air	95%	40%
Ionizing radiation HH	kg PM2.5 eq	3.38E-03				
Ionizing radiation HH	kg PM2.5 eq	3.16E-03	Carbon-14	Air	93%	93%
Ionizing radiation HH	kg PM2.5 eq	1.90E-04	Radon-222	Air	99%	6%
Photochemical ozone formation	kg NMVOC eq	1.13E-04				
Photochemical ozone formation	kg NMVOC eq	8.77E-05	Nitrogen oxides	Air	77%	77%
Photochemical ozone formation	kg NMVOC eq	1.23E-05	NMVOC, non-methane volatile organic compounds, unspecified origin	Air	88%	11%
Photochemical ozone formation	kg NMVOC eq	1.04E-05	Sulfur dioxide	Air	97%	9%
Acidification	mol H+ eq	2.42E-04				
Acidification	mol H+ eq	1.67E-04	Sulfur dioxide	Air	69%	69%
Acidification	mol H+ eq	6.49E-05	Nitrogen oxides	Air	96%	27%
Terrestrial eutrophication	mol N eq	4.18E-04				
Terrestrial eutrophication	mol N eq	3.74E-04	Nitrogen oxides	Air	89%	89%
Terrestrial eutrophication	mol N eq	4.43E-05	Ammonia	Air	100%	11%
Freshwater eutrophication	kg P eq	7.43E-06				
Freshwater eutrophication	kg P eq	7.39E-06	Phosphate	Water	99%	99%
Freshwater eutrophication	kg P eq	3.28E-08	Phosphorus	Water	100%	0%
Marine eutrophication	kg N eq	3.55E-05				
Marine eutrophication	kg N eq	3.42E-05	Nitrogen oxides	Air	96%	96%
Marine eutrophication	kg N eq	7.70E-07	Nitrate	Water	99%	2%
Freshwater ecotoxicity	CTUe	9.30E-02				
Freshwater ecotoxicity	CTUe	6.18E-02	Antimony	Water	66%	66%
Freshwater ecotoxicity	CTUe	1.27E-02	Chromium VI	Water	80%	14%
Land use	kg C deficit	5.63E-02				
Land use	kg C deficit	2.99E-02	Transformation, to mineral extraction site	Raw	53%	53%
Land use	kg C deficit	-1.23E-02	Transformation, from unknown	Raw	31%	-22%
Land use	kg C deficit	1.08E-02	Transformation, to industrial area, vegetation	Raw	51%	19%
Land use	kg C deficit	1.05E-02	Transformation, to industrial area, built up	Raw	69%	19%
Land use	kg C deficit	-1.03E-02	Transformation, from mineral extraction site	Raw	51%	-18%
Land use	kg C deficit	7.64E-03	Transformation, to dump site, benthos	Raw	65%	14%
Land use	kg C deficit	6.46E-03	Transformation, to arable, non-irrigated	Raw	76%	11%
Land use	kg C deficit	-6.45E-03	Transformation, from arable, non-irrigated	Raw	65%	-11%
Land use	kg C deficit	4.29E-03	Occupation, forest, intensive, normal	Raw	72%	8%
Land use	kg C deficit	3.46E-03	Transformation, to industrial area	Raw	78%	6%
Land use	kg C deficit	2.40E-03	Transformation, to traffic area, road network	Raw	83%	4%
Water resource depletion	m3 water eq	5.00E-05				
Water resource depletion	m3 water eq	3.15E-05	Water, OECD	Air	63%	63%
Water resource depletion	m3 water eq	6.25E-06	Water, cooling, unspecified natural origin/m3	Raw	76%	12%
Water resource depletion	m3 water eq	6.15E-06	Water, DE	Air	88%	12%
Water resource depletion	m3 water eq	2.59E-06	Water, river	Raw	93%	5%
Mineral, fossil & ren resource depletion	kg Sb eq	5.12E-04				
Mineral, fossil & ren resource depletion	kg Sb eq	5.10E-04	Indium	Raw	100%	100%
Mineral, fossil & ren resource depletion	kg Sb eq	5.80E-07	Tin	Raw	100%	0%
Cumulative energy demand non renewable	MJ oil eq	6.13E-01				
Cumulative energy demand non renewable	MJ oil eq	1.70E-01	Oil, crude	Raw	28%	28%
Cumulative energy demand non renewable	MJ oil eq	1.42E-01	Gas, natural/m3	Raw	51%	23%
Cumulative energy demand non renewable	MJ oil eq	1.28E-01	Uranium	Raw	72%	21%
Cumulative energy demand non renewable	MJ oil eq	1.03E-01	Coal, hard	Raw	89%	17%
Cumulative energy demand non renewable	MJ oil eq	6.77E-02	Coal, brown	Raw	100%	11%
Cumulative energy demand renewable	MJ oil eq	3.69E+00				
Cumulative energy demand renewable	MJ oil eq	3.60E+00	Energy, solar, converted	Raw	98%	98%
Cumulative energy demand renewable	MJ oil eq	6.63E-02	Energy, potential (in hydropower reservoir), converted	Raw	99%	2%
Nuclear waste	m3 HAA eq	8.05E-11				
Nuclear waste	m3 HAA eq	8.04E-11	Volume occupied, final repository for radioactive waste	Raw	100%	100%
Nuclear waste	m3 HAA eq	2.11E-14	Volume occupied, final repository for low-active radioactive waste	Raw	100%	0%

Tab. 6.18

Identification of the most relevant elementary flows for each impact category for 1 kWh of DC electricity produced with a 3 kWp power system with micro-Si PV panels mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant elementary flows to the characterized environmental impacts is reached.

Impact category	Unit per kWh electricity	Total	Most relevant elementary flows	Compartment	Cumulative share	Share
electricity, 3 kWp installation, mounted with micro-Si PV technology, characterized		Contribution				
Climate change	kg CO2 eq	5.76E-02				
Climate change	kg CO2 eq	4.38E-02	Carbon dioxide, fossil	Air	76%	76%
Climate change	kg CO2 eq	4.31E-03	Nitrogen fluoride	Air	83%	7%
Climate change	kg CO2 eq	3.72E-03	Methane, fossil	Air	90%	6%
Ozone depletion	kg CFC-11 eq	1.17E-09				
Ozone depletion	kg CFC-11 eq	7.87E-10	Methane, bromochlorodifluoro-, Halon 1211	Air	67%	67%
Ozone depletion	kg CFC-11 eq	1.96E-10	Methane, bromotrifluoro-, Halon 1301	Air	84%	17%
Ozone depletion	kg CFC-11 eq	1.33E-10	Ethane, 1,2-dichloro-1,1,2,2-tetrafluoro-, CFC-114	Air	95%	11%
Human toxicity, cancer effects	CTUh, c	1.67E-09				
Human toxicity, cancer effects	CTUh, c	1.46E-09	Chromium VI	Water	87%	87%
Human toxicity, cancer effects	CTUh, c	1.24E-10	Chromium	Air	95%	7%
Human toxicity, non-cancer effects	CTUh, n-c	9.91E-09				
Human toxicity, non-cancer effects	CTUh, n-c	3.53E-09	Zinc	Air	36%	36%
Human toxicity, non-cancer effects	CTUh, n-c	2.21E-09	Arsenic	Water	58%	22%
Human toxicity, non-cancer effects	CTUh, n-c	1.56E-09	Mercury	Air	74%	16%
Human toxicity, non-cancer effects	CTUh, n-c	1.10E-09	Lead	Air	85%	11%
Human toxicity, non-cancer effects	CTUh, n-c	5.80E-10	Arsenic	Air	91%	6%
Human toxicity, non-cancer effects	CTUh, n-c	5.32E-10	Cadmium	Air	96%	5%
Particulate matter	kg PM2.5 eq	6.12E-05				
Particulate matter	kg PM2.5 eq	4.33E-05	Particulates, < 2.5 um	Air	71%	71%
Particulate matter	kg U235 eq	1.65E-05	Sulfur dioxide	Air	98%	27%
Ionizing radiation HH	kg PM2.5 eq	2.62E-03				
Ionizing radiation HH	kg PM2.5 eq	2.46E-03	Carbon-14	Air	94%	94%
Ionizing radiation HH	kg PM2.5 eq	1.33E-04	Radon-222	Air	99%	5%
Photochemical ozone formation	kg NMVOC eq	2.04E-04				
Photochemical ozone formation	kg NMVOC eq	1.66E-04	Nitrogen oxides	Air	81%	81%
Photochemical ozone formation	kg NMVOC eq	2.10E-05	Sulfur dioxide	Air	92%	10%
Photochemical ozone formation	kg NMVOC eq	1.40E-05	NMVOC, non-methane volatile organic compounds, unspecified origin	Air	98%	7%
Acidification	mol H+ eq	4.71E-04				
Acidification	mol H+ eq	3.40E-04	Sulfur dioxide	Air	72%	72%
Acidification	mol H+ eq	1.23E-04	Nitrogen oxides	Air	98%	26%
Terrestrial eutrophication	mol N eq	7.44E-04				
Terrestrial eutrophication	mol N eq	7.07E-04	Nitrogen oxides	Air	95%	95%
Terrestrial eutrophication	mol N eq	3.73E-05	Ammonia	Air	100%	5%
Freshwater eutrophication	kg P eq	6.87E-06				
Freshwater eutrophication	kg P eq	6.81E-06	Phosphate	Water	99%	99%
Freshwater eutrophication	kg P eq	5.23E-08	Phosphorus	Water	100%	1%
Marine eutrophication	kg N eq	6.52E-05				
Marine eutrophication	kg N eq	6.45E-05	Nitrogen oxides	Air	99%	99%
Marine eutrophication	kg N eq	2.55E-07	Ammonia	Air	99%	0%
Freshwater ecotoxicity	CTUe	9.63E-02				
Freshwater ecotoxicity	CTUe	6.12E-02	Antimony	Water	64%	64%
Freshwater ecotoxicity	CTUe	1.44E-02	Chromium VI	Water	79%	15%
Freshwater ecotoxicity	CTUe	4.74E-03	Arsenic	Water	83%	5%
Land use	kg C deficit	5.07E-02				
Land use	kg C deficit	3.33E-02	Transformation, to mineral extraction site	Raw	66%	66%
Land use	kg C deficit	-1.17E-02	Transformation, from unknown	Raw	43%	-23%
Land use	kg C deficit	-9.77E-03	Transformation, from mineral extraction site	Raw	23%	-19%
Land use	kg C deficit	8.68E-03	Transformation, to dump site, benthos	Raw	40%	17%
Land use	kg C deficit	6.45E-03	Transformation, to industrial area	Raw	53%	13%
Land use	kg C deficit	5.81E-03	Transformation, to arable, non-irrigated	Raw	65%	11%
Land use	kg C deficit	-5.80E-03	Transformation, from arable, non-irrigated	Raw	53%	-11%
Land use	kg C deficit	5.04E-03	Transformation, to industrial area, built up	Raw	63%	10%
Land use	kg C deficit	4.98E-03	Transformation, to dump site	Raw	73%	10%
Land use	kg C deficit	2.66E-03	Occupation, dump site	Raw	78%	5%
Land use	kg C deficit	2.20E-03	Occupation, forest, intensive, normal	Raw	82%	4%
Water resource depletion	m3 water eq	9.47E-05				
Water resource depletion	m3 water eq	4.33E-05	Water, OECD	Air	46%	46%
Water resource depletion	m3 water eq	3.78E-05	Water, CN	Air	86%	40%
Water resource depletion	m3 water eq	7.21E-06	Water, cooling, unspecified natural origin/m3	Raw	93%	8%
Mineral, fossil & ren resource depletion	kg Sb eq	1.52E-06				
Mineral, fossil & ren resource depletion	kg Sb eq	3.87E-07	Silver	Raw	26%	26%
Mineral, fossil & ren resource depletion	kg Sb eq	3.31E-07	Copper	Raw	47%	22%
Mineral, fossil & ren resource depletion	kg Sb eq	2.35E-07	Fluorspar	Raw	63%	15%
Mineral, fossil & ren resource depletion	kg Sb eq	2.29E-07	Zinc	Raw	78%	15%
Mineral, fossil & ren resource depletion	kg Sb eq	1.59E-07	Molybdenum	Raw	88%	10%
Cumulative energy demand non renewable	MJ oil eq	6.59E-01				
Cumulative energy demand non renewable	MJ oil eq	2.08E-01	Coal, hard	Raw	32%	32%
Cumulative energy demand non renewable	MJ oil eq	2.07E-01	Oil, crude	Raw	63%	31%
Cumulative energy demand non renewable	MJ oil eq	1.24E-01	Gas, natural/m3	Raw	82%	19%
Cumulative energy demand non renewable	MJ oil eq	9.03E-02	Uranium	Raw	96%	14%
Cumulative energy demand renewable	MJ oil eq	3.69E+00				
Cumulative energy demand renewable	MJ oil eq	3.60E+00	Energy, solar, converted	Raw	98%	98%
Cumulative energy demand renewable	MJ oil eq	8.02E-02	Energy, potential (in hydropower reservoir), converted	Raw	100%	2%
Nuclear waste	m3 HAA eq	5.84E-11				
Nuclear waste	m3 HAA eq	5.84E-11	Volume occupied, final repository for radioactive waste	Raw	100%	100%
Nuclear waste	m3 HAA eq	1.49E-14	Volume occupied, final repository for low-active radioactive waste	Raw	100%	0%

Tab. 6.19 Identification of the most relevant elementary flows for each impact category for 1 kWh of DC electricity produced with a 3 kWp power system with multi-Si PV panels mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant elementary flows to the characterized environmental impacts is reached.

Impact category	Unit per kWh electricity	Total	Most relevant elementary flows	Compartment	Cumulative share	Share
electricity, 3 kWp installation, mounted with multi-Si PV technology, characterized		Contribution				
Climate change	kg CO2 eq	5.69E-02				
Climate change	kg CO2 eq	4.78E-02	Carbon dioxide, fossil	Air	84%	84%
Climate change	kg CO2 eq	4.76E-03	Methane, fossil	Air	92%	8%
Ozone depletion	kg CFC-11 eq	1.77E-09				
Ozone depletion	kg CFC-11 eq	8.40E-10	Methane, bromochlorodifluoro-, Halon 1211	Air	47%	47%
Ozone depletion	kg CFC-11 eq	3.93E-10	Methane, tetrachloro-, CFC-10	Air	70%	22%
Ozone depletion	kg CFC-11 eq	1.56E-10	Ethane, 1,2-dichloro-1,1,2,2-tetrafluoro-, CFC-114	Air	78%	9%
Ozone depletion	kg CFC-11 eq	1.40E-10	Methane, dichlorodifluoro-, CFC-12	Air	86%	8%
Ozone depletion	kg CFC-11 eq	1.28E-10	Methane, bromotrifluoro-, Halon 1301	Air	94%	7%
Human toxicity, cancer effects	CTUh, c	1.42E-09				
Human toxicity, cancer effects	CTUh, c	1.16E-09	Chromium VI	Water	82%	82%
Human toxicity, cancer effects	CTUh, c	1.38E-10	Chromium	Air	91%	10%
Human toxicity, non-cancer effects	CTUh, n-c	1.05E-08				
Human toxicity, non-cancer effects	CTUh, n-c	3.32E-09	Arsenic	Water	32%	32%
Human toxicity, non-cancer effects	CTUh, n-c	1.81E-09	Mercury	Air	49%	17%
Human toxicity, non-cancer effects	CTUh, n-c	1.56E-09	Zinc	Air	64%	15%
Human toxicity, non-cancer effects	CTUh, n-c	1.50E-09	Lead	Air	78%	14%
Human toxicity, non-cancer effects	CTUh, n-c	8.62E-10	Arsenic	Air	86%	8%
Human toxicity, non-cancer effects	CTUh, n-c	8.02E-10	Cadmium	Air	94%	8%
Particulate matter	kg PM2.5 eq	6.98E-05				
Particulate matter	kg PM2.5 eq	5.03E-05	Particulates, < 2.5 um	Air	72%	72%
Particulate matter	kg U235 eq	1.80E-05	Sulfur dioxide	Air	98%	26%
Ionizing radiation HH	kg PM2.5 eq	2.90E-03				
Ionizing radiation HH	kg PM2.5 eq	2.72E-03	Carbon-14	Air	94%	94%
Ionizing radiation HH	kg PM2.5 eq	1.54E-04	Radon-222	Air	99%	5%
Photochemical ozone formation	kg NMVOC eq	1.94E-04				
Photochemical ozone formation	kg NMVOC eq	1.53E-04	Nitrogen oxides	Air	79%	79%
Photochemical ozone formation	kg NMVOC eq	2.28E-05	Sulfur dioxide	Air	90%	12%
Photochemical ozone formation	kg NMVOC eq	1.43E-05	NMVOC, non-methane volatile organic compounds, unspecified origin	Air	98%	7%
Acidification	mol H+ eq	4.91E-04				
Acidification	mol H+ eq	3.68E-04	Sulfur dioxide	Air	75%	75%
Acidification	mol H+ eq	1.13E-04	Nitrogen oxides	Air	98%	23%
Terrestrial eutrophication	mol N eq	6.96E-04				
Terrestrial eutrophication	mol N eq	6.53E-04	Nitrogen oxides	Air	94%	94%
Terrestrial eutrophication	mol N eq	4.31E-05	Ammonia	Air	100%	6%
Freshwater eutrophication	kg P eq	7.98E-06				
Freshwater eutrophication	kg P eq	7.88E-06	Phosphate	Water	99%	99%
Freshwater eutrophication	kg P eq	9.08E-08	Phosphorus	Water	100%	1%
Marine eutrophication	kg N eq	6.31E-05				
Marine eutrophication	kg N eq	5.96E-05	Nitrogen oxides	Air	94%	94%
Marine eutrophication	kg N eq	2.80E-06	Nitrate	Water	99%	4%
Freshwater ecotoxicity	CTUe	1.02E-01				
Freshwater ecotoxicity	CTUe	6.32E-02	Antimony	Water	62%	62%
Freshwater ecotoxicity	CTUe	1.15E-02	Chromium VI	Water	73%	11%
Freshwater ecotoxicity	CTUe	7.08E-03	Zinc	Water	80%	7%
Land use	kg C deficit	6.75E-02				
Land use	kg C deficit	2.76E-02	Transformation, to mineral extraction site	Raw	41%	41%
Land use	kg C deficit	1.70E-02	Transformation, to arable, non-irrigated	Raw	66%	25%
Land use	kg C deficit	-1.70E-02	Transformation, from arable, non-irrigated	Raw	41%	-25%
Land use	kg C deficit	-1.64E-02	Transformation, from unknown	Raw	17%	-24%
Land use	kg C deficit	1.14E-02	Transformation, to industrial area, vegetation	Raw	33%	17%
Land use	kg C deficit	1.03E-02	Transformation, to industrial area, built up	Raw	49%	15%
Land use	kg C deficit	-8.93E-03	Transformation, from mineral extraction site	Raw	35%	-13%
Land use	kg C deficit	7.93E-03	Transformation, to industrial area	Raw	47%	12%
Land use	kg C deficit	7.00E-03	Transformation, to dump site, benthos	Raw	58%	10%
Land use	kg C deficit	6.28E-03	Transformation, to dump site	Raw	67%	9%
Land use	kg C deficit	5.05E-03	Occupation, forest, intensive, normal	Raw	74%	7%
Land use	kg C deficit	4.17E-03	Transformation, to traffic area, road network	Raw	81%	6%
Water resource depletion	m3 water eq	6.84E-05				
Water resource depletion	m3 water eq	3.52E-05	Water, OECD	Air	51%	51%
Water resource depletion	m3 water eq	2.47E-05	Water, cooling, unspecified natural origin/m3	Raw	88%	36%
Mineral, fossil & ren resource depletion	kg Sb eq	1.93E-05				
Mineral, fossil & ren resource depletion	kg Sb eq	1.67E-05	Silver	Raw	86%	86%
Mineral, fossil & ren resource depletion	kg Sb eq	7.06E-07	Tellurium	Raw	90%	4%
Cumulative energy demand non renewable	MJ oil eq	7.09E-01				
Cumulative energy demand non renewable	MJ oil eq	2.69E-01	Coal, hard	Raw	38%	38%
Cumulative energy demand non renewable	MJ oil eq	1.55E-01	Oil, crude	Raw	60%	22%
Cumulative energy demand non renewable	MJ oil eq	1.51E-01	Gas, natural/m3	Raw	81%	21%
Cumulative energy demand non renewable	MJ oil eq	1.04E-01	Uranium	Raw	96%	15%
Cumulative energy demand renewable	MJ oil eq	3.71E+00				
Cumulative energy demand renewable	MJ oil eq	3.60E+00	Energy, solar, converted	Raw	97%	97%
Cumulative energy demand renewable	MJ oil eq	8.58E-02	Energy, potential (in hydropower reservoir), converted	Raw	99%	2%
Nuclear waste	m3 HAA eq	6.64E-11				
Nuclear waste	m3 HAA eq	6.64E-11	Volume occupied, final repository for radioactive waste	Raw	100%	100%
Nuclear waste	m3 HAA eq	1.64E-14	Volume occupied, final repository for low-active radioactive waste	Raw	100%	0%

Tab. 6.20 Identification of the most relevant elementary flows for each impact category for 1 kWh of DC electricity with a 3 kWp power system with mono-Si PV panels mounted on a slanted roof. The green shaded cells indicate that the threshold of a cumulative 80 % contribution of the most relevant elementary flows to the characterized environmental impacts is reached.

Impact category	Unit per kWh electricity	Total	Most relevant elementary flows	Compartment	Cumulative share	Share
electricity, 3 kWp installation, mounted with mono-Si PV technology, characterized		Contribution				
Climate change	kg CO2 eq	9.14E-02				
Climate change	kg CO2 eq	7.76E-02	Carbon dioxide, fossil	Air	85%	85%
Climate change	kg CO2 eq	9.21E-03	Methane, fossil	Air	95%	10%
Ozone depletion	kg CFC-11 eq	2.40E-09				
Ozone depletion	kg CFC-11 eq	1.26E-09	Methane, bromochlorodifluoro-, Halon 1211	Air	53%	53%
Ozone depletion	kg CFC-11 eq	5.54E-10	Methane, tetrachloro-, CFC-10	Air	76%	23%
Ozone depletion	kg CFC-11 eq	2.04E-10	Ethane, 1,2-dichloro-1,1,2,2-tetrafluoro-, CFC-114	Air	84%	8%
Ozone depletion	kg CFC-11 eq	1.36E-10	Methane, bromotrifluoro-, Halon 1301	Air	90%	6%
Human toxicity, cancer effects	CTUh, c	1.56E-09				
Human toxicity, cancer effects	CTUh, c	1.25E-09	Chromium VI	Water	80%	80%
Human toxicity, cancer effects	CTUh, c	1.45E-10	Chromium	Air	89%	9%
Human toxicity, non-cancer effects	CTUh, n-c	1.35E-08				
Human toxicity, non-cancer effects	CTUh, n-c	5.17E-09	Arsenic	Water	38%	38%
Human toxicity, non-cancer effects	CTUh, n-c	2.59E-09	Mercury	Air	58%	19%
Human toxicity, non-cancer effects	CTUh, n-c	1.82E-09	Zinc	Air	71%	14%
Human toxicity, non-cancer effects	CTUh, n-c	1.58E-09	Lead	Air	83%	12%
Human toxicity, non-cancer effects	CTUh, n-c	8.91E-10	Arsenic	Air	90%	7%
Human toxicity, non-cancer effects	CTUh, n-c	7.99E-10	Cadmium	Air	95%	6%
Particulate matter	kg PM2.5 eq	1.34E-04				
Particulate matter	kg PM2.5 eq	1.01E-04	Particulates, < 2.5 um	Air	75%	75%
Particulate matter	kg PM2.5 eq	3.09E-05	Sulfur dioxide	Air	98%	23%
Ionizing radiation HH	kg PM2.5 eq	3.79E-03				
Ionizing radiation HH	kg PM2.5 eq	3.53E-03	Carbon-14	Air	93%	93%
Ionizing radiation HH	kg PM2.5 eq	2.09E-04	Radon-222	Air	99%	6%
Photochemical ozone formation	kg NMVOC eq	3.24E-04				
Photochemical ozone formation	kg NMVOC eq	2.59E-04	Nitrogen oxides	Air	80%	80%
Photochemical ozone formation	kg NMVOC eq	3.87E-05	Sulfur dioxide	Air	92%	12%
Photochemical ozone formation	kg NMVOC eq	1.91E-05	NMVOC, non-methane volatile organic compounds, unspecified origin	Air	98%	6%
Acidification	mol H+ eq	8.26E-04				
Acidification	mol H+ eq	6.25E-04	Sulfur dioxide	Air	76%	76%
Acidification	mol H+ eq	1.91E-04	Nitrogen oxides	Air	99%	23%
Terrestrial eutrophication	mol N eq	1.14E-03				
Terrestrial eutrophication	mol N eq	1.10E-03	Nitrogen oxides	Air	96%	96%
Terrestrial eutrophication	mol N eq	4.22E-05	Ammonia	Air	100%	4%
Freshwater eutrophication	kg P eq	8.62E-06				
Freshwater eutrophication	kg P eq	8.56E-06	Phosphate	Water	99%	99%
Freshwater eutrophication	kg P eq	5.50E-08	Phosphorus	Water	100%	1%
Marine eutrophication	kg N eq	1.13E-04				
Marine eutrophication	kg N eq	1.01E-04	Nitrogen oxides	Air	89%	89%
Marine eutrophication	kg N eq	1.15E-05	Nitrate	Water	99%	10%
Freshwater ecotoxicity	CTUe	1.06E-01				
Freshwater ecotoxicity	CTUe	6.38E-02	Antimony	Water	60%	60%
Freshwater ecotoxicity	CTUe	1.23E-02	Chromium VI	Water	72%	12%
Freshwater ecotoxicity	CTUe	7.65E-03	Arsenic	Water	79%	7%
Freshwater ecotoxicity	CTUe	5.29E-03	Zinc	Water	84%	5%
Land use	kg C deficit	8.48E-02				
Land use	kg C deficit	3.02E-02	Transformation, to mineral extraction site	Raw	36%	36%
Land use	kg C deficit	-2.22E-02	Transformation, from unknown	Raw	9%	-26%
Land use	kg C deficit	1.68E-02	Transformation, to arable, non-irrigated	Raw	29%	20%
Land use	kg C deficit	-1.68E-02	Transformation, from arable, non-irrigated	Raw	9%	-20%
Land use	kg C deficit	1.46E-02	Transformation, to industrial area	Raw	27%	17%
Land use	kg C deficit	1.18E-02	Transformation, to dump site	Raw	41%	14%
Land use	kg C deficit	1.14E-02	Transformation, to industrial area, vegetation	Raw	54%	13%
Land use	kg C deficit	1.03E-02	Transformation, to industrial area, built up	Raw	66%	12%
Land use	kg C deficit	8.93E-03	Transformation, to dump site, benthos	Raw	77%	11%
Land use	kg C deficit	-8.29E-03	Transformation, from mineral extraction site	Raw	67%	-10%
Land use	kg C deficit	7.03E-03	Occupation, forest, intensive, normal	Raw	75%	8%
Land use	kg C deficit	5.94E-03	Occupation, dump site	Raw	82%	7%
Land use	kg C deficit	4.61E-03	Transformation, to traffic area, road network	Raw	88%	5%
Water resource depletion	m3 water eq	1.12E-04				
Water resource depletion	m3 water eq	5.34E-05	Water, cooling, unspecified natural origin/m3	Raw	48%	48%
Water resource depletion	m3 water eq	4.94E-05	Water, OECD	Air	92%	44%
Mineral, fossil & ren resource depletion	kg Sb eq	1.26E-05				
Mineral, fossil & ren resource depletion	kg Sb eq	1.07E-05	Silver	Raw	85%	85%
Mineral, fossil & ren resource depletion	kg Sb eq	4.53E-07	Tellurium	Raw	89%	4%
Cumulative energy demand non renewable	MJ oil eq	1.08E+00				
Cumulative energy demand non renewable	MJ oil eq	5.16E-01	Coal, hard	Raw	48%	48%
Cumulative energy demand non renewable	MJ oil eq	2.16E-01	Gas, natural/m3	Raw	68%	20%
Cumulative energy demand non renewable	MJ oil eq	1.67E-01	Oil, crude	Raw	83%	15%
Cumulative energy demand non renewable	MJ oil eq	1.40E-01	Uranium	Raw	96%	13%
Cumulative energy demand renewable	MJ oil eq	3.75E+00				
Cumulative energy demand renewable	MJ oil eq	3.60E+00	Energy, solar, converted	Raw	96%	96%
Cumulative energy demand renewable	MJ oil eq	1.19E-01	Energy, potential (in hydropower reservoir), converted	Raw	99%	3%
Nuclear waste	m3 HAA eq	8.85E-11				
Nuclear waste	m3 HAA eq	8.84E-11	Volume occupied, final repository for radioactive waste	Raw	100%	100%
Nuclear waste	m3 HAA eq	2.16E-14	Volume occupied, final repository for low-active radioactive waste	Raw	100%	0%

6.7 Matching of the ILCD method with elementary flows

The completeness of the PEF impact assessment was evaluated based on the matching of the ILCD method with the elementary flows. The inventories of all life cycle stages of photovoltaic electricity production (product, construction, use, end-of-life, recycling) rely on a total of 765 elementary flows. The ILCD impact assessment method takes 468 elementary flows into account, which corresponds to a share of 61 %. The remaining elementary flows, which make up a share of 39 % in the total number of elementary flows, are not considered in the weighted results. Several reasons can be distinguished for an elementary flow not being included in the weighted results:

- 274 elementary flows are not considered in any of the impact categories defined in the ILCD impact assessment method.
- 11 elementary flows are included in the impact categories defined in the ILCD impact assessment method but have a characterization factor equal to 0.
- 7 elementary flows are included in the additional impact categories cumulative energy demand renewable (4), cumulative energy demand non-renewable (1) and nuclear waste (2). The elementary flows in these impact categories are characterized but not included in the normalized and weighted results.
- 5 elementary flows are included in the interim impact category ionizing radiation E. The elementary flows in these impact categories are characterized (not shown in the results) but not included in the normalized and weighted results.

The elementary flows not matched by any of the default, additional and interim impact categories include a variety of substances. For instance, evaporation of water from Croatia or from Serbia and Montenegro, sum parameters such as emissions of aromatic or chlorinated hydrocarbons to air as well as the extraction of raw materials such as gravel and gypsum are not assessed in the ILCD impact assessment method.

7. Benchmark

In this study five distinctly different PV technologies are assessed. Unlike many other PEF pilots, the product analyzed is a capital good which is used to produce a product for final consumption, namely electricity. Electricity is produced by many different power plant technologies such as thermal power plants (lignite, hard coal, heavy fuel oil, natural and industrial gas, nuclear, biomass), hydroelectric power plants and wind power plants.

Furthermore, the environmental performance of PV electricity is not only dependent on the manufacturing of panels, mounting structures and electrical equipment (and the respective supply chains) but also on the place and orientation of the PV panels. The annual yield of a panel installed may vary between (on the average) about 830 kWh/kWp if installed in Sweden to nearly 1'500 kWh/kWp if the same panel is being installed in Portugal. Furthermore, deviating from the optimal orientation may lead to a reduction in annual yield of 25 % and even more.

These two aspects should be kept in mind when discussing possible approaches to define a benchmark for PV electricity. Instead of proposing preliminary but tangible benchmark values, three alternative possible approaches on how to define benchmarks in this PEF pilot are described, namely the electricity benchmark, the technology benchmark and the personal budget benchmark.

The *electricity benchmark* is derived from a comparison of the environmental impacts of PV electricity with the environmental impacts of electricity produced with other technologies. Electricity production with fossil and nuclear energy are in the focus of such a comparison because hard coal, lignite and nuclear power plants need to be replaced to achieve an environmentally benign electricity market.

Reduction factors, specific to each of the impact category indicators, could be defined relative to the current environmental performance of current electricity production. The level of current environmental performance could be identified by either using the maximum impacts of individual technologies or the mix of power plant technologies based on non-renewable energy resources and operated within EU28. Policy based, impact specific reduction factors could be applied on the values of these reference technologies. If for instance an 80 % reduction in greenhouse gas emissions is required to comply with the 2 °C target, a climate change benchmark of PV electricity could be 20 % of the greenhouse gas emissions of 1 kWh of fossil based electricity.

This approach is rather straightforward but needs life cycle assessment data and models comparable to the data established within this PEF pilot. Although such data are readily available, the models would need to be established first, if one aims at a perfect fit. However, existing LCA models of electricity production may serve this purpose because the benchmark needs not be defined to the decimal point.

The *technology benchmarks* are separately defined per individual PV technology. The technology specific benchmark could be defined in a way that a certain percentage of all panels of a certain technology installed in Europe comply with it. Hence, the benchmark would put emphasis on an appropriate geographic location and an appropriate orientation of the panel, rather than on the environmental performance of panel manufacture and construction. Installing panels in northern European countries with annual yields below the average would require PV systems with lower environmental impacts and vice versa.

The definition of these benchmarks is rather straightforward. It starts from the premise that the average environmental performance of PV electricity as quantified in this screening LCA is appropriate with respect to the EU green products policy. The incentive is more on the optimal use of the product PV panel rather than on optimizing manufacturing processes and material efficiencies.

The *personal budget benchmark* starts from the premise that a certain share of the personal annual budget of environmental impacts shall be available for electricity consumption. The personal annual budget of environmental impacts is derived from the normalization values published by the European Union (Benini et al. 2014, p. 92). The share attributable to electricity consumption may be defined globally (same share for all impact categories) or individually per impact category. It may consider reduction requirements or reflect the current situation.

This approach faces two challenges. On one hand it might be difficult (and probably arbitrary) to define an environmental budget for the average annual per capita consumption of electricity. On the other hand, the data used to determine the annual per capita environmental impacts is based on domestic impacts only. Environmental impacts caused by processes, which predominantly occur outside Europe may lead to substantial mismatch between the personal budget and the impacts caused by PV electricity. This drawback can be overcome by using either consumption based environmental impacts to quantify the personal budget or by using data on global environmental impacts.

Benchmarking the environmental impacts of PV electricity on the level of (equally) weighted single score results is not recommended. One single substance (indium resource) within one single indicator (abiotic resource depletion) caused by one technology (CIS) is heavily dominating the results.

8. Summary and Conclusion

This study encompasses the life cycle assessment of direct current electricity produced with five different PV technologies (CdTe, CIS, micromorphous-Si, multicrystalline-Si, monocrystalline-Si) on three different mounting types (integrated in roof, mounted on roof and open ground) and two system

scales (3 kWp residential scale and 570 kWp large scale). The PV technologies differ in material consumption and module efficiencies but also in their environmental performance. The following life cycle stages are included in the product system of PV electricity: product stage, construction stage, operation stage and end-of-life stage. Potential benefits for recycling are quantified and allocated 50:50 to the electricity production on one hand and to the products made of recycled material, on the other hand. The production of the inverter is not part of the base case, but its influence is assessed in a sensitivity analysis.

Depending on the PV technology the weighted environmental impacts vary and range from $8.37 \cdot 10^{-6}$ pt/kWh (CdTe PV modules integrated in roof) to $3.57 \cdot 10^{-4}$ pt/kWh (CIS PV panels on open ground system). Within each technology, the integrated systems cause the lowest impacts per kWh of electricity produced, curtesy followed by the mounted systems. The open ground systems cause the highest environmental impact of the systems analysed. These differences are due to the land use, the mounting system and the electric installation. The ranking of the environmental performance of the different PV technologies for the mounted residential-scale systems is: the smallest impacts are caused by electricity production with CdTe PV modules, followed by micromorphous-Si panels, monocrystalline-Si panels, multicrystalline-Si panels and finally CIS PV panels.

Regarding the relative importance of the different life cycle stages, it becomes clear that the product stage and the construction stage (mounting and installation) cause the major part of the environmental impacts. The share of these life cycle stages varies between the different PV technologies, whereby the product stage is always more important than the construction stage when considering the weighted results of the residential scale PV systems mounted on a slanted roof. The environmental impacts during the use phase are negligible with the renewable cumulative energy demand (harvesting of solar energy for electricity production) being the only exception. The end-of-life stage has a share of up to 10 % in the total impacts with the transport of waste materials and the electricity consumed in the recycling process being most important. The potential benefits due to recycling vary strongly depending on the impact category considered and are typically in the range of -5 % to -15 %.

The most important impact categories are mineral, fossil and renewable resource depletion, human toxicity (cancer and non cancer effects), freshwater ecotoxicity, particulate matter potential and acidification potential. With regard to the mineral, fossil and renewable resource depletion impacts, the supply chains of semiconductor materials (cadmium, tellurium, indium), silver (mainly used in the metallization paste for multicrystalline-Si and monocrystalline-Si PV modules), copper (mainly used in the electric installation) and zinc (used in various processes such as secondary aluminium production) are identified as process hotspots. Human toxicity cancer effects are caused by the disposal of redmud from bauxite digestion (supply chain of primary aluminium) and by the disposal of slag generated in the production of unalloyed electric steel. The substance hotspots in this impact category are chromium VI emitted to water and chromium emissions to air. Both of these elementary flows are primarily associated with the supply chain of steel production. The non cancer effects on human toxicity are mainly caused by the production of primary copper and zinc, by indium rich leaching residues as well as by the disposal of hard coal ash. The most relevant elementary flows are zinc and mercury emitted to air in the process of unalloyed electric steel production. Another substance hotspot identified in the impact category human toxicity non cancer effects is the emission of arsenic to water. The processes causing the major part of these emissions are natural gas extraction and the beneficiation of iron ore. The most relevant processes and elementary flows in the impact category freshwater ecotoxicity are identical for the five PV technologies analysed, namely the waste incineration of plastic components of the electric installation and the disposal of redmud from bauxite digestion. Antimony emissions to water occur during the extraction of natural gas and chromium VI is

emitted to water primarily during the beneficiation of iron ore. The particulate matter impacts are especially relevant for the silicon-based PV technologies and dominated by electricity production in hard coal power plants. The emissions of PM_{2.5} and sulfur dioxide to air mainly occur during aluminium electrolysis and flat glass production. The production of MG-silicon is another important process with regard to emissions of sulfur dioxide to air. The acidification potential is among the five most relevant impact categories for the CdTe, CIS and monocrystalline-Si PV technologies. In this impact category, the operation of transoceanic freight ships, the production of flat glass and the electricity generation in hard coal power plants are identified as process hotspots. The most relevant elementary flows are emissions of sulfur dioxide and nitrogen oxides to air.

The analyses revealed that the narrow part of the supply chains of PV module production (supply chain of silicon, cadmium, tellurium, indium) is hardly responsible for emissions to the environment or extraction of resources (Indium resource consumption being one of the few exceptions). Hence, the potential for further reducing the environmental impacts lies in increasing material and energy efficiency of the PV modules and PV systems and in particular in purchasing electricity from less polluting power sources.

Data quality was assessed according to the requirements of the European Commission (2013). The minimum “fair” required data rating was achieved and surpassed for all data used. In fact all datasets used score good quality or higher. The main information gaps occur with regard to the environmental efficiency of panel and module production in Asia as well as with regard to the end-of-life treatment efforts and emissions of all types of panels except CdTe. These gaps are closed by assuming the same environmental efficiency of panel production (same process specific emissions) but adjusting the electricity mix used in manufacturing to the average Chinese and Malaysian situation. Data about end-of-life treatment of CdTe PV modules is extrapolated to the treatment of all other technologies. The estimated recycling efforts have turned out to be conservative compared to data provided by recyclers taking part in the PV CYCLE programme. Real data about panel manufacture in Asia and about end-of-life treatment of CIS, micromorphous-Si, multicrystalline-Si and monocrystalline-Si PV panels would further enhance the quality of future assessments.

The main limitations are given by limited data availability of direct, process-specific emissions, in particular regarding processes taking place in the Asian region, and scarce data available regarding the mass and energy flows of the end-of-life stage due to yet too little volumes and experiences with industrialized panel takeback and treatment (except CdTe). A more technical limitation of this screening study is that the most relevant elementary flows identified in the hotspot analysis could not be matched to their respective most relevant processes due to the lack of a suitable function to perform this step in an efficient way.

Benchmarks can be defined using different approaches. It is yet too early to establish tangible rigid values. First a discussion is needed within the Technical Secretariat and with the European Commission about the purpose and about the intended effects as well as the unintended potential effects of such benchmark values.

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ANNEX I

A.1. Data quality assessment

A.1.1. Overview

This chapter describes the detailed data quality assessment according to the European Commission (European Commission 2013). In the screening step a minimum “fair” quality data rating is required for data contributing to at least 90 % of the impact estimated for each EF impact category, as assessed via a qualitative expert judgment.

In subchapter A.1.2 the data quality assessment of the electricity production with CdTe PV panels is described, while subchapters A.1.3 to A.1.6 describe those of CIS, micromorphous-Si, multicrystalline-Si and monocrystalline-Si PV technology.

All datasets used scored good quality or higher. A minimum “fair” quality was met and excelled in all cases. The following Tab. A. 1 shows an overview of the data quality rating achieved for all life cycle stages of the solar electricity production and all PV technologies for the mounted installation.

Tab. A. 1 Overview of data quality (scores and level) of electricity production of all PV technologies per life cycle stage, mounted on a slanted roof top, residential scale. Regarded in this overview are the processes causing 90 % of the impact per impact category. Data quality scores were assessed according to requirements of the European Commission (2013)

electricity production with:	Product stage		Construction stage		Use stage		End-of-life stage		Potential benefits due to recycling	
	data quality scores	data quality level	data quality scores	data quality level	data quality scores	data quality level	data quality scores	data quality level	data quality scores	data quality level
CdTe PV technology	1.4	Excellent data quality	1.3	Excellent data quality	2.2	Good data quality	1.7	Very good data quality	1.7	Very good data quality
CIS PV technology	2.0	Very good data quality	1.3	Excellent data quality	2.2	Good data quality	2.5	Good data quality	2.5	Good data quality
micro-Si PV technology	1.3	Excellent data quality	1.3	Excellent data quality	2.2	Good data quality	2.5	Good data quality	2.5	Good data quality
multi-Si PV technology	2.5	Good data quality	1.3	Good data quality	2.2	Good data quality	2.5	Good data quality	2.5	Good data quality
mono-Si PV technology	2.5	Good data quality	1.3	Good data quality	2.2	Good data quality	2.5	Good data quality	2.5	Good data quality

A.1.2. Data quality assessment of electricity production with CdTe PV panels

The following Tab. A. 2 lists the data quality assessment according to PEF guideline’s requirements of the CdTe PV panel at the regional storage (product stage). The PV panel produced in Malaysia is most important as it has the largest share in the regional mix. The data quality assessment for the Malaysian CdTe PV panel is found in Tab. A. 3. The processes contributing to 90 % of the impacts per impact category all hold good or better data quality.

Tab. A. 2 data quality assessment according to the European Commission (2013) of the CdTe PV panel mix at the regional storage (product stage) of electricity production with CdTe PV technology

	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	
Technology	C	M	TIR	TeR	GR	P	$DQR = (C+M+TIR+TeR+GR+P)/6$		
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Formula 1	PEF guide Table 6	90% contribution
Process									
photovoltaic laminate, CdTe, at plant, US	2	2	1	1	1	2	1.5	Excellent data quality	yes
photovoltaic laminate, CdTe, at plant, MY	2	2	1	1	1	2	1.5	Excellent data quality	yes
Transport, transoceanic freight ship	2	2	4	1	1	2	2.0	Very good data quality	-
Transport, freight, rail	2	2	4	1	1	2	2.0	Very good data quality	-
Transport, lorry>16t, fleet average	2	2	3	1	1	2	1.8	Very good data quality	-

Tab. A. 3 data quality assessment according to the European Commission (2013) of the CdTe PV panel produced in Malaysia

	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	
Technology	C	M	TIR	TeR	GR	P	$DQR = (C+M+TIR+TeR+GR+P)/6$		
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Formula 1	PEF guide Table 6	90% contribution
Process									
photovoltaic laminate, CdTe, at plant, MY	2	2	1	1	1	2	1.5	Excellent data quality	yes
electricity, medium voltage, at grid, MY U	2	2	1	1	1	2	1.5	Excellent data quality	yes
photovoltaic panel factory CdTe/p/US	2	2	1	1	1	2	1.5	Excellent data quality	yes
tap water, at user	2	2	2	1	5	2	2.3	Good data quality	yes
Tempering, flat glass	2	2	4	1	5	2	2.7	Good data quality	yes
Copper, at regional storage	2	2	3	1	5	2	2.5	Good data quality	yes
Solar glass, low-iron, at regional storage	2	2	4	1	5	2	2.7	Good data quality	yes
Flat glass, uncoated, at plant	2	2	3	1	5	2	2.5	Good data quality	yes
Glass fibre reinforced plastic, polyamide, injection moulding, at plant	2	2	4	1	5	2	2.7	Good data quality	yes
Ethylvinylacetate, foil, at plant	2	2	4	1	5	2	2.7	Good data quality	yes
Cadmium telluride, semiconductor-grade, at plant	2	2	3	1	5	2	2.5	Good data quality	yes
Cadmium sulphide, semiconductor-grade, at plant	2	2	3	1	5	2	2.5	Good data quality	yes
Corrugated board, mixed fibre, single wall, at plant	2	2	3	1	5	2	2.5	Good data quality	yes
Transport, transoceanic freight ship	2	2	4	1	1	2	2.0	Very good data quality	yes
disposal, plastics, mixture, 15.3% water, to municipal incineration	2	2	1	1	5	2	2.2	Good data quality	yes
Nitric acid, 50% in H2O, at plant	2	2	4	1	5	2	2.7	Good data quality	-
Sulphuric acid, liquid, at plant	2	2	4	1	5	2	2.7	Good data quality	-
Silica sand, at plant	2	2	4	1	5	2	2.7	Good data quality	-
Sodium chloride, powder, at plant	2	2	4	1	5	2	2.7	Good data quality	-
Hydrogen peroxide, 50% in H2O, at plant	2	2	4	1	5	2	2.7	Good data quality	-
Isopropanol, at plant	2	2	4	1	5	2	2.7	Good data quality	-
Sodium hydroxide, 50% in H2O, production mix, at plant	2	2	4	1	5	2	2.7	Good data quality	-
Chemicals inorganic, at plant	2	2	4	1	5	3	2.8	Good data quality	-
Chemicals organic, at plant	2	2	4	1	2	3	2.3	Good data quality	-
Nitrogen, liquid, at plant	2	2	4	1	5	2	2.7	Good data quality	-
Helium, at plant	2	2	3	1	5	2	2.5	Good data quality	-
Silicone product, at plant	2	2	3	1	5	2	2.5	Good data quality	-
Transport, lorry>16t, fleet average	2	2	3	1	5	2	2.5	Good data quality	-
Transport, freight, rail	2	2	4	1	5	2	2.7	Good data quality	-
disposal, municipal solid waste, 22.9% water, to municipal incineration	2	2	1	1	5	2	2.2	Good data quality	-

The data quality of the construction stage is shown in Tab. A. 4. The data contributing to 90 % of the impacts per impact category fulfills the PEF requirements and it is caused by the electric installation and the mounting structure.

Tab. A. 4 data quality assessment according to the European Commission (2013) of the construction stage of electricity production with CdTe PV technology

	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	
Technology	C	M	TIR	TeR	GR	P	$DQR = (C+M+TIR+TeR+GR+P)/6$		
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Formula 1	PEF guide Table 6	90% contribution
Process									
electrical installation	2	2	1	1	3	2	1.8	Very good data quality	yes
mounting structure	2	2	1	1	1	2	1.5	Excellent data quality	yes
electricity	2	2	1	1	1	2	1.5	Excellent data quality	-
transport lorry >16 t	2	2	4	1	1	2	2.0	Very good data quality	-

The data quality of the construction stage is shown in Tab. A. 5. The data contributing to 90 % of the impacts per impact category fulfills the PEF requirements. The electricity for mounting is the most important parameter regarding data quality of the use stage.

Tab. A. 5 data quality assessment according to the European Commission (2013) of the use stage of electricity production with CdTe PV technology

	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	
Technology	C	M	TIR	TeR	GR	P	$DQR = (C+M+TIR+TeR+GR+P)/6$		
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Formula 1	PEF guide Table 6	90% contribution
process									
electricity production	2	2	1	1	1	2	1.5	Excellent data quality	yes
water	2	2	2	1	1	2	1.7	Very good data quality	-
waste water treatment	2	2	2	1	3	2	2.0	Very good data quality	-

The data quality assessment for the end-of-life stage of the CdTe PV system is described in Tab. A. 6. The processes contributing to 90 % of the impacts per impact category all hold good or better data quality. The electricity consumption, the lorry transport and the consumption of hydrogen peroxide are the parameters most important.

Tab. A. 6 data quality assessment according to the European Commission (2013) of the end-of-life stage of electricity production with CdTe PV technology

	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	
Technology	C	M	TIR	TeR	GR	P	DQR = (C+M+ TIR + TeR +GR+ P)/6		
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Formula 1	PEF guide Table 6	90% contribution
Process									
Transport, lorry>16t, fleet average	2	2	3	1	1	1	1.7	Very good data quality	yes
electricity, medium voltage, at grid	2	2	1	1	3	1	1.7	Very good data quality	yes
Sulphuric acid, liquid, at plant	2	2	4	1	1	1	1.8	Very good data quality	yes
Hydrogen peroxide, 50% in H2O, at plant	2	2	4	1	1	1	1.8	Very good data quality	yes
Sodium hydroxide, 50% in H2O, production mix, at plant	2	2	4	1	1	1	1.8	Very good data quality	yes
Treatment, PV cell production effluent, to wastewater treatment, class 3	2	2	2	1	3	1	1.8	Very good data quality	yes
disposal, plastics, mixture, 15.3% water, to municipal incineration	2	2	1	1	3	1	1.7	Very good data quality	yes
CdTe PV module takeback + recycling	2	2	1	1	1	1	1.3	Excellent data quality	-
Water, deionised, at plant	2	2	4	1	3	1	2.2	Good data quality	-
Disposal, inert waste, 5% water, to inert material landfill	2	2	4	1	3	1	2.2	Good data quality	-

The data quality assessment for potential benefits for recycling of CdTe PV panels is shown in Tab. A. 7. The processes contributing to 90 % of the impacts per impact category all have good or better data quality. The primary and secondary aluminium as well as the primary copper are the parameter most sensitive regarding data quality.

Tab. A. 7 data quality assessment according to the European Commission (2013) of the potential benefits for recycling (50/50) from CdTe panel recycling

	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	
Technology	C	M	TIR	TeR	GR	P	DQR = (C+M+ TIR + TeR +GR+ P)/6		
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Formula 1	PEF guide Table 6	90% contribution
Process									
Avoided burden from recycling, CdTe PV module, mounted construction	2	2	1	1	1	2	1.5	Excellent data quality	yes
Cadmium sludge, from zinc electrolysis, at plant	2	2	3	1	1	2	1.8	Very good data quality	yes
Copper telluride cement, from copper production	2	2	3	1	1	2	1.8	Very good data quality	yes
Soda, powder, at plant	2	2	4	1	1	2	2.0	Very good data quality	yes
natural gas, high pressure, at consumer	2	2	1	1	1	2	1.5	Excellent data quality	yes
Heavy fuel oil, at regional storage	2	2	4	1	1	2	2.0	Very good data quality	yes
Copper, at regional storage	2	2	3	1	1	2	1.8	Very good data quality	yes
Copper, secondary, at refinery	2	2	3	1	1	2	1.8	Very good data quality	yes
Aluminium, primary, at plant	2	2	4	1	1	2	2.0	Very good data quality	yes
Aluminium, secondary, from old scrap, at plant	2	2	4	1	1	2	2.0	Very good data quality	yes
Pig iron, at plant	2	2	3	1	1	2	1.8	Very good data quality	yes
Silica sand, at plant	2	2	4	1	3	2	2.3	Good data quality	-
Limestone, milled, packed, at plant	2	2	4	1	3	2	2.3	Good data quality	-

A.1.3. Data quality assessment of electricity production with CIS PV panels

Tab. A. 8 lists the data quality assessment according to PEF guideline's requirements of the CIS PV panel at the regional storage (product stage). The PV panel production takes place in Germany. The laminate production and the aluminium for framing are the parameter most important in terms of sourcing 90 % of the impacts. The data quality assessment for the laminate CIS is found in Tab. A. 9. The processes contributing to 90 % of the impacts per impact category all hold good or better data quality.

Tab. A. 8 data quality assessment according to the European Commission (2013) of the CIS PV panel mix at the regional storage (product stage) of electricity production with CIS PV technology

	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	
Technology	C	M	TIR	TeR	GR	P	$DQR = (C+M+TIR+TeR+GR+P)/6$		
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Formula 1	PEF guide Table 6	90% contribution
Process									
photovoltaic laminate, CIS, at plant, DE	2	2	2	1	1	2	1.7	Very good quality	yes
Aluminium alloy, AlMg3, at plant	2	2	4	1	2	2	2.2	"Good quality"	yes
photovoltaic panel, CIS, at plant, DE	2	2	2	1	1	2	1.7	Very good quality	-
Light fuel oil, burned in industrial furnace 1MW, non-modulating	2	2	3	1	2	2	2.0	Very good quality	-
Glass fibre reinforced plastic, polyamide, injection moulding, at plant	2	2	4	1	2	2	2.2	"Good quality"	-
Transport, lorry>16t, fleet average	2	2	3	1	2	2	2.0	Very good quality	-
Transport, freight, rail	2	2	4	1	2	2	2.2	"Good quality"	-
disposal, plastics, mixture, 15.3% water, to municipal incineration	2	2	1	1	3	2	1.8	Very good quality	-

The processes contributing most to 90 % of the environmental impacts are the electricity consumption, solar glass and the consumption of indium.

Tab. A. 9 data quality assessment according to the European Commission (2013) of the CIS laminate produced in Germany

	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	
Technology	C	M	TIR	TeR	GR	P	$DQR = (C+M+TIR+TeR+GR+P)/6$		
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Formula 1	PEF guide Table 6	90% contribution
Process									
electricity, medium voltage, at grid/kWh/DE U	2	2	1	1	1	2	1.5	"Excellent quality"	yes
tap water, at user/kg/RER U	2	2	2	1	2	2	1.8	Very good quality	yes
Tempering, flat glass/RER U	2	2	4	1	2	2	2.2	"Good quality"	yes
Copper, at regional storage/RER U	2	2	3	1	2	2	2.0	Very good quality	yes
Aluminium, production mix, at plant/RER U	2	2	4	1	2	2	2.2	"Good quality"	yes
Silicone product, at plant/RER U	2	2	3	1	2	2	2.0	Very good quality	yes
Molybdenum, at regional storage/RER U	2	2	4	1	2	2	2.2	"Good quality"	yes
Tin, at regional storage/RER U	2	2	4	1	2	2	2.2	"Good quality"	yes
Solar glass, low-iron, at regional storage/RER U	2	2	4	1	2	2	2.2	"Good quality"	yes
Photovoltaic panel factory/GLO/I U	2	2	3	1	2	2	2.0	Very good quality	yes
polyvinylbutyral foil, at plant/RER U	2	2	2	1	2	2	1.8	Very good quality	yes
Hydrochloric acid, 30% in H2O, at plant/RER U	2	2	4	1	2	2	2.2	"Good quality"	yes
EUR-flat pallet/RER U	2	2	4	1	2	2	2.2	"Good quality"	yes
Transport, lorry>16t, fleet average/RER U	2	2	3	1	2	2	2.0	Very good quality	yes
Flat glass, uncoated, at plant/RER U	2	2	3	1	2	2	2.0	Very good quality	yes
Indium, at regional storage/RER U	2	2	3	1	2	2	2.0	Very good quality	yes
Ethylvinylacetate, foil, at plant/RER U	2	2	4	1	2	2	2.2	"Good quality"	yes
Polyethylene terephthalate, granulate, amorphous, at plant/RER U	2	2	5	1	2	2	2.3	"Good quality"	yes
Polyphenylene sulfide, at plant/GLO U	2	2	4	1	2	2	2.2	"Good quality"	yes
Nitrogen, liquid, at plant/RER U	2	2	4	1	2	2	2.2	"Good quality"	yes
Transport, freight, rail/RER U	2	2	4	1	2	2	2.2	"Good quality"	yes
disposal, plastics, mixture, 15.3% water, to municipal incineration/kg/CH U	2	2	1	1	3	2	1.8	Very good quality	yes
Treatment, sewage, unpolluted, to wastewater treatment, class 3/CH U	2	2	2	1	3	2	2.0	Very good quality	-
Diode, unspecified, at plant/GLO U	2	2	4	1	2	2	2.2	"Good quality"	-
Selenium, at plant/RER U	2	2	4	1	2	2	2.2	"Good quality"	-
Wire drawing, copper/RER U	2	2	2	1	2	2	1.8	Very good quality	-
Polyethylene, HDPE, granulate, at plant/RER U	2	2	4	1	2	2	2.2	"Good quality"	-
Hydrogen peroxide, 50% in H2O, at plant/RER U	2	2	4	1	2	2	2.2	"Good quality"	-
Urea, as N, at regional storehouse/RER U	2	2	5	1	2	2	2.3	"Good quality"	-
Disposal, waste, Si waferprod., inorg. 9.4% water, to residual material landfill/CH U	2	2	4	1	3	2	2.3	"Good quality"	-
photovoltaic laminate, CIS, at plant/m2/DE/I U	2	2	2	1	1	2	1.7	Very good quality	-
Cadmium sulphide, semiconductor-grade, at plant/US U	2	2	3	1	5	2	2.5	"Good quality"	-
Gallium, semiconductor-grade, at regional storage/RER U	2	2	3	1	2	2	2.0	Very good quality	-
Flux, wave soldering, at plant/GLO U	2	2	4	1	2	2	2.2	"Good quality"	-
Zinc oxide, at plant/RER U	2	2	4	1	2	2	2.2	"Good quality"	-
Argon, liquid, at plant/RER U	2	2	1	1	2	2	1.7	Very good quality	-
Butyl acrylate, at plant/RER U	2	2	4	1	2	2	2.2	"Good quality"	-
Diborane, at plant/GLO U	2	2	2	1	2	2	1.8	Very good quality	-
Sulphuric acid, liquid, at plant/RER U	2	2	4	1	2	2	2.2	"Good quality"	-
Hydrogen sulphide, H2S, at plant/RER U	2	2	4	1	2	2	2.2	"Good quality"	-
Sodium hydroxide, 50% in H2O, production mix, at plant/RER U	2	2	4	1	2	2	2.2	"Good quality"	-
Ammonia, liquid, at regional storehouse/RER U	2	2	4	1	2	2	2.2	"Good quality"	-
Disposal, inert waste, 5% water, to inert material landfill/CH U	2	2	4	1	3	2	2.3	"Good quality"	-
disposal, glass, 0% water, to municipal incineration/kg/CH U	2	2	4	1	3	2	2.3	"Good quality"	-

The data quality of the construction stage is shown in Tab. A. 10. The data contributing to 90 % of the impacts per impact category fulfills the PEF requirements and it is caused by the electric installation and the mounting structure.

Tab. A. 10 data quality assessment according to the European Commission (2013) of the construction stage of electricity production with CIS PV technology

Technology	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	
	C	M	TIR	TeR	GR	P	$DQR = (C+M+TIR+TeR+GR+P)/6$		
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Formula 1	PEF guide Table 6	90% contribution
Process									
electrical installation	2	2	1	1	3	2	1.8	Very good quality	yes
mounting structure	2	2	1	1	1	2	1.5	"Excellent quality"	yes
electricity	2	2	1	1	1	2	1.5	"Excellent quality"	-
transport lorry >16 t	2	2	4	4	1	2	2.0	Very good quality	-

The data quality of the construction stage is shown in Tab. A. 11. The data contributing to 90 % of the impacts per impact category fulfills the PEF requirements. The electricity for mounting is the parameter most important regarding causing 90 % of the impacts of the use stage.

Tab. A. 11 data quality assessment according to the European Commission (2013) of the use stage of electricity production with CIS PV technology

Technology	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	
	C	M	TIR	TeR	GR	P	$DQR = (C+M+TIR+TeR+GR+P)/6$		
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Formula 1	PEF guide Table 6	90% contribution
process									
electricity production	2	2	1	1	1	2	1.5	"Excellent quality"	yes
water	2	2	2	1	1	2	1.7	Very good quality	-
waste water treatment	2	2	2	1	3	2	2.0	Very good quality	-

The data quality assessment for the end-of-life stage of the CIS PV system is described in Tab. A. 12. The processes contributing to 90 % of the impacts per impact category all hold good data quality. The electricity consumption, the lorry transport and the consumption of hydrogen peroxide seem to be the parameter most important.

Tab. A. 12 data quality assessment according to the European Commission (2013) of the end-of-life stage of electricity production with CIS PV technology

Technology	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	
	C	M	TIR	TeR	GR	P	$DQR = (C+M+TIR+TeR+GR+P)/6$		
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Formula 1	PEF guide Table 6	90% contribution
Process									
Transport, lorry >16t, fleet average	2	2	3	1	1	2	1.8	Very good quality	yes
electricity, medium voltage, production ENTSO, at grid	2	2	1	1	1	2	1.5	"Excellent quality"	yes
Hydrogen peroxide, 50% in H2O, at plant	2	2	4	1	1	2	2.0	Very good quality	yes
Sodium hydroxide, 50% in H2O, production mix, at plant	2	2	4	1	1	2	2.0	Very good quality	yes
Treatment, PV cell production effluent, to wastewater treatment, class 3	2	2	2	1	3	2	2.0	Very good quality	yes
disposal, plastics, mixture, 15.3% water, to municipal incineration	2	2	1	1	3	2	1.8	Very good quality	yes
Disposal, inert waste, 5% water, to inert material landfill	2	2	4	1	3	2	2.3	"Good quality"	-
Sulphuric acid, liquid, at plant	2	2	4	1	1	2	2.0	Very good quality	-
Water, deionised, at plant	2	2	4	1	3	2	2.3	"Good quality"	-

The data quality assessment for potential benefits for recycling of CIS PV panels is shown in Tab. A. 13. The processes contributing to 90 % of the impacts per impact category all have good data quality. The primary and secondary aluminium as well as the primary copper are the parameters most important in terms of sourcing 90 % of the impacts.

Tab. A. 13 data quality assessment according to the European Commission (2013) of the potential benefits for recycling (50/50) from CIS panel recycling

	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	90% contribution
Technology	C	M	TIR	TeR	GR	P	DQR = (C+M+ TIR + TeR +GR+ P)/6		
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Formula 1	PEF guide Table 6	
Process	2	2	4	1	1	3	2.2	"Good quality"	
Aluminium, primary, at plant	2	2	4	1	1	3	2.2	"Good quality"	
Aluminium, secondary, from old scrap, at plant	2	2	3	1	1	3	2.0	Very good quality	
Copper, at regional storage	2	2	4	1	1	2	2.0	Very good quality	
Soda, powder, at plant	2	2	1	1	1	3	1.7	Very good quality	
natural gas, high pressure, at consumer	2	2	4	1	1	3	2.2	"Good quality"	
Heavy fuel oil, at regional storage	2	2	3	1	1	3	2.0	Very good quality	
Copper, secondary, at refinery	2	2	1	1	1	2	1.5	"Excellent quality"	
Pig iron, at plant	2	2	4	1	3	2	2.3	"Good quality"	
Avoided burden from recycling, CIS PV module, mounted construction	2	2	4	1	3	2	2.3	"Good quality"	
Limestone, milled, packed, at plant	2	2	4	1	3	2	2.3	"Good quality"	
Silica sand, at plant	2	2	4	1	3	2	2.3	"Good quality"	

A.1.4. Data quality assessment of electricity production with micromorphous-Si PV panels

Tab. A. 14 lists the data quality assessment according to PEF guideline's requirements of the micromorphous-Si PV laminate (product stage). The production takes place in China. The processes contributing to 90 % of the impacts per impact category all hold good or better data quality and are the electricity-, the solar glass- and the aluminium consumption.

Tab. A. 14 data quality assessment according to the European Commission (2013) of the micro-Si PV panel mix at the regional storage (product stage) of electricity production with CIS PV technology

	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	
Technology	C	M	TiR	TeR	GR	P	DQR = (C+M+ TiR + TeR +GR+ P)/6		
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Formula 1	PEF guide Table 6	90% contribution
Process	5	5	5	5	5	5			
photovoltaic panel, micro-Si, at plant	2	2	1	1	1	2	1.5	Excellent data quality	yes
Solar glass, low-iron, at regional storage	2	2	4	1	5	2	2.7	Good data quality	yes
Silver, at regional storage	2	2	3	1	5	2	2.5	Good data quality	yes
Cable, three-conductor cable, at plant	2	2	3	1	2	2	2.0	Good data quality	yes
Copper, at regional storage	2	2	3	1	5	2	2.5	Good data quality	yes
electricity, high voltage, at grid	2	2	1	1	1	2	1.5	Excellent data quality	yes
Nitrogen, liquid, at plant	2	2	4	1	5	2	2.7	Good data quality	yes
silane, at plant	2	2	1	1	5	2	2.2	Good data quality	yes
nitrogen trifluoride, at plant	2	2	1	1	5	2	2.2	Good data quality	yes
natural gas, high pressure, at consumer	2	2	1	1	5	2	2.2	Good data quality	yes
Aluminium, production mix, at plant	2	2	4	1	5	2	2.7	Good data quality	yes
Transport, lorry 16-32t, EURO3	2	2	3	1	5	2	2.5	Good data quality	yes
Solar collector factory	2	2	4	1	5	2	2.7	Good data quality	yes
disposal, glass, 0% water, to municipal incineration	2	2	1	1	5	2	2.2	Good data quality	yes
Bisphenol A, powder, at plant	2	2	4	1	5	2	2.7	Good data quality	-
Wire drawing, copper	2	2	2	1	5	2	2.3	Good data quality	-
Adhesive for metals, at plant	2	2	3	1	5	2	2.5	Good data quality	-
Ethylvinylacetate, foil, at plant	2	2	4	1	5	2	2.7	Good data quality	-
Polyethylene, HDPE, granulate, at plant	2	2	2	1	5	2	2.3	Good data quality	-
Injection moulding	2	2	4	1	5	2	2.7	Good data quality	-
Silicone product, at plant	2	2	3	1	5	2	2.5	Good data quality	-
Zeolite, slurry, 50% in H2O, at plant	2	2	4	1	5	2	2.7	Good data quality	-
Compressed air, average generation, >30kW, 7 bar gauge, at	2	2	3	1	5	2	2.5	Good data quality	-
Water, decarbonised, at plant	2	2	4	1	5	2	2.7	Good data quality	-
Water, deionised, at plant	2	2	4	1	5	2	2.7	Good data quality	-
Hydrogen, liquid, at plant	2	2	4	1	5	2	2.7	Good data quality	-
Phosphane, at plant	2	2	3	1	2	2	2.0	Good data quality	-
Trimethyl borate, at plant	2	2	3	1	2	2	2.0	Good data quality	-
Carbon dioxide liquid, at plant	2	2	4	1	5	2	2.7	Good data quality	-
Argon, liquid, at plant	2	2	4	1	5	2	2.7	Good data quality	-
Diborane, at plant	2	2	2	1	2	2	1.8	Very good data quality	-
Oxygen, liquid, at plant	2	2	4	1	5	2	2.7	Good data quality	-
Chemicals organic, at plant	2	2	4	1	2	2	2.2	Good data quality	-
Adhesive for metals, at plant	2	2	3	1	5	2	2.5	Good data quality	-
Transport, lorry 20-28t, fleet average	2	2	3	1	5	2	2.5	Good data quality	-
Transport, transoceanic freight ship	2	2	4	1	1	2	2.0	Good data quality	-
Metal working machine, unspecified, at plant	2	2	3	1	5	2	2.5	Good data quality	-
Treatment, sewage, unpolluted, from residence, to wastewa	2	2	2	1	5	2	2.3	Good data quality	-
Disposal, solvents mixture, 16.5% water, to hazardous waste	2	2	2	1	5	2	2.3	Good data quality	-
disposal, rubber, unspecified, 0% water, to municipal incine	2	2	1	1	5	2	2.2	Good data quality	-

The data quality of the construction stage is shown in Tab. A. 15. The data contributing to 90 % of the impacts per impact category fulfills the PEF requirements and it is caused by the electric installation and the mounting structure.

Tab. A. 15 data quality assessment according to the European Commission (2013) of the construction stage of electricity production with micro-Si PV technology

	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	
Technology	C	M	TiR	TeR	GR	P	DQR = (C+M+ TiR + TeR +GR+ P)/6		
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Formula 1	PEF guide Table 6	90% contribution
Process	5	5	5	5	5	5			
electrical installation	2	2	1	1	3	2	1.8	Very good data quality	yes
mounting structure	2	2	1	1	1	2	1.5	Excellent data quality	yes
electricity	2	2	1	1	1	2	1.5	Excellent data quality	-
Transport, lorry 20-28t, fleet average	2	2	4	1	1	2	2.0	Good data quality	-

The data quality of the construction stage is shown in Tab. A. 16. The data contributing to 90 % of the impacts per impact category fulfills the PEF requirements. The electricity for mounting is the parameter most important in terms of causing 90 % of the impacts of the use stage.

Tab. A. 16 data quality assessment according to the European Commission (2013) of the use stage of electricity production with micro-Si PV technology

Technology	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	90% contribution
	C	M	TiR	TeR	GR	P	$DQR = (C+M+TiR+TeR+GR+P)/6$ PEF guide Formula 1	PEF guide Table 6	
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5			
process									
electricity production	2	2	1	1	1	2	1.5	Excellent data quality	yes
water	2	2	2	1	1	2	1.7	Very good data quality	-
waste water treatment	2	2	2	1	3	2	2.0	Good data quality	-

The data quality assessment for the end-of-life stage of the micromorphous-Si PV system is described in Tab. A. 17. The processes contributing to 90 % of the impacts per impact category all hold good data quality. The electricity consumption, the lorry transport and the consumption of hydrogen peroxide are the parameters most important.

Tab. A. 17 data quality assessment according to the European Commission (2013) of the end-of-life stage of electricity production with micro-Si PV technology

Technology	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	90% contribution
	C	M	TiR	TeR	GR	P	$DQR = (C+M+TiR+TeR+GR+P)/6$ PEF guide Formula 1	PEF guide Table 6	
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5			
Process									
Transport, lorry >16t, fleet average	2	2	3	1	1	2	1.8	Very good data quality	yes
electricity, medium voltage, production ENTSO, at grid	2	2	1	1	1	2	1.5	Excellent data quality	yes
Hydrogen peroxide, 50% in H2O, at plant	2	2	4	1	1	2	2.0	Very good data quality	yes
Sodium hydroxide, 50% in H2O, production mix, at plant	2	2	4	1	1	2	2.0	Very good data quality	yes
Treatment, PV cell production effluent, to wastewater treatment	2	2	2	1	3	2	2.0	Very good data quality	yes
disposal, plastics, mixture, 15.3% water, to municipal incineration	2	2	1	1	3	2	1.8	Very good data quality	yes
Disposal, inert waste, 5% water, to inert material landfill	2	2	4	1	3	2	2.3	Good data quality	-
Sulphuric acid, liquid, at plant	2	2	4	1	1	2	2.0	Very good data quality	-
Water, deionised, at plant	2	2	4	1	3	2	2.3	Good data quality	-

The data quality assessment for potential benefits for recycling of micromorphous-Si PV panels is shown in Tab. A. 18. The processes contributing to 90 % of the impacts per impact category all have good data quality. The primary and secondary aluminium as well as the primary copper are the parameters most important in terms of sourcing 90 % of the impacts.

Tab. A. 18 data quality assessment according to the European Commission (2013) of the potential benefits for recycling (50/50) from micro-Si panel recycling

	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	
Technology	C	M	TiR	TeR	GR	P	$DQR = (C+M+TiR+TeR+GR+P)/6$		
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Formula 1	PEF guide Table 6	90% contribution
Process	5	5	5	5	5	5			
Aluminium, primary, at plant	2	2	4	1	1	3	2.2	Good data quality	yes
Aluminium, secondary, from old scrap, at plant	2	2	4	1	1	3	2.2	Good data quality	yes
Copper, at regional storage	2	2	3	1	1	3	2.0	Very good data quality	yes
Soda, powder, at plant	2	2	4	1	1	2	2.0	Very good data quality	yes
natural gas, high pressure, at consumer	2	2	1	1	1	3	1.7	Very good data quality	yes
Heavy fuel oil, at regional storage	2	2	4	1	1	3	2.2	Good data quality	yes
Copper, secondary, at refinery	2	2	3	1	1	3	2.0	Very good data quality	yes
Pig iron, at plant	2	2	3	1	2	3	2.2	Good data quality	yes
Avoided burden from recycling, micro-Si PV module, mounted	2	2	1	1	1	2	1.5	Excellent data quality	yes
Limestone, milled, packed, at plant	2	2	4	1	3	2	2.3	Good data quality	-
Silica sand, at plant	2	2	4	1	3	2	2.3	Good data quality	-

A.1.5. Data quality assessment of electricity production with multicrystalline-Si PV panels

Tab. A. 19 lists the data quality assessment according to PEF guideline's requirements of the multicrystalline-Si PV panel at regional storage (product stage). The production takes place in China, Europe and Asia/Pacific. The Chinese production has the main contribution to 90 % of the impacts. However the processes contributing to 90 % of the impacts per impact category all hold good data quality. The data quality assessment for the Chinese multicrystalline-Si PV panel is found in Tab. A. 20. The processes contributing to 90 % of the impacts per impact category all hold good or better data quality.

Tab. A. 19 data quality assessment according to the European Commission (2013) of the multicrystalline-Si PV panel mix at the regional storage (product stage) of electricity production with multicrystalline PV technology

	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	
Technology	C	M	TiR	TeR	GR	P	$DQR = (C+M+TiR+TeR+GR+P)/6$		
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Formula 1	PEF guide Table 6	90% contribution
Process	5	5	5	5	5	5			
Panel / China	2	2	2	1	5	3	2.5	fair quality or better	yes
Panel / Europe	2	2	2	1	5	3	2.5	fair quality or better	yes
Panel / Asia-Pacific	2	2	2	1	5	3	2.5	fair quality or better	yes
Transport	2	2	2	1	5	3	2.5	fair quality or better	-
Direct emissions	2	2	2	1	5	3	2.5	fair quality or better	-

Tab. A. 20 data quality assessment according to the European Commission (2013) of the multicrystalline PV panel produced in China

	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	
Technology	C	M	TiR	TeR	GR	P	$DQR = (C+M+ TiR + TeR +GR+ P)/6$		
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Formula 1	PEF guide Table 6	90% contribution
Process	5	5	5	5	5	5			
Solar cells	2	2	2	1	5	3	2.5	fair quality or better	yes
Factory	2	2	2	1	5	3	2.5	fair quality or better	yes
Aluminum frame	2	2	2	1	5	3	2.5	fair quality or better	yes
Interconnection: Cu, Sn, Pb, wire drawing	2	2	2	1	5	3	2.5	fair quality or better	yes
Packaging	2	2	2	1	5	3	2.5	fair quality or better	yes
Glass and tempering	2	2	2	1	5	3	2.5	fair quality or better	yes
Junction box: plastic, bypass diode, silicone	2	2	2	1	5	3	2.5	fair quality or better	-
Disposal and waste treatment	2	2	2	1	5	3	2.5	fair quality or better	-
Transport	2	2	2	1	5	3	2.5	fair quality or better	-
Encapsulation: EVA + PE protection foil	2	2	2	1	5	3	2.5	fair quality or better	-
Electricity	2	2	2	1	5	3	2.5	fair quality or better	-
Backsheet: PVF / PET	2	2	2	1	5	3	2.5	fair quality or better	-
Direct emissions	2	2	2	1	5	3	2.5	fair quality or better	-
Chemicals	2	2	2	1	5	3	2.5	fair quality or better	-
Water	2	2	2	1	5	3	2.5	fair quality or better	-
Diesel	2	2	2	1	5	3	2.5	fair quality or better	-

The data quality of the construction stage is shown in Tab. A. 21. The data contributing to 90 % of the impacts per impact category fulfills the PEF requirements and it is caused by the electric installation and the mounting structure.

Tab. A. 21 data quality assessment according to the European Commission (2013) of the construction stage of electricity production with multicrystalline-Si PV technology

	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	
Technology	C	M	TiR	TeR	GR	P	$DQR = (C+M+ TiR + TeR +GR+ P)/6$		
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Formula 1	PEF guide Table 6	90% contribution
Process	5	5	5	5	5	5			
electrical installation	2	2	1	1	3	2	1.8	very good quality	yes
mounting structure	2	2	1	1	1	2	1.5	excellent quality	yes
electricity	2	2	1	1	1	2	1.5	excellent quality	-
Transport, lorry 20-28t, fleet average	2	2	4	1	1	2	2.0	good quality	-

The data quality of the construction stage is shown in Tab. A. 22. The data contributing to 90 % of the impacts per impact category fulfills the PEF requirements. The electricity for mounting is the parameter most important in terms of causing 90 % of the impacts of the use stage.

Tab. A. 22 data quality assessment according to the European Commission (2013) of the use stage of electricity production with multicrystalline PV technology

	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	
Technology	C	M	TiR	TeR	GR	P	$DQR = (C+M+TiR+TeR+GR+P)/6$		
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Formula 1	PEF guide Table 6	90% contribution
process									
electricity production	2	2	1	1	1	2	1.5	excellent quality	yes
water	2	2	2	1	1	2	1.7	very good quality	-
waste water treatment	2	2	2	1	3	2	2.0	good quality	-

The data quality assessment for the end-of-life stage of the multicrystalline-Si PV system is described in Tab. A. 23. The processes contributing to 90 % of the impacts per impact category all hold good data quality. The electricity consumption, the lorry transport and the consumption of hydrogen peroxide are the parameters most important.

Tab. A. 23 data quality assessment according to the European Commission (2013) of the end-of-life stage of electricity production with multicrystalline PV technology

	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	
Technology	C	M	TiR	TeR	GR	P	$DQR = (C+M+TiR+TeR+GR+P)/6$		
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Formula 1	PEF guide Table 6	90% contribution
Process									
Transport, lorry >16t, fleet average	2	2	3	1	1	2	1.8	very good quality	yes
electricity, medium voltage, production ENT	2	2	1	1	1	2	1.5	excellent quality	yes
Hydrogen peroxide, 50% in H ₂ O, at plant	2	2	4	1	1	2	2.0	very good quality	yes
Sodium hydroxide, 50% in H ₂ O, production r	2	2	4	1	1	2	2.0	very good quality	yes
Treatment, PV cell production effluent, to w	2	2	2	1	3	2	2.0	very good quality	yes
disposal, plastics, mixture, 15.3% water, to	2	2	1	1	3	2	1.8	very good quality	yes
Disposal, inert waste, 5% water, to inert ma	2	2	4	1	3	2	2.3	good quality	-
Sulphuric acid, liquid, at plant	2	2	4	1	1	2	2.0	very good quality	-
Water, deionised, at plant	2	2	4	1	3	2	2.3	good quality	-

The data quality assessment for potential benefits for recycling of multicrystalline-Si PV panels is shown in Tab. A. 24. The processes contributing to 90 % of the impacts per impact category all have good data quality. The primary and secondary aluminium as well as the primary copper are the parameters most important in terms of sourcing 90 % of the impacts.

Tab. A. 24 data quality assessment according to the European Commission (2013) of the potential benefits for recycling (50/50) from multicrystalline -Si panel recycling

	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	
Technology	C	M	TiR	TeR	GR	P	$DQR = (C+M+ TiR + TeR +GR+ P)/6$		
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Formula 1	PEF guide Table 6	90% contribution
Process									
Aluminium, primary, at plant	2	2	4	1	1	3	2.2	good quality	yes
Aluminium, secondary, from old scrap, at pl	2	2	4	1	1	3	2.2	good quality	yes
Copper, at regional storage	2	2	3	1	1	3	2.0	very good quality	yes
Soda, powder, at plant	2	2	4	1	1	2	2.0	very good quality	yes
natural gas, high pressure, at consumer	2	2	1	1	1	3	1.7	very good quality	yes
Heavy fuel oil, at regional storage	2	2	4	1	1	3	2.2	good quality	yes
Copper, secondary, at refinery	2	2	3	1	1	3	2.0	very good quality	yes
Pig iron, at plant	2	2	3	1	2	3	2.2	good quality	yes
Avoided burden from recycling, multi-Si PV r	2	2	1	1	1	2	1.5	excellent quality	yes
Limestone, milled, packed, at plant	2	2	4	1	3	2	2.3	good quality	-
Silica sand, at plant	2	2	4	1	3	2	2.3	good quality	-

A.1.6. Data quality assessment of electricity production with monocrystalline-Si PV panels

Tab. A. 25 lists the data quality assessment according to PEF guideline's requirements of the monocrystalline-Si PV panel at regional storage (product stage). The production takes place in China, Europe and Asia/Pacific. The Chinese production has the main contribution to 90 % of the impacts. However the processes contributing to 90 % of the impacts per impact category all hold good data quality. The data quality assessment for the Chinese monocrystalline-Si PV panel is found in Tab. A. 26. The processes contributing to 90 % of the impacts per impact category all hold good or better data quality.

Tab. A. 25 data quality assessment according to the European Commission (2013) of the monocrystalline-Si PV panel mix at the regional storage (product stage) of electricity production with monocrystalline-Si PV technology

	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	
	C	M	TiR	TeR	GR	P	$DQR = (C+M+ TiR + TeR +GR+ P)/6$		
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Formula 1	PEF guide Table 6	
Panel / China	2	2	2	1	5	3	2.5	fair quality or better	yes
Panel / Europe	2	2	2	1	5	3	2.5	fair quality or better	yes
Panel / Asia-Pacific	2	2	2	1	5	3	2.5	fair quality or better	yes
Transport	2	2	2	1	5	3	2.5	fair quality or better	-
Direct emissions	2	2	2	1	5	3	2.5	fair quality or better	-

Tab. A. 26 data quality assessment according to the European Commission (2013) of the monocrystalline-Si PV panel produced in China

	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	
	C	M	TIR	TeR	GR	P	$DQR = (C+M+TIR+TeR+GR+P)/6$		
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Formula 1	PEF guide Table 6	
Solar cells	2	2	2	1	5	3	2.5	fair quality or better	yes
Interconnection: Cu, Sn, Pb, wire drawing	2	2	2	1	5	3	2.5	fair quality or better	yes
Factory	2	2	2	1	5	3	2.5	fair quality or better	yes
Aluminum frame	2	2	2	1	5	3	2.5	fair quality or better	yes
Glass and tempering	2	2	2	1	5	3	2.5	fair quality or better	-
Packaging	2	2	2	1	5	3	2.5	fair quality or better	-
Junction box: plastic, bypass diode, silicone	2	2	2	1	5	3	2.5	fair quality or better	-
Disposal and waste treatment	2	2	2	1	5	3	2.5	fair quality or better	-
Transport	2	2	2	1	5	3	2.5	fair quality or better	-
Backsheet: PVF / PET	2	2	2	1	5	3	2.5	fair quality or better	-
Encapsulation: EVA + PE protection foil	2	2	2	1	5	3	2.5	fair quality or better	-
Electricity	2	2	2	1	5	3	2.5	fair quality or better	-
Chemicals	2	2	2	1	5	3	2.5	fair quality or better	-
Direct emissions	2	2	2	1	5	3	2.5	fair quality or better	-
Water	2	2	2	1	5	3	2.5	fair quality or better	-
Diesel	2	2	2	1	5	3	2.5	fair quality or better	-

The data quality of the construction stage is shown in Tab. A. 27. The data contributing to 90 % of the impacts per impact category fulfills the PEF requirements and it is caused by the electric installation and the mounting structure.

Tab. A. 27 data quality assessment according to the European Commission (2013) of the construction stage of electricity production with monocrystalline-Si PV technology

	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	
Technology	C	M	TIR	TeR	GR	P	$DQR = (C+M+TIR+TeR+GR+P)/6$		
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Formula 1	PEF guide Table 6	90% contribution
Process									
electrical installation	2	2	1	1	3	2	1.8	very good quality	yes
mounting structure	2	2	1	1	1	2	1.5	excellent quality	yes
electricity	2	2	1	1	1	2	1.5	excellent quality	-
Transport, lorry 20-28t, fleet average	2	2	1	1	1	2	1.5	excellent quality	-

The data quality of the construction stage is shown in Tab. A. 28. The data contributing to 90 % of the impacts per impact category fulfills the PEF requirements. The electricity for mounting is the parameter most important in terms of causing 90 % of the impacts of the use stage.

Tab. A. 28 data quality assessment according to the European Commission (2013) of the use stage of electricity production with monocrystalline-Si PV technology

	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	
Technology	C	M	TiR	TeR	GR	P	$DQR = (C+M+ TiR + TeR +GR+ P)/6$		
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Formula 1	PEF guide Table 6	90% contribution
process									
electricity production	2	2	1	1	1	2	1.5	excellent quality	yes
water	2	2	2	1	1	2	1.7	very good quality	-
waste water treatment	2	2	2	1	3	2	2.0	good quality	-

The data quality assessment for the end-of-life stage of the monocrystalline-Si PV system is described in Tab. A. 29. The processes contributing to 90 % of the impacts per impact category all hold good data quality. The electricity consumption, the lorry transport and the consumption of hydrogen peroxide are the parameters most important.

Tab. A. 29 data quality assessment according to the European Commission (2013) of the end-of-life stage of electricity production with monocrystalline-Si PV technology

	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	
Technology	C	M	TiR	TeR	GR	P	$DQR = (C+M+ TiR + TeR +GR+ P)/6$		
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Formula 1	PEF guide Table 6	90% contribution
Process									
Transport, lorry >16t, fleet average	2	2	3	1	1	2	1.8	very good quality	yes
electricity, medium voltage, production ENT	2	2	1	1	1	2	1.5	excellent quality	yes
Hydrogen peroxide, 50% in H2O, at plant	2	2	4	1	1	2	2.0	very good quality	yes
Sodium hydroxide, 50% in H2O, production r	2	2	4	1	1	2	2.0	very good quality	yes
Treatment, PV cell production effluent, to w	2	2	2	1	3	2	2.0	very good quality	yes
disposal, plastics, mixture, 15.3% water, to	2	2	1	1	3	2	1.8	very good quality	yes
Disposal, inert waste, 5% water, to inert ma	2	2	4	1	3	2	2.3	good quality	-
Sulphuric acid, liquid, at plant	2	2	4	1	1	2	2.0	very good quality	-
Water, deionised, at plant	2	2	4	1	3	2	2.3	good quality	-

The data quality assessment for potential benefits for recycling of monocrystalline-Si PV panels is shown in Tab. A. 30. The processes contributing to 90 % of the impacts per impact category all have good data quality. The primary and secondary aluminium as well as the primary copper are the parameters most important in terms of sourcing 90 % of the impacts.

Tab. A. 30 data quality assessment according to the European Commission (2013) of the potential benefits for recycling (50/50) from monocrystalline-Si panel recycling

	Completeness	Methodological appropriateness and consistency	Time-related representativeness	Technological representativeness	Geographical representativeness	Precision / uncertainty	Data Quality Rating of the dataset	Data quality level	
Technology	C	M	TiR	TeR	GR	P	$DQR = (C+M+ TiR + TeR +GR+ P)/6$		
	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Table 5	PEF guide Formula 1	PEF guide Table 6	
Process									90% contribution
Aluminium, primary, at plant	2	2	4	1	1	3	2.2	good quality	yes
Aluminium, secondary, from old scrap, at pl	2	2	4	1	1	3	2.2	good quality	yes
Copper, at regional storage	2	2	3	1	1	3	2.0	very good quality	yes
Soda, powder, at plant	2	2	4	1	1	2	2.0	very good quality	yes
natural gas, high pressure, at consumer	2	2	1	1	1	3	1.7	very good quality	yes
Heavy fuel oil, at regional storage	2	2	4	1	1	3	2.2	good quality	yes
Copper, secondary, at refinery	2	2	3	1	1	3	2.0	very good quality	yes
Pig iron, at plant	2	2	3	1	2	3	2.2	good quality	yes
Avoided burden from recycling, mono-Si PV	2	2	1	1	1	2	1.5	excellent quality	yes
Limestone, milled, packed, at plant	2	2	4	1	3	2	2.3	good quality	-
Silica sand, at plant	2	2	4	1	3	2	2.3	good quality	-

ANNEX II

In the following sections the environmental impacts of electricity produced with the five different photovoltaic panel technologies cadmium-telluride, copper indium selenide, micromorphous silicon, multicrystalline silicon and monocrystalline silicon are shown. The unit of analysis is defined as 1 kWh (kilowatt hour) of DC electricity generated by a PV module. The reference flow is the PV module, measured in kWp (kilowatt peak), the maximum power output under standard conditions of a module. The subchapters 5.5 to 5.9 in the main part of the report describe the environmental impacts of producing 1 kWh electricity with the five PV panel technologies used in residential scale photovoltaic power plants mounted on slanted roofs. Here are the environmental impacts of electricity produced with residential scale photovoltaic power plants integrated in slanted roofs and with large scale open ground installations described.

The overall results of all impact categories, all technologies and all mounting systems are described in Tab. 5.1 in the main report. The following subchapters A.1 to A.5 describe the detailed results of all five PV technologies for the integrated and the open ground system.

A.1. Results CdTe PV technology

A.1.1. Overview

In section A.1.2 the results prior to normalization are assessed. In the sections A.1.3 and A.1.4 the normalized and the weighted results of electricity production with CdTe PV panels, integrated in a slanted roof, residential scale and on a large scale open ground system are described.

A.1.2. Results prior to normalization

The results prior to normalization of the production of 1 kWh of electricity with CdTe PV panels integrated in a slanted roof in a 3 kWp installation system and on a 570 kWp open ground system are summarized in Tab. A. 31 and Tab. A. 32. The life cycle stages analyzed are the product stage, the construction stage, the operation stage and the end-of-life stage. Net benefits caused by end-of-life recycling are allocated to the production of electricity with a share of 50 %. The other half is allocated to the life cycle of products potentially using the secondary products (see also subchapter 3.5).

Tab. A. 31 Environmental impact results (characterized) of 1 kWh of electricity produced with a 3 kWp power system with CdTe PV panels integrated in a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, integrated with CdTe PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change	kg CO2 eq	1.35E-02	9.60E-03	3.45E-06	1.63E-03	2.47E-02	-2.73E-03	2.20E-02
Ozone depletion	kg CFC-11 eq	6.73E-10	2.29E-10	2.13E-13	5.24E-11	9.55E-10	-6.43E-11	8.90E-10
Human toxicity, cancer effects	CTUh, c	8.10E-11	4.59E-10	5.25E-13	3.42E-11	5.75E-10	-1.47E-10	4.27E-10
Human toxicity, non-cancer effects	CTUh, n-c	8.36E-10	4.11E-09	8.15E-12	1.03E-10	5.06E-09	-2.53E-10	4.81E-09
Particulate matter	kg PM2.5 eq	6.25E-06	6.22E-06	1.89E-09	3.70E-07	1.28E-05	-1.90E-06	1.09E-05
Ionizing radiation HH	kBq U235 eq	2.84E-04	7.41E-04	6.40E-07	9.43E-05	1.12E-03	-2.04E-04	9.16E-04
Photochemical ozone formation	kg NMVOC eq	6.14E-05	2.53E-05	1.25E-08	5.09E-06	9.18E-05	-5.91E-06	8.59E-05
Acidification	mol H+ eq	1.14E-04	6.47E-05	2.59E-08	5.38E-06	1.84E-04	-1.93E-05	1.65E-04
Terrestrial eutrophication	mol N eq	2.41E-04	8.30E-05	6.78E-08	1.85E-05	3.42E-04	-2.38E-05	3.19E-04
Freshwater eutrophication	kg P eq	5.32E-07	3.88E-06	4.57E-09	1.22E-07	4.54E-06	-1.02E-06	3.52E-06
Marine eutrophication	kg N eq	2.14E-05	6.92E-06	1.00E-07	1.83E-06	3.03E-05	-1.74E-06	2.85E-05
Freshwater ecotoxicity	CTUe	3.80E-03	7.40E-02	2.03E-05	1.12E-03	7.90E-02	-2.75E-03	7.62E-02
Land use	kg C deficit	1.75E-02	9.01E-03	1.42E-05	1.32E-03	2.78E-02	-2.36E-03	2.55E-02
Water resource depletion	m3 water eq	4.18E-06	1.31E-05	9.14E-08	1.09E-06	1.84E-05	-4.53E-06	1.39E-05
Mineral, fossil & ren resource depletion	kg Sb eq	6.57E-06	5.11E-07	2.83E-11	6.06E-09	7.08E-06	-2.85E-06	4.23E-06
Cumulative energy demand non renewable	MJ oil eq	1.92E-01	1.39E-01	5.29E-05	2.15E-02	3.53E-01	-3.22E-02	3.21E-01
Cumulative energy demand renewable	MJ oil eq	8.05E-03	2.32E-02	3.60E+00	1.04E-03	3.63E+00	-7.58E-03	3.62E+00
Nuclear waste	m3 HAA eq	7.25E-12	1.63E-11	1.19E-14	2.49E-12	2.60E-11	-4.53E-12	2.15E-11

The following Fig. A. 1 shows the share per life cycle stage per impact category of electricity production with CdTe PV panels integrated in a slanted roof. The potential benefits for recycling are in relation to the sum of the life cycle stages production to end-of-life.

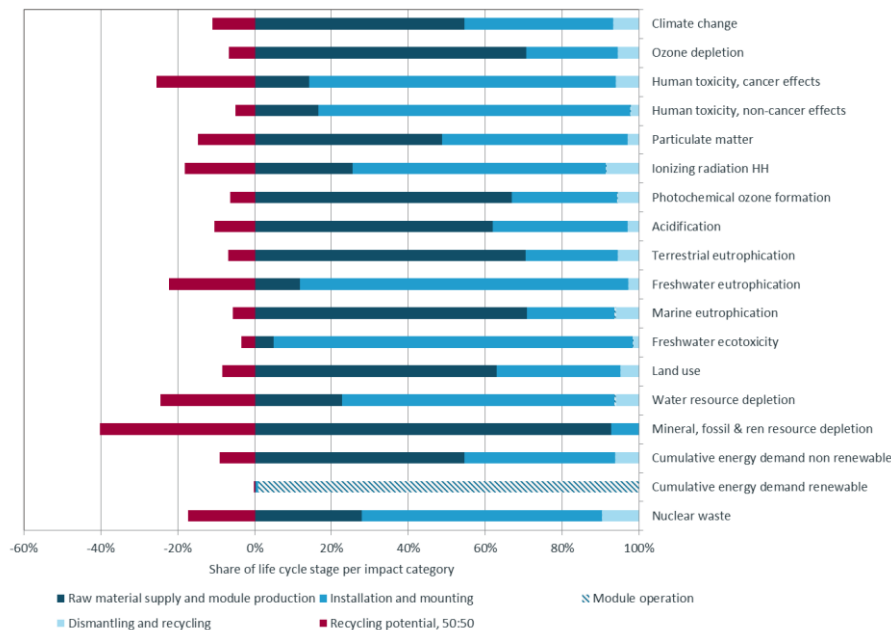


Fig. A. 1 Environmental impact results (characterized, indexed to 100 %) of 1 kWh of electricity produced with a 3 kWp power system with CdTe PV panels integrated in a slanted roof. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.

Tab. A. 32 lists the environmental impacts per impact category (characterized results) for 1 kWh of electricity produced with CdTe PV panels on an open ground system. Fig. A. 2 shows the relative impact per life cycle stage and per impact category.

Tab. A. 32 Environmental impact results (characterized) of 1 kWh of electricity produced with a 570 kWp power system with CdTe PV panels on an open ground system.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 570 kWp installation, open ground with CdTe PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change	kg CO ₂ eq	1.35E-02	1.52E-02	3.47E-06	1.63E-03	3.03E-02	-4.94E-03	2.54E-02
Ozone depletion	kg CFC-11 eq	6.73E-10	4.83E-10	2.14E-13	5.24E-11	1.21E-09	-9.79E-11	1.11E-09
Human toxicity, cancer effects	CTUh, c	8.10E-11	1.56E-09	5.25E-13	3.42E-11	1.68E-09	-2.48E-10	1.43E-09
Human toxicity, non-cancer effects	CTUh, n-c	8.36E-10	7.15E-09	8.15E-12	1.03E-10	8.09E-09	7.96E-11	8.17E-09
Particulate matter	kg PM _{2.5} eq	6.25E-06	8.62E-06	1.90E-09	3.70E-07	1.52E-05	-2.80E-06	1.24E-05
Ionizing radiation HH	kBq U235 eq	2.84E-04	1.34E-03	6.41E-07	9.43E-05	1.72E-03	-3.62E-04	1.36E-03
Photochemical ozone formation	kg NMVOC eq	6.14E-05	3.77E-05	1.25E-08	5.09E-06	1.04E-04	-9.78E-06	9.44E-05
Acidification	mol H ⁺ eq	1.14E-04	8.81E-05	2.61E-08	5.38E-06	2.07E-04	-2.49E-05	1.83E-04
Terrestrial eutrophication	mol N eq	2.41E-04	1.69E-04	6.81E-08	1.85E-05	4.28E-04	-3.72E-05	3.91E-04
Freshwater eutrophication	kg P eq	5.32E-07	1.69E-06	4.57E-09	1.22E-07	2.35E-06	-4.28E-07	1.92E-06
Marine eutrophication	kg N eq	2.14E-05	1.15E-05	1.00E-07	1.83E-06	3.48E-05	-2.99E-06	3.19E-05
Freshwater ecotoxicity	CTUe	3.80E-03	3.88E-02	2.04E-05	1.12E-03	4.38E-02	-3.28E-03	4.05E-02
Land use	kg C deficit	1.75E-02	8.91E+00	1.34E-05	1.32E-03	8.93E+00	-4.01E-03	8.92E+00
Water resource depletion	m ³ water eq	4.18E-06	2.33E-05	9.14E-08	1.09E-06	2.86E-05	-7.90E-06	2.07E-05
Mineral, fossil & ren resource depletion	kg Sb eq	6.57E-06	4.97E-07	3.40E-11	6.06E-09	7.07E-06	-2.75E-06	4.32E-06
Cumulative energy demand non renewable	MJ oil eq	1.92E-01	2.18E-01	5.31E-05	2.15E-02	4.31E-01	-6.28E-02	3.69E-01
Cumulative energy demand renewable	MJ oil eq	8.05E-03	4.02E-02	3.60E+00	1.04E-03	3.65E+00	-1.30E-02	3.64E+00
Nuclear waste	m ³ HAA eq	7.25E-12	2.94E-11	1.19E-14	2.49E-12	3.92E-11	-8.07E-12	3.11E-11

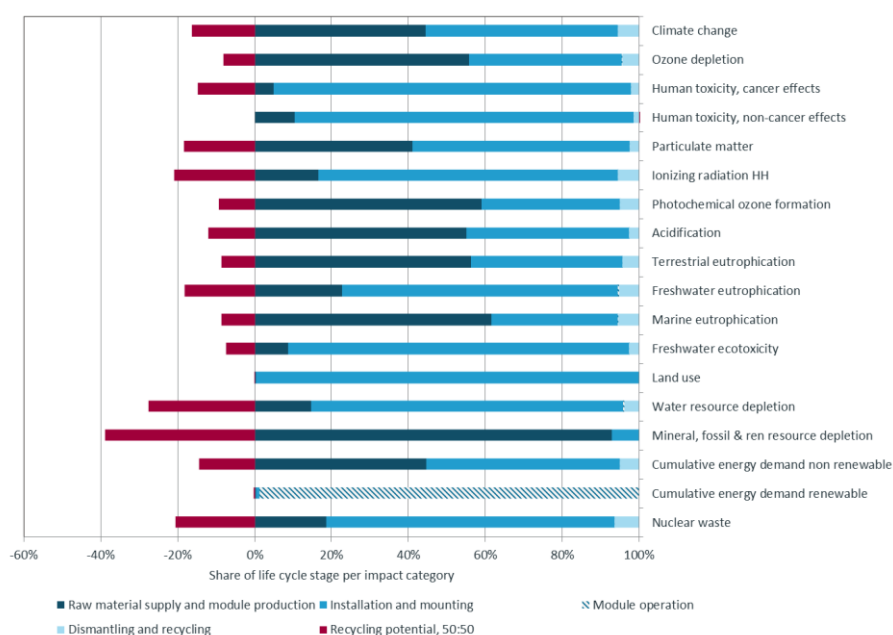


Fig. A. 2 Environmental impact results (characterized, indexed to 100 %) of 1 kWh of electricity produced with a 570 kWp power system with CdTe PV panels on an open ground system. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.

A.1.3. Results normalized

The normalized environmental impact results of the electricity production with CdTe PV panels integrated in a slanted roof and with an open ground system are listed in Tab. A. 33 and Tab. A. 34, respectively. The three additional impact categories cumulative energy demand non-renewable, cumulative energy demand renewable and nuclear waste are not subject to normalization.

The results show that the production of 1 kWh electricity with the mounted CdTe PV system contributes the largest share to the mineral, fossil and renewable resource depletion potential of an average European citizen. An average European person consumes around 5'000 kWh of electricity.

Assuming it would all be produced with such a 3 kWp CdTe PV system, this would sum up to 21 % of the present budget of resource depletion. The most important impact category for the 570 kWp CdTe open ground system is land use. If an average European person consumed electricity exclusively from this PV system, 60 % of the present land use budget per person would be spent on electricity production.

Tab. A. 33 Environmental impact results (normalized) of 1 kWh of electricity produced with a 3 kWp power system with CdTe PV panels integrated in a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, integrated with CdTe PV technology, normalized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change		1.46E-06	1.04E-06	3.74E-10	1.77E-07	2.68E-06	-2.96E-07	2.38E-06
Ozone depletion		3.12E-08	1.06E-08	9.85E-12	2.42E-09	4.42E-08	-2.98E-09	4.12E-08
Human toxicity, cancer effects		2.20E-06	1.24E-05	1.42E-08	9.26E-07	1.56E-05	-3.99E-06	1.16E-05
Human toxicity, non-cancer effects		1.57E-06	7.71E-06	1.53E-08	1.93E-07	9.49E-06	-4.75E-07	9.02E-06
Particulate matter		1.64E-06	1.64E-06	4.97E-10	9.74E-08	3.38E-06	-5.01E-07	2.88E-06
Ionizing radiation HH		2.51E-07	6.56E-07	5.66E-10	8.34E-08	9.91E-07	-1.80E-07	8.10E-07
Photochemical ozone formation		1.94E-06	7.98E-07	3.93E-10	1.61E-07	2.90E-06	-1.86E-07	2.71E-06
Acidification		2.41E-06	1.37E-06	5.48E-10	1.14E-07	3.89E-06	-4.08E-07	3.48E-06
Terrestrial eutrophication		1.37E-06	4.72E-07	3.85E-10	1.05E-07	1.95E-06	-1.35E-07	1.81E-06
Freshwater eutrophication		3.59E-07	2.62E-06	3.08E-09	8.24E-08	3.07E-06	-6.88E-07	2.38E-06
Marine eutrophication		1.27E-06	4.10E-07	5.94E-09	1.08E-07	1.79E-06	-1.03E-07	1.69E-06
Freshwater ecotoxicity		4.35E-07	8.47E-06	2.32E-09	1.28E-07	9.04E-06	-3.14E-07	8.72E-06
Land use		2.34E-07	1.20E-07	1.90E-10	1.76E-08	3.72E-07	-3.16E-08	3.41E-07
Water resource depletion		5.14E-08	1.61E-07	1.12E-09	1.33E-08	2.26E-07	-5.57E-08	1.71E-07
Mineral, fossil & ren resource depletion		6.50E-05	5.06E-06	2.80E-10	6.00E-08	7.01E-05	-2.83E-05	4.19E-05

Tab. A. 34 Environmental impact results (normalized) of 1 kWh of electricity produced with a 570 kWp power system with CdTe PV panels on an open ground system.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 570 kWp installation, open ground with CdTe PV technology, normalized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change		1.46E-06	1.65E-06	3.76E-10	1.77E-07	3.29E-06	-5.36E-07	2.75E-06
Ozone depletion		3.12E-08	2.23E-08	9.89E-12	2.42E-09	5.59E-08	-4.53E-09	5.14E-08
Human toxicity, cancer effects		2.20E-06	4.24E-05	1.42E-08	9.26E-07	4.55E-05	-6.73E-06	3.88E-05
Human toxicity, non-cancer effects		1.57E-06	1.34E-05	1.53E-08	1.93E-07	1.52E-05	1.49E-07	1.53E-05
Particulate matter		1.64E-06	2.27E-06	5.00E-10	9.74E-08	4.01E-06	-7.38E-07	3.27E-06
Ionizing radiation HH		2.51E-07	1.19E-06	5.67E-10	8.34E-08	1.52E-06	-3.20E-07	1.20E-06
Photochemical ozone formation		1.94E-06	1.19E-06	3.95E-10	1.61E-07	3.29E-06	-3.09E-07	2.98E-06
Acidification		2.41E-06	1.86E-06	5.51E-10	1.14E-07	4.39E-06	-5.27E-07	3.86E-06
Terrestrial eutrophication		1.37E-06	9.60E-07	3.87E-10	1.05E-07	2.43E-06	-2.11E-07	2.22E-06
Freshwater eutrophication		3.59E-07	1.14E-06	3.09E-09	8.24E-08	1.59E-06	-2.89E-07	1.30E-06
Marine eutrophication		1.27E-06	6.80E-07	5.94E-09	1.08E-07	2.06E-06	-1.77E-07	1.88E-06
Freshwater ecotoxicity		4.35E-07	4.44E-06	2.33E-09	1.28E-07	5.01E-06	-3.76E-07	4.63E-06
Land use		2.34E-07	1.19E-04	1.68E-10	1.76E-08	1.19E-04	-5.35E-08	1.19E-04
Water resource depletion		5.14E-08	2.86E-07	1.12E-09	1.33E-08	3.52E-07	-9.70E-08	2.55E-07
Mineral, fossil & ren resource depletion		6.50E-05	4.92E-06	3.43E-10	6.00E-08	7.00E-05	-2.72E-05	4.27E-05

A.1.4. Results weighted

Tab. A. 35 shows the weighted environmental impact results (equal weighting) of the electricity production with CdTe PV panels in an integrated roof system whereas Tab. A. 36 shows the results for electricity production on a 570 kWp open ground system. The blue bars within column “Total” illustrate the importance of each impact category in the total of impacts. The three additional impact

categories cumulative energy demand non-renewable, cumulative energy demand renewable and nuclear waste are not included in the weighted results.

The environmental impacts of the production of 1 kWh electricity with a 3 kWp power system with CdTe PV panels integrated in a slanted roof, are mainly driven by mineral, fossil and renewable resource depletion (47 %), human toxicity cancer effects (13 %). The impact categories human toxicity non cancer effects and freshwater ecotoxicity contribute 10 % each to the total weighted environmental impacts (Fig. A. 3).

Tab. A. 35 Environmental impact results (weighted) of 1 kWh of electricity produced with a 3 kWp power system with CdTe PV panels integrated in a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, integrated with CdTe PV technology, weighted		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Total	Pt	5.35E-06	2.87E-06	3.01E-09	1.51E-07	8.37E-06	-2.38E-06	5.99E-06
Climate change	Pt	9.73E-08	6.94E-08	2.49E-11	1.18E-08	1.79E-07	-1.97E-08	1.59E-07
Ozone depletion	Pt	2.08E-09	7.07E-10	6.57E-13	1.62E-10	2.95E-09	-1.99E-10	2.75E-09
Human toxicity, cancer effects	Pt	1.46E-07	8.29E-07	9.48E-10	6.17E-08	1.04E-06	-2.66E-07	7.72E-07
Human toxicity, non-cancer effects	Pt	1.05E-07	5.14E-07	1.02E-09	1.29E-08	6.33E-07	-3.16E-08	6.01E-07
Particulate matter	Pt	1.10E-07	1.09E-07	3.32E-11	6.50E-09	2.25E-07	-3.34E-08	1.92E-07
Ionizing radiation HH	Pt	1.67E-08	4.37E-08	3.78E-11	5.56E-09	6.60E-08	-1.20E-08	5.40E-08
Photochemical ozone formation	Pt	1.29E-07	5.32E-08	2.62E-11	1.07E-08	1.93E-07	-1.24E-08	1.81E-07
Acidification	Pt	1.61E-07	9.12E-08	3.65E-11	7.58E-09	2.59E-07	-2.72E-08	2.32E-07
Terrestrial eutrophication	Pt	9.12E-08	3.15E-08	2.57E-11	6.99E-09	1.30E-07	-9.03E-09	1.21E-07
Freshwater eutrophication	Pt	2.39E-08	1.75E-07	2.06E-10	5.49E-09	2.05E-07	-4.59E-08	1.59E-07
Marine eutrophication	Pt	8.45E-08	2.73E-08	3.96E-10	7.21E-09	1.19E-07	-6.85E-09	1.13E-07
Freshwater ecotoxicity	Pt	2.90E-08	5.65E-07	1.55E-10	8.53E-09	6.02E-07	-2.10E-08	5.81E-07
Land use	Pt	1.56E-08	8.03E-09	1.26E-11	1.17E-09	2.48E-08	-2.11E-09	2.27E-08
Water resource depletion	Pt	3.43E-09	1.07E-08	7.49E-11	8.89E-10	1.51E-08	-3.71E-09	1.14E-08
Mineral, fossil & ren resource depletion	Pt	4.33E-06	3.37E-07	1.86E-11	4.00E-09	4.67E-06	-1.88E-06	2.79E-06

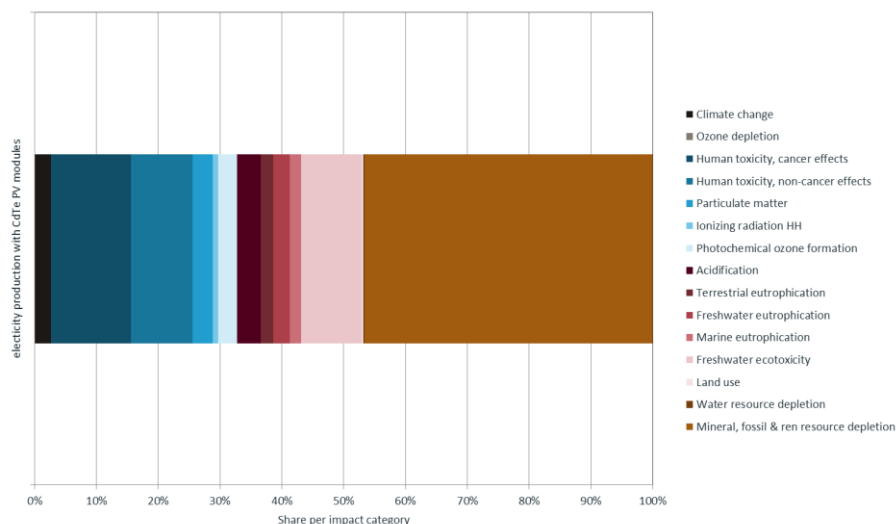


Fig. A. 3 Share per impact category on the weighted result for the production of 1 kWh of electricity produced with a 3 kWp power system with CdTe PV panels integrated in a slanted roof.

The electricity production with CdTe PV panels on an open ground system has its main impact in the categories land use (50 %), followed by mineral, fossil and renewable resource depletion potential (18 %), human toxicity cancer effects (16 %) and human toxicity non cancer effects (6 %). Fig. A. 4 illustrates this.

Tab. A. 36 Environmental impact results (weighted) of 1 kWh of electricity produced with a 570 kWp power system with CdTe PV panels on an open ground system.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 570 kWp installation, open ground with CdTe PV technology, weighted		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Total	Pt	5.35E-06	1.30E-05	3.02E-09	1.51E-07	1.85E-05	-2.50E-06	1.60E-05
Climate change	Pt	9.73E-08	1.10E-07	2.51E-11	1.18E-08	2.19E-07	-3.57E-08	1.84E-07
Ozone depletion	Pt	2.08E-09	1.49E-09	6.59E-13	1.62E-10	3.73E-09	-3.02E-10	3.43E-09
Human toxicity, cancer effects	Pt	1.46E-07	2.83E-06	9.49E-10	6.17E-08	3.03E-06	-4.49E-07	2.59E-06
Human toxicity, non-cancer effects	Pt	1.05E-07	8.94E-07	1.02E-09	1.29E-08	1.01E-06	9.96E-09	1.02E-06
Particulate matter	Pt	1.10E-07	1.51E-07	3.33E-11	6.50E-09	2.67E-07	-4.92E-08	2.18E-07
Ionizing radiation HH	Pt	1.67E-08	7.93E-08	3.78E-11	5.56E-09	1.02E-07	-2.14E-08	8.03E-08
Photochemical ozone formation	Pt	1.29E-07	7.93E-08	2.64E-11	1.07E-08	2.19E-07	-2.06E-08	1.99E-07
Acidification	Pt	1.61E-07	1.24E-07	3.67E-11	7.58E-09	2.92E-07	-3.51E-08	2.57E-07
Terrestrial eutrophication	Pt	9.12E-08	6.40E-08	2.58E-11	6.99E-09	1.62E-07	-1.41E-08	1.48E-07
Freshwater eutrophication	Pt	2.39E-08	7.61E-08	2.06E-10	5.49E-09	1.06E-07	-1.93E-08	8.64E-08
Marine eutrophication	Pt	8.45E-08	4.53E-08	3.96E-10	7.21E-09	1.37E-07	-1.18E-08	1.26E-07
Freshwater ecotoxicity	Pt	2.90E-08	2.96E-07	1.55E-10	8.53E-09	3.34E-07	-2.51E-08	3.09E-07
Land use	Pt	1.56E-08	7.94E-06	1.17E-11	1.17E-09	7.96E-06	-3.57E-09	7.95E-06
Water resource depletion	Pt	3.43E-09	1.91E-08	7.49E-11	8.89E-10	2.34E-08	-6.47E-09	1.70E-08
Mineral, fossil & ren resource depletion	Pt	4.33E-06	3.28E-07	2.26E-11	4.00E-09	4.67E-06	-1.82E-06	2.85E-06

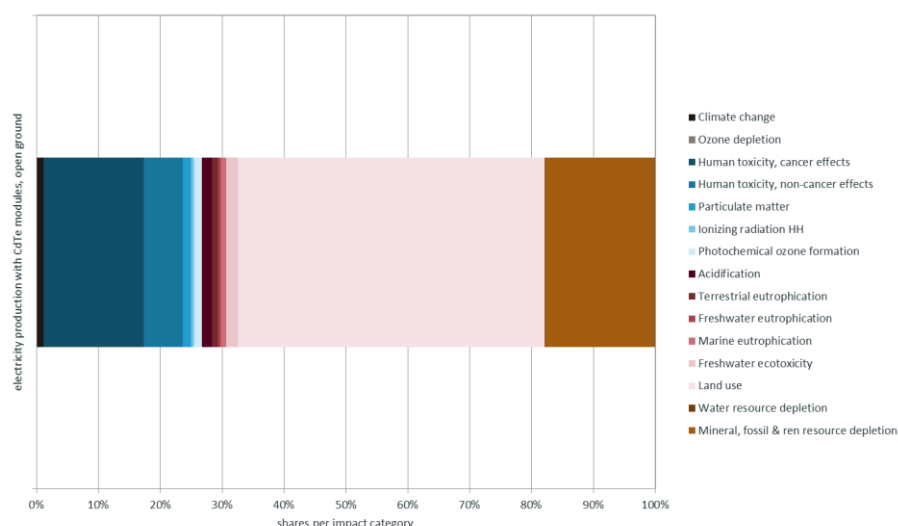


Fig. A. 4 Share per impact category on the weighted result for the production of 1 kWh of electricity produced with a 570 kWp power system with CdTe PV panels on an open ground system.

A.2. Results CIS PV technology

A.2.1. Overview

In section A.2.2 the results prior to normalization are assessed. In the sections A.2.3 and A.2.4 the normalized and the weighted results of electricity production with CIS PV panels, integrated in a slanted roof, residential scale and of a large scale open ground system are described.

A.2.2. Results prior to normalization

The results prior to normalization of the production of 1 kWh of electricity with CIS PV panels integrated in a slanted roof in a 3 kWp installation system and on a 570 kWp open ground system are summarized in Tab. A. 37 and Tab. A. 38. The life cycle stages analyzed are the product stage, the construction stage, the operation stage and the end-of-life stage. Net benefits caused by end-of-life

recycling are allocated to the production of electricity with a share of 50 %. The other half is allocated to the life cycle of products potentially using the secondary products (see also subchapter 3.5).

Tab. A. 37 Environmental impact results (characterized) of 1 kWh of electricity produced with a 3 kWp power system with CIS PV panels integrated in a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, integrated with CIS PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change	kg CO ₂ eq	2.63E-02	1.20E-02	4.46E-06	2.55E-03	4.09E-02	-4.15E-03	3.68E-02
Ozone depletion	kg CFC-11 eq	1.36E-09	2.92E-10	2.75E-13	8.65E-11	1.74E-09	-9.35E-11	1.64E-09
Human toxicity, cancer effects	CTUh, c	3.14E-10	5.81E-10	6.80E-13	5.97E-11	9.55E-10	-2.50E-10	7.05E-10
Human toxicity, non-cancer effects	CTUh, n-c	4.07E-09	4.48E-09	1.06E-11	1.79E-10	8.74E-09	-7.35E-11	8.67E-09
Particulate matter	kg PM _{2.5} eq	9.44E-06	7.42E-06	2.45E-09	7.74E-07	1.76E-05	-2.68E-06	1.50E-05
Ionizing radiation HH	kBq U235 eq	2.10E-03	9.48E-04	8.28E-07	2.83E-04	3.33E-03	-3.47E-04	2.99E-03
Photochemical ozone formation	kg NMVOC eq	7.90E-05	3.04E-05	1.62E-08	9.26E-06	1.19E-04	-8.57E-06	1.10E-04
Acidification	mol H ⁺ eq	1.40E-04	7.50E-05	3.36E-08	1.14E-05	2.26E-04	-2.58E-05	2.01E-04
Terrestrial eutrophication	mol N eq	2.99E-04	9.96E-05	8.79E-08	3.34E-05	4.32E-04	-3.27E-05	4.00E-04
Freshwater eutrophication	kg P eq	2.46E-06	3.97E-06	5.92E-09	1.39E-07	6.58E-06	-1.02E-06	5.56E-06
Marine eutrophication	kg N eq	2.55E-05	8.40E-06	1.30E-07	3.30E-06	3.73E-05	-2.53E-06	3.48E-05
Freshwater ecotoxicity	CTUe	1.19E-02	7.63E-02	2.63E-05	1.98E-03	9.02E-02	-3.77E-03	8.64E-02
Land use	kg C deficit	4.21E-02	1.12E-02	1.84E-05	2.34E-03	5.56E-02	-3.28E-03	5.23E-02
Water resource depletion	m ³ water eq	2.14E-05	1.68E-05	1.19E-07	1.46E-06	3.98E-05	-7.75E-06	3.20E-05
Mineral, fossil & ren resource depletion	kg Sb eq	5.12E-04	5.30E-07	-4.70E-10	1.04E-08	5.12E-04	-2.36E-08	5.12E-04
Cumulative energy demand non renewable	MJ oil eq	4.05E-01	1.74E-01	6.84E-05	3.62E-02	6.16E-01	-5.08E-02	5.65E-01
Cumulative energy demand renewable	MJ oil eq	4.15E-02	2.97E-02	3.60E+00	2.36E-03	3.67E+00	-1.29E-02	3.66E+00
Nuclear waste	m ³ HAA eq	5.18E-11	2.08E-11	1.54E-14	5.91E-12	7.86E-11	-7.75E-12	7.08E-11

The following Fig. A. 5 shows the share per life cycle stage per impact category of electricity production with CIS PV panels integrated in a slanted roof. The potential benefits for recycling are in relation to the sum of the life cycle stages production to end-of-life.

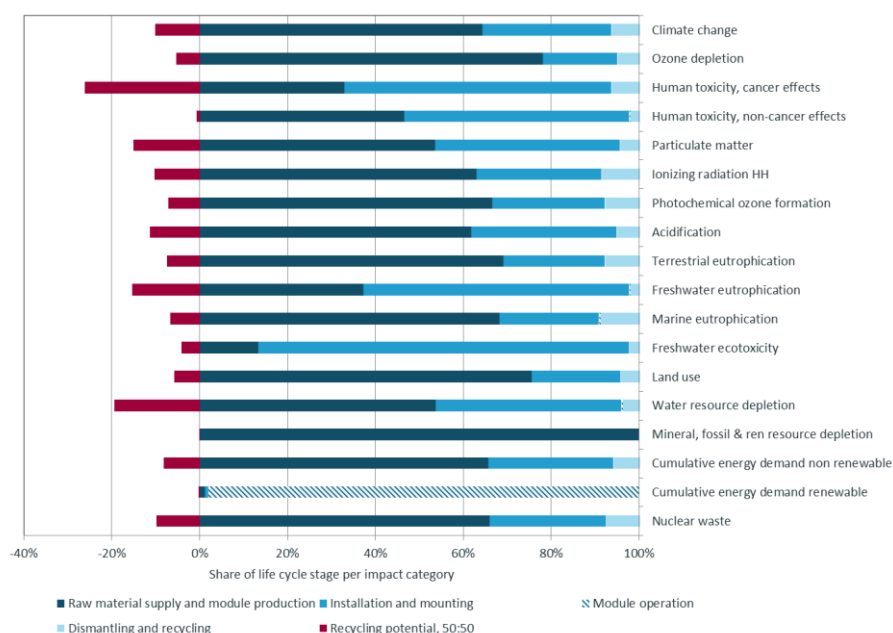


Fig. A. 5 Environmental impact results (characterized, indexed to 100 %) of 1 kWh of electricity produced with a 3 kWp power system with CIS PV panels integrated in a slanted roof. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.

Tab. A. 38 lists the environmental impacts per impact category (characterized results) for 1 kWh of electricity produced with CIS PV panels on an open ground system. Fig. A. 6 shows the relative impact per life cycle stage and per impact category.

Tab. A. 38 Environmental impact results (characterized) of 1 kWh of electricity produced with a 570 kWp power system with CIS PV panels on an open ground system.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 570 kWp installation, open ground with CIS PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change	kg CO ₂ eq	2.63E-02	1.97E-02	4.50E-06	2.55E-03	4.85E-02	-7.07E-03	4.15E-02
Ozone depletion	kg CFC-11 eq	1.36E-09	6.24E-10	2.77E-13	8.65E-11	2.07E-09	-1.38E-10	1.93E-09
Human toxicity, cancer effects	CTUh, c	3.14E-10	2.02E-09	6.81E-13	5.97E-11	2.40E-09	-3.85E-10	2.01E-09
Human toxicity, non-cancer effects	CTUh, n-c	4.07E-09	9.04E-09	1.06E-11	1.79E-10	1.33E-08	2.87E-10	1.36E-08
Particulate matter	kg PM _{2.5} eq	9.44E-06	1.10E-05	2.46E-09	7.74E-07	2.12E-05	-3.96E-06	1.73E-05
Ionizing radiation HH	kBq U235 eq	2.10E-03	1.74E-03	8.31E-07	2.83E-04	4.12E-03	-5.56E-04	3.57E-03
Photochemical ozone formation	kg NMVOC eq	7.90E-05	4.81E-05	1.62E-08	9.26E-06	1.36E-04	-1.40E-05	1.22E-04
Acidification	mol H ⁺ eq	1.40E-04	1.12E-04	3.38E-08	1.14E-05	2.63E-04	-3.47E-05	2.28E-04
Terrestrial eutrophication	mol N eq	2.99E-04	2.16E-04	8.82E-08	3.34E-05	5.49E-04	-5.15E-05	4.98E-04
Freshwater eutrophication	kg P eq	2.46E-06	1.91E-06	5.92E-09	1.39E-07	4.51E-06	-4.41E-07	4.07E-06
Marine eutrophication	kg N eq	2.55E-05	1.47E-05	1.30E-07	3.30E-06	4.36E-05	-4.26E-06	3.94E-05
Freshwater ecotoxicity	CTUe	1.19E-02	4.56E-02	2.64E-05	1.98E-03	5.96E-02	-4.68E-03	5.49E-02
Land use	kg C deficit	4.21E-02	1.16E+01	1.65E-05	2.34E-03	1.16E+01	-5.52E-03	1.16E+01
Water resource depletion	m ³ water eq	2.14E-05	3.01E-05	1.19E-07	1.46E-06	5.31E-05	-1.22E-05	4.09E-05
Mineral, fossil & ren resource depletion	kg Sb eq	5.12E-04	6.08E-07	-1.30E-11	1.04E-08	5.12E-04	9.00E-08	5.12E-04
Cumulative energy demand non renewable	MJ oil eq	4.05E-01	2.80E-01	6.89E-05	3.62E-02	7.22E-01	-9.13E-02	6.30E-01
Cumulative energy demand renewable	MJ oil eq	4.15E-02	5.21E-02	3.60E+00	2.36E-03	3.70E+00	-2.01E-02	3.68E+00
Nuclear waste	m ³ HAA eq	5.18E-11	3.81E-11	1.54E-14	5.91E-12	9.58E-11	-1.24E-11	8.34E-11

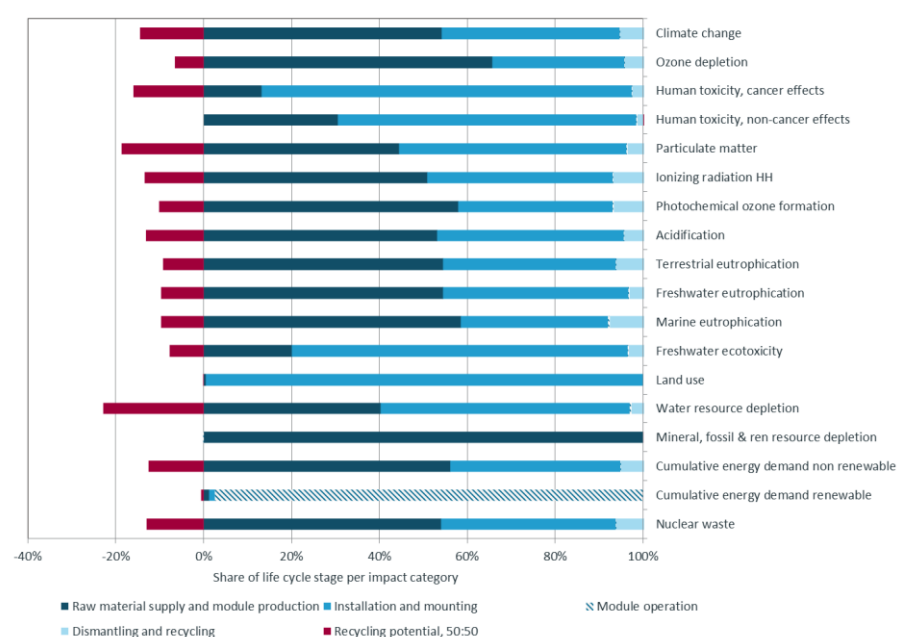


Fig. A. 6 Environmental impact results (characterized, indexed to 100 %) of 1 kWh of electricity produced with a 570 kWp power system with CIS PV panels on an open ground system. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.

A.2.3. Results normalized

The normalized environmental impact results of the electricity production with CIS PV panels integrated in a slanted roof and with an open ground system are listed in Tab. A. 39 and Tab. A. 40, respectively. The three additional impact categories cumulative energy demand non-renewable, cumulative energy demand renewable and nuclear waste are not subject to normalization.

The results show that the production of 1 kWh electricity contributes the largest share to the mineral, fossil and renewable resource depletion potential of an average European citizen. An average European person consumes around 5'000 kWh of electricity. Assuming it would all be produced with such a 3 kWp CIS PV system or a 570 kWp open ground system, this would sum up to 2'540 % of the present budget of resource depletion. In other words the present budget of resource depletion would be surmounted by a factor of 25. This large contribution to the mineral resource depletion is mainly caused by the indium consumption in the panel production.

Tab. A. 39 Environmental impact results (normalized) of 1 kWh of electricity produced with a 3 kWp power system with CIS PV panels integrated in a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, integrated with CIS PV technology, normalized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change		2.86E-06	1.30E-06	4.84E-10	2.76E-07	4.44E-06	-4.50E-07	3.99E-06
Ozone depletion		6.29E-08	1.35E-08	1.28E-11	4.00E-09	8.05E-08	-4.33E-09	7.61E-08
Human toxicity, cancer effects		8.52E-06	1.57E-05	1.84E-08	1.62E-06	2.59E-05	-6.77E-06	1.91E-05
Human toxicity, non-cancer effects		7.63E-06	8.41E-06	1.98E-08	3.36E-07	1.64E-05	-1.38E-07	1.63E-05
Particulate matter		2.49E-06	1.95E-06	6.45E-10	2.04E-07	4.64E-06	-7.05E-07	3.94E-06
Ionizing radiation HH		1.86E-06	8.39E-07	7.33E-10	2.51E-07	2.95E-06	-3.07E-07	2.64E-06
Photochemical ozone formation		2.49E-06	9.60E-07	5.10E-10	2.92E-07	3.75E-06	-2.70E-07	3.48E-06
Acidification		2.96E-06	1.59E-06	7.11E-10	2.40E-07	4.79E-06	-5.46E-07	4.24E-06
Terrestrial eutrophication		1.70E-06	5.66E-07	5.00E-10	1.90E-07	2.46E-06	-1.86E-07	2.27E-06
Freshwater eutrophication		1.66E-06	2.69E-06	4.00E-09	9.42E-08	4.44E-06	-6.89E-07	3.75E-06
Marine eutrophication		1.51E-06	4.97E-07	7.69E-09	1.95E-07	2.21E-06	-1.50E-07	2.06E-06
Freshwater ecotoxicity		1.37E-06	8.73E-06	3.01E-09	2.27E-07	1.03E-05	-4.31E-07	9.89E-06
Land use		5.62E-07	1.49E-07	2.46E-10	3.13E-08	7.43E-07	-4.39E-08	6.99E-07
Water resource depletion		2.63E-07	2.06E-07	1.46E-09	1.80E-08	4.89E-07	-9.52E-08	3.93E-07
Mineral, fossil & ren resource depletion		5.06E-03	5.24E-06	-4.55E-09	1.03E-07	5.07E-03	-2.34E-07	5.07E-03

Tab. A. 40 Environmental impact results (normalized) of 1 kWh of electricity produced with a 570 kWp power system with CIS PV panels on an open ground system.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 570 kWp installation, open ground with CIS PV technology, normalized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change		2.86E-06	2.13E-06	4.88E-10	2.76E-07	5.27E-06	-7.67E-07	4.50E-06
Ozone depletion		6.29E-08	2.89E-08	1.28E-11	4.00E-09	9.59E-08	-6.40E-09	8.95E-08
Human toxicity, cancer effects		8.52E-06	5.48E-05	1.84E-08	1.62E-06	6.50E-05	-1.04E-05	5.45E-05
Human toxicity, non-cancer effects		7.63E-06	1.70E-05	1.98E-08	3.36E-07	2.49E-05	5.38E-07	2.55E-05
Particulate matter		2.49E-06	2.89E-06	6.48E-10	2.04E-07	5.58E-06	-1.04E-06	4.54E-06
Ionizing radiation HH		1.86E-06	1.54E-06	7.35E-10	2.51E-07	3.65E-06	-4.92E-07	3.16E-06
Photochemical ozone formation		2.49E-06	1.52E-06	5.12E-10	2.92E-07	4.30E-06	-4.40E-07	3.86E-06
Acidification		2.96E-06	2.36E-06	7.15E-10	2.40E-07	5.56E-06	-7.33E-07	4.83E-06
Terrestrial eutrophication		1.70E-06	1.23E-06	5.01E-10	1.90E-07	3.12E-06	-2.92E-07	2.83E-06
Freshwater eutrophication		1.66E-06	1.29E-06	4.00E-09	9.42E-08	3.05E-06	-2.98E-07	2.75E-06
Marine eutrophication		1.51E-06	8.70E-07	7.70E-09	1.95E-07	2.58E-06	-2.52E-07	2.33E-06
Freshwater ecotoxicity		1.37E-06	5.22E-06	3.02E-09	2.27E-07	6.81E-06	-5.36E-07	6.28E-06
Land use		5.62E-07	1.54E-04	2.23E-10	3.13E-08	1.55E-04	-7.38E-08	1.55E-04
Water resource depletion		2.63E-07	3.70E-07	1.46E-09	1.80E-08	6.52E-07	-1.50E-07	5.02E-07
Mineral, fossil & ren resource depletion		5.06E-03	6.02E-06	-1.97E-10	1.03E-07	5.07E-03	8.91E-07	5.07E-03

A.2.4. Results weighted

Tab. A. 41 shows the weighted environmental impact results (equal weighting) of the electricity production with CIS PV panels in an integrated roof system whereas Tab. A. 42 shows the results for electricity production on a 570 kWp open ground system. The blue bars within column "Total"

illustrate the importance of each impact category in the total of impacts. The three additional impact categories cumulative energy demand non-renewable, cumulative energy demand renewable and nuclear waste are not included in the weighted results.

The environmental impacts of the production of 1 kWh electricity with a 3 kWp power system with CIS PV panels integrated in a slanted roof, are mainly driven by mineral, fossil and renewable resource depletion (99 %), see also Fig. A. 7.

Tab. A. 41 Environmental impact results (weighted) of 1 kWh of electricity produced with a 3 kWp power system with CIS PV panels integrated in a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, integrated with CIS PV technology, weighted		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Total	Pt	3.40E-04	3.26E-06	3.57E-09	2.72E-07	3.44E-04	-7.35E-07	3.43E-04
Climate change	Pt	1.90E-07	8.70E-08	3.23E-11	1.84E-08	2.96E-07	-3.00E-08	2.66E-07
Ozone depletion	Pt	4.20E-09	9.01E-10	8.50E-13	2.67E-10	5.36E-09	-2.89E-10	5.08E-09
Human toxicity, cancer effects	Pt	5.68E-07	1.05E-06	1.23E-09	1.08E-07	1.73E-06	-4.52E-07	1.27E-06
Human toxicity, non-cancer effects	Pt	5.09E-07	5.61E-07	1.32E-09	2.24E-08	1.09E-06	-9.20E-09	1.08E-06
Particulate matter	Pt	1.66E-07	1.30E-07	4.30E-11	1.36E-08	3.10E-07	-4.70E-08	2.63E-07
Ionizing radiation HH	Pt	1.24E-07	5.60E-08	4.88E-11	1.67E-08	1.97E-07	-2.05E-08	1.76E-07
Photochemical ozone formation	Pt	1.66E-07	6.40E-08	3.40E-11	1.95E-08	2.50E-07	-1.80E-08	2.32E-07
Acidification	Pt	1.97E-07	1.06E-07	4.74E-11	1.60E-08	3.19E-07	-3.64E-08	2.83E-07
Terrestrial eutrophication	Pt	1.13E-07	3.77E-08	3.33E-11	1.26E-08	1.64E-07	-1.24E-08	1.51E-07
Freshwater eutrophication	Pt	1.11E-07	1.79E-07	2.67E-10	6.28E-09	2.96E-07	-4.59E-08	2.50E-07
Marine eutrophication	Pt	1.01E-07	3.31E-08	5.13E-10	1.30E-08	1.47E-07	-1.00E-08	1.37E-07
Freshwater ecotoxicity	Pt	9.11E-08	5.82E-07	2.01E-10	1.51E-08	6.88E-07	-2.87E-08	6.59E-07
Land use	Pt	3.75E-08	9.95E-09	1.64E-11	2.09E-09	4.95E-08	-2.93E-09	4.66E-08
Water resource depletion	Pt	1.75E-08	1.38E-08	9.71E-11	1.20E-09	3.26E-08	-6.34E-09	2.62E-08
Mineral, fossil & ren resource depletion	Pt	3.38E-04	3.50E-07	-3.14E-10	6.88E-09	3.38E-04	-1.56E-08	3.38E-04

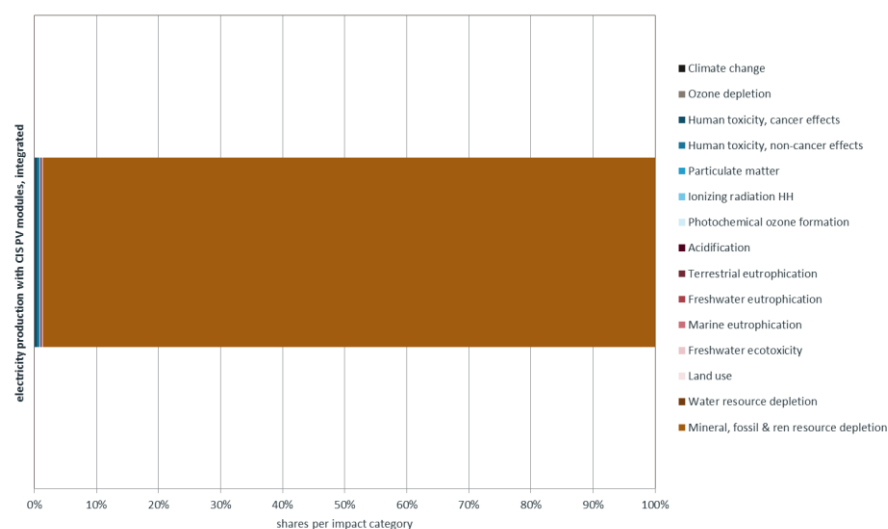


Fig. A. 7 Share per impact category on the weighted result for the production of 1 kWh of electricity produced with a 3 kWp power system with CIS PV panels integrated in a slanted roof.

The electricity production with CIS PV panels on an open ground system has its main impact in the mineral, fossil and renewable resource depletion potential (95 %), followed by land use (3 %) and human toxicity cancer effects (1 %). Fig. A. 8 illustrates this.

Tab. A. 42 Environmental impact results (weighted) of 1 kWh of electricity produced with a 570 kWp power system with CIS PV panels on an open ground system.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 570 kWp installation, open ground with CIS PV technology, weighted		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Total	Pt	3.40E-04	1.68E-05	3.88E-09	2.72E-07	3.57E-04	-9.40E-07	3.56E-04
Climate change	Pt	1.90E-07	1.42E-07	3.25E-11	1.84E-08	3.51E-07	-5.12E-08	3.00E-07
Ozone depletion	Pt	4.20E-09	1.93E-09	8.55E-13	2.67E-10	6.39E-09	-4.27E-10	5.96E-09
Human toxicity, cancer effects	Pt	5.68E-07	3.66E-06	1.23E-09	1.08E-07	4.33E-06	-6.96E-07	3.64E-06
Human toxicity, non-cancer effects	Pt	5.09E-07	1.13E-06	1.32E-09	2.24E-08	1.66E-06	3.59E-08	1.70E-06
Particulate matter	Pt	1.66E-07	1.93E-07	4.32E-11	1.36E-08	3.72E-07	-6.95E-08	3.03E-07
Ionizing radiation HH	Pt	1.24E-07	1.03E-07	4.90E-11	1.67E-08	2.43E-07	-3.28E-08	2.10E-07
Photochemical ozone formation	Pt	1.66E-07	1.01E-07	3.42E-11	1.95E-08	2.87E-07	-2.94E-08	2.58E-07
Acidification	Pt	1.97E-07	1.58E-07	4.76E-11	1.60E-08	3.71E-07	-4.89E-08	3.22E-07
Terrestrial eutrophication	Pt	1.13E-07	8.20E-08	3.34E-11	1.26E-08	2.08E-07	-1.95E-08	1.89E-07
Freshwater eutrophication	Pt	1.11E-07	8.59E-08	2.67E-10	6.28E-09	2.03E-07	-1.99E-08	1.83E-07
Marine eutrophication	Pt	1.01E-07	5.80E-08	5.13E-10	1.30E-08	1.72E-07	-1.68E-08	1.55E-07
Freshwater ecotoxicity	Pt	9.11E-08	3.48E-07	2.01E-10	1.51E-08	4.54E-07	-3.57E-08	4.19E-07
Land use	Pt	3.75E-08	1.03E-05	1.49E-11	2.09E-09	1.03E-05	-4.92E-09	1.03E-05
Water resource depletion	Pt	1.75E-08	2.47E-08	9.71E-11	1.20E-09	4.35E-08	-9.99E-09	3.35E-08
Mineral, fossil & ren resource depletion	Pt	3.38E-04	4.01E-07	-7.79E-12	6.88E-09	3.38E-04	5.94E-08	3.38E-04

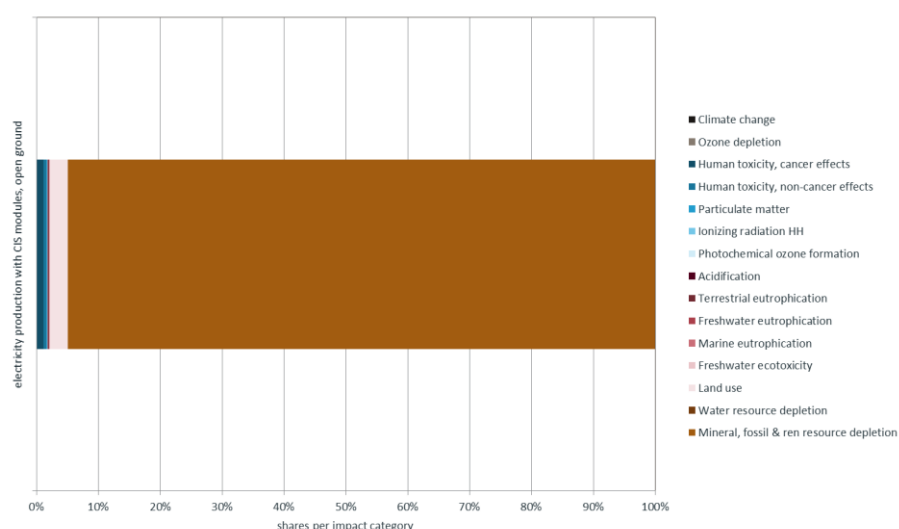


Fig. A. 8 Share per impact category on the weighted result for the production of 1 kWh of electricity produced with a 570 kWp power system with CIS PV panels on an open ground system.

A.3. Results micromorphous-Si PV technology

A.3.1. Overview

In section A.3.2 the results prior to normalization are assessed. In the sections A.3.3 and A.3.4 the normalized and the weighted results of electricity production with micromorphous-Si PV panels, integrated in a slanted roof, residential scale and of a large scale open ground system are described.

A.3.2. Results prior to normalization

The results prior to normalization of the production of 1 kWh of electricity with micromorphous-Si PV panels integrated in a slanted roof in a 3 kWp installation system and on a 570 kWp open ground system are summarized in Tab. A. 43 and Tab. A. 44. The life cycle stages analyzed are the product stage, the construction stage, the operation stage and the end-of-life stage. Net benefits caused by end-of-life recycling are allocated to the production of electricity with a share of 50 %. The other half is

allocated to the life cycle of products potentially using the secondary products (see also subchapter 3.5).

Tab. A. 43 Environmental impact results (characterized) of 1 kWh of electricity produced with a 3 kWp power system with micro-Si PV panels integrated in a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, integrated with micro-Si PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change	kg CO ₂ eq	4.09E-02	1.29E-02	4.81E-06	2.56E-03	5.64E-02	-4.70E-03	5.17E-02
Ozone depletion	kg CFC-11 eq	7.50E-10	3.14E-10	2.98E-13	8.70E-11	1.15E-09	-1.06E-10	1.05E-09
Human toxicity, cancer effects	CTUh, c	4.00E-10	6.23E-10	7.35E-13	6.01E-11	1.08E-09	-2.83E-10	8.01E-10
Human toxicity, non-cancer effects	CTUh, n-c	4.63E-09	4.61E-09	1.14E-11	1.80E-10	9.43E-09	-3.27E-11	9.40E-09
Particulate matter	kg PM _{2.5} eq	5.00E-05	7.84E-06	2.61E-09	7.79E-07	5.86E-05	-2.98E-06	5.56E-05
Ionizing radiation HH	kBq U235 eq	1.20E-03	1.02E-03	8.95E-07	2.85E-04	2.50E-03	-3.94E-04	2.11E-03
Photochemical ozone formation	kg NMVOC eq	1.67E-04	3.22E-05	1.74E-08	9.32E-06	2.08E-04	-9.55E-06	1.99E-04
Acidification	mol H ⁺ eq	3.62E-04	7.86E-05	3.61E-08	1.14E-05	4.52E-04	-2.84E-05	4.23E-04
Terrestrial eutrophication	mol N eq	6.15E-04	1.05E-04	9.46E-08	3.36E-05	7.54E-04	-3.64E-05	7.18E-04
Freshwater eutrophication	kg P eq	1.84E-06	4.01E-06	6.39E-09	1.40E-07	6.00E-06	-1.03E-06	4.96E-06
Marine eutrophication	kg N eq	5.48E-05	8.90E-06	1.40E-07	3.32E-06	6.72E-05	-2.83E-06	6.43E-05
Freshwater ecotoxicity	CTUe	1.39E-02	7.70E-02	2.84E-05	1.99E-03	9.30E-02	-4.15E-03	8.88E-02
Land use	kg C deficit	3.51E-02	1.19E-02	1.99E-05	2.36E-03	4.94E-02	-3.67E-03	4.58E-02
Water resource depletion	m ³ water eq	6.33E-05	1.81E-05	1.28E-07	1.47E-06	8.30E-05	-8.79E-06	7.42E-05
Mineral, fossil & ren resource depletion	kg Sb eq	9.53E-07	5.39E-07	4.78E-11	1.05E-08	1.50E-06	-1.35E-08	1.49E-06
Cumulative energy demand non renewable	MJ oil eq	4.31E-01	1.87E-01	7.39E-05	3.64E-02	6.54E-01	-5.75E-02	5.96E-01
Cumulative energy demand renewable	MJ oil eq	3.65E-02	3.20E-02	3.60E+00	2.38E-03	3.67E+00	-1.46E-02	3.66E+00
Nuclear waste	m ³ HAA eq	2.67E-11	2.24E-11	1.66E-14	5.95E-12	5.51E-11	-8.81E-12	4.63E-11

The following Fig. A. 9 shows the share per life cycle stage per impact category of electricity production with micromorphous-Si PV panels integrated in a slanted roof. The potential benefits for recycling are in relation to the sum of the life cycle stages production to end-of-life.

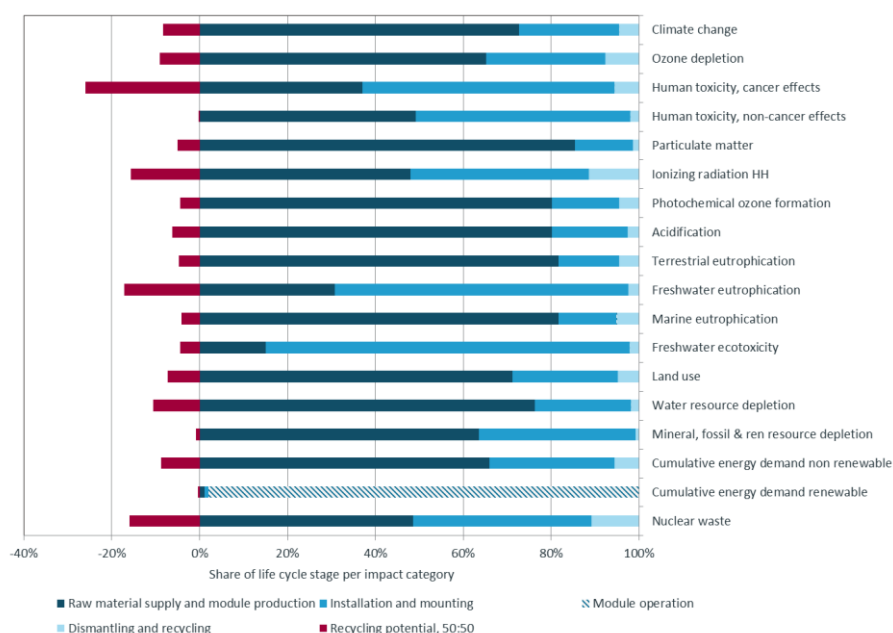


Fig. A. 9 Environmental impact results (characterized, indexed to 100 %) of 1 kWh of electricity produced with a 3 kWp power system with micro-Si PV panels integrated in a slanted roof. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.

Tab. A. 44 lists the environmental impacts per impact category (characterized results) for 1 kWh of electricity produced with micromorphous-Si PV panels on an open ground system. Fig. A. 10 shows the relative impact per life cycle stage and per impact category.

Tab. A. 44 Environmental impact results (characterized) of 1 kWh of electricity produced with a 570 kWp power system with micro-Si PV panels on an open ground system.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 570 kWp installation, open ground with micro-Si PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change	kg CO2 eq	4.09E-02	2.12E-02	4.85E-06	2.56E-03	6.47E-02	-7.85E-03	5.68E-02
Ozone depletion	kg CFC-11 eq	7.50E-10	6.74E-10	2.99E-13	8.70E-11	1.51E-09	-1.54E-10	1.36E-09
Human toxicity, cancer effects	CTUh, c	4.00E-10	2.18E-09	7.35E-13	6.01E-11	2.64E-09	-4.30E-10	2.22E-09
Human toxicity, non-cancer effects	CTUh, n-c	4.63E-09	9.70E-09	1.14E-11	1.80E-10	1.45E-08	3.35E-10	1.49E-08
Particulate matter	kg PM2.5 eq	5.00E-05	1.18E-05	2.66E-09	7.79E-07	6.26E-05	-4.39E-06	5.82E-05
Ionizing radiation HH	kBq U235 eq	1.20E-03	1.88E-03	8.97E-07	2.85E-04	3.36E-03	-6.20E-04	2.74E-03
Photochemical ozone formation	kg NMVOC eq	1.67E-04	5.17E-05	1.75E-08	9.32E-06	2.28E-04	-1.54E-05	2.12E-04
Acidification	mol H+ eq	3.62E-04	1.20E-04	3.65E-08	1.14E-05	4.93E-04	-3.83E-05	4.55E-04
Terrestrial eutrophication	mol N eq	6.15E-04	2.33E-04	9.52E-08	3.36E-05	8.81E-04	-5.69E-05	8.25E-04
Freshwater eutrophication	kg P eq	1.84E-06	1.98E-06	6.40E-09	1.40E-07	3.97E-06	-4.61E-07	3.51E-06
Marine eutrophication	kg N eq	5.48E-05	1.58E-05	1.40E-07	3.32E-06	7.41E-05	-4.72E-06	6.93E-05
Freshwater ecotoxicity	CTUe	1.39E-02	4.80E-02	2.85E-05	1.99E-03	6.39E-02	-5.19E-03	5.87E-02
Land use	kg C deficit	3.51E-02	1.25E+01	1.86E-05	2.36E-03	1.25E+01	-6.10E-03	1.25E+01
Water resource depletion	m3 water eq	6.33E-05	3.25E-05	1.28E-07	1.47E-06	9.75E-05	-1.36E-05	8.38E-05
Mineral, fossil & ren resource depletion	kg Sb eq	9.53E-07	6.48E-07	4.90E-11	1.05E-08	1.61E-06	1.03E-07	1.72E-06
Cumulative energy demand non renewable	MJ oil eq	4.31E-01	3.02E-01	7.44E-05	3.64E-02	7.70E-01	-1.01E-01	6.68E-01
Cumulative energy demand renewable	MJ oil eq	3.65E-02	5.62E-02	3.60E+00	2.38E-03	3.70E+00	-2.24E-02	3.67E+00
Nuclear waste	m3 HAA eq	2.67E-11	4.11E-11	1.67E-14	5.95E-12	7.38E-11	-1.39E-11	5.99E-11

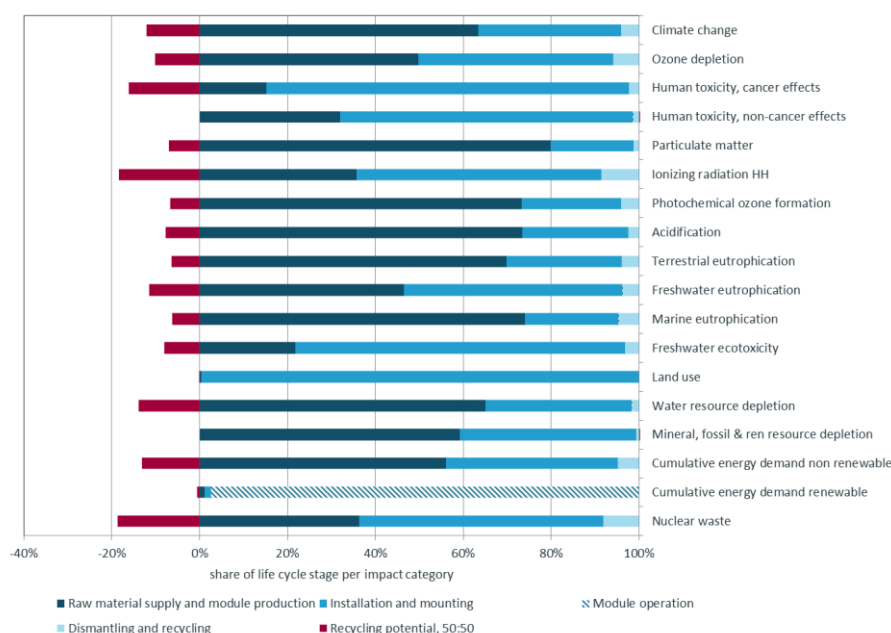


Fig. A. 10 Environmental impact results (characterized, indexed to 100 %) of 1 kWh of electricity produced with a 570 kWp power system with micro-Si PV panels on an open ground system. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.

A.3.3. Results normalized

The normalized environmental impact results of the electricity production with micromorphous-Si PV panels integrated in a slanted roof and with an open ground system are listed in Tab. A. 45 and Tab. A. 46, respectively. The three additional impact categories cumulative energy demand non-renewable, cumulative energy demand renewable and nuclear waste are not subject to normalization.

The results show that the production of 1 kWh electricity with the integrated micro-Si PV system contributes the most to the impact category human toxicity cancer effects. An average European person consumes around 5'000 kWh of electricity. Assuming it would all be produced with such a

3 kWp micromorphous-Si PV system, this would sum up to 11 % of the present budget of the impacts in the category human toxicity, cancer effects. The impact category land use is the most important for the 570 kWp open ground micro-Si PV system. If an average European person consumed electricity exclusively from this PV system, 84 % of the present land use budget per person would be spent on electricity production.

Tab. A. 45 Environmental impact results (normalized) of 1 kWh of electricity produced with a 3 kWp power system with micro-Si PV panels integrated in a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, integrated with micro-Si PV technology, normalized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change		4.44E-06	1.40E-06	5.21E-10	2.78E-07	6.11E-06	-5.09E-07	5.61E-06
Ozone depletion		3.47E-08	1.45E-08	1.38E-11	4.03E-09	5.33E-08	-4.89E-09	4.84E-08
Human toxicity, cancer effects		1.08E-05	1.69E-05	1.99E-08	1.63E-06	2.94E-05	-7.67E-06	2.17E-05
Human toxicity, non-cancer effects		8.68E-06	8.66E-06	2.14E-08	3.38E-07	1.77E-05	-6.13E-08	1.76E-05
Particulate matter		1.31E-05	2.06E-06	6.86E-10	2.05E-07	1.54E-05	-7.83E-07	1.46E-05
Ionizing radiation HH		1.06E-06	9.03E-07	7.92E-10	2.52E-07	2.22E-06	-3.49E-07	1.87E-06
Photochemical ozone formation		5.26E-06	1.02E-06	5.48E-10	2.94E-07	6.57E-06	-3.01E-07	6.27E-06
Acidification		7.65E-06	1.66E-06	7.63E-10	2.42E-07	9.55E-06	-6.00E-07	8.95E-06
Terrestrial eutrophication		3.49E-06	5.98E-07	5.38E-10	1.91E-07	4.28E-06	-2.07E-07	4.08E-06
Freshwater eutrophication		1.25E-06	2.71E-06	4.32E-09	9.49E-08	4.05E-06	-6.97E-07	3.35E-06
Marine eutrophication		3.24E-06	5.26E-07	8.31E-09	1.96E-07	3.97E-06	-1.67E-07	3.81E-06
Freshwater ecotoxicity		1.59E-06	8.81E-06	3.25E-09	2.28E-07	1.06E-05	-4.75E-07	1.02E-05
Land use		4.70E-07	1.59E-07	2.65E-10	3.15E-08	6.61E-07	-4.91E-08	6.12E-07
Water resource depletion		7.78E-07	2.22E-07	1.57E-09	1.81E-08	1.02E-06	-1.08E-07	9.12E-07
Mineral, fossil & ren resource depletion		9.44E-06	5.33E-06	4.73E-10	1.04E-07	1.49E-05	-1.33E-07	1.47E-05

Tab. A. 46 Environmental impact results (normalized) of 1 kWh of electricity produced with a 570 kWp power system with micro-Si PV panels on an open ground system.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 570 kWp installation, open ground with micro-Si PV technology, normalized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change		4.44E-06	2.30E-06	5.26E-10	2.78E-07	7.02E-06	-8.52E-07	6.16E-06
Ozone depletion		3.47E-08	3.12E-08	1.38E-11	4.03E-09	7.00E-08	-7.12E-09	6.28E-08
Human toxicity, cancer effects		1.08E-05	5.92E-05	1.99E-08	1.63E-06	7.17E-05	-1.16E-05	6.00E-05
Human toxicity, non-cancer effects		8.68E-06	1.82E-05	2.14E-08	3.38E-07	2.72E-05	6.29E-07	2.79E-05
Particulate matter		1.31E-05	3.11E-06	7.00E-10	2.05E-07	1.65E-05	-1.15E-06	1.53E-05
Ionizing radiation HH		1.06E-06	1.66E-06	7.94E-10	2.52E-07	2.97E-06	-5.49E-07	2.43E-06
Photochemical ozone formation		5.26E-06	1.63E-06	5.53E-10	2.94E-07	7.19E-06	-4.87E-07	6.70E-06
Acidification		7.65E-06	2.54E-06	7.71E-10	2.42E-07	1.04E-05	-8.10E-07	9.62E-06
Terrestrial eutrophication		3.49E-06	1.32E-06	5.41E-10	1.91E-07	5.01E-06	-3.23E-07	4.68E-06
Freshwater eutrophication		1.25E-06	1.34E-06	4.32E-09	9.49E-08	2.69E-06	-3.12E-07	2.37E-06
Marine eutrophication		3.24E-06	9.35E-07	8.31E-09	1.96E-07	4.38E-06	-2.79E-07	4.10E-06
Freshwater ecotoxicity		1.59E-06	5.49E-06	3.26E-09	2.28E-07	7.31E-06	-5.94E-07	6.72E-06
Land use		4.70E-07	1.67E-04	2.40E-10	3.15E-08	1.67E-04	-8.16E-08	1.67E-04
Water resource depletion		7.78E-07	4.00E-07	1.57E-09	1.81E-08	1.20E-06	-1.67E-07	1.03E-06
Mineral, fossil & ren resource depletion		9.44E-06	6.42E-06	4.86E-10	1.04E-07	1.60E-05	1.02E-06	1.70E-05

A.3.4. Results weighted

Tab. A. 47 shows the weighted environmental impact results (equal weighting) of the electricity production with micromorphous-Si PV panels in an integrated roof system whereas Tab. A. 48 shows the results for electricity production on a 570 kWp open ground system. The blue bars within column “Total” illustrate the importance of each impact category in the total of impacts. The three additional

impact categories cumulative energy demand non-renewable, cumulative energy demand renewable and nuclear waste are not included in the weighted results.

The environmental impacts of the production of 1 kWh electricity with a 3 kWp power system with micromorphous-Si PV panels integrated in a slanted roof, are mainly driven by human toxicity cancer effects (19 %), followed by human toxicity non cancer effects (15 %), the depletion of mineral, fossil and renewable resources and particulate matter emissions (both 13 %). The shares of the different impact categories on the weighted results are shown in Fig. A. 11.

Tab. A. 47 Environmental impact results (weighted) of 1 kWh of electricity produced with a 3 kWp power system with micro-Si PV panels integrated in a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, integrated with micro-Si PV technology, weighted		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Total	Pt	4.76E-06	3.40E-06	4.22E-09	2.74E-07	8.43E-06	-8.07E-07	7.63E-06
Climate change	Pt	2.96E-07	9.31E-08	3.48E-11	1.85E-08	4.08E-07	-3.40E-08	3.74E-07
Ozone depletion	Pt	2.32E-09	9.68E-10	9.20E-13	2.69E-10	3.55E-09	-3.26E-10	3.23E-09
Human toxicity, cancer effects	Pt	7.22E-07	1.13E-06	1.33E-09	1.09E-07	1.96E-06	-5.11E-07	1.45E-06
Human toxicity, non-cancer effects	Pt	5.79E-07	5.77E-07	1.43E-09	2.25E-08	1.18E-06	-4.09E-09	1.18E-06
Particulate matter	Pt	8.77E-07	1.38E-07	4.57E-11	1.37E-08	1.03E-06	-5.22E-08	9.76E-07
Ionizing radiation HH	Pt	7.07E-08	6.02E-08	5.28E-11	1.68E-08	1.48E-07	-2.33E-08	1.25E-07
Photochemical ozone formation	Pt	3.51E-07	6.77E-08	3.65E-11	1.96E-08	4.38E-07	-2.01E-08	4.18E-07
Acidification	Pt	5.10E-07	1.11E-07	5.09E-11	1.61E-08	6.37E-07	-4.00E-08	5.97E-07
Terrestrial eutrophication	Pt	2.33E-07	3.99E-08	3.59E-11	1.27E-08	2.86E-07	-1.38E-08	2.72E-07
Freshwater eutrophication	Pt	8.30E-08	1.80E-07	2.88E-10	6.32E-09	2.70E-07	-4.64E-08	2.24E-07
Marine eutrophication	Pt	2.16E-07	3.51E-08	5.54E-10	1.31E-08	2.65E-07	-1.12E-08	2.54E-07
Freshwater ecotoxicity	Pt	1.06E-07	5.88E-07	2.17E-10	1.52E-08	7.09E-07	-3.16E-08	6.77E-07
Land use	Pt	3.13E-08	1.06E-08	1.77E-11	2.10E-09	4.40E-08	-3.27E-09	4.08E-08
Water resource depletion	Pt	5.19E-08	1.48E-08	1.05E-10	1.21E-09	6.80E-08	-7.20E-09	6.08E-08
Mineral, fossil & ren resource depletion	Pt	6.29E-07	3.56E-07	3.16E-11	6.92E-09	9.92E-07	-8.89E-09	9.83E-07

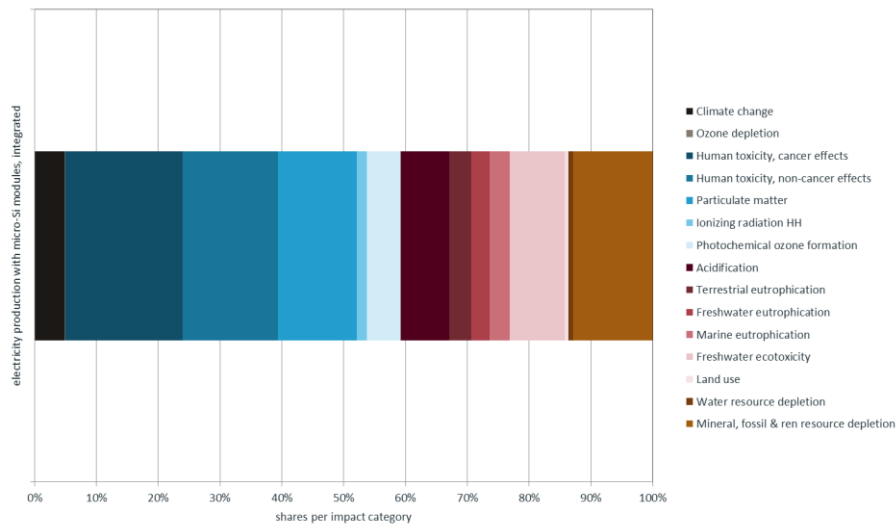


Fig. A. 11 Share per impact category on the weighted result for the production of 1 kWh of electricity produced with a 3 kWp power system with micro-Si PV panels integrated in a slanted roof.

The electricity production with micromorphous-Si PV panels on an open ground system has its main impact in the land use (50 %), followed by human toxicity cancer effects (18 %) and human toxicity non cancer effects (8 %). Fig. A. 12 illustrates this.

Tab. A. 48 Environmental impact results (weighted) of 1 kWh of electricity produced with a 570 kWp power system with micro-Si PV panels on an open ground system.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 570 kWp installation, open ground with micro-Si PV technology, weighted		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Total	Pt	4.76E-06	1.81E-05	4.23E-09	2.74E-07	2.31E-05	-1.04E-06	2.21E-05
Climate change	Pt	2.96E-07	1.53E-07	3.51E-11	1.85E-08	4.68E-07	-5.68E-08	4.11E-07
Ozone depletion	Pt	2.32E-09	2.08E-09	9.23E-13	2.69E-10	4.66E-09	-4.75E-10	4.19E-09
Human toxicity, cancer effects	Pt	7.22E-07	3.95E-06	1.33E-09	1.09E-07	4.78E-06	-7.76E-07	4.00E-06
Human toxicity, non-cancer effects	Pt	5.79E-07	1.21E-06	1.43E-09	2.25E-08	1.82E-06	4.19E-08	1.86E-06
Particulate matter	Pt	8.77E-07	2.07E-07	4.66E-11	1.37E-08	1.10E-06	-7.70E-08	1.02E-06
Ionizing radiation HH	Pt	7.07E-08	1.11E-07	5.29E-11	1.68E-08	1.98E-07	-3.66E-08	1.62E-07
Photochemical ozone formation	Pt	3.51E-07	1.09E-07	3.69E-11	1.96E-08	4.79E-07	-3.25E-08	4.47E-07
Acidification	Pt	5.10E-07	1.69E-07	5.14E-11	1.61E-08	6.95E-07	-5.40E-08	6.41E-07
Terrestrial eutrophication	Pt	2.33E-07	8.82E-08	3.61E-11	1.27E-08	3.34E-07	-2.15E-08	3.12E-07
Freshwater eutrophication	Pt	8.30E-08	8.94E-08	2.88E-10	6.32E-09	1.79E-07	-2.08E-08	1.58E-07
Marine eutrophication	Pt	2.16E-07	6.24E-08	5.54E-10	1.31E-08	2.92E-07	-1.86E-08	2.74E-07
Freshwater ecotoxicity	Pt	1.06E-07	3.66E-07	2.17E-10	1.52E-08	4.87E-07	-3.96E-08	4.48E-07
Land use	Pt	3.13E-08	1.11E-05	1.63E-11	2.10E-09	1.12E-05	-5.44E-09	1.11E-05
Water resource depletion	Pt	5.19E-08	2.66E-08	1.05E-10	1.21E-09	7.98E-08	-1.11E-08	6.87E-08
Mineral, fossil & ren resource depletion	Pt	6.29E-07	4.28E-07	3.24E-11	6.92E-09	1.06E-06	6.83E-08	1.13E-06

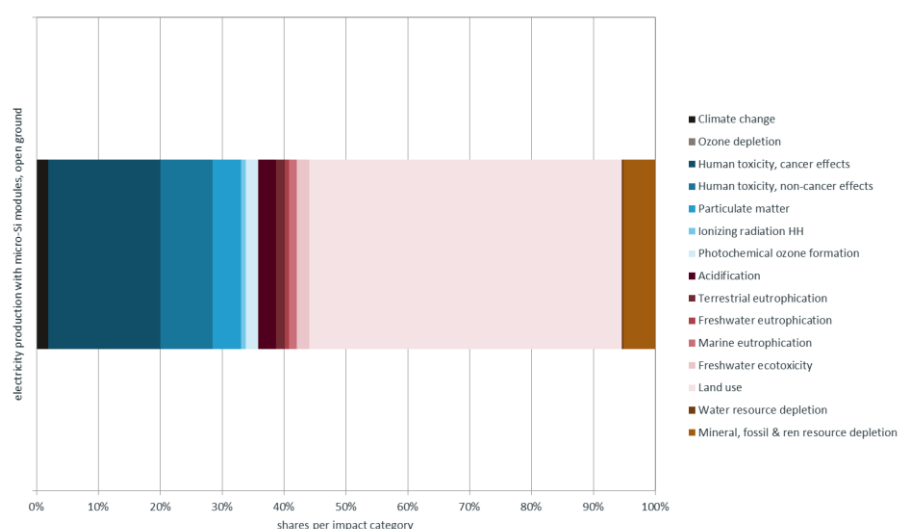


Fig. A. 12 Share per impact category on the weighted result for the production of 1 kWh of electricity produced with a 570 kWp power system with micro-Si PV panels on an open ground system.

A.4. Results multicrystalline-Si PV technology

A.4.1. Overview

In section A.4.2 the results prior to normalization are assessed. In the sections A.4.3 and A.4.4 the normalized and the weighted results of electricity production with multicrystalline-Si PV panels, integrated in a slanted roof, residential scale and of a large scale open ground system are described.

A.4.2. Results prior to normalization

The results prior to normalization of the production of 1 kWh of electricity with multicrystalline-Si PV panels integrated in a slanted roof in a 3 kWp installation system and on a 570 kWp open ground system are summarized in Tab. A. 49 and Tab. A. 50. The life cycle stages analyzed are the product stage, the construction stage, the operation stage and the end-of-life stage. Net benefits caused by end-of-life recycling are allocated to the production of electricity with a share of 50 %. The other half is

allocated to the life cycle of products potentially using the secondary products (see also subchapter 3.5).

Tab. A. 49 Environmental impact results (characterized) of 1 kWh of electricity produced with a 3 kWp power system with multi-Si PV panels integrated in a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, integrated with multi-Si PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change	kg CO ₂ eq	4.52E-02	9.19E-03	3.25E-06	1.41E-03	5.58E-02	-2.82E-03	5.29E-02
Ozone depletion	kg CFC-11 eq	1.48E-09	2.18E-10	2.02E-13	4.78E-11	1.74E-09	-6.24E-11	1.68E-09
Human toxicity, cancer effects	CTUh, c	5.38E-10	4.39E-10	4.99E-13	3.30E-11	1.01E-09	-1.83E-10	8.28E-10
Human toxicity, non-cancer effects	CTUh, n-c	6.12E-09	4.05E-09	7.75E-12	9.89E-11	1.03E-08	-1.44E-10	1.01E-08
Particulate matter	kg PM _{2.5} eq	6.14E-05	6.02E-06	1.74E-09	4.28E-07	6.78E-05	-1.96E-06	6.59E-05
Ionizing radiation HH	kBq U235 eq	1.91E-03	7.07E-04	6.08E-07	1.57E-04	2.77E-03	-2.49E-04	2.52E-03
Photochemical ozone formation	kg NMVOC eq	1.66E-04	2.43E-05	1.18E-08	5.12E-06	1.95E-04	-6.30E-06	1.89E-04
Acidification	mol H ⁺ eq	4.05E-04	6.29E-05	2.44E-08	6.29E-06	4.75E-04	-1.97E-05	4.55E-04
Terrestrial eutrophication	mol N eq	5.97E-04	7.97E-05	6.42E-08	1.85E-05	6.95E-04	-2.34E-05	6.71E-04
Freshwater eutrophication	kg P eq	3.11E-06	3.87E-06	4.35E-09	7.71E-08	7.06E-06	-9.88E-07	6.07E-06
Marine eutrophication	kg N eq	5.51E-05	6.63E-06	9.55E-08	1.82E-06	6.37E-05	-1.84E-06	6.18E-05
Freshwater ecotoxicity	CTUe	2.42E-02	7.37E-02	1.93E-05	1.09E-03	9.90E-02	-2.92E-03	9.60E-02
Land use	kg C deficit	5.63E-02	8.62E-03	1.35E-05	1.29E-03	6.62E-02	-2.27E-03	6.39E-02
Water resource depletion	m ³ water eq	4.71E-05	1.25E-05	8.70E-08	8.09E-07	6.04E-05	-5.59E-06	5.48E-05
Mineral, fossil & ren resource depletion	kg Sb eq	1.88E-05	5.07E-07	1.47E-11	5.76E-09	1.93E-05	-4.30E-08	1.93E-05
Cumulative energy demand non renewable	MJ oil eq	5.46E-01	1.33E-01	5.00E-05	2.00E-02	7.00E-01	-3.52E-02	6.65E-01
Cumulative energy demand renewable	MJ oil eq	7.20E-02	2.21E-02	3.60E+00	1.31E-03	3.70E+00	-9.33E-03	3.69E+00
Nuclear waste	m ³ HAA eq	4.42E-11	1.55E-11	1.13E-14	3.27E-12	6.30E-11	-5.58E-12	5.74E-11

The following Fig. A. 13 shows the share per life cycle stage per impact category of electricity production with multicrystalline-Si PV panels integrated in a slanted roof. The potential benefits for recycling are in relation to the sum of the life cycle stages production to end-of-life.

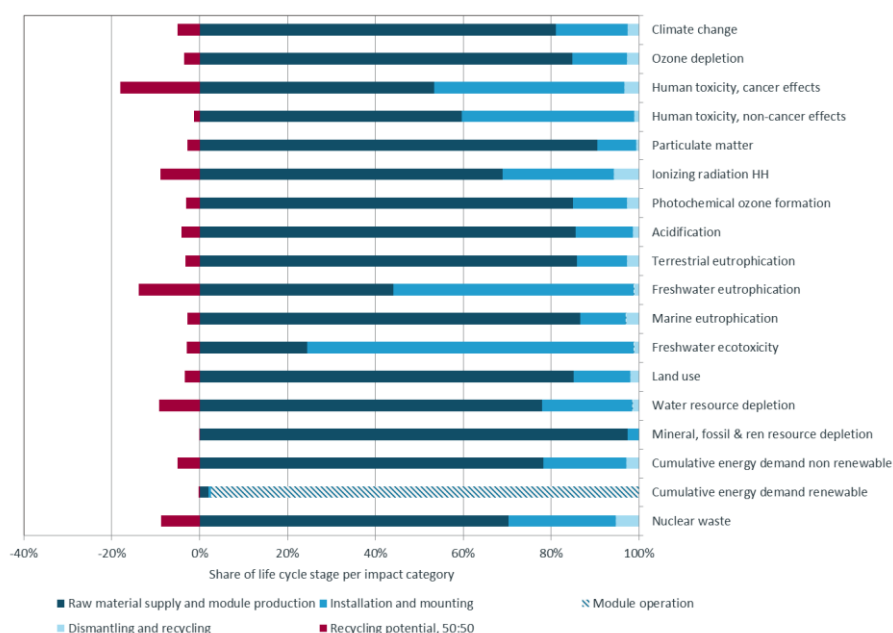


Fig. A. 13 Environmental impact results (characterized, indexed to 100 %) of 1 kWh of electricity produced with a 3 kWp power system with multi-Si PV panels integrated in a slanted roof. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.

Tab. A. 50 lists the environmental impacts per impact category (characterized results) for 1 kWh of electricity produced with multicrystalline-Si PV panels on an open ground system. Fig. A. 14 shows the relative impact per life cycle stage and per impact category.

Tab. A. 50 Environmental impact results (characterized) of 1 kWh of electricity produced with a 570 kWp power system with multi-Si PV panels on an open ground system.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 570 kWp installation, open ground with multi-Si PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change	kg CO2 eq	4.52E-02	1.45E-02	3.30E-06	1.41E-03	6.11E-02	-4.96E-03	5.61E-02
Ozone depletion	kg CFC-11 eq	1.48E-09	4.59E-10	2.03E-13	4.78E-11	1.98E-09	-9.50E-11	1.89E-09
Human toxicity, cancer effects	CTUh, c	5.38E-10	1.49E-09	5.00E-13	3.30E-11	2.06E-09	-2.81E-10	1.78E-09
Human toxicity, non-cancer effects	CTUh, n-c	6.12E-09	6.84E-09	7.76E-12	9.89E-11	1.31E-08	1.92E-10	1.33E-08
Particulate matter	kg PM2.5 eq	6.14E-05	8.23E-06	1.80E-09	4.28E-07	7.01E-05	-2.82E-06	6.72E-05
Ionizing radiation HH	kBq U235 eq	1.91E-03	1.28E-03	6.10E-07	1.57E-04	3.35E-03	-4.03E-04	2.94E-03
Photochemical ozone formation	kg NMVOC eq	1.66E-04	3.59E-05	1.19E-08	5.12E-06	2.07E-04	-1.00E-05	1.97E-04
Acidification	mol H+ eq	4.05E-04	8.42E-05	2.48E-08	6.29E-06	4.96E-04	-2.50E-05	4.71E-04
Terrestrial eutrophication	mol N eq	5.97E-04	1.61E-04	6.48E-08	1.85E-05	7.76E-04	-3.62E-05	7.40E-04
Freshwater eutrophication	kg P eq	3.11E-06	1.65E-06	4.35E-09	7.71E-08	4.84E-06	-3.83E-07	4.46E-06
Marine eutrophication	kg N eq	5.51E-05	1.09E-05	9.55E-08	1.82E-06	6.80E-05	-3.05E-06	6.49E-05
Freshwater ecotoxicity	CTUe	2.42E-02	3.77E-02	1.94E-05	1.09E-03	6.30E-02	-3.41E-03	5.96E-02
Land use	kg C deficit	5.63E-02	8.49E+00	1.24E-05	1.29E-03	8.54E+00	-3.86E-03	8.54E+00
Water resource depletion	m3 water eq	4.71E-05	2.22E-05	8.71E-08	8.09E-07	7.01E-05	-8.85E-06	6.13E-05
Mineral, fossil & ren resource depletion	kg Sb eq	1.88E-05	4.79E-07	3.06E-11	5.76E-09	1.93E-05	5.94E-08	1.93E-05
Cumulative energy demand non renewable	MJ oil eq	5.46E-01	2.07E-01	5.05E-05	2.00E-02	7.74E-01	-6.50E-02	7.09E-01
Cumulative energy demand renewable	MJ oil eq	7.20E-02	3.83E-02	3.60E+00	1.31E-03	3.71E+00	-1.46E-02	3.70E+00
Nuclear waste	m3 HAA eq	4.42E-11	2.80E-11	1.13E-14	3.27E-12	7.55E-11	-9.01E-12	6.65E-11

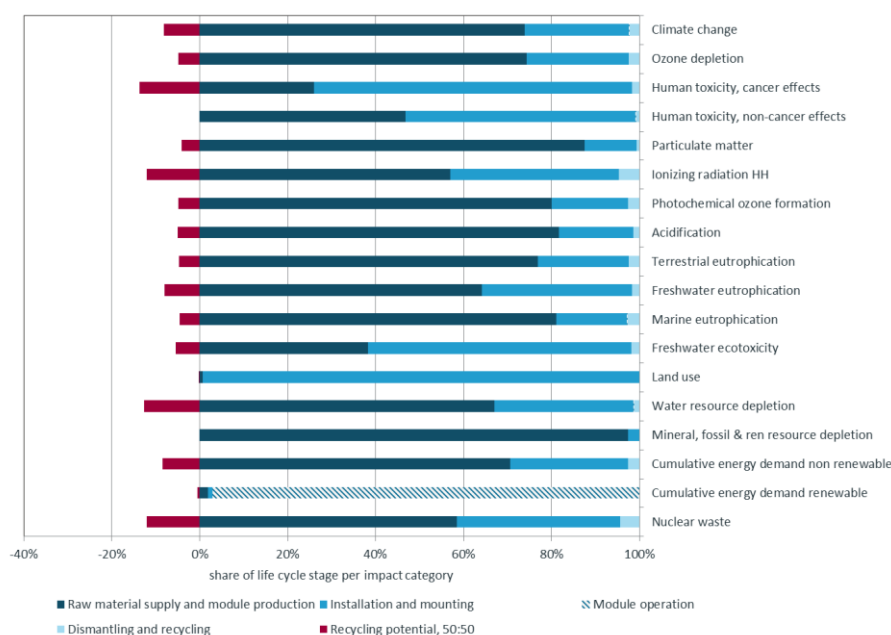


Fig. A. 14 Environmental impact results (characterized, indexed to 100 %) of 1 kWh of electricity produced with a 570 kWp power system with multi-Si PV panels on an open ground system. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.

A.4.3. Results normalized

The normalized environmental impact results of the electricity production with multicrystalline-Si PV panels integrated in a slanted roof and on an open ground system are listed in Tab. A. 51 and Tab. A. 52, respectively. The three additional impact categories cumulative energy demand non-renewable, cumulative energy demand renewable and nuclear waste are not subject to normalization.

The results show that the production of 1 kWh electricity contributes the largest share to the mineral, fossil and renewable resource depletion potential of an average European citizen. An average European person consumes around 5'000 kWh of electricity. Assuming it would all be produced with

such a 3 kWp multicrystalline-Si PV system or a 570 kWp open ground system, this would sum up to 95 % or 96 % of the present budget of resource depletion, respectively.

Tab. A. 51 Environmental impact results (normalized) of 1 kWh of electricity produced with a 3 kWp power system with multi-Si PV panels integrated in a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, integrated with multi-Si PV technology, normalized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change		4.90E-06	9.97E-07	3.53E-10	1.53E-07	6.05E-06	-3.05E-07	5.74E-06
Ozone depletion		6.83E-08	1.01E-08	9.35E-12	2.21E-09	8.06E-08	-2.89E-09	7.77E-08
Human toxicity, cancer effects		1.46E-05	1.19E-05	1.35E-08	8.94E-07	2.74E-05	-4.97E-06	2.24E-05
Human toxicity, non-cancer effects		1.15E-05	7.60E-06	1.45E-08	1.86E-07	1.93E-05	-2.69E-07	1.90E-05
Particulate matter		1.62E-05	1.58E-06	4.59E-10	1.13E-07	1.79E-05	-5.16E-07	1.73E-05
Ionizing radiation HH		1.69E-06	6.25E-07	5.38E-10	1.39E-07	2.45E-06	-2.20E-07	2.23E-06
Photochemical ozone formation		5.23E-06	7.66E-07	3.71E-10	1.62E-07	6.15E-06	-1.99E-07	5.96E-06
Acidification		8.57E-06	1.33E-06	5.15E-10	1.33E-07	1.00E-05	-4.17E-07	9.62E-06
Terrestrial eutrophication		3.39E-06	4.53E-07	3.65E-10	1.05E-07	3.95E-06	-1.33E-07	3.81E-06
Freshwater eutrophication		2.10E-06	2.62E-06	2.94E-09	5.21E-08	4.77E-06	-6.68E-07	4.10E-06
Marine eutrophication		3.26E-06	3.92E-07	5.65E-09	1.08E-07	3.77E-06	-1.09E-07	3.66E-06
Freshwater ecotoxicity		2.77E-06	8.43E-06	2.21E-09	1.25E-07	1.13E-05	-3.34E-07	1.10E-05
Land use		7.53E-07	1.15E-07	1.80E-10	1.73E-08	8.85E-07	-3.04E-08	8.55E-07
Water resource depletion		5.78E-07	1.53E-07	1.07E-09	9.93E-09	7.42E-07	-6.87E-08	6.74E-07
Mineral, fossil & ren resource depletion		1.86E-04	5.02E-06	1.42E-10	5.70E-08	1.91E-04	-4.26E-07	1.91E-04

Tab. A. 52 Environmental impact results (normalized) of 1 kWh of electricity produced with a 570 kWp power system with multi-Si PV panels on an open ground system.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 570 kWp installation, open ground with multi-Si PV technology, normalized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change		4.90E-06	1.57E-06	3.58E-10	1.53E-07	6.63E-06	-5.38E-07	6.09E-06
Ozone depletion		6.83E-08	2.13E-08	9.42E-12	2.21E-09	9.18E-08	-4.40E-09	8.74E-08
Human toxicity, cancer effects		1.46E-05	4.04E-05	1.36E-08	8.94E-07	5.59E-05	-7.63E-06	4.82E-05
Human toxicity, non-cancer effects		1.15E-05	1.28E-05	1.46E-08	1.86E-07	2.45E-05	3.61E-07	2.49E-05
Particulate matter		1.62E-05	2.17E-06	4.74E-10	1.13E-07	1.84E-05	-7.43E-07	1.77E-05
Ionizing radiation HH		1.69E-06	1.13E-06	5.40E-10	1.39E-07	2.96E-06	-3.56E-07	2.60E-06
Photochemical ozone formation		5.23E-06	1.13E-06	3.76E-10	1.62E-07	6.52E-06	-3.16E-07	6.20E-06
Acidification		8.57E-06	1.78E-06	5.24E-10	1.33E-07	1.05E-05	-5.28E-07	9.96E-06
Terrestrial eutrophication		3.39E-06	9.13E-07	3.68E-10	1.05E-07	4.41E-06	-2.06E-07	4.20E-06
Freshwater eutrophication		2.10E-06	1.12E-06	2.94E-09	5.21E-08	3.27E-06	-2.59E-07	3.01E-06
Marine eutrophication		3.26E-06	6.46E-07	5.65E-09	1.08E-07	4.02E-06	-1.81E-07	3.84E-06
Freshwater ecotoxicity		2.77E-06	4.32E-06	2.22E-09	1.25E-07	7.21E-06	-3.90E-07	6.82E-06
Land use		7.53E-07	1.13E-04	1.70E-10	1.73E-08	1.14E-04	-5.16E-08	1.14E-04
Water resource depletion		5.78E-07	2.72E-07	1.07E-09	9.93E-09	8.62E-07	-1.09E-07	7.53E-07
Mineral, fossil & ren resource depletion		1.86E-04	4.74E-06	3.08E-10	5.70E-08	1.91E-04	5.88E-07	1.91E-04

A.4.4. Results weighted

Tab. A. 53 shows the weighted environmental impact results (equal weighting) of the electricity production with multicrystalline-Si PV panels in an integrated roof system whereas Tab. A. 54 shows the results for electricity production on a 570 kWp open ground system. The blue bars within column “Total” illustrate the importance of each impact category in the total of impacts. The three additional impact categories cumulative energy demand non-renewable, cumulative energy demand renewable and nuclear waste are not included in the weighted results.

The environmental impacts of the production of 1 kWh electricity with a 3 kWp power system with multicrystalline-Si PV panels integrated in a slanted roof, are mainly driven by mineral, fossil and renewable resource depletion (64 %), followed by human toxicity cancer effects (8 %), human toxicity non cancer effects and particulate matter emissions (both 6 %). The shares of the individual impact categories on the weighted results are illustrated in Fig. A. 15.

Tab. A. 53 Environmental impact results (weighted) of 1 kWh of electricity produced with a 3 kWp power system with multi-Si PV panels integrated in a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, integrated with multi-Si PV technology, weighted		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Total	Pt	1.74E-05	2.80E-06	2.86E-09	1.50E-07	2.04E-05	-5.78E-07	1.98E-05
Climate change	Pt	3.26E-07	6.65E-08	2.35E-11	1.02E-08	4.03E-07	-2.04E-08	3.83E-07
Ozone depletion	Pt	4.55E-09	6.74E-10	6.23E-13	1.48E-10	5.38E-09	-1.93E-10	5.18E-09
Human toxicity, cancer effects	Pt	9.72E-07	7.94E-07	9.02E-10	5.96E-08	1.83E-06	-3.31E-07	1.50E-06
Human toxicity, non-cancer effects	Pt	7.65E-07	5.07E-07	9.70E-10	1.24E-08	1.29E-06	-1.80E-08	1.27E-06
Particulate matter	Pt	1.08E-06	1.06E-07	3.07E-11	7.51E-09	1.19E-06	-3.44E-08	1.16E-06
Ionizing radiation HH	Pt	1.13E-07	4.17E-08	3.59E-11	9.24E-09	1.64E-07	-1.47E-08	1.49E-07
Photochemical ozone formation	Pt	3.48E-07	5.11E-08	2.47E-11	1.08E-08	4.10E-07	-1.33E-08	3.97E-07
Acidification	Pt	5.71E-07	8.87E-08	3.44E-11	8.86E-09	6.69E-07	-2.78E-08	6.41E-07
Terrestrial eutrophication	Pt	2.26E-07	3.02E-08	2.43E-11	6.99E-09	2.63E-07	-8.87E-09	2.54E-07
Freshwater eutrophication	Pt	1.40E-07	1.74E-07	1.96E-10	3.47E-09	3.18E-07	-4.45E-08	2.74E-07
Marine eutrophication	Pt	2.18E-07	2.61E-08	3.77E-10	7.19E-09	2.51E-07	-7.27E-09	2.44E-07
Freshwater ecotoxicity	Pt	1.84E-07	5.62E-07	1.47E-10	8.35E-09	7.55E-07	-2.23E-08	7.33E-07
Land use	Pt	5.02E-08	7.68E-09	1.20E-11	1.15E-09	5.90E-08	-2.03E-09	5.70E-08
Water resource depletion	Pt	3.86E-08	1.02E-08	7.13E-11	6.62E-10	4.95E-08	-4.58E-09	4.49E-08
Mineral, fossil & ren resource depletion	Pt	1.24E-05	3.35E-07	9.86E-12	3.80E-09	1.27E-05	-2.84E-08	1.27E-05

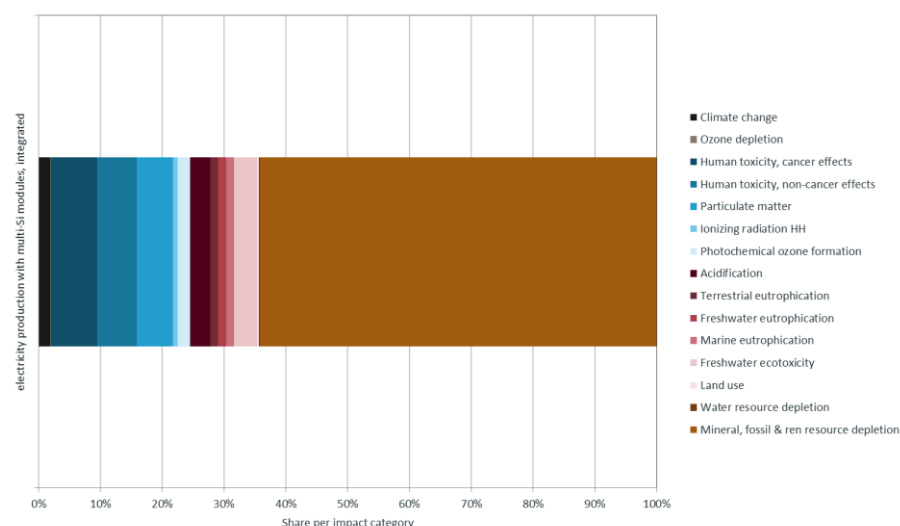


Fig. A. 15 Share per impact category on the weighted result for the production of 1 kWh of electricity produced with a 3 kWp power system with multi-Si PV panels integrated in a slanted roof.

The electricity production with multicrystalline-Si PV panels on an open ground system has its main impact in the mineral, fossil and renewable resource depletion (44 %), followed by land use (26 %), human toxicity cancer effects (11 %) and human toxicity non cancer effects (6 %). Fig. A. 16 illustrates this.

Tab. A. 54 Environmental impact results (weighted) of 1 kWh of electricity produced with a 570 kWp power system with multi-Si PV panels on an open ground system.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 570 kWp installation, open ground with multi-Si PV technology, weighted		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Total	Pt	1.74E-05	1.24E-05	2.87E-09	1.50E-07	3.00E-05	-6.90E-07	2.93E-05
Climate change	Pt	3.26E-07	1.05E-07	2.39E-11	1.02E-08	4.42E-07	-3.59E-08	4.06E-07
Ozone depletion	Pt	4.55E-09	1.42E-09	6.28E-13	1.48E-10	6.12E-09	-2.93E-10	5.83E-09
Human toxicity, cancer effects	Pt	9.72E-07	2.69E-06	9.04E-10	5.96E-08	3.72E-06	-5.08E-07	3.22E-06
Human toxicity, non-cancer effects	Pt	7.65E-07	8.56E-07	9.71E-10	1.24E-08	1.63E-06	2.41E-08	1.66E-06
Particulate matter	Pt	1.08E-06	1.44E-07	3.16E-11	7.51E-09	1.23E-06	-4.95E-08	1.18E-06
Ionizing radiation HH	Pt	1.13E-07	7.55E-08	3.60E-11	9.24E-09	1.97E-07	-2.38E-08	1.74E-07
Photochemical ozone formation	Pt	3.48E-07	7.54E-08	2.51E-11	1.08E-08	4.35E-07	-2.11E-08	4.14E-07
Acidification	Pt	5.71E-07	1.19E-07	3.50E-11	8.86E-09	6.99E-07	-3.52E-08	6.64E-07
Terrestrial eutrophication	Pt	2.26E-07	6.09E-08	2.45E-11	6.99E-09	2.94E-07	-1.37E-08	2.80E-07
Freshwater eutrophication	Pt	1.40E-07	7.45E-08	1.96E-10	3.47E-09	2.18E-07	-1.72E-08	2.01E-07
Marine eutrophication	Pt	2.18E-07	4.31E-08	3.77E-10	7.19E-09	2.68E-07	-1.20E-08	2.56E-07
Freshwater ecotoxicity	Pt	1.84E-07	2.88E-07	1.48E-10	8.35E-09	4.81E-07	-2.60E-08	4.55E-07
Land use	Pt	5.02E-08	7.56E-06	1.12E-11	1.15E-09	7.61E-06	-3.44E-09	7.61E-06
Water resource depletion	Pt	3.86E-08	1.81E-08	7.13E-11	6.62E-10	5.74E-08	-7.25E-09	5.02E-08
Mineral, fossil & ren resource depletion	Pt	1.24E-05	3.16E-07	2.07E-11	3.80E-09	1.27E-05	3.92E-08	1.28E-05

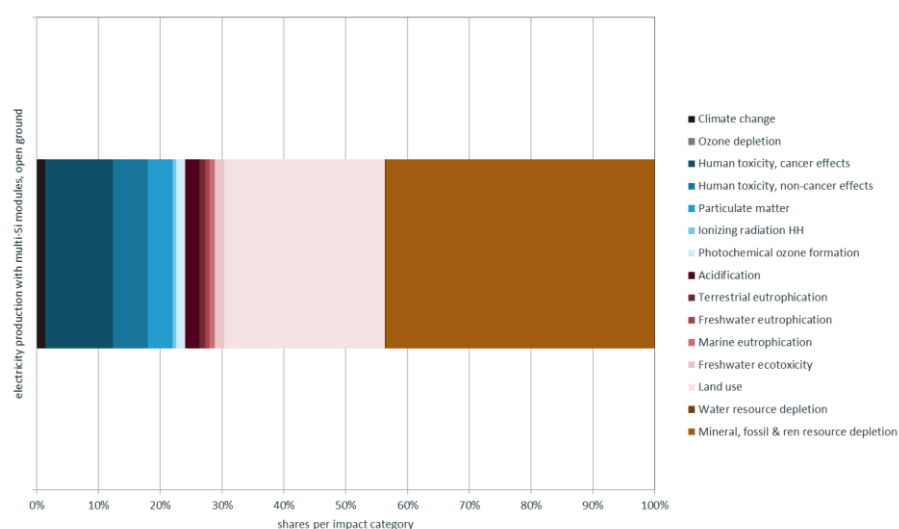


Fig. A. 16 Share per impact category on the weighted result for the production of 1 kWh of electricity produced with a 570 kWp power system with multi-Si PV panels on an open ground system.

A.5. Results monocrystalline-Si PV technology

A.5.1. Overview

In section A.5.2 the results prior to normalization are assessed. In the sections A.5.3 and A.5.4 the normalized and the weighted results of electricity production with monocrystalline-Si PV panels, integrated in a slanted roof, residential scale and of a large scale open ground system are described.

A.5.2. Results prior to normalization

The results prior to normalization of the production of 1 kWh of electricity with monocrystalline-Si PV panels integrated in a slanted roof in a 3 kWp installation system and on a 570 kWp open ground system are summarized in Tab. A. 55 and Tab. A. 56. The life cycle stages analyzed are the product stage, the construction stage, the operation stage and the end-of-life stage. Net benefits caused by end-of-life recycling are allocated to the production of electricity with a share of 50 %. The other half is

allocated to the life cycle of products potentially using the secondary products (see also subchapter 3.5).

Tab. A. 55 Environmental impact results (characterized) of 1 kWh of electricity produced with a 3 kWp power system with mono-Si PV panels integrated in a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, integrated with mono-Si PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change	kg CO ₂ eq	8.00E-02	8.99E-03	3.13E-06	1.43E-03	9.04E-02	-2.74E-03	8.77E-02
Ozone depletion	kg CFC-11 eq	2.12E-09	2.13E-10	1.96E-13	4.87E-11	2.38E-09	-6.08E-11	2.32E-09
Human toxicity, cancer effects	CTUh, c	7.09E-10	4.29E-10	4.86E-13	3.36E-11	1.17E-09	-1.79E-10	9.93E-10
Human toxicity, non-cancer effects	CTUh, n-c	9.12E-09	4.02E-09	7.54E-12	1.01E-10	1.33E-08	-1.49E-10	1.31E-08
Particulate matter	kg PM _{2.5} eq	1.26E-04	5.92E-06	1.63E-09	4.36E-07	1.32E-04	-1.92E-06	1.30E-04
Ionizing radiation HH	kBq U235 eq	2.82E-03	6.89E-04	5.91E-07	1.59E-04	3.67E-03	-2.42E-04	3.43E-03
Photochemical ozone formation	kg NMVOC eq	2.96E-04	2.39E-05	1.13E-08	5.21E-06	3.25E-04	-6.17E-06	3.19E-04
Acidification	mol H ⁺ eq	7.42E-04	6.21E-05	2.34E-08	6.40E-06	8.10E-04	-1.94E-05	7.91E-04
Terrestrial eutrophication	mol N eq	1.05E-03	7.85E-05	6.21E-08	1.88E-05	1.14E-03	-2.29E-05	1.12E-03
Freshwater eutrophication	kg P eq	3.77E-06	3.86E-06	4.23E-09	7.85E-08	7.71E-06	-9.87E-07	6.72E-06
Marine eutrophication	kg N eq	1.05E-04	6.52E-06	9.29E-08	1.86E-06	1.14E-04	-1.80E-06	1.12E-04
Freshwater ecotoxicity	CTUe	2.90E-02	7.35E-02	1.88E-05	1.11E-03	1.04E-01	-2.87E-03	1.01E-01
Land use	kg C deficit	7.39E-02	8.44E-03	1.31E-05	1.32E-03	8.37E-02	-2.22E-03	8.15E-02
Water resource depletion	m ³ water eq	9.15E-05	1.21E-05	8.47E-08	8.23E-07	1.05E-04	-5.44E-06	9.91E-05
Mineral, fossil & ren resource depletion	kg Sb eq	1.21E-05	5.06E-07	2.00E-11	5.87E-09	1.26E-05	-4.44E-08	1.25E-05
Cumulative energy demand non renewable	MJ oil eq	9.21E-01	1.30E-01	4.82E-05	2.04E-02	1.07E+00	-3.43E-02	1.04E+00
Cumulative energy demand renewable	MJ oil eq	1.13E-01	2.16E-02	3.60E+00	1.33E-03	3.74E+00	-9.09E-03	3.73E+00
Nuclear waste	m ³ HAA eq	6.70E-11	1.51E-11	1.10E-14	3.33E-12	8.54E-11	-5.43E-12	8.00E-11

The following Fig. A. 17 shows the share per life cycle stage per impact category of electricity production with monocrystalline-Si PV panels integrated in a slanted roof. The potential benefits for recycling are in relation to the sum of the life cycle stages production to end-of-life.

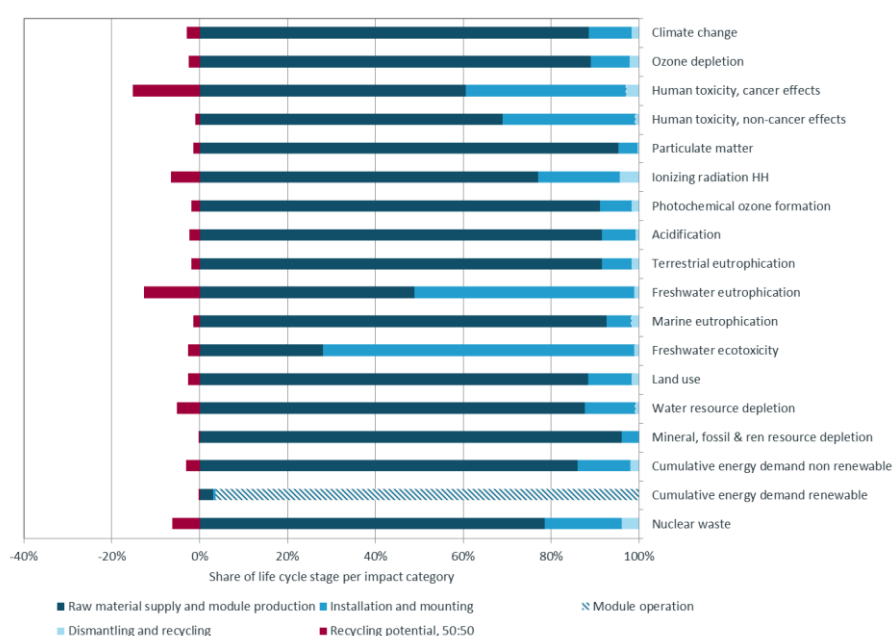


Fig. A. 17 Environmental impact results (characterized, indexed to 100 %) of 1 kWh of electricity produced with a 3 kWp power system with mono-Si PV panels integrated in a slanted roof. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.

Tab. A. 56 lists the environmental impacts per impact category (characterized results) for 1 kWh of electricity produced with monocrystalline-Si PV panels on an open ground system. Fig. A. 18 shows the relative impact per life cycle stage and per impact category.

Tab. A. 56 Environmental impact results (characterized) of 1 kWh of electricity produced with a 570 kWp power system with mono-Si PV panels on an open ground system.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 570 kWp installation, open ground with mono-Si PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change	kg CO ₂ eq	8.00E-02	1.41E-02	3.21E-06	1.43E-03	9.56E-02	-4.83E-03	9.07E-02
Ozone depletion	kg CFC-11 eq	2.12E-09	4.47E-10	1.98E-13	4.87E-11	2.62E-09	-9.25E-11	2.52E-09
Human toxicity, cancer effects	CTUh, c	7.09E-10	1.45E-09	4.87E-13	3.36E-11	2.19E-09	-2.74E-10	1.92E-09
Human toxicity, non-cancer effects	CTUh, n-c	9.12E-09	6.68E-09	7.55E-12	1.01E-10	1.59E-08	1.85E-10	1.61E-08
Particulate matter	kg PM _{2.5} eq	1.26E-04	8.03E-06	1.75E-09	4.36E-07	1.34E-04	-2.75E-06	1.31E-04
Ionizing radiation HH	kBq U235 eq	2.82E-03	1.25E-03	5.94E-07	1.59E-04	4.23E-03	-3.92E-04	3.84E-03
Photochemical ozone formation	kg NMVOC eq	2.96E-04	3.50E-05	1.16E-08	5.21E-06	3.36E-04	-9.77E-06	3.27E-04
Acidification	mol H ⁺ eq	7.42E-04	8.22E-05	2.41E-08	6.40E-06	8.30E-04	-2.43E-05	8.06E-04
Terrestrial eutrophication	mol N eq	1.05E-03	1.57E-04	6.30E-08	1.88E-05	1.22E-03	-3.53E-05	1.19E-03
Freshwater eutrophication	kg P eq	3.77E-06	1.64E-06	4.24E-09	7.85E-08	5.48E-06	-3.79E-07	5.10E-06
Marine eutrophication	kg N eq	1.05E-04	1.07E-05	9.30E-08	1.86E-06	1.18E-04	-2.97E-06	1.15E-04
Freshwater ecotoxicity	CTUe	2.90E-02	3.72E-02	1.89E-05	1.11E-03	6.73E-02	-3.33E-03	6.40E-02
Land use	kg C deficit	7.39E-02	8.26E+00	1.22E-05	1.32E-03	8.34E+00	-3.76E-03	8.33E+00
Water resource depletion	m ³ water eq	9.15E-05	2.16E-05	8.48E-08	8.23E-07	1.14E-04	-8.62E-06	1.05E-04
Mineral, fossil & ren resource depletion	kg Sb eq	1.21E-05	4.69E-07	3.15E-11	5.87E-09	1.25E-05	5.72E-08	1.26E-05
Cumulative energy demand non renewable	MJ oil eq	9.21E-01	2.02E-01	4.91E-05	2.04E-02	1.14E+00	-6.33E-02	1.08E+00
Cumulative energy demand renewable	MJ oil eq	1.13E-01	3.73E-02	3.60E+00	1.33E-03	3.75E+00	-1.42E-02	3.74E+00
Nuclear waste	m ³ HAA eq	6.70E-11	2.73E-11	1.10E-14	3.33E-12	9.76E-11	-8.77E-12	8.88E-11

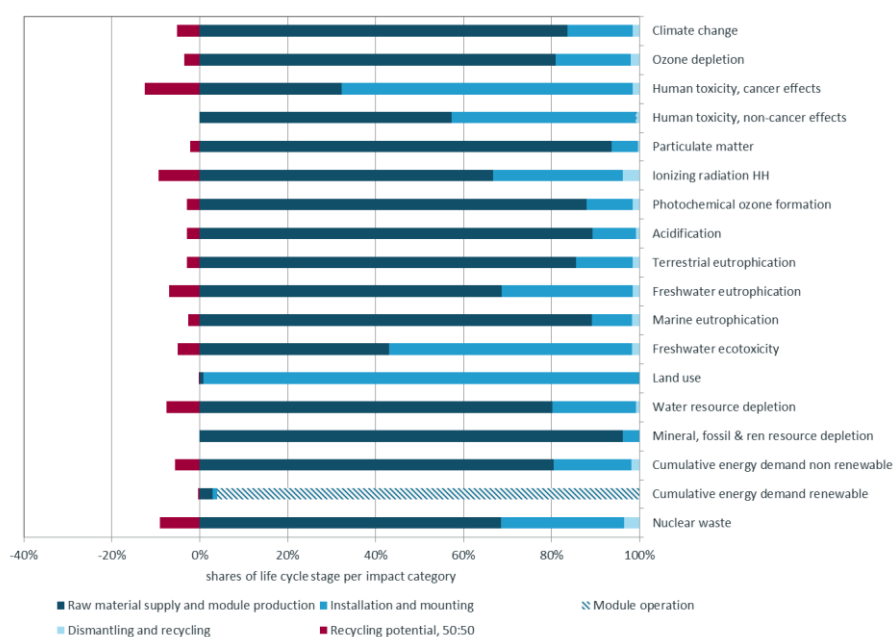


Fig. A. 18 Environmental impact results (characterized, indexed to 100 %) of 1 kWh of electricity produced with a 570 kWp power system with mono-Si PV panels on an open ground system. The potential benefits due to recycling are illustrated relative to the overall environmental impacts from production to end-of-life.

A.5.3. Results normalized

The normalized environmental impact results of the electricity production with monocrystalline-Si PV panels integrated in a slanted roof and on an open ground system are listed in Tab. A. 57 and Tab. A. 58, respectively. The three additional impact categories cumulative energy demand non-renewable, cumulative energy demand renewable and nuclear waste are not subject to normalization.

The results show that the production of 1 kWh electricity contributes the largest share to the mineral, fossil and renewable resource depletion potential of an average European citizen. An average European person consumes around 5'000 kWh of electricity. Assuming it would all be produced with such a 3 kWp monocrystalline-Si PV system or a 570 kWp open ground system, this would sum up to 62 % of the present budget of mineral, fossil and renewable resource depletion.

Tab. A. 57 Environmental impact results (normalized) of 1 kWh of electricity produced with a 3 kWp power system with mono-Si PV panels integrated in a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, integrated with mono-Si PV technology, normalized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change		8.67E-06	9.75E-07	3.39E-10	1.55E-07	9.80E-06	-2.97E-07	9.51E-06
Ozone depletion		9.81E-08	9.87E-09	9.07E-12	2.25E-09	1.10E-07	-2.81E-09	1.07E-07
Human toxicity, cancer effects		1.92E-05	1.16E-05	1.32E-08	9.11E-07	3.18E-05	-4.85E-06	2.69E-05
Human toxicity, non-cancer effects		1.71E-05	7.54E-06	1.42E-08	1.89E-07	2.49E-05	-2.80E-07	2.46E-05
Particulate matter		3.31E-05	1.56E-06	4.29E-10	1.15E-07	3.48E-05	-5.06E-07	3.43E-05
Ionizing radiation HH		2.50E-06	6.10E-07	5.23E-10	1.41E-07	3.25E-06	-2.14E-07	3.04E-06
Photochemical ozone formation		9.35E-06	7.54E-07	3.57E-10	1.64E-07	1.03E-05	-1.95E-07	1.01E-05
Acidification		1.57E-05	1.31E-06	4.94E-10	1.35E-07	1.71E-05	-4.10E-07	1.67E-05
Terrestrial eutrophication		5.95E-06	4.46E-07	3.52E-10	1.07E-07	6.51E-06	-1.30E-07	6.38E-06
Freshwater eutrophication		2.54E-06	2.61E-06	2.86E-09	5.30E-08	5.21E-06	-6.67E-07	4.54E-06
Marine eutrophication		6.23E-06	3.86E-07	5.50E-09	1.10E-07	6.73E-06	-1.07E-07	6.63E-06
Freshwater ecotoxicity		3.32E-06	8.41E-06	2.15E-09	1.27E-07	1.19E-05	-3.28E-07	1.15E-05
Land use		9.88E-07	1.13E-07	1.75E-10	1.76E-08	1.12E-06	-2.97E-08	1.09E-06
Water resource depletion		1.12E-06	1.49E-07	1.04E-09	1.01E-08	1.28E-06	-6.69E-08	1.22E-06
Mineral, fossil & ren resource depletion		1.19E-04	5.01E-06	2.00E-10	5.81E-08	1.24E-04	-4.40E-07	1.24E-04

Tab. A. 58 Environmental impact results (normalized) of 1 kWh of electricity produced with a 570 kWp power system with mono-Si PV panels on an open ground system.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 570 kWp installation, open ground with mono-Si PV technology, normalized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change		8.67E-06	1.53E-06	3.48E-10	1.55E-07	1.04E-05	-5.24E-07	9.84E-06
Ozone depletion		9.81E-08	2.07E-08	9.16E-12	2.25E-09	1.21E-07	-4.28E-09	1.17E-07
Human toxicity, cancer effects		1.92E-05	3.93E-05	1.32E-08	9.11E-07	5.94E-05	-7.42E-06	5.20E-05
Human toxicity, non-cancer effects		1.71E-05	1.25E-05	1.42E-08	1.89E-07	2.99E-05	3.46E-07	3.02E-05
Particulate matter		3.31E-05	2.11E-06	4.59E-10	1.15E-07	3.53E-05	-7.24E-07	3.46E-05
Ionizing radiation HH		2.50E-06	1.10E-06	5.26E-10	1.41E-07	3.74E-06	-3.47E-07	3.40E-06
Photochemical ozone formation		9.35E-06	1.10E-06	3.66E-10	1.64E-07	1.06E-05	-3.08E-07	1.03E-05
Acidification		1.57E-05	1.74E-06	5.09E-10	1.35E-07	1.76E-05	-5.15E-07	1.70E-05
Terrestrial eutrophication		5.95E-06	8.91E-07	3.58E-10	1.07E-07	6.95E-06	-2.01E-07	6.75E-06
Freshwater eutrophication		2.54E-06	1.10E-06	2.86E-09	5.30E-08	3.71E-06	-2.56E-07	3.45E-06
Marine eutrophication		6.23E-06	6.31E-07	5.50E-09	1.10E-07	6.98E-06	-1.76E-07	6.80E-06
Freshwater ecotoxicity		3.32E-06	4.25E-06	2.16E-09	1.27E-07	7.70E-06	-3.81E-07	7.32E-06
Land use		9.88E-07	1.10E-04	1.66E-10	1.76E-08	1.11E-04	-5.02E-08	1.11E-04
Water resource depletion		1.12E-06	2.65E-07	1.04E-09	1.01E-08	1.40E-06	-1.06E-07	1.29E-06
Mineral, fossil & ren resource depletion		1.19E-04	4.64E-06	3.04E-10	5.81E-08	1.24E-04	5.66E-07	1.25E-04

A.5.4. Results weighted

Tab. A. 59 shows the weighted environmental impact results (equal weighting) of the electricity production with monocrystalline-Si PV panels in an integrated roof system whereas Tab. A. 60 shows the results for electricity production on a 570 kWp open ground system. The blue bars within column “Total” illustrate the importance of each impact category in the total of impacts. The three additional

impact categories cumulative energy demand non-renewable, cumulative energy demand renewable and nuclear waste are not included in the weighted results.

The environmental impacts of the production of 1 kWh electricity with a 3 kWp power system with monocrystalline-Si PV panels integrated in a slanted roof, are mainly driven by mineral, fossil and renewable resource depletion (44 %) followed by the impacts of particulate matter emissions (12 %) and human toxicity cancer and non-cancer effects (10 % and 9 %, respectively). The shares of the individual impact categories on the weighted results are illustrated in Fig. A. 19.

Tab. A. 59 Environmental impact results (weighted) of 1 kWh of electricity produced with a 3 kWp power system with mono-Si PV panels integrated in a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, integrated with mono-Si PV technology, weighted		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Total	Pt	1.64E-05	2.77E-06	2.78E-09	1.53E-07	1.93E-05	-5.68E-07	1.87E-05
Climate change	Pt	5.78E-07	6.50E-08	2.26E-11	1.04E-08	6.54E-07	-1.98E-08	6.34E-07
Ozone depletion	Pt	6.54E-09	6.58E-10	6.04E-13	1.50E-10	7.35E-09	-1.88E-10	7.16E-09
Human toxicity, cancer effects	Pt	1.28E-06	7.75E-07	8.78E-10	6.07E-08	2.12E-06	-3.23E-07	1.79E-06
Human toxicity, non-cancer effects	Pt	1.14E-06	5.03E-07	9.44E-10	1.26E-08	1.66E-06	-1.87E-08	1.64E-06
Particulate matter	Pt	2.21E-06	1.04E-07	2.86E-11	7.64E-09	2.32E-06	-3.37E-08	2.28E-06
Ionizing radiation HH	Pt	1.67E-07	4.07E-08	3.49E-11	9.41E-09	2.17E-07	-1.43E-08	2.02E-07
Photochemical ozone formation	Pt	6.23E-07	5.03E-08	2.38E-11	1.10E-08	6.84E-07	-1.30E-08	6.71E-07
Acidification	Pt	1.05E-06	8.75E-08	3.30E-11	9.02E-09	1.14E-06	-2.73E-08	1.11E-06
Terrestrial eutrophication	Pt	3.97E-07	2.97E-08	2.35E-11	7.12E-09	4.34E-07	-8.68E-09	4.25E-07
Freshwater eutrophication	Pt	1.70E-07	1.74E-07	1.91E-10	3.54E-09	3.47E-07	-4.44E-08	3.03E-07
Marine eutrophication	Pt	4.15E-07	2.57E-08	3.67E-10	7.32E-09	4.49E-07	-7.11E-09	4.42E-07
Freshwater ecotoxicity	Pt	2.21E-07	5.60E-07	1.43E-10	8.50E-09	7.90E-07	-2.19E-08	7.68E-07
Land use	Pt	6.59E-08	7.53E-09	1.17E-11	1.17E-09	7.46E-08	-1.98E-09	7.26E-08
Water resource depletion	Pt	7.49E-08	9.95E-09	6.94E-11	6.74E-10	8.56E-08	-4.46E-09	8.12E-08
Mineral, fossil & ren resource depletion	Pt	7.96E-06	3.34E-07	1.32E-11	3.87E-09	8.30E-06	-2.93E-08	8.27E-06

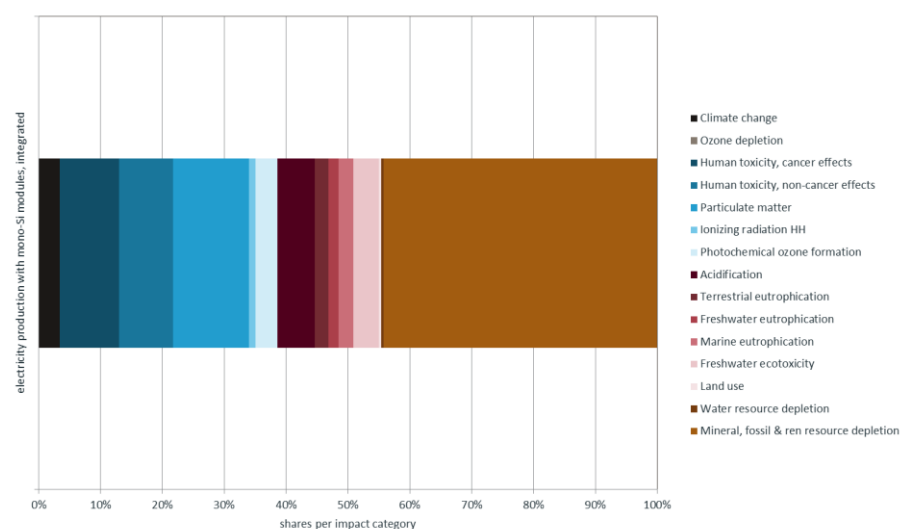


Fig. A. 19 Share per impact category on the weighted result for the production of 1 kWh of electricity produced with a 3 kWp power system with mono-Si PV panels integrated in a slanted roof.

The electricity production with monocrystalline-Si PV panels on an open ground system has its main impact in the mineral, fossil and renewable resource depletion potential (30 %), followed by land use (26 %), human toxicity cancer effects (12 %), particulate matter emissions (8 %) and human toxicity non cancer effects (7 %). Fig. A. 20 illustrates this.

Tab. A. 60 Environmental impact results (weighted) of 1 kWh of electricity produced with a 570 kWp power system with mono-Si PV panels on an open ground system.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 570 kWp installation, open ground with mono-Si PV technology, weighted		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Total	Pt	1.64E-05	1.21E-05	2.80E-09	1.53E-07	2.86E-05	-6.74E-07	2.79E-05
Climate change	Pt	5.78E-07	1.02E-07	2.32E-11	1.04E-08	6.91E-07	-3.49E-08	6.56E-07
Ozone depletion	Pt	6.54E-09	1.38E-09	6.11E-13	1.50E-10	8.07E-09	-2.85E-10	7.79E-09
Human toxicity, cancer effects	Pt	1.28E-06	2.62E-06	8.80E-10	6.07E-08	3.96E-06	-4.95E-07	3.47E-06
Human toxicity, non-cancer effects	Pt	1.14E-06	8.35E-07	9.45E-10	1.26E-08	1.99E-06	2.31E-08	2.01E-06
Particulate matter	Pt	2.21E-06	1.41E-07	3.07E-11	7.64E-09	2.35E-06	-4.83E-08	2.31E-06
Ionizing radiation HH	Pt	1.67E-07	7.35E-08	3.50E-11	9.41E-09	2.50E-07	-2.31E-08	2.26E-07
Photochemical ozone formation	Pt	6.23E-07	7.36E-08	2.44E-11	1.10E-08	7.08E-07	-2.05E-08	6.87E-07
Acidification	Pt	1.05E-06	1.16E-07	3.39E-11	9.02E-09	1.17E-06	-3.43E-08	1.14E-06
Terrestrial eutrophication	Pt	3.97E-07	5.94E-08	2.39E-11	7.12E-09	4.63E-07	-1.34E-08	4.50E-07
Freshwater eutrophication	Pt	1.70E-07	7.36E-08	1.91E-10	3.54E-09	2.47E-07	-1.71E-08	2.30E-07
Marine eutrophication	Pt	4.15E-07	4.21E-08	3.67E-10	7.32E-09	4.65E-07	-1.17E-08	4.53E-07
Freshwater ecotoxicity	Pt	2.21E-07	2.83E-07	1.44E-10	8.50E-09	5.13E-07	-2.54E-08	4.88E-07
Land use	Pt	6.59E-08	7.36E-06	1.08E-11	1.17E-09	7.43E-06	-3.35E-09	7.43E-06
Water resource depletion	Pt	7.49E-08	1.77E-08	6.94E-11	6.74E-10	9.33E-08	-7.06E-09	8.63E-08
Mineral, fossil & ren resource depletion	Pt	7.96E-06	3.10E-07	2.06E-11	3.87E-09	8.28E-06	3.77E-08	8.31E-06

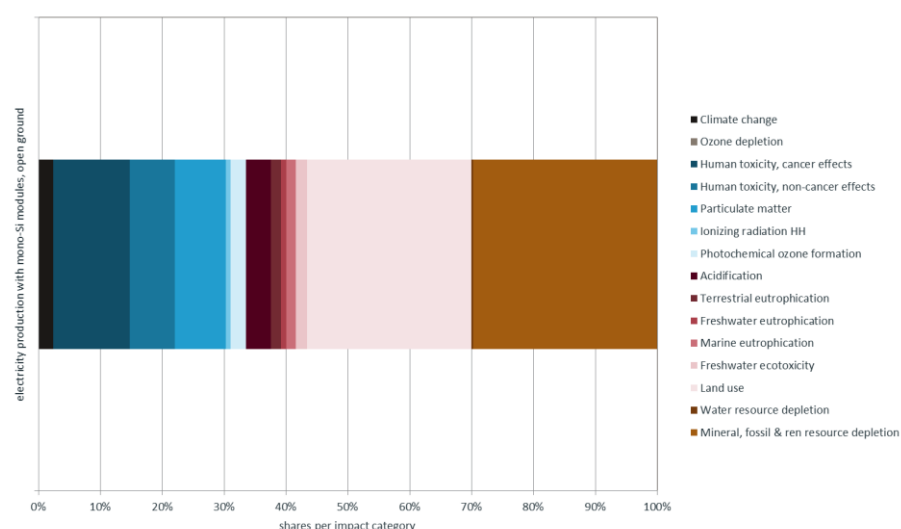


Fig. A. 20 Share per impact category on the weighted result for the production of 1 kWh of electricity produced with a 570 kWp power system with mono-Si PV panels on an open ground system.

A.6. Results prior to normalization of electricity production with average PV module

The following Tab. A. 61 shows the shares per impact category per PV technology of the electricity production with the average PV module. All life cycle stages are included. Tab. A. 62 shows the environmental impacts (characterized) per life cycle stage of the electricity generated by a 3 kWp residential scale power system with average PV modules.

Tab. A. 61 Environmental impact results (characterized per technology) of 1 kWh of electricity produced with a 3 kWp power system with average PV panels mounted on a slanted roof.

Impact category	Unit per kWh electricity	Total all life cycle stages, benefits for recycling included					Total
electricity, 3 kWp installation, mounted with average PV technology, characterized		CdTe	CIS	micro-Si	multi-Si	mono-Si	Average PV panels
Climate change	kg CO ₂ eq	1.38E-03	1.29E-03	2.33E-03	2.39E-02	3.54E-02	6.43E-02
Ozone depletion	kg CFC-11 eq	5.90E-11	5.97E-11	5.00E-11	7.77E-10	9.55E-10	1.90E-09
Human toxicity, cancer effects	CTUh, c	4.16E-11	3.53E-11	5.07E-11	4.73E-10	4.88E-10	1.09E-09
Human toxicity, non-cancer effects	CTUh, n-c	3.27E-10	3.21E-10	4.48E-10	4.74E-09	5.44E-09	1.13E-08
Particulate matter	kg PM _{2.5} eq	7.04E-07	5.34E-07	2.52E-06	2.98E-05	5.27E-05	8.63E-05
Ionizing radiation HH	kBq U235 eq	6.44E-05	1.10E-04	1.02E-04	1.18E-03	1.43E-03	2.89E-03
Photochemical ozone formation	kg NMVOC eq	5.49E-06	3.91E-06	9.04E-06	8.57E-05	1.29E-04	2.34E-04
Acidification	mol H+ eq	1.07E-05	7.22E-06	1.94E-05	2.07E-04	3.21E-04	5.66E-04
Terrestrial eutrophication	mol N eq	2.05E-05	1.43E-05	3.28E-05	3.06E-04	4.56E-04	8.30E-04
Freshwater eutrophication	kg P eq	2.26E-07	1.98E-07	2.28E-07	2.77E-06	2.74E-06	6.16E-06
Marine eutrophication	kg N eq	1.84E-06	1.25E-06	2.94E-06	2.82E-05	4.55E-05	7.97E-05
Freshwater ecotoxicity	CTUe	4.92E-03	3.11E-03	4.11E-03	4.40E-02	4.13E-02	9.75E-02
Land use	kg C deficit	1.65E-03	1.87E-03	2.11E-03	2.92E-02	3.32E-02	6.80E-02
Water resource depletion	m ³ water eq	9.98E-07	1.21E-06	3.47E-06	2.56E-05	4.08E-05	7.21E-05
Mineral, fossil & ren resource depletion	kg Sb eq	2.69E-07	1.80E-05	6.85E-08	8.71E-06	5.08E-06	3.21E-05
Cumulative energy demand non renewable	MJ oil eq	2.01E-02	1.97E-02	2.67E-02	2.99E-01	4.19E-01	7.84E-01
Cumulative energy demand renewable	MJ oil eq	2.29E-01	1.29E-01	1.65E-01	1.67E+00	1.51E+00	3.70E+00
Nuclear waste	m ³ HAA eq	1.50E-12	2.59E-12	2.23E-12	2.69E-11	3.32E-11	6.64E-11

Tab. A. 62 Environmental impact results (characterized) of 1 kWh of electricity produced with a 3 kWp power system with average PV panels mounted on a slanted roof.

Impact category	Unit per kWh electricity	Product stage	Construction stage	Use stage	End-of-life stage	Total, recycling benefits excluded	Potential benefits due to recycling	Total, recycling benefits included
electricity, 3 kWp installation, mounted with average PV technology, characterized		Raw material supply and module production	Installation and mounting	Module operation	Dismantling and recycling	Total all life stages, recycling benefits excluded	Recycling potential, 50:50	Total all life cycle stages, benefits for recycling included
Climate change	kg CO ₂ eq	5.64E-02	9.91E-03	3.33E-03	1.52E-03	6.78E-02	-3.54E-03	6.43E-02
Ozone depletion	kg CFC-11 eq	1.65E-09	2.76E-10	2.07E-13	5.16E-11	1.98E-09	-7.57E-11	1.90E-09
Human toxicity, cancer effects	CTUh, c	5.64E-10	7.10E-10	5.12E-13	3.55E-11	1.31E-09	-2.22E-10	1.09E-09
Human toxicity, non-cancer effects	CTUh, n-c	6.86E-09	4.41E-09	7.96E-12	1.06E-10	1.14E-08	-1.10E-10	1.13E-08
Particulate matter	kg PM _{2.5} eq	8.16E-05	6.65E-06	1.77E-09	4.55E-07	8.87E-05	-2.36E-06	8.63E-05
Ionizing radiation HH	kBq U235 eq	2.15E-03	8.75E-04	6.24E-07	1.64E-04	3.19E-03	-3.05E-04	2.89E-03
Photochemical ozone formation	kg NMVOC eq	2.09E-04	2.69E-05	1.20E-08	5.49E-06	2.41E-04	-7.74E-06	2.34E-04
Acidification	mol H+ eq	5.12E-04	7.04E-05	2.49E-08	6.69E-06	5.89E-04	-2.31E-05	5.66E-04
Terrestrial eutrophication	mol N eq	7.47E-04	9.16E-05	6.57E-08	1.98E-05	8.58E-04	-2.86E-05	8.30E-04
Freshwater eutrophication	kg P eq	3.13E-06	3.95E-06	4.46E-09	8.56E-08	7.17E-06	-1.01E-06	6.16E-06
Marine eutrophication	kg N eq	7.23E-05	7.67E-06	9.80E-08	1.96E-06	8.20E-05	-2.28E-06	7.97E-05
Freshwater ecotoxicity	CTUe	2.40E-02	7.57E-02	1.98E-05	1.18E-03	1.01E-01	-3.37E-03	9.75E-02
Land use	kg C deficit	5.95E-02	9.92E-03	1.38E-05	1.39E-03	7.08E-02	-2.84E-03	6.80E-02
Water resource depletion	m ³ water eq	6.22E-05	1.57E-05	8.93E-08	8.85E-07	7.89E-05	-6.81E-06	7.21E-05
Mineral, fossil & ren resource depletion	kg Sb eq	3.18E-05	5.17E-07	2.29E-12	6.20E-09	3.23E-05	-2.08E-07	3.21E-05
Cumulative energy demand non renewable	MJ oil eq	6.65E-01	1.42E-01	5.12E-05	2.16E-02	8.29E-01	-4.47E-02	7.84E-01
Cumulative energy demand renewable	MJ oil eq	8.19E-02	2.79E-02	3.60E+00	1.39E-03	3.71E+00	-1.13E-02	3.70E+00
Nuclear waste	m ³ HAA eq	5.06E-11	1.92E-11	1.16E-14	3.45E-12	7.33E-11	-6.82E-12	6.64E-11

A.7. Environmental impacts of waste material disposal

The cradle to gate environmental impacts of the disposal of waste materials are neglected in the end-of-life stage of PV electricity. A fraction of less than 1 % of the total environmental impacts per kWh of DC electricity produced is caused by the disposal of waste materials. It is assumed that copper telluride and cadmium sludge are disposed of in underground deposits, whereas the remaining materials (flat glass, solar glass, copper, aluminium, AlMg₃ alloy and steel) are landfilled as inert materials. The relative significance of the disposal of waste materials is highest in the case of a CdTe PV system mounted on a slanted roof among the PV systems analysed. The characterized results of

the disposal of waste materials in relation to the environmental impacts of 1 kWh electricity generated by a mounted CdTe PV system are compiled in Tab. A. 63.

Tab. A. 63 Environmental impact results (characterized, shown as percentages of the characterized environmental impacts of 1 kWh electricity generated) of the disposal of waste materials for a 3 kWp power system with CdTe PV panels mounted on a slanted roof.

Burdens from waste disposal relative to total environmental impacts of electricity production - CdTe mounted on roof 3kWp - characterized		Flat Glass	Solar Glass	Copper	Aluminium	AlMg3 alloy	Steel	Copper telluride	Cadmium
		Inert material landfill	Inert material landfill	Inert material landfill	Inert material landfill	Inert material landfill	Inert material landfill	Underground deposit	Underground deposit
Climate change	%	0.031%	0.032%	0.001%	0.007%	0.000%	0.003%	0.003%	0.003%
Ozone depletion	%	0.071%	0.072%	0.002%	0.015%	0.000%	0.008%	0.001%	0.001%
Human toxicity, cancer effects	%	0.009%	0.010%	0.000%	0.002%	0.000%	0.001%	0.005%	0.004%
Human toxicity, non-cancer effects	%	0.008%	0.008%	0.000%	0.002%	0.000%	0.001%	0.002%	0.001%
Particulate matter	%	0.045%	0.046%	0.001%	0.010%	0.000%	0.005%	0.003%	0.003%
Ionizing radiation HH	%	0.019%	0.019%	0.000%	0.004%	0.000%	0.002%	0.002%	0.002%
Photochemical ozone formation	%	0.091%	0.093%	0.002%	0.020%	0.000%	0.010%	0.002%	0.002%
Acidification	%	0.033%	0.034%	0.001%	0.007%	0.000%	0.004%	0.002%	0.002%
Terrestrial eutrophication	%	0.082%	0.084%	0.002%	0.018%	0.000%	0.009%	0.002%	0.002%
Freshwater eutrophication	%	0.002%	0.002%	0.000%	0.000%	0.000%	0.000%	0.001%	0.001%
Marine eutrophication	%	0.085%	0.086%	0.002%	0.018%	0.000%	0.009%	0.002%	0.002%
Freshwater ecotoxicity	%	0.004%	0.004%	0.000%	0.001%	0.000%	0.000%	0.001%	0.001%
Land use	%	0.667%	0.681%	0.018%	0.143%	0.000%	0.072%	0.008%	0.007%
Water resource depletion	%	0.017%	0.017%	0.000%	0.004%	0.000%	0.002%	0.001%	0.001%
Mineral, fossil & ren resource depletion	%	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%
Cumulative energy demand non renewable	%	0.060%	0.062%	0.002%	0.013%	0.000%	0.007%	0.003%	0.003%
Cumulative energy demand renewable	%	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%
Nuclear waste	%	0.014%	0.015%	0.000%	0.003%	0.000%	0.002%	0.002%	0.001%